



CCNAv7: Introduction to Network (ITN)

Companion Guide



Introduction to Networks Companion Guide (CCNAv7)

Cisco Networking Academy

Cisco Press

Introduction to Networks Companion Guide (CCNAv7)

Cisco Networking Academy

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Command Syntax Conventions

The conventions used to present command syntax in this book are the same conventions used in the IOS Command Reference. The Command Reference describes these conventions as follows:

- **Boldface** indicates commands and keywords that are entered literally as shown. In actual configuration examples and output (not general command syntax), boldface indicates commands that are manually input by the user (such as a **show** command).
- *Italic* indicates arguments for which you supply actual values.
- Vertical bars (|) separate alternative, mutually exclusive elements.
- Square brackets ([]) indicate an optional element.
- Braces ({ }) indicate a required choice.
- Braces within brackets ([{ }]) indicate a required choice within an optional element.

Introduction

Introduction to Networks Companion Guide (CCNAv7) is the official supplemental textbook for the Cisco Network Academy CCNA Introduction to Networks Version 7 course. Cisco Networking Academy is a comprehensive program that delivers information technology skills to students around the world. The curriculum emphasizes real-world practical application and provides opportunities to gain the skills and hands-on experience needed to design, install, operate, and maintain networks in small business, medium-sized business as well as enterprise and service provider environments.

This book provides a ready reference that explains the same networking concepts, technologies, protocols, and devices as the online curriculum. This book emphasizes key topics, terms, and activities and provides some alternative explanations and examples to supplement the course. You can use the online curriculum as directed by your instructor and then use this *Companion Guide*'s study tools to help solidify your understanding of all the topics.

Who Should Read This Book

The book, like the course it accompanies, is designed as an introduction to data network technology for those pursuing careers as network professionals as well as those who need an introduction to network technology for professional growth. Topics are presented concisely, starting with the most fundamental concepts and progressing to a comprehensive understanding of network communication. The content of this text provides the foundation for additional Cisco Networking Academy courses and preparation for the CCNA certification.

Book Features

The educational features of this book focus on supporting topic coverage, readability, and practice of the course material to facilitate your full understanding of the course material.

Topic Coverage

The following list gives you a thorough overview of the features provided in each chapter so that you can make constructive use of your study time:

- **Objectives:** Listed at the beginning of each chapter, the objectives reference the core concepts covered in the chapter. The objectives match the objectives stated in the corresponding chapters of the online curriculum; however, the question format in the *Companion Guide* encourages you to think about finding the answers as you read the chapter.

- **Notes:** These are short sidebars that point out interesting facts, timesaving methods, and important safety issues.
- **Summary:** At the end of each chapter is a summary of the chapter's key concepts. It provides a synopsis of the chapter and serves as a study aid.
- **Practice:** At the end of chapter is a full list of all the labs, class activities, and Packet Tracer activities to refer to at study time.

Readability

The following features are provided to help you understand networking vocabulary:

- **Key terms:** Each chapter begins with a list of key terms, along with a page-number reference to find the term used inside the chapter. The terms are listed in the order in which they are explained in the chapter. This handy reference allows you to find a term, flip to the page where the term appears, and see the term used in context. The Key Terms Glossary defines all the key terms.
- **Key Terms Glossary:** This book contains an all-new Key Terms Glossary that defines more than 1000 terms.

Practice

Practice makes perfect. This *Companion Guide* offers you ample opportunities to put what you learn into practice. You will find the following features valuable and effective in reinforcing the instruction that you receive:

- **Check Your Understanding questions and answer key:** Review questions are presented at the end of each chapter as a self-assessment tool. These questions match the style of questions in the online course. Appendix A, “Answers to ‘Check Your Understanding’ Questions,” provides an answer key to all the questions and includes an explanation of each answer.
- **Labs and activities:** Throughout each chapter, you are directed back to the online course to take advantage of the activities provided to reinforce concepts. In addition, at the end of each chapter is a “Practice” section that lists all the labs and activities to provide practice with the topics introduced in this chapter.
- **Page references to online course:** After most headings is a number in parentheses—for example, (1.1.2). This number refers to the page number in the online course so that you can easily jump to that spot online to view a video, practice an activity, perform a lab, or review a topic.



Interactive
Graphic

Video



About Packet Tracer Software and Activities

Interspersed throughout the chapters, you'll find a few Cisco Packet Tracer activities. Packet Tracer allows you to create networks, visualize how packets flow in a network, and use basic testing tools to determine whether a network would work. When you see this icon, you can use Packet Tracer with the listed file to perform a task suggested in this book. The activity files are available in the online course. Packet Tracer software is available only through the Cisco Networking Academy website. Ask your instructor for access to Packet Tracer.

How This Book Is Organized

This book corresponds closely to the Cisco Networking Academy CCNA IT Essential v7 course and is divided into 17 chapters, one appendix, and a glossary of key terms:

- **Chapter 1, “Networking Today”:** This chapter introduces the concept of a network and provides an overview of the different types of networks encountered. It examines how networks impact the way we work, learn, and play. This chapter also examines recent trends in networks, such as video, cloud computing, and BYOD and how to help ensure robust, reliable, secure networks to support these trends.
- **Chapter 2, “Basic Switch and End Device Configuration”:** This chapter introduces the operating system used with most Cisco devices: Cisco IOS. The basic purpose and functions of IOS are described, as are methods to access IOS. The chapter also describes how to maneuver through the IOS command-line interface as well as basic IOS device configuration.
- **Chapter 3, “Protocols and Models”:** This chapter examines the importance of rules or protocols for network communication. It explores the OSI reference model and the TCP/IP communication suite and examines how these models provide the necessary protocols to allow communication to occur on a modern converged network.
- **Chapter 4, “Physical Layer”:** This chapter introduces the lowest layer of the OSI model: the physical layer. This chapter explains the transmission of bits over the physical medium.
- **Chapter 5, “Number Systems”:** This chapter explains how to convert between decimal, binary, and hexadecimal number systems. Understanding these number systems is essential to understanding IPv4, IPv6, and Ethernet MAC addressing.

- **Chapter 6, “Data Link Layer”:** This chapter discusses how the data link layer prepares network layer packets for transmission, controls access to the physical media, and transports data across various media. This chapter includes a description of the encapsulation protocols and processes that occur as data travels across the LAN and the WAN.
- **Chapter 7, “Ethernet Switching”:** This chapter examines the functionality of the Ethernet LAN protocols. It explores how Ethernet functions, including how devices use Ethernet MAC addresses to communicate in a multiaccess network. The chapter discusses how Ethernet switches build MAC address tables and forward Ethernet frames.
- **Chapter 8, “Network Layer”:** This chapter introduces the function of the network layer—routing—and the basic device that performs this function—the router. It presents important routing concepts related to addressing, path determination, and data packets for both IPv4 and IPv6. The chapter also introduces how routers perform packet forwarding, static and dynamic routing, and the IP routing table.
- **Chapter 9, “Address Resolution”:** This chapter discusses how host computers and other end devices determine the Ethernet MAC address for a known IPv4 or IPv6 address. This chapter examines the ARP protocol for IPv4 address resolution and the Neighbor Discovery Protocol for IPv6.
- **Chapter 10, “Basic Router Configuration”:** This chapter explains how to configure a Cisco router, including IPv4 and IPv6 addressing on an interface.
- **Chapter 11, “IPv4 Addressing”:** This chapter focuses on IPv4 network addressing, including the types of addresses and address assignment. It describes how to use subnet masks to determine the number of subnetworks and hosts in a network. It examines how to improve network performance by optimally dividing the IPv4 address space based on network requirements. It explores the calculation of valid host addresses and the determination of both subnet and broadcast addresses.
- **Chapter 12, “IPv6 Addressing”:** This chapter focuses on IPv6 network addressing, including IPv6 address representation, types of addresses, and the structure of different types of IPv6 address. The chapter introduces the different methods that an end device can receive an IPv6 address automatically.
- **Chapter 13, “ICMP”:** This chapter introduces Internet Control Message Protocol (ICMP) tools, such as **ping** and **trace**.

- **Chapter 14, “Transport Layer”:** This chapter introduces Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) and examines how each of these protocols transports information across the network. It explores how TCP uses segmentation, the three-way handshake, and expectational acknowledgments to ensure reliable delivery of data. It also examines the best-effort delivery mechanism provided by UDP and describes when its use would be preferred over the use of TCP.
- **Chapter 15, “Application Layer”:** This chapter introduces some protocols of the TCP/IP application layer, which also relates to the top three layers of the OSI model. The chapter focuses on the role of the application layer and how the applications, services, and protocols in the application layer make robust communication across data networks possible. This will be demonstrated by examining some key protocols and services, including HTTP, HTTPS, DNS, DHCP, SMTP/POP, and FTP.
- **Chapter 16, “Network Security Fundamentals”:** This chapter introduces network security threats and vulnerabilities. Various network attacks and mitigation techniques are discussed, along with how to secure network devices.
- **Chapter 17, “Build a Small Network”:** This chapter reexamines the various components in a small network and describes how they work together to allow network growth. It examines network configuration and troubleshooting issues, along with different troubleshooting methodologies.
- **Appendix A, “Answers to ‘Check Your Understanding’ Questions”:** This appendix lists the answers to the “Check Your Understanding” review questions that are included at the end of each chapter.
- **Key Terms Glossary:** The Key Terms Glossary provides definitions for all the key terms identified in each chapter.

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Networking Today

Objectives

Upon completion of this chapter, you will be able to answer the following questions:

- How do networks affect our daily lives?
- How are host and network devices used?
- What are network representations, and how are they used in network topologies?
- What are the characteristics of common types of networks?
- How do LANs and WANs interconnect to the internet?
- What are the four basic requirements of a reliable network?
- How do trends such as BYOD, online collaboration, video, and cloud computing change the way we interact?
- What are some basic security threats and solutions for all networks?
- What employment opportunities are available in the networking field?

Key Terms

This chapter uses the following key terms. You can find the definitions in the glossary at the end of the book.

server page 4

client page 4

end device page 6

intermediary device page 6

topology page 10

small office and home office (SOHO) networks page 12

local-area networks (LANs) page 13

wide-area networks (WANs) page 13

internet page 15

intranet page 16

extranet page 16

internet service provider (ISP) page 17

digital subscriber line (DSL) page 18

cellular connection page 18

satellite connection page 19

dialup telephone connection page 19

converged data network page 21

fault-tolerant network page 24

scalable network page 24

quality of service (QoS) page 25

confidentiality page 27

integrity page 27

availability page 27

bring your own device (BYOD) page 28

cloud computing page 29

powerline networking page 31

*wireless internet service provider
(WISP) page 32*

Introduction (1.0)

Congratulations! This chapter starts you on your path to a successful career in information technology by giving you a foundational understanding of the creation, operation, and maintenance of networks. As a bonus, you get to dive into networking simulations using Packet Tracer. We promise you will really enjoy it!

Networks Affect Our Lives (1.1)

Networks are all around us. They provide us with a way to communicate and share information and resources with individuals in the same location or around the world. Networks require an extensive array of technologies and procedures that can readily adapt to varying conditions and requirements.

Networks Connect Us (1.1.1)

Among all of the essentials for human existence, the need to interact with others ranks just below our need to sustain life. Communication is almost as important to us as our reliance on air, water, food, and shelter.

In today's world, through the use of networks, we are connected as never before. People with ideas can communicate instantly with others to make those ideas reality. News events and discoveries are known worldwide in seconds. Individuals can even connect and play games with friends physically separated by oceans and continents.

Video

Video—The Cisco Networking Academy Learning Experience (1.1.2)

World changers aren't born. They are made. Since 1997 Cisco Networking Academy has been working toward a single goal: educating and building the skills of the next generation of talent required for the digital economy. Refer to the online course to view this video.

No Boundaries (1.1.3)

Advancements in networking technologies are perhaps the most significant changes in the world today. They are helping to create a world in which national borders, geographic distances, and physical limitations become less relevant and present ever-diminishing obstacles.

The internet has changed the manner in which our social, commercial, political, and personal interactions occur. The immediate nature of communications over the internet encourages the creation of global communities. Global communities allow for social interaction that is independent of location or time zone.

The creation of online communities for the exchange of ideas and information has the potential to increase productivity opportunities around the globe.

The cloud lets us store documents and pictures and access them anywhere, anytime. So whether we are on a train, in a park, or standing on top of a mountain, we can seamlessly access our cloud-stored data and applications on any device.

Network Components (1.2)

Many different components are required to enable a network to provide services and resources. These various components work together to ensure that resources are delivered in an efficient manner to those requiring the services.

Host Roles (1.2.1)

If you want to be part of a global online community, your computer, tablet, or smart-phone must first be connected to a network. That network must be connected to the internet. This section discusses the parts of a network. See if you recognize these components in your own home or school network!

Any computer that is connected to a network and that participates directly in network communication is classified as a host. Hosts can be called end devices. Some hosts are also called clients. However, the term *host* specifically refers to a device on a network that is assigned a number for communication purposes. This number, which identifies the host within the particular network, is called the Internet Protocol (IP) address. An IP address identifies the host and the network to which the host is attached.

Servers are computers with software that allows them to provide information, such as email or web pages, to other end devices on the network. Each service requires separate server software. For example, a server requires web server software in order to provide web services to the network. A computer with server software can simultaneously provide services to many different clients.

As mentioned earlier, a client is a type of host. **Clients** have software for requesting and displaying the information obtained from the server, as shown in Figure 1-1.



Figure 1-1 A Client and a Server

An example of client software is a web browser, such as Chrome or Firefox. A single computer can also run multiple types of client software. For example, a user can

check email and view a web page while instant messaging and listening to an audio stream. Table 1-1 lists three common types of server software.

Table 1-1 Common Server Software

Software Type	Description
Email	An email server runs email server software. Clients use mail client software, such as Microsoft Outlook, to access email on the server.
Web	A web server runs web server software. Clients use browser software, such as Windows Internet Explorer, to access web pages on the server.
File	A file server stores corporate and user files in a central location. The client devices access these files with client software such as Windows File Explorer.

Peer-to-Peer (1.2.2)

Client and server software usually run on separate computers, but it is also possible for one computer to be used for both roles at the same time. In small businesses and homes, many computers function as both servers and clients on the network. This type of network, called a peer-to-peer network, is shown in Figure 1-2.

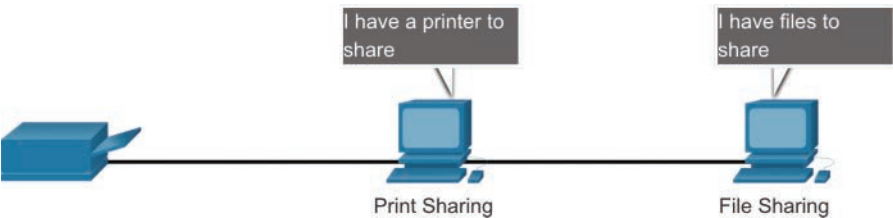


Figure 1-2 Peer-to-Peer Network

Table 1-2 outlines the advantages and disadvantages of peer-to-peer networking.

Table 1-2 Peer-to-Peer Networking Advantages and Disadvantages

Advantages	Disadvantages
Easy to set up	No centralized administration
Less complex	Not as secure
Lower cost because network devices and dedicated servers may not be required	Not scalable
Can be used for simple tasks such as transferring files and sharing printers	All devices may act as both clients and servers, which can slow their performance

End Devices (1.2.3)

The network devices that people are most familiar with are end devices. To distinguish one end device from another, each end device on a network has an address. When an *end device* initiates communication, it uses the address of the destination end device to specify where to deliver the message.

An end device is either the source or destination of a message transmitted over the network, as shown in Figure 1-3.

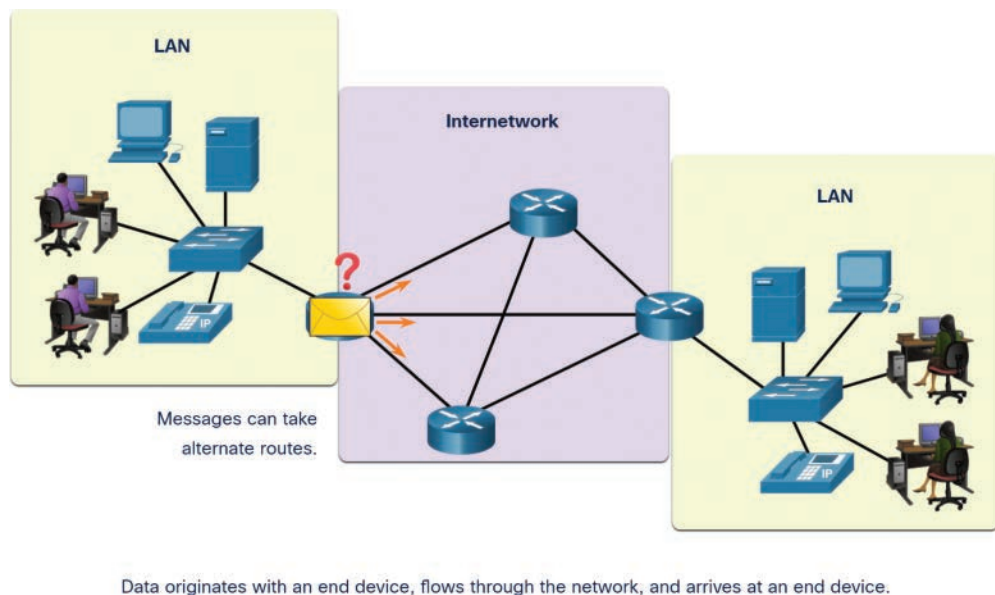


Figure 1-3 Data Flow Through a Network

Intermediary Devices (1.2.4)

Intermediary devices connect individual end devices to a network. They can connect multiple individual networks to form an internetwork. These intermediary devices provide connectivity and ensure that data flows across the network.

Intermediary devices use the destination end device address, in conjunction with information about the network interconnections, to determine the path that messages should take through the network. Figure 1-4 shows examples of the most common intermediary devices.

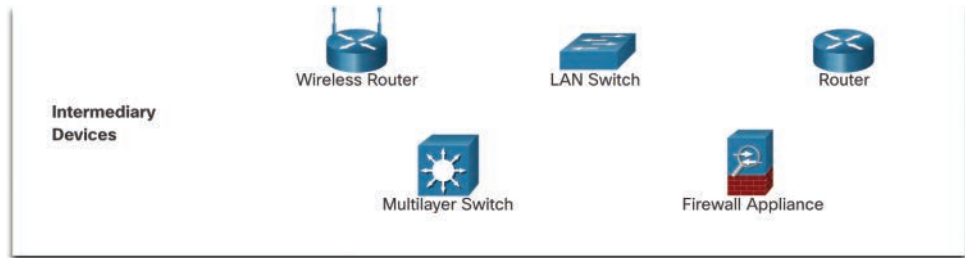


Figure 1-4 Intermediary Devices

Intermediary network devices perform some or all of these functions:

- Regenerate and retransmit communication signals
- Maintain information about what pathways exist through the network and internetwork
- Notify other devices about errors and communication failures
- Direct data along alternate pathways when there is a link failure
- Classify and direct messages according to priorities
- Permit or deny the flow of data, based on security settings

Note

Figure 1-4 does not show any legacy Ethernet hubs. An Ethernet hub is also known as a multiport repeater. Repeaters regenerate and retransmit communication signals. Notice that every intermediary device performs the function of a repeater.

Network Media (1.2.5)

Communication transmits across a network on media. The media provide the channel over which a message travels from source to destination.

Modern networks primarily use three types of media to interconnect devices, as shown in Figure 1-5:

- **Metal wires within cables:** Data is encoded into electrical impulses.
- **Glass or plastic fibers within cables (fiber-optic cable):** Data is encoded into pulses of light.
- **Wireless transmission:** Data is encoded via modulation of specific frequencies of electromagnetic waves.

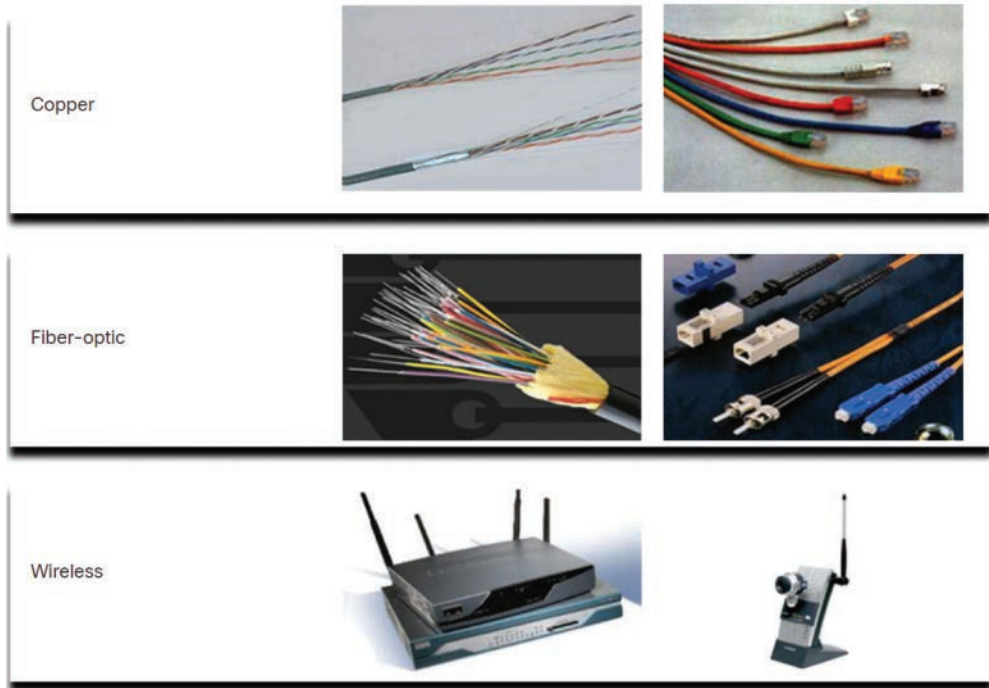


Figure 1-5 Network Media

Different types of network media have different features and benefits. Not all network media have the same characteristics, and they are not all appropriate for the same purpose.

**Interactive
Graphic**

Check Your Understanding—Network Components (1.2.6)

Refer to the online course to complete this activity.

Network Representations and Topologies (1.3)

A network's infrastructure is documented using commonly used symbols to represent devices and different types of diagrams to represent the interconnection of these devices in the network. Understanding these symbols and diagrams is an important aspect of understanding network communications.

Network Representations (1.3.1)

Network architects and administrators must be able to show what their networks look like. They need to be able to easily see which components connect to other components, where they are located, and how they are connected. Diagrams of networks

often use symbols, like those shown in Figure 1-6, to represent the different devices and connections in a network.

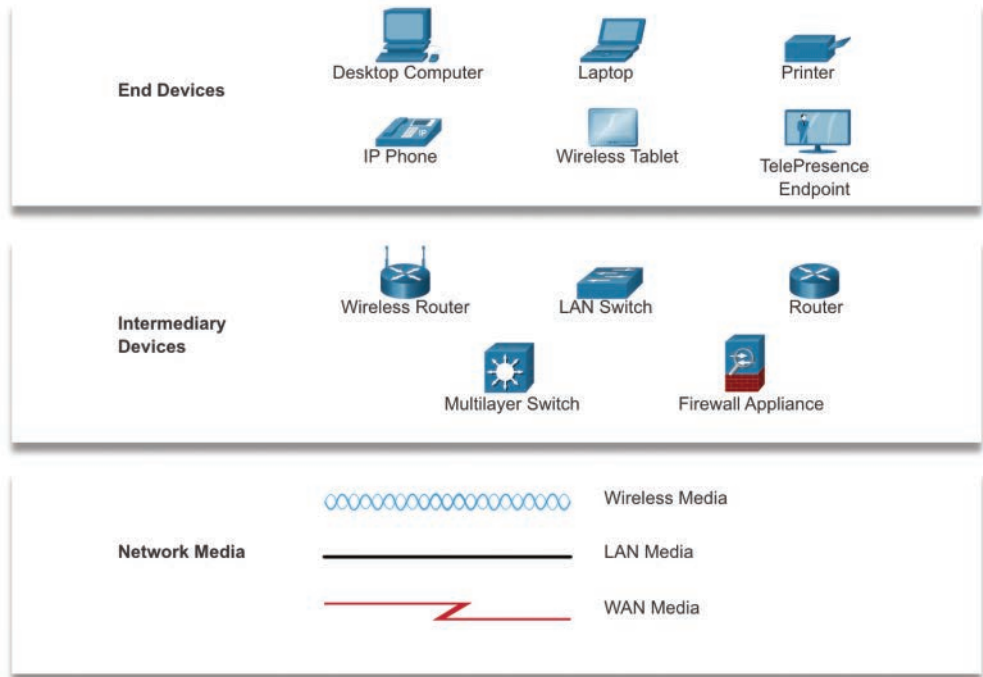


Figure 1-6 Network Symbols for Topology Diagrams

A diagram provides an easy way to understand how devices connect in a network. This type of “picture” of a network is known as a *topology diagram*. The ability to recognize the logical representations of the physical networking components is critical to being able to visualize the organization and operation of a network.

In addition to these representations, specialized terminology is used to describe how each of these devices and media connect to each other:

- **Network interface card (NIC):** A NIC physically connects an end device to a network.
- **Physical port:** A port is a connector or an outlet on a networking device where a medium connects to an end device or another networking device.
- **Interface:** An interface is a specialized port on a networking device that connects to a network. Because routers connect networks, the ports on a router are referred to as *network interfaces*.

Note

Often, the terms *port* and *interface* are used interchangeably.

Topology Diagrams (1.3.2)

Topology diagrams are mandatory documentation for anyone working with a network. Such a diagram provides a visual map of how the network is connected. There are two types of topology diagrams: physical and logical.

Physical Topology Diagrams

A physical topology diagram illustrates the physical locations of intermediary devices and cable installation, as shown in Figure 1-7. You can see that the rooms in which these devices are located are labeled in this physical topology.

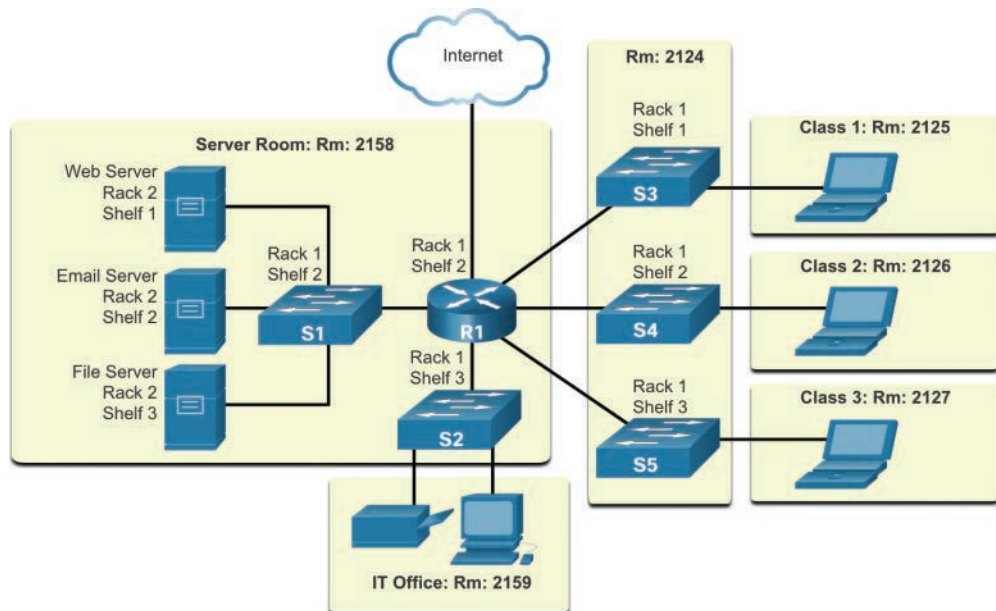


Figure 1-7 Physical Topology Example

Logical Topology Diagrams

A logical topology diagram illustrates devices, ports, and the addressing scheme of a network, as shown in Figure 1-8. You can see which end devices are connected to which intermediary devices and what media are being used.

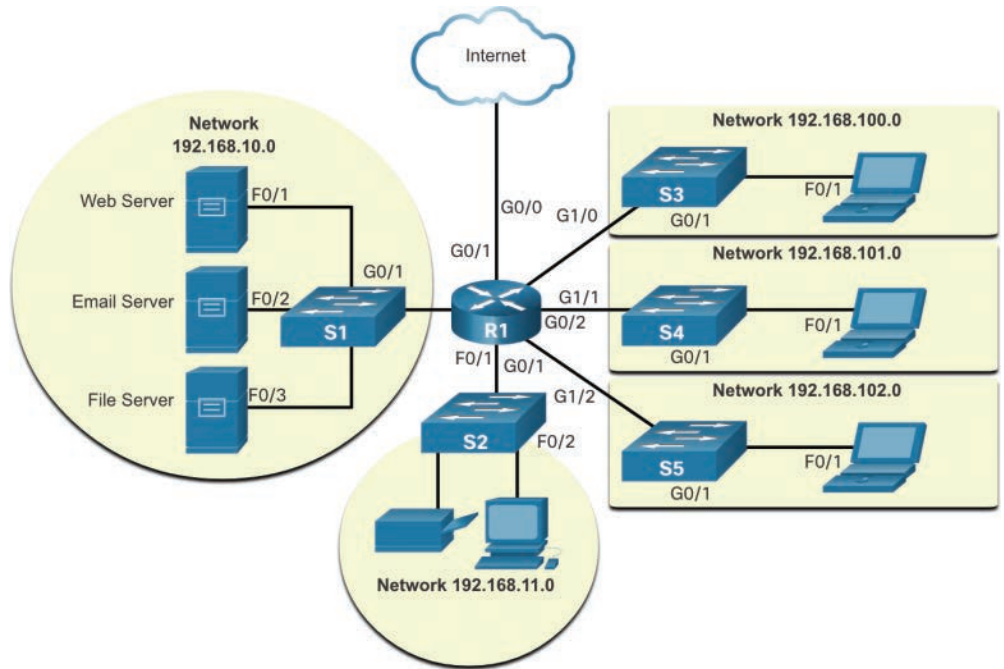


Figure 1-8 Logical Topology Example

The topologies shown in physical and logical diagrams are appropriate for your level of understanding at this point in the course. Search the internet for “network topology diagrams” to see some more complex examples. If you add the word “Cisco” to your search phrase, you will find many topologies using icons that are similar to what you have seen in these figures.

**Interactive
Graphic**

Check Your Understanding—Network Representations and Topologies (1.3.3)

Refer to the online course to complete this activity.

Common Types of Networks (1.4)

Networks can be categorized in various ways, including by size, by location, or by function. No matter the type of network being discussed, the underlying principles apply to all types of networks.

Networks of Many Sizes (1.4.1)

Now that you are familiar with the components that make up networks and their representations in physical and logical topologies, you are ready to learn about the many different types of networks.

Networks come in all sizes. They range from simple networks consisting of two computers to networks connecting millions of devices.

Simple home networks let you share resources, such as printers, documents, pictures, and music, among a few local end devices.

Small office and home office (SOHO) networks allow people to work from home or a remote office. Many self-employed workers use these types of networks to advertise and sell products, order supplies, and communicate with customers.

Businesses and large organizations use networks to provide consolidation, storage, and access to information on network servers. Networks provide email, instant messaging, and collaboration among employees. Many organizations use a network connection to the internet to provide products and services to customers.

The internet is the largest network in existence. In fact, the term *internet* means a “network of networks.” The internet is a collection of interconnected private and public networks.

In small businesses and homes, many computers function as both servers and clients on the network. This type of network is called a peer-to-peer network. There are networks of varying sizes that can be categorized in various ways, including the following:

- **Small home networks:** Small home networks connect a few computers to each other and to the internet.
- **SOHO networks:** A SOHO network allows computers in a home office or a remote office to connect to a corporate network or access centralized, shared resources.
- **Medium to large networks:** Medium to large networks, such as those used by corporations and schools, can have many locations with hundreds or thousands of interconnected hosts.
- **Worldwide networks:** The internet is a network of networks that connects hundreds of millions of computers worldwide.

LANs and WANs (1.4.2)

Network infrastructures vary greatly in terms of

- Size of the area covered
- Number of users connected

- Number and types of services available
- Area of responsibility

The two most common types of network infrastructures are *local-area networks (LANs)* and *wide-area networks (WANs)*. A LAN is a network infrastructure that provides access to users and end devices in a small geographic area. A LAN is typically used in a department within an enterprise, a home, or a small business network. A WAN is a network infrastructure that provides access to other networks over a wide geographic area, which is typically owned and managed by a larger corporation or a telecommunications service provider. Figure 1-9 shows LANs connected to a WAN.

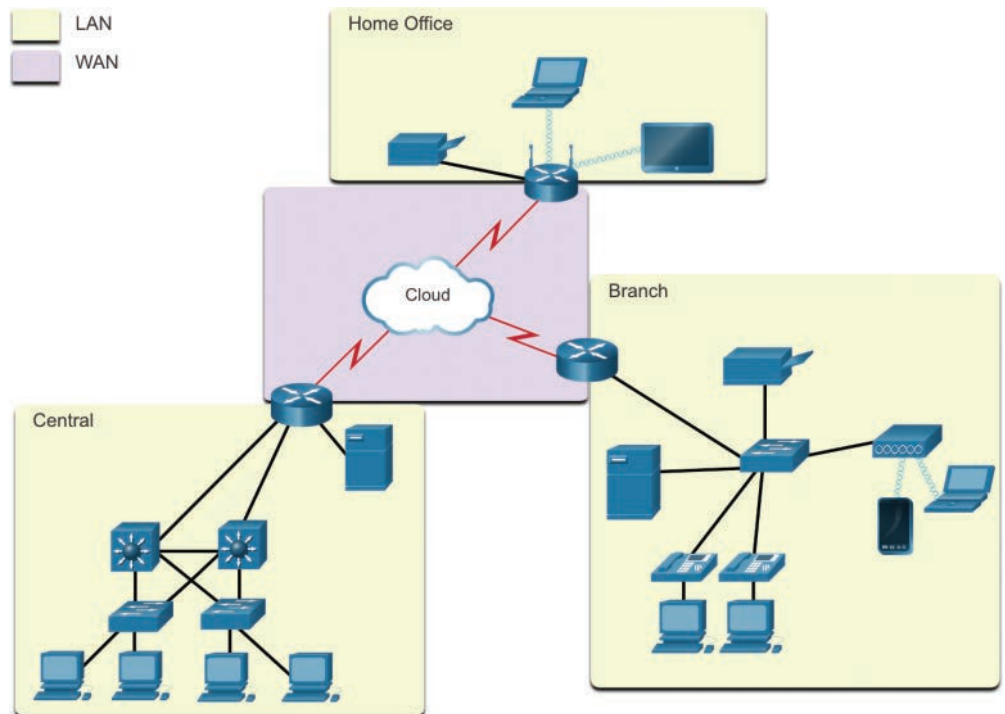


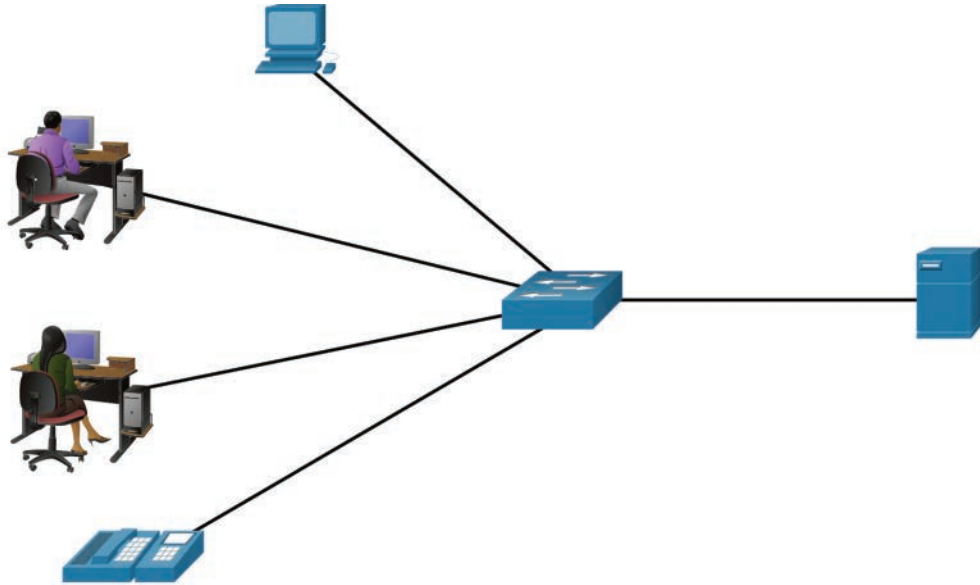
Figure 1-9 Example of Connected LANs and WANs

LANs

A LAN is a network infrastructure that spans a small geographic area. LANs have specific characteristics:

- LANs interconnect end devices in a limited area such as a home, school, office building, or campus.

- A LAN is usually administered by a single organization or individual. Administrative control is enforced at the network level and governs the security and access control policies.
- LANs provide high-speed bandwidth to internal end devices and intermediary devices, as shown Figure 1-10.



A network serving a home, small building, or a small campus is considered a LAN.

Figure 1-10 Example of a LAN

WANs

Figure 1-11 shows a WAN that interconnects two LANs. A WAN is a network infrastructure that spans a wide geographic area. WANs are typically managed by service providers (SPs) or internet service providers (ISPs).

WANs have specific characteristics:

- WANs interconnect LANs over wide geographic areas such as between cities, states, provinces, countries, or continents.
- WANs are usually administered by multiple service providers.
- WANs typically provide slower-speed links between LANs.

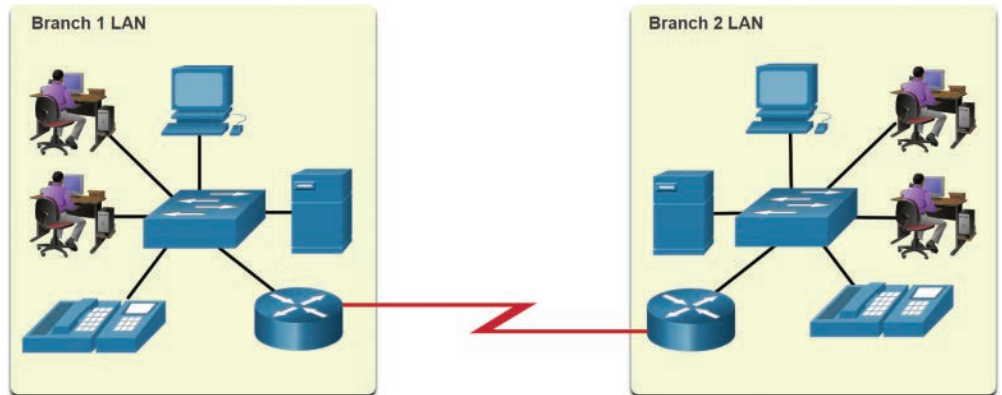


Figure 1-11 Example of a WAN Link

The Internet (1.4.3)

The *internet* is a worldwide collection of interconnected networks (*internetworks*, or *internet* for short). Figure 1-12 shows one way to view the internet as a collection of interconnected LANs and WANs.

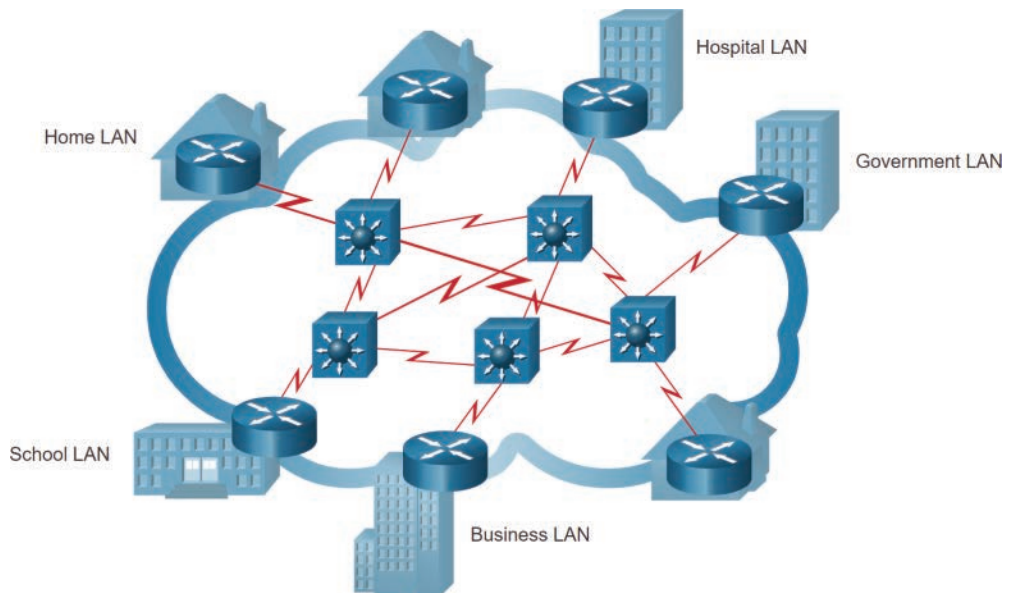


Figure 1-12 Example of a View of the Internet

Some of the LAN examples in Figure 1-12 are connected to each other through a WAN connection. WANs are then connected to each other. The WAN connection lines (which look like lightning bolts) represent the varieties of ways we connect networks. WANs can connect through copper wires, fiber-optic cables, and wireless transmissions (not shown).

The internet is not owned by any individual or group. Ensuring effective communication across this diverse infrastructure requires the application of consistent and commonly recognized technologies and standards as well as the cooperation of many network administration agencies. Organizations have been developed to help maintain the structure and standardization of internet protocols and processes. These organizations include the Internet Engineering Task Force (IETF), Internet Corporation for Assigned Names and Numbers (ICANN), and the Internet Architecture Board (IAB), among many others.

Intranets and Extranets (1.4.4)

Two other terms are similar to the term internet: intranet and extranet.

The term *intranet* is often used to refer to a private connection of LANs and WANs that belongs to an organization. An intranet is designed to be accessible only by the organization's members, employees, or others with authorization.

An organization may use an *extranet* to provide secure and safe access to individuals who work for a different organization but require access to the organization's data. Here are some examples of extranets:

- A company that is providing access to outside suppliers and contractors
- A hospital that is providing a booking system to doctors so they can make appointments for their patients
- A local education office that is providing budget and personnel information to the schools in its district

Figure 1-13 illustrates the levels of access that different groups have to a company intranet, a company extranet, and the internet.

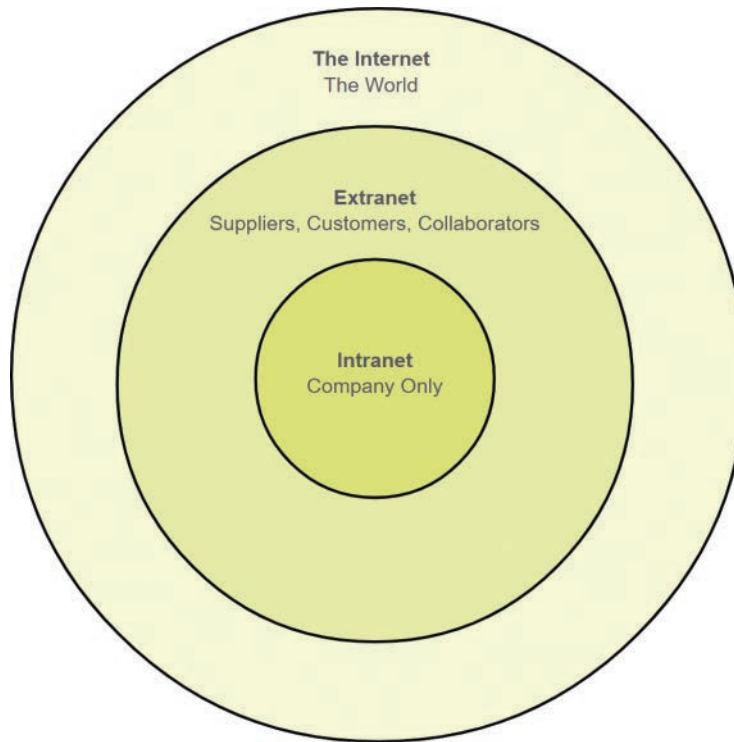


Figure 1-13 Levels of Access from Intranet to Internet

**Interactive
Graphic**

Check Your Understanding—Common Types of Networks (1.4.5)

Refer to the online course to complete this activity.

Internet Connections (1.5)

End devices such as computers and smartphones connect to a network in a variety of ways, using both wired and wireless means. These same types of connections are used to interconnect intermediary devices.

Internet Access Technologies (1.5.1)

Now you have a basic understanding of what makes up a network and the different types of networks. How do you actually connect users and organizations to the internet? As you may already know, there are many different ways to do this.

Home users, remote workers, and small offices typically require a connection to an *internet service provider (ISP)* to access the internet. Connection options vary

greatly between ISPs and in different geographic locations. However, popular choices include broadband cable, broadband digital subscriber line (DSL), wireless WANs, and mobile services.

Organizations usually need access to other corporate sites as well as the internet. Fast connections are required to support business services such as IP phones, video conferencing, and data center storage. ISPs offer business-class interconnections. Popular business-class services include business DSL, leased lines, and Metro Ethernet.

Home and Small Office Internet Connections (1.5.2)

Figure 1-14 illustrates common connection options for small office and home office users:

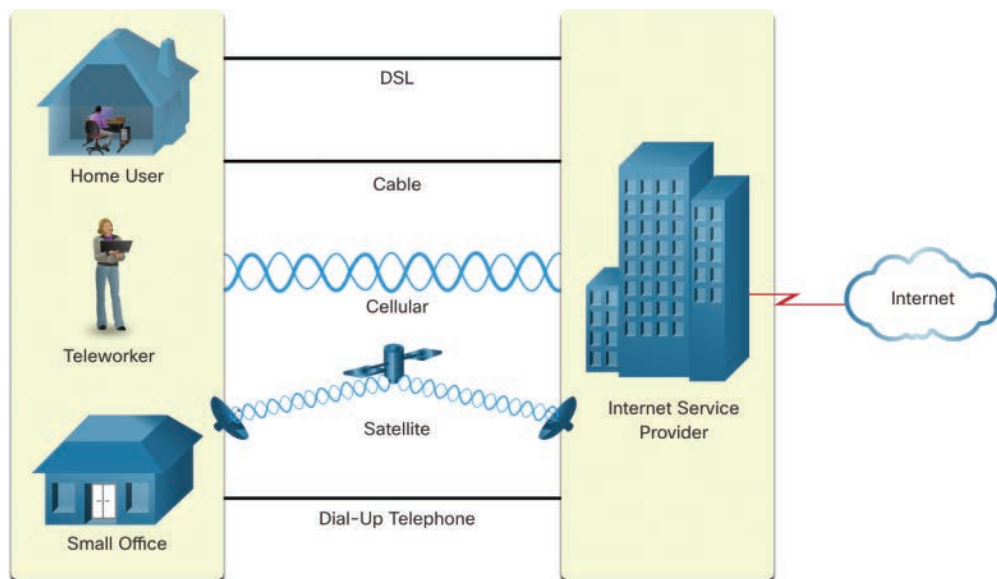


Figure 1-14 Small Office and Home Office Connection Options

- **Cable connection:** With this type of connection, typically offered by cable television service providers, the internet data signal transmits on the same cable that delivers cable television. This connection type provides a high-bandwidth, high-availability, and an always-on connection to the internet.
- **Digital subscriber line (DSL):** DSL provides high bandwidth, high availability, and an always-on connection to the internet. DSL runs over a telephone line. In general, small office and home office users connect using asymmetrical DSL (ADSL), which means that the download speed is faster than the upload speed.
- **Cellular connection:** Cellular internet access uses a cellphone network to connect. Wherever you can get a cellular signal, you can get cellular internet

access. Performance is limited by the capabilities of the phone or other device and the cell tower to which it is connected.

- **Satellite connection:** The availability of satellite internet access is a benefit in areas that would otherwise have no internet connectivity at all. A satellite dish must have a clear line of sight to the satellite.
- **Dialup telephone connection:** This is an inexpensive option that uses any phone line and a modem. The low bandwidth provided by a dialup modem connection is not sufficient for large data transfers, although it is useful for mobile access while traveling.

The choice of connection varies depending on geographic location and service provider availability.

Businesses Internet Connections (1.5.3)

Corporate connection options differ from home user options. Businesses may require higher bandwidth, dedicated bandwidth, and managed services. Connection options that are available differ depending on the type of service providers located nearby.

Figure 1-15 illustrates common connection options for businesses:

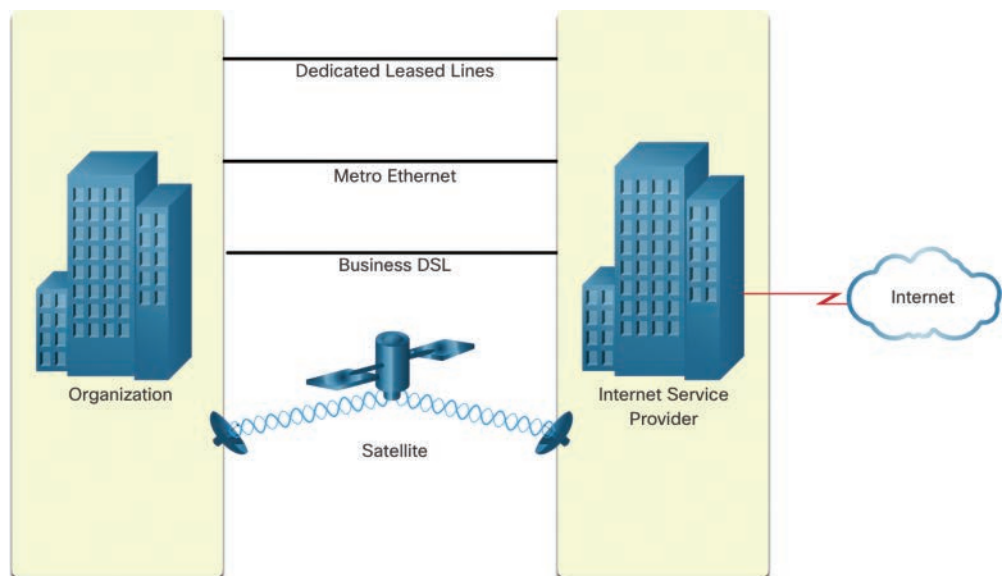


Figure 1-15 Business Connection Options

- **Dedicated leased lines:** Leased lines are reserved circuits within a service provider's network that connect geographically separated offices for private voice and/or data networking. The circuits are rented at a monthly or yearly rate.

- **Metro Ethernet:** This is sometimes known as Ethernet WAN. In this chapter, we will refer to it as Metro Ethernet. Metro Ethernet can be used to extend LAN access technology into the WAN. Ethernet is a LAN technology you will learn about in a later chapter.
- **Business DSL:** Business DSL is available in various formats. A popular choice is symmetric DSL (SDSL), which is similar to the consumer version of DSL but provides uploads and downloads at the same high speeds.
- **Satellite:** Satellite service can provide a connection when a wired solution is not available.

The choice of connection varies depending on geographic location and service provider availability.

The Converging Network (1.5.4)

Consider a school built 30 years ago. Back then, some classrooms were cabled for the data network, telephone network, and video network for televisions. These separate networks could not communicate with each other. Each network used different technologies to carry communication signals. Each network had its own set of rules and standards to ensure successful communication. Multiple services ran on multiple networks, as shown in Figure 1-16.

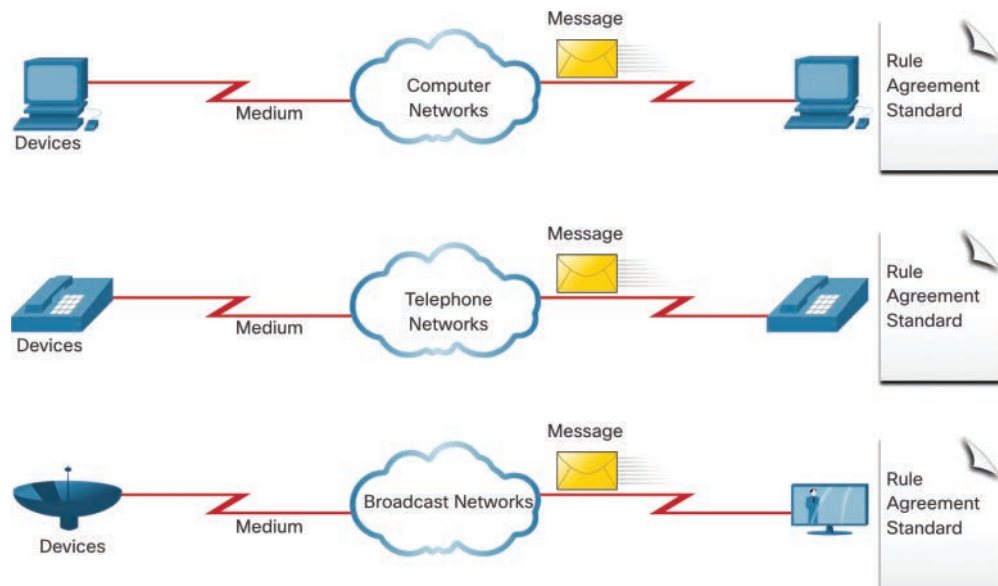


Figure 1-16 Traditional Networks

Today, separate data, telephone, and video networks have converged. Unlike dedicated networks, converged networks are capable of delivering data, voice, and video between many different types of devices over the same network infrastructure. This network infrastructure uses the same set of rules, agreements, and implementation standards. *Converged data networks* carry multiple services on one network, as shown in Figure 1-17.

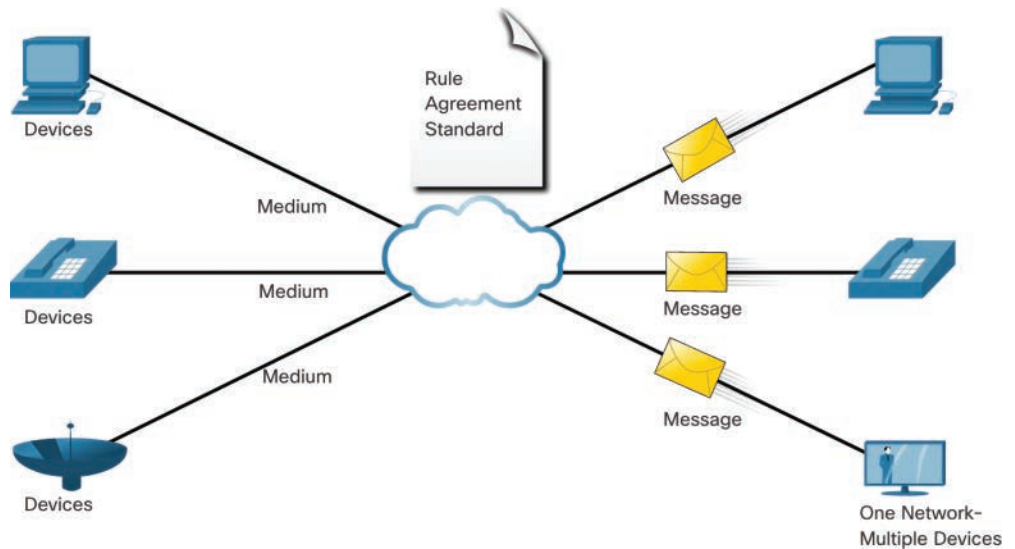


Figure 1-17 Converged Network

Video

Video—Download and Install Packet Tracer (1.5.5)

This video shows you how to download and install Packet Tracer, which you can use to simulate the creation and testing of networks on your computer. Packet Tracer is a fun, take-home, flexible software program that will give you the opportunity to use the network representations and theories that you have just learned to build network models and explore relatively complex LANs and WANs.

Students commonly use Packet Tracer to

- Prepare for a certification exam
- Practice what they learn in networking courses
- Sharpen their skills for a job interview
- Examine the impact of adding new technologies into existing network designs
- Build their skills for jobs in the Internet of Things
- Compete in global design challenges (such as at the 2017 PT 7 Design Challenge on Facebook)

Packet Tracer is an essential learning tool used in many Cisco Networking Academy courses.

To obtain and install a copy of Cisco Packet Tracer, follow these steps:



- Step 1.** Log in to your Cisco Networking Academy “I’m Learning” page.
- Step 2.** Select Resources.
- Step 3.** Select Download Packet Tracer.
- Step 4.** Select the version of Packet Tracer you require.
- Step 5.** Save the file to your computer.
- Step 6.** Launch the Packet Tracer installation program.

Refer to the online course to view this video.

Video

Video—Getting Started in Cisco Packet Tracer (1.5.6)

Packet Tracer is a tool that allows you to simulate real networks. It provides three main features:

- You can add devices and connect them via cables or wirelessly
- You can select, delete, inspect, label, and group components within a network

You can manage a network by opening an existing/sample network, saving your current network, and modifying your user profile or preferences

If you have used any program such as a word processor or spreadsheet, you are already familiar with the File menu commands located in the top menu bar. The Open, Save, Save As, and Exit commands work as they would for any program, but there are two commands that are special to Packet Tracer:

- The Open Samples command displays a directory of prebuilt examples of features and configurations for various network and Internet of Things devices included within Packet Tracer.
- The Exit and Logout command removes the registration information for this copy of Packet Tracer and requires the next user of this copy of Packet Tracer to go through the login procedure again.

Refer to the online course to view this video.

Packet Tracer
Activity**Packet Tracer—Network Representation (1.5.7)**

In this activity, you will explore how Packet Tracer serves as a modeling tool for network representations.

Reliable Networks (1.6)

A network is a platform for distributing a wide range of services to end users in a reliable, efficient, and secure manner.

Network Architecture (1.6.1)

Have you ever been busy working online only to have “the internet go down”? As you know by now, the internet did not go down, but it is possible to lose your connection to it—and that can be very frustrating. With so many people in the world relying on network access to work and learn, it is imperative that networks be reliable. In this context, *reliability* means more than your connection to the internet. This section focuses on the four aspects of network reliability.

The role of networks has changed. What was once a data-only network is now a system that enables connections between people, devices, and information in a media-rich, converged network environment. For networks to function efficiently and grow in this type of environment, networks must be built on a standard network architecture.

Networks also support a wide range of applications and services. They must operate over the many different types of cables and devices that make up the physical infrastructure. The term *network architecture*, in this context, refers to the technologies that support the infrastructure and the programmed services and rules, or protocols, that move data across the network.

As networks evolve, there are four basic characteristics that network architects must address to meet user expectations:

- Fault tolerance
- Scalability
- Quality of service (QoS)
- Security

Fault Tolerance (1.6.2)

A *fault-tolerant network* is a network that limits the number of devices affected by a failure. It is built to allow quick recovery when a failure occurs. These networks depend on multiple paths between the source and destination of a message. If one path fails, the messages are instantly sent over a different link. Having multiple paths to a destination is known as *redundancy*.

Implementing a packet-switched network is one way to provide redundancy. Packet switching splits traffic into packets that are routed over a shared network. A single message, such as an email or a video stream, is broken into multiple message blocks, called *packets*. Each packet has the necessary addressing information of the source and destination of the message. The routers within the network switch the packets based on the condition of the network at that moment. This means that all the packets in a single message could take very different paths to the same destination. In Figure 1-18, the user is unaware and unaffected by the router that is dynamically changing the route when a link fails.

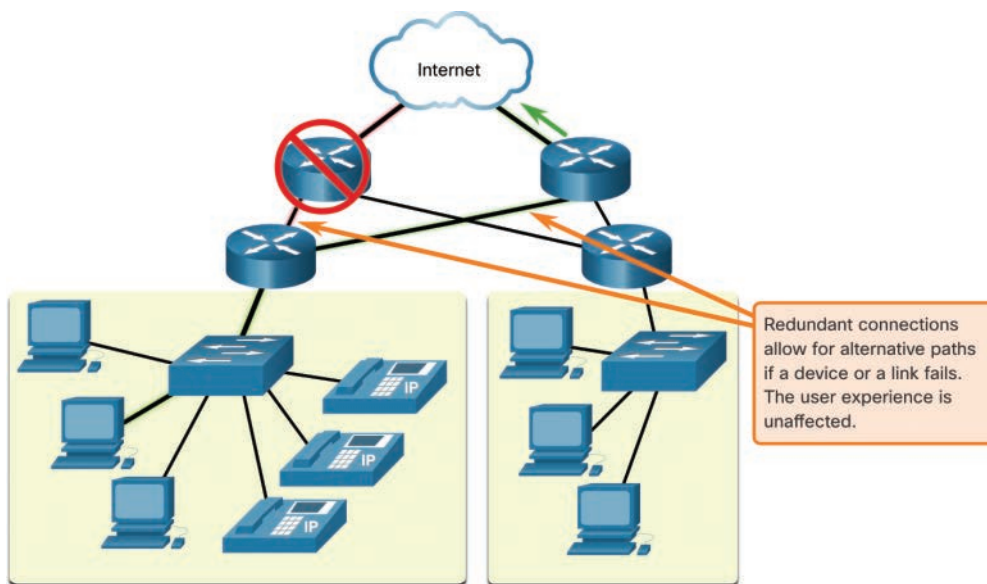


Figure 1-18 Fault-Tolerant Design

Scalability (1.6.3)

A *scalable network* expands quickly to support new users and applications. It does this without degrading the performance of services that are being accessed by existing users. Figure 1-19 shows how a new network is easily added to an existing

network. These networks are scalable because the designers have followed accepted standards and protocols. Because of these standards and protocols, software and hardware vendors can focus on improving products and services without having to design a new set of rules for operating within the network.

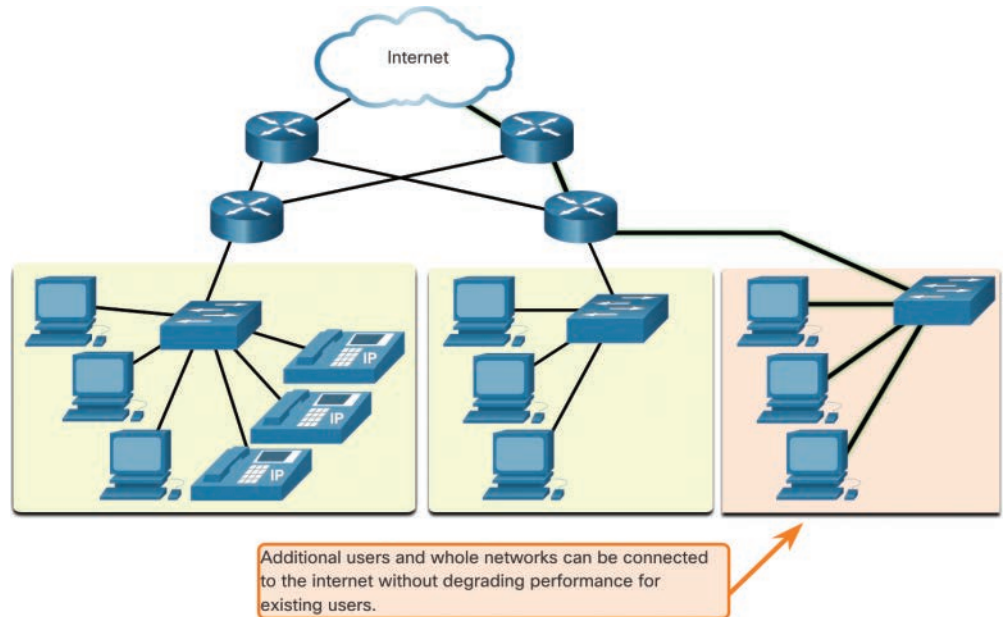


Figure 1-19 Scalable Design

Quality of Service (1.6.4)

Quality of service (QoS) is an increasing requirement in networks today. New applications available to users over networks, such as voice and live video transmissions, create higher expectations for the quality of the delivered services. Have you ever tried to watch a video and experienced constant breaks and pauses? As data, voice, and video content continue to converge onto the same network, QoS becomes a primary mechanism for managing congestion and ensuring reliable delivery of content to all users.

Congestion occurs when the demand for bandwidth exceeds the amount available. Network bandwidth is measured in the number of bits that can be transmitted in a single second, or bits per second (bps). When simultaneous communications are attempted across a network, the demand for network bandwidth can exceed its availability, creating network congestion.

When the volume of traffic is greater than what can be transported across the network, devices hold the packets in memory until resources become available to transmit them. In Figure 1-20, one user is requesting a web page, and another is on a phone call. With a QoS policy in place, the router can manage the flow of data and voice traffic and give priority to voice communication if the network experiences congestion.

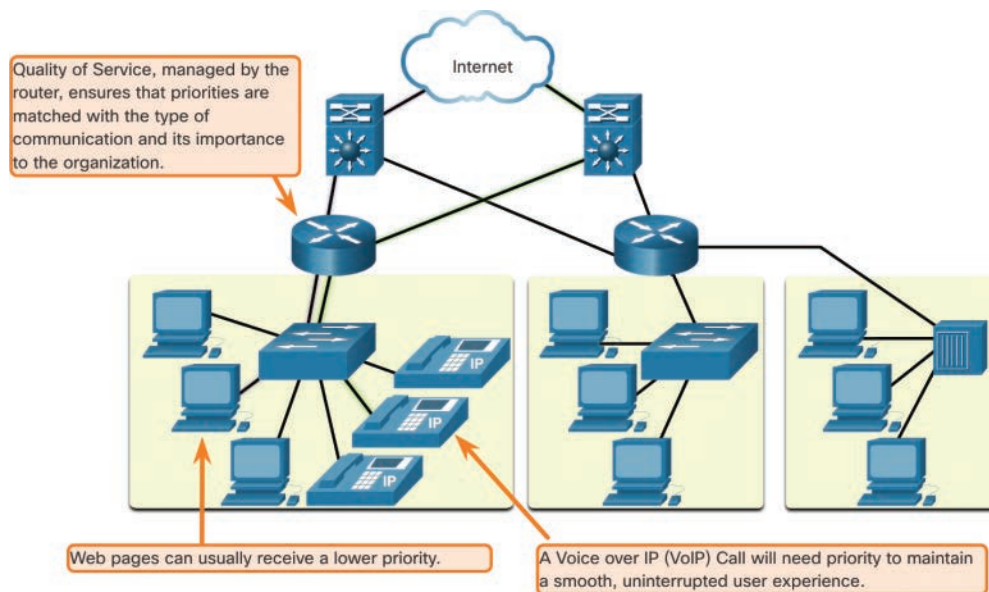


Figure 1-20 QoS Design

Network Security (1.6.5)

The network infrastructure, the services, and the data contained on network-attached devices are crucial personal and business assets. Network administrators must address two types of network security concerns: network infrastructure security and information security.

Securing the network infrastructure involves physically securing devices that provide network connectivity and preventing unauthorized access to the management software that resides on them, as shown in Figure 1-21.

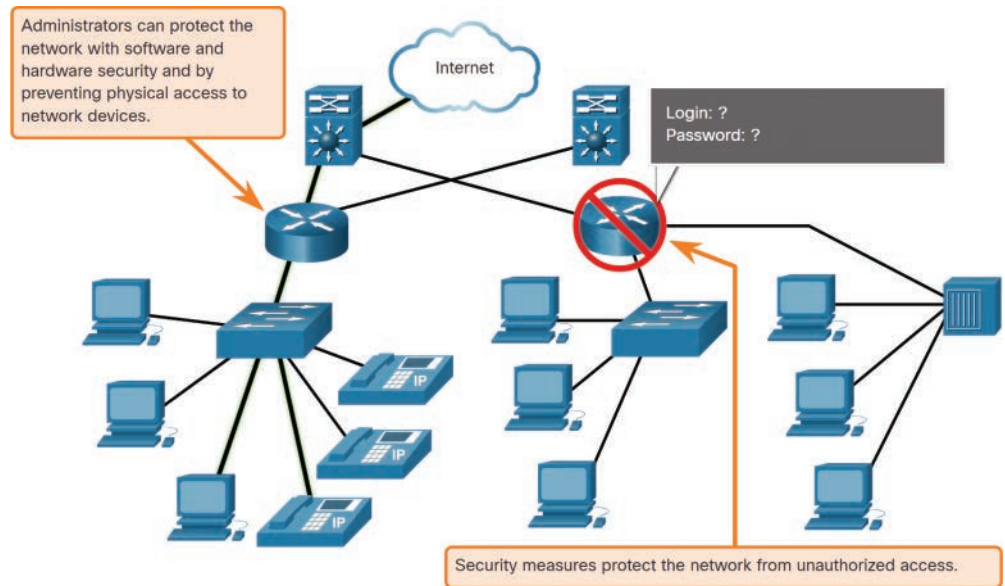


Figure 1-21 Security Design

Network administrators must also protect the information contained within the packets being transmitted over the network, as well as the information stored on network-attached devices. In order to achieve the goals of network security, there are three primary requirements:

- **Confidentiality:** Data confidentiality means that only the intended and authorized recipients can access and read data.
- **Integrity:** Data integrity assures users that the information has not been altered in transmission, from origin to destination.
- **Availability:** Data availability assures users of timely and reliable access to data services for authorized users.

**Interactive
Graphic**

Check Your Understanding—Reliable Networks (1.6.6)

Refer to the online course to complete this activity.

Network Trends (1.7)

The network environment continues to evolve, providing new experiences and opportunities for end users. The network is now capable of delivering services and applications in a manner that was once only a dream.

Recent Trends (1.7.1)

You know a lot about networks now, including what they are made of, how they connect us, and what is needed to keep them reliable. But networks, like everything else, continue to change. You, as a NetAcad student, need to know about a few trends in networking.

As new technologies and end-user devices come to market, businesses and consumers must continue to adjust to the ever-changing network environment. Several networking trends affect organizations and consumers:

- Bring your own device (BYOD)
- Online collaboration
- Video communications
- Cloud computing

Bring Your Own Device (BYOD) (1.7.2)

The concept of any device, for any content, in any manner is a major global trend that requires significant changes to the way we use devices and safely connect them to networks. *Bring your own device (BYOD)* gives end users the freedom to use personal tools to access information and communicate across a business or campus network. With the growth of consumer devices and the related drop in cost, employees and students may have advanced computing and networking devices for personal use. These include laptops, notebooks, tablets, smartphones, and e-readers. Such devices may be purchased by the company or school, purchased by the individual, or both.

BYOD refers to any device, with any ownership, used anywhere.

Online Collaboration (1.7.3)

Individuals want to connect to a network not only for access to data applications but also to collaborate with one another. *Collaboration* is defined as “the act of working with another or others on a joint project.” Collaboration tools, such as Cisco Webex (see Figure 1-22), enable employees, students, teachers, customers, and partners to instantly connect, interact, and achieve their objectives.

Collaboration is a critical and strategic priority that organizations are using to remain competitive. Collaboration is also a priority in education. Students need to collaborate to assist each other in learning, to develop the team skills used in the workforce, and to work together on team-based projects.



Figure 1-22 Cisco Webex Interface

Cisco Webex Teams is a multifunctional collaboration tool that lets you send instant messages to one or more people, post images, and post videos and links. Each team “space” maintains a history of everything that is posted there.

Video Communications (1.7.4)

Another facet of networking that is critical to a communication and collaboration effort is video. Video is used for communication, collaboration, and entertainment. Video calls can be made to and from anyone with an internet connection, regardless of where they are located.

Video conferencing is a powerful tool for communicating with others, both locally and globally. Video is becoming a critical requirement for effective collaboration as organizations extend across geographic and cultural boundaries.

Video

Video—Cisco Webex for Huddles (1.7.5)

Refer to the online course to view this video.

Cloud Computing (1.7.6)

Cloud computing is one of the ways that we access and store data. Cloud computing allows us to store personal files—even back up an entire drive—on servers over the internet. Applications such as word processing and photo editing can be accessed using the cloud.

For businesses, cloud computing extends the capabilities of IT without requiring investment in new infrastructure, training for new personnel, or licensing of new software. These services are available on demand and delivered economically to any device that is anywhere in the world without compromising security or function.

Cloud computing is possible because of data centers. Data centers are facilities used to house computer systems and associated components. A data center can occupy one room of a building, one or more floors, or an entire warehouse-sized building. Data centers are typically very expensive to build and maintain. For this reason, only large organizations use privately built data centers to house their data and provide services to users. Smaller organizations that cannot afford to maintain their own private data centers can reduce the overall cost of ownership by leasing server and storage services from a larger data center organization in the cloud.

For security, reliability, and fault tolerance, cloud providers often store data in distributed data centers. Instead of storing all the data of a person or an organization in one data center, it is stored in multiple data centers in different locations.

As described in Table 1-3, there are four primary types of clouds: public clouds, private clouds, hybrid clouds, and community clouds.

Table 1-3 Cloud Types

Cloud Type	Description
Public	Cloud-based applications and services offered in a public cloud are made available to the general population. Services may be free or offered on a pay-per-use model, such as paying for online storage. A public cloud uses the internet to provide services.
Private	Cloud-based applications and services offered in a private cloud are intended for a specific organization or entity, such as a government. A private cloud can be set up using the organization's private network, although this can be expensive to build and maintain. A private cloud can also be managed by an outside organization with strict access security.
Hybrid	A hybrid cloud is made up of two or more clouds (for example, part private and part public), where each part remains a distinct object but the two are connected using a single architecture. Individuals on a hybrid cloud would be able to have degrees of access to various services based on user access rights.
Community	A community cloud is created for exclusive use by specific entities or organizations. The differences between public clouds and community clouds are the functional needs that have been customized for the community. For example, healthcare organizations must remain compliant with policies and laws (such as HIPAA) that require special authentication and confidentiality. Community clouds are used by multiple organizations that have similar needs and concerns. A community cloud is similar to a public cloud environment but with set levels of security, privacy, and even regulatory compliance, as in a private cloud.

Technology Trends in the Home (1.7.7)

Networking trends are not only affecting the way we communicate at work and at school but also changing many aspects of the home. The newest home trends include smart home technology.

Smart home technology is being integrated into everyday appliances, which can connect with other devices to make the appliances more “smart,” or automated. For example, you could prepare food and place it in the oven for cooking prior to leaving the house for the day. You would program your smart oven for the food you want it to cook, and the oven would be connected to your calendar of events so that it could determine what time you should be available to eat and adjust start times and length of cooking accordingly. It could even adjust cooking times and temperatures based on changes in schedule. In addition, a smartphone or tablet connection would let you connect to the oven directly to make any desired adjustments. When the food is ready, the oven could send an alert message to you (or someone you specify) to indicate that the food is done.

Smart home technology is being developed for all rooms in a house. Smart home technology will become more common as home networking and high-speed internet technology expand.

Powerline Networking (1.7.8)

Powerline networking for home networks uses existing electrical wiring to connect devices, as shown in Figure 1-23.

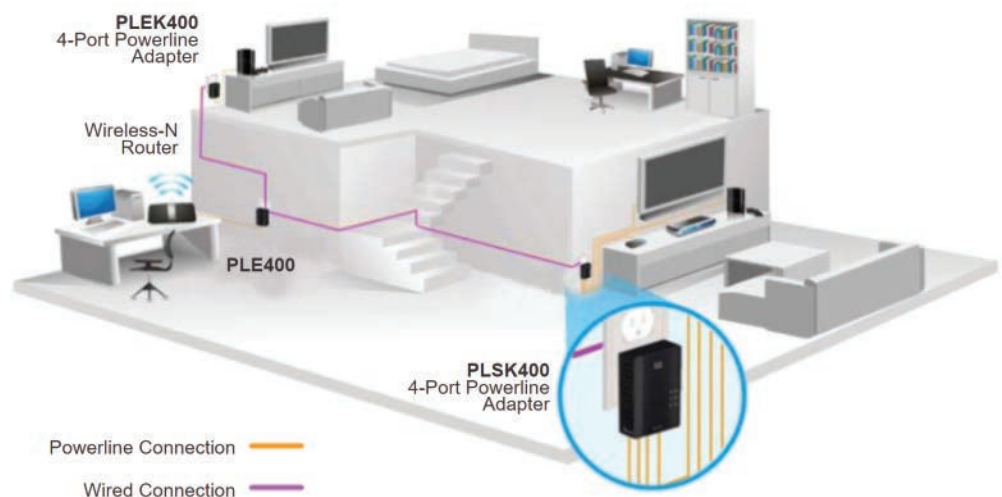


Figure 1-23 Powerline Networking Adapters

Using a standard powerline adapter, devices can connect to the LAN wherever there is an electrical outlet. No data cables need to be installed, and little to no additional electricity is used. Using the same wiring that delivers electricity, powerline networking sends data on certain frequencies.

Powerline networking is especially useful when wireless access points cannot reach all the devices in the home. Powerline networking is not a substitute for dedicated cabling in data networks. However, it is an alternative when data network cables or wireless communications are not possible or effective.

Wireless Broadband (1.7.9)

In many areas where cable and DSL are not available, wireless may be used to connect to the internet.

Wireless Internet Service Providers

A *wireless internet service provider (WISP)* is an ISP that connects subscribers to a designated access point or hotspot using wireless technologies similar to those found in home wireless local-area networks (WLANs). WISPs are most commonly found in rural environments where DSL or cable services are not available.

Although a separate transmission tower may be installed for the antenna, typically the antenna is attached to an existing elevated structure, such as a water tower or a radio tower. A small dish or antenna is installed on the subscriber's roof, in range of the WISP transmitter. The subscriber's access unit is connected to the wired network inside the home. From the perspective of the home user, the setup is not much different from that of DSL or cable service. The main difference is that the connection from the home to the ISP is wireless instead of a physical cable.

Wireless Broadband Service

Another wireless solution for homes and small businesses is wireless broadband. This solution uses the same cellular technology as a smartphone. An antenna is installed outside the house, providing either wireless or wired connectivity for devices in the home. In many areas, home wireless broadband is competing directly with DSL and cable services.

Interactive
Graphic

Check Your Understanding—Network Trends (1.7.10)

Refer to the online course to complete this activity.

Network Security (1.8)

Security is a critical component in designing, implementing, and maintaining networks. Network engineers and administrators must always consider security risks and employ the proper mitigation methods before deploying any type of network service.

Security Threats (1.8.1)

You have, no doubt, heard or read news stories about a company network being breached, giving threat actors access to the personal information of thousands of customers. Network security is always going to be a top priority of administrators.

Network security is an integral part of computer networking, regardless of whether the network is a home network with a single connection to the internet or a corporate network with thousands of users. Network security must consider the environment, as well as the tools and requirements of the network. It must be able to secure data while still allowing for the quality of service that users expect of the network.

Securing a network involves using protocols, technologies, devices, tools, and techniques to protect data and mitigate threats. Threat vectors may be external or internal. Many external network security threats today originate from the internet.

There are several common external threats to networks:

- **Viruses, worms, and Trojan horses:** These contain malicious software or code running on a user device.
- **Spyware and adware:** These types of software are installed on a user's device to secretly collect information about the user.
- **Zero-day attacks:** Also called zero-hour attacks, these attacks occur on the first day that a vulnerability becomes known.
- **Threat actor attacks:** In these attacks, malicious persons attack user devices or network resources.
- **Denial-of-service attacks:** These attacks slow or crash applications and processes on a network device.
- **Data interception and theft:** This type of attack involves capturing private information from an organization's network.
- **Identity theft:** This type of attack involves stealing the login credentials of a user in order to access private data.

It is important to consider internal threats. Many studies have shown that the most common data breaches are related to internal network users. Such breaches may

be attributed to lost or stolen devices, accidental misuse by employees, or even malicious employees. BYOD strategies make corporate data especially vulnerable. Therefore, when developing a security policy, it is important to address both external and internal security threats, as shown in Figure 1-24.

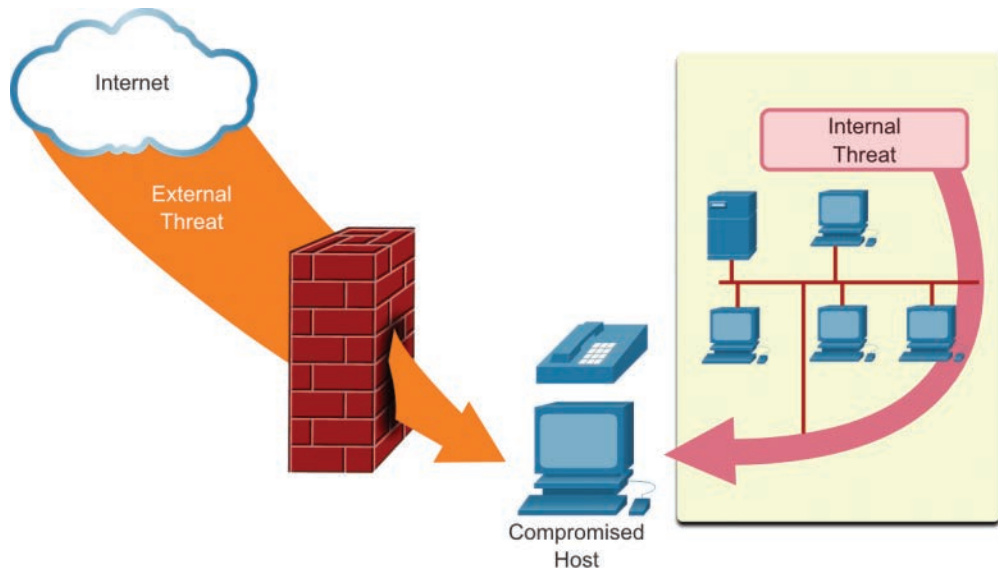


Figure 1-24 External and Internal Threats

Security Solutions (1.8.2)

No single solution can protect a network from the variety of threats that exist. For this reason, security should be implemented in multiple layers, using more than one security solution. If one security component fails to identify and protect a network, others may succeed.

A home network security implementation is usually rather basic. Typically, you implement it on the end devices, as well as at the point of connection to the internet. You may even be able to rely on contracted services from the ISP.

These are the basic security components for a home or small office network:

- **Antivirus and antispyware:** These applications help to protect end devices from becoming infected with malicious software.
- **Firewall filtering:** Firewall filtering blocks unauthorized access into and out of the network. This may include a host-based firewall system that prevents unauthorized access to the end device or a basic filtering service on the home router to prevent unauthorized access from the outside world into the network.

In contrast, the network security implementation for a corporate network usually consists of many components built into the network to monitor and filter traffic. Ideally, all components work together to minimize maintenance and improve security. Larger networks and corporate networks use antivirus, antispyware, and firewall filtering, and they also have additional security requirements:

- **Dedicated firewall systems:** These provide more advanced firewall capabilities that can filter large amounts of traffic with more granularity.
- **Access control lists (ACL):** ACLs further filter access and traffic forwarding based on IP addresses and applications.
- **Intrusion prevention systems (IPS):** These systems identify fast-spreading threats, such as zero-day or zero-hour attacks.
- **Virtual private networks (VPN):** These networks provide secure access into an organization for remote workers.

Network security requirements must consider the environment as well as the various applications and computing requirements. Both home and business environments must be able to secure their data while still allowing for the quality of service that users expect of each technology. In addition, any security solution that is implemented must be adaptable to the growing needs of the network and changing trends.

The study of network security threats and mitigation techniques starts with a clear understanding of the underlying switching and routing infrastructure used to organize network services.

**Interactive
Graphic**

Check Your Understanding—Network Security (1.8.3)

Refer to the online course to complete this activity.

The IT Professional (1.9)

The role of an IT professional is constantly evolving. An IT professional is always learning in an ever-changing environment.

CCNA (1.9.1)

As a NetAcad student, you may already have a career in IT, or you may be educating yourself to prepare for your career. In either case, it is good to know about the skills needed for the various jobs that are available in IT.

Network engineers are more vital today than ever before, and their roles and required skills are constantly evolving. The Cisco Certified Network Associate (CCNA) certification demonstrates that you have knowledge of foundational technologies and ensures that you stay relevant with the skills needed for the adoption of next-generation technologies.

The requirements of the CCNA for networking engineers have been consolidated and updated to three courses and one exam covering the fundamental topics for all network technologies. This new CCNA focuses on IP foundation and security topics, along with wireless, virtualization, automation, and network programmability.

Cisco offers new DevNet certifications at the associate, specialist, and professional levels to validate your software development skills.

In addition, you can obtain specialist certifications (such as Cisco Enterprise Advanced Infrastructure Specialist certification) to validate your skills in line with your job role and interests.

You can start where you want. There are no prerequisites to start earning your associate-, specialist-, professional-, or expert-level certifications. Continuing education credits for recertification and ongoing development are now available for CCNA certification.

Networking Jobs (1.9.2)

Your CCNA certification will prepare you for a variety of jobs in today's market. At www.netacad.com you can click the Careers menu and then select Employment opportunities. You can find employment opportunities where you live by using the Talent Bridge Matching Engine. Use this new program to search for jobs with Cisco and Cisco partners and distributors seeking Cisco Networking Academy students and alumni.

You can also search for IT jobs using online search engines such as Indeed, Glassdoor, and Monster. Use search terms such as *IT*, *network administrator*, *network architect*, and *computer systems administrator*. You can also search using the term *Cisco CCNA*.



Lab—Research IT and Networking Job Opportunities (1.9.3)

In this lab, you will complete the following objectives: Part 1: Research Job Opportunities; Part 2: Reflect on Research.

Summary (1.10)

The following is a summary of the topics in the chapter and their corresponding online modules.

Networks Affect Our Lives

In today's world, through the use of networks, we are connected as never before. People with ideas can communicate instantly with others to make those ideas reality. The creation of online communities for the exchange of ideas and information has the potential to increase productivity opportunities around the globe. The cloud enables us to store documents and pictures and access them anywhere, anytime.

Network Components

All computers that are connected to networks and participate directly in network communication are classified as hosts. Hosts can be called end devices, and some hosts are also called clients. Many computers function as servers and clients on a network called a peer-to-peer network. An end device is either the source or destination of a message transmitted over a network. Intermediary devices connect the individual end devices to the network and can connect multiple individual networks to form an internetwork. Intermediary devices use the destination end device address, in conjunction with information about the network interconnections, to determine the path that messages should take through the network. The media provide the channel over which the message travels from source to destination.

Network Representations and Topologies

Diagrams of networks often use symbols to represent the different devices and connections that make up a network. A topology diagram provides an easy way to understand how devices connect in a large network. A physical topology diagram illustrates the physical locations of intermediary devices and cable installation. A logical topology diagram illustrates devices, ports, and the addressing scheme of a network.

Common Types of Networks

A small home network connects a few computers to each other and to the internet. A small office/home office (SOHO) network allows computers in a home office or a remote office to connect to a corporate network or access centralized shared resources. Medium to large networks, such as those used by corporations and schools, can have many locations with hundreds or thousands of interconnected

hosts. The internet is a network of networks that connects hundreds of millions of computers worldwide. The two most common network infrastructures are local-area networks (LANs), and wide-area networks (WANs). A LAN is a network infrastructure that spans a small geographic area. A WAN is a network infrastructure that spans a wide geographic area. An intranet is a private connection of LANs and WANs that belongs to an organization. An organization may use an extranet to provide secure and safe access to individuals who work for a different organization but require access to the organization's data.

Internet Connections

SOHO internet connections include cable, DSL, cellular, satellite, and dialup telephone. Business internet connections include dedicated leased lines, Metro Ethernet, business DSL, and satellite. The choice of connection varies depending on geographic location and service provider availability. Traditional separate networks used different technologies, rules, and standards. Converged networks deliver data, voice, and video between many different types of devices over the same network infrastructure. This network infrastructure uses the same set of rules, agreements, and implementation standards. Packet Tracer is a flexible software program that lets you use network representations and theories to build network models and explore relatively complex LANs and WANs.

Reliable Networks

The term *network architecture* refers to the technologies that support the infrastructure and the programmed services and rules, or protocols, that move data across a network. As networks evolve, we have learned that there are four basic characteristics that network architects must address to meet user expectations: fault tolerance, scalability, quality of service (QoS), and security. A fault-tolerant network limits the number of devices affected by a failure. Redundancy means having multiple paths to a destination. A scalable network expands quickly to support new users and applications. Networks are scalable because the designers follow accepted standards and protocols. QoS is a primary mechanism for managing congestion and ensuring reliable delivery of content to users. Network administrators must address two types of network security concerns: network infrastructure security and information security. To achieve the goals of network security, there are three primary requirements: confidentiality, integrity, and availability.

Network Trends

Several recent networking trends affect organizations and consumers: bring your own device (BYOD), online collaboration, video communications, and cloud computing.

BYOD refers to any device, with any ownership, used anywhere. Collaboration tools such as Cisco WebEx enable employees, students, teachers, customers, and partners to instantly connect, interact, and achieve their objectives. Video is used for communication, collaboration, and entertainment. Video calls can be made to and from anyone with an internet connection, regardless of where they are located. Cloud computing allows us to store personal files—even back up an entire drive—on servers over the internet. Applications such as word processing and photo editing can be accessed using the cloud. There are four primary types of clouds: public clouds, private clouds, hybrid clouds, and custom clouds. Smart home technology is being developed for all rooms in a house. Smart home technology will become more common as home networking and high-speed internet technology expand. Using the same wiring that delivers electricity, powerline networking sends data on certain frequencies. A wireless internet service provider (WISP) is an ISP that connects subscribers to a designated access point or hotspot using wireless technologies similar to those found in home wireless local-area networks (WLANs).

Network Security

There are several common external threats to networks:

- Viruses, worms, and Trojan horses
- Spyware and adware
- Zero-day attacks
- Threat actor attacks
- Denial-of-service attacks
- Data interception and theft
- Identity theft

These are the basic security components for a home or small office network:

- Antivirus and antispyware
- Firewall filtering

Larger networks and corporate networks use antivirus, antispyware, and firewall filtering, and they also have additional security requirements:

- Dedicated firewall systems
- Access control lists (ACL)
- Intrusion prevention systems (IPS)
- Virtual private networks (VPN)

The IT Professional

The Cisco Certified Network Associate (CCNA) certification demonstrates that you have knowledge of foundational technologies and ensures that you stay relevant with skills needed for the adoption of next-generation technologies. Your CCNA certification will prepare you for a variety of jobs in today's market. At www.netacad.com you can click the Careers menu and then select Employment opportunities. You can find employment opportunities where you live by using the Talent Bridge Matching Engine to search for jobs with Cisco as well as Cisco partners and distributors seeking Cisco Networking Academy students and alumni.

Practice

The following activities provide practice with the topics introduced in this chapter. The lab is available in the companion *Introduction to Networks Labs & Study Guide (CCNAv7)* (ISBN 9780136634454). The Packet Tracer activity instructions are also provided in the *Labs & Study Guide*. The PKA files are available in the online course.



Lab

Lab 1.9.3: Research IT and Networking Job Opportunities



Packet Tracer Activity

Packet Tracer 1.5.7: Network Representation

Check Your Understanding Questions

Complete all the review questions listed here to test your understanding of the topics and concepts in this chapter. The appendix “Answers to ‘Check Your Understanding’ Questions” lists the answers.

1. During a routine inspection, a technician discovered that software that was installed on a computer was secretly collecting data about websites that were visited by users of the computer. Which type of threat is affecting this computer?
 - a. DoS attack
 - b. identity theft
 - c. spyware
 - d. zero-day attack

2. Which term refers to a network that provides secure access to the corporate offices for suppliers, customers, and collaborators?
 - a. internet
 - b. intranet
 - c. extranet
 - d. extended net
3. A large corporation has modified its network to allow users to access network resources from their personal laptops and smartphones. Which networking trend does this describe?
 - a. cloud computing
 - b. online collaboration
 - c. bring your own device
 - d. video conferencing
4. What is an ISP?
 - a. It is a standards body that develops cabling and wiring standards for networking.
 - b. It is a protocol that establishes how computers in a local network communicate.
 - c. It is an organization that enables individuals and businesses to connect to the internet.
 - d. It is a networking device that combines the functionality of several different networking devices in one.
5. For which of the following would the use of a WISP be recommended?
 - a. an internet café in a city
 - b. a farm in a rural area without wired broadband access
 - c. any home with multiple wireless devices
 - d. an apartment in a building with cable access to the internet
6. What characteristic of a network enables it to quickly grow to support new users and applications without impacting the performance of the service being delivered to existing users?
 - a. reliability
 - b. scalability
 - c. quality of service
 - d. accessibility

7. A college is building a new dormitory on its campus. Workers are digging in the ground to install a new water pipe for the dormitory. A worker accidentally damages a fiber-optic cable that connects two of the existing dormitories to the campus data center. Although the cable has been cut, students in the dormitories experience only a very short interruption of network services. What characteristic of the network is described here?
- a. quality of service
 - b. scalability
 - c. security
 - d. fault tolerance
 - e. integrity
8. What are two characteristics of a scalable network? (Choose two.)
- a. easily overloaded with increased traffic
 - b. grows in size without impacting existing users
 - c. is not as reliable as a small network
 - d. suitable for modular devices that allow for expansion
 - e. offers limited number of applications
9. Which device performs the function of determining the path that messages should take through internetworks?
- a. a router
 - b. a firewall
 - c. a web server
 - d. a DSL modem
10. Which two internet connection options do not require that physical cables be run to a building? (Choose two.)
- a. DSL
 - b. cellular
 - c. satellite
 - d. dialup
 - e. dedicated leased line

11. What type of network must a home user access in order to do online shopping?
 - a. an intranet
 - b. the internet
 - c. an extranet
 - d. a local area network
12. How does BYOD change the way businesses implement networks?
 - a. BYOD requires organizations to purchase laptops rather than desktops.
 - b. BYOD users are responsible for their own network security, thus reducing the need for organizational security policies.
 - c. BYOD devices are more expensive than devices that are purchased by an organization.
 - d. BYOD provides flexibility in where and how users can access network resources.
13. An employee wants to access the network of an organization remotely, in the safest possible way. What network feature would allow an employee to gain secure remote access to a company network?
 - a. ACL
 - b. IPS
 - c. VPN
 - d. BYOD
14. What is the internet?
 - a. It is a network based on Ethernet technology.
 - b. It provides network access for mobile devices.
 - c. It provides connections through interconnected global networks.
 - d. It is a private network for an organization with LAN and WAN connections.
15. What are two functions of end devices on a network? (Choose two.)
 - a. They originate the data that flows through the network.
 - b. They direct data over alternate paths in the event of a link failure.
 - c. They filter the flow of data to enhance security.
 - d. They are the interface between humans and the communications network.
 - e. They provide the channel over which the network message travels.

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Basic Switch and End Device Configuration

Objectives

Upon completion of this chapter, you will be able to answer the following questions:

- How do you access a Cisco IOS device for configuration purposes?
- How do you navigate Cisco IOS to configure network devices?
- What is the command structure of Cisco IOS software?
- How do you configure a Cisco IOS device using the CLI?
- How do you use IOS commands to save the running configuration?
- How do devices communicate across network media?
- How do you configure a host device with an IP address?
- How do you verify connectivity between two end devices?

Key Terms

This chapter uses the following key terms. You can find the definitions in the glossary at the end of the book.

kernel page 46

shell page 46

command-line interface (CLI) page 46

graphical user interface (GUI) page 46

Cisco Internetwork Operating System (IOS) page 48

firmware page 48

console page 49

Secure Shell (SSH) page 50

Telnet page 50

user EXEC mode page 53

privileged EXEC mode page 53

global configuration mode page 53

ping page 57

traceroute page 57

virtual terminal (vty) page 64

nonvolatile random-access memory page 67

random-access memory page 67

IPv4 address page 72

subnet mask page 72

IPv6 address page 72

switch virtual interface (SVI) page 74

Dynamic Host Configuration Protocol (DHCP) page 76

Domain Name System (DNS) page 76

Introduction (2.0)

As part of your career in networking, you might have to set up a new network or maintain and upgrade an existing one. In either case, you'll configure switches and end devices so that they are secure and perform effectively based on your requirements.

Out of the box, switches and end devices come with some general configuration. But for your particular network, switches and end devices require your specific information and instructions. In this module, you will learn how to access Cisco IOS network devices. You will learn basic configuration commands and use them to configure and verify a Cisco IOS device and an end device with an IP address.

Of course, there is much more to network administration, but none of that can happen until switches and end devices are configured. Let's get started!

Cisco IOS Access (2.1)

This section introduces the operating system used in most Cisco devices.

Operating Systems (2.1.1)

Every end device and network device must have an operating system (OS). As shown in Figure 2-1, the portion of the OS that interacts directly with computer hardware is known as the *kernel*. The portion that interfaces with applications and the user is known as the *shell*. The user can interact with the shell by using a *command-line interface (CLI)* or a *graphical user interface (GUI)*:

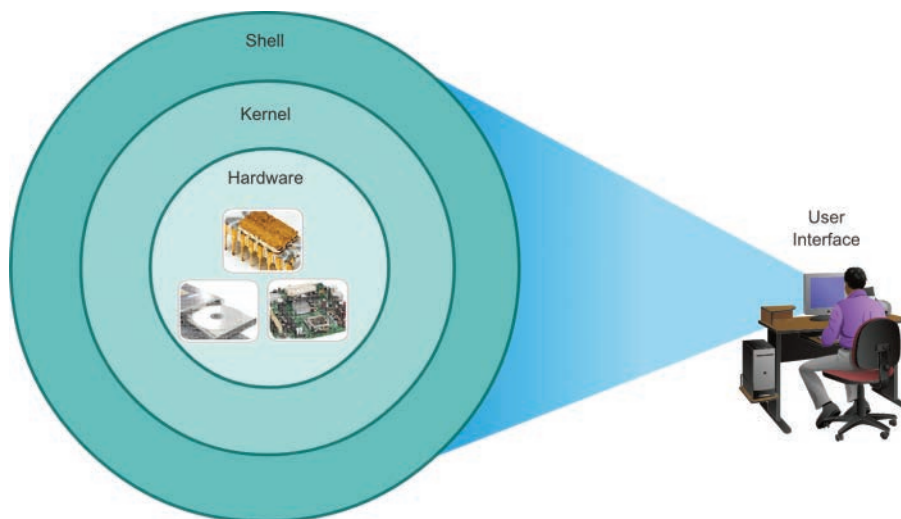


Figure 2-1 Shell, Kernel, and Hardware

- **Shell:** The shell is the user interface that allows users to request specific tasks from the computer. These requests can be made either through the CLI or a GUI.
- **Kernel:** The kernel communicates between the hardware and software of a computer and manages how hardware resources are used to meet software requirements.
- **Hardware:** Hardware is the physical part of a computer, including underlying electronics.

When using a CLI, the user interacts directly with the system in a text-based environment by entering commands on the keyboard at a command prompt, as shown in Example 2-1. The system executes the command and often provides textual output. Operating the CLI requires very little overhead; however, it does require that the user have knowledge of the underlying command structure that controls the system.

Example 2-1 CLI Example

```
analyst@secOps ~]$ ls
Desktop Downloads lab.support.files second_drive
[analyst@secOps ~]$
```

GUI (2.1.2)

A GUI such as Windows, macOS, Linux KDE, Apple iOS, or Android allows a user to interact with a system using an environment of graphical icons, menus, and windows. Compared to a CLI, the Windows 10 GUI example in Figure 2-2 is more user friendly and requires less knowledge of the underlying command structure that controls the system. For this reason, most users rely on GUI environments.

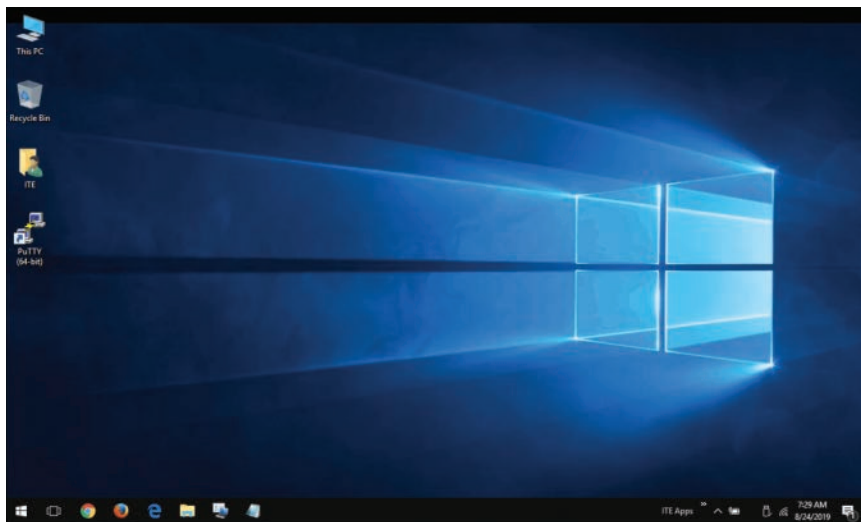


Figure 2-2 Windows 10 GUI

However, GUIs may not always be able to provide all the features available with the CLI. GUIs can also fail, crash, or simply not operate as specified. For these reasons, network devices are typically accessed through a CLI. The CLI is less resource intensive and very stable compared to a GUI.

The family of network operating systems used on many Cisco devices is called *Cisco Internetwork Operating System (IOS)*. Cisco IOS is used on many Cisco routers and switches, regardless of the type or size of the device. Each device type (for example, router or switch) uses a different version of Cisco IOS. Other Cisco operating systems include IOS XE, IOS XR, and NX-OS.

Note

The operating system on home routers is usually called *firmware*. The most common method for configuring a home router is by using a web browser-based GUI.

Purpose of an OS (2.1.3)

Network operating systems are similar to PC operating systems. Through a GUI, a PC operating system enables a user to do the following:

- Use a mouse to make selections and run programs
- Enter text and text-based commands
- View output on a monitor

A CLI-based network operating system (such as Cisco IOS on a switch or router) enables a network technician to do the following:

- Use a keyboard to run CLI-based network programs
- Use a keyboard to enter text and text-based commands
- View output on a monitor

Cisco networking devices run particular versions of Cisco IOS. The IOS version depends on the type of device being used and the required features. While every device comes with a default IOS and feature set, it is possible to upgrade the IOS version or feature set to obtain additional capabilities.

Figure 2-3 lists IOS software releases for a Cisco Catalyst 2960 switch.

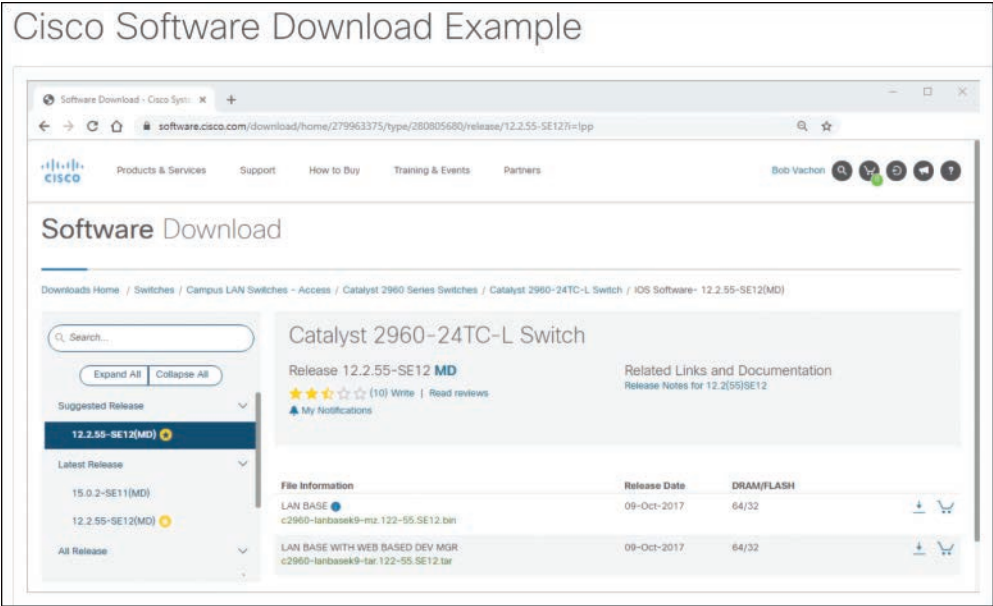


Figure 2-3 Cisco Software Download Web Page

Access Methods (2.1.4)

A switch forwards traffic by default and does not need to be explicitly configured to operate. For example, two configured hosts connected to the same new switch would be able to communicate.

Regardless of the default behavior of a new switch, all switches should be configured and secured. Table 2-1 lists three access methods for configuring and securing a switch.

Table 2-1 Cisco Device Access Methods

Method	Description
<i>Console</i>	This is a physical management port that provides <i>out-of-band</i> access to a Cisco device. <i>Out-of-band</i> access refers to access through a dedicated management channel that is used for device maintenance purposes only. The advantage of using a console port is that the device is accessible even if no networking services are configured, such as when performing the initial configuration. A console connection requires a computer running terminal emulation software and a special console cable to connect to the device.

Method	Description
<i>Secure Shell (SSH)</i>	SSH is an in-band and recommended method for remotely establishing a secure CLI connection through a virtual interface over a network. Unlike a console connection, an SSH connection requires active networking services on the device, including an active interface configured with an address. Most versions of Cisco IOS include an SSH server and an SSH client that can be used to establish SSH sessions with other devices.
<i>Telnet</i>	Telnet is an insecure, in-band method of remotely establishing a CLI session through a virtual interface over a network. Unlike SSH, Telnet does not provide a secure, encrypted connection, and it should be used only in a lab environment. User authentication, passwords, and commands are sent over the network in plaintext. The best practice is to use SSH instead of Telnet. Cisco IOS includes both a Telnet server and a Telnet client.

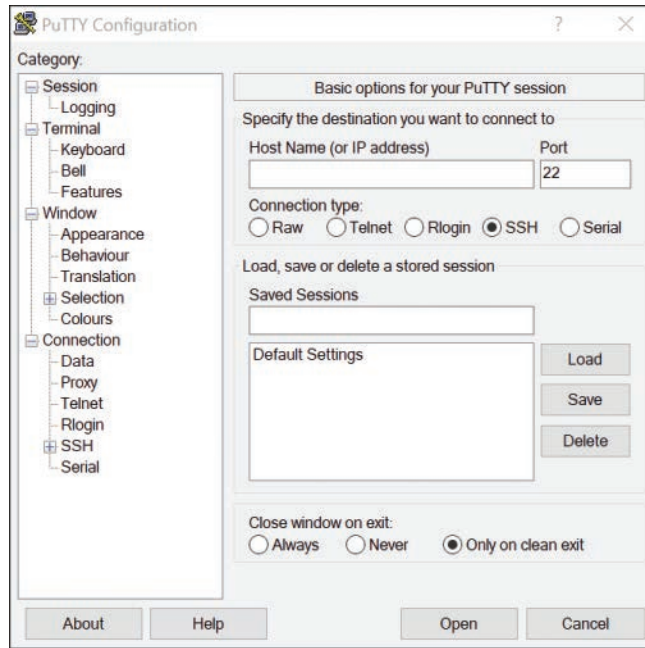
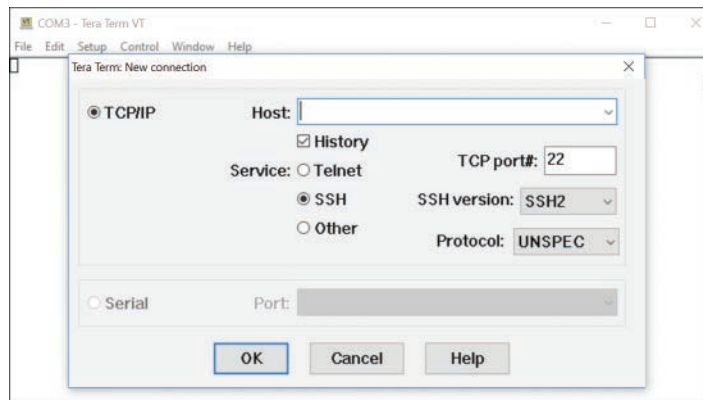
Note

Some devices, such as routers, may also support a legacy auxiliary port that was used to establish a CLI session remotely over a telephone connection using a modem. Similar to a console connection, the AUX port is out-of-band and does not require networking services to be configured or available.

Terminal Emulation Programs (2.1.5)

There are several terminal emulation programs you can use to connect to a networking device either with a serial connection over a console port or with an SSH/Telnet connection. These programs allow you to enhance your productivity by adjusting window sizes, changing font sizes, and changing color schemes.

Figure 2-4 through Figure 2-6 show the GUIs for three popular terminal emulation programs: PuTTY, Tera Term, and SecureCRT.

**Figure 2-4** PuTTY**Figure 2-5** Tera Term

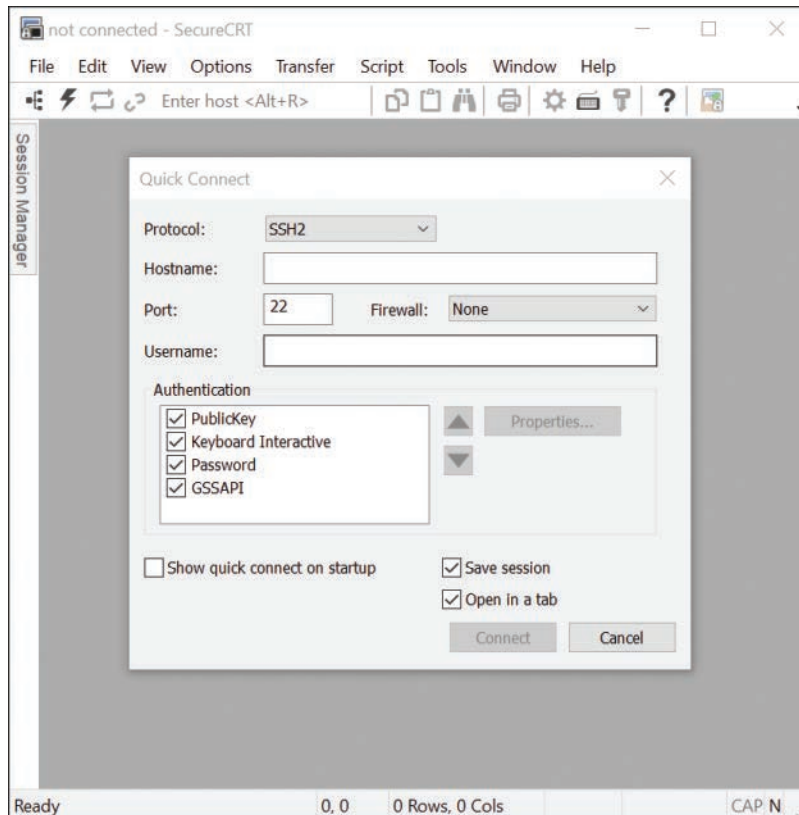


Figure 2-6 SecureCRT

**Interactive
Graphic**

Check Your Understanding—Cisco IOS Access (2.1.6)

Refer to the online course to complete this activity.

IOS Navigation (2.2)

This section introduces the fundamentals of some of the modes of Cisco IOS.

Primary Command Modes (2.2.1)

In the previous section, you learned that every network device requires an OS and that these devices can be configured using the CLI or a GUI. Using the CLI may provide a network administrator with more precise control and flexibility than using the GUI. This section discusses using the CLI to navigate the Cisco IOS.

As a security feature, the Cisco IOS software separates management access into the following two command modes:

- **User EXEC mode:** This mode has limited capabilities but is useful for basic operations. It allows only a limited number of basic monitoring commands and does not allow the execution of any commands that might change the configuration of the device. User EXEC mode is identified by the CLI prompt that ends with the > symbol.
- **Privileged EXEC mode:** To execute configuration commands, a network administrator must access privileged EXEC mode. Higher configuration modes, such as global configuration mode, can be reached only from privileged EXEC mode. Privileged EXEC mode can be identified by the prompt ending with the # symbol.

Table 2-2 summarizes the two modes and displays the default CLI prompts for a Cisco switch and router.

Table 2-2 IOS Command Modes

Command Mode	Description	Default Device Prompts
User EXEC mode	This mode allows access to only a limited number of basic monitoring commands. It is often referred to as “view-only” mode.	Switch> Router>
Privileged EXEC mode	This mode allows access to all commands and features. The user can use any monitoring commands and can execute configuration and management commands.	Switch# Router#

Configuration Mode and Subconfiguration Modes (2.2.2)

To configure a device, a user must enter *global configuration mode*, which is commonly called *global config mode*.

From global config mode, a user can make CLI configuration changes that affect the operation of the device as a whole. Global configuration mode is identified by a prompt that ends with (config)# after the device name, such as **Switch(config)#**.

Global configuration mode is accessed before other specific configuration modes. From global config mode, the user can enter different subconfiguration modes.

Each of these modes allows the configuration of a particular part or function of the IOS device. Two common subconfiguration modes are

- **Line configuration mode:** Used to configure console, SSH, Telnet, or AUX access.
- **Interface configuration mode:** Used to configure a switch port or router network interface.

When the CLI is used, the mode is identified by the command-line prompt that is unique to that mode. By default, every prompt begins with the device name. Following the name, the remainder of the prompt indicates the mode. For example, the default prompt for line configuration mode is **Switch(config-line)#**, and the default prompt for interface configuration mode is **Switch(config-if)#**.

Video

Video—IOS CLI Primary Command Modes (2.2.3)

Refer to the online course to view this video.

Navigate Between IOS Modes (2.2.4)

Various commands are used to move in and out of command prompts. To move from user EXEC mode to privileged EXEC mode, use the **enable** command. Use the **disable** privileged EXEC mode command to return to user EXEC mode.

Note

Privileged EXEC mode is sometimes called *enable mode*.

To move in and out of global configuration mode, use the **configure terminal** privileged EXEC mode command. To return to the privileged EXEC mode, enter the **exit** global config mode command.

There are many different subconfiguration modes. As shown in Example 2-2, to enter line subconfiguration mode, you use the **line** command followed by the management line type and number you wish to access. Use the **exit** command to exit a subconfiguration mode and return to global configuration mode.

Example 2-2 Entering a Subconfiguration Mode

```
Switch(config)# line console 0
Switch(config-line)# exit
Switch(config)#
```

To move from any subconfiguration mode of the global configuration mode to the mode one step above it in the hierarchy of modes, enter the **exit** command.

To move from any subconfiguration mode to the privileged EXEC mode, enter the **end** command, as shown in Example 2-3, or enter the key combination Ctrl+Z.

Example 2-3 Moving Directly Back to Privileged EXEC Mode

```
Switch(config-line)# end  
Switch#
```

You can also move directly from one subconfiguration mode to another. In Example 2-4, notice that after an interface is selected, the command prompt changes from (config-line)# to (config-if)#.

Example 2-4 Moving Between Subconfiguration Modes

```
Switch(config-line)# interface FastEthernet 0/1  
Switch(config-if)#
```

Video

Video—Navigate Between IOS Modes (2.2.5)

Refer to the online course to view this video.

A Note About Syntax Checker Activities (2.2.6)

When you are learning how to modify device configurations, you might want to start in a safe, non-production environment before using real equipment. NetAcad provides a variety of simulation tools to help build your configuration and troubleshooting skills. These simulation tools typically do not have all the functionality of real equipment. One such tool is Syntax Checker. With Syntax Checker, you are given a set of instructions to enter a specific set of commands. You cannot progress in Syntax Checker unless the exact and full command is entered as specified. More advanced simulation tools, such as Packet Tracer, let you enter abbreviated commands, much as you would do on real equipment.

Interactive Graphic

Syntax Checker—Navigate Between IOS Modes (2.2.7)

Use the Syntax Checker activity to navigate between IOS command lines on a switch. Refer to the online course to complete this activity.

**Interactive
Graphic****Check Your Understanding—IOS Navigation (2.2.8)**

Refer to the online course to complete this activity.

The Command Structure (2.3)

Cisco IOS, like other operating systems, uses commands that have a specific structure. To configure an IOS device, a network technician needs to understand this structure. This section introduces the IOS command structure.

Basic IOS Command Structure (2.3.1)

This section covers the basic structure of commands for Cisco IOS. A network administrator must know the basic IOS command structure to be able to use the CLI for device configuration.

A Cisco IOS device supports many commands. Each IOS command has a specific format, or syntax, and can be executed only in the appropriate mode. The general syntax for a command, shown in Figure 2-7, is the command followed by any appropriate keywords and arguments:

- **Keyword:** This is a specific parameter defined in the operating system (in Figure 2-7, **ip protocols**).
- **Argument:** This is not predefined; it is a value or variable defined by the user (in Figure 2-7, **192.168.10.5**).

After entering each complete command, including any keywords and arguments, press the Enter key to submit the command to the command interpreter.

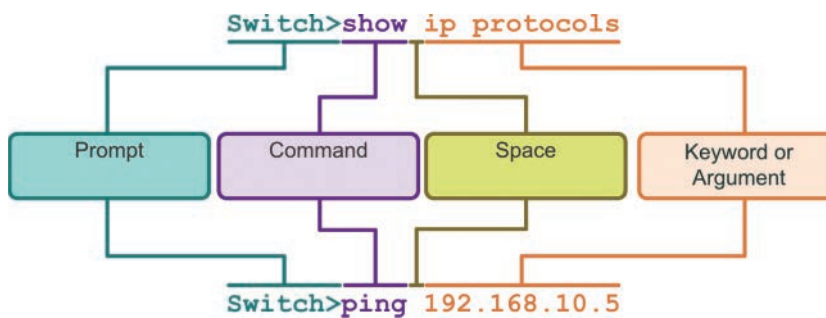


Figure 2-7 Command Syntax

IOS Command Syntax Check (2.3.2)

A command might require one or more arguments. To determine the keywords and arguments required for a command, refer to the command syntax. The syntax provides the pattern, or format, that must be used when entering a command.

As identified in Table 2-3, boldface text indicates commands and keywords that are entered as shown. Italic text indicates an argument for which the user provides the value.

Table 2-3 Command Syntax Conventions

Convention	Description
boldface	Boldface text indicates commands and keywords that you enter literally as shown.
<i>italics</i>	Italic text indicates arguments for which you supply values.
[x]	Square brackets indicate an optional element (keyword or argument).
{x}	Braces indicate a required element (keyword or argument).
[x {y z }]	Braces and vertical lines within square brackets indicate a required choice within an optional element. Spaces are used to clearly delineate parts of the command.

For instance, the syntax for the **description** command is **description** *string*. The argument is a *string* value provided by the user. The **description** command is typically used to identify the purpose of an interface. For example, the command **description Connects to the main headquarter office switch** describes the location of the other device at the end of the connection.

The following examples demonstrate conventions used to document and use IOS commands:

- **ping** *ip-address*: The command is **ping**, and the user-defined argument is the IP address of the destination device (for example, **ping 10.10.10.5**).
- **traceroute** *ip-address*: The command is **traceroute**, and the user-defined argument is the IP address of the destination device (for example, **traceroute 192.168.254.254**).

If a command is complex and has multiple arguments, you might see it represented like this:

```
Switch(config-if)# switchport port-security aging { static | time time | type
{absolute | inactivity}}
```

The command is typically followed by a detailed description of the command and each argument.

The Cisco IOS Command Reference is the ultimate source of information for a particular IOS command.

IOS Help Features (2.3.3)

IOS has two forms of help available: context-sensitive help and command syntax check.

Context-sensitive help enables you to quickly find answers to questions such as these:

- Which commands are available in each command mode?
- Which commands start with specific characters or groups of characters?
- Which arguments and keywords are available for particular commands?

To access context-sensitive help, simply enter a question mark, `?`, at the CLI.

Command syntax check verifies that the user has entered a valid command. When a command is entered, the command-line interpreter evaluates the command from left to right. If the interpreter understands the command, the requested action is executed, and the CLI returns to the appropriate prompt. However, if the interpreter cannot understand the command being entered, it provides feedback describing what is wrong with the command.

Video

Video—Context Sensitive Help and Command Syntax Check (2.3.4)

Refer to the online course to view this video.

Hot Keys and Shortcuts (2.3.5)

The IOS CLI provides hot keys and shortcuts that make configuring, monitoring, and troubleshooting easier.

Commands and keywords can be shortened to the minimum number of characters that identify a unique selection. For example, the **configure** command can be shortened to **conf** because **configure** is the only command that begins with **conf**. An even shorter version, **con**, will not work because more than one command begins with **con**. Keywords can also be shortened.

Table 2-4 lists keystrokes that enhance command-line editing.

Table 2-4 Command-Line Editing Shortcuts

Keystroke	Description
Tab	Completes a partial command name entry.
Backspace	Erases the character to the left of the cursor.
Ctrl+D	Erases the character at the cursor.
Ctrl+K	Erases all characters from the cursor to the end of the command line.
Esc+D	Erases all characters from the cursor to the end of the word.
Ctrl+U or Ctrl+X	Erases all characters from the cursor back to the beginning of the command line.
Ctrl+W	Erases the word to the left of the cursor.
Ctrl+A	Moves the cursor to the beginning of the line.
Left Arrow or Ctrl+B	Moves the cursor one character to the left.
Esc+B	Moves the cursor back one word to the left.
Esc+F	Moves the cursor forward one word to the right.
Right Arrow or Ctrl+F	Moves the cursor one character to the right.
Ctrl+E	Moves the cursor to the end of command line.
Up Arrow or Ctrl+P	Recalls the commands in the history buffer, beginning with the most recent commands.
Ctrl+R or Ctrl+I or Ctrl+L	Redisplays the system prompt and command line after a console message is received.

Note

While the Delete key typically deletes the character to the right of the prompt, the IOS command structure does not recognize the Delete key.

When a command's output produces more text than can be displayed in a terminal window, IOS displays a --More-- prompt. Table 2-5 describes the keystrokes that can be used when this prompt is displayed.

Table 2-5 Keystrokes to Use After a --More-- Prompt

Keystroke	Description
Enter key	Displays the next line.
Spacebar	Displays the next screen.
Any other key	Ends the display string, returning to privileged EXEC mode.

Table 2-6 lists keystrokes used to exit an operation.

Table 2-6 Keystrokes to Exit an Operation

Keystroke	Description
Ctrl+C	When in any configuration mode, ends the configuration mode and returns to privileged EXEC mode. When in setup mode, aborts to the command prompt.
Ctrl+Z	When in any configuration mode, ends the configuration mode and returns to privileged EXEC mode.
Ctrl+Shift+6	All-purpose break sequence used to abort DNS lookups, traceroutes, pings, and so on.

Video

Video—Hot Keys and Shortcuts (2.3.6)

Refer to the online course to view this video.

Packet Tracer
Activity

Packet Tracer—Navigate the IOS (2.3.7)

In this activity, you will practice skills necessary for navigating Cisco IOS, including different user access modes, various configuration modes, and common commands used on a regular basis. You will also practice accessing the context-sensitive help by configuring the `clock` command.



Lab—Navigate the IOS by Using Tera Term for Console Connectivity (2.3.8)

In this lab, you will complete the following objectives:

- Part 1: Access a Cisco Switch Through the Serial Console Port
- Part 2: Display and Configure Basic Device Settings
- Part 3: (Optional) Access a Cisco Router Using a Mini-USB Console Cable

Basic Device Configuration (2.4)

Before devices can be used in a network, they need to be configured. This section introduces the basic configuration of Cisco IOS devices.

Device Names (2.4.1)

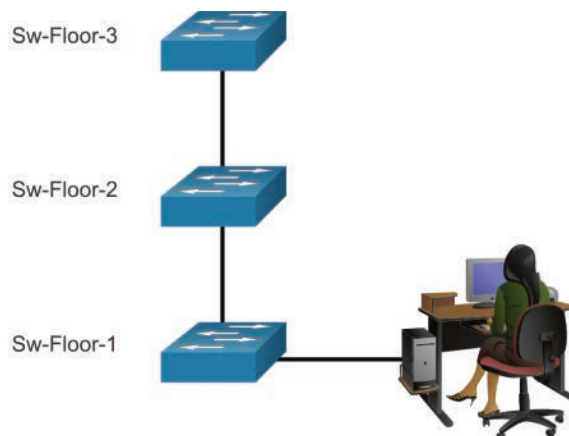
You have learned a great deal about Cisco IOS, navigating IOS, and the command structure. Now, you are ready to configure devices! The first configuration command on any device should be to give it a unique device name or hostname. By default, every device is assigned a factory default name. For example, a Cisco IOS switch is named “Switch.”

The problem is that if all switches in a network were left with their default names, it would be difficult to identify a specific device. For instance, how would you know that you were connected to the right device when accessing it remotely using SSH? The hostname provides confirmation that you are connected to the correct device.

The default name should be changed to something more descriptive. Choosing names wisely makes it easier to remember, document, and identify network devices. Here are some important naming guidelines for hosts:

- Start with a letter
- Include no spaces
- End with a letter or digit
- Use only letters, digits, and dashes
- Be fewer than 64 characters in length

An organization must choose a naming convention that makes it easy and intuitive to identify a specific device. The hostnames used in the device IOS preserve capitalization and lowercase characters. For example, Figure 2-8 shows that three switches, spanning three different floors, are interconnected together in a network. The naming convention that was used incorporates the location and the purpose of each device. Network documentation should explain how these names are chosen so additional devices can be named accordingly.



When network devices are named, they are easy to identify for configuration purposes.

Figure 2-8 Example of Naming Multiple Devices for Easy Identification

When the naming convention has been identified, the next step is to use the CLI to apply names to the devices. As shown in Example 2-5, from the privileged EXEC mode, access the global configuration mode by entering the **configure terminal** command. Notice the change in the command prompt.

Example 2-5 Configuring a Hostname

```
Switch# configure terminal
Switch(config)# hostname Sw-Floor-1
Sw-Floor-1(config)#
```

From global configuration mode, enter the command **hostname** followed by the name of the switch and press Enter. Notice the change in the command prompt name.

Note

To return the switch to the default prompt, use the **no hostname** global config command.

Always make sure the documentation is updated each time a device is added or modified. Identify each device in the documentation by its location, purpose, and address.

Password Guidelines (2.4.2)

The use of weak or easily guessed passwords continues to be the biggest security concern of organizations. Network devices, including home wireless routers, should always have passwords configured to limit administrative access.

Cisco IOS can be configured to use hierarchical mode passwords to allow different access privileges on a network device.

All networking devices should limit administrative access by having privileged EXEC, user EXEC, and remote Telnet access secured with passwords. In addition, all passwords should be encrypted, and legal notifications should be provided.

When choosing passwords, use strong passwords that are not easily guessed. These are some key points to consider when choosing passwords:

- Use passwords that are more than eight characters in length.
- Use a combination of uppercase and lowercase letters, numbers, special characters, and/or numeric sequences.
- Avoid using the same password for multiple devices.
- Do not use common words because they are easily guessed.

Use an internet search to find a password generator. Many of them allow you to set the length, character set, and other parameters.

Note

Most of the labs in this course use simple passwords such as **cisco** or **class**. These passwords are considered weak and easily guessable and should be avoided in production environments. We only use these passwords for convenience in a classroom setting or to illustrate configuration examples.

Configure Passwords (2.4.3)

When you initially connect to a device, you are in user EXEC mode. This mode is secured using the console.

To secure user EXEC mode access, enter line console configuration mode by using the **line console 0** global configuration command, as shown in Example 2-6. The zero is used to represent the first (and in most cases the only) console interface. Next, specify the user EXEC mode password by using the **password password** command. Finally, enable user EXEC access by using the **login** command.

Example 2-6 Setting a Console Password

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line console 0
Sw-Floor-1(config-line)# password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

Console access now requires a password before allowing access to the user EXEC mode.

To have administrator access to all IOS commands, including those for configuring a device, you must gain privileged EXEC mode access. It is the most important access method because it provides complete access to a device.

To secure privileged EXEC access, use the **enable secret** *password* global config command, as shown in Example 2-7.

Example 2-7 Setting a Privileged EXEC Password

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# enable secret class
Sw-Floor-1(config)# exit
Sw-Floor-1#
```

Virtual terminal (vty) lines enable remote access using Telnet or SSH to the device. Many Cisco switches support up to 16 vty lines, numbered 0 to 15.

To secure vty lines, enter line vty mode by using the **line vty 0 15** global config command. Next, specify the vty password by using the **password** *password* command. Finally, enable vty access by using the **login** command.

Example 2-8 shows an example of securing the vty lines on a switch.

Example 2-8 Setting a Remote Access Password

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line vty 0 15
Sw-Floor-1(config-line)# password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

Encrypt Passwords (2.4.4)

The startup-config and running-config files display most passwords in plaintext. This is a security threat because anyone who has access to these files can discover the passwords.

To encrypt all plaintext passwords, use the **service password-encryption** global config command, as shown in Example 2-9.

Example 2-9 Encrypting Passwords

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# service password-encryption
Sw-Floor-1(config)#
```

This command applies weak encryption to all unencrypted passwords. This encryption applies only to passwords in the configuration file, not to passwords as they are sent over the network. The purpose of this command is to keep unauthorized individuals from viewing passwords in the configuration file.

Use the **show running-config** command to verify that passwords are now encrypted, as shown in Example 2-10.

Example 2-10 Verifying That Passwords Are Encrypted

```
Sw-Floor-1(config)# end
Sw-Floor-1# show running-config
!
!
<Output omitted>
!
line con 0
password 7 094F471A1A0A
login
!
line vty 0 4
password 7 03095A0F034F38435B49150A1819
login
line vty 5 15
password 7 094F471A1A0A
login
!
!
!
end
```

Banner Messages (2.4.5)

Although requiring passwords is one way to keep unauthorized personnel out of a network, it is vital to provide a method for declaring that only authorized personnel should attempt to access a device. To do this, you can add a banner to the device output. Banners can be an important part of the legal process in the event that someone is prosecuted for breaking into a device. Some legal systems do not allow prosecution, or even the monitoring of users, unless a notification is visible.

To create a banner message of the day on a network device, use the **banner motd # the message of the day #** global config command. The # in the command syntax is called the *delimiting character*. It is entered before and after the message. The delimiting character can be any character as long as it does not occur in the message.

For this reason, symbols such as the # are often used. After the command is executed, the banner will be displayed on all subsequent attempts to access the device until the banner is removed.

Example 2-11 shows the steps to configure the banner on Sw-Floor-1.

Example 2-11 Configuring a Banner

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# banner motd #Authorized Access Only#
```

Video**Video—Secure Administrative Access to a Switch (2.4.6)**

Refer to the online course to view this video.

**Interactive
Graphic****Syntax Checker—Basic Device Configuration (2.4.7)**

Secure management access to a switch as follows:

- Assign a device name.
- Secure user EXEC mode access.
- Secure privileged EXEC mode access.
- Secure vty access.
- Encrypt all plaintext passwords.
- Display a login banner.

Refer to the online course to complete this activity.

**Interactive
Graphic****Check Your Understanding—Basic Device Configuration (2.4.8)**

Refer to the online course to complete this activity.

Save Configurations (2.5)

Configuration changes to Cisco IOS-based devices are made to the running configuration. This working configuration should be backed up to support network recovery. This section examines some of the methods used to back up and restore the running configuration on Cisco IOS devices.

Configuration Files (2.5.1)

You now know how to perform basic configuration on a switch, including setting passwords and banner messages. This section shows you how to save your configurations.

Two system files store the device configuration:

- **startup-config:** This is the saved configuration file that is stored in *nonvolatile random-access memory (NVRAM)*. It contains all the commands that will be used by the device upon startup or reboot. Flash does not lose its contents when the device is powered off.

Note

Routers save the configuration file in a single location known as startup-config in NVRAM. Many switches save the configuration in two connected files: startup-config in NVRAM and the config.txt file in flash memory.

- **running-config:** This is stored in *random-access memory (RAM)*. It reflects the current configuration. Modifying a running configuration affects the operation of a Cisco device immediately. RAM is volatile memory. It loses all of its content when the device is powered off or restarted.

The **show running-config** privileged EXEC mode command is used to view the running config. As shown in Example 2-12, the command lists the complete configuration currently stored in RAM.

Example 2-12 Verifying the Configuration Stored in RAM

```
Sw-Floor-1# show running-config
Building configuration...
Current configuration : 1351 bytes
!
! Last configuration change at 00:01:20 UTC Mon Mar 1 1993
!
version 15.0
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname Sw-Floor-1
!
```

To view the startup configuration file, use the **show startup-config** privileged EXEC command.

If power to a device is lost, or if a device is restarted, all configuration changes will be lost unless they have been saved. To save changes made to the running configuration to the startup configuration file, use the **copy running-config startup-config** privileged EXEC mode command.

Alter the Running Configuration (2.5.2)

If changes made to the running config do not have the desired effect and the running-config has not yet been saved, you can restore the device to its previous configuration. Remove the changed commands individually or reload the device by using the **reload** privileged EXEC mode command to restore the startup-config.

The downside to using the **reload** command to remove an unsaved running config is the brief amount of time the device will be offline, causing network downtime.

When a reload is initiated, IOS detects that the running config has changes that were not saved to the startup configuration. A prompt appears, asking whether to save the changes. To discard the changes, enter **n** or **no**.

Alternatively, if undesired changes were saved to the startup config, it may be necessary to clear all the configurations. This requires erasing the startup config and restarting the device. The startup config is removed by using the **erase startup-config** privileged EXEC mode command. After the command is issued, the switch prompts you for confirmation. Press Enter to accept.

After removing the startup config from NVRAM, reload the device to remove the current running config file from RAM. On reload, a switch loads the default startup config that originally shipped with the device.

Video

Video—Alter the Running Configuration (2.5.3)

Refer to the online course to view this video.

Capture Configuration to a Text File (2.5.4)

Configuration files can be saved and archived to a text document. The following sequence of steps ensures that a working copy of the configuration file is available for editing or reuse later.

For example, assume that a switch has been configured, and the running config has been saved on the device. Follow these steps:



- Step 1.** Open terminal emulation software, such as PuTTY (see Figure 2-9) or Tera Term, that is already connected to a switch.

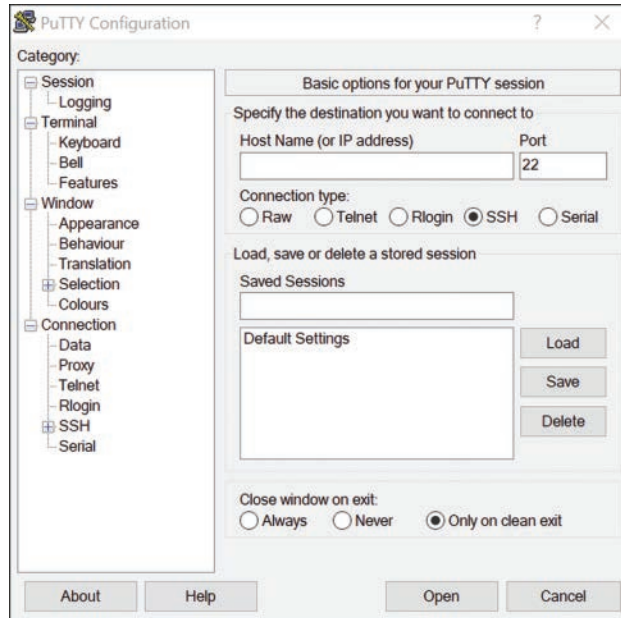


Figure 2-9 PuTTY Startup Screen

Step 2. Enable logging in the terminal software and assign a name and file location for saving the log file. Figure 2-10 shows that **All session output** will be captured to the file specified (that is, MySwitchLogs).

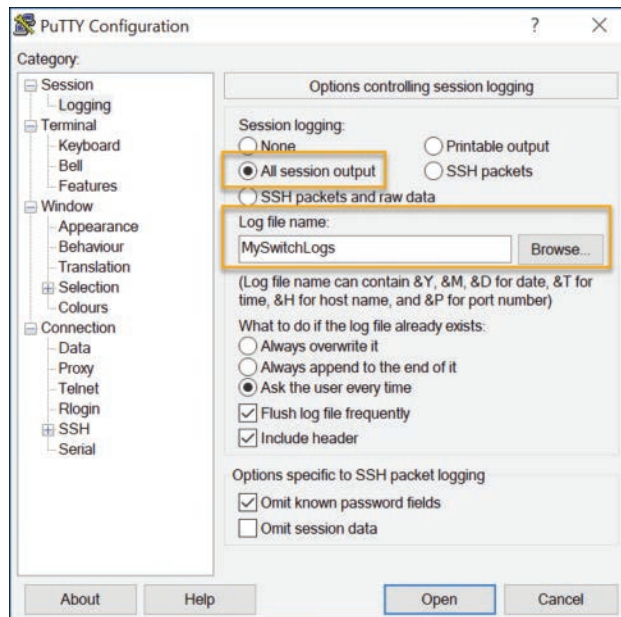


Figure 2-10 Setting PuTTY to Log a Session to a Text File

- Step 3.** Execute the **show running-config** command, as shown in Example 2-13, or the **show startup-config** command at the privileged EXEC prompt. Text displayed in the terminal window is then placed in the chosen file.

Example 2-13 Displaying and Logging a Configuration to a Text File

```
Sw-Floor-1# show running-config
Building configuration...
```

- Step 4.** Disable logging in the terminal software. Figure 2-11 shows how to disable logging by choosing the **None** session logging option.

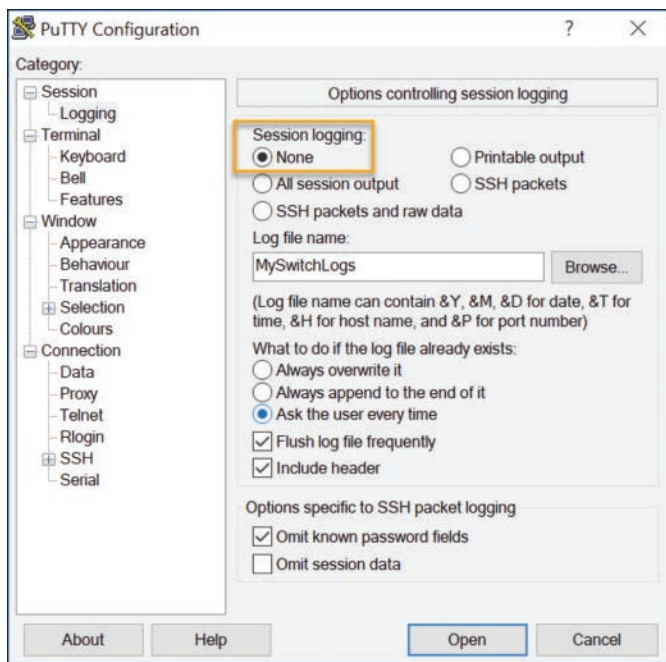


Figure 2-11 Turning Off Session Logging

The text file created can be used as a record of how the device is currently implemented. The file might need to be edited before being used to restore a saved configuration to a device.

To restore a configuration file to a device:



- Step 1.** Enter global configuration mode on the device.
- Step 2.** Copy and paste the text file into the terminal window connected to the switch.

The text in the file is applied as commands in the CLI and becomes the running configuration on the device. This is a convenient method of manually configuring a device.

Packet Tracer
Activity

Packet Tracer—Configure Initial Switch Settings (2.5.5)

In this activity, you will perform basic switch configurations. You will secure access to the CLI and console ports using encrypted and plaintext passwords. You will learn how to configure messages for users logging in to the switch. These banners will also be used to warn unauthorized users that access is prohibited.

Ports and Addresses (2.6)

For devices to communicate on a network, each device must have addressing information applied. This section introduces IP addresses, interfaces, and ports.

IP Addresses (2.6.1)

Congratulations, you have performed a basic device configuration! Of course, the fun is not over yet. If you want your end devices to communicate with each other, you must ensure that each of them has an appropriate IP address and is correctly connected. You will learn about IP addresses, device ports, and the media used to connect devices in this section.

The use of IP addresses is the primary means of enabling devices to locate one another and establish end-to-end communication on the internet. Each end device on a network must be configured with an IP address. Examples of end devices include

- Computers (workstations, laptops, file servers, web servers)
- Network printers
- VoIP phones
- Security cameras
- Smartphones
- Mobile handheld devices (such as wireless barcode scanners)

The structure of an *IPv4 address* is called dotted-decimal notation; with this notation, an address is represented using four decimal numbers between 0 and 255. IPv4 addresses are assigned to individual devices connected to a network.

Note

IP in this course refers to both the IPv4 and IPv6 protocols. IPv6 is the most recent version of IP and is replacing the more common IPv4.

With an IPv4 address, a *subnet mask* is also necessary. An IPv4 subnet mask is a 32-bit value that differentiates the network portion of the address from the host portion. Coupled with the IPv4 address, the subnet mask determines to which subnet a device is a member.

The example in Figure 2-12 displays the IPv4 address (192.168.1.10), subnet mask (255.255.255.0), and default gateway (192.168.1.1) assigned to a host. The default gateway address is the IP address of the router that the host will use to access remote networks, including the internet.

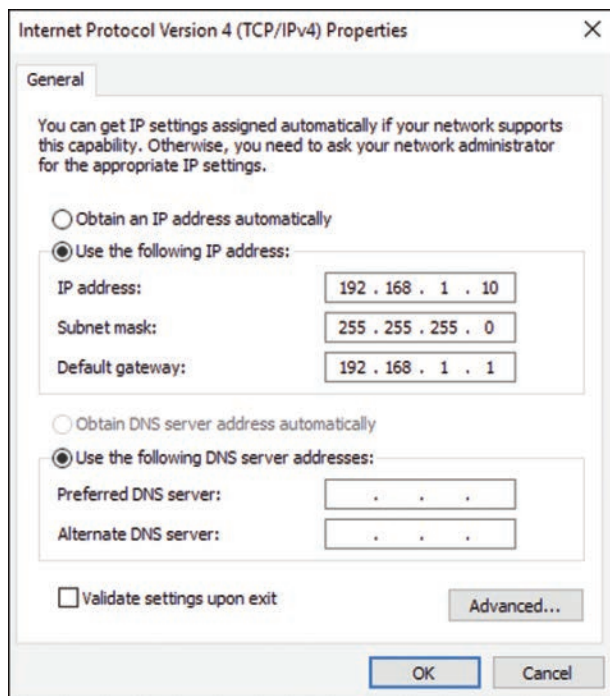


Figure 2-12 Configuring or Verifying IPv4 Addressing on a Windows Host

An *IPv6 address* is 128 bits in length and written as a string of hexadecimal values, as shown in Figure 2-13. Every 4 bits is represented by a single hexadecimal digit, for a total of 32 hexadecimal values. Groups of four hexadecimal digits are separated by a colon (:). IPv6 addresses are not case-sensitive and can be written in either lower-case or uppercase.

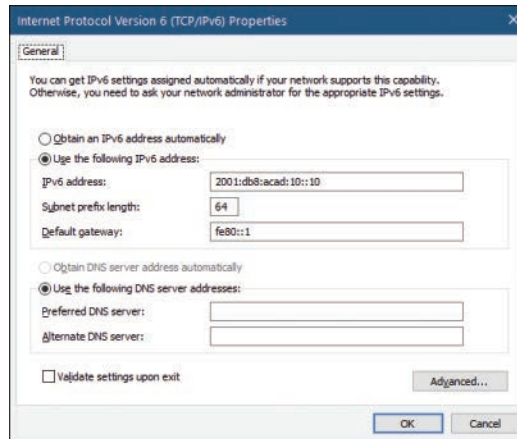


Figure 2-13 Configuring or Verifying IPv6 Addressing on a Windows Host

Interfaces and Ports (2.6.2)

Network communications depend on end-user device interfaces, networking device interfaces, and the cables that connect them. Each physical interface has specifications, or standards, that define it. A cable connecting to the interface must be designed to match the physical standards of the interface. Types of network media include twisted-pair copper cables, fiber-optic cables, coaxial cables, and wireless, as shown in Figure 2-14.

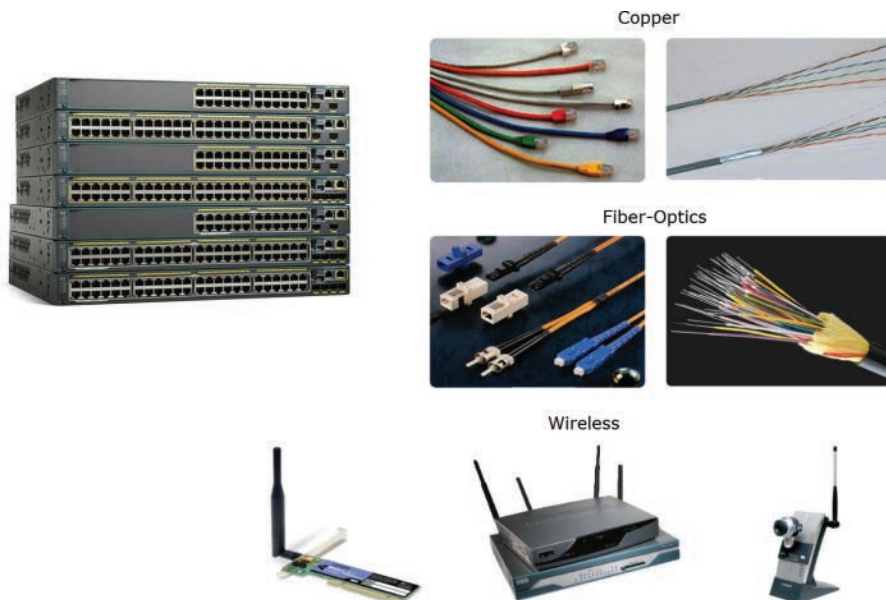


Figure 2-14 Types of Network Media

Different types of network media have different features and benefits. Not all network media have the same characteristics. Not all media are appropriate for the same purpose. These are some of the differences between various types of media:

- Distance the media can successfully carry signals
- Environment in which the media is to be installed
- Amount of data and the speed at which it must be transmitted
- Cost of the media and installation

Not only does each link on the internet require a specific network media type, but each link also requires a particular network technology. For example, Ethernet is the most common local-area network (LAN) technology used today. Ethernet ports are found on end-user devices, switch devices, and other networking devices that can physically connect to a network by using a cable.

Cisco IOS Layer 2 switches have physical ports for devices to connect. These ports do not support Layer 3 IP addresses. Therefore, a switch has one or more *switch virtual interfaces (SVIs)*. These are virtual interfaces because there is no physical hardware on the device associated with it; an SVI is created in software.

The virtual interface lets you remotely manage a switch over a network by using IPv4 and IPv6. Each switch comes with one SVI appearing in the default configuration “out of the box.” The default SVI is interface VLAN 1.

Note

A Layer 2 switch does not need an IP address. The IP address assigned to the SVI is used to remotely access the switch. An IP address is not necessary for the switch to perform its operations.

Interactive Graphic

Check Your Understanding—Ports and Addresses (2.6.3)

Refer to the online course to complete this activity.

Configure IP Addressing (2.7)

This section introduces how IP addresses are applied to end devices and to an Ethernet switch for remote access.

Manual IP Address Configuration for End Devices (2.7.1)

Much as you need a telephone number to text or call a friend, an end device in a network needs an IP address in order to communicate with other devices on the network. In this section, you will see how to implement basic connectivity by configuring IP addressing on switches and PCs.

IPv4 address information can be entered into end devices manually or automatically using Dynamic Host Configuration Protocol (DHCP).

To manually configure an IPv4 address on a Windows host, open the **Control Panel > Network Sharing Center > Change adapter settings** and choose the adapter. Next, right-click and select **Properties** to display the Ethernet Properties dialog, as shown in Figure 2-15.

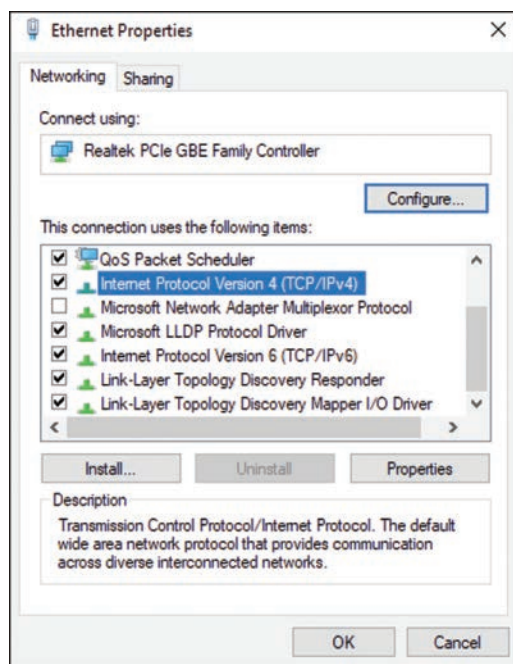


Figure 2-15 Accessing IPv4 Properties on a Windows Host

Highlight **Internet Protocol Version 4 (TCP/IPv4)** and click **Properties** to open the Internet Protocol Version 4 (TCP/IPv4) Properties window. Configure the IPv4 address and subnet mask information, as well as the default gateway, as shown in Figure 2-16.

Note

IPv6 addressing and configuration options are similar to the IPv4 options.

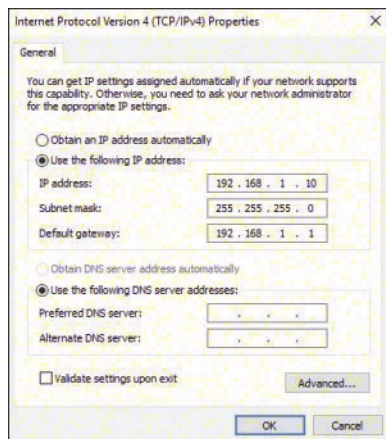


Figure 2-16 Manually Configuring IPv4 Addressing on a Windows Host

Note

The DNS server addresses are the IPv4 and IPv6 addresses of the Domain Name System (DNS) servers, which are used to translate IP addresses to domain names, such as www.cisco.com.

Automatic IP Address Configuration for End Devices (2.7.2)

End devices typically default to using *Dynamic Host Configuration Protocol (DHCP)* for automatic IPv4 address configuration. DHCP is a technology that is used in almost every network. The best way to understand why DHCP is so popular is by considering all the extra work that would have to take place without it.

In a network, DHCP enables automatic IPv4 address configuration for every end device that is DHCP enabled. Imagine the amount of time it would take if every time you connected to the network, you had to manually enter the IPv4 address, the subnet mask, the default gateway, and the *Domain Name System (DNS)* server. Multiply that by every user and every device in an organization, and you see the problem. Manual configuration also increases the chance of misconfiguration by duplicating another device's IPv4 address.

As shown in Figure 2-17, to configure DHCP on a Windows PC, you only need to select **Obtain an IP address automatically** and **Obtain DNS server address automatically**. Your PC then searches out a DHCP server and is assigned the address settings necessary to communicate on the network.

Note

IPv6 uses DHCPv6 and SLAAC (stateless address autoconfiguration) for dynamic address allocation.



Figure 2-17 Setting a Windows Host to Obtain IPv4 Addressing Automatically

**Interactive
Graphic**

Syntax Checker—Verify Windows PC IP Configuration (2.7.3)

It is possible to display the IP configuration settings on a Windows PC by using the **ipconfig** command at the command prompt. The output will show the IPv4 address, subnet mask, and gateway information received from the DHCP server. Enter the **ipconfig** command to display the IP configuration on a Windows PC. Refer to the online course to complete this activity.

Switch Virtual Interface Configuration (2.7.4)

To access a switch remotely, an IP address and a subnet mask must be configured on the SVI. To configure an SVI on a switch, use the **interface vlan 1** global configuration command (where vlan 1 is not an actual physical interface but a virtual one), as shown in Example 2-14. Next, assign an IPv4 address by using the **ip address ip-address subnet-mask** interface configuration command. Finally, enable the virtual interface by using the **no shutdown** interface configuration command. Much like a Windows host, a switch configured with an IPv4 address typically also needs to have a default gateway assigned. You can assign the default gateway by using the **ip default-gateway ip-address** global configuration command, where *ip-address* is the IPv4 address of the local router on the network.

After these commands are configured, the switch has all the IPv4 elements ready for communication over the network.

Example 2-14 Configuring the SVI on a Switch

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# interface vlan 1
Sw-Floor-1(config-if)# ip address 192.168.1.20 255.255.255.0
Sw-Floor-1(config-if) ip default-gateway 192.168.1.1
Sw-Floor-1(config-if)# no shutdown
```

**Interactive
Graphic****Syntax Checker—Configure a Switch Virtual Interface (2.7.5)**

Refer to the online course to complete this activity.

**Packet Tracer
Activity****Packet Tracer—Implement Basic Connectivity (2.7.6)**

In this activity, you will first perform basic switch configurations. Then you will implement basic connectivity by configuring IP addressing on switches and PCs. When the IP addressing configuration is complete, you will use various **show** commands to verify configurations and use the **ping** command to verify basic connectivity between devices.

Verify Connectivity (2.8)

A principal troubleshooting technique is to verify the logical connectivity between two or more IPv4 devices. This section references the video activities that demonstrate how to verify connectivity.

Video**Video Activity—Test the Interface Assignment (2.8.1)**

In the previous section, you implemented basic connectivity by configuring IP addressing on switches and PCs. Then you verified your configurations and connectivity to verify that the configurations were working.

You will continue this process in this section. Using the CLI, you will verify the interfaces and the addresses of the switches and routers in your network. In the same way that you use commands and utilities like **ipconfig** to verify the network configuration of a PC host, you use particular commands to verify the interfaces and address settings of intermediary devices such as switches and routers.

Refer to the online course to view this video.

Video**Video Activity—Test End-to-End Connectivity (2.8.2)**

The **ping** command can be used to test connectivity to another device on the network or a website on the internet.

Summary (2.9)

The following is a summary of the topics in the chapter and their corresponding online modules.

Cisco IOS Access

Every end device and network device requires an operating system (OS). The user can interact with the shell by using a keyboard and a command-line interface (CLI) to run CLI-based network programs, use a keyboard to enter text and text-based commands, and view output on a monitor.

As a security feature, the Cisco IOS software separates management access into the two command modes: user EXEC mode and privileged EXEC mode.

IOS Navigation

Global configuration mode is accessed before other specific configuration modes. From global config mode, the user can enter different subconfiguration modes. Each of these modes allows the configuration of a particular part or function of the IOS device. Two common subconfiguration modes are line configuration mode and interface configuration mode. To move in and out of global configuration mode, use the **configure terminal** privileged EXEC mode command. To return to the privileged EXEC mode, enter the **exit** global config mode command.

The Command Structure

Each IOS command has a specific format, or syntax, and can be executed only in the appropriate mode. The general syntax for a command is the command followed by any appropriate keywords and arguments. IOS has two forms of help available: context-sensitive help and command syntax check.

Basic Device Configuration

The first configuration command on any device should be to give it a unique device name or hostname. Network devices should always have passwords configured to limit administrative access. Cisco IOS can be configured to use hierarchical mode passwords to allow different access privileges for a network device. It is important to configure and encrypt all passwords. It is also important to provide a method for declaring that only authorized personnel should attempt to access a device by adding a banner to the device output.

Save Configurations

Two system files store a device's configuration: startup-config and running-config. It is possible to alter running configuration files if they have not been saved. It is also possible to save configuration files and archived them to text documents.

Ports and Addresses

IP addresses enable devices to locate one another and establish end-to-end communication on the internet. Each end device on a network must be configured with an IP address. The structure of an IPv4 address is called dotted-decimal notation; with this notation, an address is represented using four decimal numbers between 0 and 255.

Configure IP Addressing

IPv4 address information can be entered into end devices manually or automatically using Dynamic Host Configuration Protocol (DHCP). In a network, DHCP enables automatic IPv4 address configuration for every end device that is DHCP enabled. To access a switch remotely, an IP address and a subnet mask must be configured on the SVI. To configure an SVI on a switch, use the **interface vlan 1** global configuration command (where vlan 1 is not an actual physical interface but a virtual one).

Verify Connectivity

In the same way that you use commands and utilities to verify a PC host's network configuration, you also use commands to verify the interfaces and address settings of intermediary devices such as switches and routers. The **show ip interface brief** command verifies the condition of a switch interface. The **ping** command can be used to test connectivity to another device on the network or a website on the internet.

Packet Tracer
 Activity

Packet Tracer—Basic Switch and End Device Configuration (2.9.1)

As a recently hired LAN technician, you have been asked by your network manager to demonstrate your ability to configure a small LAN. Your tasks include configuring initial settings on two switches by using Cisco IOS and configuring IP address parameters on host devices to provide end-to-end connectivity. You are to use two switches and two hosts on a cabled and powered network.



Lab—Basic Switch and End Device Configuration (2.9.2)

In this lab, you will complete the following objectives:

- Part 1: Set Up the Network Topology

- Part 2: Configure PC Hosts
 - Part 3: Configure and Verify Basic Switch Settings
-

Practice

The following activities provide practice with the topics introduced in this chapter. The lab is available in the companion *Introduction to Networks Labs & Study Guide (CCNAv7)* (ISBN 9780136634454). The Packet Tracer activity instructions are also provided in the *Labs & Study Guide*. The PKA files are available in the online course.



Labs

Lab 2.3.8: Navigate the IOS by Using Tera Term for Console Connectivity

Lab 2.9.2: Basic Switch and End Device Configuration



Packet Tracer Activities

Packet Tracer 2.3.7: Navigate the IOS

Packet Tracer 2.5.5: Configure Initial Switch Settings

Packet Tracer 2.7.6: Implement Basic Connectivity

Packet Tracer 2.9.1: Basic Switch and End Device Configuration

Check Your Understanding Questions

Complete all the review questions listed here to test your understanding of the topics and concepts in this chapter. The appendix “Answers to ‘Check Your Understanding’ Questions” lists the answers.

1. Which statement is true about the running configuration file in a Cisco IOS device?
 - a. It affects the operation of the device immediately when modified.
 - b. It is stored in NVRAM.
 - c. It should be deleted using the **erase running-config** command.
 - d. It is automatically saved when the router reboots.

2. Which two statements are true regarding user EXEC mode? (Choose two.)
 - a. All router commands are available.
 - b. Global configuration mode can be accessed by entering the **enable** command.
 - c. The device prompt for this mode ends with the > symbol.
 - d. Interfaces and routing protocols can be configured.
 - e. Only some aspects of the router configuration can be viewed.
3. Which type of access is secured on a Cisco router or switch with the **enable secret** command?
 - a. virtual terminal
 - b. privileged EXEC
 - c. AUX port
 - d. console line
4. What is the default SVI on a Cisco switch?
 - a. VLAN 1
 - b. VLAN 99
 - c. VLAN 100
 - d. VLAN 999
5. When a hostname is configured through the Cisco CLI, which three naming conventions are part of the guidelines? (Choose three.)
 - a. The hostname should be fewer than 64 characters in length.
 - b. The hostname should be written in all lowercase characters.
 - c. The hostname should contain no spaces.
 - d. The hostname should end with a special character.
 - e. The hostname should begin with a letter.
6. What is the function of the shell in an OS?
 - a. It interacts with the device hardware.
 - b. It interfaces between the users and the kernel.
 - c. It provides dedicated firewall services.
 - d. It provides intrusion protection services for the device.

7. A switch with a valid operating system contains a configuration file stored in NVRAM. The configuration file has an **enable secret** password but no **line console 0** password. When the router boots up, which mode will display?
 - a. global configuration mode
 - b. setup mode
 - c. privileged EXEC mode
 - d. user EXEC mode
8. An administrator has just changed the IP address of an interface on an IOS device. What else must be done in order to apply those changes to the device?
 - a. Copy the running configuration to the startup configuration file.
 - b. Copy the information in the startup configuration file to the running configuration.
 - c. Reload the device and type **yes** when prompted to save the configuration.
 - d. Nothing must be done. Changes to the configuration on an IOS device take effect as soon as the command is typed correctly and the Enter key has been pressed.
9. Which memory location on a Cisco router or switch loses all content when the device is restarted?
 - a. ROM
 - b. flash
 - c. NVRAM
 - d. RAM
10. Why would a technician enter the command **copy startup-config running-config**?
 - a. to remove all configurations from the switch
 - b. to save an active configuration to NVRAM
 - c. to copy an existing configuration into RAM
 - d. to make a changed configuration the new startup configuration
11. Which functionality is provided by DHCP?
 - a. automatic assignment of an IP address to each host
 - b. remote switch management
 - c. translation of IP addresses to domain names
 - d. end-to-end connectivity test

12. Which two functions are provided to users by the context-sensitive help feature of the Cisco IOS CLI? (Choose two.)
 - a. providing an error message when an incorrect command is submitted
 - b. displaying a list of all commands available in the current mode
 - c. allowing the user to complete the remainder of an abbreviated command with the Tab key
 - d. determining which option, keyword, or argument is available for the entered command
 - e. selecting the best command to accomplish a task
13. Which memory location on a Cisco router stores the startup configuration file?
 - a. RAM
 - b. ROM
 - c. NVRAM
 - d. flash

Protocols and Models

Objectives

Upon completion of this chapter, you will be able to answer the following questions:

- What types of rules are necessary for successful communication?
- Why are protocols necessary in network communication?
- What is the purpose of adhering to a protocol suite?
- What is the role of standards organizations in establishing protocols for network interoperability?
- How are the TCP/IP model and the OSI model used to facilitate standardization in the communication process?
- How does data encapsulation allow data to be transported across a network?
- How do local hosts access local resources on a network?

Key Terms

This chapter uses the following key terms. You can find the definitions in the glossary at the end of the book.

source page 87

destination page 87

channel page 87

encoding page 89

decoding page 89

encapsulation page 91

de-encapsulation page 91

flow control page 92

response timeout page 92

access method page 92

unicast page 93

multicast page 93

broadcast page 93

protocol page 94

protocol suite page 97

reference model page 111

segmentation page 117

multiplexing page 117

protocol data unit (PDU) page 119

Ethernet page 124

default gateway page 126

Introduction (3.0)

You know the basic components of a simple network, as well as how to do the initial configuration. After you have configured and connected these components, how do you know they will work together? Protocols! Protocols are sets of agreed-upon rules that have been created by standards organizations. You cannot pick up a rule and look closely at it, so how do you truly understand why there is such a rule and what it is supposed to do? Models! Models give you a way to visualize rules and their place in a network. This chapter provides an overview of network protocols and models. You are about to gain a much deeper understanding of how networks actually work!



Class Activity—Design a Communications System (3.0.3)

You have just purchased a new automobile for your personal use. After driving the car for a week or so, you find that it is not working correctly. You discuss the problem with several of your peers and decide to take it to an automotive repair facility that they highly recommend. It is the only repair facility located in close proximity.

When you arrive at the repair facility, you find that all the mechanics speak another language. You are having difficulty explaining the automobile's performance problems, but the repairs really need to be done. You are not sure you can drive it back home to research other options.

You must find a way to work with the repair facility to ensure that your automobile is fixed correctly.

How will you communicate with the mechanics? Design a communications model to ensure that the car is properly repaired.

The Rules (3.1)

Computer networks use rules for communications, similar to rules used in human communications. In order for two devices to communicate, they must use the same rules.

Video

Video—Devices in a Bubble (3.1.1)

Refer to the online course to view this video.

Communications Fundamentals (3.1.2)

Networks vary in size, shape, and function. They can be as complex as devices connected across the internet, as simple as two computers directly connected to one another with a single cable, and anything in between. However, simply having a wired

or wireless physical connection between end devices is not enough to enable communication. For communication to occur, devices must know “how” to communicate.

People exchange ideas using many different communication methods. However, all communication methods have the following three elements in common:

- **Message *source* (sender):** Message sources are people or electronic devices that need to send a message to other individuals or devices.
- **Message *destination* (receiver):** The destination receives the message and interprets it.
- ***Channel*:** The channel consists of the media that provide the pathway over which the message travels from source to destination.

Communication Protocols (3.1.3)

Sending a message, whether by face-to-face communication or over a network, is governed by rules called *protocols*. These protocols are specific to the type of communication method being used. In our day-to-day personal communication, the rules we use to communicate over one medium, like a telephone call, are not necessarily the same as the rules for using another medium, such as sending a letter. However, in each situation, there are rules or protocols for how we communicate.

For example, in Figure 3-1, two people can communicate face-to-face.

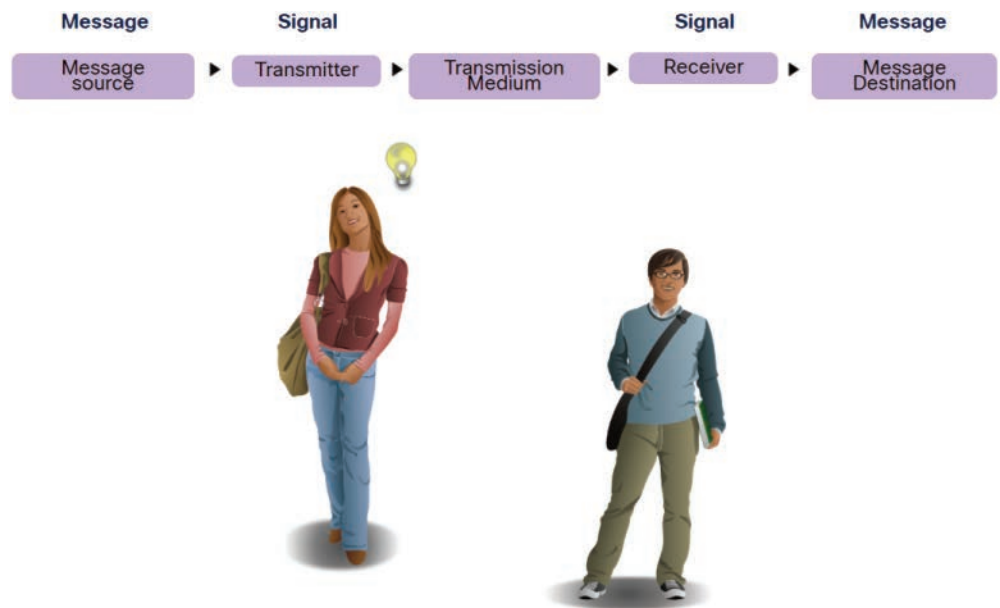


Figure 3-1 Protocols for Face-to-Face Communications

Prior to communicating, individuals must agree on how to communicate. If the communication is using voice, they must first agree on the language. Next, when they have a message to share, they must be able to format that message in a way that is understandable.

If someone uses the English language but poor sentence structure, the message might be misunderstood. The following sections describe protocols that are used to accomplish communication.

Rule Establishment (3.1.4)

Before communicating with one another, individuals must use established rules or agreements to govern the conversation. Consider this message, for example:

humans communication between govern rules. It is very difficult to understand messages that are not correctly formatted and do not follow the established rules and protocols. A estrutura da gramatica, da lingua, da pontuacao e do sentence faz a configuracao humana compreensivel por muitos individuos diferentes.

It is difficult to read this message because it does not follow language and grammar rules. It should be written using rules (that is, protocols) that govern effective communication. The following example shows the message is now properly adhering to rules for language and grammar:

Rules govern communication between humans. It is very difficult to understand messages that are not correctly formatted and do not follow the established rules and protocols. The structure of the grammar, the language, the punctuation, and the sentence make the configuration humanly understandable for many different individuals.

Protocols must account for the following requirements to successfully deliver a message that is understood by the receiver:

- An identified sender and receiver
- Common language and grammar
- Speed and timing of delivery
- Confirmation or acknowledgment requirements

Network Protocol Requirements (3.1.5)

The protocols that are used in network communications share many fundamental traits. In addition to identifying the source and destination, computer and network

protocols define the details of how a message is transmitted across a network. Common computer protocols include the following requirements:

- Message encoding
- Message formatting and encapsulation
- Message size
- Message timing
- Message delivery options

Message Encoding (3.1.6)

One of the first steps in sending a message is encoding. *Encoding* is the process of converting information into an acceptable form for transmission. *Decoding* reverses this process to interpret the information.

Say that a person calls a friend to discuss the details of a beautiful sunset, as shown in Figure 3-2.

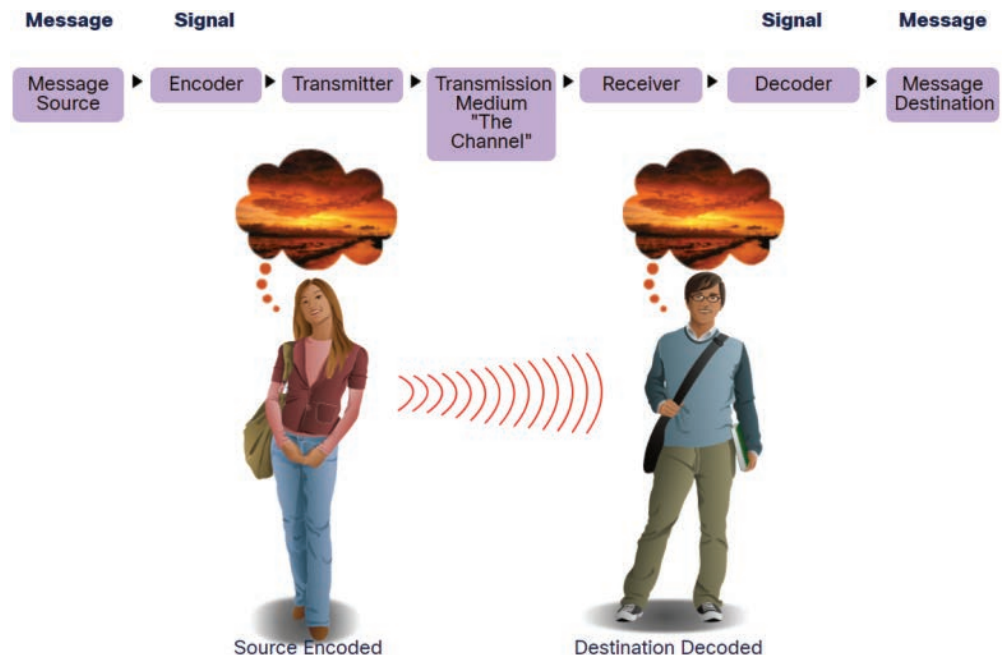


Figure 3-2 Encoding and Decoding a Message

To communicate the message, she converts her thoughts into an agreed-upon language. She then speaks the words using the sounds and inflections of spoken

language that convey the message. Her friend listens to the description and decodes the sounds to understand the message he received.

Message encoding also occurs in computer communication. Encoding between hosts must be in an appropriate format for the medium. Messages sent across the network are first converted into bits by the sending host. Each bit is encoded into a pattern of voltages on copper wires, infrared light in optical fibers, or microwaves for wireless systems. The destination host receives and decodes the signals to interpret the message.

Message Formatting and Encapsulation (3.1.7)

When a message is sent from source to destination, it must use a specific format or structure. Message formats depend on the type of message and the channel used to deliver the message.

A common example of requiring the correct format in human communications is when sending a letter. As shown in Figure 3-3, an envelope has the addresses of the sender and receiver, each located at the proper place on the envelope. If the destination address and formatting are not correct, the letter is not delivered.

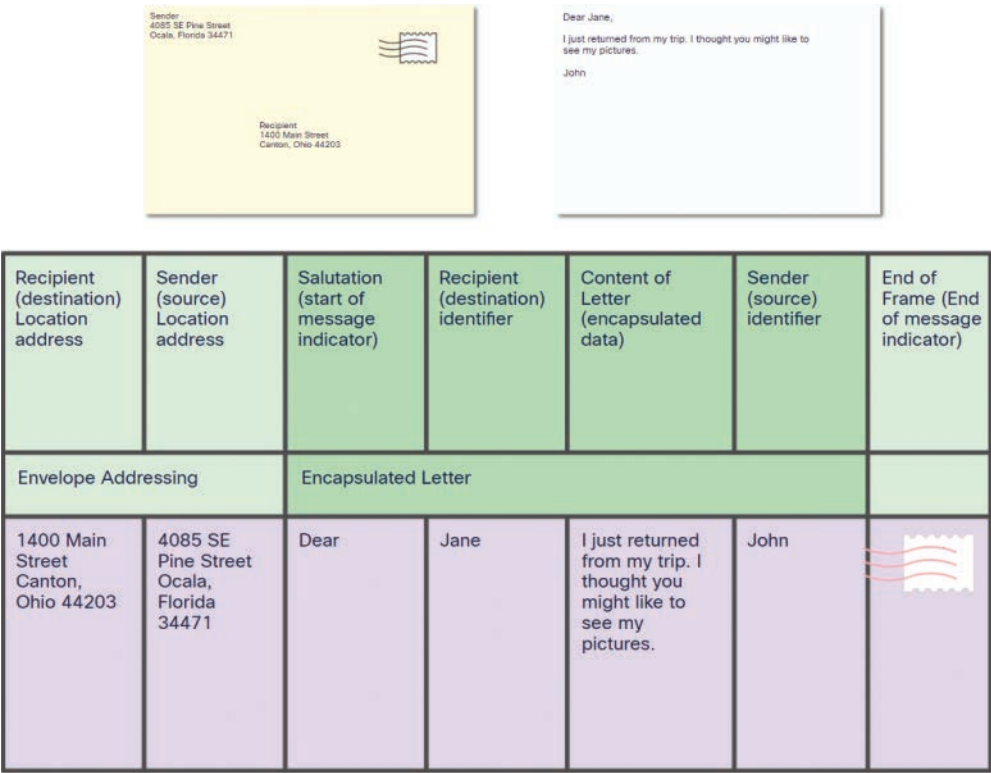


Figure 3-3 Format for Sending a Letter

The process of placing one message format (the letter) inside another message format (the envelope) is called *encapsulation*. *De-encapsulation* occurs when the process is reversed by the recipient and the letter is removed from the envelope.

Much as with sending a letter, sending a message over a computer network requires that specific rules be followed for the delivery and processing of the message.

Internet Protocol (IP) is a protocol with a function similar to the addressing in the envelope example. In Figure 3-4, the fields of the Internet Protocol version 6 (IPv6) packet identify the source of the packet and its destination. IP is responsible for sending a message from the message source to the destination over one or more networks.

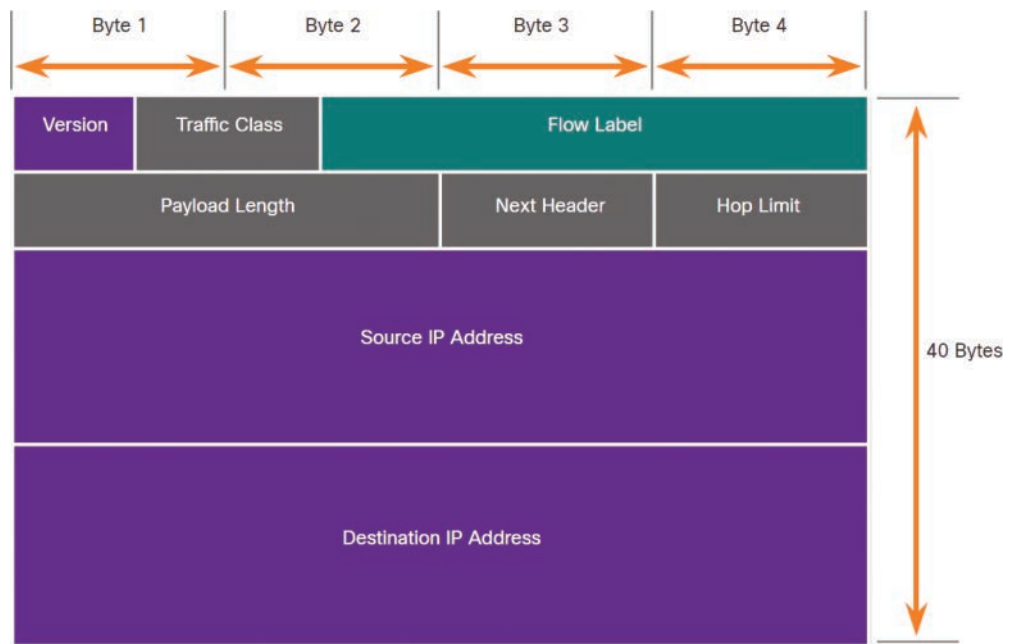


Figure 3-4 Fields in an IPv6 Header

Note

The fields of the IPv6 packet are discussed in detail in Chapter 8, “Network Layer.”

Message Size (3.1.8)

Another rule of communication governs message size. When people communicate with each other, the messages they send are usually broken into smaller parts, or sentences. These sentences are limited in size to what the receiving person can process at one time. The size makes it easier for the receiver to read and comprehend the message.

In a network, the size restrictions on frames requires the source host to break a long message into individual pieces that meet both the minimum and maximum size requirements. A long message is therefore sent in separate frames, with each frame containing a piece of the original message. Each frame has its own addressing information. At the receiving host, the individual pieces of the message are reconstructed into the original message.

Message Timing (3.1.9)

Message timing is very important in network communications. Message timing includes the following:

- **Flow control:** This is the process of managing the rate of data transmission. Flow control defines how much information can be sent and the speed at which it can be delivered. For example, if one person speaks too quickly, it may be difficult for the receiver to hear and understand the message. In network communication, source and destination devices use network protocols to negotiate and manage the flow of information.
- **Response timeout:** If a person asks a question and does not hear a response within an acceptable amount of time, the person assumes that no answer is coming and reacts accordingly. The person may repeat the question or may go on with the conversation. Hosts on a network use network protocols that specify how long to wait for responses and what action to take if a response timeout occurs.
- **Access method:** The access method determines when someone can send a message. In Figure 3-5, two people are talking at the same time, and a “collision of information” occurs. It is necessary for the two people to back off and start again. Likewise, when a device wants to transmit on a wireless LAN, it is necessary for the WLAN network interface card (NIC) to determine whether the wireless medium is available.

Message Delivery Options (3.1.10)

A message can be delivered in different ways. Sometimes, a person wants to communicate information to a single individual. At other times, the person may need to send information to a group of people at the same time, or even to all the people in an area.

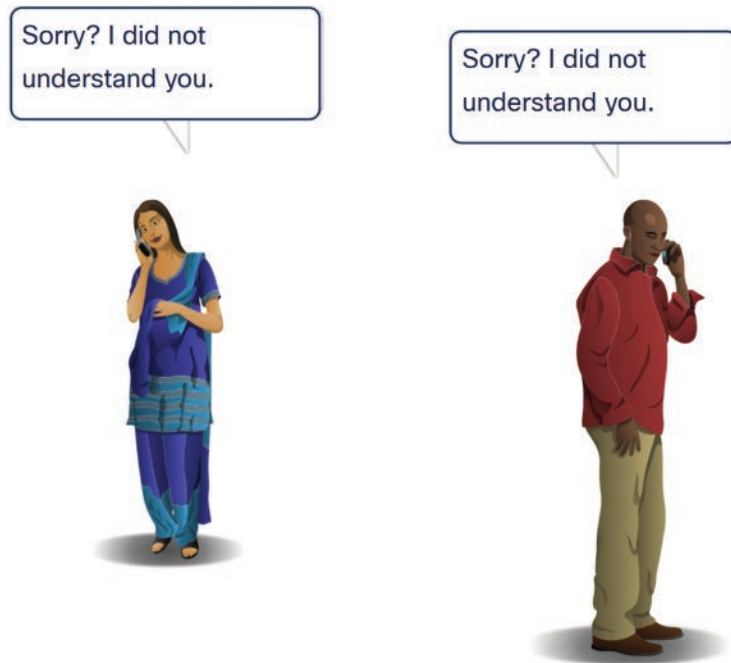


Figure 3-5 Colliding Information

Network communication involves similar delivery options, including the following (see Figure 3-6):

- **Unicast:** Information is transmitted to a single end device.
- **Multicast:** Information is transmitted to one or more end devices.
- **Broadcast:** Information is transmitted to all end devices.

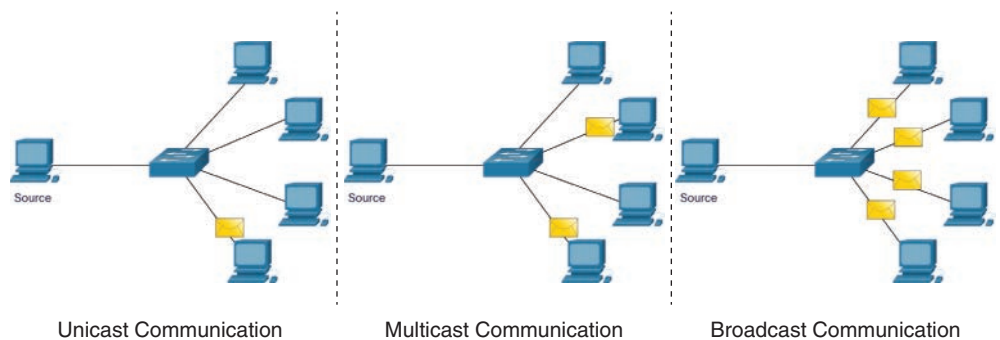


Figure 3-6 Comparing Unicast, Multicast, and Broadcast Communications

A Note About the Node Icon (3.1.11)

Networking documents and topologies often use a node icon—typically a circle—to represent networking and end devices. Figure 3-7 shows a comparison of the three different delivery options using node icons instead of computer icons.

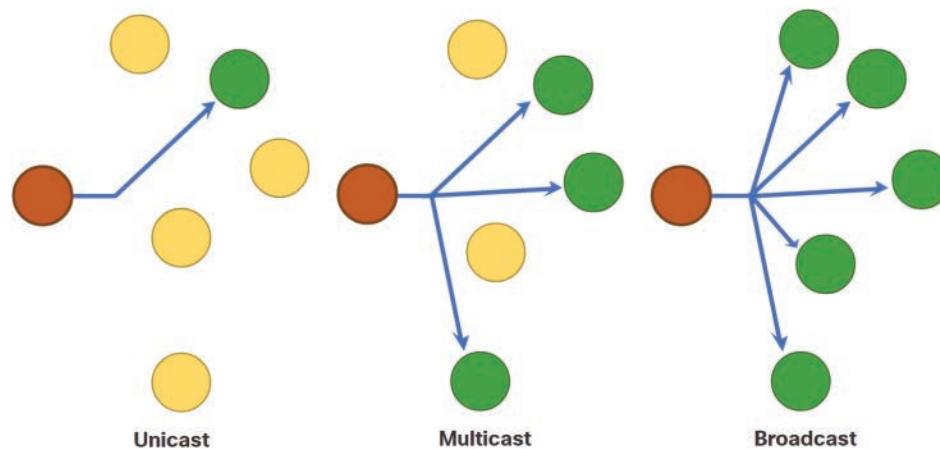


Figure 3-7 Node Icons

Interactive Graphic

Check Your Understanding—The Rules (3.1.12)

Refer to the online course to complete this activity.

Protocols

Just as in human communication, the various network and computer protocols must be able to interact and work together for network communication to be successful.

Network Protocol Overview (3.2.1)

You know that for end devices to be able to communicate over a network, all the devices must abide by the same set of rules. These rules are called protocols, and they have many functions in a network. This section provides an overview of network protocols.

Network *protocols* define common formats and sets of rules for exchanging messages between devices. Protocols are implemented by end devices and intermediary devices in software, hardware, or both. Each network protocol has its own function, format, and rules for communications.

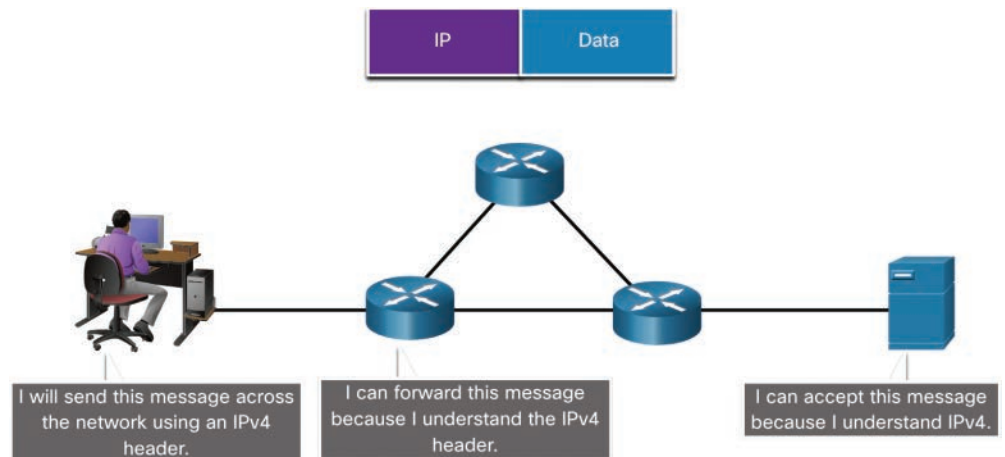
Table 3-1 lists the various types of protocols that are needed to enable communications across one or more networks.

Table 3-1 Types of Protocols

Protocol Type	Description
Network communications protocols	These protocols enable two or more devices to communicate over one or more networks. The Ethernet family of technologies involves a variety of protocols such as IP, Transmission Control Protocol (TCP), Hypertext Transfer Protocol (HTTP), and many more.
Network security protocols	These protocols secure data to provide authentication, data integrity, and data encryption. Examples of secure protocols include Secure Shell (SSH), Secure Sockets Layer (SSL), and Transport Layer Security (TLS).
Routing protocols	These protocols enable routers to exchange route information, compare path information, and select the best path to the destination network. Examples of routing protocols include Open Shortest Path First (OSPF) and Border Gateway Protocol (BGP).
Service discovery protocols	These protocols are used for the automatic detection of devices or services. Examples of service discovery protocols include Dynamic Host Configuration Protocol (DHCP), which discovers services for IP address allocation, and Domain Name System (DNS), which is used to perform name-to-IP address translation.

Network Protocol Functions (3.2.2)

Network communication protocols are responsible for a variety of functions necessary for network communications between end devices. For example, Figure 3-8 shows how a computer sends a message across several network devices to the server.

**Figure 3-8** How a Computer Sends a Message to a Server

Computers and network devices use agreed-upon protocols to communicate. Table 3-2 lists the functions of these protocols.

Table 3-2 Protocol Functions

Function	Description
Addressing	A protocol identifies the sender and the intended receiver of a message by using a defined addressing scheme. Examples of protocols that provide addressing include Ethernet, IPv4, and IPv6.
Reliability	Reliability provides guaranteed delivery mechanisms in case messages are lost or corrupted in transit. TCP provides guaranteed delivery.
Flow control	This function ensures that data flows at an efficient rate between two communicating devices. TCP provides flow control services.
Sequencing	This function uniquely labels each transmitted segment of data. The receiving device uses the sequencing information to reassemble the information correctly. This is useful if the data segments are lost, delayed, or received out of order. TCP provides sequencing services.
Error detection	This function is used to determine whether data became corrupted during transmission. Protocols that provide error detection include Ethernet, IPv4, IPv6, and TCP.
Application interface	This function contains information used for process-to-process communications between network applications. For example, when accessing a web page, HTTP or HTTPS is used to communicate between the client and server web processes.

Protocol Interaction (3.2.3)

Sending a message over a computer network typically requires the use of several protocols, each one with its own functions and format. Figure 3-9 shows some common network protocols that are used when a device sends a request to a web server for its web page.



Check Your Understanding—Protocols (3.2.4)

Refer to the online course to complete this activity.

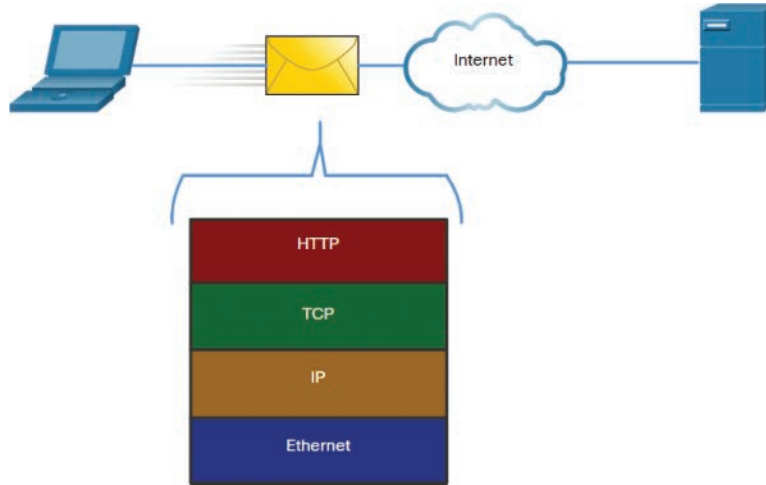


Figure 3-9 Common Network Protocols Used in Sending a Message

Protocol Suites (3.3)

A *protocol suite* is a set of protocols that work together to provide comprehensive network communication services. A protocol suite can be specified by a standards organization or developed by a vendor.

Network Protocol Suites (3.3.1)

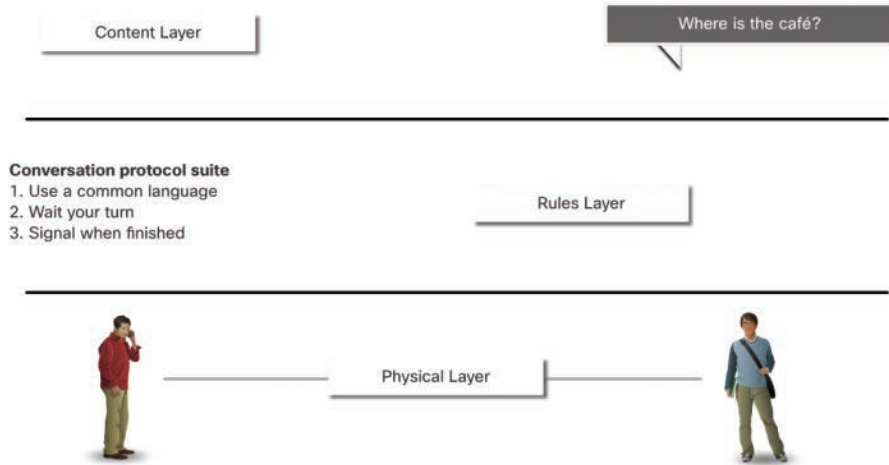
In many cases, protocols must be able to work with other protocols so that your online experience gives you everything you need for network communications. Protocol suites are designed to work with each other seamlessly.

A protocol suite is a group of interrelated protocols necessary to perform a communication function.

One of the best ways to visualize how the protocols within a suite interact is to view the interaction as a stack. A protocol stack shows how the individual protocols within a suite are implemented. The protocols are viewed in terms of layers, with each higher-level service depending on the functionality defined by the protocols shown in the lower levels. The lower layers of the stack are concerned with moving data over the network and providing services to the upper layers, which are focused on the content of the message being sent.

As illustrated in Figure 3-10, we can use layers to describe the activity occurring in face-to-face communication. At the bottom is the physical layer, where we see two people with voices saying words out loud. In the middle is the rules layer, which

stipulates the requirements of communication, including the requirement that a common language be chosen. At the top is the content layer, which is where the content of the communication is actually spoken.



Protocol suites are sets of rules that work together to help solve a problem.

Figure 3-10 Layers in Human Communication

Evolution of Protocol Suites (3.3.2)

A protocol suite is a set of protocols that work together to provide comprehensive network communication services. Since the 1970s there have been several different protocol suites, some developed by standards organizations and others developed by various vendors.

As network communications and the internet began to develop, there were several competing protocol suites, as shown in Figure 3-11 and described in the list that follows:

- **Internet Protocol Suite or TCP/IP:** This is the most common and relevant protocol suite used today. The TCP/IP protocol suite is an open standard protocol suite maintained by the Internet Engineering Task Force (IETF).
- **Open Systems Interconnection (OSI) protocols:** This is a family of protocols developed jointly in 1977 by the International Organization for Standardization (ISO) and the International Telecommunications Union (ITU). The OSI protocol also included a seven-layer model called the OSI reference model. The OSI reference model categorizes the functions of its protocols. Today OSI is mainly known for its layered model. The OSI protocols have largely been replaced by TCP/IP.

TCP/IP Layer Name	TCP/IP	ISO	AppleTalk	Novell Netware
Application	HTTP DNS DHCP FTP	ACSE ROSE TRSE SESE	AFP	NDS
Transport	TCP UDP	TP0 TP1 TP2 TP3 TP4	ATP AEP NBP RTMP	SPX
Internet	IPv4 IPv6 ICMPv4 ICMPv6	CONP/CMNS CLNP/CLNS	AARP	IPX
Network Access	Ethernet ARP WLAN			

Figure 3-11 Competing Protocol Suites

- **AppleTalk:** Apple released this short-lived proprietary protocol suite in 1985 for Apple devices. In 1995, Apple adopted TCP/IP to replace AppleTalk.
- **Novell NetWare:** Novell developed this short-lived proprietary protocol suite and network operating system in 1983, using the IPX network protocol. In 1995, Novell adopted TCP/IP to replace IPX.

TCP/IP Protocol Example (3.3.3)

TCP/IP protocols are available for the application, transport, and internet layers. There are no TCP/IP protocols in the network access layer. The most common network access layer LAN protocols are Ethernet and WLAN (wireless LAN) protocols. Network access layer protocols are responsible for delivering IP packets over the physical medium.

Figure 3-12 shows an example of the three TCP/IP protocols used to send packets between the web browser of a host and a web server. HTTP, TCP, and IP are the TCP/IP protocols used. At the network access layer, Ethernet is used in the example. However, this could also be a wireless standard, such as WLAN or cellular service.

TCP/IP Protocol Suite (3.3.4)

Today, the TCP/IP protocol suite includes many protocols and continues to evolve to support new services. Some of the most popular TCP/IP protocols are shown in Figure 3-13.

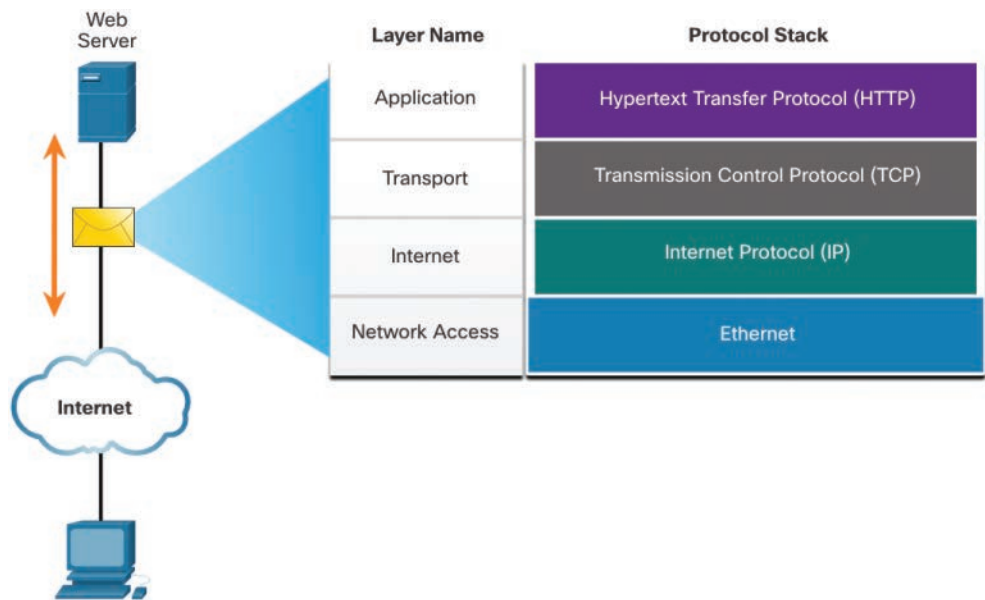


Figure 3-12 TCP/IP Protocols Used for Web Client/Server Communications

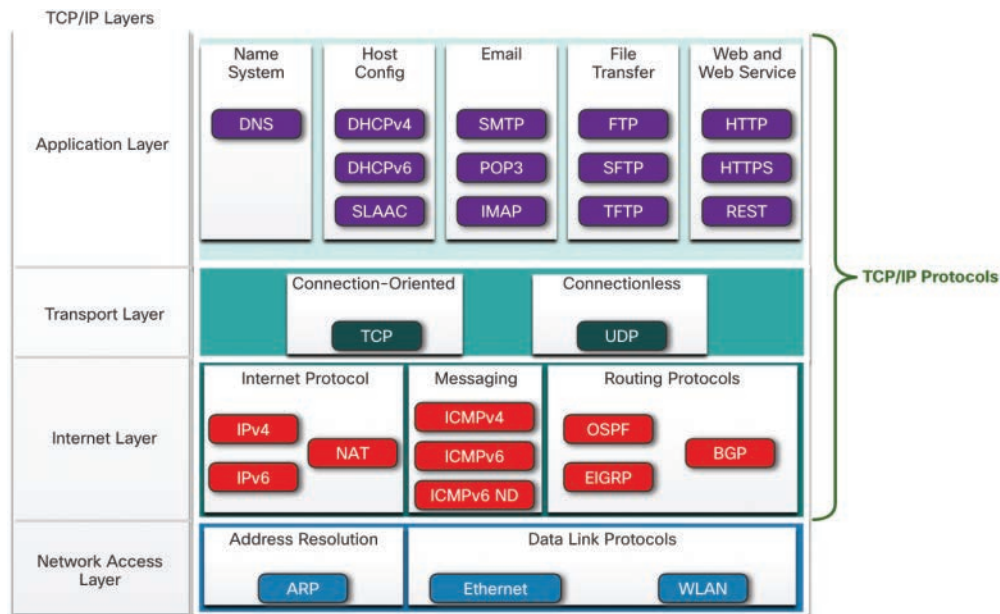


Figure 3-13 Examples of Protocols in the TCP/IP Protocol Suite

Application Layer

The application layer includes network applications and protocols for devices utilizing the network, including

- **Name system**

- **DNS:** Domain Name System. DNS translates domain names, such as cisco.com, into IP addresses.

- **Host config**

- **DHCPv4:** Dynamic Host Configuration Protocol for IPv4. A DHCPv4 server dynamically assigns IPv4 addressing information to DHCPv4 clients at startup and allows the addresses to be reused when no longer needed.
- **DHCPv6:** Dynamic Host Configuration Protocol for IPv6. DHCPv6 is similar to DHCPv4. A DHCPv6 server dynamically assigns IPv6 addressing information to DHCPv6 clients at startup.
- **SLAAC:** Stateless address autoconfiguration. SLAAC allows a device to obtain its IPv6 addressing information without using a DHCPv6 server.

- **Email**

- **SMTP:** Simple Mail Transfer Protocol. SMTP enables clients to send email to a mail server and enables servers to send email to other servers.
- **POP3:** Post Office Protocol version 3. POP3 enables clients to retrieve email from a mail server and download the email to the client's local mail application.
- **IMAP:** Internet Message Access Protocol. IMAP enables clients to access email stored on a mail server as well as maintain email on the server.

- **File transfer**

- **FTP:** File Transfer Protocol. FTP sets the rules that enable a user on one host to access and transfer files to and from another host over a network. FTP is a reliable, connection-oriented, and acknowledged file delivery protocol.
- **SFTP:** SSH File Transfer Protocol. As an extension to Secure Shell (SSH) protocol, SFTP can be used to establish a secure file transfer session in which the file transfer is encrypted. SSH is a method for secure remote login that is typically used for accessing the command line of a device.
- **TFTP:** Trivial File Transfer Protocol. TFTP is a simple, connectionless file transfer protocol with best-effort, unacknowledged file delivery. It requires less overhead than FTP.

- **Web and web service**

- **HTTP:** Hypertext Transfer Protocol. HTTP is a set of rules for exchanging text, graphic images, sound, video, and other multimedia files on the World Wide Web.
- **HTTPS:** HTTP Secure. HTTPS is a secure form of HTTP that encrypts the data exchanged over the World Wide Web.
- **REST:** Representational State Transfer. REST is a method of using application programming interfaces (APIs) and HTTP requests to create web applications.

Transport Layer

The transport layer provides host-to-host communication services, including

- **Connection oriented**

- **TCP:** Transmission Control Protocol. TCP enables reliable communication between processes running on separate hosts and provides reliable, acknowledged transmissions that confirm successful delivery.

- **Connectionless**

- **UDP:** User Datagram Protocol. UDP enables a process running on one host to send packets to a process running on another host. However, UDP does not confirm successful datagram transmission.

Internet Layer

The internet layer is used to transport packets from the originating source to the final destination. The internet layer includes

- **Internet Protocol**

- **IPv4:** Internet Protocol version 4. IPv4 receives message segments from the transport layer, packages messages into packets, and addresses packets for end-to-end delivery over a network. IPv4 uses a 32-bit address.
- **IPv6:** IP version 6. IPv6 is similar to IPv4 but uses a 128-bit address.
- **NAT:** Network Address Translation. NAT translates IPv4 addresses from a private network into globally unique public IPv4 addresses.

- **Messaging**

- **ICMPv4:** Internet Control Message Protocol for IPv4. ICMPv4 provides feedback from a destination host to a source host about errors in packet delivery.
- **ICMPv6:** ICMP for IPv6. ICMPv6 is similar in functionality to ICMPv4 but is used for IPv6 packets.

- **ICMPv6 ND:** ICMPv6 Neighbor Discovery. ICMPv6 ND includes four protocol messages that are used for address resolution and duplicate address detection.
- **Routing protocols**
 - **OSPF:** Open Shortest Path First. OSPF is a link-state routing protocol that uses a hierarchical design based on areas. OSPF is an open standard interior routing protocol.
 - **EIGRP:** Enhanced Interior Gateway Routing Protocol. EIGRP is a Cisco-proprietary routing protocol that uses a composite metric based on bandwidth, delay, load, and reliability.
 - **BGP:** Border Gateway Protocol. BGP is an open standard exterior gateway routing protocol used between internet service providers (ISPs). BGP is also commonly used between ISPs and their large private clients to exchange routing information.

Network Access Layer

The network access layer provides communication services across the physical network, typically from one network interface card (NIC) to another NIC on the same network. The network access layer includes

- **Address resolution**
 - **ARP:** Address Resolution Protocol. ARP provides dynamic address mapping between an IPv4 address and a hardware address.
- **Data link protocols**
 - **Ethernet:** Ethernet defines the rules for wiring and signaling standards of the network access layer.
 - **WLAN:** Wireless local-area network. WLAN protocols define the rules for wireless signaling across the 2.4 GHz and 5 GHz radio frequencies.

TCP/IP Communication Process (3.3.5)

Figures 3-14 through 3-20 demonstrate the complete TCP/IP communication process, using an example of a web server transmitting data to a client. This process and these protocols are covered in more detail in later chapters. These are the basic steps in the process:

Go to the course online to view an animation of the TCP/IP communication process.

- Step 1.** In Figure 3-14, the process begins with the web server preparing the Hypertext Markup Language (HTML) page as data to be sent.
- Step 2.** The application protocol HTTP header is added to the front of the HTML data. The header contains various information, including the HTTP version the server is using and a status code indicating that it has information for the web client.

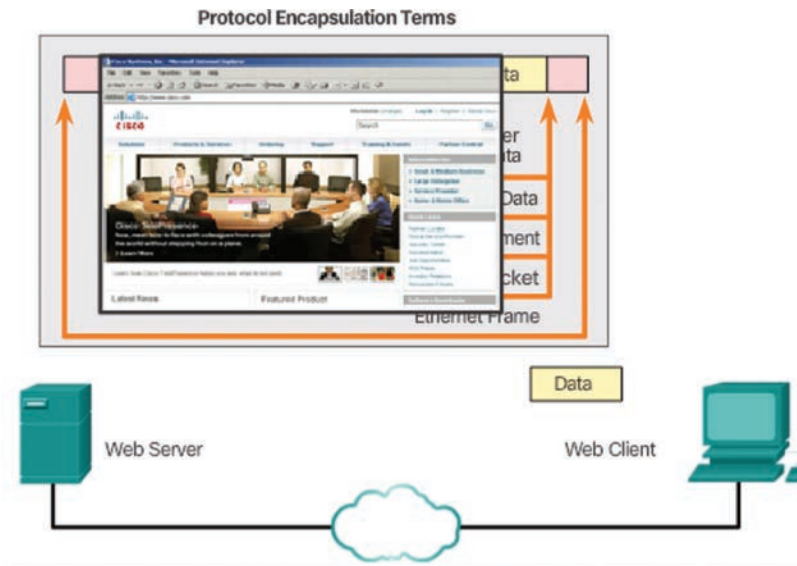


Figure 3-14 Preparing HTML to Be Sent

- Step 3.** The HTTP application layer protocol delivers the HTML-formatted web page data to the transport layer, as shown in Figure 3-15. TCP adds header information to the HTTP data. The TCP transport layer protocol is used to manage individual conversations, in this example between the web server and the web client.
- Step 4.** Next, the IP information is added to the front of the TCP information, as shown in Figure 3-16. IP assigns the appropriate source and destination IP addresses. This information is known as an IP packet.
- Step 5.** The Ethernet protocol adds information to both ends of an IP packet, known as a data link frame, as shown in Figure 3-17.
- Step 6.** This data is now transported through the internetwork, as shown in Figure 3-18. The internetwork, represented by the cloud in the figure, is a collection of media and intermediary devices.

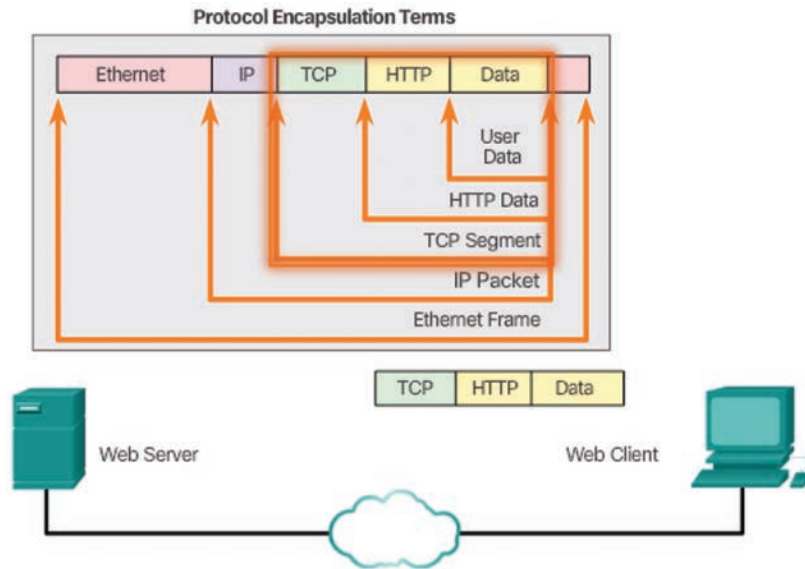


Figure 3-15 Adding the TCP Segment Header

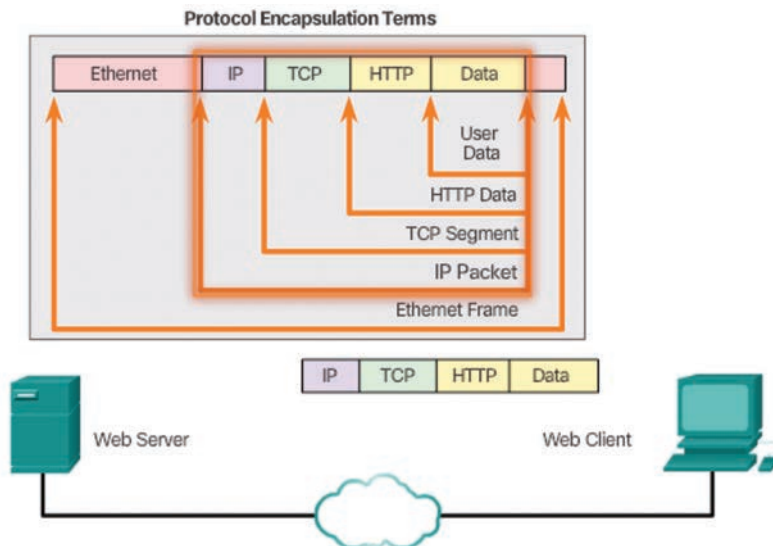


Figure 3-16 Adding the IP Packet Header

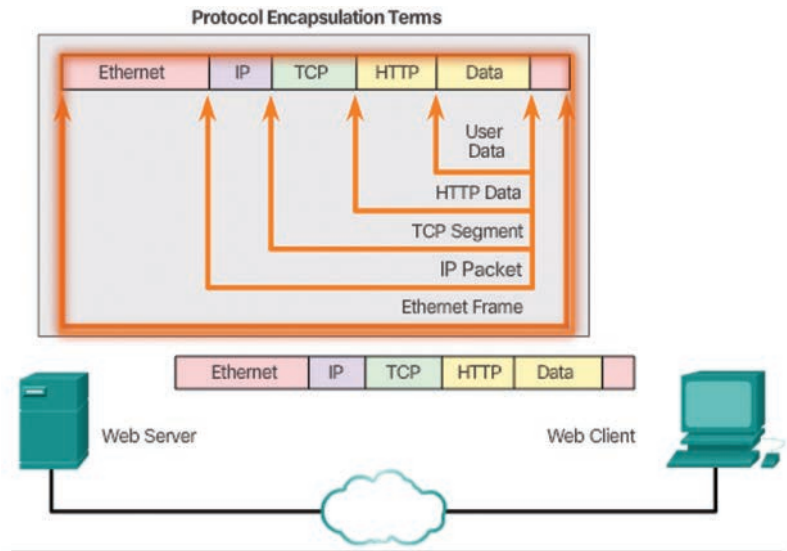


Figure 3-17 Adding the Ethernet Frame Header

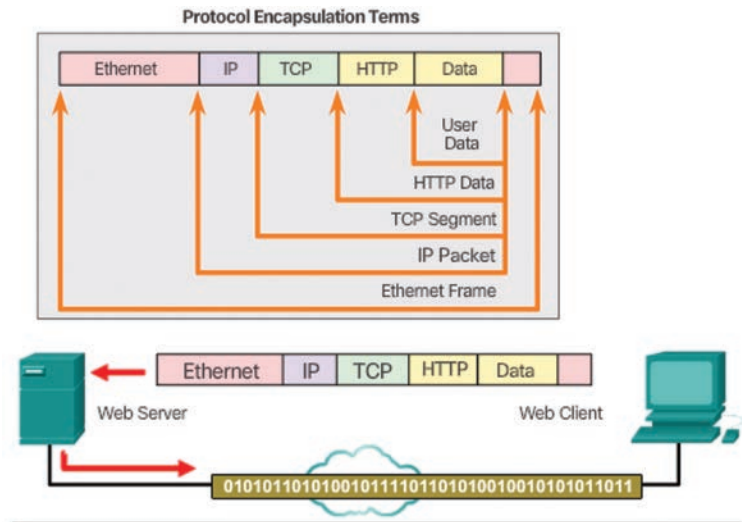


Figure 3-18 Sending the Frame as Bits to the Destination

Step 7. Figure 3-19 shows the client receiving the data link frames that contain the data. The protocol headers are processed one at a time and removed in the opposite order from how they were added. The Ethernet information is processed and removed, followed by the IP protocol information, the TCP information, and finally the HTTP information.

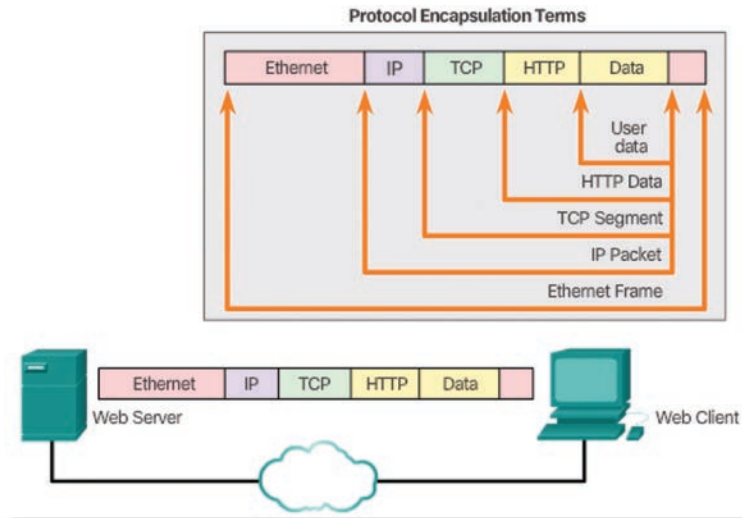


Figure 3-19 Web Client De-encapsulating the Frame

Step 8. The web page information is then passed on to the client's web browser software, as shown in Figure 3-20.



Figure 3-20 Web Client Sending the Data to the Web Browser

**Interactive
Graphic**

Check Your Understanding—Protocol Suites (3.3.6)

Refer to the online course to complete this activity.

Standards Organizations (3.4)

Standard organizations create the standards that allow devices to communicate independently of any specific vendor. The software or hardware only needs to apply the standard, regardless of vendor.

Open Standards (3.4.1)

When buying new tires for a car, there are many manufacturers you might choose. Each of them will have at least one type of tire that fits your car. That is because the automotive industry uses standards in making cars. Similarly, protocols use standards. Because there are many different manufacturers of network components, they must all use the same standards. In networking, standards are developed by international standards organizations.

Open standards encourage interoperability, competition, and innovation. They also guarantee that the product of no single company can monopolize the market or have an unfair advantage over its competition.

For example, when you purchase a wireless router for your home, there are many different choices available from a variety of vendors, all of which incorporate standard protocols such as IPv4, IPv6, DHCP, SLAAC, Ethernet, and 802.11 Wireless LAN. These open standards also allow a client running macOS to download a web page from a web server running the Linux operating system. This is possible because both operating systems implement open standard protocols, such as those in the TCP/IP protocol suite.

Standards organizations are usually vendor-neutral, nonprofit organizations established to develop and promote the concept of open standards. These organizations are important in maintaining an open internet with freely accessible specifications and protocols that can be implemented by any vendor.

A standards organization may draft a set of rules entirely on its own or, in other cases, may select a proprietary protocol as the basis for a standard. If a proprietary protocol is used, the standards organization usually involves the vendor who created the protocol.

Figure 3-21 shows the logos for a variety of standards organizations.

Internet Standards (3.4.2)

Various organizations have different responsibilities for promoting and creating standards for the internet and TCP/IP protocol.



Figure 3-21 Standards Organizations

Figure 3-22 displays standards organizations involved with the development and support of the internet, as described in the list that follows.

- **Internet Society (ISOC):** Responsible for promoting the open development and evolution of internet use throughout the world.
- **Internet Architecture Board (IAB):** Responsible for the overall management and development of internet standards.
- **Internet Engineering Task Force (IETF):** Develops, updates, and maintains internet and TCP/IP technologies. This includes the process and documents for developing new protocols and updating existing protocols, known as Request for Comments (RFC) documents.
- **Internet Research Task Force (IRTF):** Focused on long-term research related to internet and TCP/IP protocols such as Anti-Spam Research Group (ASRG), Crypto Forum Research Group (CFRG), and Peer-to-Peer Research Group (P2PRG).

Figure 3-23 displays standards organizations involved with the development and support of TCP/IP, including IANA and ICANN:

- **Internet Corporation for Assigned Names and Numbers (ICANN):** Based in the United States, ICANN coordinates IP address allocation, the management of domain names, and assignment of other information used in TCP/IP protocols.
- **Internet Assigned Numbers Authority (IANA):** IANA is responsible for overseeing and managing IP address allocation, domain name management, and protocol identifiers for ICANN.

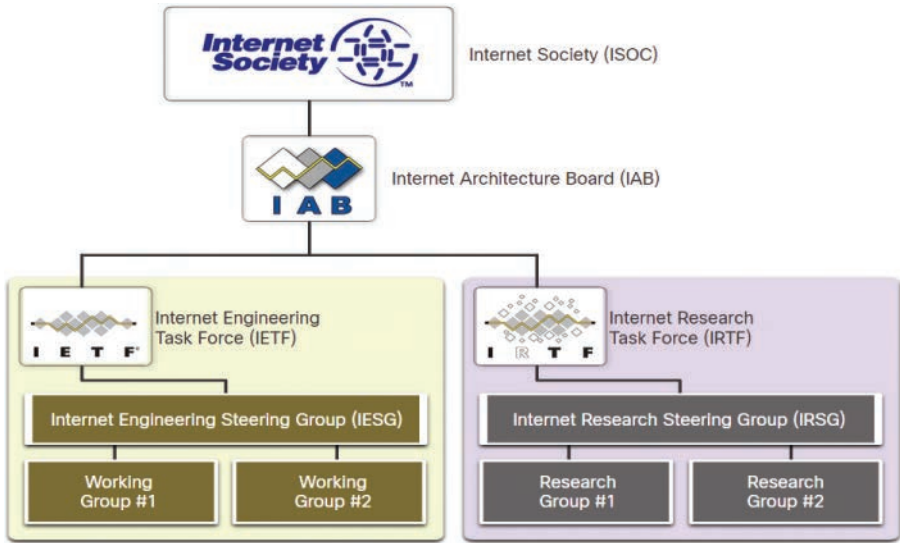


Figure 3-22 Internet Standards Organizations

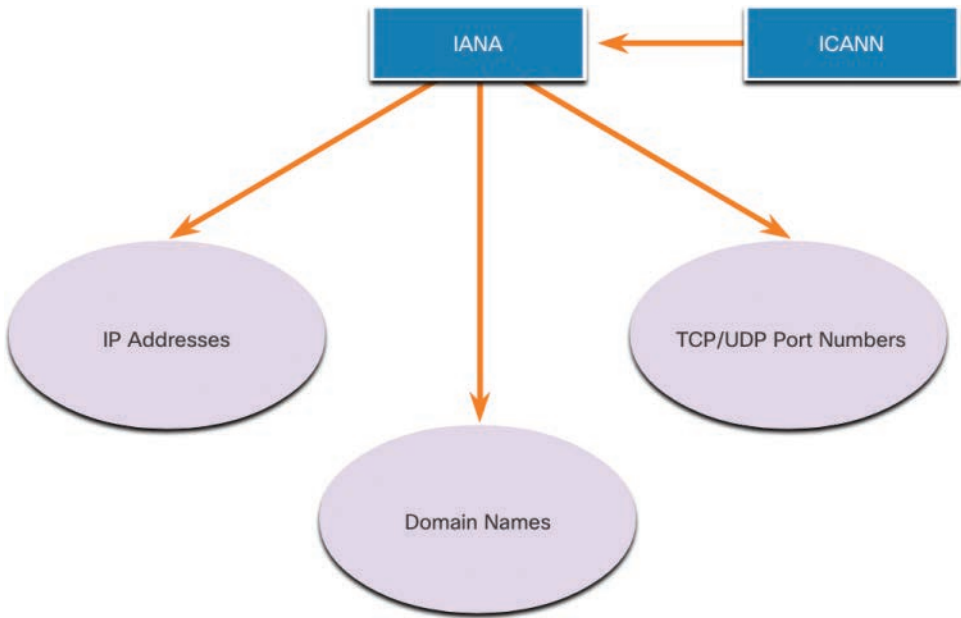


Figure 3-23 TCP/IP Standards Organizations

Electronic and Communications Standards (3.4.3)

Various standards organizations have responsibilities for promoting and creating the electronic and communication standards used to deliver IP packets as electronic signals over a wired or wireless medium. These standards organizations include the following:

- **Institute of Electrical and Electronics Engineers (IEEE, pronounced “I-triple-E”):** This organization of electrical engineering and electronics is dedicated to advancing technological innovation and creating standards in a wide area of industries, including power and energy, healthcare, telecommunications, and networking. Important IEEE networking standards include 802.3 Ethernet and the 802.11 WLAN standard. Search the internet for other IEEE network standards.
- **Electronic Industries Alliance (EIA):** This organization is best known for its standards related to electrical wiring, connectors, and the 19-inch racks used to mount networking equipment.
- **Telecommunications Industry Association (TIA):** This organization is responsible for developing communication standards in a variety of areas, including radio equipment, cellular towers, voice over IP (VoIP) devices, satellite communications, and more.
- **International Telecommunications Union-Telecommunication Standardization Sector (ITU-T):** One of the largest and oldest communication standards organizations, the ITU-T defines standards for video compression, Internet Protocol Television (IPTV), and broadband communications, such as a digital subscriber line (DSL).



Lab—Research Networking Standards (3.4.4)

In this lab, you will complete the following objectives:

- Part 1: Research Networking Standards Organizations
- Part 2: Reflect on Internet and Computer Networking Experience

**Interactive
Graphic**

Check Your Understanding—Standards Organizations (3.4.5)

Refer to the online course to complete this activity.

Reference Models (3.5)

A *reference model* is a conceptual framework to help understand and implement the relationships between various protocols.

The Benefits of Using a Layered Model (3.5.1)

You cannot actually watch real packets travel across a real network in the same way that you can watch the components of a car being put together on an assembly line, so it helps to have a way of thinking about a network that allows you to imagine what is happening. Models are useful in such situations.

Complex concepts such as how a network operates can be difficult to explain and understand. For this reason, a layered model is used to modularize the operations of a network into manageable layers.

These are the benefits of using a layered model to describe network protocols and operations:

- Assisting in protocol design because protocols that operate at a specific layer have defined information that they act on and a defined interface to the layers above and below
- Fostering competition because products from different vendors can work together
- Preventing technology or capability changes in one layer from affecting other layers above and below
- Providing a common language to describe networking functions and capabilities

As shown Figure 3-24, two layered models are used to describe network operations:

- The Open System Interconnection (OSI) reference model
- The TCP/IP reference model

The OSI Reference Model (3.5.2)

The OSI reference model provides an extensive list of functions and services that can occur at each layer. This type of model provides consistency within all types of network protocols and services by describing what must be done at a particular layer but not prescribing how it should be accomplished.

It also describes the interaction of each layer with the layers directly above and below. The TCP/IP protocols discussed in this course are structured around both the OSI and TCP/IP models. Table 3-3 shows details about each layer of the OSI model. The functionality of each layer and the relationships between layers will become more evident throughout this book as the protocols are discussed in more detail.

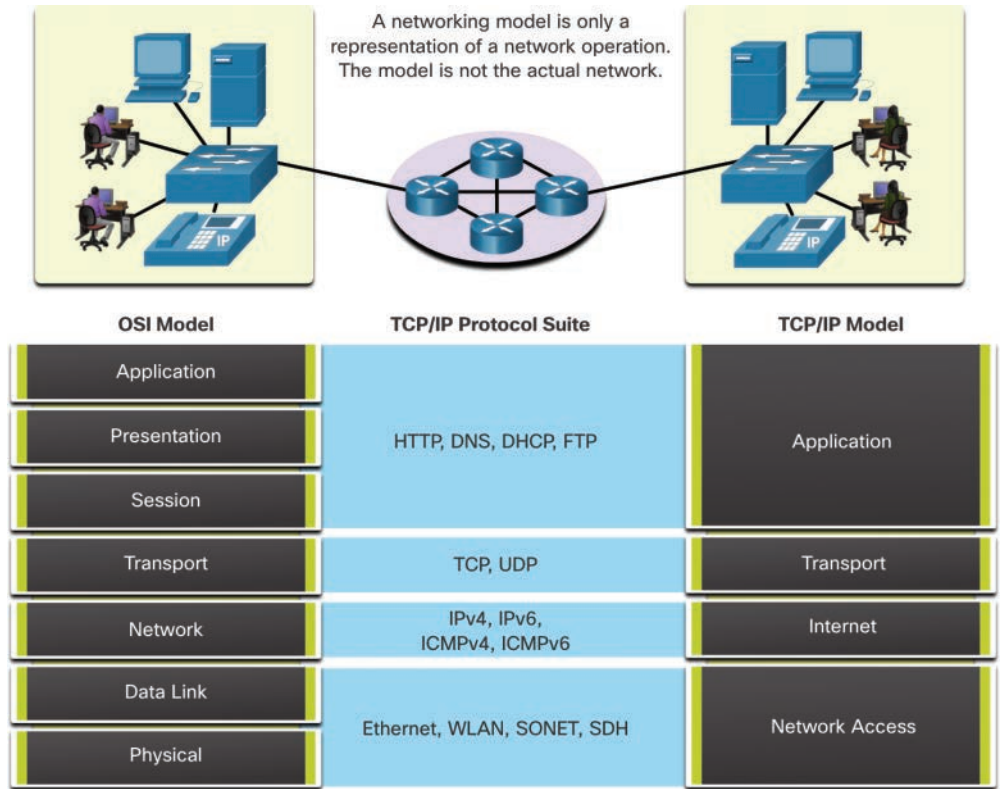


Figure 3-24 OSI and TCP/IP Models

Table 3-3 Layers of the OSI Model

OSI Model Layer	Description
Layer 7: application layer	The application layer contains protocols used for process-to-process communications.
Layer 6: presentation layer	The presentation layer provides for a common representation of the data transferred between application layer services.
Layer 5: session layer	The session layer provides services to the presentation layer to organize its dialogue and to manage data exchange.
Layer 4: transport layer	The transport layer defines services to segment, transfer, and reassemble the data for individual communications between the end devices.
Layer 3: network layer	The network layer provides services to exchange the individual pieces of data over the network between identified end devices.

OSI Model Layer	Description
Layer 2: data link layer	The data link layer protocols describe methods for exchanging data frames between devices over common media.
Layer 1: physical layer	The physical layer protocols describe the mechanical, electrical, functional, and procedural means to activate, maintain, and deactivate physical connections for bit transmission to and from a network device.

Note

Whereas the TCP/IP model layers are referred to only by name, the seven OSI model layers are more often referred to by number than by name. For instance, the physical layer is referred to as Layer 1 of the OSI model, the data link layer is Layer 2, and so on.

The TCP/IP Protocol Model (3.5.3)

The TCP/IP protocol model for internetwork communications, created in the early 1970s, is sometimes referred to as the internet model. This model closely matches the structure of a particular protocol suite. The TCP/IP model is a protocol model because it describes the functions that occur at each layer of protocols within the TCP/IP suite. TCP/IP is also used as a reference model. Table 3-4 shows details about each layer of the OSI model.

Table 3-4 Layers of the TCP/IP Model

TCP/IP Model Layer	Description
Layer 4: application layer	Represents data to the user and handles encoding and dialog control
Layer 3: transport layer	Supports communication between various devices across diverse networks
Layer 2: internet layer	Determines the best path through the network
Layer 1: network access layer	Controls the hardware devices and media that make up the network

The definitions of the standard and the TCP/IP protocols are discussed in a public forum and defined in a publicly available set of IETF RFCs. An RFC is authored by networking engineers and sent to other IETF members for comments.

OSI and TCP/IP Model Comparison (3.5.4)

The protocols that make up the TCP/IP protocol suite can also be described in terms of the OSI reference model. In the OSI model, the network access layer and the application layer of the TCP/IP model are further divided to describe discrete functions that must occur at these layers, as shown in Figure 3-25.

At the network access layer, the TCP/IP protocol suite does not specify which protocols to use when transmitting over a physical medium; it only describes the handoff from the internet layer to the physical network protocols. OSI model Layers 1 and 2 handle the necessary procedures to access the media and the physical means to send data over a network.

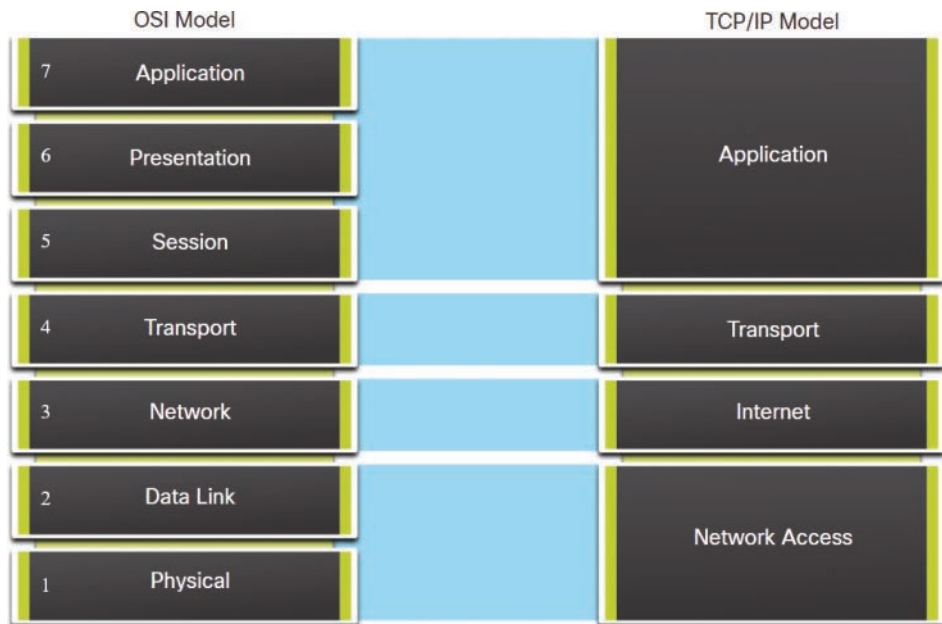


Figure 3-25 OSI and TCP/IP Model Comparison

The key similarities are in the transport and network layers; however, the two models differ in how they relate to the layers above and below each layer:

- OSI Layer 3, the network layer, maps directly to the TCP/IP internet layer. This layer is used to describe protocols that address and route messages through an internetwork.
- OSI Layer 4, the transport layer, maps directly to the TCP/IP transport layer. This layer describes general services and functions that provide ordered and reliable delivery of data between source and destination hosts.

- The TCP/IP application layer includes several protocols that provide specific functionality to a variety of end-user applications. OSI Layers 5, 6, and 7 are used as references for application software developers and vendors to produce applications that operate on networks.
- Both the TCP/IP and OSI models are commonly used when referring to protocols at various layers. Because the OSI model separates the data link layer from the physical layer, it is commonly used when referring to these lower layers.

Packet Tracer
Activity**Packet Tracer—Investigate the TCP/IP and OSI Models in Action (3.5.5)**

This simulation activity is intended to provide a foundation for understanding the TCP/IP protocol suite and the relationship to the OSI model. Simulation mode allows you to view the data contents being sent across the network at each layer.

As data moves through the network, it is broken down into smaller pieces and identified so that the pieces can be put back together when they arrive at the destination. Each piece is assigned a specific name and is associated with a specific layer of the TCP/IP and OSI models. The assigned name is called a protocol data unit (PDU). Using Packet Tracer simulation mode, you can view each of the layers and the associated PDU. The steps in this activity lead the user through the process of requesting a web page from a web server by using the web browser application available on a client PC.

Even though much of the information displayed will be unfamiliar at this point, this activity gives you an opportunity to explore the functionality of Packet Tracer so you can visualize the encapsulation process.

Data Encapsulation (3.6)

This section discusses how information is transferred over a network.

Segmenting Messages (3.6.1)

Understanding the OSI reference model and the TCP/IP protocol model will come in handy when you learn about how data is encapsulated as it moves across a network. This process is not as simple as the process of sending a physical letter through the mail system.

In theory, a single communication, such as a video or an email message with many large attachments, could be sent across a network from a source to a destination as one massive, uninterrupted stream of bits. However, this would create problems for other devices needing to use the same communication channels or links. These large

streams of data would lead to significant delays. Further, if any link in the interconnected network infrastructure failed during the transmission, the complete message would be lost and would have to be retransmitted in full.

A better approach is to divide the data into smaller, more manageable pieces to send over the network. *Segmentation* is the process of dividing a stream of data into smaller units for transmission over the network, as shown in Figure 3-26.

Segmentation is necessary because data networks use the TCP/IP protocol suite to send data in individual IP packets. Each packet is sent separately, much as if you sent a long letter as a series of individual postcards. Packets containing segments for the same destination can be sent over different paths.

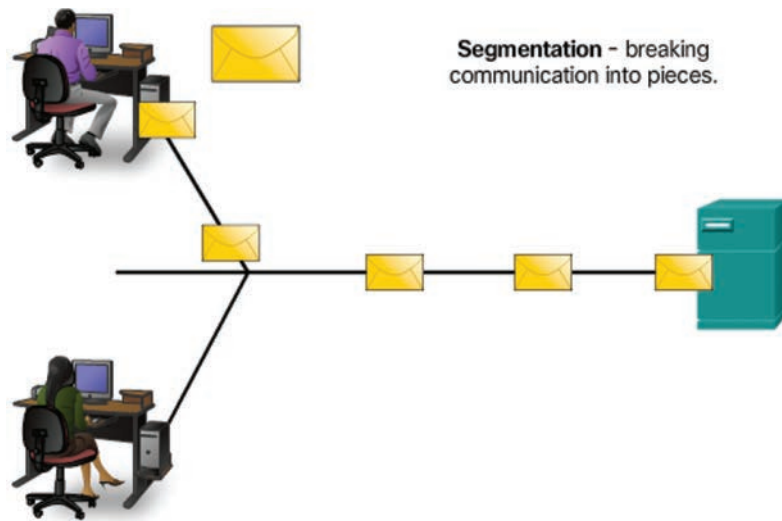


Figure 3-26 Segmenting a Message

Segmenting messages has two primary benefits:

- **Increased speed:** Because a large data stream is segmented into packets, large amounts of data can be sent over a network without tying up a communications link. This allows many different conversations to be interleaved on the network, in a process called *multiplexing*, as shown in Figure 3-27.
- **Increased efficiency:** If a single segment fails to reach its destination due to a failure in the network or network congestion, only that segment—rather than the entire data stream—needs to be retransmitted.

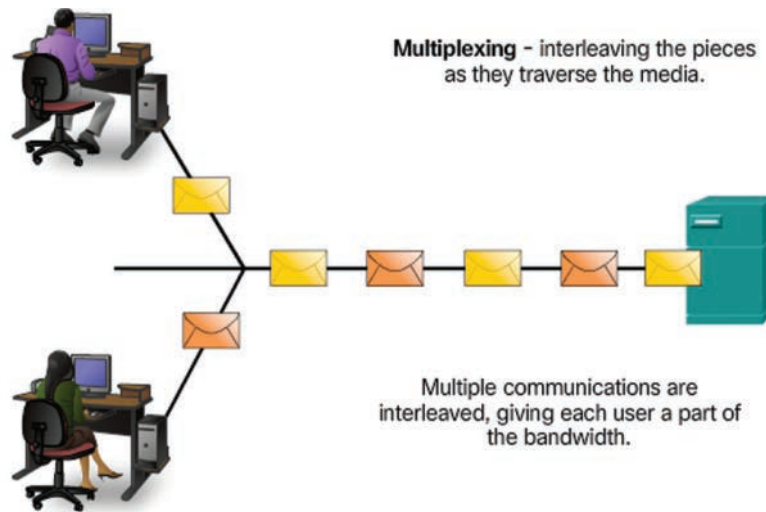


Figure 3-27 Multiplexing Multiple Messages

Sequencing (3.6.2)

The challenge in using segmentation and multiplexing to transmit messages across a network is the level of complexity added to the process. Imagine if you had to send a 100-page letter, but each envelope could hold only 1 page; you would need 100 envelopes, and each of them would need to be addressed individually. It is possible that the 100-page letter in 100 different envelopes would arrive out of order. Consequently, the information in each envelope would need to include a sequence number to ensure that the receiver could reassemble the pages in the proper order.

In network communications, each segment of a message must go through a similar process to ensure that it gets to the correct destination and can be reassembled into the content of the original message, as shown in Figure 3-28. TCP is responsible for sequencing the individual segments.

Protocol Data Units (3.6.3)

As application data is passed down the protocol stack on its way to be transmitted across the network media, various protocol information is added at each level. This process is known as *encapsulation*.

Note

Although the UDP PDU is called a datagram, IP packets are sometimes also referred to as IP datagrams.

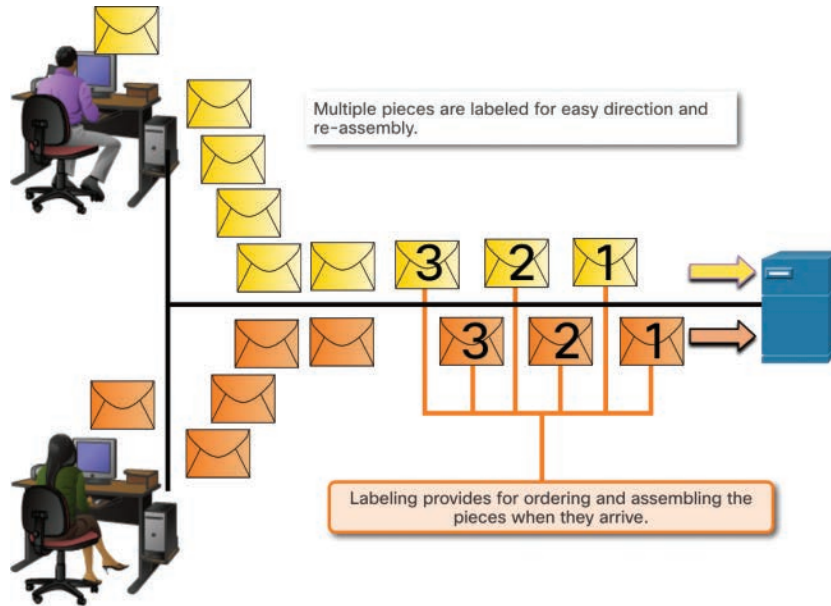


Figure 3-28 Labeling Segments for Reassembly

The form that a piece of data takes at any layer is called a *protocol data unit (PDU)*. During encapsulation, each layer encapsulates the PDU that it receives from the layer above in accordance with the protocol being used. At each stage of the process, a PDU has a different name to reflect its function. Although there is no universal naming convention for PDUs, in this book, the PDUs are named according to the protocols of the TCP/IP suite. The PDUs for the various forms of data are shown in Figure 3-29.

The following is a list of the PDUs, starting from the application layer:

- **Data:** The general term for the PDU used at the application layer
- **Segment:** Transport layer PDU
- **Packet:** Network layer PDU
- **Frame:** Data link layer PDU
- **Bits:** Physical layer PDU, used when physically transmitting data over the medium

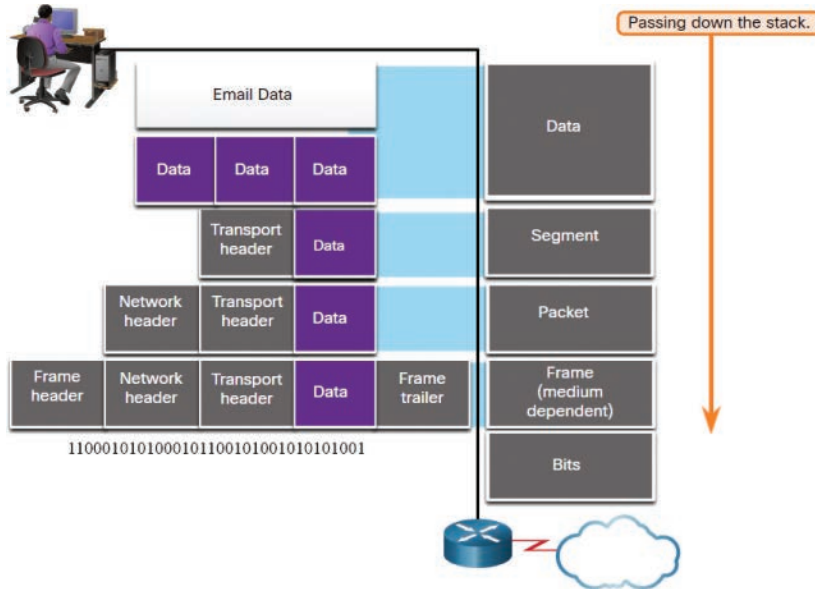


Figure 3-29 Encapsulation

Note

If the transport header is TCP, then the PDU is a segment. If the transport header is UDP, then it is a datagram.

Encapsulation Example (3.6.4)

When messages are being sent on a network, the encapsulation process works from top to bottom. At each layer, the upper-layer information is considered data within the encapsulated protocol. For example, in Figure 3-29, the transport layer header (for example, TCP), together with the original data, is considered data within the lower network layer of the IP packet. The network layer prepends a network layer protocol (IP). In other words, the TCP segment is considered the data portion of the network layer or IP packet.

De-encapsulation Example (3.6.5)

The encapsulation process is reversed at the receiving host; this is known as de-encapsulation. De-encapsulation is the process used by a receiving device to remove one or more of the protocol headers. The data is de-encapsulated as it moves up the stack toward the end-user application. As each layer receives the data from a lower layer, it processes the header for that layer. After processing the header, the header is removed, and the data portion is handed off to a higher-layer protocol.

**Interactive
Graphic****Check Your Understanding—Data Encapsulation (3.6.6)**

Refer to the online course to complete this activity.

Data Access (3.7)

Before you can access a network resource, the data must be encapsulated with the correct destination addresses and must also contain proper source addressing information to allow the destination device to reply. Accessing a local network resource requires two types of addresses that have different roles.

Addresses (3.7.1)

As you just learned, it is necessary to segment messages in a network. But those segmented messages cannot go anywhere if they are not addressed properly. This section provides an overview of network addresses. You will also get a chance to use the Wireshark tool, which will help you “view” network traffic.

The network layer and data link layer are responsible for delivering data from a source device to a destination device. As shown Figure 3-30, protocols at both layers contain source and destination addresses, but these addresses have different purposes:

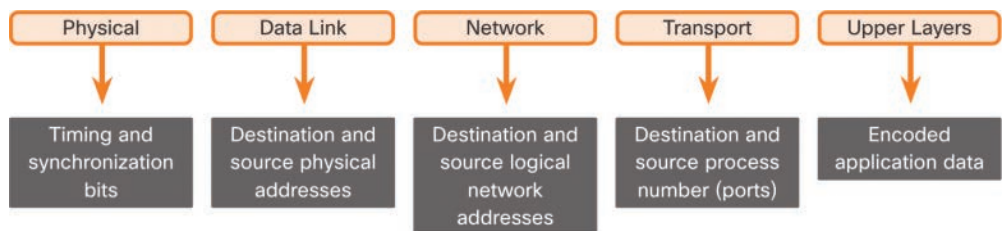


Figure 3-30 Network Addresses and Data Link Addresses

- **Network layer source and destination addresses:** Responsible for delivering an IP packet from the original source to the final destination, which may be on the same network or on a remote network.
- **Data link layer source and destination addresses:** Responsible for delivering the data link frame from one network interface card (NIC) to another NIC on the same network.

Layer 3 Logical Address (3.7.2)

An IP address is the network layer, or Layer 3, logical address used to deliver an IP packet from the original source to the final destination, as shown in Figure 3-31.

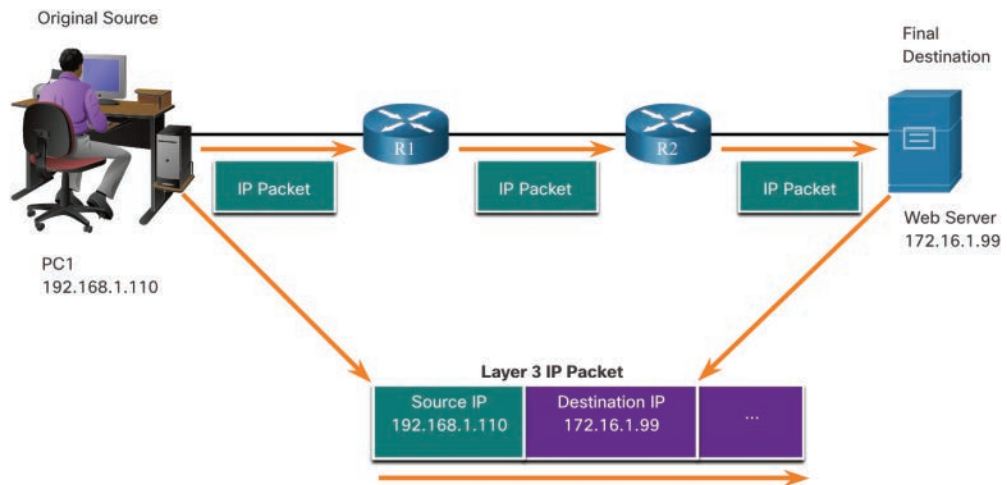


Figure 3-31 Layer 3 Network Addresses

An IP packet contains two IP addresses:

- **Source IP address:** The IP address of the sending device, which is the original source of the packet
- **Destination IP address:** The IP address of the receiving device, which is the final destination of the packet

The IP addresses indicate the original source IP address and the final destination IP address. This is true whether the source and destination are on the same IP network or different IP networks.

An IP address contains two parts:

- **Network portion (IPv4) or prefix (IPv6):** The leftmost part of the address indicates the network of which the IP address is a member. All devices on the same network have the same network portion of the address.
- **Host portion (IPv4) or interface ID (IPv6):** After the network portion, the remaining part of the address is the host portion, which identifies a specific device on the network. This portion is unique for each device or interface on the network.

Note

The subnet mask (IPv4) or prefix length (IPv6) is used to distinguish the network portion of an IP address from the host portion.

Devices on the Same Network (3.7.3)

Let's look at an example of a client computer, PC1, communicating with an FTP server on the same IP network:

- **Source IPv4 address:** This is the IPv4 address of the sending device, the client computer PC1 192.168.1.110.
- **Destination IPv4 address:** This is the IPv4 address of the receiving device, FTP server 192.168.1.9.

Notice in Figure 3-32 that the network portions of both the source IPv4 address and the destination IPv4 address are on the same network. The network portion of the source IPv4 address and the network portion of the destination IPv4 address are the same; therefore, the source and destination are on the same network.

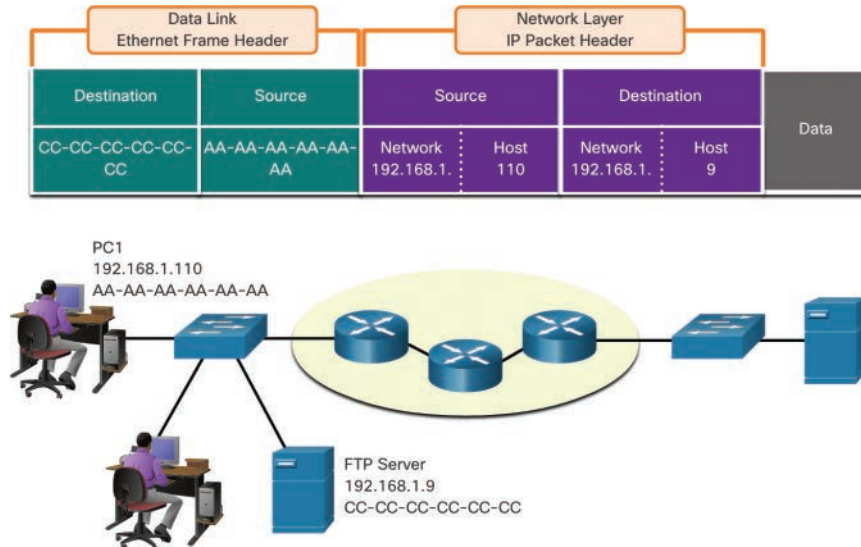


Figure 3-32 Network Layer Addresses on the Same Network

Role of the Data Link Layer Addresses: Same IP Network (3.7.4)

When the sender and receiver of an IP packet are on the same network, the data link frame is sent directly to the receiving device. On an *Ethernet* network, the data link addresses are known as Ethernet Media Access Control (MAC) addresses, as highlighted in Figure 3-33.

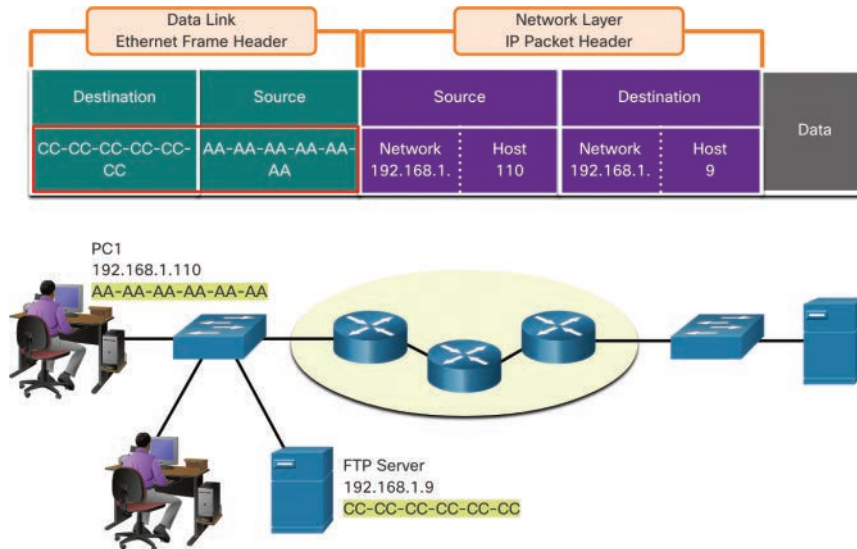


Figure 3-33 Data Link Layer Addresses on the Same Network

MAC addresses are physically embedded on Ethernet NICs. There are source and destination MAC addresses:

- **Source MAC address:** This is the data link address, or the Ethernet MAC address, of the device that sends the data link frame with the encapsulated IP packet. In this example, the MAC address of the Ethernet NIC of PC1 is AA-AA-AA-AA-AA-AA, written in hexadecimal notation.
- **Destination MAC address:** When the receiving device is on the same network as the sending device, this is the data link address of the receiving device. In this example, the destination MAC address is the MAC address of the FTP server, CC-CC-CC-CC-CC-CC, written in hexadecimal notation.

The frame with the encapsulated IP packet can be transmitted from PC1 directly to the FTP server.

Devices on a Remote Network (3.7.5)

What are the roles of the network layer address and the data link layer address when a device is communicating with a device on a remote network? In the example shown in Figure 3-33, a client computer, PC1, is communicating with a server, named Web Server, on a different IP network.

Role of the Network Layer Addresses (3.7.6)

When the sender of a packet is on a different network from the receiver, the source and destination IP addresses represent hosts on different networks. This is indicated by the network portion of the IP address of the destination host.

- **Source IPv4 address:** The IPv4 address of the sending device, the client computer PC1: 192.168.1.110
- **Destination IPv4 address:** The IPv4 address of the receiving device, the server Web Server: 172.16.1.99

Notice in Figure 3-34 that the network portion of the source IPv4 address and destination IPv4 address are on different networks.

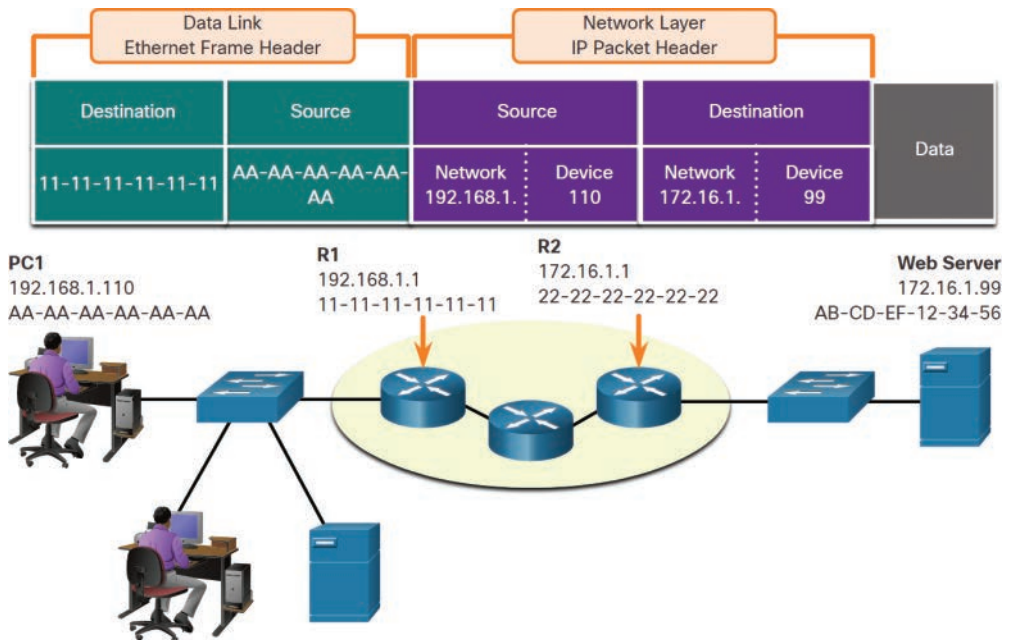


Figure 3-34 Network Layer Addresses on Different Networks

Role of the Data Link Layer Addresses: Different IP Networks (3.7.7)

When the sender and receiver of an IP packet are on different networks, the Ethernet data link frame cannot be sent directly to the destination host because the host is not directly reachable in the network of the sender. The Ethernet frame must be sent to another device, known as the *default gateway*. In our example, the default gateway is R1. R1 has an Ethernet data link address that is on the same network as PC1. This allows PC1 to reach the router directly.

- **Source MAC address:** The source MAC address is the Ethernet MAC address of the sending device, PC1. The MAC address of the Ethernet interface of PC1 is AA-AA-AA-AA-AA-AA.
- **Destination MAC address:** When the receiving device, the destination IP address, is on a different network from the sending device, the sending device uses the Ethernet MAC address of the default gateway or router. In this example, the destination MAC address is the MAC address of the R1 Ethernet interface, 11-11-11-11-11-11. This is the interface that is attached to the same network as PC1, as shown in Figure 3-35.

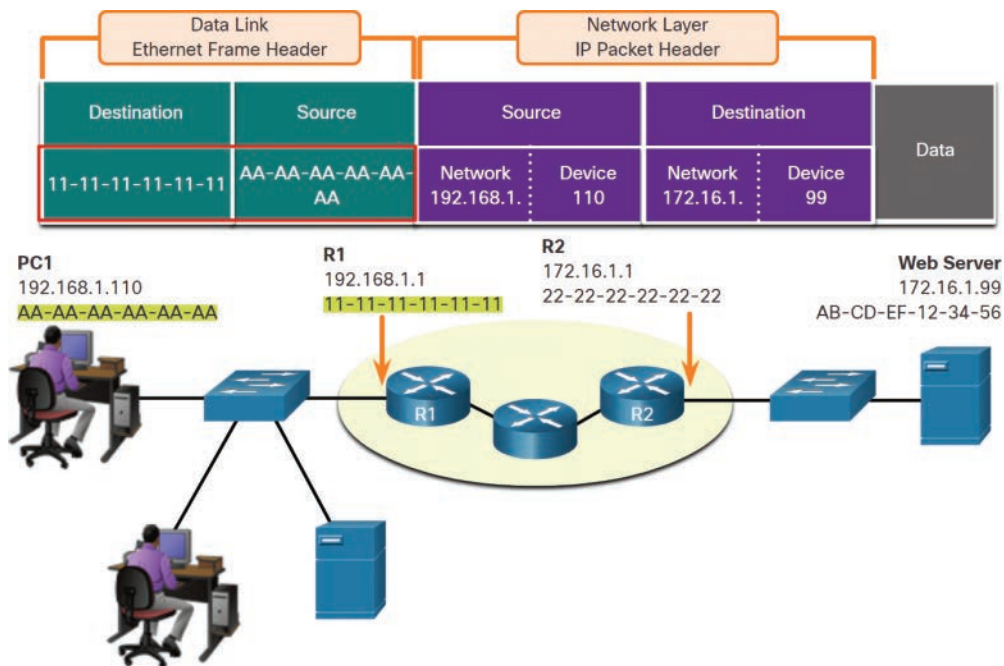


Figure 3-35 Data Link Layer Addresses on Different Networks

The Ethernet frame with the encapsulated IP packet can now be transmitted to R1. R1 forwards the packet to the destination, Web Server. This might mean that R1 forwards the packet to another router or directly to Web Server if the destination is on a network connected to R1.

It is important that the IP address of the default gateway be configured on each host on the local network. All packets to destinations on remote networks are sent to the default gateway. Ethernet MAC addresses and the default gateway are discussed in more detail in Chapters 7, 8, and 9.

Data Link Addresses (3.7.8)

The data link layer (Layer 2) physical address has a unique role. The purpose of the data link address is to deliver the data link frame from one network interface to another network interface on the same network.

Before an IP packet can be sent over a wired or wireless network, it must be encapsulated in a data link frame so it can be transmitted over the physical medium. The process is illustrated in Figures 3-36 through 3-38.

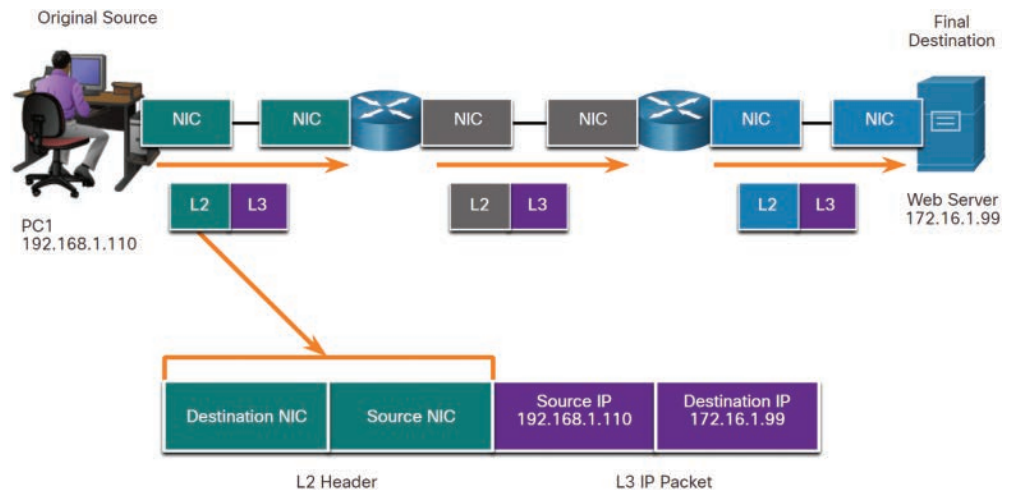


Figure 3-36 Layer 2 Data Link Addresses: First Hop

As the IP packet travels from host to router, from router to router, and finally from router to host, at each point along the way, the IP packet is encapsulated in a new data link frame. Each data link frame contains the source data link address of the NIC sending the frame and the destination data link address of the NIC receiving the frame.

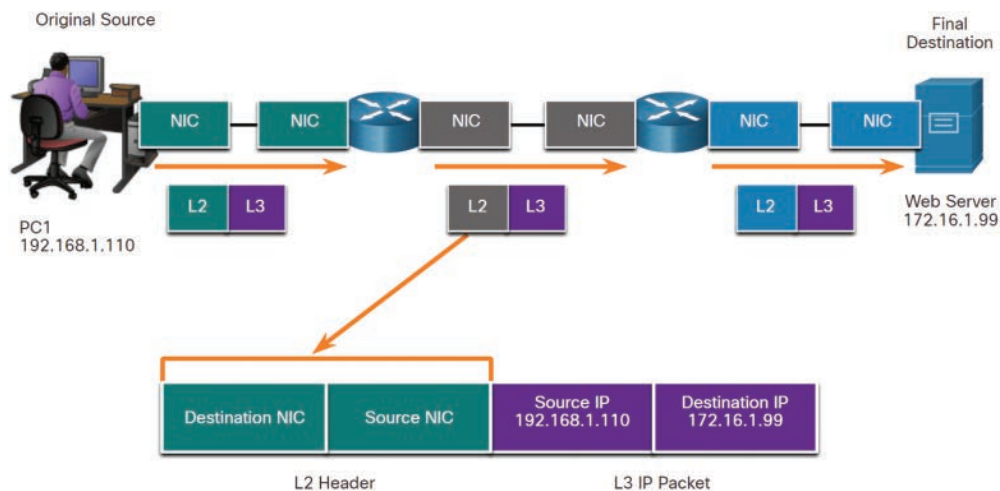


Figure 3-37 Layer 2 Data Link Addresses: Second Hop

The Layer 2 (data link) protocol is only used to deliver the packet from NIC to NIC on the same network. The router removes the Layer 2 information as it is received on one NIC and adds new data link information before forwarding out the exit NIC on its way toward the final destination.

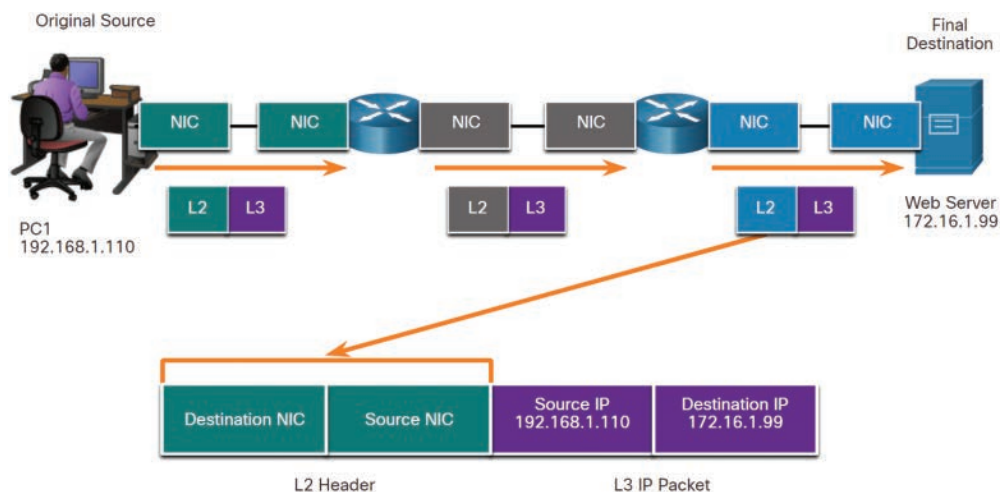


Figure 3-38 Layer 2 Data Link Addresses: Third Hop

The IP packet is encapsulated in a data link frame that contains the following data link information:

- **Source data link address:** The physical address of the NIC that is sending the data link frame.

- **Destination data link address:** The physical address of the NIC that is receiving the data link frame. This address is either the next hop router or the address of the final destination device.

**Lab—Install Wireshark (3.7.9)**

Wireshark is a software protocol analyzer, or “packet sniffer” application, used for network troubleshooting, analysis, software and protocol development, and education. Wireshark is used throughout the course to demonstrate network concepts. In this lab, you will download and install Wireshark.

**Lab—Use Wireshark to View Network Traffic (3.7.10)**

In this lab, you will use Wireshark to capture and analyze traffic.

**Interactive
Graphic****Check Your Understanding—Data Access (3.7.11)**

Refer to the online course to complete this activity.

Summary (3.8)

The following is a summary of the topics in the chapter and their corresponding online modules.

The Rules

All communication methods have three elements in common: message source (sender), message destination (receiver), and channel. Sending a message is governed by rules called *protocols*. Protocols must include an identified sender and receiver, common language and grammar, speed and timing of delivery, and confirmation or acknowledgment requirements. Common computer protocols include these requirements: message encoding, formatting and encapsulation, size, timing, and delivery options. Encoding is the process of converting information into another acceptable form for transmission. Decoding reverses this process to interpret the information. Message formats depend on the type of message and the channel used to deliver the message. Message timing includes flow control, response timeout, and access method. Message delivery options include unicast, multicast, and broadcast.

Protocols

Protocols are implemented by end devices and intermediary devices in software, hardware, or both. A message sent over a computer network typically requires the use of several protocols, each one with its own functions and format. Each network protocol has its own function, format, and rules for communications. The Ethernet family of protocols includes IP, TCP, HTTP, and many more. Protocols such as SSH, SSL, and TLS secure data to provide authentication, data integrity, and data encryption. Protocols such as OSPF and BGP enable routers to exchange route information, compare path information, and select the best path to the destination network. Protocols such as DHCP and DNS are used for the automatic detection of devices or services. Computers and network devices use agreed-upon protocols that provide functions such as addressing, reliability, flow control, sequencing, error-detection, and an application interface.

Protocol Suites

A protocol suite is a group of interrelated protocols necessary to perform a communication function. A protocol stack shows how the individual protocols within a suite are implemented. Since the 1970s, there have been several different protocol suites, some developed by standards organizations and others developed by various vendors.

TCP/IP protocols are available for the application, transport, and internet layers. TCP/IP is the protocol suite used by today's networks and internet. TCP/IP offers two important aspects to vendors and manufacturers: an open standard protocol suite and a standards-based protocol suite. The TCP/IP protocol suite communication process enables processes such as a web server encapsulating and sending a web page to a client, as well as the client de-encapsulating the web page for display in a web browser.

Standards Organizations

Open standards encourage interoperability, competition, and innovation. Standards organizations are usually vendor-neutral, nonprofit organizations established to develop and promote the concept of open standards. Various organizations have different responsibilities for promoting and creating standards for the internet, including ISOC, IAB, IETF, and IRTF. Standards organizations that develop and support TCP/IP include ICANN and IANA. Electronic and communications standards organizations include IEEE, EIA, TIA, and ITU-T.

Reference Models

The two reference models that are used to describe network operations are OSI and TCP/IP. The OSI model has seven layers:

- Layer 7: application layer
- Layer 6: presentation layer
- Layer 5: session layer
- Layer 4: transport layer
- Layer 3: network layer
- Layer 2: data link layer
- Layer 1: physical layer

The TCP/IP model has four layers:

- Layer 4: application layer
- Layer 3: transport layer
- Layer 2: internet layer
- Layer 1: network access layer

Data Encapsulation

Segmenting messages has two primary benefits:

- By sending smaller individual pieces from source to destination, many different conversations can be interleaved on the network. This is called *multiplexing*.
- Segmentation can increase the efficiency of network communications. If part of a message fails to make it to the destination, only the missing parts need to be retransmitted.

TCP is responsible for sequencing the individual segments of a message. The form that a piece of data takes at any layer is called a *protocol data unit (PDU)*. During encapsulation, each layer encapsulates the PDU that it receives from the layer above in accordance with the protocol being used. When sending messages on a network, the encapsulation process works from top to bottom. This process is reversed at the receiving host and is known as *de-encapsulation*. De-encapsulation is the process used by a receiving device to remove one or more of the protocol headers. The data is de-encapsulated as it moves up the stack toward the end-user application.

Data Access

The network layer and data link layer are responsible for delivering data from a source device to a destination device. Protocols at both layers contain source and destination addresses, but their addresses have different purposes:

- **Network layer source and destination addresses:** Responsible for delivering the IP packet from the original source to the final destination, which may be on the same network or on a remote network.
- **Data link layer source and destination addresses:** Responsible for delivering the data link frame from one network interface card (NIC) to another NIC on the same network.

The IP addresses indicate the original source IP address and final destination IP address. An IP address contains two parts: the network portion (IPv4) or prefix (IPv6) and the host portion (IPv4) or interface ID (IPv6). When the sender and receiver of an IP packet are on the same network, the data link frame is sent directly to the receiving device. On an Ethernet network, the data link addresses are known as Ethernet Media Access Control (MAC) addresses. When the sender of a packet is on a different network from the receiver, the source and destination IP addresses represent hosts on different networks. The Ethernet frame must be sent to another device, known as a router or default gateway.

Practice

The following activities provide practice with the topics introduced in this chapter. The lab is available in the companion *Introduction to Networks Labs & Study Guide (CCNAv7)* (ISBN 9780136634454). The Packet Tracer activity instructions are also provided in the *Labs & Study Guide*. The PKA files are available in the online course.



Labs

Lab 3.4.4: Research Networking Standards

Lab 3.7.9: Install Wireshark

Lab 3.7.10: Use Wireshark to View Network Traffic



Packet Tracer Activity

Packet Tracer 3.5.5: Investigate the TCP-IP and OSI Models in Action

Check Your Understanding Questions

Complete all the review questions listed here to test your understanding of the topics and concepts in this chapter. The appendix “Answers to ‘Check Your Understanding’ Questions” lists the answers.

1. Which of the following are standards organizations? (Choose three.)
 - a. IANA
 - b. TCP/IP
 - c. IEEE
 - d. IETF
 - e. OSI
 - f. MAC
2. What type of communication sends a message to all devices on a local area network?
 - a. broadcast
 - b. multicast
 - c. unicast
 - d. allcast

3. In computer communication, what is the purpose of message encoding?
 - a. to convert information to the appropriate form for transmission
 - b. to interpret information
 - c. to break large messages into smaller frames
 - d. to negotiate correct timing for successful communications
4. Which message delivery option is used when all devices need to receive the same message simultaneously?
 - a. duplex
 - b. unicast
 - c. multicast
 - d. broadcast
5. What are two benefits of using a layered network model? (Choose two.)
 - a. It assists in protocol design.
 - b. It speeds up packet delivery.
 - c. It prevents designers from creating their own model.
 - d. It prevents technology in one layer from affecting other layers.
 - e. It ensures that a device at one layer can function at the next higher layer.
6. What is the purpose of protocols in data communications?
 - a. specifying the bandwidth of the channel or medium for each type of communication
 - b. specifying the device operating systems that will support the communication
 - c. providing the rules required for a specific type of communication to occur
 - d. dictating the content of a message sent during communication
7. Which logical address is used for delivery of data to a remote network?
 - a. destination MAC address
 - b. destination IP address
 - c. destination port number
 - d. source MAC address
 - e. source IP address

8. What is the general term that is used to describe a piece of data at any layer of a networking model?
 - a. frame
 - b. packet
 - c. protocol data unit
 - d. segment
9. Which two protocols function at the internet layer? (Choose two.)
 - a. POP
 - b. BOOTP
 - c. ICMP
 - d. IP
 - e. Ethernet
10. Which layer of the OSI model defines services to segment and reassemble data for individual communications between end devices?
 - a. application
 - b. presentation
 - c. session
 - d. transport
 - e. network
11. Which type of communication sends a message to a group of host destinations simultaneously?
 - a. broadcast
 - b. multicast
 - c. unicast
 - d. anycast
12. What process is used to receive transmitted data and convert it into a readable message?
 - a. access control
 - b. decoding
 - c. encapsulation
 - d. flow control

13. What is done to an IP packet before it is transmitted over the physical medium?
 - a. It is tagged with information, guaranteeing reliable delivery.
 - b. It is segmented into smaller individual pieces.
 - c. It is encapsulated into a TCP segment.
 - d. It is encapsulated in a Layer 2 frame.
14. What process is used to place one message inside another message for transfer from a source to a destination?
 - a. access control
 - b. decoding
 - c. encapsulation
 - d. flow control
15. A web client is sending a request for a web page to a web server. From the perspective of the client, what is the correct order of the protocol stack that is used to prepare the request for transmission?
 - a. HTTP, IP, TCP, Ethernet
 - b. HTTP, TCP, IP, Ethernet
 - c. Ethernet, TCP, IP, HTTP
 - d. Ethernet, IP, TCP, HTTP

Physical Layer

Objectives

Upon completion of this chapter, you will be able to answer the following questions:

- What are the purpose and functions of the physical layer in a network?
- What are the characteristics of the physical layer?
- What are the basic characteristics of copper cabling?
- How is UTP cable used in Ethernet networks?
- What is fiber-optic cable, and what are its main advantages over other media?
- How do you connect devices using wired and wireless media?

Key Terms

This chapter uses the following key terms. You can find the definitions in the glossary at the end of the book.

wireless access point (AP) page 138

network interface card (NIC) page 139

*International Organization for
Standardization (ISO)* page 141

*Telecommunications Industry Association/
Electronic Industries Association
(TIA/EIA)* page 141

*International Telecommunication Union
(ITU)* page 141

*American National Standards Institute
(ANSI)* page 141

*Institute of Electrical and Electronics
Engineers (IEEE)* page 141

encoding page 142

Manchester encoding page 143

bandwidth page 145

throughput page 146

goodput page 146

electromagnetic interference (EMI) page 147

crosstalk page 147

unshielded twisted-pair (UTP) page 148

shielded twisted-pair (STP) page 150

coaxial cable page 151

fiber-optic cable page 152

Wi-Fi page 165

Bluetooth page 166

WiMAX page 166

Zigbee page 166

Introduction (4.0)

The physical layer of the OSI model sits at the bottom of the stack. It is part of the network access layer of the TCP/IP model. Without the physical layer, you would not have a network. This chapter explains, in detail, the three ways to connect to the physical layer. Packet Tracer activities and a lab will give you the confidence you need to cable up your own network! Let's get busy!

Purpose of the Physical Layer (4.1)

All data being transferred over a network must be represented on a medium by the sending node and interpreted on a medium by the receiving node. The physical layer is responsible for these functions. This section explores the physical layer.

The Physical Connection (4.1.1)

Whether connecting to a local printer in a home or to a website in another country, before any network communications can occur, a physical connection to a local network must be established. A physical connection can be a wired connection using a cable or a wireless connection using radio waves.

The type of physical connection used depends on the setup of the network. For example, in many corporate offices, employees have desktop or laptop computers that are physically connected, via cable, to a shared switch. This type of setup is a wired network. Data is transmitted through a physical cable.

In addition to wired connections, many businesses also offer wireless connections for laptops, tablets, and smartphones. With wireless devices, data is transmitted using radio waves. Wireless connectivity is common as individuals and businesses alike know its advantages. Devices on a wireless network must be connected to a *wireless access point (AP)* or wireless router like the one shown in Figure 4-1.



Figure 4-1 Wireless Router

The numbers in Figure 4-1 indicate the components of an access point:

1. The wireless antennas (which are embedded inside the router version shown in Figure 4-1)
2. Several Ethernet switchports
3. An internet port

Much like corporate offices, most homes offer both wired and wireless network connectivity. Figure 4-2 shows a home router and a laptop connecting to the local-area network (LAN).



Figure 4-2 Wired Connection to a Wireless Router

A *network interface card (NIC)* connects a device to a network. Ethernet NICs are used for wired connections, as shown in Figure 4-3, whereas wireless local-area network (WLAN) NICs are used for wireless. An end-user device may include one or both types of NIC. A network printer, for example, may only have an Ethernet NIC, in which case it must connect to the network using an Ethernet cable. Other devices, such as tablets and smartphones, might only contain a WLAN NIC and must use a wireless connection.

Not all physical connections are equal, in terms of the performance level, when connecting to a network.

The Physical Layer (4.1.2)

The OSI physical layer provides the means to transport the bits that make up a data link layer frame across the network media. This layer accepts a complete frame from the data link layer and encodes it as a series of signals that are transmitted to the local media. The encoded bits that comprise a frame are received by either an end device or an intermediate device.

Figure 4-4 show an example of the encapsulation process. In the last part of this process, the bits are sent over the physical medium. The physical layer encodes the frames and creates the electrical, optical, or radio wave signals that represent the bits in each frame. These signals are then sent over the media, one at a time.



Figure 4-3 Wired Connection Using an Ethernet NIC

The destination node physical layer retrieves these individual signals from the media, restores them to their bit representations, and passes the bits up to the data link layer as a complete frame.

Interactive
Graphic

Refer to the online course to view an animation of the encapsulation process.

Interactive
Graphic

Check Your Understanding—Purpose of the Physical Layer (4.1.3)

Refer to the online course to complete this activity.

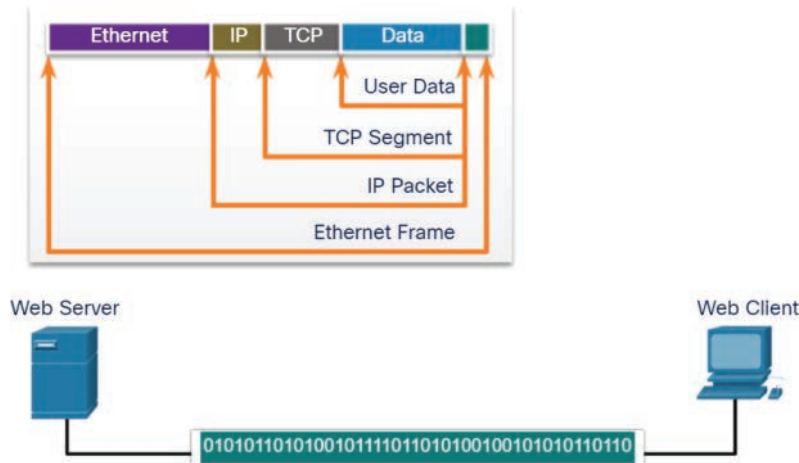


Figure 4-4 Bits Transported over the Medium

Physical Layer Characteristics (4.2)

At the foundation of network communications is the physical layer, Layer 1. This section examines standards and components that make up the physical layer.

Physical Layer Standards (4.2.1)

The previous section provides a high-level overview of the physical layer and its place in a network. This section dives a bit deeper into the specifics of the physical layer. It looks at the components and the media used to build a network, as well as the standards that are required so that everything works together.

The protocols and operations of the upper OSI layers are performed using software designed by software engineers and computer scientists. The services and protocols in the TCP/IP suite are defined by the Internet Engineering Task Force (IETF).

The physical layer consists of electronic circuitry, media, and connectors developed by engineers. Therefore, it is appropriate that the standards governing this hardware are defined by the relevant electrical and communications engineering organizations.

Many different international and national organizations, regulatory government organizations, and private companies are involved in establishing and maintaining physical layer standards. For instance, the physical layer hardware, media, encoding, and signaling standards are defined and governed by the following standards organizations (see Figure 4-5):

- *International Organization for Standardization (ISO)*
- *Telecommunications Industry Association/Electronic Industries Association (TIA/EIA)*
- *International Telecommunication Union (ITU)*
- *American National Standards Institute (ANSI)*
- *Institute of Electrical and Electronics Engineers (IEEE)*
- National telecommunications regulatory authorities, including the Federal Communications Commission (FCC) in the United States and the European Telecommunications Standards Institute (ETSI)

In addition to these, there are many regional cabling standards groups that develop local specifications, such as CSA (Canadian Standards Association), CENELEC (European Committee for Electrotechnical Standardization), and JSA/JIS (Japanese Standards Association).

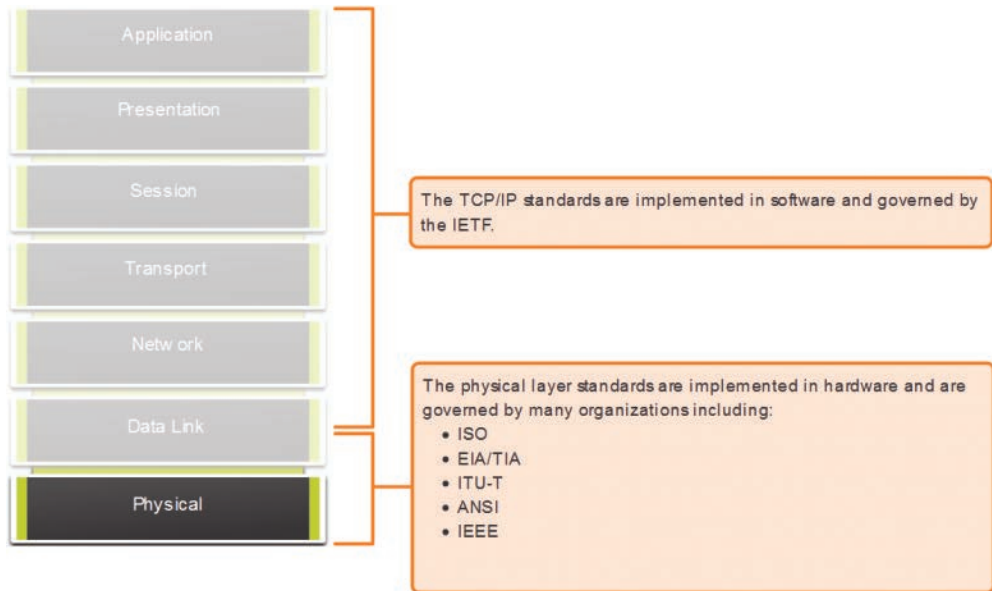


Figure 4-5 Physical Layer Standards Organizations

Physical Components (4.2.2)

The physical layer standards address three functional areas:

- Physical components
- Encoding
- Signaling

The physical components are the electronic hardware devices, media, and other connectors that transmit the signals representing bits. Hardware components such as NICs, interfaces and connectors, and cables (including cable materials and cable designs) are all specified in standards associated with the physical layer. The various ports and interfaces on a Cisco 1941 router are also examples of physical components with specific connectors and pinouts based on standards.

Encoding (4.2.3)

Encoding, or line encoding, is a method of converting a stream of data bits into a predefined “code.” Codes are groupings of bits used to provide a predictable pattern that can be recognized by both a sender and a receiver. In other words, encoding is a method or pattern used to represent digital information. This is similar to how Morse code encodes a message using a series of dots and dashes.

For example, *Manchester encoding* represents a high- to low-voltage transition as a 0 bit and a low- to high-voltage transition as a 1 bit. An example of Manchester encoding is illustrated in Figure 4-6. The transition occurs at the middle of each bit period. This type of encoding is used in 10 Mbps Ethernet. Faster data rates require more complex encoding. Manchester encoding is used in older Ethernet standards, such as 10BASE-T. Ethernet 100BASE-TX uses 4B/5B encoding, and 1000BASE-T uses 8B/10B encoding.

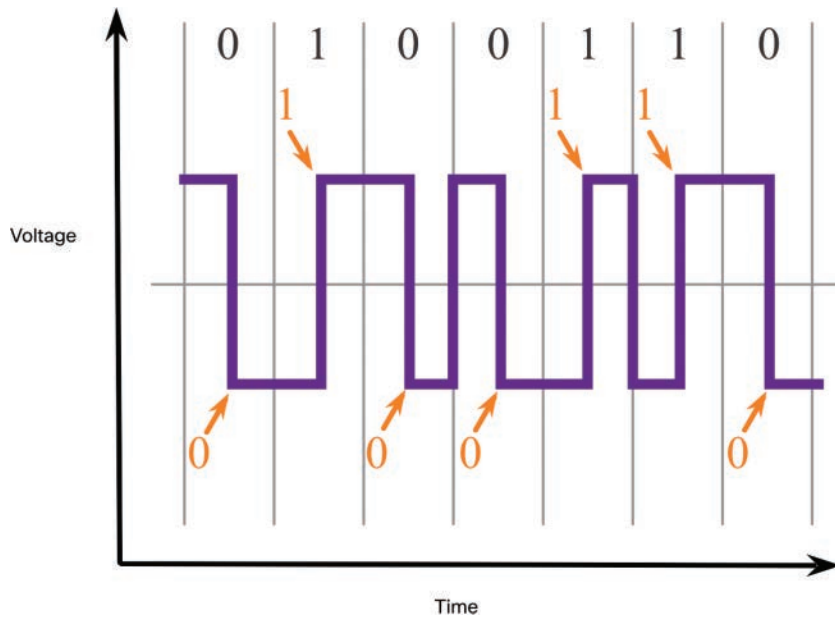


Figure 4-6 Manchester Encoding

Signaling (4.2.4)

The physical layer must generate the electrical, optical, or wireless signals that represent the 1s and 0s on the media. The way that bits are represented is called the *signaling method*. The physical layer standards must define what type of signal represents a 1 and what type of signal represents a 0. This can be as simple as a change in the level of an electrical signal or optical pulse. For example, a long pulse might represent a 1, whereas a short pulse might represent a 0. This is similar to the signaling method used in Morse code, which may use a series of on/off tones, lights, or clicks to send text over telephone wires or between ships at sea.

Figures 4-7 through 4-9 show illustrations of signaling for copper cable, fiber-optic cable, and wireless media.

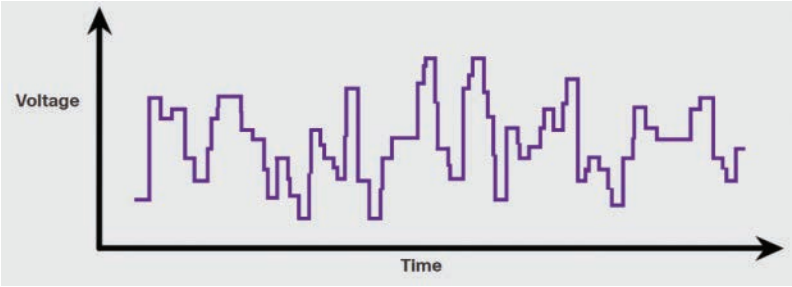


Figure 4-7 Electrical Signals over Copper Cable

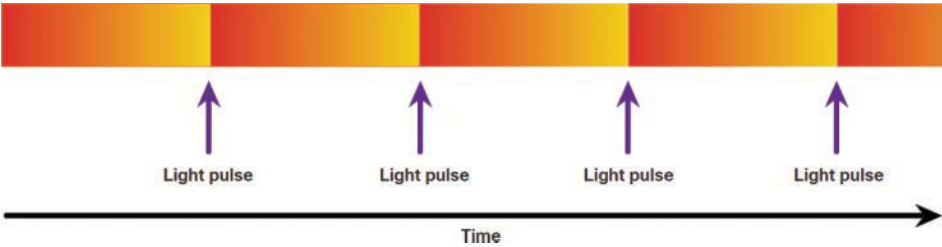


Figure 4-8 Light Pulses over Fiber-Optic Cable

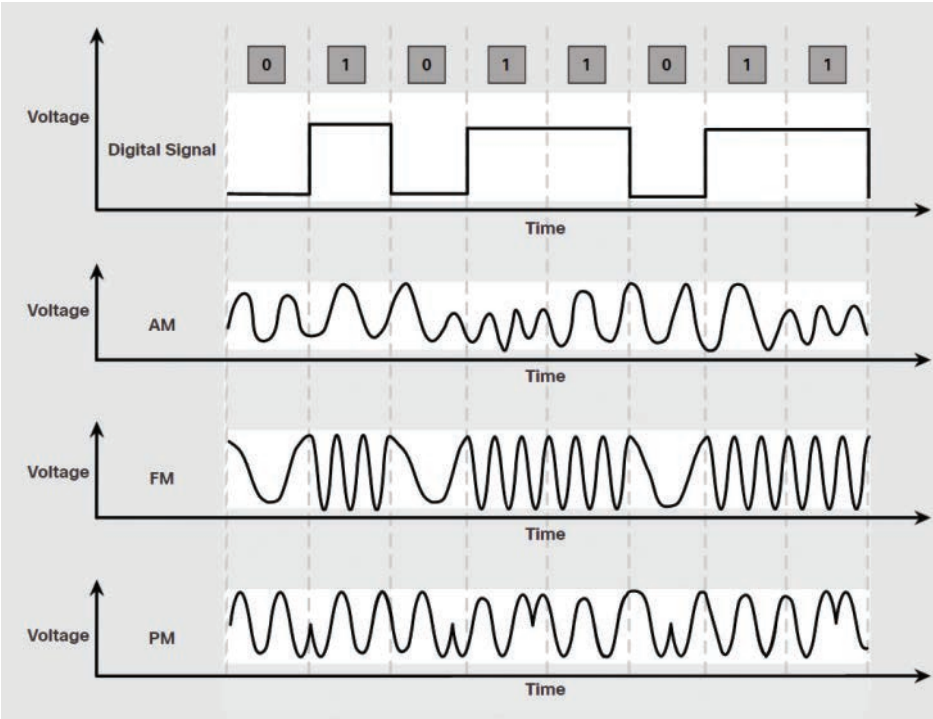


Figure 4-9 Microwave Signals over Wireless

Bandwidth (4.2.5)

Different physical media support the transfer of bits at different rates. Data transfer is usually discussed in terms of bandwidth. *Bandwidth* is the capacity at which a medium can carry data. Digital bandwidth measures the amount of data that can flow from one place to another in a given amount of time. Bandwidth is typically measured in kilobits per second (Kbps), megabits per second (Mbps), or gigabits per second (Gbps). Bandwidth is sometimes thought of as the speed that bits travel, but this is not accurate. For example, in both 10 Mbps and 100 Mbps Ethernet, the bits are sent at the speed of electricity. The difference between 10 Mbps and 100 Mbps Ethernet is the number of bits transmitted per second.

A combination of factors determines the practical bandwidth of a network:

- The properties of the physical media
- The technologies chosen for signaling and detecting network signals

Physical media properties, current technologies, and the laws of physics all play roles in determining the available bandwidth.

Table 4-1 shows the commonly used units of measure for bandwidth.

Table 4-1 Bandwidth Units

Unit of Bandwidth	Abbreviation	Equivalence
Bits per second	bps	1 bps = fundamental unit of bandwidth
Kilobits per second	Kbps	1 Kbps = 1000 bps = 10^3 bps
Megabits per second	Mbps	1 Mbps = 1,000,000 bps = 10^6 bps
Gigabits per second	Gbps	1 Gbps = 1,000,000,000 bps = 10^9 bps
Terabits per second	Tbps	1 Tbps = 1,000,000,000,000 bps = 10^{12} bps

Bandwidth Terminology (4.2.6)

Terms used to measure the quality of bandwidth include

- Latency
- Throughput
- Goodput

Latency

Latency refers to the amount of time, including delays, for data to travel from one point to another.

In an internetwork, or a network with multiple segments, throughput cannot be faster than the slowest link in the path from source to destination. Even if all, or most, of the segments have high bandwidth, it takes only one segment in the path with low throughput to create a bottleneck in the throughput of the entire network.

Throughput

Throughput is the measure of the transfer of bits across the media over a given period of time.

Due to a number of factors, throughput usually does not match the specified bandwidth in physical layer implementations. Throughput is usually lower than the bandwidth. Many factors that influence throughput, including the following:

- The amount of traffic
- The type of traffic
- The latency created by the number of network devices encountered between source and destination

There are many online speed tests that can reveal the throughput of an internet connection.

Goodput

There is a third measurement to assess the transfer of usable data: goodput. *Goodput* is the measure of usable data transferred over a given period of time. Goodput is throughput minus traffic overhead for establishing sessions, acknowledgments, encapsulation, and retransmitted bits. Goodput is always lower than throughput, which is generally lower than the bandwidth.

Interactive
Graphic

Check Your Understanding—Physical Layer Characteristics (4.2.7)

Refer to the online course to complete this activity.

Copper Cabling (4.3)

One of the oldest and most used media for communications is copper cabling. This section examines the characteristics and use of copper media in data networks.

Characteristics of Copper Cabling (4.3.1)

Copper cabling is the most common type of cabling used in networks today. In fact, copper cabling is not just one type of cable. There are three different types of copper cabling that are each used in specific situations.

Networks use copper cabling because it is inexpensive and easy to install, and it has low resistance to electrical current. However, copper cabling is limited by distance and signal interference.

Data is transmitted on copper cables as electrical pulses. A detector in the network interface of a destination device must receive a signal that can be successfully decoded to match the signal sent. However, the farther the signal travels, the more it deteriorates. This is referred to as *signal attenuation*. For this reason, all copper media must follow strict distance limitations, as specified by the guiding standards.

The timing and voltage values of electrical pulses are also susceptible to interference from two sources:

- **Electromagnetic interference (EMI) or radio frequency interference (RFI):** EMI and RFI signals can distort and corrupt the data signals being carried by copper media. Potential sources of EMI and RFI include radio waves and electromagnetic devices, such as fluorescent lights or electric motors.
- **Crosstalk:** Crosstalk is a disturbance caused by the electric or magnetic fields of a signal on one wire to the signal in an adjacent wire. In telephone circuits, crosstalk can result in hearing part of another voice conversation from an adjacent circuit. Specifically, when an electrical current flows through a wire, it creates a small, circular magnetic field around the wire, which can be picked up by an adjacent wire.

Figure 4-10 shows how data transmission can be affected by interference:

1. A pure digital signal is transmitted.
2. On the medium, there is an interference signal.
3. The digital signal is corrupted by the interference signal.
4. The receiving computer reads a changed signal. Notice that a 0 bit is now interpreted as a 1 bit.

To counter the negative effects of EMI and RFI, some types of copper cables are wrapped in metallic shielding and require proper grounding connections.

To counter the negative effects of crosstalk, some types of copper cables have opposing circuit wire pairs twisted together, which effectively cancels the crosstalk.

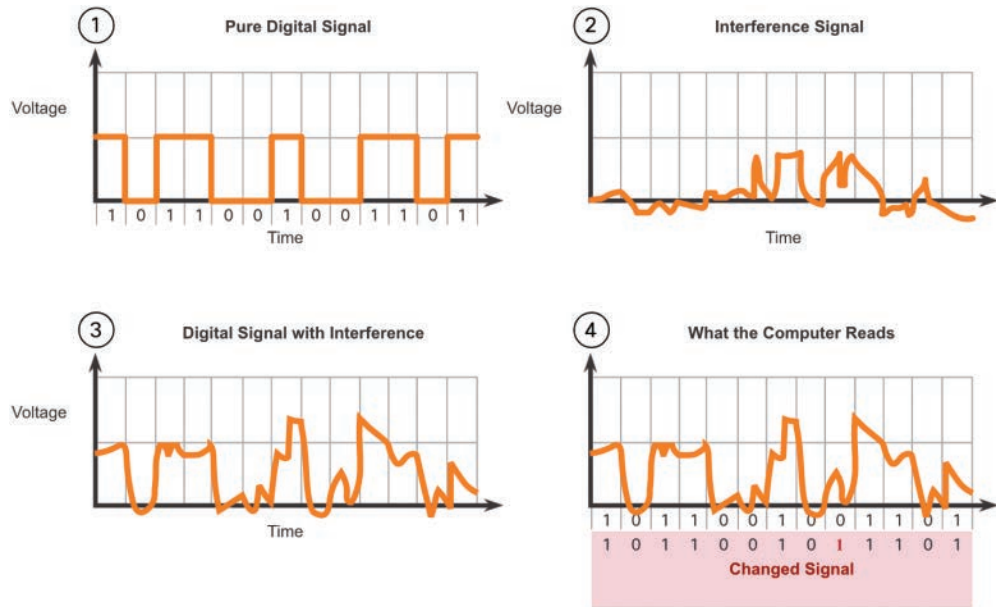


Figure 4-10 Effect of Interference on Data Transmission

The susceptibility of copper cables to electronic noise can also be limited by

- Selecting the cable type or category most suited to a given networking environment
- Designing a cable infrastructure to avoid known and potential sources of interference in the building structure
- Using cabling techniques that include the proper handling and termination of the cables

Types of Copper Cabling (4.3.2)

There are three main types of copper media used in networking, as shown in Figure 4-11.

Unshielded Twisted-Pair (UTP) (4.3.3)

Unshielded twisted-pair (UTP) cabling is the most common networking medium. UTP cabling, terminated with RJ-45 connectors, is used for interconnecting network hosts with intermediary networking devices, such as switches and routers.



Unshielded Twisted-Pair (UTP) Cable



Shielded Twisted-Pair (STP) Cable



Coaxial Cable

Figure 4-11 Types of Copper Cable

In LANs, UTP cable consists of four pairs of color-coded wires that have been twisted together and then encased in a flexible plastic sheath that protects against minor physical damage, as shown in Figure 4-12. The twisting of wires helps protect against signal interference from other wires.

The color codes identify the individual pairs and wires and aid in cable termination.

The numbers in Figure 4-12 identify some key features of UTP cable:

1. The outer jacket protects the copper wires from physical damage.
2. Twisted pairs protect the signal from interference.
3. Color-coded plastic insulation electrically isolates wires from each other and identifies the pairs.

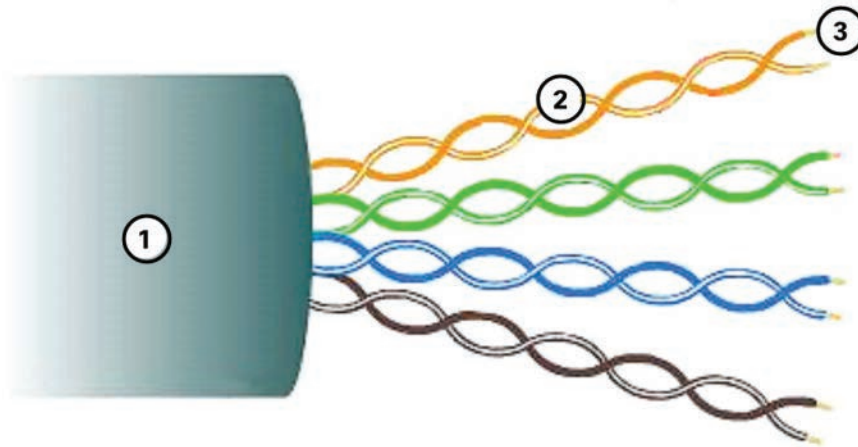


Figure 4-12 UTP Cable

Shielded Twisted-Pair (STP) (4.3.4)

Shielded twisted-pair (STP) provides better noise protection than UTP cabling. However, compared to UTP cable, STP cable is significantly more expensive and difficult to install. Like UTP cable, STP uses RJ-45 connectors.

STP cables combine the techniques of shielding to counter EMI and RFI and wire twisting to counter crosstalk. To gain the full benefit of the shielding, STP cables are terminated with special shielded STP data connectors. If the cable is improperly grounded, the shield may act as an antenna and pick up unwanted signals.

The STP cable shown in Figure 4-13 uses four pairs of wires, each wrapped in a foil shield, and then wrapped in an overall metallic braid or foil.

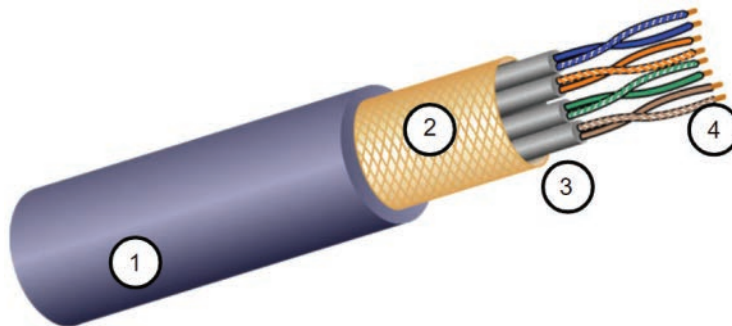


Figure 4-13 STP Cable

The numbers in Figure 4-13 identify some key features of STP cable:

1. Outer jacket
2. Braided or foil shield
3. Foil shields
4. Twisted pairs

Coaxial Cable (4.3.5)

Coaxial cable, or coax for short, gets its name from the fact that there are two conductors that share the same axis. As shown in Figure 4-14, coaxial cable consists of the following:

1. The entire cable is covered with a cable jacket to prevent minor physical damage.
2. The insulating material is surrounded by a woven copper braid, or metallic foil, that acts as the second wire in the circuit and as a shield for the inner conductor. This second layer, or shield, also reduces the amount of outside electromagnetic interference.
3. A layer of flexible plastic insulation surrounds a copper conductor.
4. A copper conductor is used to transmit the electronic signals.

Different types of connectors are used with coax cable. Figure 4-14 shows the Bayonet Neill–Concelman (BNC), N type, and F type connectors.



Figure 4-14 Coaxial Cable and Connectors

The numbers in Figure 4-14 identify some key features of coaxial cable:

1. Outer jacket
2. Braided copper shielding
3. Plastic insulation
4. Copper conductor

Although UTP cable has essentially replaced coaxial cable in modern Ethernet installations, the coaxial cable design is used in the following situations:

- **Wireless installations:** Coaxial cables attach antennas to wireless devices. The coaxial cable carries radio frequency (RF) energy between the antennas and the radio equipment.
- **Cable internet installations:** Cable service providers provide internet connectivity to their customers by replacing portions of the coaxial cable and supporting amplification elements with *fiber-optic cable*. However, the wiring inside the customer's premises is still coax cable.

**Interactive
Graphic**

Check Your Understanding—Copper Cabling (4.3.6)

Refer to the online course to complete this activity.

UTP Cabling (4.4)

Copper media has some inherent issues. Twisting the internal pairs of the copper media, as is done in UTP, is a low-cost solution to improve cabling performance. This section further explores UTP cabling.

Properties of UTP Cabling (4.4.1)

In the previous section, you learned a bit about unshielded twisted-pair (UTP) copper cabling. Because UTP cabling is the standard for use in LANs, this section goes into detail about its advantages and limitations, as well as what can be done to avoid problems.

When used as a networking medium, UTP cabling consists of four pairs of color-coded copper wires that have been twisted together and then encased in a flexible plastic sheath. Its small size can be advantageous during installation.

UTP cable does not use shielding to counter the effects of EMI and RFI. Instead, cable designers have discovered other ways to limit the negative effect of crosstalk:

- **Cancellation:** Designers now pair wires in a circuit. When two wires with magnetic fields exactly opposite each other are placed close together in an electrical circuit, the two magnetic fields cancel each other out and also cancel out any outside EMI and RFI signals.
- **Varying the number of twists per wire pair:** To further enhance the cancellation effect of paired circuit wires, designers vary the number of twists of each wire pair in a cable. UTP cable must follow precise specifications governing how many twists or braids are permitted per meter (3.28 feet) of cable. Notice in Figure 4-15 that the bottom pair (which is orange/orange white, though you can't see that in this book) is twisted less than the pair just above it (which is a blue/blue white pair). Each colored pair is twisted a different number of times.

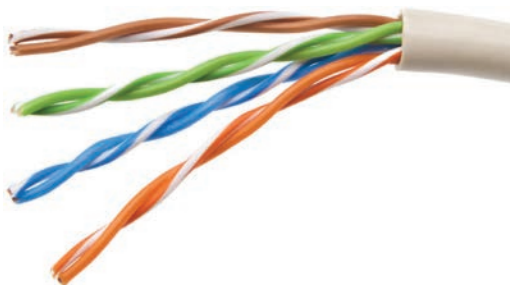


Figure 4-15 Different Number of Twists in Each UTP Pair

UTP cable relies solely on the cancellation effect produced by the twisted wire pairs to limit signal degradation and effectively provide self-shielding for wire pairs within the network media.

UTP Cabling Standards and Connectors (4.4.2)

UTP cabling conforms to the standards established jointly by the TIA/EIA. Specifically, TIA/EIA-568 stipulates the commercial cabling standards for LAN installations and is the standard most commonly used in LAN cabling environments. Some of the elements defined are as follows:

- Cable types
- Cable lengths
- Connectors
- Cable termination
- Methods of testing cable

The electrical characteristics of copper cabling are defined by the Institute of Electrical and Electronics Engineers (IEEE). IEEE rates UTP cabling according to its performance. Cables are placed into categories based on their ability to carry various bandwidth rates. For example, Category 5 cable is used commonly in 100BASE-TX Fast Ethernet installations. Other categories include Enhanced Category 5 cable, Category 6, and Category 6a.

Cables in higher categories are designed and constructed to support higher data rates. As new gigabit speed Ethernet technologies are being developed and adopted, Category 5e is now the minimally acceptable cable type, with Category 6 being the recommended type for new building installations.

Figure 4-16 shows three categories of UTP cable:

- Category 3 was originally used for voice communication over voice lines but later began to be used for data transmission.
- Category 5 and 5e are used for data transmission. Category 5 supports 100 Mbps, and Category 5e supports 1000 Mbps.
- Category 6 has an added separator between each wire pair to support higher speeds. Category 6 supports up to 10 Gbps.
- Category 6a is similar to Category 6 with improved crosstalk characteristics to allow for longer distances.
- Category 7 also supports 10 Gbps.
- Category 8 supports 40 Gbps.

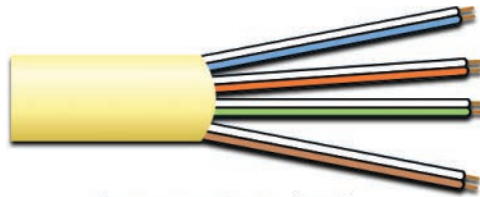
Some manufacturers are making cables exceeding the TIA/EIA Category 6a specifications and refer to these as Category 7.

UTP cable is usually terminated with RJ-45 connectors. The TIA/EIA-568 standard describes the wire color codes and pin assignments (pinouts) for Ethernet cables.

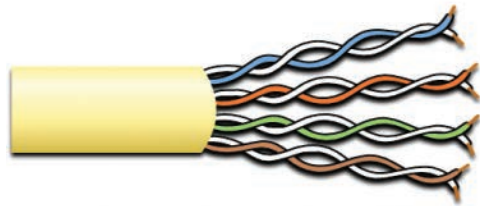
As shown in Figure 4-17, the RJ-45 connector is the male component, crimped at the end of the cable.

The socket, shown in Figure 4-18, is the female component of a network device, wall, cubicle partition outlet, or patch panel.

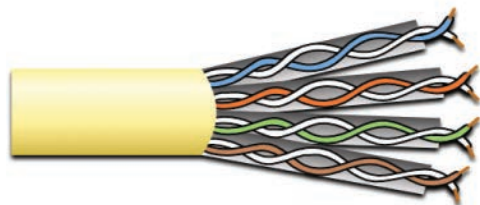
When terminated improperly, a cable is a potential source of physical layer performance degradation. Figure 4-19 shows an example of a badly terminated UTP cable. This bad connector has wires that are exposed, untwisted, and not entirely covered by the sheath.



Category 3 Cable (UTP)



Category 5 and 5e Cable (UTP)



Category 6 Cable (UTP)

Figure 4-16 Categories of UTP



Figure 4-17 RJ-45 UTP Plugs



Figure 4-18 RJ-45 UTP Sockets

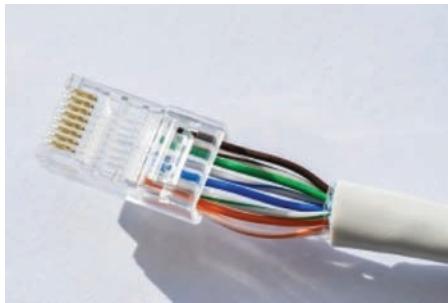


Figure 4-19 Poorly Terminated UTP Cable

Figure 4-20 shows a properly terminated UTP cable. It is a good connector with wires that are untwisted only to the extent necessary to attach the connector.



Figure 4-20 Properly Terminated UTP Cable

Note

Improper cable termination can impact transmission performance.

Straight-Through and Crossover UTP Cables (4.4.3)

Different situations may require UTP cables to be wired according to different wiring conventions. This means that the individual wires in the cable have to be connected in different orders to different sets of pins in the RJ-45 connectors.

The following are the main cable types that are created by using specific wiring conventions:

- **Ethernet straight-through:** This is the most common type of networking cable, commonly used to interconnect a host to a switch and a switch to a router.
- **Ethernet crossover:** This cable is used to interconnect similar devices—for example, to connect a switch to a switch, a host to a host, or a router to a router. However, crossover cables are now considered legacy as NICs use medium-dependent interface crossover (auto-MDIX) to automatically detect the cable type and make the internal connection.

Note

Another type of cable is a rollover cable, which is a Cisco-proprietary cable. It is used to connect a workstation to a router or switch console port.

Incorrectly using a crossover or straight-through cable between devices may not damage the devices, but connectivity and communication between the devices will not take place. This is a common error, and checking that the device connections are correct should be the first troubleshooting action if connectivity is not achieved.

Figure 4-21 identifies the individual wire pairs for the T568A and T568B standards.

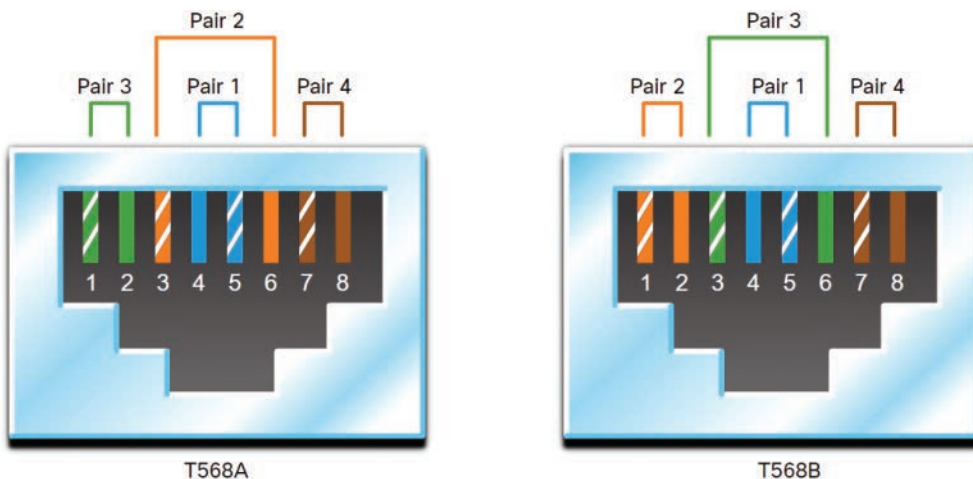


Figure 4-21 T568A and T568B Standards

Table 4-2 shows the UTP cable type, related standards, and typical applications of these cables.

Table 4-2 Cable Types and Standards

Cable Type	Standard	Application
Ethernet straight-through	Both ends T568A or both ends T568B	Connects a network host to a network device such as a switch or hub
Ethernet crossover	One end T568A, other end T568B	Connects two network hosts Connects two network intermediary devices (switch to switch or router to router)
Rollover	Cisco proprietary	Connects a workstation serial port to a router console port, using an adapter



Activity—Cable Pinouts (4.4.4)

For this activity, correctly order the wire colors to a TIA/EIA cable pinout. Select a wire sheath color by clicking it. Then click a wire to apply that sheath to it. Refer to the online course to complete this activity.

Fiber-Optic Cabling (4.5)

Networking media selection is being driven by the growing needs for network bandwidth. The distance and performance of fiber-optic cable make it a good media choice for supporting these network needs. This section examines the characteristics of fiber-optic cabling use in data networks.

Properties of Fiber-Optic Cabling (4.5.1)

As you have learned, fiber-optic cabling is a type of cabling used in many networks today. Because it is expensive, it is not as commonly used at the various types of copper cabling. But fiber-optic cabling has certain properties that make it the best option in certain situations, as discussed in this section.

Optical fiber cable transmits data over longer distances and at higher bandwidths than any other networking media. Fiber-optic cable can transmit signals with less attenuation than copper wire, and it is completely immune to EMI and RFI. Optical fiber is commonly used to interconnect network devices.

Optical fiber contains a flexible but extremely thin, transparent strand of very pure glass, not much bigger than a human hair. Bits are encoded on the fiber as light impulses. The fiber-optic cable acts as a waveguide, or “light pipe,” to transmit light between the two ends with minimal loss of signal.

As an analogy, consider an empty paper towel roll with the inside coated like a mirror. Imagine that it is 1000 meters in length, and a small laser pointer is used to send Morse code signals at the speed of light. Essentially, that is how a fiber-optic cable operates, except that it is smaller in diameter and uses sophisticated light technologies.

Types of Fiber Media (4.5.2)

Fiber-optic cables are broadly classified into two types: single-mode fiber (SMF) and multimode fiber (MMF).

Single-Mode Fiber

SMF consists of a very small core and uses expensive laser technology to send a single ray of light, as shown in Figure 4-22. SMF is popular in long-distance situations spanning hundreds of kilometers, such as those required in long-haul telephony and cable TV applications.

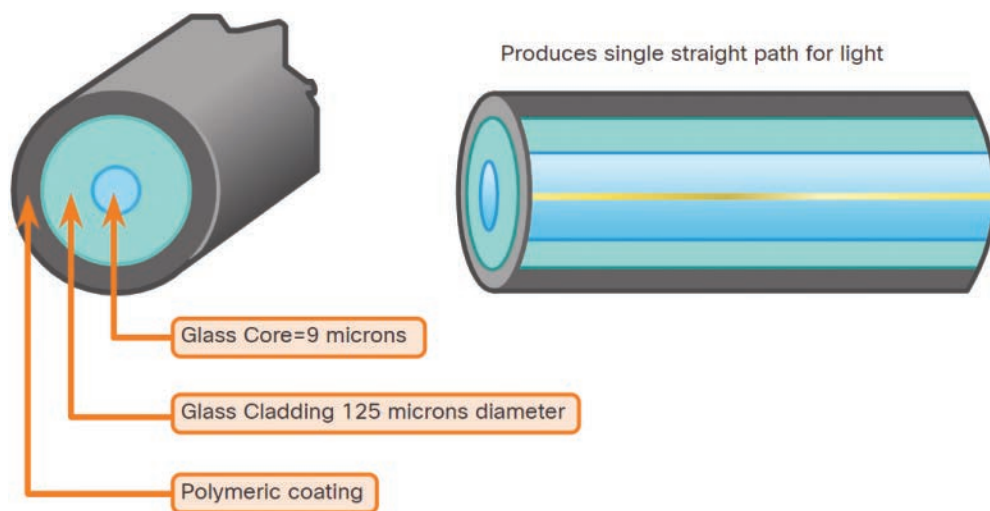


Figure 4-22 Single-Mode Fiber

Multimode Fiber

MMF consists of a larger core and uses LED emitters to send light pulses. Specifically, light from an LED enters the multimode fiber at different angles, as shown in Figure 4-23. It is popular in LANs because they can be powered by low-cost LEDs. It provides bandwidth up to 10 Gbps over link lengths of up to 550 meters.

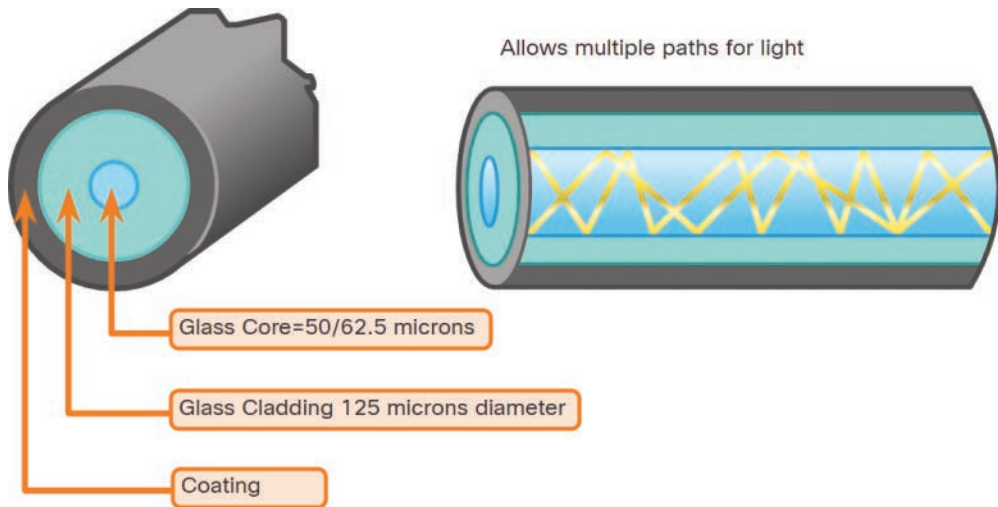


Figure 4-23 Multimode Fiber

One of the main differences between MMF and SMF is the amount of dispersion. *Dispersion* refers to the spreading out of a light pulse over time. Increased dispersion means increased loss of signal strength. MMF has greater dispersion than SMF; this is why MMF can travel only up to 550 meters before signal loss occurs.

Fiber-Optic Cabling Usage (4.5.3)

Fiber-optic cabling is now being used in four types of industry:

- **Enterprise networks:** Fiber is used for backbone cabling applications and for interconnecting infrastructure devices.
- **Fiber-to-the-home (FTTH):** Fiber is used to provide always-on broadband services to homes and small businesses.
- **Long-haul networks:** Service providers use fiber to connect countries and cities.
- **Submarine cable networks:** Fiber is used to provide reliable high-speed, high-capacity solutions capable of surviving in harsh undersea environments at up to transoceanic distances. Search the internet for “submarine cables telegeography map” to view various maps online.

Our focus in this book is the use of fiber within the enterprise.

Fiber-Optic Connectors (4.5.4)

A fiber-optic connector terminates the end of an optical fiber. A variety of fiber-optic connectors are available. The main differences among them are their dimensions and the methods of coupling. Businesses decide on the types of connectors that will be used, based on their equipment.

Note

Some switches and routers have ports that support fiber-optic connectors through a small form-factor pluggable (SFP) transceiver. Search the internet for various types of SFPs.

The straight-tip (ST) connector (see Figure 4-24) was one of the first connector types used. The connector locks securely with a “twist-on/twist-off” bayonet-style mechanism.



Figure 4-24 Straight-Tip (ST) Connectors

Subscriber connector (SC) connectors (see Figure 4-25) are sometimes referred to as square connectors or standard connectors. They are widely adopted LAN and WAN connectors that use a push/pull mechanism to ensure positive insertion. This connector type is used with multimode and single-mode fiber.



Figure 4-25 Subscriber Connector (SC) Connectors

The Lucent Connector (LC) simplex connector (see Figure 4-26) is a smaller version of the SC connector. LC connectors are sometimes called little connectors or local connectors, and their popularity is quickly growing due to their smaller size.

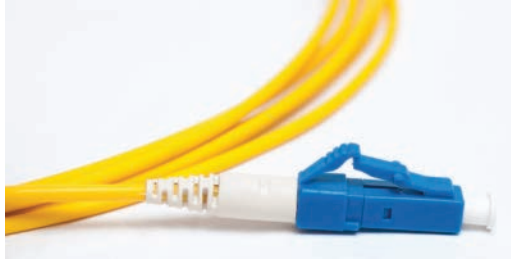


Figure 4-26 Lucent Connector (LC) Simplex Connector

A duplex multimode LC connector (see Figure 4-27) is similar to an LC simplex connector but uses a duplex connector.

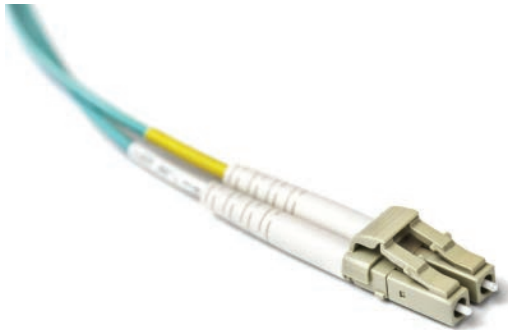


Figure 4-27 Duplex Multimode LC Connector

Until recently, light could travel in only one direction over optical fiber. Two fibers were required to support full-duplex operation. Therefore, two optical fiber cables needed to be bundled with fiber-optic patch cables and terminated with a pair of standard, single-fiber connectors. Some fiber connectors accept both the transmitting and receiving fibers in a single connector known as a duplex connector (shown in Figure 4-27). BX standards such as 100BASE-BX use different wavelengths for sending and receiving over a single fiber.

Fiber Patch Cords (4.5.5)

Fiber patch cords are required for interconnecting infrastructure devices. The use of color distinguishes between single-mode and multimode patch cords. A single-mode

fiber cable has a yellow jacket, and a multimode fiber cable has an orange (or aqua) jacket.

Figure 4-28 shows four types of fiber patch cords.

Note

A fiber cable should be protected with a small plastic cap when not in use.



Figure 4-28 Fiber Patch Cords

Fiber Versus Copper (4.5.6)

There are many advantages to using fiber-optic cable compared to using copper cable. Table 4-3 highlights some of the differences between these cable types.

At present, in most enterprise environments, optical fiber is primarily used as back-bone cabling for high-traffic, point-to-point connections between data distribution facilities. It is also used to interconnect buildings in multi-building campuses. Because fiber-optic cables do not conduct electricity and have low signal loss, they are well suited for these uses.

Table 4-3 UTP and Fiber-Optic Cabling Comparison

Implementation Issue	UTP Cabling	Fiber-Optic Cabling
Bandwidth supported	10 Mbps–10 Gbps	10 Mbps–100 Gbps
Distance	Relatively short (1–100 meters)	Relatively long (1–100,000 meters)
Immunity to EMI and RFI	Low	High (completely immune)
Immunity to electrical hazards	Low	High (completely immune)
Media and connector costs	Lowest	Highest
Installation skills required	Lowest	Highest
Safety precautions	Lowest	Highest



Check Your Understanding—Fiber-Optic Cabling (4.5.7)

Refer to the online course to complete this activity.

Wireless Media (4.6)

As more mobile devices are being used, wireless networking is growing in demand. This section explores wireless media characteristic and uses.

Properties of Wireless Media (4.6.1)

You might be reading this book or taking the accompanying course using a tablet or a smartphone. This is only possible due to the use of wireless media to connect to the physical layer of a network.

Wireless media carry electromagnetic signals that represent the binary digits of data communications using radio wave or microwave frequencies.

Wireless media provide the greatest mobility options of all media. Wireless is now the primary way users connect to home and enterprise networks, and the number of wireless-enabled devices continues to increase.

These are some of the limitations of wireless:

- **Coverage area:** Wireless data communication technologies work well in open environments. However, certain construction materials used in buildings and structures, as well as the local terrain, can limit the effective coverage.
- **Interference:** Wireless is susceptible to interference and can be disrupted by such common devices as household cordless phones, some types of fluorescent lights, microwave ovens, and other wireless communications.
- **Security:** Wireless communication coverage requires no access to a physical strand of cable. Therefore, devices and users not authorized for access to the network can gain access to the transmission. Network security is a major component of wireless network administration.
- **Shared medium:** WLANs operate in half-duplex, which means only one device can send or receive at a time. The wireless medium is shared among all wireless users. Many users accessing the WLAN simultaneously results in reduced bandwidth for each user.

Although wireless is increasing in popularity for desktop connectivity, copper and fiber are the most popular physical layer media for deployment of intermediary network devices, such as routers and switches.

Types of Wireless Media (4.6.2)

The IEEE and telecommunications industry standards for wireless data communications cover both the data link and physical layers. In each of these standards, physical layer specifications are applied to areas such as the following:

- Data-to-radio signal encoding
- Frequency and power of transmission
- Signal reception and decoding requirements
- Antenna design and construction

These are the wireless standards:

- **Wi-Fi (IEEE 802.11):** Wi-Fi is a wireless LAN (WLAN) technology that uses a contention-based protocol known as Carrier Sense Multiple Access/Collision Avoidance (CSMA/CA). The wireless NIC must first listen before transmitting to determine if the radio channel is clear. If another wireless device is transmitting,

then the NIC must wait until the channel is clear. Wi-Fi is a trademark of the Wi-Fi Alliance. Wi-Fi is used with certified WLAN devices based on the IEEE 802.11 standards.

- **Bluetooth (IEEE 802.15):** Bluetooth is a wireless personal area network (WPAN) standard that uses a device pairing process to communicate over distances from 1 to 100 meters.
- **WiMAX (IEEE 802.16):** Worldwide Interoperability for Microwave Access (WiMAX) is a wireless standard that uses a point-to-multipoint topology to provide wireless broadband access.
- **Zigbee (IEEE 802.15.4):** Zigbee is a specification used for low-data-rate, low-power communications. It is intended for applications that require short range, low data rates, and long battery life. Zigbee is typically used in industrial and Internet of Things (IoT) environments, such as for wireless light switches and medical device data collection.

Note

Other wireless technologies, such as cellular and satellite communications, can also provide data network connectivity. However, these wireless technologies are beyond the scope of this chapter.

Wireless LAN (4.6.3)

A common wireless data implementation involves enabling devices to connect wirelessly over a LAN. In general, a WLAN requires the following network devices:

- **Wireless access points (APs):** Wireless APs concentrate the wireless signals from users and connect to the existing copper-based network infrastructure, such as Ethernet. Home and small-business wireless routers integrate the functions of a router, switch, and access point into one device, as shown in Figure 4-29.
- **Wireless NIC adapters:** These adapters provide wireless communication capability to network hosts.

As the technology has developed, a number of WLAN Ethernet-based standards have emerged. When purchasing wireless devices, it is important to ensure compatibility and interoperability in a network.

The benefits of wireless data communications technologies are evident—especially the savings on costly premises wiring and the convenience of host mobility. However, network administrators must develop and apply stringent security policies and processes to protect WLANs from unauthorized access and damage.



Figure 4-29 Cisco Meraki MX64W

**Interactive
Graphic**

Check Your Understanding—Wireless Media (4.6.4)

Refer to the online course to complete this activity.

**Packet Tracer
Activity**

Packet Tracer—Connect a Wired and Wireless LAN (4.6.5)

When working in Packet Tracer, a lab environment, or a corporate setting, you should know how to select the appropriate cable and know how to properly connect devices. In this activity you will examine device configurations in Packet Tracer, select the proper cable based on the configuration, and connect the devices. In this activity you will also explore the physical view of a network in Packet Tracer.



Lab—View Wired and Wireless NIC Information (4.6.6)

In this lab, you will complete the following objectives:

- Part 1: Identify and Work with PC NICs
 - Part 2: Identify and Use the System Tray Network Icons
-

**Packet Tracer
Activity**

Packet Tracer—Connect the Physical Layer (4.7.1)

In this activity, you will explore the different options available on internetworking devices. You will also be required to determine which options provide the necessary connectivity when connecting multiple devices. Finally, you will add the correct modules and connect the devices.

Summary (4.7)

The following is a summary of the topics in the chapter and their corresponding online modules.

Purpose of the Physical Layer

Before any network communications can occur, a physical connection to a local network must be established. A physical connection can be a wired connection using a cable or a wireless connection using radio waves. A network interface card (NIC) connects a device to a network. Ethernet NICs are used for wired connections, and WLAN (wireless local-area network) NICs are used for wireless. The OSI physical layer provides the means to transport the bits that make up a data link layer frame across the network media. The physical layer accepts a complete frame from the data link layer and encodes it as a series of signals that are transmitted onto the local media. The encoded bits that comprise a frame are received by either an end device or an intermediary device.

Physical Layer Characteristics

The physical layer consists of electronic circuitry, media, and connectors developed by engineers. The physical layer standards address three functional areas: physical components, encoding, and signaling. Bandwidth is the capacity at which a medium can carry data. Digital bandwidth measures the amount of data that can flow from one place to another in a given amount of time. Throughput is a measure of the transfer of bits across the media over a given period of time and is usually lower than bandwidth. Latency refers to the amount of time, including delays, for data to travel from one given point to another. Goodput is a measure of usable data transferred over a given period of time. The physical layer produces the representation and groupings of bits for each type of media as follows:

- **Copper cable:** The signals are patterns of electrical pulses.
- **Fiber-optic cable:** The signals are patterns of light.
- **Wireless:** The signals are patterns of microwave transmissions.

Copper Cabling

Networks use copper media because it is inexpensive and easy to install, and it has low resistance to electrical current. However, copper media is limited by distance and signal interference. The timing and voltage values of the electrical pulses are also susceptible to interference from two sources: EMI and crosstalk. Three types of copper

cabling are UTP, STP, and coaxial cable (coax). UTP has an outer jacket to protect the copper wires from physical damage, twisted pairs to protect the signal from interference, and color-coded plastic insulation that electrically isolates wires from each other and identifies each pair. STP cable uses four pairs of wires, each wrapped in a foil shield, and the four pairs are then wrapped in an overall metallic braid or foil. Coaxial cable, or coax for short, gets its name from the fact that it has two conductors that share the same axis. Coax is used to attach antennas to wireless devices. Cable internet providers use coax inside their customers' premises.

UTP Cabling

UTP cabling consists of four pairs of color-coded copper wires that have been twisted together and then encased in a flexible plastic sheath. UTP cable does not use shielding to counter the effects of EMI and RFI. Instead, cable designers have discovered other ways they can limit the negative effect of crosstalk: by using cancellation and by varying the number of twists per wire pair. UTP cabling conforms to the standards established jointly by the TIA/EIA. The electrical characteristics of copper cabling are defined by the Institute of Electrical and Electronics Engineers (IEEE). UTP cable is usually terminated with RJ-45 connectors. The main cable types created by using specific wiring conventions are Ethernet straight-through and Ethernet crossover. Cisco has a proprietary UTP cable called a rollover cable that connects a workstation to a router console port.

Fiber-Optic Cabling

Optical fiber cable transmits data over longer distances and at higher bandwidths than any other networking media. Fiber-optic cable can transmit signals with less attenuation than copper wire and is completely immune to EMI and RFI. Optical fiber contains a flexible but extremely thin, transparent strand of very pure glass, not much bigger than a human hair. Bits are encoded on the fiber as light impulses. Fiber-optic cabling is now being used in enterprise networks, FTTH, long-haul networks, and submarine cable networks. There are four types of fiber-optic connectors: ST, SC, LC, and duplex multimode LC. Fiber-optic patch cords include SC-SC multimode, LC-LC single-mode, ST-LC multimode, and SC-ST single-mode. In most enterprise environments, optical fiber is primarily used as backbone cabling for high-traffic point-to-point connections between data distribution facilities and for interconnecting buildings in multi-building campuses.

Wireless Media

Wireless media carry electromagnetic signals that represent the binary digits of data communications using radio wave or microwave frequencies. Wireless does have

some limitations, including coverage area, interference, security, and the problems that occur with any shared medium. Wireless standards include Wi-Fi (IEEE 802.11), Bluetooth (IEEE 802.15), WiMAX (IEEE 802.16), and Zigbee (IEEE 802.15.4). A wireless LAN (WLAN) requires a wireless AP and wireless NIC adapters.

Practice

The following activities provide practice with the topics introduced in this chapter. The lab is available in the companion *Introduction to Networks Labs & Study Guide (CCNAv7)* (ISBN 9780136634454). The Packet Tracer activity instructions are also provided in the *Labs & Study Guide*. The PKA files are available in the online course.



Lab

Lab 4.6.6: View Wired and Wireless NIC Information



Packet Tracer Activities

Packet Tracer 4.6.5: Connect a Wired and Wireless LAN

Packet Tracer 4.7.1: Connect the Physical Layer

Check Your Understanding Questions

Complete all the review questions listed here to test your understanding of the topics and concepts in this chapter. The appendix “Answers to ‘Check Your Understanding’ Questions” lists the answers.

1. What is the purpose of the OSI physical layer?
 - a. controlling access to media
 - b. transmitting bits across the local media
 - c. performing error detection on received frames
 - d. exchanging frames between nodes over physical network media

2. Why are two strands of fiber used for a single fiber-optic connection?
 - a. The two strands allow the data to travel for longer distances without degrading.
 - b. The two strands prevent crosstalk from causing interference on the connection.
 - c. The two strands increase the speed at which the data can travel.
 - d. The two strands allow for full-duplex connectivity.
3. Which of the following describes crosstalk?
 - a. the distortion of a network signal due to fluorescent lighting
 - b. the distortion of a transmitted message due to signals carried in adjacent wires
 - c. the weakening of a network signal over long cable lengths
 - d. the loss of wireless signal over excessive distance from an access point
4. Which procedure is used to reduce the effect of crosstalk in copper cables?
 - a. requiring proper grounding connections
 - b. twisting opposing-circuit wire pairs together
 - c. wrapping a bundle of wires with metallic shielding
 - d. designing a cable infrastructure to avoid crosstalk interference
 - e. avoiding sharp bends during installation
5. Which type of UTP cable is used to connect a PC to a switch port?
 - a. console
 - b. rollover
 - c. crossover
 - d. straight-through
6. What is the definition of bandwidth?
 - a. the speed of bits across media over a given period of time
 - b. the speed at which bits travel on a network
 - c. the amount of data that can flow in a given amount of time
 - d. the measure of usable data transferred over a given period of time

7. Which statement correctly describes frame encoding?
 - a. It uses the characteristic of one wave to modify another wave.
 - b. It transmits data signals along with a clock signal that occurs at evenly spaced time durations.
 - c. It generates the electrical, optical, or wireless signals that represent the binary numbers of the frame.
 - d. It converts bits into a predefined code in order to provide a predictable pattern to help distinguish data bits from control bits.
8. Which of the following is a characteristic of UTP cabling?
 - a. cancellation
 - b. cladding
 - c. immunity to electrical hazards
 - d. woven copper braid or metallic foil
9. A wireless LAN is being deployed inside the new one-room office that is occupied by the park ranger. The office is located at the highest part of the national park. After network testing is complete, the technicians report that the wireless LAN signal is occasionally affected by some type of interference. What is a possible cause of the signal distortion?
 - a. the microwave oven
 - b. the large number of trees surrounding the office
 - c. the elevated location where the wireless LAN was installed
 - d. the number of wireless devices that are used in the wireless LAN
10. What is indicated by the term *throughput*?
 - a. the guaranteed data transfer rate offered by an ISP
 - b. the capacity of a particular medium to carry data
 - c. the measure of the usable data transferred across media over a given period of time
 - d. the measure of bits transferred across media over a given period of time
11. What is one advantage of using fiber-optic cabling rather than copper cabling?
 - a. It is usually cheaper than copper cabling.
 - b. It can be installed around sharp bends.
 - c. It is easier to terminate and install than copper cabling.
 - d. It is able to carry signals much farther than copper cabling.

12. Which standards organization oversees development of wireless LAN standards?
 - a. IANA
 - b. IEEE
 - c. ISO
 - d. TIA
13. A network administrator is designing a new network infrastructure that includes both wired and wireless connectivity. In which situation would a wireless connection be recommended?
 - a. The end-user device has only an Ethernet NIC.
 - b. The end-user device requires a dedicated connection due to performance requirements.
 - c. The end-user device needs mobility when connecting to the network.
 - d. The end-user device area has a high concentration of RFI.
14. A network administrator is troubleshooting connectivity issues on a server. Using a tester, the administrator notices that the signals generated by the server NIC are distorted and not usable. In which layer of the OSI model does the error occur?
 - a. presentation layer
 - b. network layer
 - c. physical layer
 - d. data link layer
15. What type of cable is used to connect a workstation serial port to a Cisco router console port?
 - a. crossover
 - b. rollover
 - c. straight-through
 - d. coaxial

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Number Systems

Objectives

Upon completion of this chapter, you will be able to answer the following questions:

- How do you convert numbers between decimal and binary systems?
- How do you convert numbers between decimal and hexadecimal systems?

Key Terms

This chapter uses the following key terms. You can find the definitions in the glossary at the end of the book.

dotted decimal page 176

hexadecimal page 176

binary page 176

Introduction (5.0)

Guess what? This is the 32-bit binary IPv4 address of a computer in a network: 11000000.10101000.00001010.00001010. This is the IPv4 address for the same computer in *dotted decimal*: 192.168.10.10. Which one would you rather work with? Addresses get even more complicated with IPv6, where addresses are 128 bits! To make these addresses more manageable, IPv6 uses the hexadecimal system, which includes the numbers 0 to 9 and the letters A to F.

As a network administrator, you must know how to convert binary addresses into dotted decimal and dotted decimal addresses into *binary*. You also need to know how to convert dotted decimal into *hexadecimal* and vice versa. (*Hint*: You still need your binary conversion skills when converting between dotted decimal and hexadecimal.)

They may seem complicated, but these conversions are not that hard when you learn a few tricks. The module that corresponds to this chapter contains an activity called the Binary Game, which will really help you get started. Why wait?

Binary Number System (5.1)

IPv4 addresses are 32-bit addresses expressed in decimal notation. This section discusses the binary number system along with the conversion between the binary and decimal number systems.

Binary and IPv4 Addresses (5.1.1)

IPv4 addresses begin as binary, series of only 1s and 0s. These are difficult to manage, so network administrators convert them to decimal. This section shows you a few ways to do this.

Binary is a numbering system that consists of the digits 0 and 1, called *bits*. In contrast, the decimal numbering system consists of the digits 0 to 9.

It is important to understand binary because hosts, servers, and network devices use binary addressing. Specifically, they use binary IPv4 addresses, as shown in Figure 5-1, to identify each other.

Each address consists of a string of 32 bits, divided into four sections called *octets*. Each octet contains 8 bits (or 1 byte), and each pair of octets is separated with a dot. For example, PC1 in Figure 5-1 is assigned IPv4 address 11000000.10101000.00001010.00001010. Its default gateway address would be that of the R1 Gigabit Ethernet interface: 11000000.10101000.00001010.00000001.

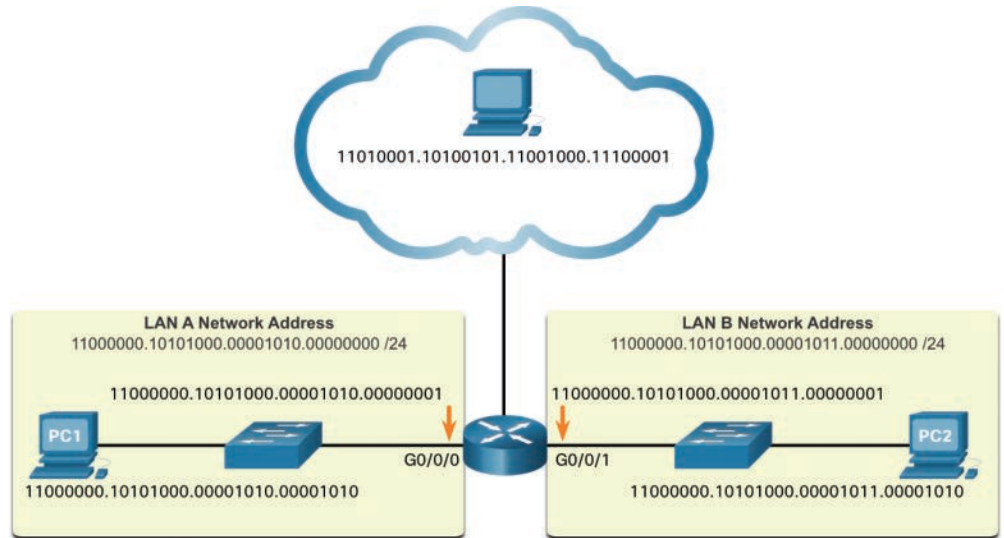


Figure 5-1 IPv4 Address in Binary Format

Binary works well with hosts and network devices. However, it is very challenging for humans to work with. For ease of use by people, IPv4 addresses are commonly expressed in dotted decimal notation. Figure 5-2 shows the same network as Figure 5-1, but here PC1 is assigned the IPv4 address 192.168.10.10, and its default gateway address is 192.168.10.1.

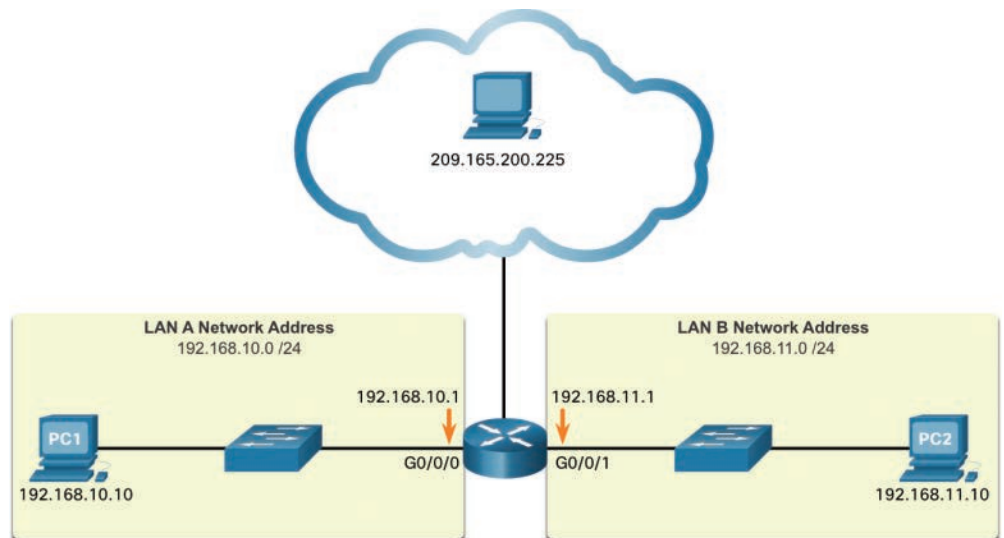


Figure 5-2 IPv4 Addresses in Dotted Decimal Format

For a solid understanding of network addressing, it is necessary to know binary addressing and gain practical skills converting between binary and dotted decimal IPv4 addresses. The following sections cover how to convert between base 2 (binary) and base 10 (decimal) numbering systems.



Video—Converting Between Binary and Decimal Numbering Systems (5.1.2)

Refer to the online course to view this video.

Binary Positional Notation (5.1.3)

Learning to convert binary to decimal requires an understanding of positional notation. *Positional notation* means that a digit represents different values depending on the “position” the digit occupies in the sequence of numbers. You already know the most common numbering system, the decimal (base 10) notation system.

The decimal positional notation system operates as described in Table 5-1.

Table 5-1 Decimal Positional Notation

Radix	10	10	10	10
Position in number	3	2	1	0
Calculate	(10^3)	(10^2)	(10^1)	(10^0)
Positional value	1000	100	10	1

The following bullets describe the rows in Table 5-1:

- Row 1 (radix) is the number base. Decimal notation is based on 10, so the radix is 10.
- Row 2 (position in number) considers the position of the decimal number starting with, from right to left, 0 (1st position), 1 (2nd position), 2 (3rd position), 3 (4th position). These numbers also represent the exponential values used to calculate the positional value in the 4th row.
- Row 3 (calculate) calculates the positional value by taking the radix and raising it by the exponential value of its position in row 2.

Note

n^0 is = 1.

- Row 4 (positional value) represents units of thousands, hundreds, tens, and ones.

To use the positional system, match a given number to its positional value. The example in Table 5-2 illustrates how positional notation is used with the decimal number 1234.

Table 5-2 Example of Using Decimal Positional Notation

	Thousands	Hundreds	Tens	Ones
Positional Value	1000	100	10	1
Decimal Number (1234)	1	2	3	4
Calculate	1×1000	2×100	3×10	4×1
Add Them Up...	1000	+ 200	+ 30	+ 4
Result	1234			

In contrast, the binary positional notation operates as described in Table 5-3.

Table 5-3 Binary Positional Notation

Radix	2	2	2	2	2	2	2	2
Position in number	7	6	5	4	3	2	1	0
Calculate	(2^7)	(2^6)	(2^5)	(2^4)	(2^3)	(2^2)	(2^1)	(2^0)
Positional value	128	64	32	16	8	4	2	1

The following bullets describe the rows in Table 5-3:

- Row 1 (radix) is the number base. Binary notation is based on 2, so the radix is 2.
- Row 2 (position in number) considers the position of the binary number starting with, from right to left, 0 (1st position), 1 (2nd position), 2 (3rd position), 3 (4th position). These numbers also represent the exponential value use to calculate the positional value in the 4th row.
- Row 3 (calculate) calculates the positional value by taking the radix and raising it by the exponential value of its position in row 2.

Note

n^0 is = 1.

- Row 4 (positional value) represents units of ones, twos, fours, eights, and so on.

The example in Table 5-4 illustrates how a binary number 11000000 corresponds to the number 192. If the binary number had been 10101000, then the corresponding decimal number would be 168.

Table 5-4 Example of Using Binary Positional Notation

Positional Value	128	64	32	16	8	4	2	1
Binary Number (11000000)	1	1	0	0	0	0	0	0
Calculate	1×128	1×64	0×32	0×16	0×8	0×4	0×2	0×1
Add Them Up...	128	+ 64	+ 0	+ 0	+ 0	+ 0	+ 0	+ 0
Result	192							



Check Your Understanding—Binary Number System (5.1.4)

Refer to the online course to complete this activity.

Convert Binary to Decimal (5.1.5)

To convert a binary IPv4 address to its dotted decimal equivalent, divide the IPv4 address into four 8-bit octets. Then apply the binary positional value to the first octet binary number and calculate accordingly.

For example, say that 11000000.10101000.00001011.00001010 is the binary IPv4 address of a host. To convert the binary address to decimal, start with the first octet, as shown in Table 5-5. Enter the 8-bit binary number under the positional value of row 1 and then calculate to produce the decimal number 192. This number goes into the first octet of the dotted decimal notation.

Table 5-5 Converting 11000000 to Decimal

Positional Value	128	64	32	16	8	4	2	1
Binary Number (11000000)	1	1	0	0	0	0	0	0
Calculate	1×128	1×64	0×32	0×16	0×8	0×4	0×2	0×1
Add Them Up...	128	+ 64	+ 0	+ 0	+ 0	+ 0	+ 0	+ 0
Result	192							

Next, convert the second octet of 10101000 as shown in Table 5-6. The resulting decimal value is 168, and it goes into the second octet.

Table 5-6 Converting 10101000 to Decimal

Positional Value	128	64	32	16	8	4	2	1
Binary Number (10101000)	1	0	1	0	1	0	0	0
Calculate	1×128	0×64	1×32	0×16	1×8	0×4	0×2	0×1
Add Them Up...	128	+ 0	+ 32	+ 0	+ 8	+ 0	+ 0	+ 0
Result	168							

Convert the third octet of 00001011 as shown in Table 5-7.

Table 5-7 Converting 00001011 to Decimal

Positional Value	128	64	32	16	8	4	2	1
Binary Number (00001011)	0	0	0	0	1	0	1	1
Calculate	0×128	0×64	0×32	0×16	1×8	0×4	1×2	1×1
Add Them Up...	0	+ 0	+ 0	+ 0	+ 8	+ 0	+ 2	+ 1
Result	11							

Convert the fourth octet of 00001010 as shown in Table 5-8. This completes the IP address and produces the dotted decimal result 192.168.11.10.

Table 5-8 Converting 00001010 to Decimal

Positional Value	128	64	32	16	8	4	2	1
Binary Number (00001010)	0	0	0	0	1	0	1	0
Calculate	0×128	0×64	0×32	0×16	1×8	0×4	1×2	0×1
Add Them Up...	0	+ 0	+ 0	+ 0	+ 8	+ 0	+ 2	+ 0
Result	10							

Interactive Graphic

Activity—Binary to Decimal Conversions (5.1.6)

This activity allows you to practice 8-bit binary to decimal conversion as much as necessary. We recommend that you work with this tool until you are able to do the conversion without error. Convert the binary number shown in the octet to its decimal value.

Refer to the online course to complete this activity.

Decimal to Binary Conversion (5.1.7)

It is also necessary to understand how to convert a dotted decimal IPv4 address to binary. A useful tool is the binary positional value table. Figures 5-3 through 5-10 show examples of this type of table. The following steps walk through how to use this table to do the conversion:



- Step 1.** In Figure 5-3, is the decimal number of the octet (n) equal to or greater than the most significant bit (128)?
- If no, then enter binary 0 in the 128 positional value.
 - If yes, then add a binary 1 in the 128 positional value and subtract 128 from the decimal number.

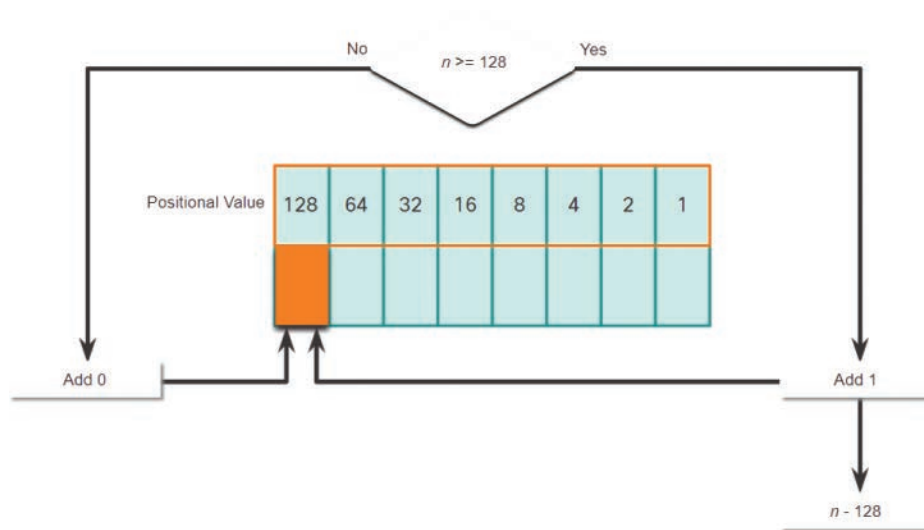


Figure 5-3 128 Positional Value

- Step 2.** In Figure 5-4, is the decimal number of the octet (n) equal to or greater than the next most significant bit (64)?
- If no, then enter binary 0 in the 64 positional value.
 - If yes, then add a binary 1 in the 64 positional value and subtract 64 from the decimal number.
- Step 3.** In Figure 5-5, is the decimal number of the octet (n) equal to or greater than the next most significant bit (32)?
- If no, then enter binary 0 in the 32 positional value.
 - If yes, then add a binary 1 in the 32 positional value and subtract 32 from the decimal number.

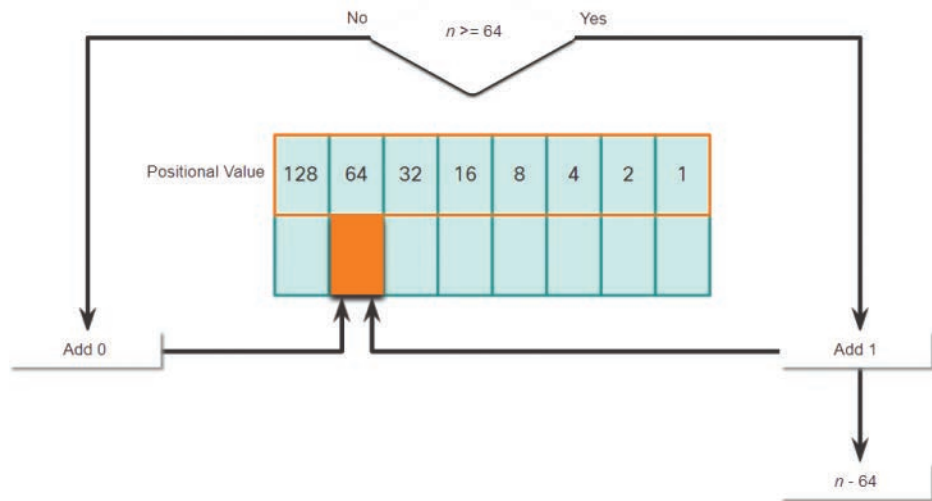


Figure 5-4 64 Positional Value

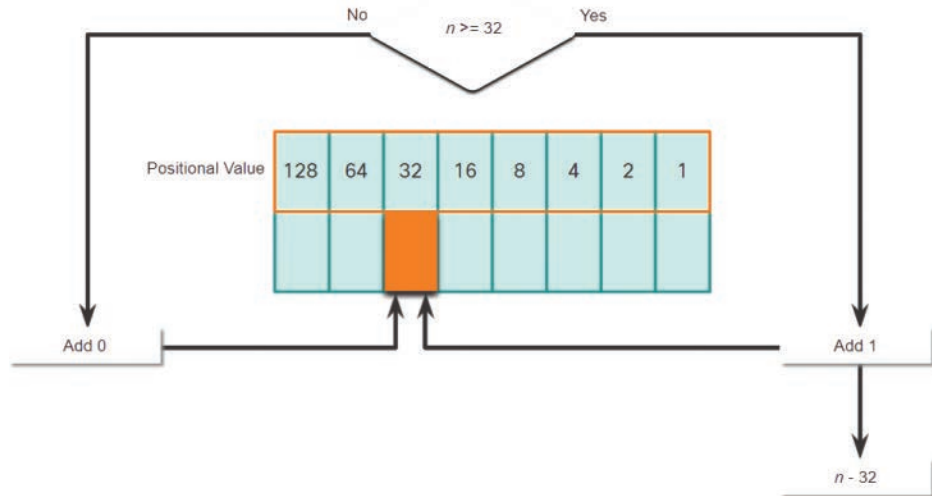


Figure 5-5 32 Positional Value

Step 4. In Figure 5-6, is the decimal number of the octet (n) equal to or greater than the next most significant bit (16)?

- If no, then enter binary 0 in the 16 positional value.
- If yes, then add a binary 1 in the 16 positional value and subtract 16 from the decimal number.

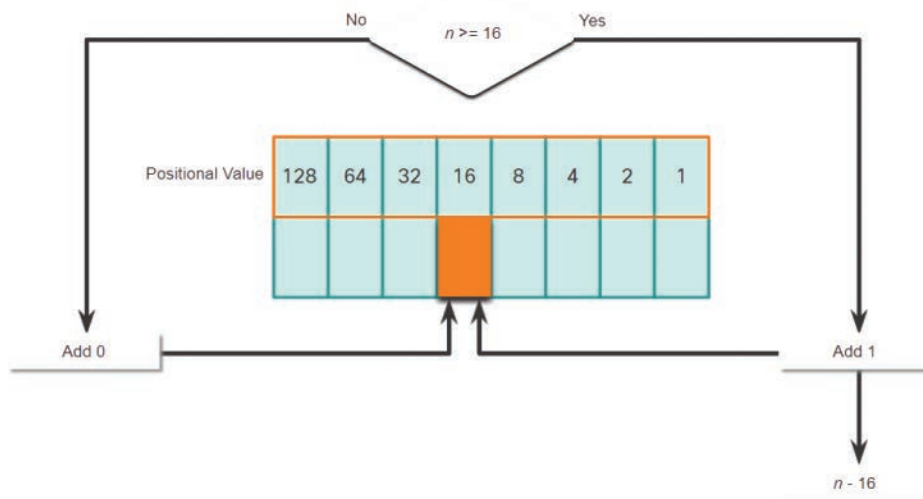


Figure 5-6 16 Positional Value

- Step 5.** In Figure 5-7, is the decimal number of the octet (n) equal to or greater than the next most significant bit (8)?
- If no, then enter binary 0 in the 8 positional value.
 - If yes, then add a binary 1 in the 8 positional value and subtract 8 from the decimal number.

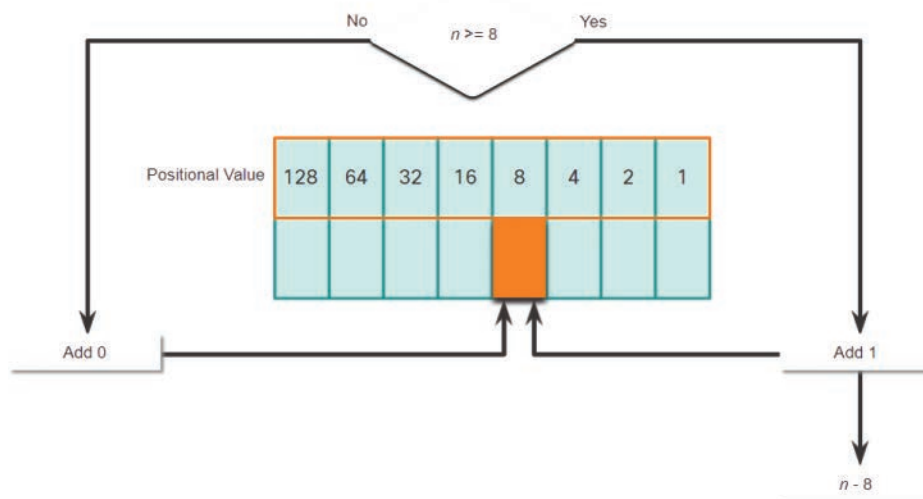


Figure 5-7 8 Positional Value

- Step 6.** In Figure 5-8, is the decimal number of the octet (n) equal to or greater than the next most significant bit (4)?
- If no, then enter binary 0 in the 4 positional value.
 - If yes, then add a binary 1 in the 4 positional value and subtract 4 from the decimal number.

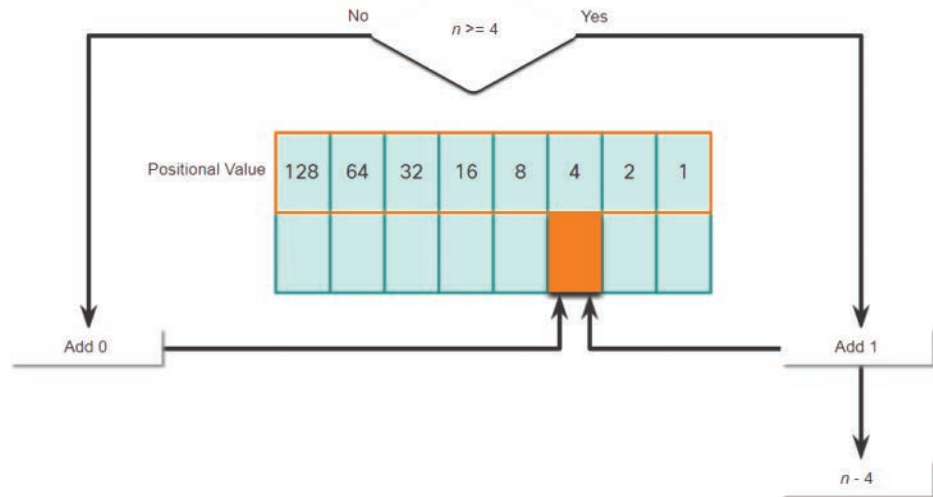


Figure 5-8 4 Positional Value

- Step 7.** In Figure 5-9, is the decimal number of the octet (n) equal to or greater than the next most significant bit (2)?
- If no, then enter binary 0 in the 2 positional value.
 - If yes, then add a binary 1 in the 2 positional value and subtract 2 from the decimal number.
- Step 8.** In Figure 5-10, is the decimal number of the octet (n) equal to or greater than the last most significant bit (1)?
- If no, then enter binary 0 in the 1 positional value.
 - If yes, then add a binary 1 in the 1 positional value and subtract 1 from the last decimal number.

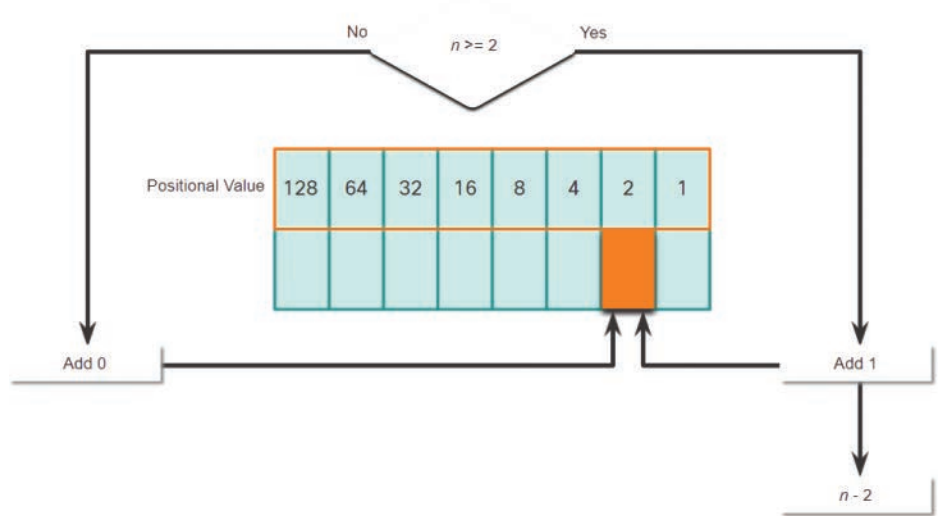


Figure 5-9 2 Positional Value

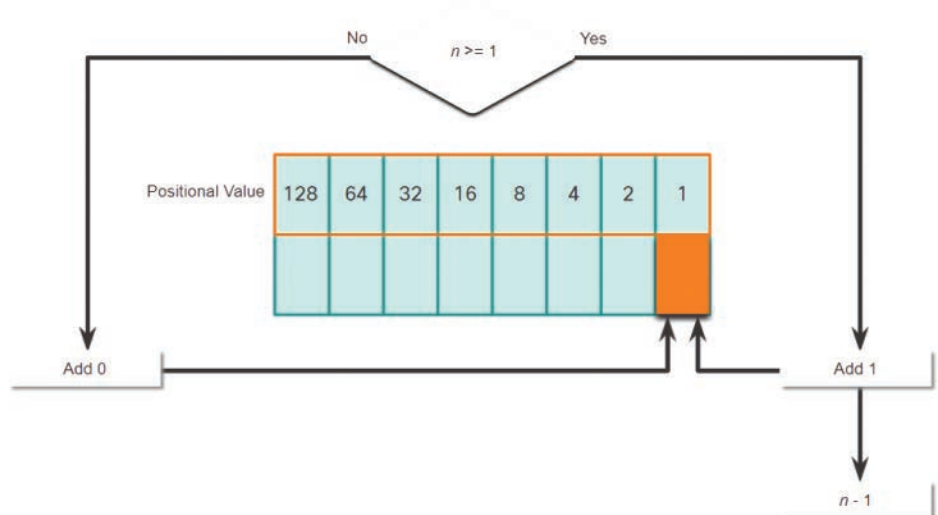


Figure 5-10 1 Positional Value

Decimal to Binary Conversion Example (5.1.8)

To help understand the process of converting from decimal to binary, consider the IP address 192.168.11.10.

The first octet number, 192, is converted to binary using the previously explained positional notation process.

It is possible to bypass the process of subtraction with easier or smaller decimal numbers. For instance, notice that it is fairly easy to calculate the third octet converted to a binary number without actually going through the subtraction process ($8 + 2 = 10$). The binary value of the third octet is 00001010.

The fourth octet is 11 ($8 + 2 + 1$). The binary value of the fourth octet is 00001011.

Converting between binary and decimal may seem challenging at first, but with practice, it should become easier over time.

Figures 5-11 through 5-21 illustrate the steps to convert the IP address 192.168.10.11 into binary:

- Step 1.** In Figure 5-11, is the first octet number 192 equal to or greater than the high-order bit 128?
- Yes it is, so add a 1 to the high-order positional value to represent 128.
 - Subtract 128 from 192 to produce a remainder of 64.

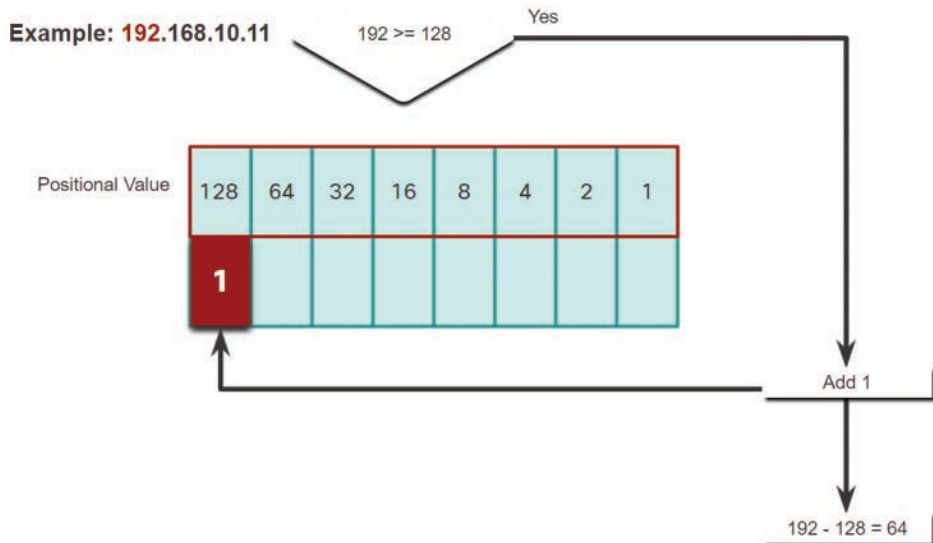


Figure 5-11 Step 1

- Step 2.** In Figure 5-12, is the remainder 64 equal to or greater than the next high-order bit 64?

- It is equal, so add a 1 to next high-order positional value.

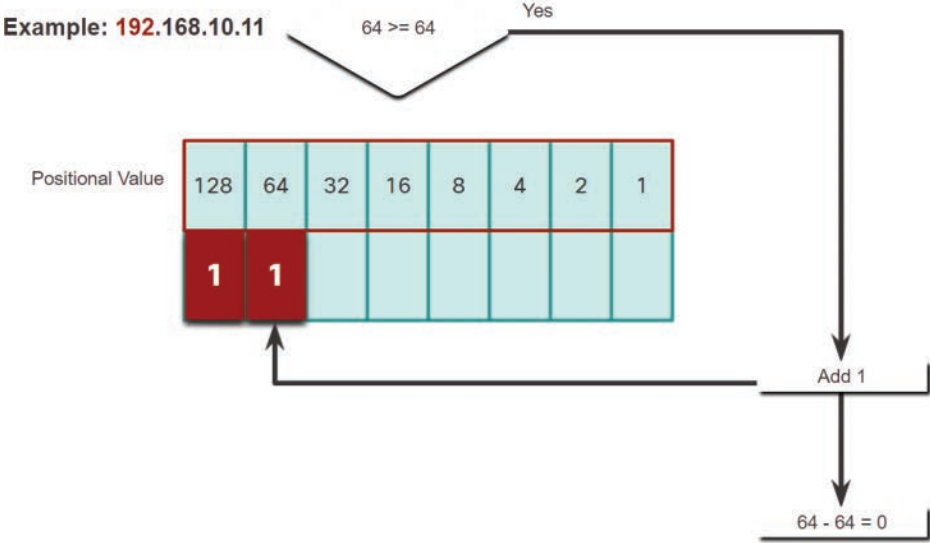


Figure 5-12 Step 2

Step 3. In Figure 5-13, since there is no remainder, enter binary 0 in the remaining positional values.

- The binary value of the first octet is **11000000**.

Example: 192.168.10.11

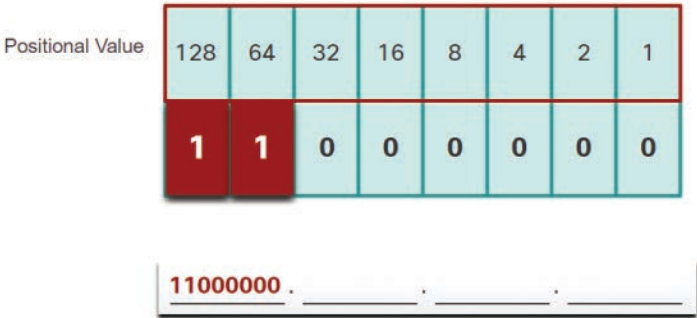


Figure 5-13 Step 3

Step 4. In Figure 5-14, is the second octet number 168 equal to or greater than the high-order bit 128?

- Yes it is, so add a 1 to the high-order positional value to represent 128.
- Subtract 128 from 168 to produce a remainder of 40.

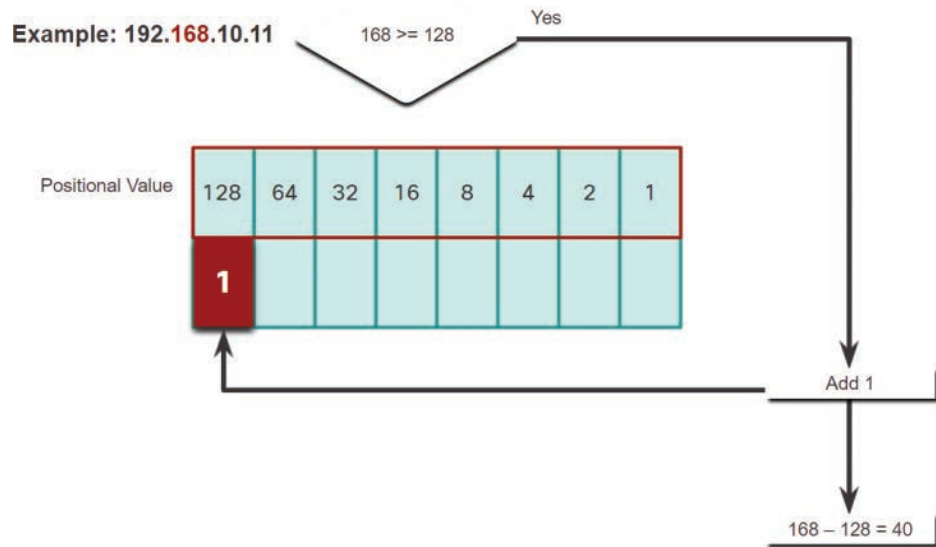


Figure 5-14 Step 4

Step 5. In Figure 5-15, is the remainder **40** equal to or greater than the next high-order bit **64**?

- No it is not, so enter a binary **0** in the positional value.

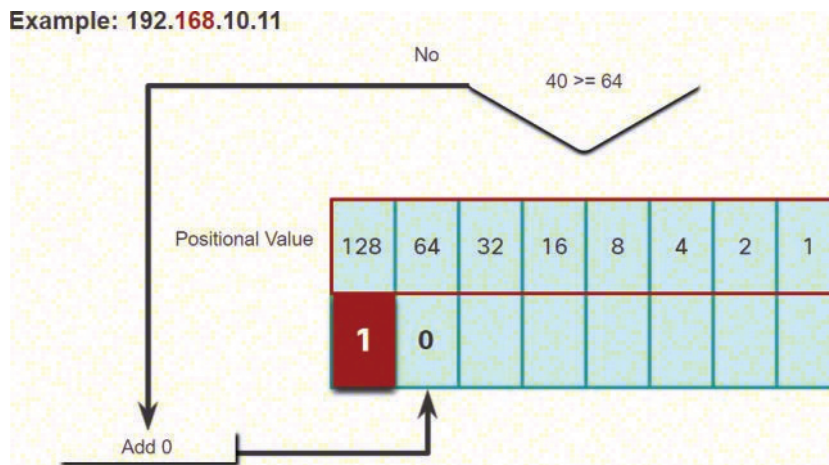


Figure 5-15 Step 5

Step 6. In Figure 5-16, is the remainder **40** equal to or greater than the next high-order bit **32**?

- Yes it is, so add a **1** to the high-order positional value to represent **32**.
- Subtract **32** from **40** to produce a remainder of **8**.

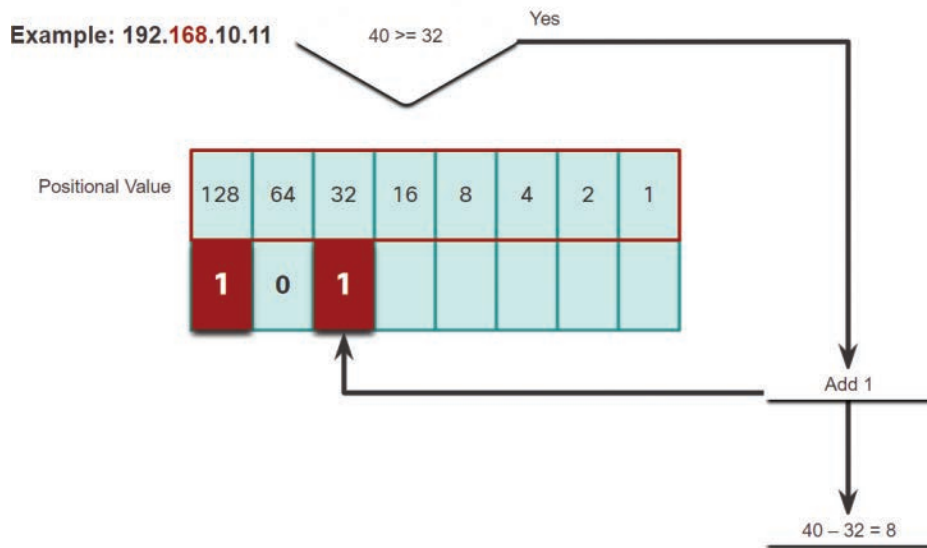


Figure 5-16 Step 6

Step 7. In Figure 5-17, is the remainder 8 equal to or greater than the next high-order bit 16?

- No it is not, so enter a binary 0 in the positional value.

Example: 192.168.10.11

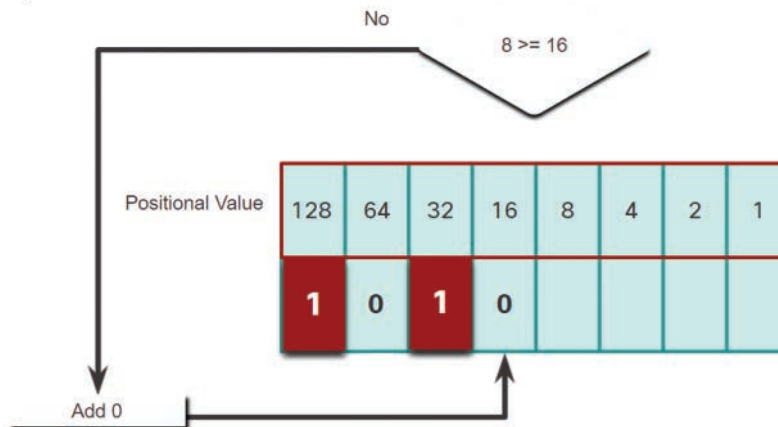


Figure 5-17 Step 7

Step 8. In Figure 5-18, is the remainder 8 equal to or greater than the next high-order bit 8?

- It is equal, so add a 1 to next high-order positional value.

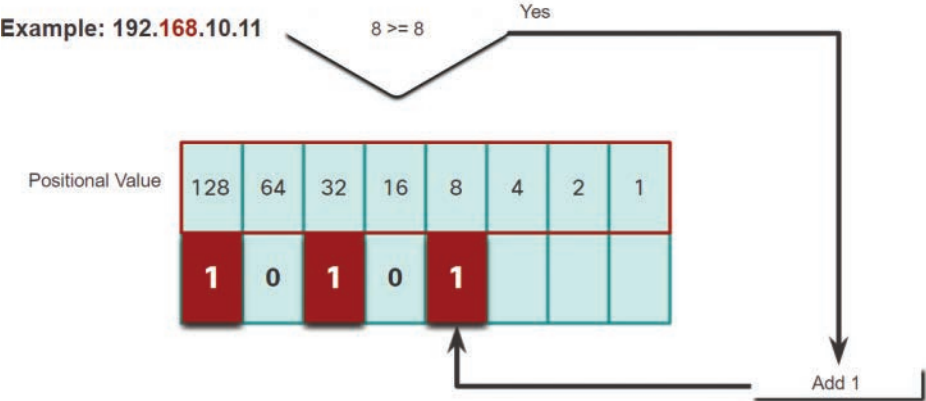


Figure 5-18 Step 8

Step 9. In Figure 5-19, since there is no remainder, enter binary 0 in the remaining positional values.

- The binary value of the second octet is 10101000.

Example: 192.**168**.10.11

Positional Value	128	64	32	16	8	4	2	1
	1	0	1	0	1	0	0	0

11000000

.

10101000

.

.

Figure 5-19 Step 9

Step 10. In Figure 5-20, the binary value of the third octet is 00001010.

Step 11. In Figure 5-21, the binary value of the fourth octet is 00001011.

Example: 192.168.10.11

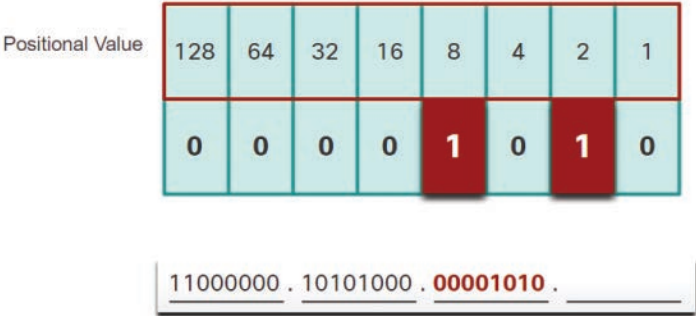


Figure 5-20 Step 10

Example: 192.168.10.11

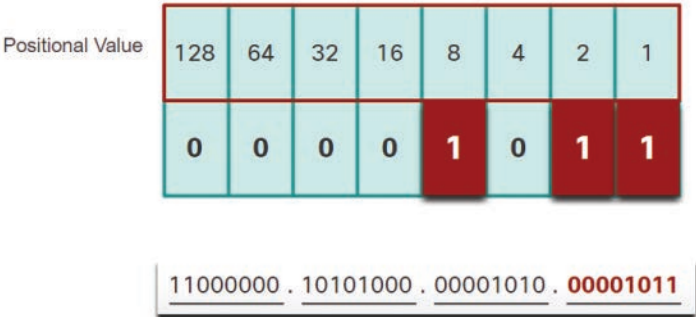


Figure 5-21 Step 11

Interactive
Graphic

Activity—Decimal to Binary Conversions (5.1.9)

This activity allows you to practice decimal conversion to 8-bit binary values. We recommend that you work with this tool until you are able to do the conversion without error. Convert the decimal number shown in the Decimal Value row to its binary bits. Refer to the online course to complete this activity.

**Interactive
Graphic**
Activity—Binary Game (5.1.10)

Playing this game is a fun way to learn binary numbers for networking.

Game link: Log in to cisco.com and download the game from <https://learningnetwork.cisco.com/docs/DOC-1803> or <https://learningnetwork.cisco.com/docs/DOC-11119>. (Create a Cisco account if you do not already have one.)

IPv4 Addresses (5.1.11)

As mentioned at the beginning of this section, routers and computers understand only binary, while humans work in decimal. It is important for you to gain a thorough understanding of these two numbering systems and how they are used in networking.

192.168.10.10 is an IP address that is assigned to a computer, as shown in Figure 5-22.

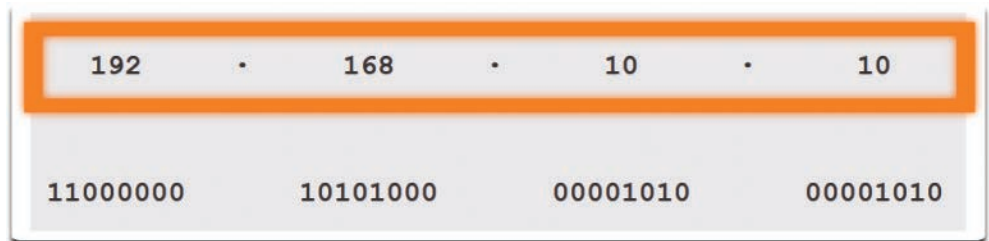


Figure 5-22 Dotted Decimal Address

This address is made up of four different octets, as shown in Figure 5-23.

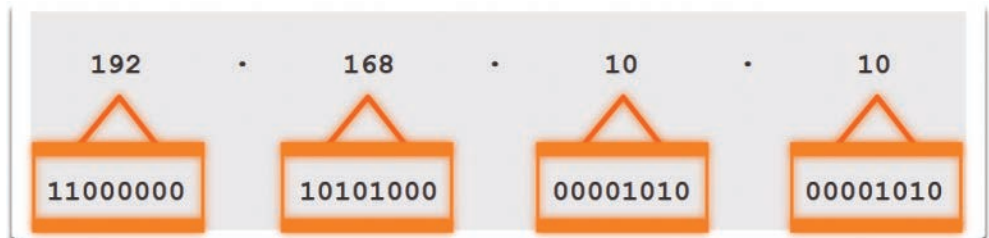


Figure 5-23 Octets

The computer stores the address as the entire 32-bit data stream, as shown in Figure 5-24.

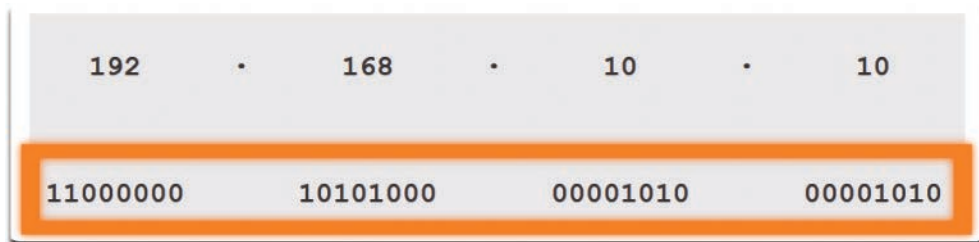


Figure 5-24 32-Bit Address

Hexadecimal Number System (5.2)

IPv6 addresses are 128-bit addresses expressed in hexadecimal notation. This section discusses the hexadecimal number system along with the conversion between the hexadecimal and decimal number systems.

Hexadecimal and IPv6 Addresses (5.2.1)

Now you know how to convert binary to decimal and decimal to binary. You need that skill to understand IPv4 addressing in a network. But you are just as likely to use IPv6 addresses in a network. To understand IPv6 addresses, you must be able to convert hexadecimal to decimal and vice versa.

Just as decimal is a base 10 number system, hexadecimal is a base 16 system. The base 16 number system uses the digits 0 to 9 and the letters A to F. Figure 5-25 shows the equivalent decimal and hexadecimal values for binary 0000 to 1111.

Binary and hexadecimal work well together because it is easier to express a value as a single hexadecimal digit than as 4 binary bits.

The hexadecimal numbering system is used in networking to represent IP version 6 (IPv6) addresses and Ethernet MAC addresses.

IPv6 addresses are 128 bits in length, and every 4 bits is represented by a single hexadecimal digit, for a total of 32 hexadecimal values. IPv6 addresses are not case-sensitive; they can be written in either lowercase or uppercase.

As shown in Figure 5-26, the preferred format for writing an IPv6 address is $x:x:x:x:x:x:x:x$, with each x consisting of four hexadecimal values. When referring to 8 bits of an IPv4 address, we use the term *octet*. In IPv6, a *hextet* is the unofficial term used to refer to a segment of 16 bits or 4 hexadecimal values. Each x is a single hextet, 16 bits, or 4 hexadecimal digits.

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Figure 5-25 Comparing Decimal, Binary, and Hexadecimal Number Systems

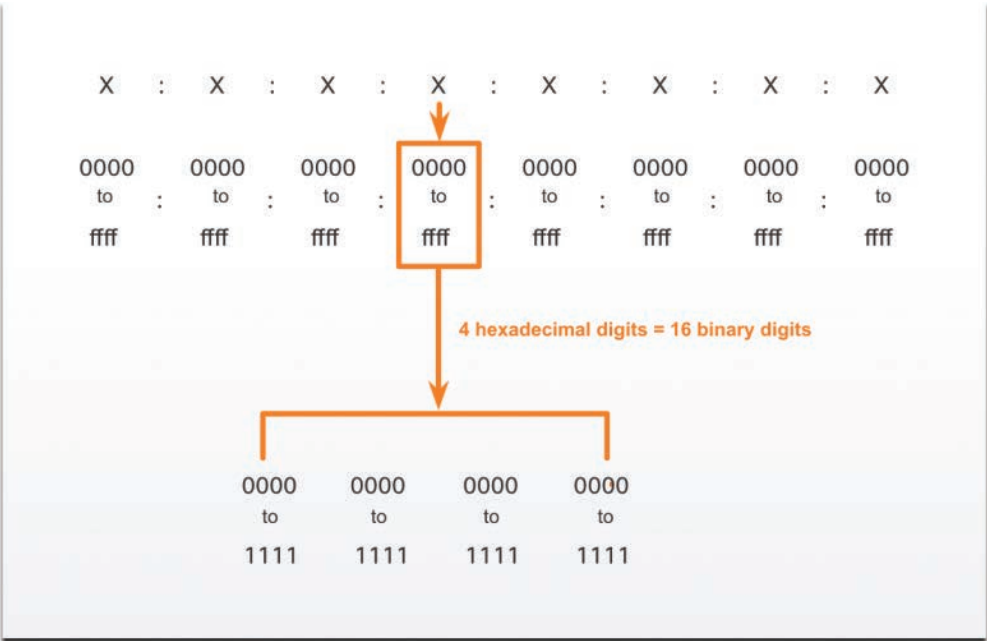


Figure 5-26 Hextets of an IPv6 Address

The sample topology in Figure 5-27 displays IPv6 hexadecimal addresses.

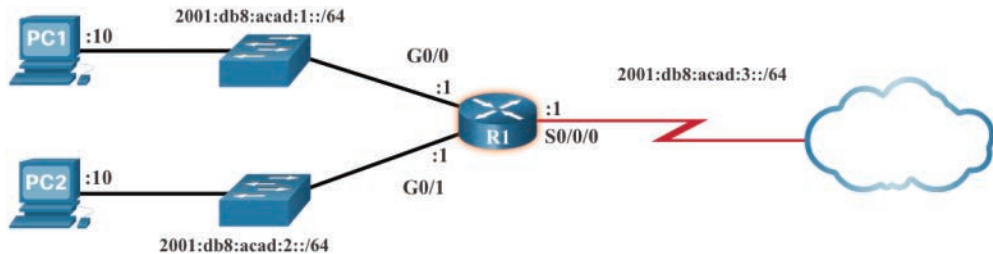


Figure 5-27 Topology with IPv6 Addresses

Video

Video—Converting Between Hexadecimal and Decimal Numbering Systems (5.2.2)

Refer to the online course to view this video.

Decimal to Hexadecimal Conversions (5.2.3)

Converting decimal numbers to hexadecimal values is a straightforward process that involves the following steps:

How To

- Step 1.** Convert the decimal number to 8-bit binary strings.
- Step 2.** Divide the binary strings into groups of four, starting from the rightmost position.
- Step 3.** Convert each set of 4 binary numbers into the equivalent hexadecimal digit.

Let's look at an example of converting the decimal number 168 to hexadecimal:

- Step 1.** 168 in binary is 10101000.
- Step 2.** 10101000 split into two groups of 4 binary digits is 1010 and 1000.
- Step 3.** 1010 is hex A, and 1000 is hex 8.

Answer: 168 is A8 in hexadecimal.

Hexadecimal to Decimal Conversion (5.2.4)

Converting hexadecimal numbers to decimal values is also a straightforward process that involves only three steps:

How To

- Step 1.** Convert the hexadecimal number to 4-bit binary strings.
- Step 2.** Create 8-bit binary grouping starting from the rightmost position.

Step 3. Convert each 8-bit binary grouping into the equivalent decimal digit.

Here is an example of the steps for converting hex **D2** to decimal:

Step 1. **D2** in 4-bit binary strings is **1101** and **0010**.

Step 2. **1101** and **0010** is **11010010** in an 8-bit grouping.

Step 3. **11010010** in binary is equivalent to **210** in decimal.

Answer: **D2** in hexadecimal is **210** in decimal.

Interactive
Graphic

Check Your Understanding—Hexadecimal Number System (5.2.5)

Refer to the online course to complete this activity.

Summary (5.3)

The following is a summary of the topics in the chapter and their corresponding online modules.

Binary Number System

Binary is a numbering system that consists of the numbers 0 and 1, called *bits*. In contrast, the decimal numbering system consists of the numbers 0 to 9. It is important to understand binary because hosts, servers, and network devices use binary addressing—specifically, binary IPv4 addresses—to identify each other. You must know binary addressing and how to convert between binary and dotted decimal IPv4 addresses. In this chapter you learned a few ways to convert decimal to binary and binary to decimal.

Hexadecimal Number System

Just as decimal is a base 10 number system, hexadecimal is a base 16 system. The base 16 number system uses the numbers 0 to 9 and the letters A to F. The hexadecimal numbering system is used in networking to represent IPv6 addresses and Ethernet MAC addresses. IPv6 addresses are 128 bits in length, and every 4 bits is represented by a single hexadecimal digit, for a total of 32 hexadecimal values. To convert hexadecimal to decimal, you must first convert the hexadecimal to binary and then convert the binary to decimal. To convert decimal to hexadecimal, you must also first convert the decimal to binary.

Practice

There are no labs or Packet Tracer activities for this chapter.

Check Your Understanding Questions

Complete all the review questions listed here to test your understanding of the topics and concepts in this chapter. The appendix “Answers to ‘Check Your Understanding’ Questions” lists the answers.

1. What is the binary representation for the decimal number 173?
 - a. 10100111
 - b. 10100101
 - c. 10101101
 - d. 10110101

2. Which address is the dotted decimal equivalent of the binary address 11101100 00010001 00001100 00001010?
 - a. 234.17.10.9
 - b. 234.16.12.10
 - c. 236.17.12.6
 - d. 236.17.12.10
3. How many binary bits exist in an IPv6 address?
 - a. 32
 - b. 48
 - c. 64
 - d. 128
 - e. 256
4. What is the binary equivalent of the decimal number 232?
 - a. 11101000
 - b. 11000110
 - c. 10011000
 - d. 11110010
5. Which two statements are correct about IPv4 and IPv6 addresses? (Choose two.)
 - a. IPv6 addresses are represented by hexadecimal numbers.
 - b. IPv4 addresses are represented by hexadecimal numbers.
 - c. IPv6 addresses are 32 bits in length.
 - d. IPv4 addresses are 32 bits in length.
 - e. IPv4 addresses are 128 bits in length.
 - f. IPv6 addresses are 64 bits in length.
6. Which IPv4 address format looks like 201.192.1.14 and was created for ease of use by people?
 - a. binary
 - b. dotted decimal
 - c. hexadecimal
 - d. ASCII

7. What is the dotted decimal representation of the IPv4 address 11001011.00000000.001110001.11010011?
 - a. 192.0.2.199
 - b. 198.51.100.201
 - c. 203.0.113.211
 - d. 209.165.201.223
8. What is the decimal equivalent of the binary number 10010101?
 - a. 149
 - b. 157
 - c. 168
 - d. 192
9. What is the decimal equivalent of the hex number 0x3F? (0x refers to hexadecimal)
 - a. 63
 - b. 77
 - c. 87
 - d. 93
10. What is the dotted decimal representation of the IPv4 address represented as the binary string 00001010.01100100.00010101.00000001?
 - a. 10.100.21.1
 - b. 10.10.20.1
 - c. 100.10.11.1
 - d. 100.21.10.1
11. What is the decimal equivalent of 0xC9? (0x refers to hexadecimal)
 - a. 185
 - b. 200
 - c. 201
 - d. 199
12. Which is a valid hexadecimal number?
 - a. f
 - b. g
 - c. h
 - d. j

- 13.** What is the binary representation of 0xCA? (0x refers to hexadecimal)
- a.** 10111010
 - b.** 11010101
 - c.** 11001010
 - d.** 11011010
- 14.** How many bits are in an IPv4 address?
- a.** 32
 - b.** 64
 - c.** 128
 - d.** 256

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Data Link Layer

Objectives

Upon completion of this chapter, you will be able to answer the following questions:

- What is the purpose and function of the data link layer in preparing communication for transmission on specific media?
- What are the characteristics of media access control methods on WAN and LAN topologies?
- What are the characteristics and functions of the data link frame?

Key Terms

This chapter uses the following key terms. You can find the definitions in the glossary at the end of the book.

Logical Link Control (LLC) page 206

Media Access Control (MAC) page 206

request for comments (RFCs) page 209

physical topology page 210

logical topology page 210

star topology page 211

extended star topology page 213

bus topology page 214

ring topology page 214

half-duplex page 215

full-duplex page 215

Carrier Sense Multiple Access/Collision Detect (CSMA/CD) page 216

Carrier Sense Multiple Access/Collision Avoidance (CSMA/CA) page 216

Introduction (6.0)

Every network has physical components and media connecting the components. Different types of media need different information about the data in order to accept it and move it across the physical network. Think of it this way: A well-hit golf ball moves through the air fast and far. It can also move through water, but it won't go as fast or as far as through air unless it is helped by a more forceful hit. This is because the golf ball is traveling through a different medium: water instead of air.

Data must have help to move across different media. The data link layer provides this help. As you might have guessed, this help differs based on a number of factors. This chapter gives you an overview of these factors, how they affect data, and the protocols designed to ensure successful delivery. Let's get started!

Purpose of the Data Link Layer (6.1)

This section introduces the role of the data link layer in sending and receiving data over the physical layer.

The Data Link Layer (6.1.1)

The data link layer of the OSI model (Layer 2), as shown in Figure 6-1, prepares network data for the physical network. The data link layer is responsible for network interface card (NIC)-to-NIC communications. The data link layer does the following:

- Enables upper layers to access the media. The upper layer protocol is completely unaware of the type of media used to forward the data.
- Accepts data, usually Layer 3 packets (that is, IPv4 or IPv6 packets), and encapsulates them into Layer 2 frames.
- Controls how data is placed and received on the media.
- Exchanges frames between endpoints over the network media.
- Receives encapsulated data, usually Layer 3 packets, and directs the data to the proper upper-layer protocol.
- Performs error detection and rejects any corrupt frame.

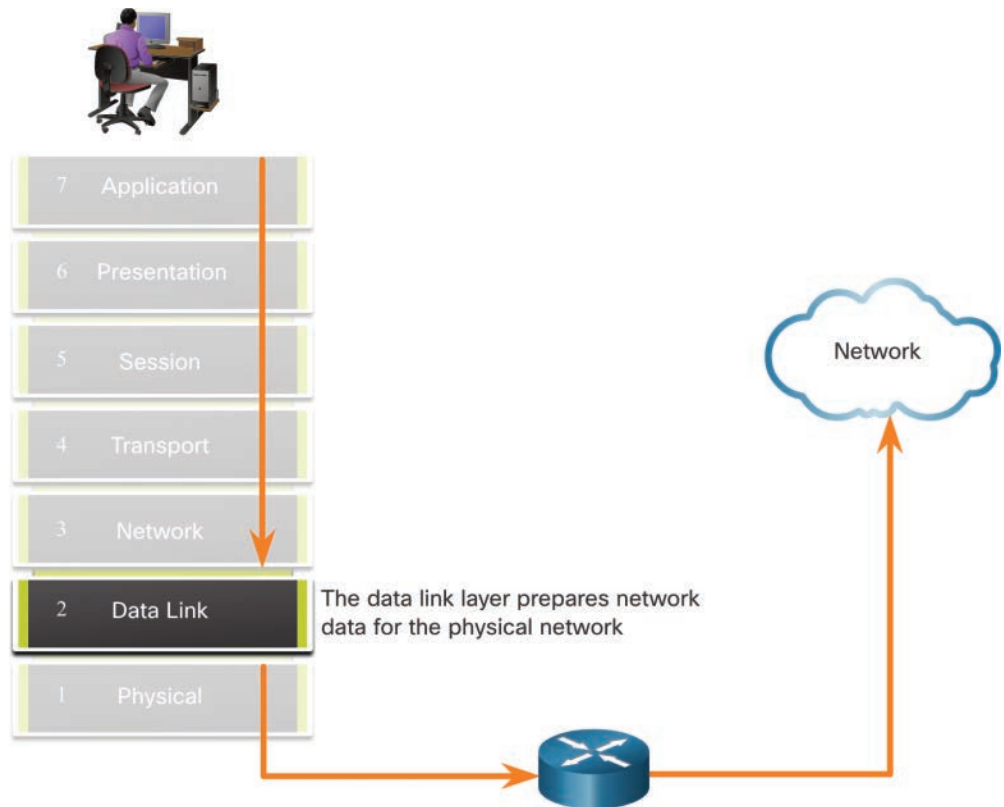


Figure 6-1 Purpose of the Data Link Layer

In computer networks, a node is a device that can receive, create, store, or forward data along a communications path. A node can be either an end device such as a laptop or mobile phone, or an intermediary device such as an Ethernet switch.

Without the data link layer, network layer protocols such as IP would have to make provisions for connecting to every type of media that could exist along a delivery path. In addition, every time a new network technology or medium was developed, IP would have to adapt.

Figure 6-2 shows an example of how the data link layer adds Layer 2 Ethernet destination and source NIC information to a Layer 3 packet. It can then convert this information to a format supported by the physical layer (that is, Layer 1).

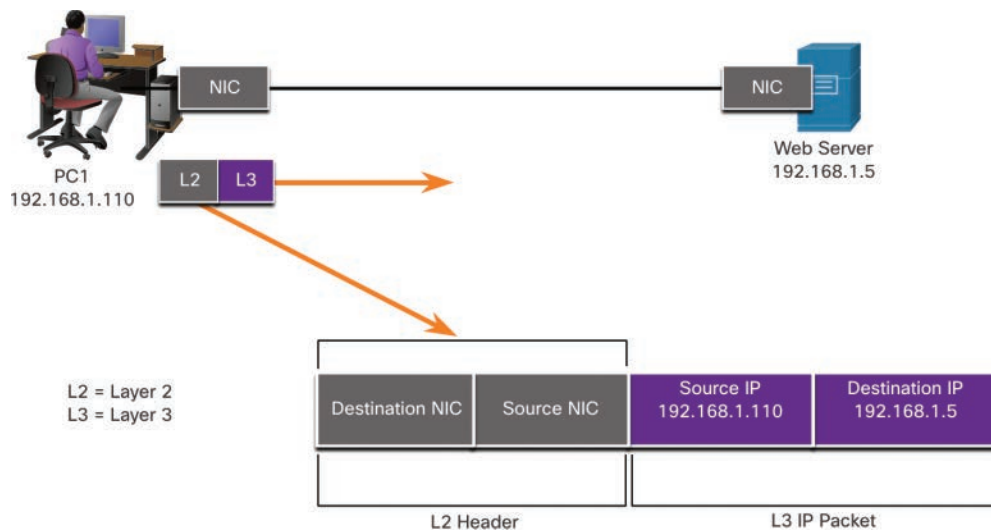


Figure 6-2 Layer 2 (Data Link Layer) Addresses

IEEE 802 LAN/MAN Data Link Sublayers (6.1.2)

IEEE 802 LAN/MAN standards are specific to Ethernet local-area networks (LANs), wireless LANs (WLANs), wireless personal-area networks (WPANs), and other types of local- and metropolitan-area networks. The IEEE 802 LAN/MAN data link layer consists of two sublayers:

- **Logical Link Control (LLC):** This IEEE 802.2 sublayer communicates between the networking software at the upper layers and the device hardware at the lower layers. It places information in the frame to identify which network layer protocol is being used for the frame. This information allows multiple Layer 3 protocols, such as IPv4 and IPv6, to use the same network interface and media.
- **Media Access Control (MAC):** This sublayer (specified in IEEE 802.3, 802.11, and 802.15), which is implemented in hardware, is responsible for data encapsulation and media access control. It provides data link layer addressing and is integrated with various physical layer technologies.

Figure 6-3 shows the two sublayers (LLC and MAC) of the data link layer.

The LLC sublayer takes the network protocol data, which is typically an IPv4 or IPv6 packet, and adds Layer 2 control information to help deliver the packet to the destination node.

The MAC sublayer controls the NIC and other hardware that is responsible for sending and receiving data on the wired or wireless LAN/MAN medium.

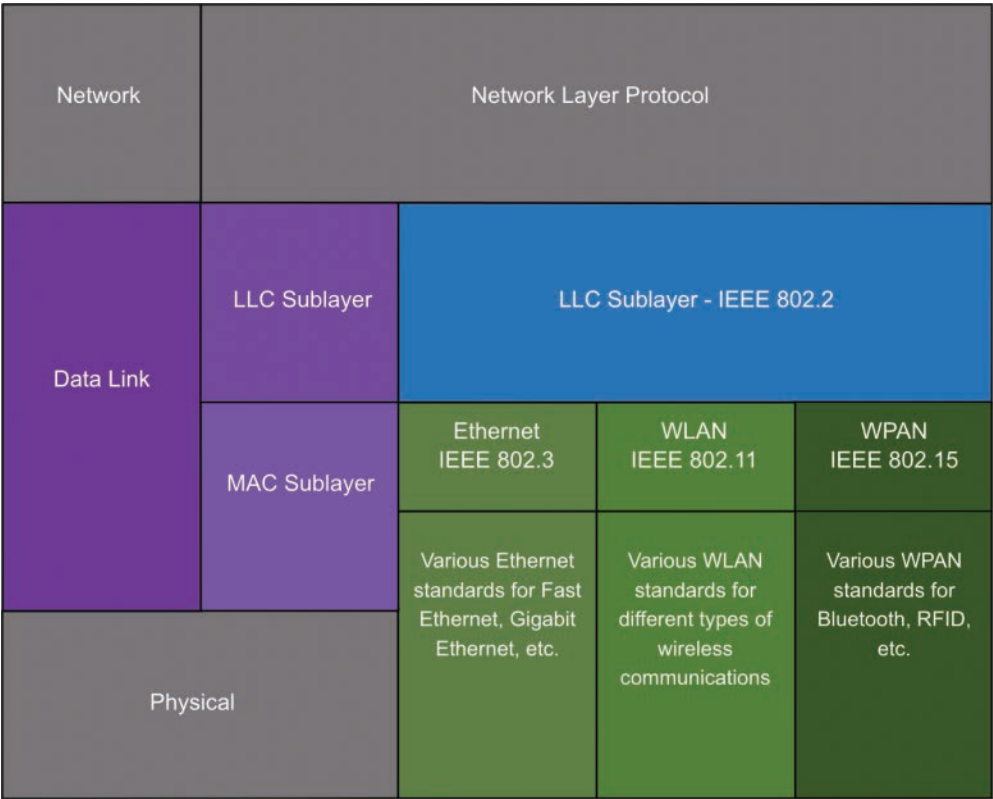


Figure 6-3 The LLC and MAC Sublayers

The MAC sublayer provides data encapsulation:

- **Frame delimiting:** The framing process provides important delimiters to identify fields within a frame. These delimiting bits provide synchronization between the transmitting and receiving nodes.
- **Addressing:** The MAC sublayer provides source and destination addressing for transporting the Layer 2 frame between devices on the same shared medium.
- **Error detection:** The MAC sublayer includes a trailer used to detect transmission errors.

The MAC sublayer also provides media access control, allowing multiple devices to communicate over a shared (half-duplex) medium. Full-duplex communications do not require access control.

Providing Access to Media (6.1.3)

Each network environment that packets encounter as they travel from a local host to a remote host can have different characteristics. For example, an Ethernet LAN

usually consists of many hosts contending for access on the network medium. The MAC sublayer resolves this. With serial links, the access method may consist of only a direct connection between two devices (usually two routers). Therefore, they do not require the techniques employed by the IEEE 802 MAC sublayer.

A router interface encapsulates a packet into the appropriate frame. A suitable media access control method is used to access each link. In any given exchange of network layer packets, there may be numerous data link layers and media transitions.

At each hop along the path, a router performs the following Layer 2 functions:

1. Accepts a frame from a medium
2. De-encapsulates the frame
3. Re-encapsulates the packet into a new frame
4. Forwards a new frame that is appropriate to the medium of that segment of the physical network

The router in Figure 6-4 has an Ethernet interface to connect to the LAN and a serial interface to connect to the WAN. As the router processes frames, it uses data link layer services to receive the frame from one medium, de-encapsulate it to the Layer 3 PDU, re-encapsulate the PDU into a new frame, and place the frame on the medium of the next link of the network.

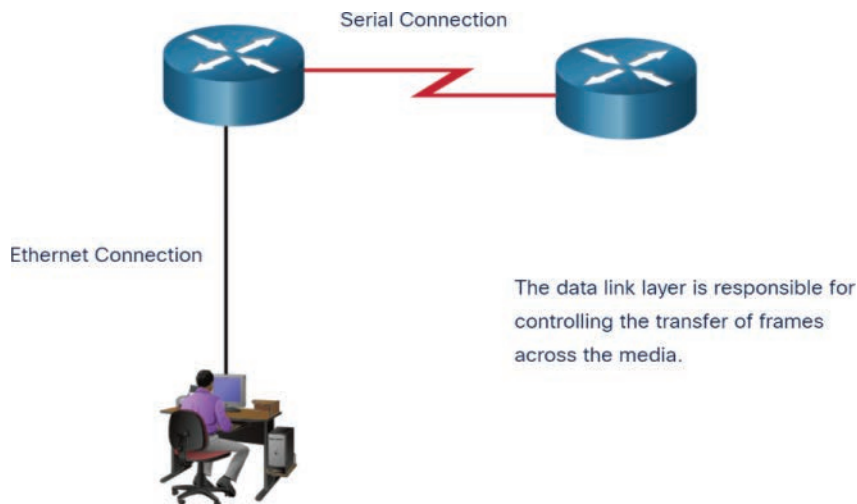


Figure 6-4 Different Data Link Frames for Different Media

Data Link Layer Standards (6.1.4)

Data link layer protocols are generally not defined by *requests for comments (RFCs)*, unlike the protocols of the upper layers of the TCP/IP suite. The Internet Engineering Task Force (IETF) maintains the functional protocols and services for the TCP/IP protocol suite in the upper layers but does not define the functions and operation of the TCP/IP network access layer.

Engineering organizations that define open standards and protocols that apply to the network access layer (that is, the OSI physical and data link layers) include the following:

- Institute of Electrical and Electronics Engineers (IEEE)
- International Telecommunication Union (ITU)
- International Organization for Standardization (ISO)
- American National Standards Institute (ANSI)

Interactive Graphic

Check Your Understanding—Purpose of the Data Link Layer (6.1.5)

Refer to the online course to complete this activity.

Topologies (6.2)

Nodes on a network can be interconnected in numerous ways. How these nodes are connected or how they communicate depends on the topology of the network. This section provides an overview of network topologies and how data access to the media is regulated.

Physical and Logical Topologies (6.2.1)

As you learned in the previous section, the data link layer prepares network data for the physical network. It must know the logical topology of a network in order to be able to determine what is needed to transfer frames from one device to another. This section explains the ways in which the data link layer works with different logical network topologies.

The topology of a network is the arrangement, or the relationship, of the network devices and the interconnections between them.

Two types of topologies are used when describing LAN and WAN networks:

- **Physical topology:** Identifies the physical connections and how end devices and intermediary devices (that is, routers, switches, and wireless access points) are interconnected. The topology may also include specific device location information, such as room number and location on the equipment rack. Physical topologies are usually point-to-point or star.
- **Logical topology:** Refers to the way a network transfers frames from one node to the next. This topology identifies virtual connections using device interfaces and Layer 3 IP addressing schemes.

The data link layer “sees” the logical topology of a network when controlling data access to the media. It is the logical topology that influences the type of network framing and media access control used.

Figure 6-5 displays a sample physical topology for a small sample network.

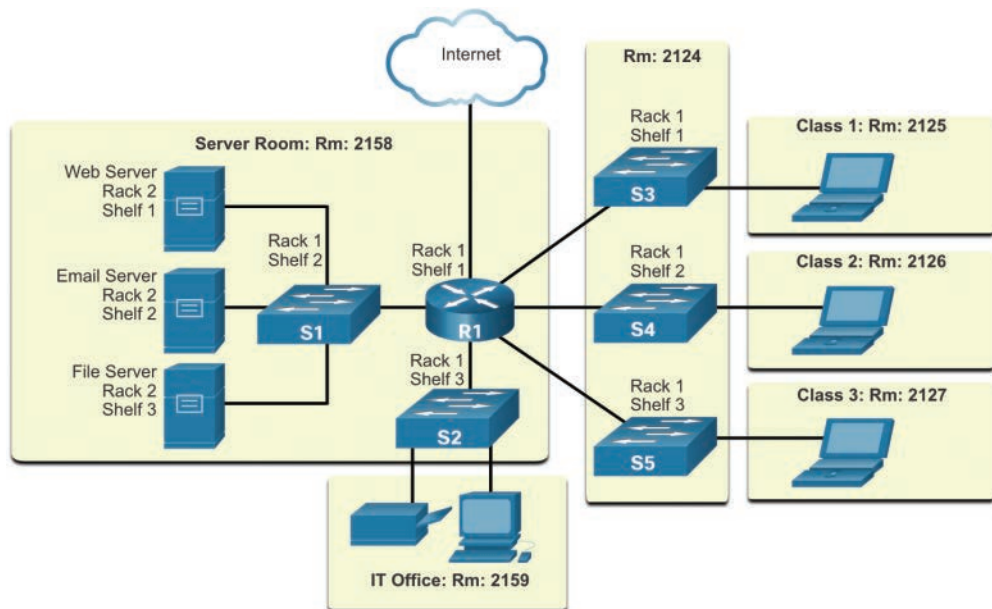


Figure 6-5 Example of a Physical Topology

Figure 6-6 displays a sample logical topology for the same network shown in Figure 6-5.

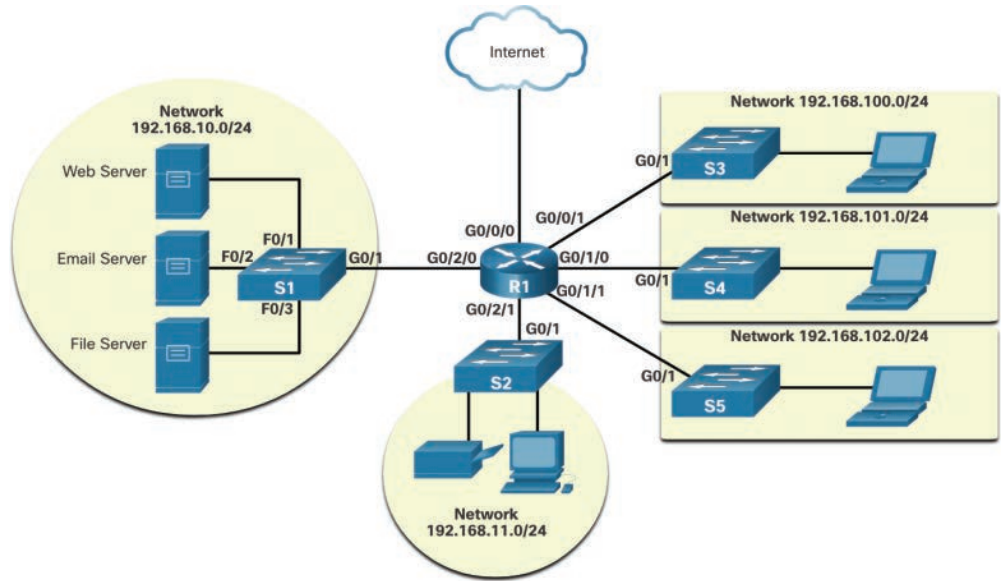


Figure 6-6 Example of a Logical Topology

WAN Topologies (6.2.2)

WANs are commonly interconnected using three common physical WAN topologies: point-to-point, hub and spoke, and mesh.

Point-to-Point

A point-to-point link (see Figure 6-7) is the simplest and most common WAN topology. It consists of a permanent link between two endpoints.



Figure 6-7 Point-to-Point Topology

Hub and Spoke

Figure 6-8 shows a WAN version of the *star topology*, in which a central site interconnects branch sites through the use of point-to-point links. In this topology, branch sites cannot exchange data with other branch sites without going through the central site.

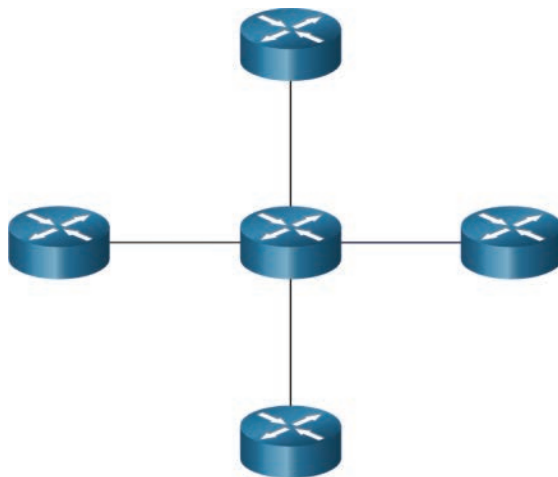


Figure 6-8 Hub and Spoke Topology

Mesh

A mesh topology (see Figure 6-9) provides high availability but requires that every end system be interconnected with every other system. Therefore, the administrative and physical costs can be significant. Each link is essentially a point-to-point link to the other node.

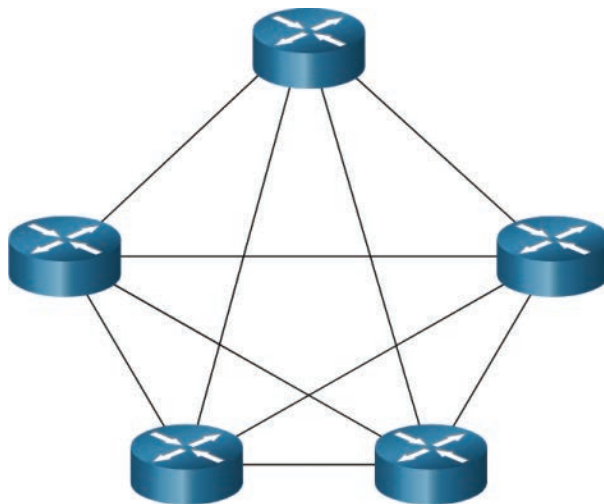


Figure 6-9 Mesh Topology

A hybrid topology is a variation or combination of any topologies. For example, a partial mesh is a hybrid topology in which some, but not all, end devices are interconnected.

Point-to-Point WAN Topology (6.2.3)

A physical point-to-point topology directly connects two nodes, as shown in Figure 6-10. In this arrangement, two nodes do not have to share the media with other hosts. In addition, when using a serial communications protocol such as Point-to-Point Protocol (PPP), a node does not have to make any determination about whether an incoming frame is destined for it or another node. Therefore, the logical data link protocols can be very simple, as all frames on the media can only travel to or from the two nodes. The node places the frames on the media at one end, and those frames are taken from the media by the node at the other end of the point-to-point circuit.



Figure 6-10 Point-to-Point WAN Topology

Note

A point-to-point connection over Ethernet requires the device to determine if the incoming frame is destined for this node.

A source node and a destination node may be indirectly connected to each other over some geographic distance using multiple intermediary devices. However, the use of physical devices in the network does not affect the logical topology, as illustrated in Figure 6-11. In Figure 6-11, adding intermediary physical connections may not change the logical topology. The logical point-to-point connection is the same.



Figure 6-11 Logical and Physical WAN Topology

LAN Topologies (6.2.4)

In multiaccess LANs, end devices (that is, nodes) are interconnected using star or extended star topologies, as shown in Figure 6-12. In this type of topology, end devices are connected to a central intermediary device—in this case, an Ethernet switch. An *extended star topology* extends the star topology by interconnecting multiple Ethernet switches. The star and extended star topologies are easy to install,

very scalable (which means it's easy to add and remove end devices), and easy to troubleshoot. Early star topologies interconnected end devices using Ethernet hubs.

At times, there may be only two devices connected on an Ethernet LAN. For example, two interconnected routers would be an example of Ethernet used on a point-to-point topology.

Legacy LAN Topologies

Early Ethernet and legacy Token Ring LAN technologies included two other types of topologies:

- **Bus topology:** In this topology, all end systems are chained to each other, and they are terminated in some form on each end. Infrastructure devices such as switches are not required to interconnect the end devices. Legacy Ethernet networks were often bus topologies using coax cables because it was inexpensive and easy to set up.
- **Ring topology:** In this topology, end systems are connected to their respective neighbors, forming a ring. The ring does not need to be terminated, unlike a bus topology. Legacy Fiber Distributed Data Interface (FDDI) and Token Ring networks used ring topologies.

Figure 6-12 illustrates how end devices are interconnected on LANs. It is common for a straight line in networking graphics to represent an Ethernet LAN, including a simple star or an extended star.

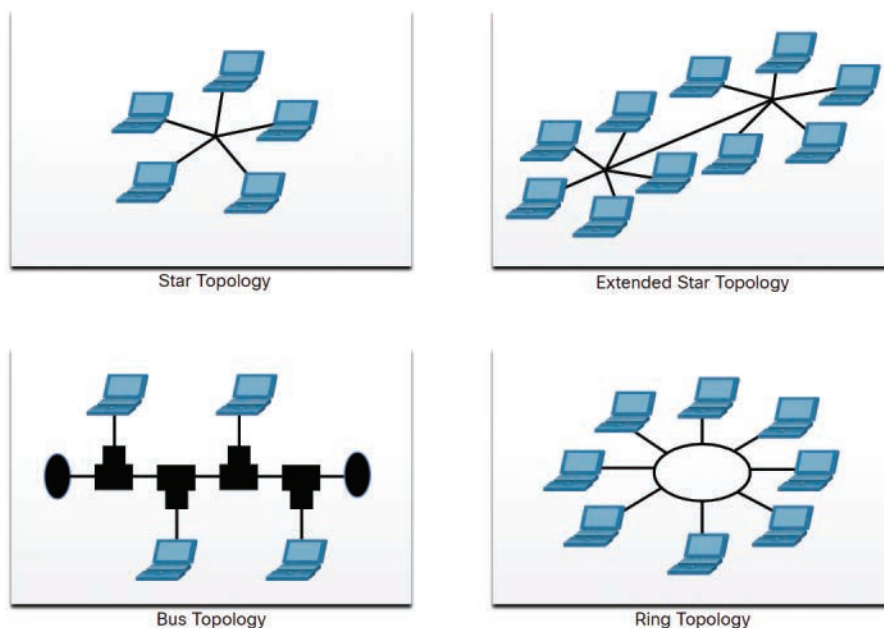


Figure 6-12 LAN Physical Topologies

Half-Duplex and Full-Duplex Communication (6.2.5)

Understanding duplex communication is important when discussing LAN topologies because it refers to the direction of data transmission between two devices. There are two common modes of duplex: half-duplex and full-duplex.

Half-Duplex Communication

In half-duplex communication, two devices can transmit and receive on the medium but cannot do so simultaneously. WLANs and legacy bus topologies with Ethernet hubs use the half-duplex mode. *Half-duplex* allows only one device to send or receive at a time on the shared medium. In Figure 6-13, the server and hub are operating in half-duplex.



Figure 6-13 Half-Duplex Communication

Full-Duplex Communication

With full-duplex communication, both devices can simultaneously transmit and receive on the shared medium. The data link layer assumes that the medium is available for transmission for both nodes at any time. Ethernet switches operate in *full-duplex* mode by default, but they can operate in half-duplex if connecting to a device such as an Ethernet hub. Figure 6-14 shows an example of full-duplex communication.

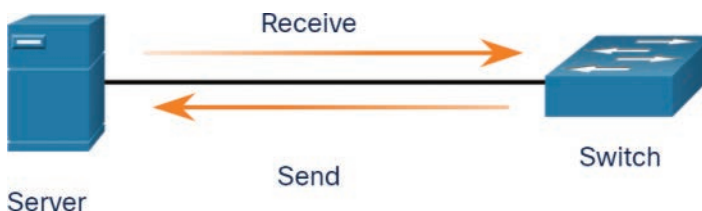


Figure 6-14 Full-Duplex Communication

In summary, half-duplex restricts the exchange of data to one direction at a time. Full-duplex allows the sending and receiving of data to happen simultaneously.

It is important that two interconnected interfaces, such as a host NIC and an interface on an Ethernet switch, operate using the same duplex mode. Otherwise, there is a duplex mismatch, which creates inefficiency and latency on the link.

Access Control Methods (6.2.6)

Ethernet LANs and WLANs are examples of multiaccess networks. A *multiaccess network* is a network that can have two or more end devices attempting to access the network simultaneously.

Some multiaccess networks require rules to govern how devices share the physical media. There are two basic access control methods for shared media:

- Contention-based access
- Controlled access

Contention-Based Access

In contention-based multiaccess networks, all nodes are operating in half-duplex, competing for the use of the medium. However, only one device can send at a time. Therefore, a process needs to be carried out if more than one device transmits at the same time. Examples of contention-based access methods include the following:

- *Carrier Sense Multiple Access/Collision Detect (CSMA/CD)* is used on legacy bus topology Ethernet LANs (see Figure 6-15).
- *Carrier Sense Multiple Access/Collision Avoidance (CSMA/CA)* is used on wireless LANs.

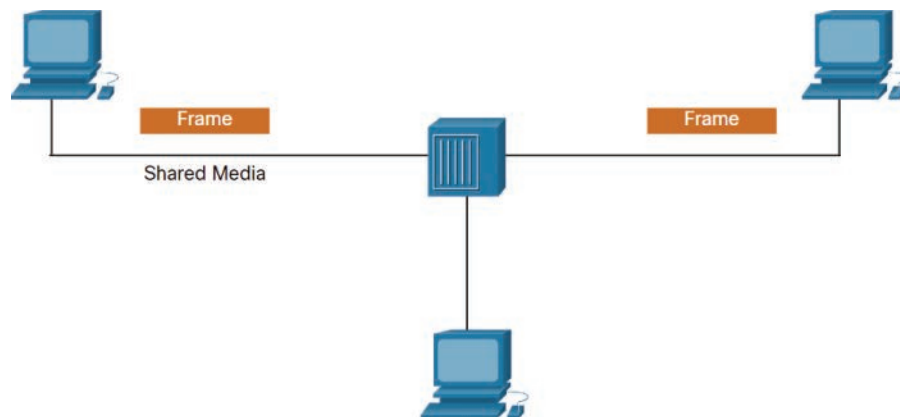
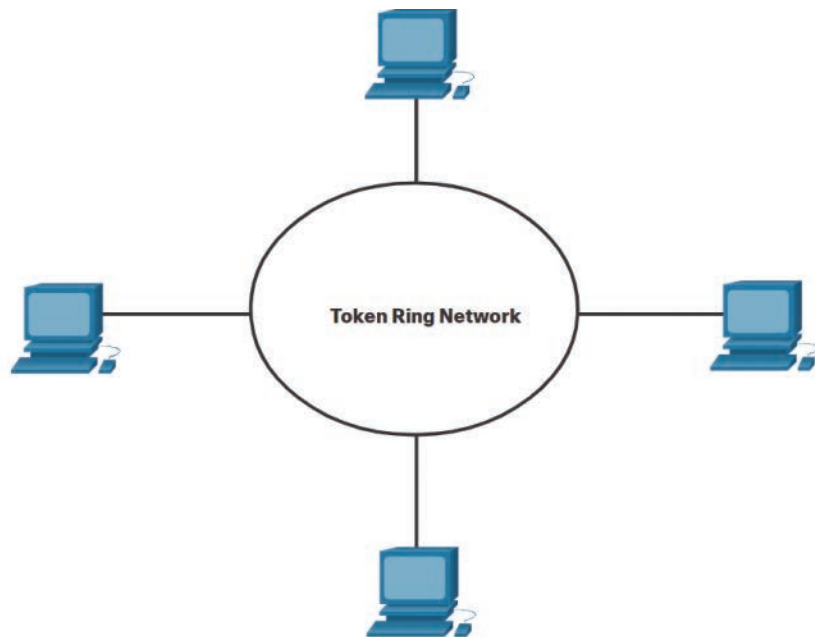


Figure 6-15 Contention-Based Access on Shared Media

Controlled Access

In a controlled multiaccess network, each node has its own time to use the medium. These deterministic legacy networks are inefficient because a device must wait its turn to access the medium. Examples of multiaccess networks that use controlled access include the following:

- Legacy Token Ring (see Figure 6-16)
- Legacy ARCNET



Each node must wait for its turn to access the network medium.

Figure 6-16 Controlled Access on Token Ring

Note

Today, Ethernet networks operate in full-duplex and do not require an access method.

Contention-Based Access—CSMA/CD (6.2.7)

Examples of contention-based access networks include the following:

- Wireless LAN (uses CSMA/CA)
- Legacy bus topology Ethernet LAN (uses CSMA/CD)
- Legacy Ethernet LAN using a hub (uses CSMA/CD)

These networks operate in half-duplex mode, meaning only one device can send or receive at a time. Therefore, a process is needed to govern when a device can send and what happens when multiple devices send at the same time.

If two devices transmit at the same time, a collision occurs. For legacy Ethernet LANs, both devices detect the collision on the network. This is the Collision Detect (CD) portion of CSMA/CD. The NIC compares data transmitted with data received, or it recognizes that the signal amplitude is higher than normal on the media. In the event of a collision, the data sent by both devices is corrupted and needs to be re-sent.

Figures 6-17 through Figure 6-19 demonstrate the CSMA/CD process in legacy Ethernet LANs that use a hub.

In Figure 6-17, PC1 has an Ethernet frame to send to PC3. The PC1 NIC needs to determine whether any device is transmitting on the medium. If it does not detect a carrier signal (in other words, if it is not receiving transmissions from another device), it assumes that the network is available to send.

The PC1 NIC sends the Ethernet frame when the medium is available.

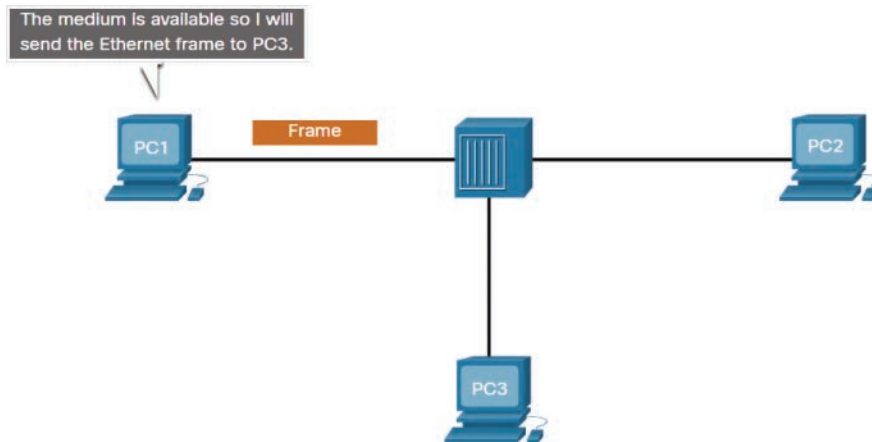


Figure 6-17 PC1 Sends a Frame

In Figure 6-18, the Ethernet hub receives and sends the frame. An Ethernet hub is also known as a *multiport repeater*. Any bits received on an incoming port are regenerated and sent out all other ports, as shown in the figure.

If another device, such as PC2, wants to transmit but is currently receiving a frame, it must wait until the channel is clear.

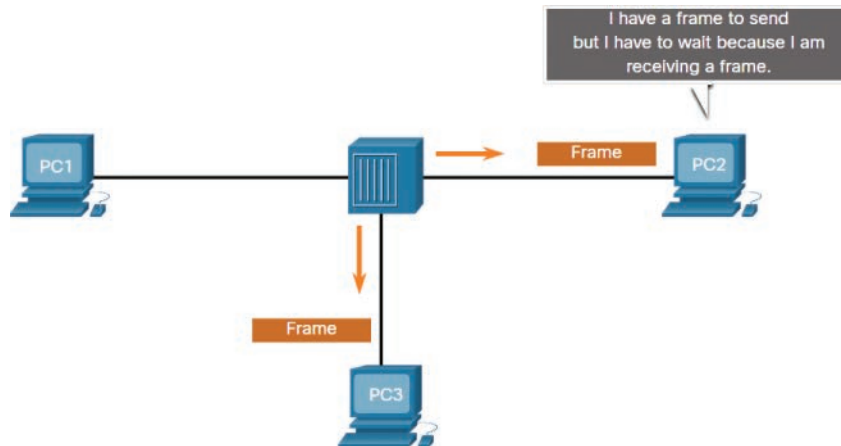


Figure 6-18 The Hub Forwards a Received Frame

In Figure 6-19, all devices attached to the hub receive the frame. However, because the frame has a destination data link address for PC3, only that device accepts and copies in the entire frame. All other device NICs ignore the frame.

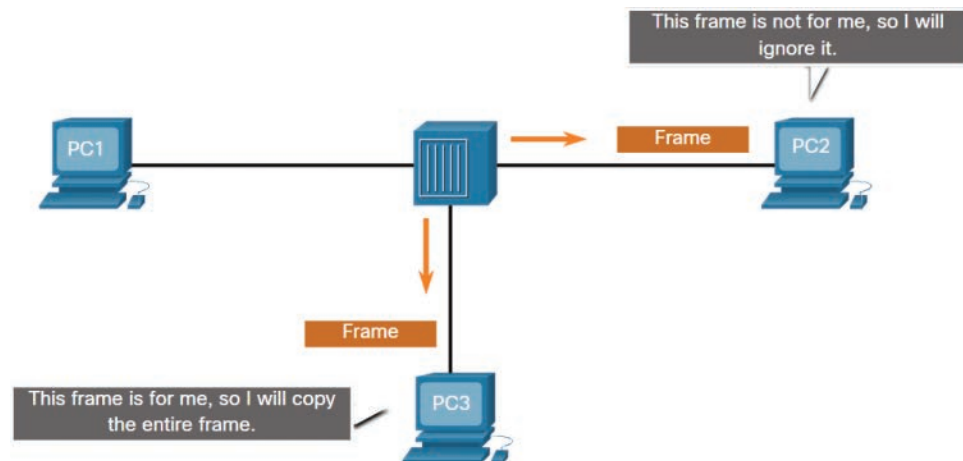


Figure 6-19 The Hub Sends the Frame

Contention-Based Access—CSMA/CA (6.2.8)

Another form of CSMA used by IEEE 802.11 WLANs is Carrier Sense Multiple Access/Collision Avoidance (CSMA/CA).

CSMA/CA uses a method similar to CSMA/CD to detect whether the medium is clear, but it uses additional techniques. In wireless environments, it may not be possible for a device to detect a collision. CSMA/CA does not detect collisions but

attempts to avoid them by waiting before transmitting. Each device that transmits includes the time duration that it needs for the transmission. All other wireless devices receive this information and know how long the medium will be unavailable.

In Figure 6-20, if host A is receiving a wireless frame from the access point, hosts B and C also see the frame and how long the medium will be unavailable.

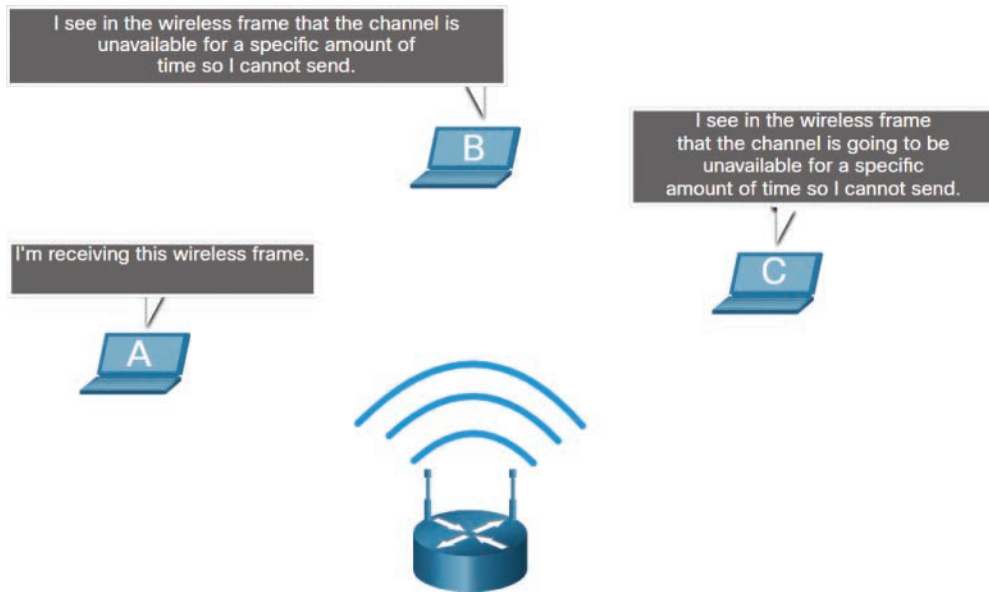


Figure 6-20 CSMA/CA

After a wireless device sends an 802.11 frame, the receiver returns an acknowledgment so that the sender knows the frame arrived.

Contention-based systems—whether Ethernet LANs using hubs or WLANs—do not scale well under heavy media use.

Note

Ethernet LANs using switches do not use a contention-based system because the switch and the host NIC operate in full-duplex mode.

Check Your Understanding—Topologies (6.2.9)

Refer to the online course to complete this activity.

Data Link Frame (6.3)

The data link layer needs to provide intelligible data between the Layer 3 of the sending host and the Layer 3 of the receiving host. To do this, the Layer 3 PDU is wrapped with a header and trailer to form the Layer 2 frame. This section examines the common elements of the frame structure and some of the commonly used data link layer protocols.

The Frame (6.3.1)

This section discusses in detail what happens to the data link frame as it moves through a network. The information appended to a frame depends on the protocol being used.

The data link layer prepares the encapsulated data (usually an IPv4 or IPv6 packet) for transport across the local media by encapsulating it with a header and a trailer to create a frame.

The data link protocol is responsible for NIC-to-NIC communications within the same network. Although there are many different data link layer protocols that describe data link layer frames, each frame type has three basic parts:

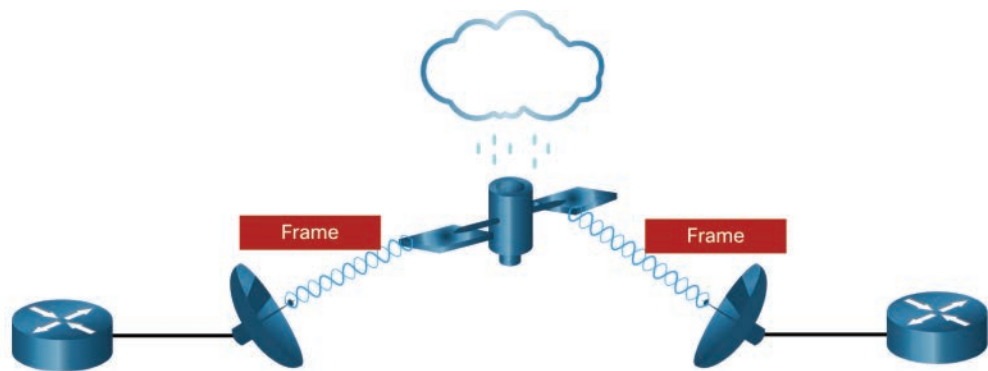
- Header
- Data
- Trailer

Unlike other encapsulation protocols, the data link layer protocols append information in the form of a trailer at the end of the frame.

All data link layer protocols encapsulate the data within the data field of the frame. However, the structure of the frame and the fields contained in the header and trailer vary according to the protocol.

There is no one frame structure that meets all data transportation needs across all types of media. Depending on the environment, the amount of control information needed in the frame varies to match the access control requirements of the media and logical topology. For example, a WLAN frame must include a procedures for collision avoidance and therefore requires additional control information compared to an Ethernet frame.

As shown in Figure 6-21, in a fragile environment, more controls are needed to ensure delivery. The header and trailer fields are larger as more control information is needed.



Greater effort is needed to ensure delivery. This means higher overhead and slower transmission rates.

Figure 6-21 Fragile Environment

Frame Fields (6.3.2)

Framing breaks a stream into decipherable groupings, with control information inserted in the header and trailer as values in different fields. This format gives the physical signals a structure that is recognized by nodes and that can be decoded into packets at the destination.

The generic frame fields are shown in Figure 6-22. Not all protocols include all these fields. The standards for a specific data link protocol define the actual frame format.

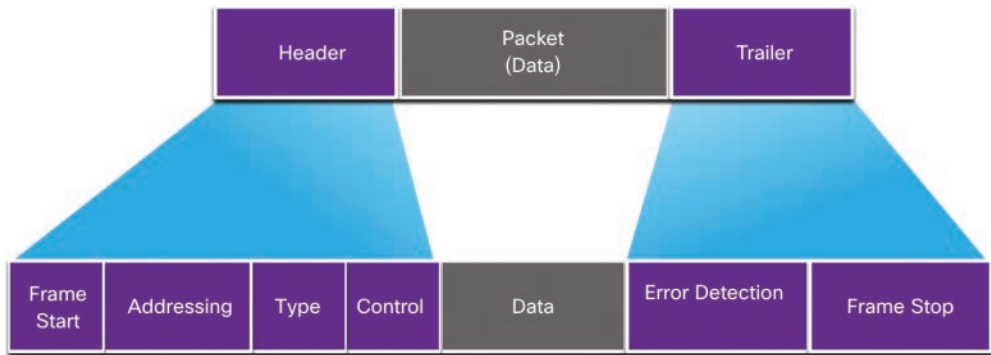


Figure 6-22 Generic Frame Format

Data link layer protocols add a trailer to the end of each frame. In a process called *error detection*, a trailer determines whether the frame arrived without error. This process places a logical or mathematical summary of the bits that comprise the frame in the trailer. The data link layer adds error detection because the signals on the

media could be subject to interference, distortion, or loss that would substantially change the bit values that those signals represent.

A transmitting node creates a logical summary of the contents of the frame, known as the cyclic redundancy check (CRC) value. This value is placed in the Frame Check Sequence (FCS) field to represent the contents of the frame. In the Ethernet trailer, the FCS provides a method for the receiving node to determine whether the frame experienced transmission errors.

Layer 2 Addresses (6.3.3)

The data link layer provides the addressing used in transporting a frame across a shared local medium. Device addresses at this layer are referred to as *physical addresses*. Data link layer addressing is contained within the frame header and specifies the frame destination node on the local network. It is typically at the beginning of the frame, so the NIC can quickly determine if it matches its own Layer 2 address before accepting the rest of the frame. The frame header may also contain the source address of the frame.

Unlike Layer 3 logical addresses, which are hierarchical, physical addresses do not indicate on what network a device is located. Rather, a physical address is unique to a specific device. A device still functions with the same Layer 2 physical address even if the device moves to another network or subnet. Therefore, Layer 2 addresses are only used to connect devices within the same shared medium on the same IP network.

Figure 6-23 through 6-25 illustrate the function of the Layer 2 and Layer 3 addresses. As an IP packet travels from host to router, router to router, and finally router to host, at each point along the way, the IP packet is encapsulated in a new data link frame. Each data link frame contains the source data link address of the NIC sending the frame and the destination data link address of the NIC receiving the frame.

In Figure 6-23, the source host encapsulates the Layer 3 IP packet in a Layer 2 frame. In the frame header, the host adds its Layer 2 address as the source and the Layer 2 address for R1 as the destination.

In Figure 6-24, R1 encapsulates the Layer 3 IP packet in a new Layer 2 frame. In the frame header, R1 adds its Layer 2 address as the source and the Layer 2 address for R2 as the destination.

In Figure 6-25, R2 encapsulates the Layer 3 IP packet in a new Layer 2 frame. In the frame header, R2 adds its Layer 2 address as the source and the Layer 2 address for the server as the destination.

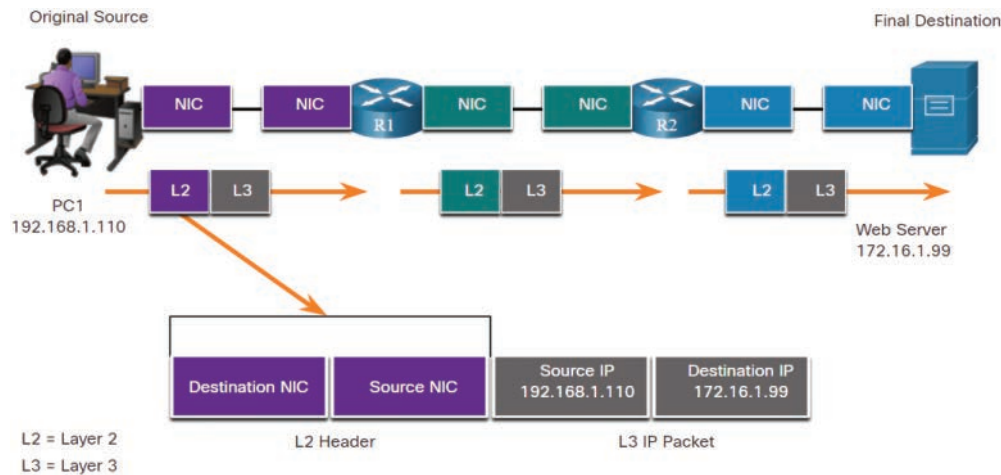


Figure 6-23 Host-to-Router Communications

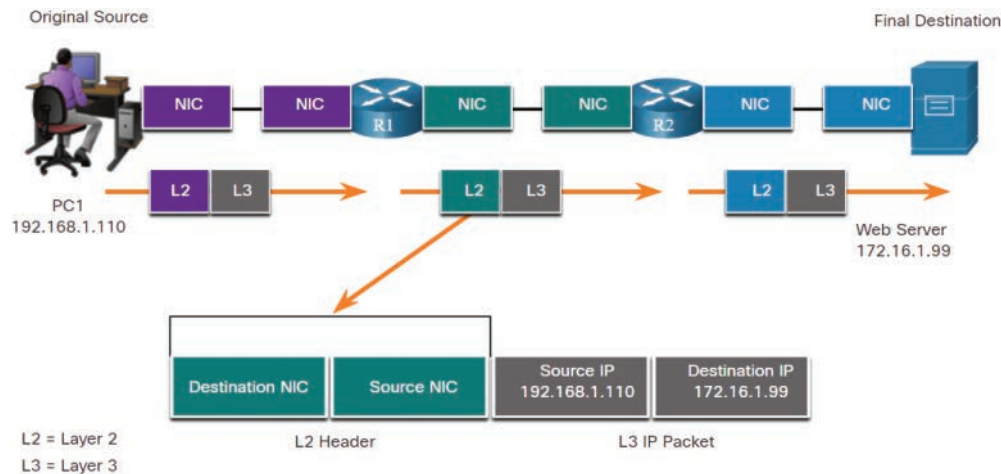


Figure 6-24 Router-to-Router Communications

The data link layer address is used only for local delivery. Addresses at this layer have no meaning beyond the local network. At Layer 3, in contrast, addresses in the packet header are carried from the source host to the destination host, regardless of the number of network hops along the route.

If the data must pass onto another network segment, an intermediary device, such as a router, is necessary. The router must accept the frame based on the physical address and de-encapsulate the frame in order to examine the hierarchical address, which is the IP address. Using the IP address, the router can determine the network location

of the destination device and the best path to reach it. When it knows where to forward the packet, the router creates a new frame for the packet, and the new frame is sent on to the next network segment toward its final destination.

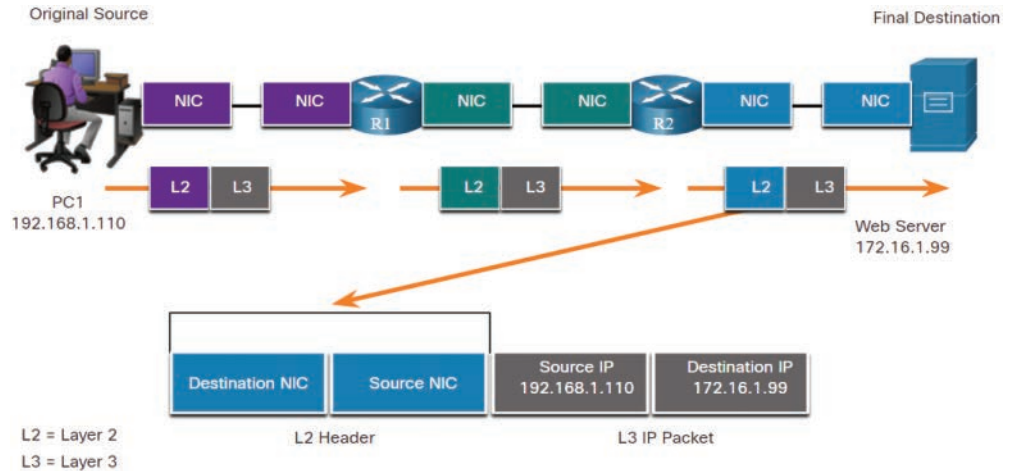


Figure 6-25 Router-to-Host Communications

LAN and WAN Frames (6.3.4)

Ethernet protocols are used by wired LANs. Wireless communications fall under the WLAN protocols (specified in IEEE 802.11), which were designed for multiaccess networks.

WANs traditionally used other types of protocols for various point-to-point, hub and spoke, and full-mesh topologies. Some of the common WAN protocols over the years have included the following:

- Point-to-Point Protocol (PPP)
- High-Level Data Link Control (HDLC)
- Frame Relay
- Asynchronous Transfer Mode (ATM)
- X.25

These Layer 2 protocols are now being replaced in the WAN by Ethernet.

In a TCP/IP network, all OSI Layer 2 protocols work with IP at OSI Layer 3. However, the Layer 2 protocol used depends on the logical topology and the physical media.

Each protocol performs media access control for specified Layer 2 logical topologies. This means that a number of different network devices can act as nodes that operate at the data link layer when implementing these protocols. These devices include the NICs on computers as well as the interfaces on routers and Layer 2 switches.

The Layer 2 protocol that is used for a particular network topology is determined by the technology used to implement that topology. The technology used is determined by the size of the network, in terms of the number of hosts and the geographic scope, as well as the services to be provided over the network.

A LAN typically uses a high-bandwidth technology capable of supporting large numbers of hosts. The relatively small geographic area of a LAN (a single building or a multi-building campus) and its high density of users make this technology cost-effective.

However, using a high-bandwidth technology is usually not cost-effective for WANs that cover large geographic areas (cities or multiple cities, for example). The cost of the long-distance physical links and the technology used to carry the signals over those distances typically results in lower bandwidth capacity.

The difference in bandwidth normally results in the use of different protocols for LANs and WANs.

Data link layer protocols include the following:

- Ethernet
- 802.11 wireless
- Point-to-Point Protocol (PPP)
- High-Level Data Link Control (HDLC)
- Frame Relay

Figure 6-26 shows examples of Layer 2 protocols.

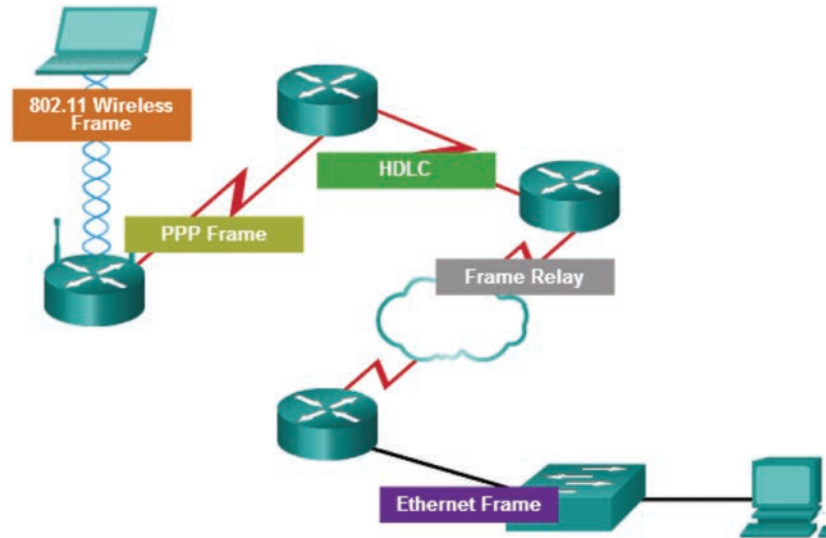


Figure 6-26 Examples of Layer 2 Protocols

**Interactive
Graphic**

Check Your Understanding—Data Link Frame (6.3.5)

Refer to the online course to complete this activity.

Summary (6.4)

The following is a summary of the topics in the chapter and their corresponding online modules.

Purpose of the Data Link Layer

The data link layer of the OSI model (Layer 2) prepares network data for the physical network. The data link layer is responsible for network interface card (NIC)-to-NIC communications. Without the data link layer, network layer protocols such as IP would have to make provisions for connecting to every type of media that could exist along a delivery path. The IEEE 802 LAN/MAN data link layer consists of two sublayers: LLC and MAC. The MAC sublayer provides data encapsulation through frame delimiting, addressing, and error detection. Router interfaces encapsulate the packet into the appropriate frame. A suitable media access control method is used to access each link. Engineering organizations that define open standards and protocols that apply to the network access layer include IEEE, ITU, ISO, and ANSI.

Topologies

The two types of topologies used in LAN and WAN networks are physical and logical. The data link layer “sees” the logical topology of a network when controlling data access to the media. The logical topology influences the type of network framing and media access control used. Three common types of physical WAN topologies are point-to-point, hub and spoke, and mesh. A physical point-to-point topology directly connects two end devices (nodes). Adding intermediate physical connections may not change the logical topology. In multiaccess LANs, nodes are interconnected using star or extended star topologies, in which nodes are connected to a central intermediary device. Physical LAN topologies include star, extended star, bus, and ring. Half-duplex communications exchange data in one direction at a time. Full-duplex sends and receives data simultaneously. Two interconnected interfaces must use the same duplex mode, or the duplex mismatch will create inefficiency and latency on the link. Ethernet LANs and WLANs are examples of multiaccess networks. A multiaccess network is a network that can have multiple nodes accessing the network simultaneously. Some multiaccess networks require rules to govern how devices share the physical media. There are two basic access control methods for shared media: contention-based access and controlled access. In contention-based multiaccess networks, all nodes operate in half-duplex. A process occurs if more than one device transmits at the same time. Examples of contention-based access methods include CSMA/CD for bus-topology Ethernet LANs and CSMA/CA for WLANs.

Data Link Frame

The data link layer prepares encapsulated data (usually an IPv4 or IPv6 packet) for transport across the local media by encapsulating it with a header and a trailer to create a frame. The data link protocol is responsible for NIC-to-NIC communications within the same network. There are many different data link layer protocols that describe data link layer frames, and each frame type has three basic parts: header, data, and trailer. Unlike other encapsulation protocols, the data link layer appends information in the trailer. There is no one frame structure that meets the needs of all data transportation across all types of media. Depending on the environment, the amount of control information needed in the frame varies to match the access control requirements of the medium and logical topology. Frame fields include frame start and stop indicator flags, addressing, type, control, data, and error detection. The data link layer provides addressing used to transport a frame across shared local medium. Device addresses at this layer are physical addresses. Data link layer addressing is contained within the frame header and specifies the frame destination node on the local network. The data link layer address is only used for local delivery. In a TCP/IP network, all OSI Layer 2 protocols work with IP at OSI Layer 3. However, the Layer 2 protocol used depends on the logical topology and the physical medium. Each protocol performs media access control for specified Layer 2 logical topologies. The Layer 2 protocol that is used for a particular network topology is determined by the technology used to implement that topology. Data link layer protocols include Ethernet, 802.11 wireless, PPP, HDLC, and Frame Relay.

Practice

There are no labs or Packet Tracer activities for this chapter.

Check Your Understanding Questions

Complete all the review questions listed here to test your understanding of the topics and concepts in this chapter. The appendix “Answers to ‘Check Your Understanding’ Questions” lists the answers.

1. What identifier is used at the data link layer to uniquely identify an Ethernet device?
 - a. IP address
 - b. MAC address
 - c. sequence number
 - d. TCP port number
 - e. UDP port number

2. Which two engineering organizations define open standards and protocols that apply to the data link layer? (Choose two.)
 - a. Institute of Electrical and Electronics Engineers (IEEE)
 - b. Internet Assigned Numbers Authority (IANA)
 - c. International Telecommunication Union (ITU)
 - d. Electronic Industries Alliance (EIA)
 - e. Internet Society (ISOC)
3. Which layer of the OSI model is responsible for specifying the encapsulation method used for specific types of media?
 - a. application
 - b. transport
 - c. data link
 - d. physical
4. What is true concerning physical and logical topologies?
 - a. The logical topology is always the same as the physical topology.
 - b. Physical topologies are concerned with how a network transfers frames.
 - c. Physical topologies display the IP addressing scheme of each network.
 - d. Logical topologies refer to how a network transfers data between devices.
5. What type of physical topology can be created by connecting all Ethernet cables to a central device?
 - a. bus
 - b. ring
 - c. star
 - d. mesh
6. A technician has been asked to develop a physical topology for a network that provides a high level of redundancy. Which physical topology requires that every node be attached to every other node on the network?
 - a. bus
 - b. hierarchical
 - c. mesh
 - d. ring
 - e. star

7. Which statement describes the half-duplex mode of data transmission?
 - a. Data transmitted over the network can flow in only one direction.
 - b. Data transmitted over the network flows in one direction at a time.
 - c. Data transmitted over the network flows in one direction to many different destinations simultaneously.
 - d. Data transmitted over the network flows in both directions at the same time.
8. Which is a function of the Logical Link Control (LLC) sublayer?
 - a. to define the media access processes that are performed by the hardware
 - b. to provide data link addressing
 - c. to identify which network layer protocol is being used
 - d. to accept segments and package them into data units called packets
9. Which data link layer media access control method does Ethernet use with legacy Ethernet hubs?
 - a. CSMA/CD
 - b. determinism
 - c. turn taking
 - d. token passing
10. What are the two sublayers of the OSI model data link layer? (Choose two.)
 - a. internet
 - b. physical
 - c. LLC
 - d. transport
 - e. MAC
 - f. network access
11. What method is used to manage contention-based access on a wireless network?
 - a. CSMA/CD
 - b. priority ordering
 - c. CSMA/CA
 - d. token passing

- 12.** What are two services performed by the data link layer of the OSI model? (Choose two.)
- a.** It determines the path used to forward packets.
 - b.** It accepts Layer 3 packet and encapsulates them into frames.
 - c.** It provides media access control and performs error detection.
 - d.** It monitors Layer 2 communication by building a MAC address table.
- 13.** What attribute of a NIC would place it at the data link layer of the OSI model?
- a.** attached Ethernet cable
 - b.** IP address
 - c.** MAC address
 - d.** RJ-45 port
 - e.** TCP/IP protocol stack
- 14.** Although CSMA/CD is still a feature of Ethernet, why is it no longer necessary?
- a.** IPv6 addresses are virtually unlimited.
 - b.** CSMA/CA is used.
 - c.** Layer 2 switches are capable of full-duplex.
 - d.** The development of half-duplex switches made CSMA/CD unnecessary.
 - e.** Gigabit Ethernet speeds make CSMA/CD unnecessary.

Ethernet Switching

Objectives

Upon completion of this chapter, you will be able to answer the following questions:

- How are the Ethernet sublayers related to the frame fields?
- What is an Ethernet MAC address?
- How does a switch build its MAC address table and forward frames?
- What are the available switch forwarding methods and port settings on Layer 2 switch ports?

Key Terms

This chapter uses the following key terms. You can find the definitions in the glossary at the end of the book.

contention-based access method page 237

collision fragment page 238

runt frame page 238

jumbo frame page 238

baby giant frame page 238

cyclic redundancy check (CRC) page 239

organizationally unique identifier (OUI) page 242

burned-in address (BIA) page 243

Address Resolution Protocol (ARP) page 245

Neighbor Discovery (ND) page 245

MAC address table page 249

unknown unicast page 250

store-and-forward switching page 254

cut-through switching page 255

fast-forward switching page 256

fragment-free switching page 256

automatic medium-dependent interface crossover (auto-MDIX) page 259

Introduction (7.0)

If you are planning to become a network administrator or a network architect, you definitely need to know about Ethernet and Ethernet switching. The two most prominent LAN technologies in use today are Ethernet and WLANs. Ethernet supports bandwidths of up to 100 Gbps, which explains its popularity. This chapter contains a lab in which you will use Wireshark to look at Ethernet frames and another lab where you will view network device MAC addresses. There are also some instructional videos to help you better understand Ethernet. By the time you have finished this chapter, you will be able to create a switched network that uses Ethernet!

Ethernet Frames (7.1)

Ethernet operates in the data link layer and the physical layer. It is a family of networking technologies that are defined in the IEEE 802.2 and 802.3 standards.

Ethernet Encapsulation (7.1.1)

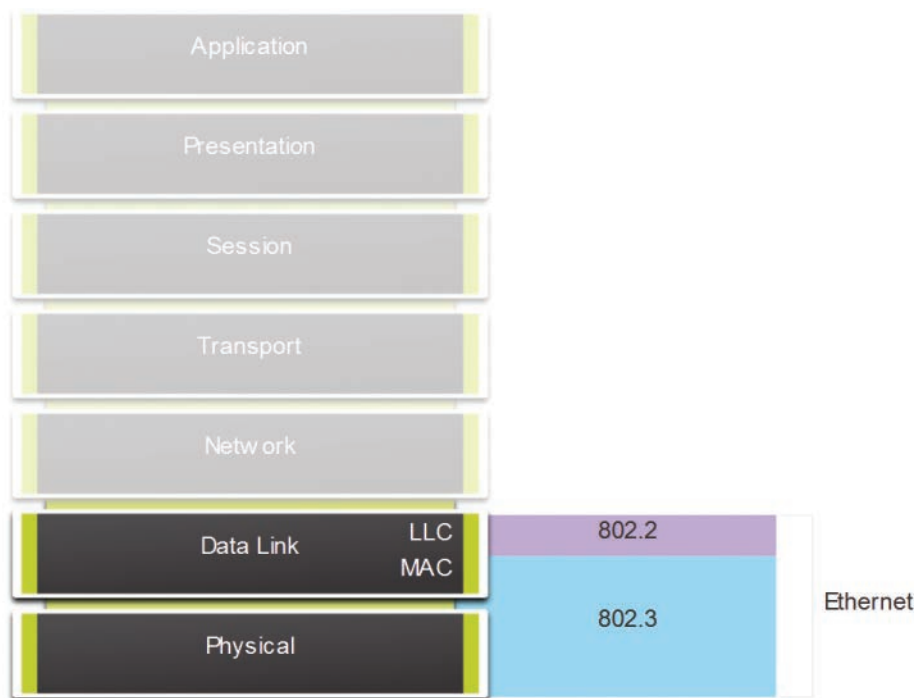
This chapter starts with a discussion of Ethernet technology, including an explanation of MAC sublayer and the Ethernet frame fields.

Two LAN technologies are used today: Ethernet and wireless LANs (WLANs). Ethernet uses wired communications, including twisted-pair, fiber-optic links, and coaxial cables.

Ethernet operates in the data link layer and the physical layer. It is a family of networking technologies defined in the IEEE 802.2 and 802.3 standards. Ethernet supports the following data bandwidths:

- 10 Mbps
- 100 Mbps
- 1000 Mbps (1 Gbps)
- 10,000 Mbps (10 Gbps)
- 40,000 Mbps (40 Gbps)
- 100,000 Mbps (100 Gbps)

As shown in Figure 7-1, Ethernet standards define both Layer 2 protocols and Layer 1 technologies.



Ethernet is defined by data link layer and physical layer protocols.

Figure 7-1 Ethernet in the OSI Model

Data Link Sublayers (7.1.2)

IEEE 802 LAN/MAN protocols, including Ethernet, use the two sublayers of the data link layer to operate: the Logical Link Control (LLC) and the Media Access Control (MAC) layers (see Figure 7-2).

Recall that the LLC and MAC sublayers have the following roles in the data link layer:

- **LLC sublayer:** This IEEE 802.2 sublayer communicates between the networking software at the upper layers and the device hardware at the lower layers. It places information in the frame to identify which network layer protocol is being used for the frame. This information allows multiple Layer 3 protocols, such as IPv4 and IPv6, to use the same network interface and media.
- **MAC sublayer:** This sublayer (specified in IEEE 802.3, 802.11, and 802.15), which is implemented in hardware, is responsible for data encapsulation and media access control. It provides data link layer addressing and is integrated with various physical layer technologies.

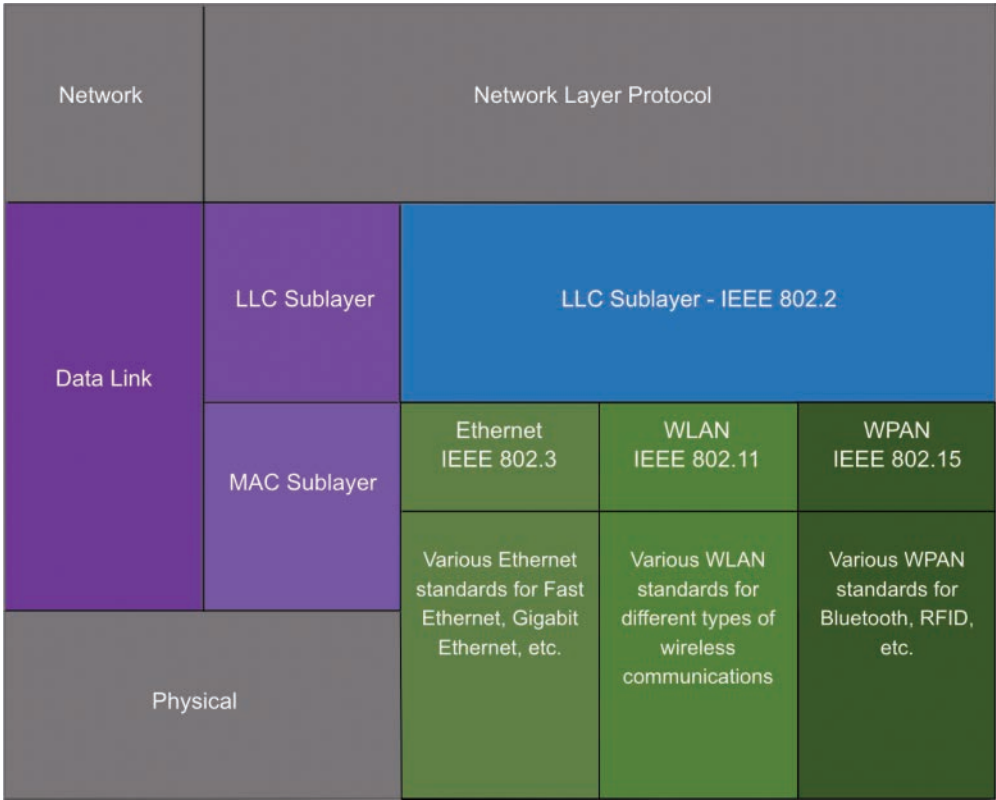


Figure 7-2 IEEE Ethernet Standards in the OSI Model

MAC Sublayer (7.1.3)

The MAC sublayer is responsible for data encapsulation and accessing the media.

Data Encapsulation

IEEE 802.3 data encapsulation includes the following:

- **Ethernet frame:** This is the internal structure of the Ethernet frame.
- **Ethernet addressing:** An Ethernet frame includes both source and destination MAC addresses to deliver the Ethernet frame from Ethernet NIC to Ethernet NIC on the same LAN.
- **Ethernet error detection:** The Ethernet frame includes a frame check sequence (FCS) trailer used for error detection.

Accessing the Media

As shown in Figure 7-3, the IEEE 802.3 MAC sublayer includes the specifications for different Ethernet communications standards over various types of media, including copper and fiber.

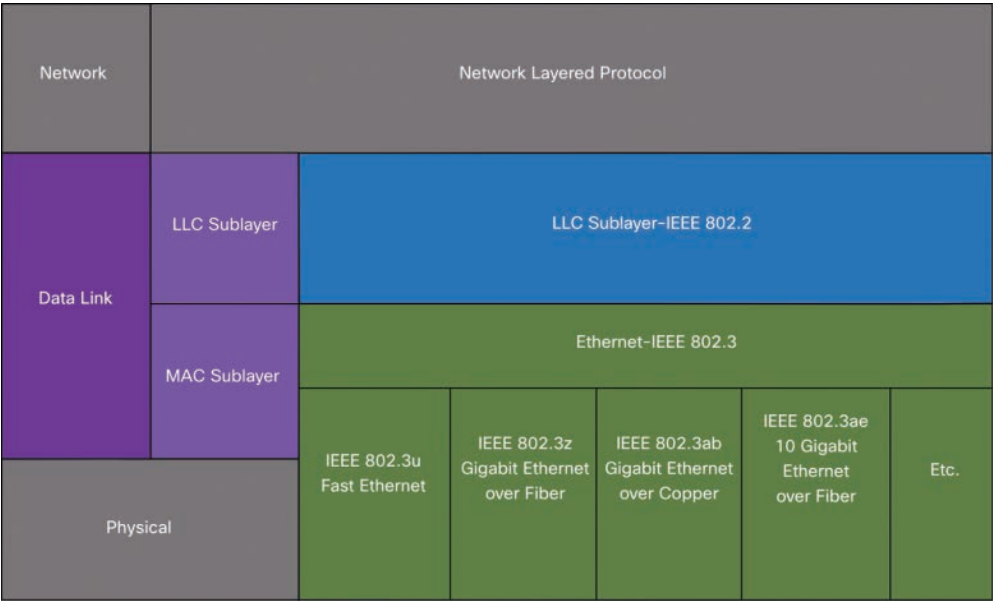


Figure 7-3 Details of the MAC Sublayer

Recall that legacy Ethernet using a bus topology or hubs is a shared, half-duplex medium. Ethernet over a half-duplex medium uses a *contention-based access method*, Carrier Sense Multiple Access/Collision Detect (CSMA/CD) to ensure that only one device is transmitting at a time. CSMA/CD allows multiple devices to share the same half-duplex medium and detects a collision when more than one device attempts to transmit simultaneously. It also provides a back-off algorithm for retransmission.

Ethernet LANs today use switches that operate in full-duplex. Full-duplex communications with Ethernet switches do not require access control through CSMA/CD.

Ethernet Frame Fields (7.1.4)

The minimum Ethernet frame size is 64 bytes, and the expected maximum is 1518 bytes. The frame size might be larger than that if additional requirements are included, such as VLAN tagging. (VLAN tagging is beyond the scope of this book.)

The frame includes all bytes from the destination MAC address field through the FCS field. The Preamble field is not included when describing the size of a frame.

Any frame less than 64 bytes in length is considered a *collision fragment* or *runt frame* and is automatically discarded by receiving stations. Frames with more than 1500 bytes of data are considered *jumbo frames* or *baby giant frames*.

If the size of a transmitted frame is less than the minimum or greater than the maximum, the receiving device drops the frame. Dropped frames are likely to result from collisions or other unwanted signals. They are considered invalid. Jumbo frames are supported by most Fast Ethernet and Gigabit Ethernet switches and NICs.

Figure 7-4 shows the fields in the Ethernet frame.

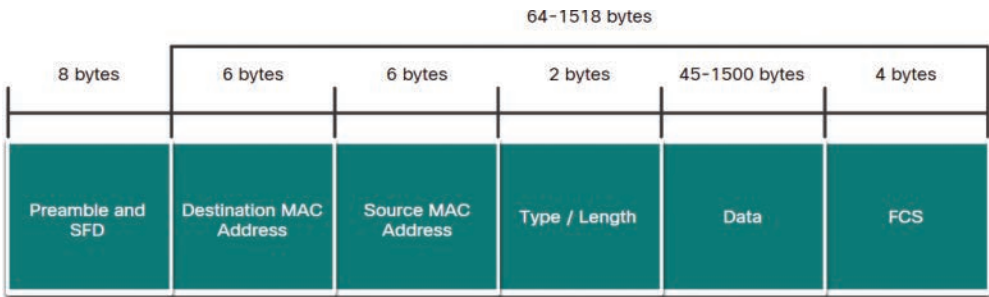


Figure 7-4 Ethernet Frame Structure and Field Size

Table 7-1 provides more information about the function of each field.

Table 7-1 Ethernet Frame Fields Detail

Field	Description
Preamble and Start Frame Delimiter fields	The preamble (7 bytes) and start frame delimiter (SFD), also called the start of frame (1 byte), fields are used for synchronization between the sending and receiving devices. These first 8 bytes of the frame are used to get the attention of the receiving nodes. Essentially, the first few bytes tell the receivers to get ready to receive a new frame.
Destination MAC Address field	This 6-byte field is the identifier for the intended recipient. Recall that Layer 2 uses this address to assist devices in determining if a frame is addressed to them. The address in a frame is compared to the MAC address in a device. If there is a match, the device accepts the frame. It can be a unicast, multicast, or broadcast address.
Source MAC Address field	This 6-byte field identifies the originating NIC or interface of the frame.

Field	Description
Type/Length field	This 2-byte field identifies the upper-layer protocol encapsulated in the Ethernet frame. Common values are, in hexadecimal, 0x800 for IPv4, 0x86DD for IPv6, and 0x806 for ARP. Note: You may also see this field referred to as EtherType, Type, or Length.
Data field	This field (which can range from 46 to 1500 bytes) contains the encapsulated data from a higher layer, which is a generic Layer 3 PDU or, more commonly, an IPv4 packet. All frames must be at least 64 bytes long. If a small packet is encapsulated, additional bits called a pad are used to increase the size of the frame to this minimum size.
Frame Check Sequence field	The frame check sequence (FCS) field (4 bytes) is used to detect errors in a frame. It uses a <i>cyclic redundancy check (CRC)</i>. The sending device includes the results of a CRC in the FCS field of the frame. The receiving device receives the frame and generates a CRC to look for errors. If the calculations match, no error occurred. Calculations that do not match indicate that the data has changed; in such a case, the frame is dropped. A change in the data could be the result of a disruption of the electrical signals that represent the bits.

Interactive Graphic

Check Your Understanding—Ethernet Switching (7.1.5)

Refer to the online course to complete this activity.



Lab—Use Wireshark to Examine Ethernet Frames (7.1.6)

In this lab, you will complete the following objectives:

- Part 1: Examine the Header Fields in an Ethernet II Frame
- Part 2: Use Wireshark to Capture and Analyze Ethernet Frames

Ethernet MAC Address (7.2)

Ethernet technology relies on MAC addresses to function. MAC addresses are used to identify the frame source and destination.

MAC Address and Hexadecimal (7.2.1)

As discussed in detail in Chapter 5, “Number Systems,” in networking, IPv4 addresses are represented using the decimal (base 10) number system and the binary (base 2) number system. IPv6 addresses and Ethernet addresses are represented using the hexadecimal (base 16) number system. To understand hexadecimal, you must first be very familiar with binary and decimal.

The hexadecimal numbering system uses the numbers 0 to 9 and the letters A to F.

An Ethernet MAC address consists of a 48-bit binary value. Hexadecimal is used to identify an Ethernet address because a single hexadecimal digit represents 4 binary bits. Therefore, a 48-bit Ethernet MAC address can be expressed using only 12 hexadecimal values.

Figure 7-5 compares the equivalent decimal and hexadecimal values for binary 0000 to 1111.

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Figure 7-5 Decimal to Binary to Hexadecimal Conversion

Given that 8 bits (1 byte) is a common binary grouping, binary 00000000 to 11111111 can be represented in hexadecimal as the range 00 to FF, as shown in the Figure 7-6.

Decimal	Binary	Hexadecimal
0	0000 0000	00
1	0000 0001	01
2	0000 0010	02
3	0000 0011	03
4	0000 0100	04
5	0000 0101	05
6	0000 0110	06
7	0000 0111	07
8	0000 1000	08
10	0000 1010	0A
15	0000 1111	0F
16	0001 0000	10
32	0010 0000	20
64	0100 0000	40
128	1000 0000	80
192	1100 0000	C0
202	1100 1010	CA
240	1111 0000	F0
255	1111 1111	FF

Figure 7-6 Selected Examples of Decimal to Binary to Hexadecimal Conversions

When using hexadecimal, leading zeros are always displayed to complete the 8-bit representation. For example, in Figure 7-6, the binary value 0000 1010 is shown to be 0A in hexadecimal.

Hexadecimal numbers are often represented by a value preceded by 0x (for example, 0x73) to distinguish between decimal and hexadecimal values in documentation.

Hexadecimal may also be represented using a subscript 16 or by using the hex number followed by an H (for example, 73H).

You might have to convert between decimal and hexadecimal values. If such conversions are required, convert the decimal or hexadecimal value to binary and then to convert the binary value to either decimal or hexadecimal as appropriate. See Chapter 5 for more information.

Ethernet MAC Address (7.2.2)

In an Ethernet LAN, every network device is connected to the same shared medium. The MAC address is used to identify the physical source and destination devices (NICs) on the local network segment. MAC addressing provides a method for device identification at the data link layer of the OSI model.

An Ethernet MAC address is a 48-bit address expressed using 12 hexadecimal digits, as shown in Figure 7-7. Because 1 byte equals 8 bits, we can also say that a MAC address is 6 bytes in length.

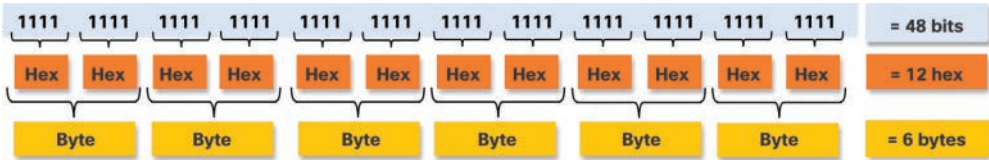


Figure 7-7 Ethernet MAC Address in Bits, Hextets, and Bytes

All MAC addresses must be unique to the Ethernet device or Ethernet interface. To ensure uniqueness, every vendor that sells Ethernet devices must register with the IEEE to obtain a unique 6-digit hexadecimal (that is, 24-bit or 3-byte) code called an *organizationally unique identifier (OUI)*.

When a vendor assigns a MAC address to a device or to an Ethernet interface, the vendor must do as follows:

- Use its assigned OUI as the first 6 hexadecimal digits.
- Assign a unique value in the last 6 hexadecimal digits.

Therefore, an Ethernet MAC address consists of a 6-digit hexadecimal vendor OUI code followed by a 6-digit hexadecimal vendor-assigned value, as shown in Figure 7-8.

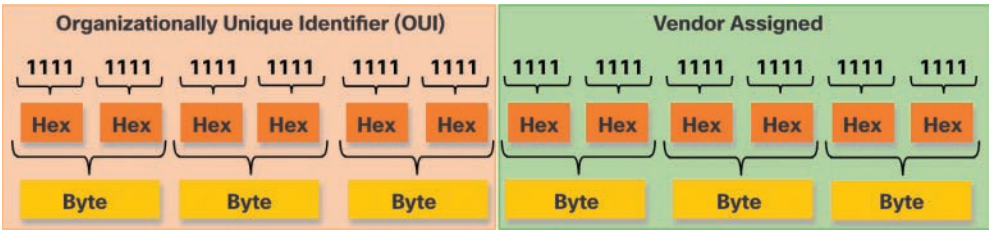


Figure 7-8 The Ethernet MAC Address Structure

For example, say that Cisco needs to assign a unique MAC address to a new device, and the IEEE has assigned Cisco the OUI 00-60-2F. Cisco would configure the device with a unique vendor code such as 3A-07-BC. Therefore, the Ethernet MAC address of that device would be 00-60-2F-3A-07-BC.

It is the responsibility of a vendor to ensure that no two of its devices are assigned the same MAC address. However, it is possible for duplicate MAC addresses to exist because of mistakes made during manufacturing, mistakes made in some virtual machine implementation methods, or modifications made using one of several

software tools. In such a case, it is necessary to modify the MAC address with a new NIC or make modifications by using software.

Frame Processing (7.2.3)

Sometimes a MAC address is referred to as a *burned-in address (BIA)* because the address is hard coded into read-only memory (ROM) on the NIC. This means that the address is permanently encoded into the ROM chip.

Note

With modern PC operating systems and NICs, it is possible to change the MAC address in software. This is useful when attempting to gain access to a network that filters based on BIA. Consequently, filtering or controlling traffic based on the MAC address is no longer as secure as it once was.

When the computer boots up, the NIC copies its MAC address from ROM into RAM. When a device is forwarding a message to an Ethernet network, as shown in Figure 7-9, the Ethernet header includes the following:

- **Source MAC address:** This is the MAC address of the source device NIC.
- **Destination MAC address:** This is the MAC address of the destination device NIC.

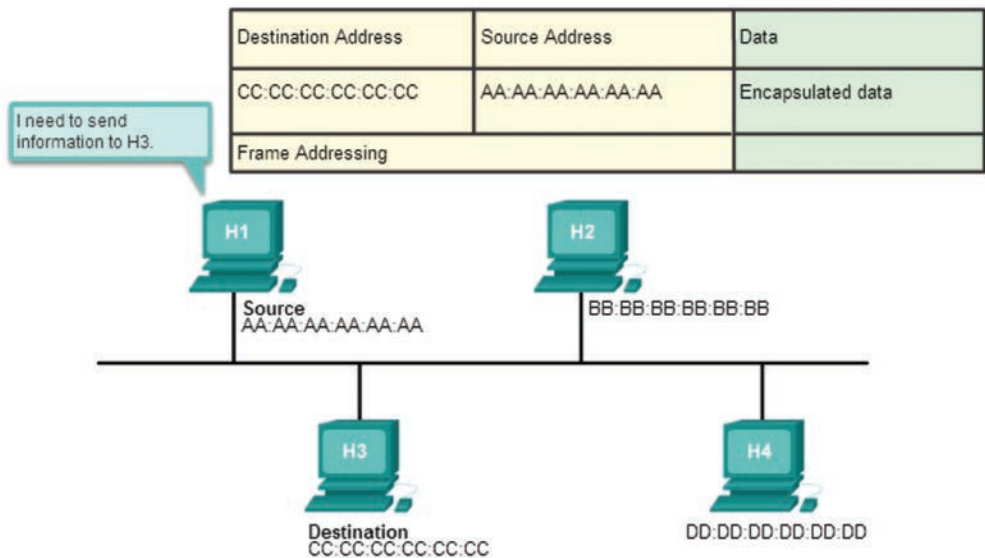


Figure 7-9 The Source Prepares a Frame to Send to the Destination

When a NIC receives an Ethernet frame, it examines the destination MAC address to see if it matches the physical MAC address that is stored in RAM. If there is no match, the device discards the frame. In Figure 7-10, H2 and H4 discard the frame. The MAC address matches for H4, so H4 passes the frame up the OSI layers, where the de-encapsulation process takes place.

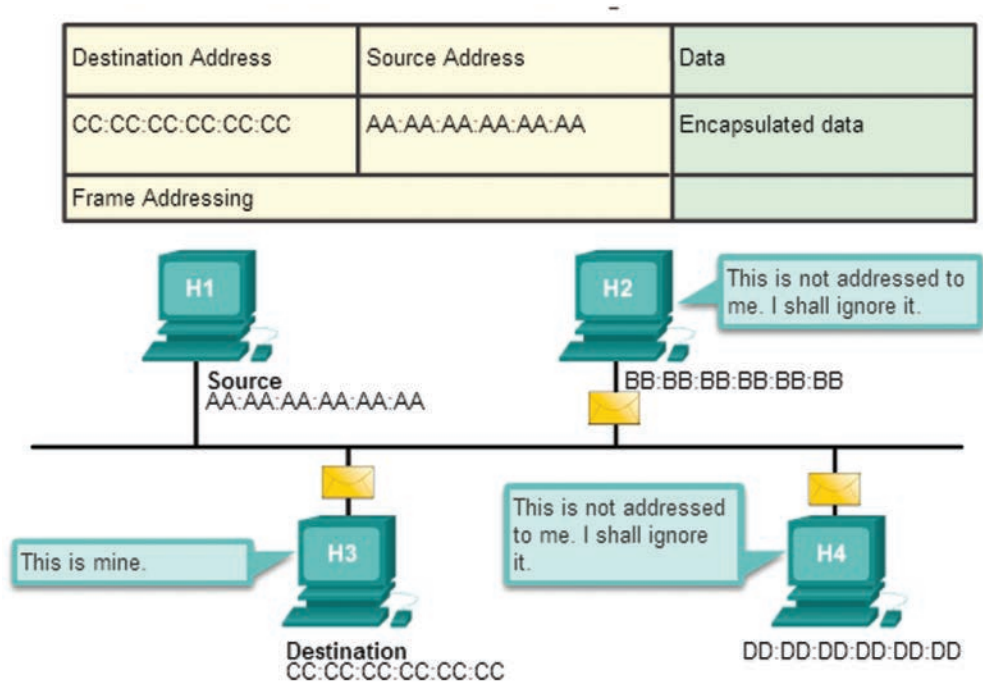


Figure 7-10 All Devices Receive the Frame, but Only the Destination Processes It

Note

Ethernet NICs also accept frames if the destination MAC address is a broadcast or a multicast group of which the host is a member.

Any device that is the source or destination of an Ethernet frame will have an Ethernet NIC and, therefore, a MAC address. This includes workstations, servers, printers, mobile devices, and routers.

Unicast MAC Address (7.2.4)

In Ethernet, different MAC addresses are used for Layer 2 unicast, broadcast, and multicast communications.

A unicast MAC address is a unique address that is used when a frame is sent from a single transmitting device to a single destination device.

In Figure 7-11, the destination MAC address and the destination IP address are both unicast.

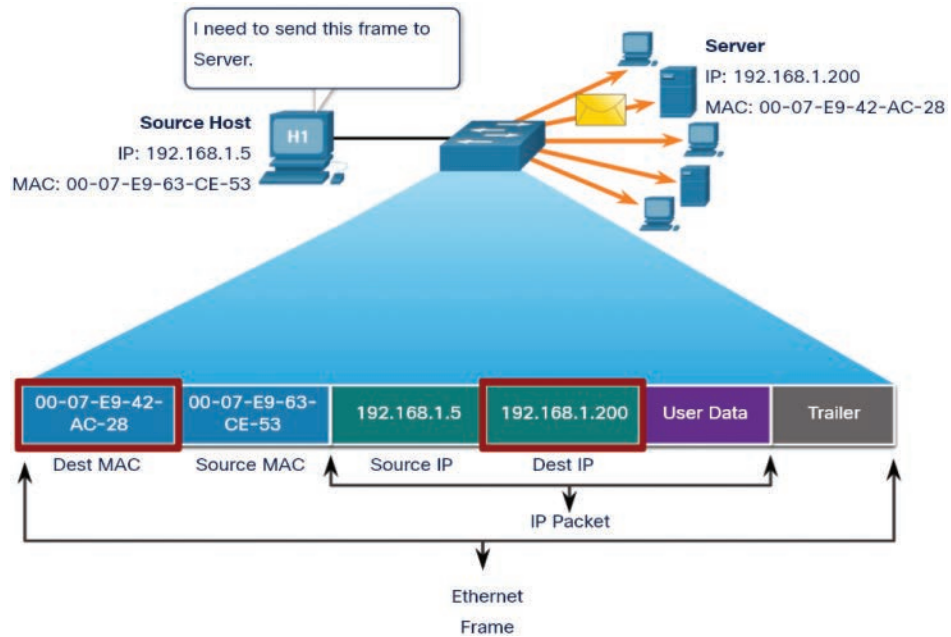


Figure 7-11 Unicast Frame Transmission

A host with IPv4 address 192.168.1.5 (source) requests a web page from the server at IPv4 unicast address 192.168.1.200. For a unicast packet to be sent and received, a destination IP address must be in the IP packet header. A corresponding destination MAC address must also be present in the Ethernet frame header. The IP address and MAC address combine to deliver data to one specific destination host.

The process that a source host uses to determine the destination MAC address associated with an IPv4 address is known as *Address Resolution Protocol (ARP)*. The process that a source host uses to determine the destination MAC address associated with an IPv6 address is known as *Neighbor Discovery (ND)*.

Note

The source MAC address must always be a unicast address.

Broadcast MAC Address (7.2.5)

An Ethernet broadcast frame is received and processed by every device on an Ethernet LAN. The features of an Ethernet broadcast are as follows:

- It has the destination MAC address FF-FF-FF-FF-FF-FF in hexadecimal (or 48 1s in binary).
- It is flooded out all Ethernet switch ports except the incoming port.
- It is not forwarded by a router.

If the encapsulated data is an IPv4 broadcast packet, this means the packet contains a destination IPv4 address that has all 1s in the host portion. This numbering in the address means that all hosts on that local network (broadcast domain) receive and process the packet.

In Figure 7-12, the destination MAC address and destination IP address are both broadcast addresses.

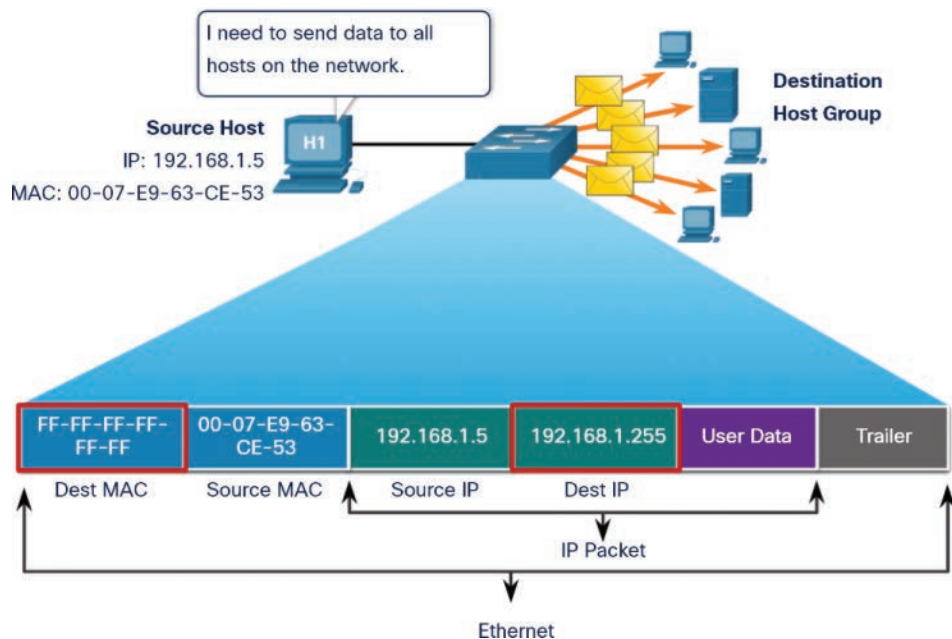


Figure 7-12 Broadcast Frame Transmission

The source host sends an IPv4 broadcast packet to all devices on its network. The IPv4 destination address is a broadcast address, 192.168.1.255. When the IPv4 broadcast packet is encapsulated in the Ethernet frame, the destination MAC address is the broadcast MAC address FF-FF-FF-FF-FF-FF in hexadecimal (or 48 1s in binary).

DHCP for IPv4 is an example of a protocol that uses Ethernet and IPv4 broadcast addresses. However, not all Ethernet broadcasts carry IPv4 broadcast packets. For example, ARP requests do not use IPv4, but the ARP message is sent as an Ethernet broadcast.

Multicast MAC Address (7.2.6)

An Ethernet multicast frame is received and processed by a group of devices on the Ethernet LAN that belong to the same multicast group. The features of an Ethernet multicast frame are as follows:

- It has destination MAC address 01-00-5E when the encapsulated data is an IPv4 multicast packet and destination MAC address 33-33 when the encapsulated data is an IPv6 multicast packet.
- There are other reserved multicast destination MAC addresses for when the encapsulated data is not IP, such as Spanning Tree Protocol (STP) and Link Layer Discovery Protocol (LLDP).
- It is flooded out all Ethernet switch ports except the incoming port, unless the switch is configured for multicast snooping.
- It is not forwarded by a router unless the router is configured to route multicast packets.

If the encapsulated data is an IP multicast packet, the devices that belong to a multicast group are assigned a multicast group IP address. The range of IPv4 multicast addresses is 224.0.0.0 to 239.255.255.255. The range of IPv6 multicast addresses begins with ff00::/8. Because a multicast address represents a group of addresses (sometimes called a host group), it can only be used as the destination of a packet. The source is always a unicast address.

As with the unicast and broadcast addresses, a multicast IP address requires a corresponding multicast MAC address to deliver frames on a local network. The multicast MAC address is associated with, and uses addressing information from, the IPv4 or IPv6 multicast address.

In Figure 7-13, the destination MAC address and destination IP address are both multicast addresses.

Routing protocols and other network protocols use multicast addressing. Applications such as video and imaging software may also use multicast addressing, although multicast applications are not as common.

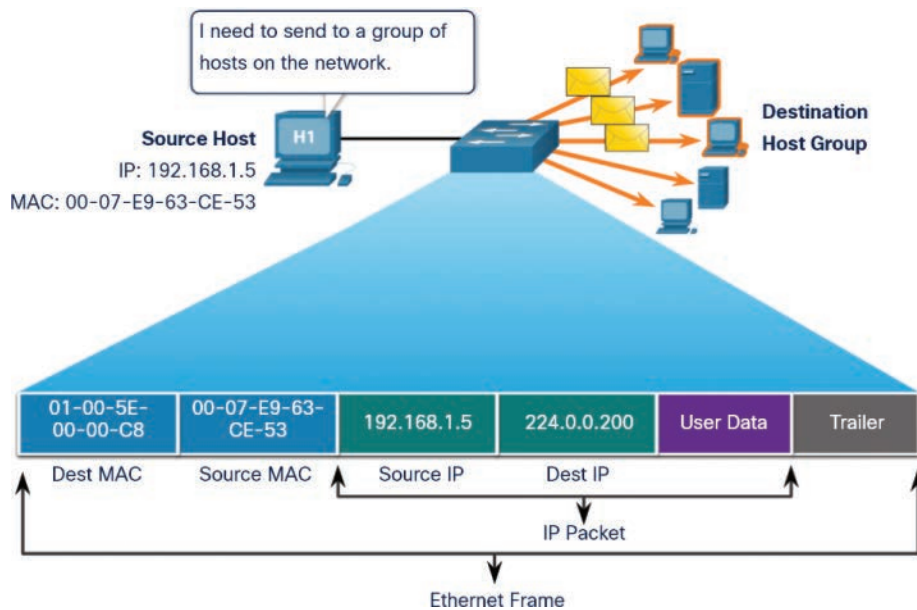


Figure 7-13 Multicast Frame Transmission



Lab—View Network Device MAC Addresses (7.2.7)

In this lab, you will complete the following objectives:

- Part 1: Set Up the Topology and Initialize Devices
- Part 2: Configure Devices and Verify Connectivity
- Part 3: Display, Describe, and Analyze Ethernet MAC Addresses

The MAC Address Table (7.3)

Compared to legacy Ethernet hubs, Ethernet switches improve efficiency and overall network performance. Although traditionally most LAN switches have operated at Layer 2 of the OSI model, an increasing number of Layer 3 switches are now being implemented. This section focuses on Layer 2 switches. Layer 3 switches are beyond the scope of this book.

Switch Fundamentals (7.3.1)

Now that you know all about Ethernet MAC addresses, it is time to talk about how a switch uses these addresses to forward (or discard) frames to other devices

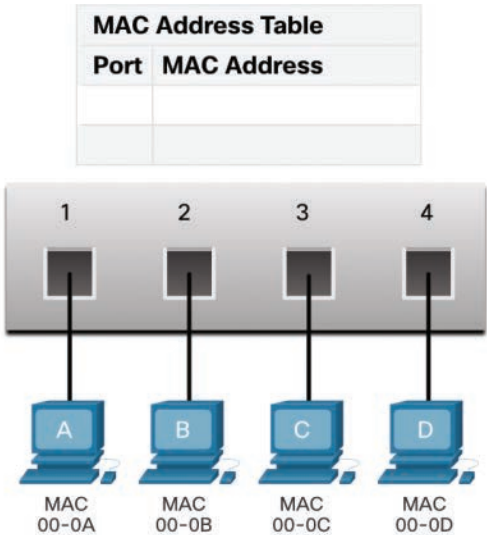
on a network. If a switch just forwarded every frame it received out all ports, your network would be so congested that it would probably come to a complete halt.

A Layer 2 Ethernet switch uses Layer 2 MAC addresses to make forwarding decisions. It is completely unaware of the data (protocol) being carried in the data portion of the frame, such as an IPv4 packet, an ARP message, or an IPv6 ND packet. The switch makes its forwarding decisions based solely on the Layer 2 Ethernet MAC addresses.

An Ethernet switch examines its *MAC address table* to make a forwarding decision for each frame. In contrast, a legacy Ethernet hub repeats bits out all ports except the incoming port. In Figure 7-14, the four-port switch was just powered on. The table shows the MAC address table, which has not yet learned the MAC addresses for the four attached PCs.

Note

MAC addresses are shortened throughout this section for demonstration purposes.



The switch MAC address table is empty.

Figure 7-14 Switch Powers Up with an Empty MAC Address Table

Note

The MAC address table is sometimes referred to as a content-addressable memory (CAM) table. While the term CAM table is fairly common, for the purposes of this course, we refer to it as a MAC address table.

Switch Learning and Forwarding (7.3.2)

A switch dynamically builds its MAC address table by examining the source MAC addresses of the frames received on a port. The switch forwards frames by searching for a match between the destination MAC address in a frame and an entry in the MAC address table.

Examine the Source MAC Address

Every frame that enters a switch is checked for new information to learn. It does this by examining the source MAC address of the frame and the port number where the frame entered the switch. If the source MAC address does not exist, it is added to the table, along with the incoming port number. If the source MAC address does exist, the switch updates the refresh timer for that entry. By default, most Ethernet switches keep an entry in the table for 5 minutes.

In Figure 7-15, for example, PC-A is sending an Ethernet frame to PC-D. The table shows that the switch adds the MAC address for PC-A to the MAC address table.

Note

If the source MAC address exists in the table but on a different port, the switch treats this as a new entry. The entry is replaced using the same MAC address but with the more current port number.

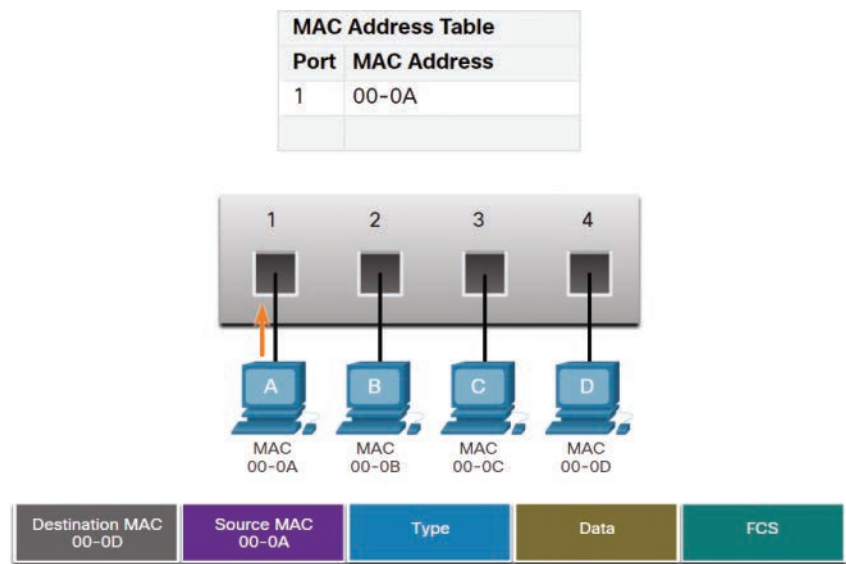
Find the Destination MAC Address

If the destination MAC address is a unicast address, the switch looks for a match between the destination MAC address of the frame and an entry in its MAC address table. If the destination MAC address is in the table, the switch forwards the frame out the specified port. If the destination MAC address is not in the table, the switch forwards the frame out all ports except the incoming port. This is called an *unknown unicast*.

As shown in Figure 7-16, the switch does not have the destination MAC address in its table for PC-D, so it sends the frame out all ports except port 1.

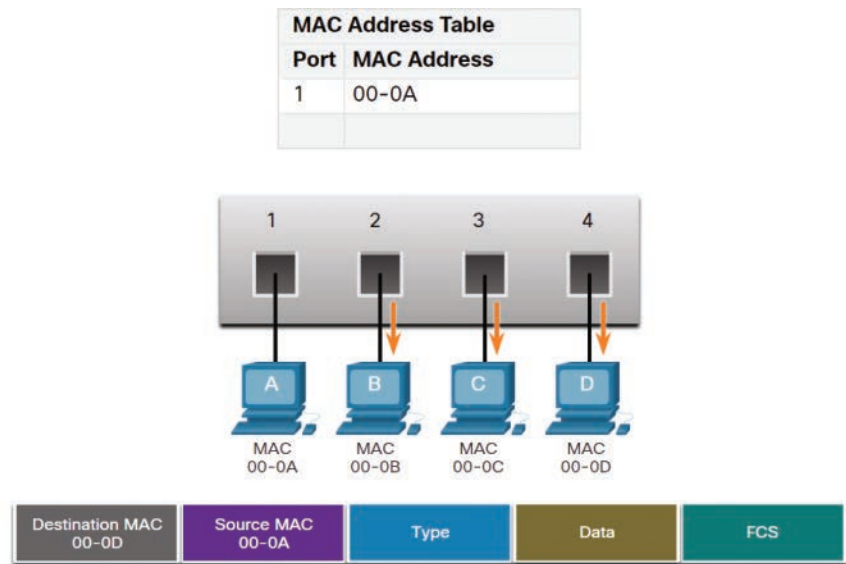
Note

If the destination MAC address is a broadcast or a multicast address, the frame is flooded out all ports except the incoming port.



- 1. PC-A sends an Ethernet frame.
- 2. The switch adds the port number and MAC address for PC-A to the MAC Address Table.

Figure 7-15 Switch Learns the MAC Address for PC-A



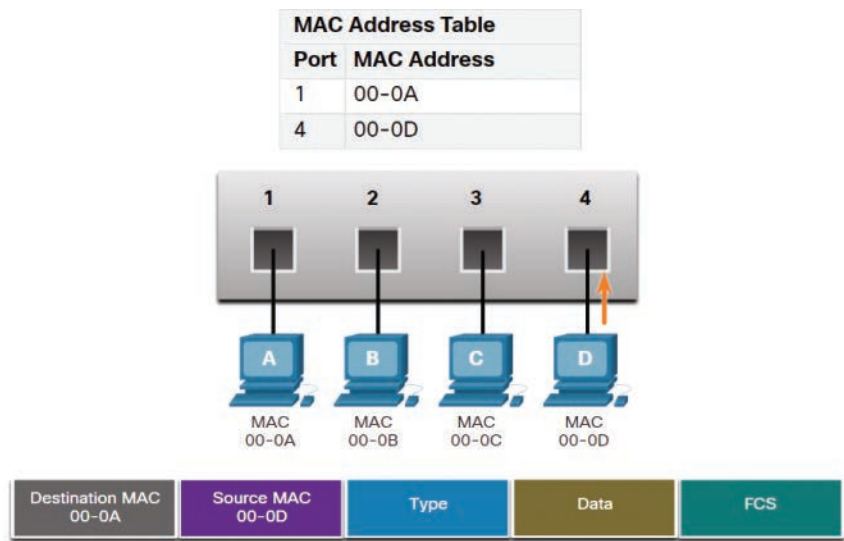
- 1. The destination MAC address is not in the table.
- 2. The switch forwards the frame out all other ports.

Figure 7-16 Switch Forwards the Frame Out All Other Ports

Filtering Frames (7.3.3)

As a switch receives frames from different devices, it is able to populate its MAC address table by examining the source MAC address of every frame. When the MAC address table of the switch contains the destination MAC address, the switch is able to filter the frame and forward out a single port.

In Figure 7-17, PC-D is replying to PC-A. The switch sees the MAC address of PC-D in the incoming frame on port 4. The switch then puts the MAC address of PC-D into the MAC address table associated with port 4.



The switch adds the port number and MAC address for PC-D to its MAC address table.

Figure 7-17 Switch Learns the MAC Address for PC-D

Next, because the switch has the destination MAC address for PC-A in the MAC address table, it sends the frame only out port 1, as shown in Figure 7-18.

Next, PC-A sends another frame to PC-D, as shown in Figure 7-19. The MAC address table already contains the MAC address for PC-A; therefore, the 5-minute refresh timer for that entry is reset. Next, because the switch table contains the destination MAC address for PC-D, it sends the frame out only port 4.



Video—MAC Address Tables on Connected Switches (7.3.4)

A switch can have multiple MAC addresses associated with a single port. This is common when the switch is connected to another switch. The switch will have a separate MAC address table entry for each frame received with a different source MAC address.

Refer to the online course to view this video.

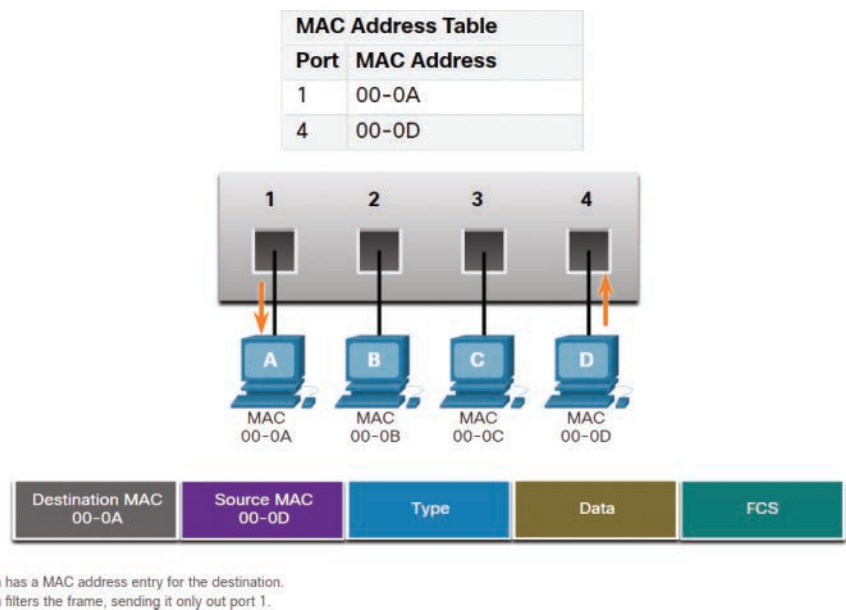


Figure 7-18 Switch Forwards the Frame Out the Port Belonging to PC-A

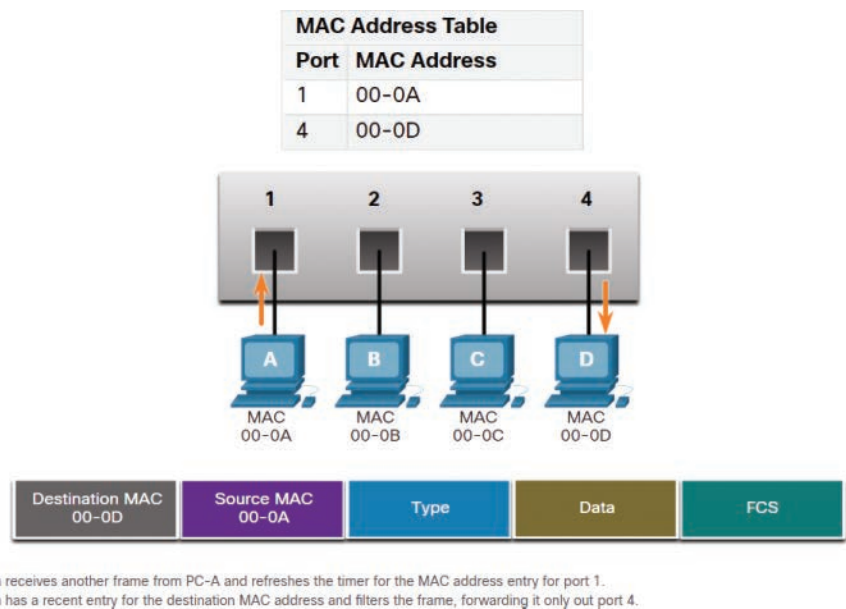


Figure 7-19 Switch Forwards the Frame Out the Port Belonging to PC-D

Video**Video—Sending the Frame to the Default Gateway (7.3.5)**

When a device has an IP address that is on a remote network, the Ethernet frame cannot be sent directly to the destination device. Instead, the Ethernet frame is sent to the MAC address of the default gateway, the router.

Refer to the online course to view this video.

Interactive Graphic**Activity—Switch It! (7.3.6)**

Use this activity to check your understanding of how a switch learns and forwards frames.

Refer to the online course to complete this activity.

**Lab—View the Switch MAC Address Table (7.3.7)**

In this lab, you will complete the following objectives:

- Part 1: Build and Configure the Network
 - Part 2: Examine the Switch MAC Address Table
-

Switch Speeds and Forwarding Methods (7.4)

Switches may have the capability to implement various forwarding methods to increase performance in a network.

Frame Forwarding Methods on Cisco Switches (7.4.1)

As you learned in the previous section, a switch uses its MAC address table to determine which port to use to forward frames. With Cisco switches, there are actually two frame forwarding methods, and there are good reasons to use one instead of the other, depending on the situation.

Switches use one of the following forwarding methods for switching data between network ports:

- ***Store-and-forward switching***: With this frame forwarding method, the switch receives the entire frame and computes the CRC. The switch uses a mathematical formula, based on the number of bits (1s) in the frame, to determine whether the received frame has an error. If the CRC is valid, the switch looks up the destination address, which determines the outgoing interface. Then the frame is forwarded out the correct port.

- **Cut-through switching:** With this frame forwarding method, the switch forwards the frame before it is entirely received. At a minimum, the destination address of the frame must be read before the frame can be forwarded.

A big advantage of store-and-forward switching is that the switch determines whether a frame has errors before propagating the frame. When an error is detected in a frame, the switch discards the frame. Discarding frames with errors reduces the amount of bandwidth consumed by corrupt data. Store-and-forward switching is required for quality of service (QoS) analysis on converged networks where frame classification for traffic prioritization is necessary. For example, voice over IP (VoIP) data streams need to have priority over web-browsing traffic.

Figure 7-20 shows the store-and-forward process.

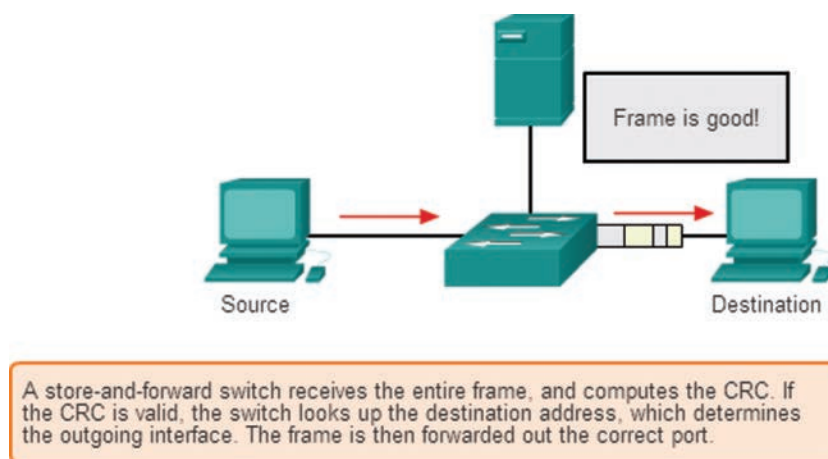
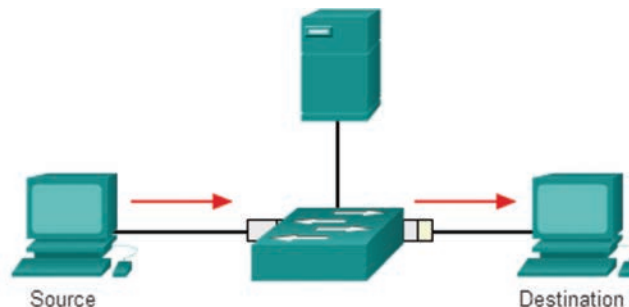


Figure 7-20 Store-and-Forward Switching

Cut-Through Switching (7.4.2)

In cut-through switching, the switch acts on the data as soon as it is received, even if the transmission is not complete. The switch buffers just enough of the frame to read the destination MAC address so that it can determine which port to use to forward the data. The destination MAC address is located in the first 6 bytes of the frame, following the preamble. The switch looks up the destination MAC address in its switching table, determines the outgoing interface port, and forwards the frame on to its destination through the designated switch port. The switch does not perform any error checking on the frame.

Figure 7-21 shows the cut-through switching process.



A cut-through switch forwards the frame before it is entirely received. At a minimum, the destination address of the frame must be read before the frame can be forwarded.

Figure 7-21 Cut-Through Switching

There are two variants of cut-through switching:

- **Fast-forward switching:** Fast-forward switching offers the lowest level of latency. With fast-forward switching, the switch immediately forwards a packet after reading the destination address. Because with fast-forward switching the switch starts forwarding before the entire packet has been received, there may be times when packets are relayed with errors. This occurs infrequently, and the destination NIC discards the faulty packet upon receipt. In fast-forward mode, latency is measured from the first bit received to the first bit transmitted. Fast-forward switching is the typical cut-through method of switching.
- **Fragment-free switching:** In fragment-free switching, the switch stores the first 64 bytes of the frame before forwarding. Fragment-free switching can be viewed as a compromise between store-and-forward switching and fast-forward switching. The reason the switch stores only the first 64 bytes of the frame is that most network errors and collisions occur during the first 64 bytes. Fragment-free switching tries to enhance fast-forward switching by performing a small error check on the first 64 bytes of the frame to ensure that a collision has not occurred before forwarding the frame. Fragment-free switching is a compromise between the high latency and high integrity of store-and-forward switching and the low latency and reduced integrity of fast-forward switching.

Some switches are configured to perform cut-through switching on a per-port basis until a user-defined error threshold is reached, and then they automatically change to store-and-forward. When the error rate falls below the threshold, the port automatically changes back to cut-through switching.

Memory Buffering on Switches (7.4.3)

An Ethernet switch may use a buffering technique to store frames before forwarding them. Buffering may also be used when the destination port is busy due to congestion. The switch stores the frame until it can be transmitted.

As shown in Table 7-2, there are two methods of memory buffering.

Table 7-2 Memory Buffering Methods

Method	Description
Port-based memory buffering	Frames are stored in queues that are linked to specific incoming and outgoing ports.
	A frame is transmitted to the outgoing port only when all the frames ahead in the queue have been successfully transmitted.
	It is possible for a single frame to delay the transmission of all the frames in memory because a destination port is busy. This delay occurs even if the other frames could be transmitted to open destination ports.
Shared memory buffering	All frames are deposited into a common memory buffer shared by all switch ports, and the amount of buffer memory required by a port is dynamically allocated.
	The frames in the buffer are dynamically linked to the destination port, enabling a packet to be received on one port and then transmitted on another port, without moving it to a different queue.

Shared memory buffering results in the ability to store larger frames with potentially fewer dropped frames. This is important with asymmetric switching, which allows for different data rates on different ports, such as when connecting a server to a 10 Gbps switch port and PCs to 1 Gbps ports.

Duplex and Speed Settings (7.4.4)

Two of the most basic settings on a switch are the bandwidth (sometimes referred to as *speed*) and duplex settings for each individual switch port. It is critical that the duplex and bandwidth settings match between the switch port and the connected devices, such as computers or other switches.

Two types of duplex settings are used for communications on an Ethernet network:

- **Full-duplex:** Both ends of the connection can send and receive simultaneously.
- **Half-duplex:** Only one end of the connection can send at a time.

Autonegotiation is an optional function on most Ethernet switches and NICs. It enables two devices to automatically negotiate the best speed and duplex capabilities. Full-duplex is chosen if both devices have the capability, along with their highest common bandwidth.

In Figure 7-22, the Ethernet NIC for PC-A can operate in full-duplex or half-duplex and at 10 Mbps or 100 Mbps. PC-A is connected to switch S1 on port 1, which can operate in full-duplex or half-duplex and at 10 Mbps, 100 Mbps, or 1000 Mbps (1 Gbps). If both devices are using autonegotiation, the operating mode is full-duplex, at 100 Mbps.

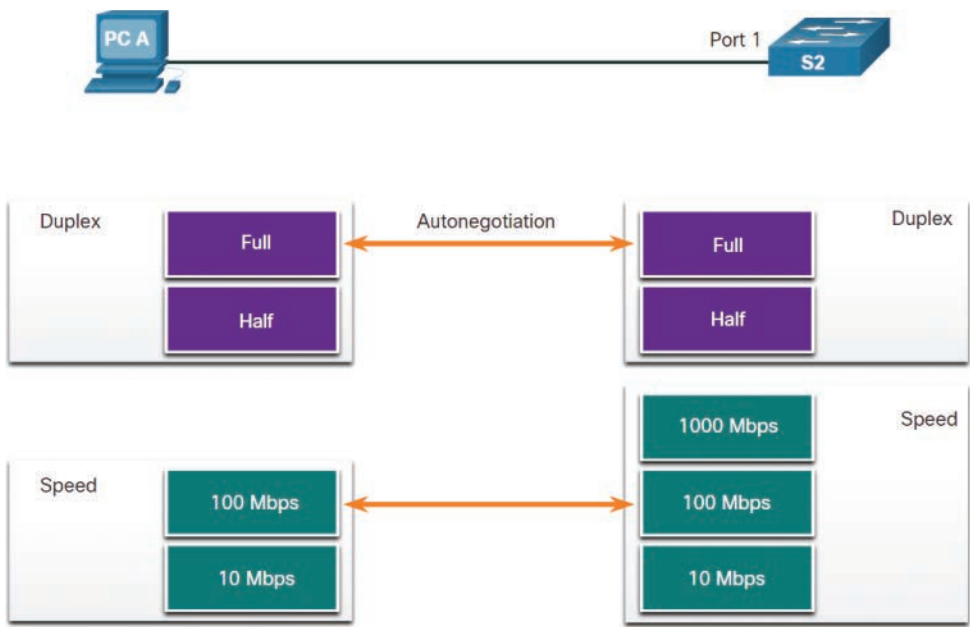


Figure 7-22 Duplex and Speed Settings

Note

Most Cisco switches and Ethernet NICs default to autonegotiation for speed and duplexing. Gigabit Ethernet ports operate only in full-duplex.

Duplex mismatch is one of the most common causes of performance issues on 10/100 Mbps Ethernet links. It occurs when one port on the link operates at half-duplex while the other port operates at full-duplex, as shown in Figure 7-23. In this scenario, S2 will continually experience collisions because S1 keeps sending frames any time it has something to send.



Figure 7-23 Duplex Mismatch

Duplex mismatch occurs when one or both ports on a link are reset, and the autonegotiation process does not result in the two link partners having the same configuration. It also can occur when users reconfigure one side of a link and forget to reconfigure the other. Both sides of a link should have autonegotiation on, or both sides should have it off. Best practice is to configure both Ethernet switch ports as full-duplex.

Auto-MDIX (7.4.5)

At one time, connections between devices required the use of either a crossover cable or a straight-through cable. The type of cable required depended on the type of interconnecting devices. For example, Figure 7-24 identifies the correct cable types required to interconnect a switch to a switch, a switch to a router, a switch to a host, or a router to a host. A crossover cable is used for connecting like devices, and a straight-through cable is used for connecting unlike devices.

Note

A direct connection between a router and a host requires a crossover connection.

Most switch devices now support the *automatic medium-dependent interface crossover (auto-MDIX)* feature. When this feature is enabled, the switch automatically detects the type of cable attached to the port and configures the interfaces accordingly. Therefore, you can use either a crossover cable or a straight-through cable for connections to a copper 10/100/1000 port on a switch, regardless of the type of device on the other end of the connection.

The auto-MDIX feature is enabled by default on switches running Cisco IOS Release 12.2(18)SE or later. However, the feature can be disabled. For this reason, you should always use the correct cable type and should not rely on the auto-MDIX feature. Auto-MDIX can be re-enabled using the `mdix auto` interface configuration command.

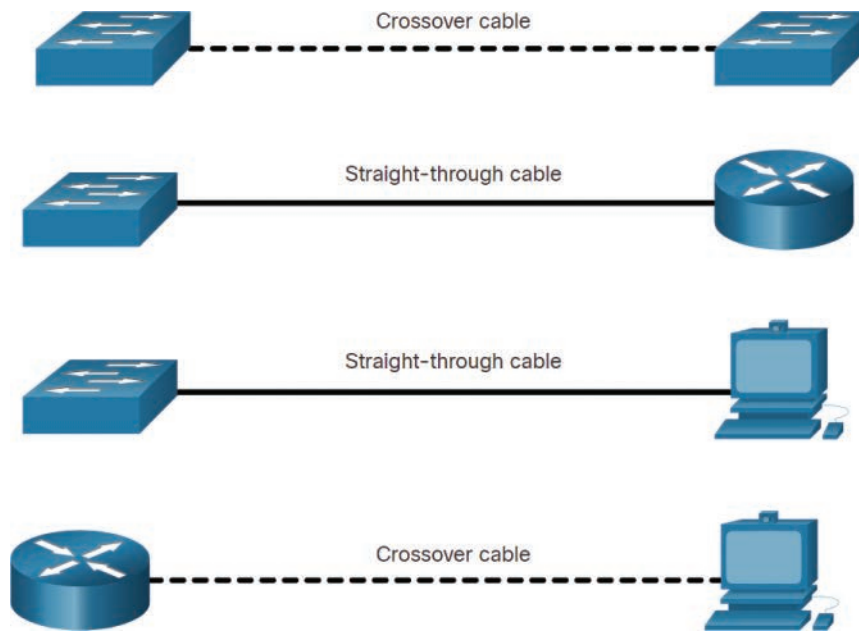


Figure 7-24 Cable Types

**Interactive
Graphic**

Check Your Understanding—Switch Speeds and Forwarding Methods (7.4.6)

Refer to the online course to complete this activity.

Summary (7.5)

The following is a summary of the topics in the chapter and their corresponding online modules.

Ethernet Frame

Ethernet operates at the data link layer and the physical layer. Ethernet standards define both the Layer 2 protocols and the Layer 1 technologies. Ethernet operates at the LLC and MAC sublayers of the data link layer. Data encapsulation includes the following: Ethernet frame, Ethernet addressing, and Ethernet error detection. Ethernet LANs use switches that operate in full-duplex. The Ethernet frame fields are Preamble and Start Frame Delimiter, Destination MAC Address, Source MAC Address, EtherType, Data, and FCS.

Ethernet MAC Address

The binary number system uses the digits 0 and 1. Decimal uses 0 through 9. Hexadecimal uses 0 through 9 and the letters A through F. The MAC address is used to identify the physical source and destination devices (NICs) on the local network segment. MAC addressing provides a method for device identification at the data link layer of the OSI model. An Ethernet MAC address is a 48-bit address expressed using 12 hexadecimal digits, or 6 bytes. An Ethernet MAC address consists of a 6-digit hexadecimal vendor OUI code followed by a 6-digit hexadecimal vendor-assigned value. When a device is forwarding a message to an Ethernet network, the Ethernet header includes the source and destination MAC addresses. In Ethernet, different MAC addresses are used for Layer 2 unicast, broadcast, and multicast communications.

The MAC Address Table

A Layer 2 Ethernet switch makes forwarding decisions based solely on Layer 2 Ethernet MAC addresses. The switch dynamically builds its MAC address table by examining the source MAC addresses of the frames received on a port. The switch forwards frames by searching for a match between the destination MAC address in the frame and an entry in the MAC address table. As a switch receives frames from different devices, it is able to populate its MAC address table by examining the source MAC address of each frame. When the MAC address table of the switch contains the destination MAC address, the switch is able to filter the frame and forward it out a single port.

Switch Speeds and Forwarding Methods

Switches use one of two forwarding methods for switching data between network ports: store-and-forward switching or cut-through switching. Two variants of cut-through switching are fast-forward and fragment-free switching. Two methods of memory buffering are port-based memory buffering and shared memory buffering. Two types of duplex settings are used for communications on an Ethernet network: full-duplex and half-duplex. Autonegotiation is an optional function on most Ethernet switches and NICs. It enables two devices to automatically negotiate the best speed and duplex capabilities. Full-duplex is chosen if both devices have the capability, and their highest common bandwidth is chosen. Most switch devices now support the automatic medium-dependent interface crossover (auto-MDIX) feature. When this feature is enabled, the switch automatically detects the type of cable attached to the port and configures the interfaces accordingly.

Practice

The following activities provide practice with the topics introduced in this chapter. The lab is available in the companion *Introduction to Networks Labs & Study Guide (CCNAv7)* (ISBN 9780136634454). The Packet Tracer activity instructions are also provided in the *Labs & Study Guide*. The PKA files are available in the online course.



Labs

Lab 7.1.6: Use Wireshark to Examine Ethernet Frames

Lab 7.2.7: View Network Device MAC Addresses

Lab 7.3.7: View the Switch MAC Address Table

Check Your Understanding Questions

Complete all the review questions listed here to test your understanding of the topics and concepts in this chapter. The appendix “Answers to ‘Check Your Understanding’ Questions” lists the answers.

1. Which network device makes forwarding decisions based only on the destination MAC address that is contained in a frame?
 - a. repeater
 - b. hub
 - c. Layer 2 switch
 - d. router
2. For which network device is the primary function to send data to a specific destination based on the information found in the MAC address table?
 - a. hub
 - b. router
 - c. Layer 2 switch
 - d. modem
3. What does the LLC sublayer do?
 - a. It performs data encapsulation.
 - b. It communicates with upper protocol layers.
 - c. It is responsible for media access control.
 - d. It adds a header and trailer to a packet to form an OSI Layer 2 PDU.
4. Which statement is true about MAC addresses?
 - a. MAC addresses are implemented by software.
 - b. A NIC needs a MAC address only if it is connected to a WAN.
 - c. The first 3 bytes are used by the vendor-assigned OUI.
 - d. The ISO is responsible for MAC address regulations.
5. What happens to a runt frame received by a Cisco Ethernet switch?
 - a. The frame is dropped.
 - b. The frame is returned to the originating network device.
 - c. The frame is broadcast to all other devices on the same network.
 - d. The frame is sent to the default gateway.
6. What are the minimum and maximum sizes of an Ethernet frame? (Choose two.)
 - a. 56 bytes
 - b. 64 bytes
 - c. 128 bytes
 - d. 1024 bytes
 - e. 1518 bytes

7. What addressing information does a switch record in order to build its MAC address table?
 - a. the destination Layer 3 addresses of incoming packets
 - b. the destination Layer 2 addresses of outgoing frames
 - c. the source Layer 3 addresses of outgoing frames
 - d. the source Layer 2 addresses of incoming frames
8. Which two characteristics describe Ethernet technology? (Choose two.)
 - a. It is supported by IEEE 802.3 standards.
 - b. It is supported by IEEE 802.5 standards.
 - c. It typically uses an average of 16 Mbps for data transfer.
 - d. It uses unique MAC addresses to ensure that data is sent to and processed by the appropriate destination.
 - e. It uses a ring topology.
9. What statement describes MAC addresses?
 - a. They are globally unique.
 - b. They are routable only within the private network.
 - c. They are added as part of a Layer 3 PDU.
 - d. They have 32-bit binary values.
10. What is the special value assigned to the first 24 bits of a multicast MAC address?
 - a. 01-5E-00
 - b. FF-00-5E
 - c. FF-FF-FF
 - d. 01-00-5E
11. What will a host on an Ethernet network do if it receives a frame with a destination MAC address that does not match its own MAC address?
 - a. It will discard the frame.
 - b. It will forward the frame to the next host.
 - c. It will remove the frame from the media.
 - d. It will strip off the data link frame to check the destination IP address.

- 12.** What is auto-MDIX?
- a.** a type of Cisco switch
 - b.** an Ethernet connector type
 - c.** a feature that automatically determines speed and duplex
 - d.** a feature that detects Ethernet cable type
- 13.** Which two functions or operations are performed by the MAC sublayer? (Choose two.)
- a.** It is responsible for media access control.
 - b.** It performs the function for NIC driver software.
 - c.** It adds a header and trailer to form an OSI Layer 2 PDU.
 - d.** It handles communication between upper and lower layers.
 - e.** It adds control information to the network protocol header.
- 14.** What type of address is 01-00-5E-0A-00-02?
- a.** an address that reaches every host inside a local subnet
 - b.** an address that reaches one specific host
 - c.** an address that reaches every host in the network
 - d.** an address that reaches a specific group of hosts

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Network Layer

Objectives

Upon completion of this chapter, you will be able to answer the following questions:

- How does the network layer use IP protocols for reliable communications?
- What is the role of the major header fields in the IPv4 packet?
- What is the role of the major header fields in the IPv6 packet?
- How do network devices use routing tables to direct packets to a destination network?
- What is the function of fields in the routing table of a router?

Key Terms

This chapter uses the following key terms. You can find the definitions in the glossary at the end of the book.

routing page 269

connectionless page 271

best effort page 271

media independent page 271

maximum transmission unit (MTU) page 274

fragmentation page 274

Internet Control Message Protocol (ICMP) page 275

Network Address Translation (NAT) page 277

loopback interface page 281

default gateway page 282

directly connected networks page 286

remote networks page 286

default route page 286

static route page 287

dynamic routing protocol page 288

Introduction (8.0)

By now you might have noticed that the modules in this course—and the chapters in this book—are progressing from the bottom up through the OSI model layers. This chapter focuses on the network layer of the OSI model, which is where communication protocols and routing protocols operate. Say you want to send an email to a friend who lives in another city—or even another country. This person is not on the same network as you. A simple switched network cannot get your message any further than the end of that network. You need some help to keep the message moving along the path to your friend's end device. To send an email (a video, or a file, and so on) to anyone who is not on your local network, you must have access to routers. To access routers, you must use network layer protocols. To help you visualize these processes, this module contains two Wireshark activities. Enjoy!

Network Layer Characteristics (8.1)

This section introduces the protocols and functions of the network layer. The function of the network layer is to facilitate the transport of data from one network to another. This section introduces the elementary functions of the network layer.

The Network Layer (8.1.1)

The network layer, or OSI Layer 3, provides services that allow end devices to exchange data across networks. As shown in Figure 8-1, IP version 4 (IPv4) and IP version 6 (IPv6) are the principal network layer communication protocols. Other network layer protocols include routing protocols such as Open Shortest Path First (OSPF) and messaging protocols such as Internet Control Message Protocol (ICMP).

To accomplish end-to-end communication across network boundaries, network layer protocols perform four basic operations:

- **Addressing of end devices:** End devices must be configured with unique IP addresses for identification on the network.
- **Encapsulation:** The network layer encapsulates the protocol data unit (PDU) from the transport layer into a packet. The encapsulation process adds IP header information, such as the IP address of the source (sending) and destination (receiving) hosts. The encapsulation process is performed by the source of the IP packet.

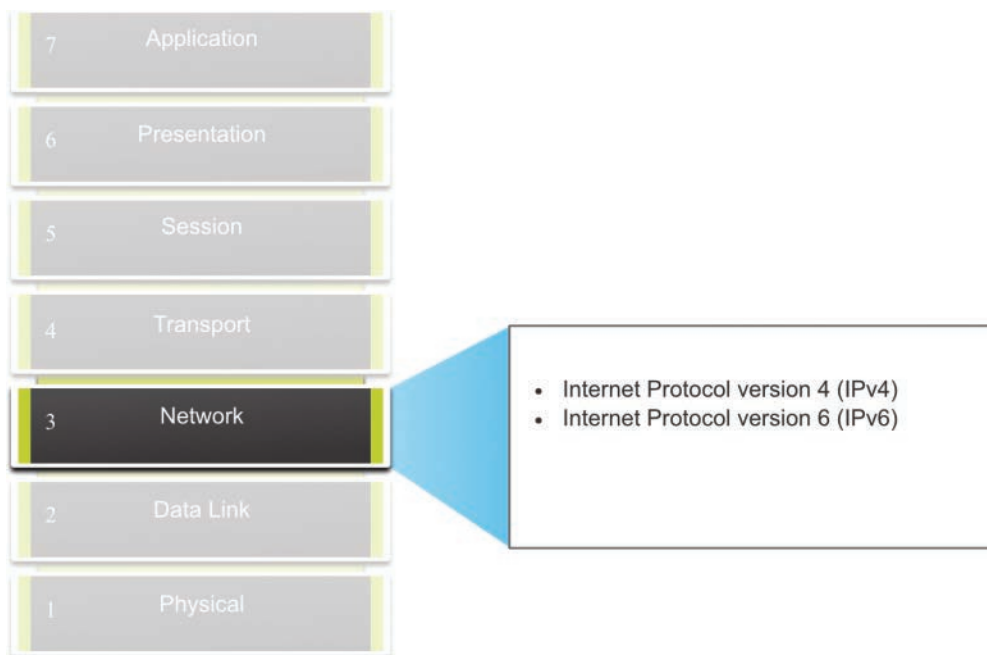


Figure 8-1 Network Layer of the OSI Model

- **Routing:** The network layer provides services to direct the packets to a destination host on another network. To travel to other networks, the packet must be processed by a router. The role of the router is to select the best path and direct packets toward the destination host in a process known as *routing*. A packet may cross many routers before reaching the destination host. Each router a packet crosses to reach the destination host is called a *hop*.
- **De-encapsulation:** When a packet arrives at the network layer of the destination host, the host checks the IP header of the packet. If the destination IP address within the header matches its own IP address, the IP header is removed from the packet. After the packet is de-encapsulated by the network layer, the resulting Layer 4 PDU is passed up to the appropriate service at the transport layer. The de-encapsulation process is performed by the destination host of the IP packet.

Unlike the transport layer (OSI Layer 4), which manages the data transport between the processes running on each host, network layer communication protocols (that is, IPv4 and IPv6) specify the packet structure and processing used to carry the data from one host to another host. Operating without regard to the data carried in each packet allows the network layer to carry packets for multiple types of communications between multiple hosts.

IP Encapsulation (8.1.2)

IP encapsulates the segment or other data from the transport layer (the layer just above the network layer) by adding an IP header. The IP header is used to deliver a packet to the destination host.

Figure 8-2 illustrates how the transport layer PDU is encapsulated by the network layer PDU to create an IP packet.

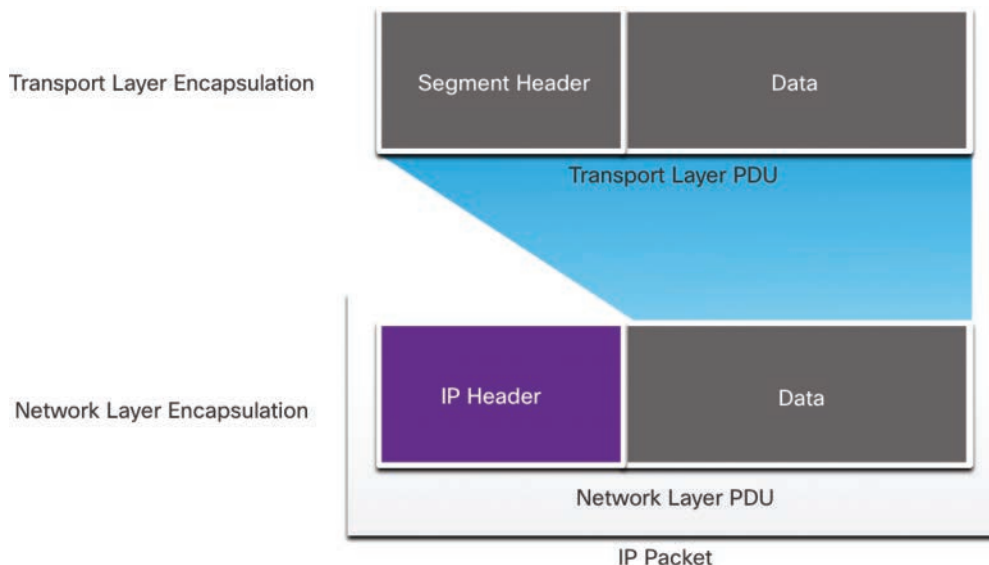


Figure 8-2 Transport Layer PDU Encapsulated in the Network Layer

The process of encapsulating data layer by layer enables the services at the different layers to develop and scale without affecting the other layers. This means the transport layer segments can be readily packaged by IPv4 or IPv6 or by any new protocol that might be developed in the future.

The IP header is examined by Layer 3 devices (that is, routers and Layer 3 switches) as it travels across a network to its destination. It is important to note that the IP addressing information remains the same from the time the packet leaves the source host until it arrives at the destination host, except when translated by a device performing Network Address Translation (NAT) for IPv4.

Note

NAT is discussed in Chapters 8 and 12.

Routers implement routing protocols to route packets between networks. The packet forwarding performed by these intermediary devices involves examining the network layer addressing in the packet header. In all cases, the data portion of the packet—that is, the encapsulated transport layer PDU or other data—remains unchanged during the network layer processes.

Characteristics of IP (8.1.3)

IP was designed as a protocol with low overhead. It provides only the functions that are necessary to deliver a packet from a source to a destination over an interconnected system of networks. The protocol was not designed to track and manage the flow of packets. These functions, if required, are performed by other protocols at other layers, primarily TCP at Layer 4.

These are the basic characteristics of IP:

- **Connectionless:** There is no connection with the destination established before sending data packets.
- **Best effort:** IP is inherently unreliable because packet delivery is not guaranteed.
- **Media independent:** Operation is independent of the medium (that is, copper, optical fiber, or wireless) carrying the data.

Connectionless (8.1.4)

IP is connectionless, meaning that no dedicated end-to-end connection is created by IP before data is sent. Connectionless communication is conceptually similar to sending a letter to someone without notifying the recipient in advance. Figure 8-3 illustrates this key point.



Figure 8-3 Letter Analogy of Connectionless Communication

Connectionless data communications work on the same principle. As shown in Figure 8-4, IP requires no initial exchange of control information to establish an end-to-end connection before packets are forwarded.

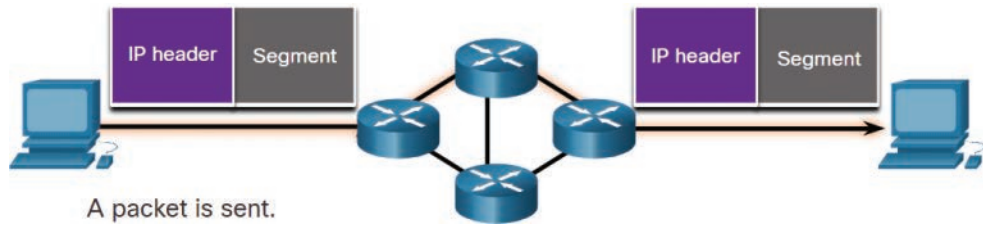


Figure 8-4 IP is Connectionless

Best Effort (8.1.5)

IP does not require additional fields in the header to maintain an established connection. This process greatly reduces the overhead of IP. However, with no pre-established end-to-end connection, senders are unaware whether destination devices are present and functional when sending packets; they also are not aware of whether the destination receives a packet or whether a destination device is able to access and read a packet.

IP does not guarantee that all packets that are sent are, in fact, received. Figure 8-5 illustrates the unreliable, or best-effort delivery, characteristic of IP. As an unreliable network layer protocol, IP does not guarantee that all sent packets will be received. Other protocols manage the process of tracking packets and ensuring their delivery.

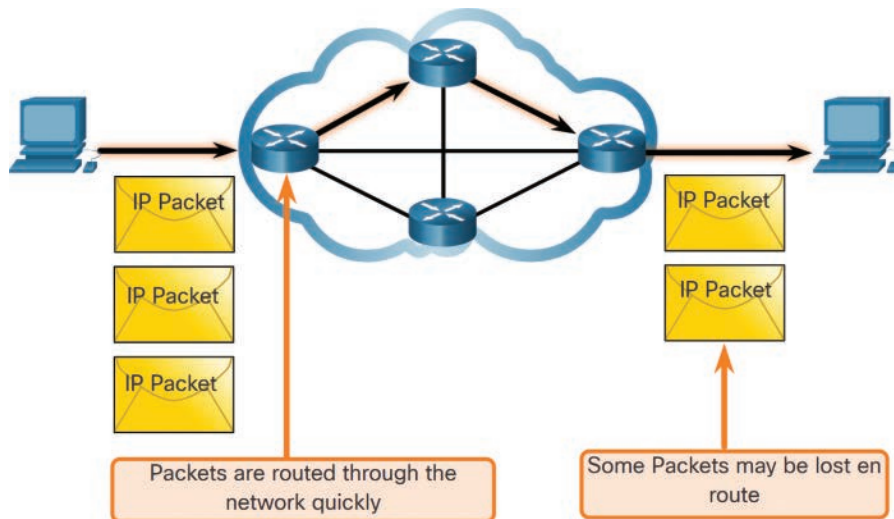


Figure 8-5 Best-Effort Delivery

Media Independent (8.1.6)

Unreliable means that IP does not have the capability to manage and recover from undelivered or corrupt packets. This is because while IP packets are sent with information about the location of delivery, they do not contain information that can be processed to inform the sender about whether delivery was successful. Packets may arrive at the destination corrupted, out of sequence, or not at all. IP provides no capability for packet retransmissions if errors occur.

If out-of-order packets are delivered, or if packets are missing, then applications using the data, or upper-layer services, must resolve these issues. This allows IP to function very efficiently. In the TCP/IP protocol suite, reliability is the role of TCP at the transport layer.

IP operates independently of the media that carry the data at lower layers of the protocol stack. As shown in Figure 8-6, IP packets can be communicated as electronic signals over copper cable, as optical signals over fiber, or wirelessly as radio signals.

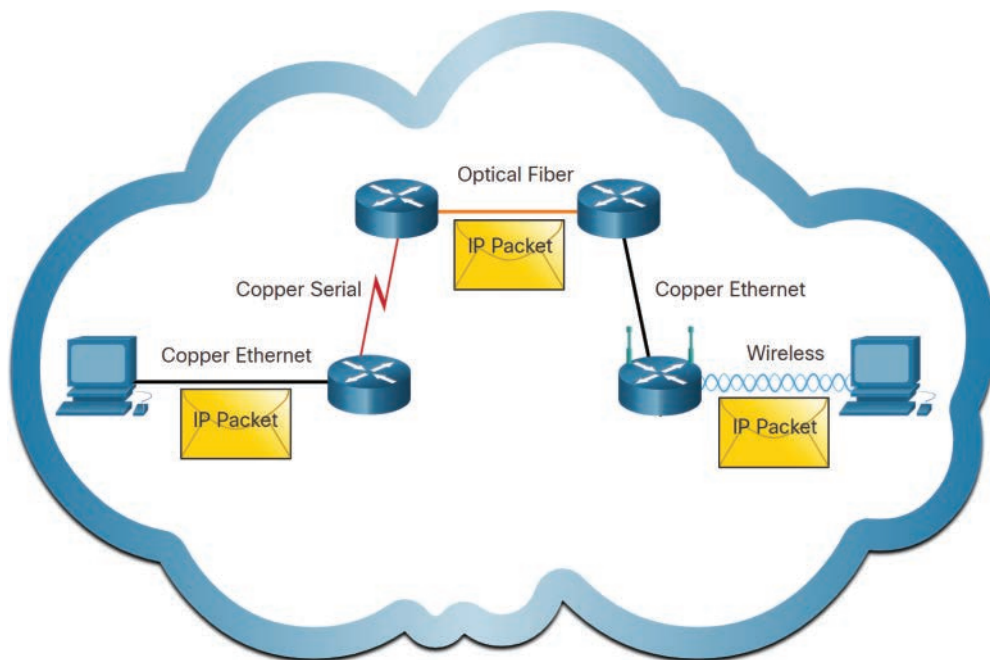


Figure 8-6 IP Packets Cross Multiple Media Types

The OSI data link layer is responsible for preparing an IP packet for transmission over the communications medium. This means that the delivery of IP packets is not limited to any particular medium.

There is, however, one major characteristic of the media that the network layer considers: the maximum size of the PDU that each medium can transport. This characteristic is referred to as the *maximum transmission unit (MTU)*. Part of the control communication between the data link layer and the network layer is the establishment of a maximum size for a packet. The data link layer passes the MTU value up to the network layer. The network layer then determines how large packets can be.

In some cases, an intermediate device, usually a router, must split up an IPv4 packet when forwarding it from one medium to another medium with a smaller MTU. This process is called *fragmenting the packet*, or *fragmentation*. Fragmentation causes latency. IPv6 packets cannot be fragmented by the router.

**Interactive
Graphic****Check Your Understanding—IP Characteristics (8.1.7)**

Refer to the online course to complete this activity.

IPv4 Packet (8.2)

The ability to provide the end-to-end transfer of data by the network layer is based on the content and interpretation of the Layer 3 header. This section examines the structure and contents of the IPv4 header.

IPv4 Packet Header (8.2.1)

IPv4 is one of the primary network layer communication protocols. The IPv4 header of a packet is used to ensure that this packet is delivered to its next stop on the way to its destination end device.

An IPv4 packet header consists of fields containing important information about the packet. These fields contain binary numbers that are examined by the Layer 3 process.

IPv4 Packet Header Fields (8.2.2)

The binary values of each IPv4 packet header field identify various settings of the IP packet. Protocol header diagrams, which are read left to right and top to bottom, provide visuals to refer to when discussing protocol fields. The IP protocol header diagram in Figure 8-7 identifies the fields of an IPv4 packet.

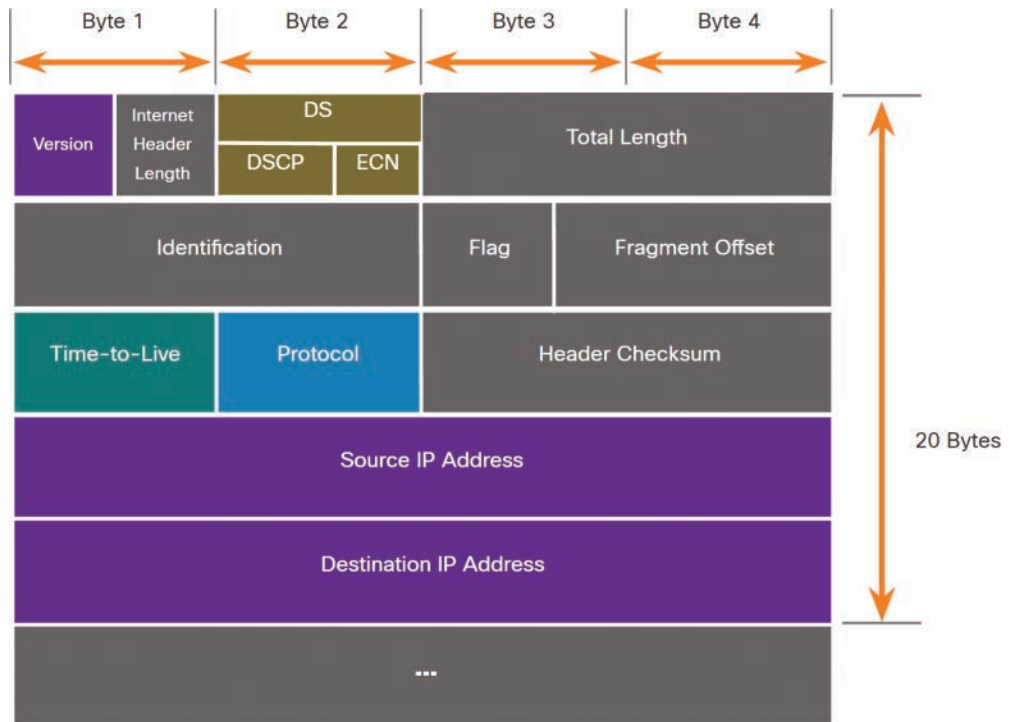


Figure 8-7 IPv4 Packet Header Fields

Significant fields in the IPv4 header include the following:

- **Version:** This field contains a 4-bit binary value set to 0100 that identifies this as an IPv4 packet.
- **Differentiated Services, or DiffServ (DS):** Formerly called the Type of Service (ToS) field, the DS field is an 8-bit field used to determine the priority of each packet. The 6 most significant bits of the DiffServ field are the differentiated services code point (DSCP) bits, and the last 2 bits are the explicit congestion notification (ECN) bits.
- **Header Checksum:** This field is used to detect corruption in the IPv4 header.
- **Time-to-Live (TTL):** The TTL field contains an 8-bit binary value that is used to limit the lifetime of a packet. The source device of the IPv4 packet sets the initial TTL value. It is decreased by 1 each time the packet is processed by a router. If the TTL field decrements to 0, the router discards the packet and sends an *Internet Control Message Protocol (ICMP)* Time Exceeded message to the source IP address. Because the router decrements the TTL of each packet, the router must also recalculate the header checksum.

- **Protocol:** This field is used to identify the next-level protocol. This 8-bit binary value indicates the data payload type that the packet is carrying, which enables the network layer to pass the data to the appropriate upper-layer protocol. Common values include ICMP (1), TCP (6), and UDP (17).
- **Source IPv4 Address:** This field contains a 32-bit binary value that represents the source IPv4 address of the packet. The source IPv4 address is always a unicast address.
- **Destination IPv4 Address:** This field contains a 32-bit binary value that represents the destination IPv4 address of the packet. The destination IPv4 address is a unicast, multicast, or broadcast address.

The two most commonly referenced fields are the Source IPv4 Address and Destination IPv4 Address fields. These fields identify where the packet is coming from and where it is going. Typically, these addresses do not change while a packet is traveling from the source to the destination.

The Internet Header Length (IHL), Total Length, and Header Checksum fields are used to identify and validate a packet.

Other fields are used to reorder a fragmented packet. Specifically, the IPv4 packet uses Identification, Flags, and Fragment Offset fields to keep track of the fragments. A router may have to fragment an IPv4 packet when forwarding it from one medium to another with a smaller MTU.

The Options and Padding fields are rarely used and are beyond the scope of this chapter.

Video

Video—Sample IPv4 Headers in Wireshark (8.2.3)

Refer to the online course to view this video.

Interactive Graphic

Check Your Understanding—IPv4 Packet (8.2.4)

Refer to the online course to complete this activity.

IPv6 Packet (8.3)

This section introduces the successor of IPv4: IPv6.

Limitations of IPv4 (8.3.1)

IPv4 is still in use today, but IPv6 will eventually replace IPv4. To better understand why you need to know about IPv6, it helps to know the limitations of IPv4 and the advantages of IPv6.

Through the years, a number of protocols and processes have been developed to address new challenges. However, even with all those changes, IPv4 still faces three major issues:

- **IPv4 address depletion:** IPv4 has a limited number of unique public addresses available. Although there are approximately 4 billion IPv4 addresses, the increasing number of new IP-enabled devices and always-on connections, as well as the potential growth in less-developed regions have increased the need for more addresses.
- **Lack of end-to-end connectivity:** *Network Address Translation (NAT)* is a technology commonly implemented within IPv4 networks. NAT provides a way for multiple devices to share a single public IPv4 address. However, because the public IPv4 address is shared, the IPv4 address of an internal network host is hidden. This can be problematic for technologies that require end-to-end connectivity.
- **Increased network complexity:** While NAT has extended the life span of IPv4, it was only meant as a transition mechanism to IPv6. NAT in its various implementations creates additional complexity in the network, creating latency and making troubleshooting more difficult.

IPv6 Overview (8.3.2)

In the early 1990s, the Internet Engineering Task Force (IETF) grew concerned about the issues with IPv4 and began to look for a replacement. This activity led to the development of IP version 6 (IPv6). IPv6 overcomes the limitations of IPv4 and is a powerful enhancement with features that better suit current and foreseeable network demands.

IPv6 provides improvements such as the following:

- **Increased address space:** IPv6 addresses are based on 128-bit hierarchical addressing, whereas IPv4 addresses have 32 bits.
- **Improved packet handling:** The IPv6 header has been simplified and has fewer fields.
- **Eliminates the need for NAT:** Thanks to the large number of public IPv6 addresses, NAT between a private IPv4 address and a public IPv4 address is not needed. This avoids some of the NAT-induced problems experienced by applications that require end-to-end connectivity.

The 32-bit IPv4 address space provides approximately 4,294,967,296 unique addresses. IPv6 address space provides 340,282,366,920,938,463,374,607,431,768,211,456, or 340 undecillion addresses. This is roughly equivalent to every grain of sand on Earth.

Figure 8-8 compares the IPv4 address space and the IPv6 address space.

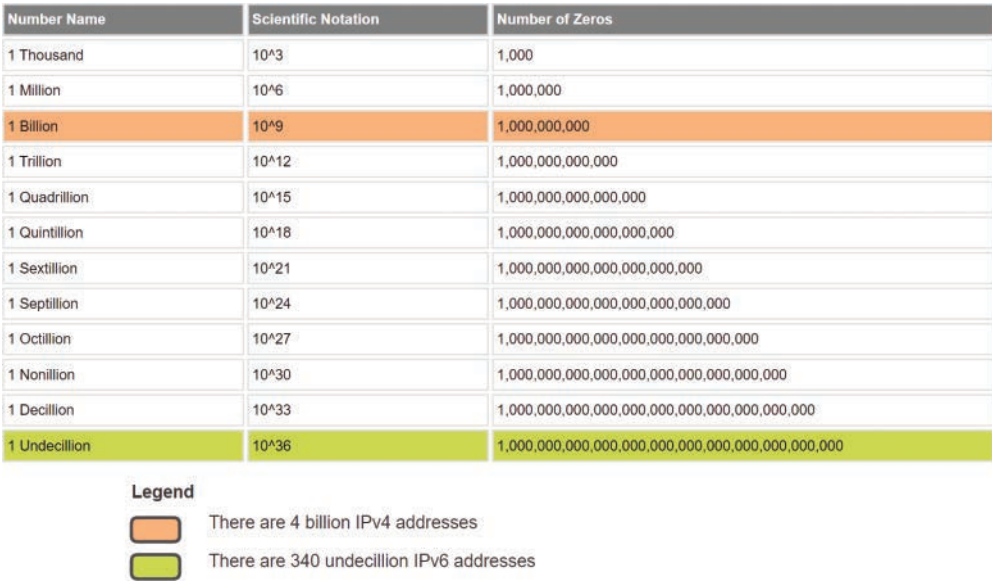


Figure 8-8 IPv4 and IPv6 Address Space

IPv4 Packet Header Fields in the IPv6 Packet Header (8.3.3)

One of the major design improvements of IPv6 over IPv4 is the simplified IPv6 header.

The IPv4 header consists of a variable-length header of 20 octets (up to 60 bytes if the Options field is used) and 12 basic header fields, not including the Options field and the Padding field.

For IPv6, some fields have remained the same, some fields have changed names and positions, and some IPv4 fields are no longer required, as highlighted in Figure 8-9.

The simplified IPv6 header shown in Figure 8-10 consists of a fixed-length header of 40 octets (largely due to the length of the source and destination IPv6 addresses).

The simplified IPv6 headers can be processed more efficiently than IPv4 headers.

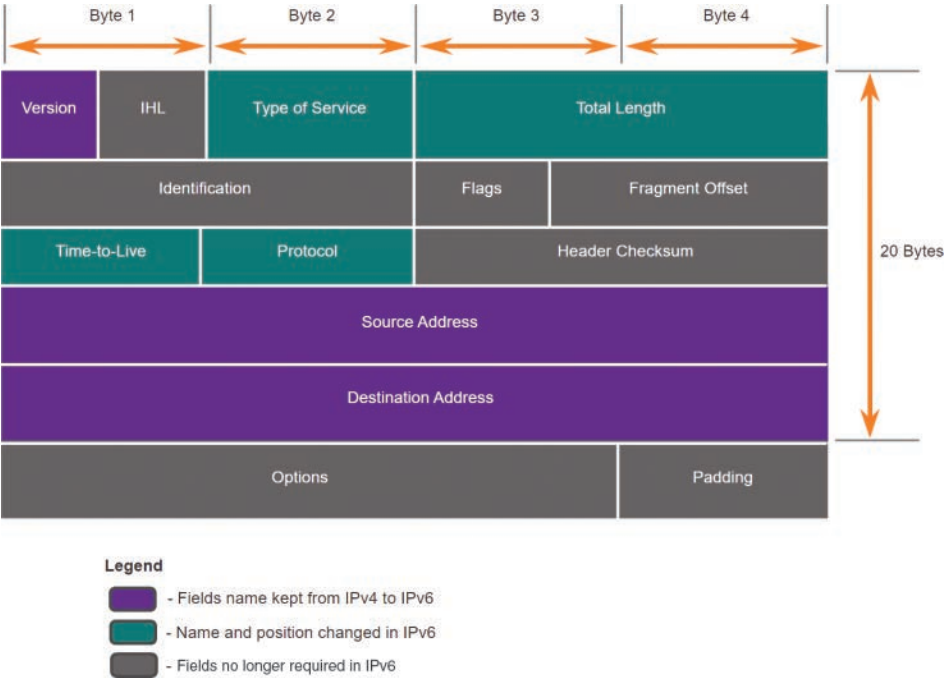


Figure 8-9 IPv4 Fields Kept, Changed, or Removed

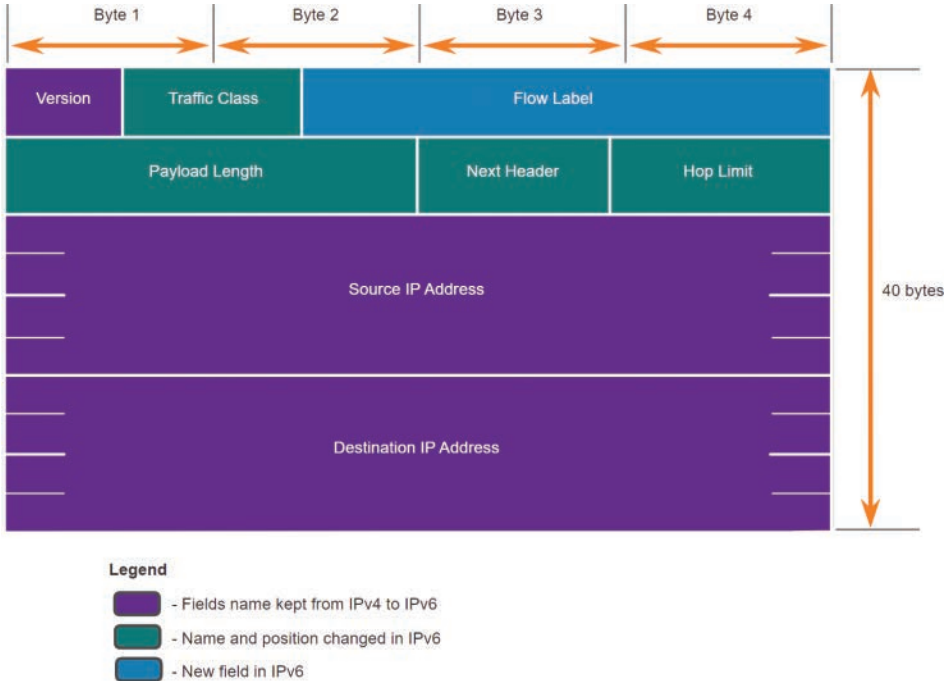


Figure 8-10 IPv6 Packet Header Fields

IPv6 Packet Header (8.3.4)

The fields for the IPv6 packet header, as shown in Figure 8-10, are as follows:

- **Version:** This field contains a 4-bit binary value set to 0110 that identifies this as an IP version 6 packet.
- **Traffic Class:** This 8-bit field is equivalent to the IPv4 Differentiated Services (DS) field.
- **Flow Label:** This 20-bit field suggests that all packets with the same flow label receive the same type of handling by routers.
- **Payload Length:** This 16-bit field indicates the length of the data portion, or payload, of the IPv6 packet. This does not include the length of the IPv6 header, which is a fixed 40-byte header.
- **Next Header:** This 8-bit field is equivalent to the IPv4 Protocol field. It indicates the data payload type that the packet is carrying, enabling the network layer to pass the data to the appropriate upper-layer protocol.
- **Hop Limit:** This 8-bit field replaces the IPv4 TTL field. This value is decremented by 1 each time a router forwards the packet. When the counter reaches 0, the packet is discarded, and an ICMPv6 Time Exceeded message is forwarded to the sending host. This message indicates that the packet did not reach its destination because the hop limit was exceeded. Unlike IPv4, IPv6 does not include an IPv6 Header Checksum field, because this function is performed at both the lower and upper layers. This means the checksum does not need to be recalculated by each router when it decrements the Hop Limit field, which also improves network performance.
- **Source IPv6 Address:** This 128-bit field identifies the IPv6 address of the sending host.
- **Destination IPv6 Address:** This 128-bit field identifies the IPv6 address of the receiving host.

An IPv6 packet may also contain extension headers (EHs), which provide optional network layer information. Extension headers are optional and are placed between the IPv6 header and the payload. EHs are used for fragmentation, security, to support mobility, and more.

Unlike with IPv4, routers do not fragment routed IPv6 packets.

Video

Video—Sample IPv6 Headers in Wireshark (8.3.5)

Refer to the online course to view this video.

**Interactive
Graphic**
Check Your Understanding—IPv6 Packet (8.3.6)

Refer to the online course to complete this activity.

How a Host Routes (8.4)

Hosts need to communicate with hosts that might be on networks other than the local network. This section examines how communication from hosts is able to reach hosts on remote networks.

Host Forwarding Decision (8.4.1)

With both IPv4 and IPv6, packets are always created at the source host. The source host must be able to direct a packet to the destination host. To do this, host end devices create their own routing table. This section discusses how end devices use routing tables.

Another role of the network layer is to direct packets between hosts. A host can send a packet to the following:

- **Itself:** A host can ping itself by sending a packet to the special IPv4 address 127.0.0.1 or the IPv6 address `::1`, which is referred to as the *loopback interface*. Pinging the loopback interface tests the TCP/IP protocol stack on the host.
- **Local host:** This is a destination host that is on the same local network as the sending host. The source and destination hosts share the same network address.
- **Remote host:** This is a destination host on a remote network. The source and destination hosts do not share the same network address.

Figure 8-11 illustrates PC1 connecting to a local host on the same network and to a remote host located on another network.

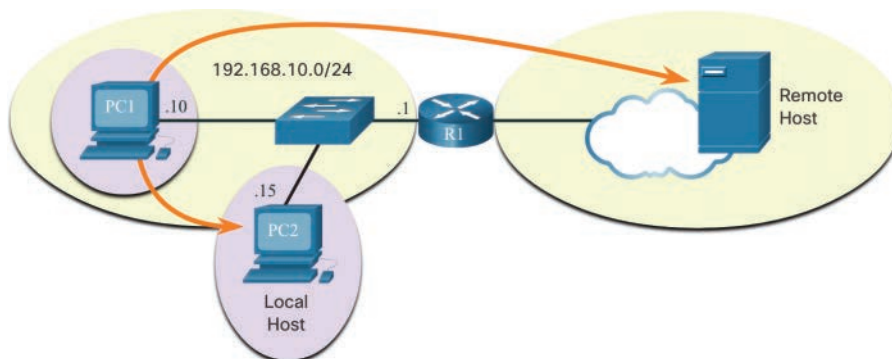


Figure 8-11 Hosts Can Connect to Local and Remote Networks

Whether a packet is destined for a local host or a remote host is determined by the source end device. The source end device determines whether the destination IP address is on the same network that the source device itself is on. The method of determination varies by IP version:

- **In IPv4:** The source device uses its own subnet mask along with its own IPv4 address and the destination IPv4 address to make this determination.
- **In IPv6:** The local router advertises the local network address (prefix) to all devices on the network.

In a home or business network, you may have several wired and wireless devices interconnected together by an intermediary device, such as a LAN switch or a wireless access point (WAP). This intermediary device provides interconnections between local hosts on the local network. Local hosts can reach each other and share information without the need for any additional devices. If a host is sending a packet to a device that is configured with the same IP network as the host device, the packet is simply forwarded out the host interface, through the intermediary device, and to the destination device directly.

Of course, in most situations, we want our devices to be able to connect beyond the local network segment, such as out to other homes, businesses, and the internet. Devices that are beyond the local network segment are known as *remote hosts*. When a source device sends a packet to a remote destination device, the help of routers and routing is needed. Routing is the process of identifying the best path to a destination. The router connected to the local network segment is referred to as the *default gateway*.

Default Gateway (8.4.2)

The *default gateway* is a network device (that is, a router, or Layer 3 switch) that can route traffic to other networks. If you use the analogy of a room for a network, then the default gateway is like a doorway. If you want to get to another room or network, you need to find the doorway.

On a network, a default gateway is usually a router with these features:

- It has a local IP address in the same address range as other hosts on the local network.
- It can accept data into the local network and forward data out of the local network.
- It routes traffic to other networks.

A default gateway is required to send traffic outside the local network. Traffic cannot be forwarded outside the local network if there is no default gateway, the default gateway address is not configured, or the default gateway is down.

A Host Routes to the Default Gateway (8.4.3)

A host routing table typically includes a default gateway. With IPv4, the host receives the IPv4 address of the default gateway either dynamically from Dynamic Host Configuration Protocol (DHCP) or configured manually. With IPv6, the router can advertise the default gateway address or the host can be configured manually.

In Figure 8-12, assume that PC1 and PC2 are configured with the IPv4 address 192.168.10.1 as the default gateway.

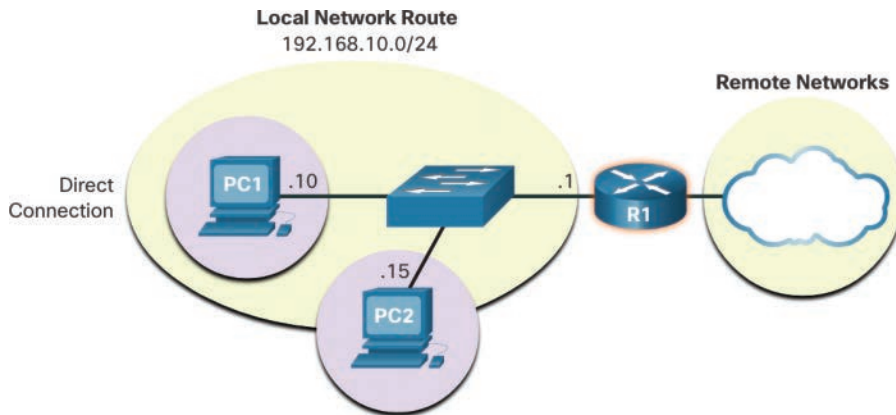


Figure 8-12 Hosts Use a Default Gateway for Remote Network Access

Having a default gateway configured creates a default route in the routing table of the PC. A default route is the route or pathway the computer takes when it tries to contact a remote network.

In Figure 8-12, PC1 and PC2 both have default routes to send all traffic destined to remote networks to R1.

Host Routing Tables (8.4.4)

On a Windows host, the `route print` or `netstat -r` command can be used to display the host routing table. Both of these commands generate the same output. The output may seem overwhelming at first, but it is fairly simple to understand.

Figure 8-13 displays a sample topology for host routes.



Figure 8-13 Host Route Topology

Example 8-1 shows the output generated by the `netstat -r` command on PC1 in Figure 8-13.

Example 8-1 IPv4 Routing Table for PC1

```
C:\Users\PC1> netstat -r

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          192.168.10.1     192.168.10.10    25
127.0.0.0                  255.0.0.0        On-link          127.0.0.1        306
127.0.0.1                  255.255.255.255  On-link          127.0.0.1        306
127.255.255.255           255.255.255.255  On-link          127.0.0.1        306
192.168.10.0               255.255.255.0    On-link          192.168.10.10    281
192.168.10.10              255.255.255.255  On-link          192.168.10.10    281
192.168.10.255             255.255.255.255  On-link          192.168.10.10    281
224.0.0.0                  240.0.0.0        On-link          127.0.0.1        306
224.0.0.0                  240.0.0.0        On-link          192.168.10.10    281
255.255.255.255           255.255.255.255  On-link          127.0.0.1        306
255.255.255.255           255.255.255.255  On-link          192.168.10.10    281
```

Note

The output in Example 8-1 displays the IPv4 route table.

The output of the `netstat -r` command or the equivalent `route print` command has three sections related to the current TCP/IP network connections:

- **Interface List:** This section lists the Media Access Control (MAC) address and assigned interface number of each network-capable interface on the host, including Ethernet, Wi-Fi, and Bluetooth adapters.
- **IPv4 Route Table:** This section lists all known IPv4 routes, including direct connections, the local network, and local default routes.
- **IPv6 Route Table:** This section lists all known IPv6 routes, including direct connections, the local network, and local default routes.



Check Your Understanding—How a Host Routes (8.4.5)

Refer to the online course to complete this activity.

Introduction to Routing (8.5)

This section introduces the role of the router in the routing process and provides an introduction the use of routing tables for forwarding packets.

Router Packet Forwarding Decision (8.5.1)

In this chapter, you have already learned about host routing tables. Most networks also contain routers, which are intermediary devices that also contain routing tables. This section covers router operations at the network layer. When a host sends a packet to another host, it consults its routing table to determine where to send the packet. If the destination host is on a remote network, the packet is forwarded to the default gateway, which is usually the local router.

What happens when a packet arrives on a router interface? The router examines the destination IP address of the packet and searches its routing table to determine where to forward the packet. The routing table contains a list of all known network addresses (prefixes) and where to forward the packet. These entries are known as *route entries* or *routes*. The router forwards a packet using the route entry that matches best (that is, is longest). Figure 8-14 illustrates this forwarding process:

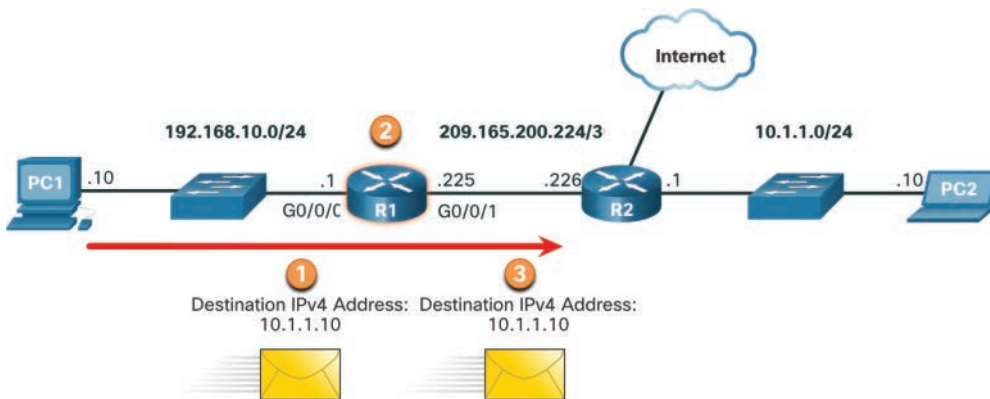


Figure 8-14 Packet Forwarding Process

- Step 1.** The packet arrives on the Gigabit Ethernet 0/0/0 interface of router R1. R1 de-encapsulates the Layer 2 Ethernet header and trailer.
- Step 2.** Router R1 examines the destination IPv4 address of the packet and searches for the best match in its IPv4 routing table. The route entry indicates that this packet is to be forwarded to router R2.
- Step 3.** Router R1 encapsulates the packet into a new Ethernet header and trailer and forwards the packet to the next hop router R2.

Table 8-1 shows the pertinent information from the R1 routing table.

Table 8-1 R1 Routing Table

Route	Next Hop or Exit Interface
192.168.10.0/24	G0/0/0
209.165.200.224/30	G0/0/1
10.1.1.0/24	Through R2
Default Route 0.0.0.0/0	Through R2

IP Router Routing Table (8.5.2)

The routing table of a router contains network route entries that list all the possible known network destinations.

The routing table stores three types of route entries:

- *Directly connected networks*: These network route entries are active router interfaces. A router adds a directly connected route when an interface is configured with an IP address and is activated. Each router interface is connected to a different network segment. In Figure 8-15, the directly connected networks in the R1 IPv4 routing table would be 192.168.10.0/24 and 209.165.200.224/30.
- *Remote networks*: These network route entries are connected to other routers. Routers learn about remote networks either by being explicitly configured by an administrator or by exchanging route information using a dynamic routing protocol. In Figure 8-15, the remote network in the R1 IPv4 routing table would be 10.1.1.0/24.
- *Default route*: Like a host, most routers also include a default route entry, a gateway of last resort. The default route is used when there is no better (longer) match in the IP routing table. In Figure 8-15, the R1 IPv4 routing table would most likely include a default route to forward all packets to router R2.

Figure 8-15 identifies the directly connected and remote networks of router R1.

In Figure 8-15, R1 has two directly connect networks:

- 192.168.10.0/24
- 209.165.200.224/30

R1 also has remote networks (that is, 10.1.1.0/24 and the internet) that it can learn about.

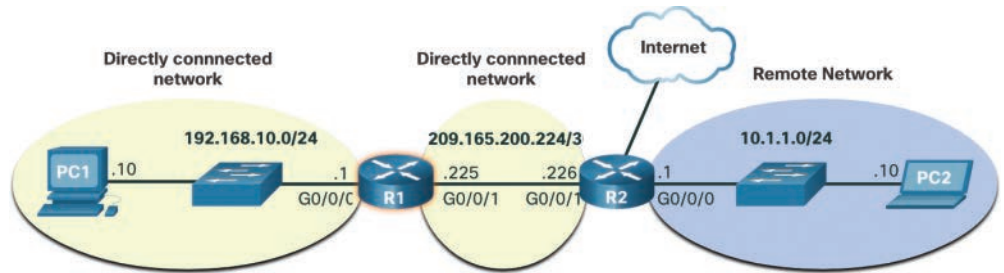


Figure 8-15 Example Topology of Directly Connected and Remote Networks

A router can learn about remote networks in one of two ways:

- **Manually:** Remote networks are manually entered into the route table using static routes.
- **Dynamically:** Remote routes are automatically learned using a dynamic routing protocol.

Static Routing (8.5.3)

Static routes are route entries that are manually configured. Figure 8-16 shows an example of a static route that was manually configured on router R1. A static route includes the remote network address and the IP address of the next hop router.

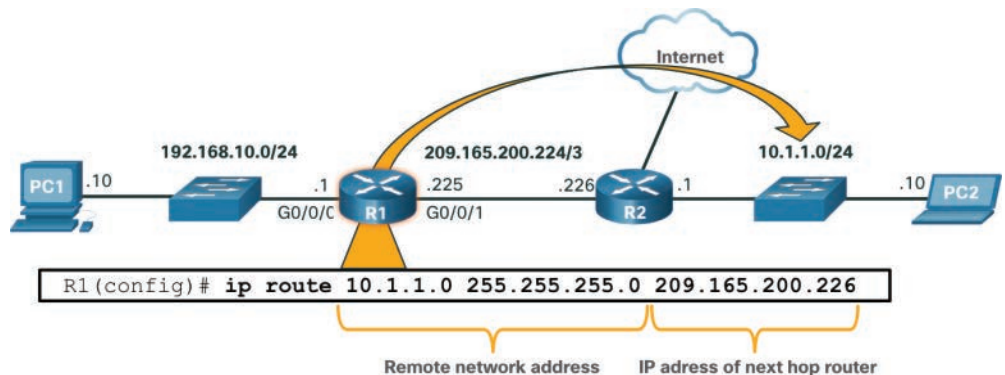


Figure 8-16 Static Routing Example

If there is a change in the network topology, a static route is not automatically updated and must be manually reconfigured. For example, in Figure 8-17, R1 has a static route to reach the 10.1.1.0/24 network via R2. If that path becomes unavailable, R1 needs to be reconfigured with a new static route to the 10.1.1.0/24 network via R3. Router R3 therefore needs to have a route entry in its routing table to send packets destined for 10.1.1.0/24 to R2.

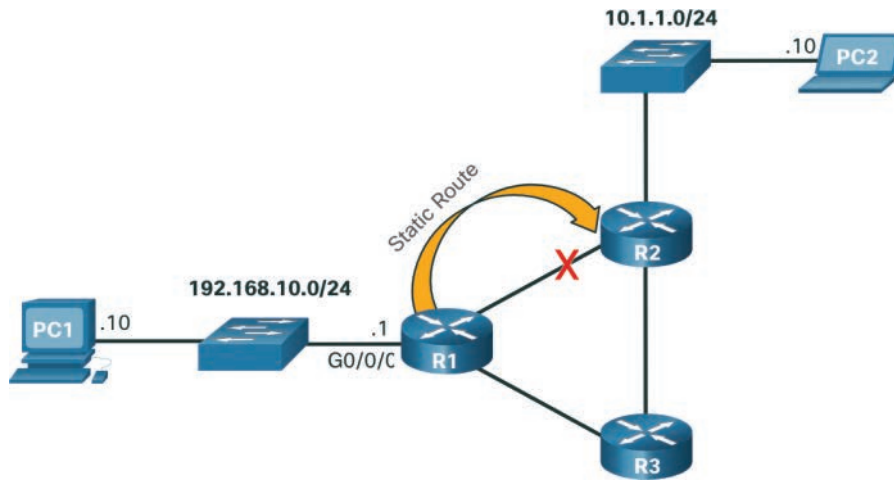


Figure 8-17 Static Routing Does Not Automatically Update to Topology Changes

Static routing has the following characteristics:

- A static route must be configured manually.
- The administrator needs to reconfigure a static route if there is a change in the topology and the static route is no longer viable.
- A static route is appropriate for a small network and when there are few or no redundant links.
- A static route is commonly used with a dynamic routing protocol for configuring a default route.

Dynamic Routing (8.5.4)

A *dynamic routing protocol* allows the routers to automatically learn about remote networks, including a default route, from other routers. Routers that use dynamic routing protocols automatically share routing information with other routers and compensate for any topology changes without involving the network administrator. If there is a change in the network topology, routers share this information using the dynamic routing protocol and automatically update their routing tables.

Dynamic routing protocols include OSPF and Enhanced Interior Gateway Routing Protocol (EIGRP). Figure 8-18 shows an example of routers R1 and R2 automatically sharing network information using the routing protocol OSPF.

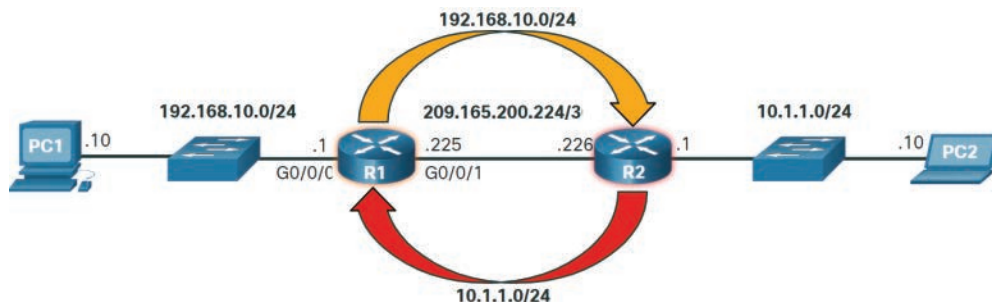


Figure 8-18 Dynamic Routing Example

Basic dynamic routing configuration only requires the network administrator to enable the directly connected networks within the dynamic routing protocol. The dynamic routing protocol automatically does the following:

- Discovers remote networks
- Maintains up-to-date routing information
- Chooses the best paths to destination networks
- Attempts to find a new best path if the current path is no longer available

When a router is manually configured with a static route or learns about a remote network dynamically using a dynamic routing protocol, the remote network address and next hop address are entered into the IP routing table. As shown Figure 8-19, if there is a change in the network topology, the routers automatically adjust and attempt to find a new best path.

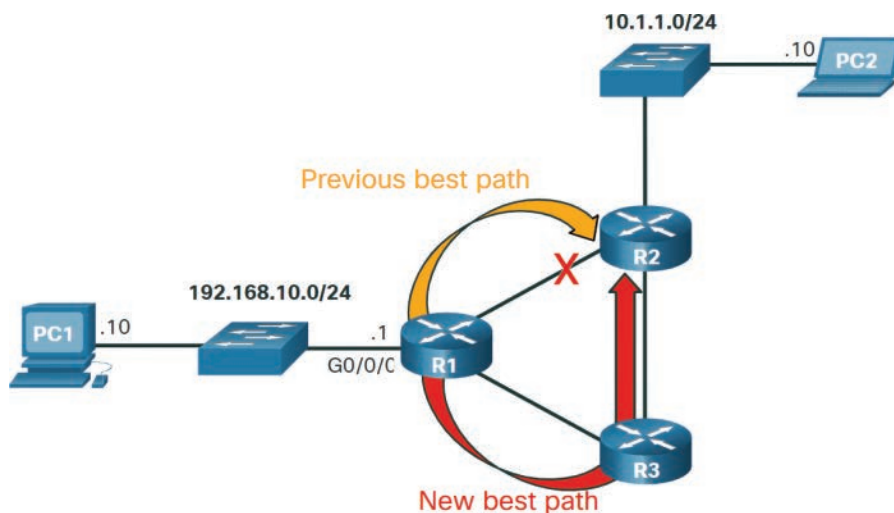


Figure 8-19 Dynamic Routing Automatically Updates to Topology Changes

Note

It is common for some routers to use a combination of both static routes and a dynamic routing protocol.

Video**Video—IPv4 Router Routing Tables (8.5.5)**

Refer to the online course to view this video.

Introduction to an IPv4 Routing Table (8.5.6)

Notice in Figure 8-20 that R2 is connected to the internet. Therefore, the administrator configured R1 with a default static route sending packets to R2 when there is no specific entry in the routing table that matches the destination IP address. R1 and R2 are also using OSPF routing to advertise directly connected networks.

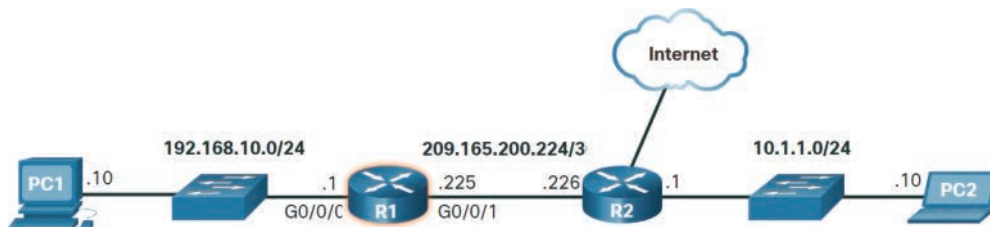


Figure 8-20 Sample Topology for IPv4 Routing Table

The `show ip route` privileged EXEC mode command is used to view the IPv4 routing table on a Cisco IOS router. Example 8-2 shows the IPv4 routing table of router R1.

Example 8-2 R1 IPv4 Routing Table

```
R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 209.165.200.226 to network 0.0.0.0
S*    0.0.0.0/0 [1/0] via 209.165.200.226, GigabitEthernet0/0/1
```

```

10.0.0.0/24 is subnetted, 1 subnets
O    10.1.1.0 [110/2] via 209.165.200.226, 00:02:45, GigabitEthernet0/0/1
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L    192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C    209.165.200.224/30 is directly connected, GigabitEthernet0/0/1
L    209.165.200.225/32 is directly connected, GigabitEthernet0/0/1
R1#

```

At the beginning of each routing table entry is a code that is used to identify the type of route or how the route was learned. Common route sources (codes) include these:

- **L:** Directly connected local interface IP address
- **C:** Directly connected network
- **S:** Static route manually configured by an administrator
- **O:** OSPF
- **D:** EIGRP

The routing table displays all of the known IPv4 destination routes for R1.

A directly connected route is automatically created when a router interface is configured with IP address information and is activated. The router adds two route entries with the codes **C** (that is, the connected network) and **L** (that is, the local interface IP address of the connected network). The route entries also identify the exit interface to use to reach the network. The two directly connected networks in this example are 192.168.10.0/24 and 209.165.200.224/30.

Routers R1 and R2 are also using the OSPF dynamic routing protocol to exchange router information. In the sample routing table, R1 has a route entry for the 10.1.1.0/24 network that it learned dynamically from router R2 thanks to the OSPF routing protocol.

A default route has a network address of all zeros. For example, say that the IPv4 network address is 0.0.0.0. A static route entry in the routing table begins with the code **S***, as highlighted in Example 8-2.

Check Your Understanding—Introduction to Routing (8.5.7)

Refer to the online course to complete this activity.

Summary (8.6)

The following is a summary of the topics in the chapter and their corresponding online modules.

Network Layer Characteristics

The network layer (OSI Layer 3) provides services to allow end devices to exchange data across networks. IPv4 and IPv6 are the principal network layer communication protocols. The network layer also includes the routing protocol OSPF and messaging protocols such as ICMP. Network layer protocols perform four basic operations: addressing end devices, encapsulation, routing, and de-encapsulation. IPv4 and IPv6 specify the packet structure and processing used to carry the data from one host to another host. IP encapsulates the transport layer segment by adding an IP header, which is used to deliver the packet to the destination host. The IP header is examined by Layer 3 devices (that is, routers) as it travels across a network to its destination. IP is connectionless, meaning that it creates no dedicated end-to-end connection before data is sent. In addition, IP is best effort, meaning that it does not guarantee that all packets that are sent are, in fact, received. Finally, IP is media independent, meaning that it operates independently of the media that carry the data at lower layers of the protocol stack.

IPv4 Packet

An IPv4 packet header consists of fields containing information about the packet. These fields contain binary numbers that are examined by the Layer 3 process. The binary values of the fields identify various settings of the IP packet. Significant fields in the IPv4 header include Version, DS, Header Checksum, TTL, Protocol, Source IPv4 Address, and Destination IPv4 Address.

IPv6 Packet

IPv6 is designed to overcome the limitations of IPv4, including IPv4 address depletion, lack of end-to-end connectivity, and increased network complexity. IPv6 increases the available address space, improves packet handling, and eliminates the need for NAT. The fields in the IPv6 packet header include Version, Traffic Class, Flow Label, Payload Length, Next Header, Hop Limit, Source IPv6 Address, and Destination IPv6 Address.

How a Host Routes

A host can send a packet to itself, to another local host, or to a remote host. In IPv4, the source device uses its own subnet mask along with its own IPv4 address and the destination IPv4 address to determine whether the destination host is on the same network. In IPv6, the local router advertises the local network address (prefix) to all devices on the network to make this determination. The default gateway is the network device (that is, router) that can route traffic to other networks. On a network, a default gateway is usually a router that has a local IP address in the same address range as other hosts on the local network, can accept data into the local network and forward data out the local network, and can route traffic to other networks. A host routing table typically includes a default gateway. In IPv4, the host may receive the IPv4 address of the default gateway dynamically through DHCP, or it may be configured manually. In IPv6, the router can advertise the default gateway address, or the host can be configured manually. On a Windows host, the **route print** or **netstat -r** command can be used to display the host routing table.

Introduction to Routing

When a host sends a packet to another host, it consults its routing table to determine where to send the packet. If the destination host is on a remote network, the packet is forwarded to the default gateway, which is usually the local router. What happens when a packet arrives on a router interface? The router examines the packet's destination IP address and searches its routing table to determine where to forward the packet. The routing table contains a list of all known network addresses (prefixes) and where to forward the packet. These entries are known as route entries, or routes. The router forwards the packet using the best (longest) matching route entry. The routing table of a router stores three types of route entries: directly connected networks, remote networks, and a default route. Routers learn about remote networks either manually or dynamically using a dynamic routing protocol. Static routes are route entries that are manually configured. A static route includes the remote network address and the IP address of the next hop router. OSPF and EIGRP are two dynamic routing protocols. The **show ip route** privileged EXEC mode command is used to view the IPv4 routing table on a Cisco IOS router. At the beginning of an IPv4 routing table is a code that is used to identify the type of route or how the route was learned. Common route sources (codes) include:

- **L**: Directly connected local interface IP address
- **C**: Directly connected network
- **S**: Static route manually configured by an administrator
- **O**: Open Shortest Path First (OSPF)
- **D**: Enhanced Interior Gateway Routing Protocol (EIGRP)

Practice

There are no labs or Packet Tracer activities for this chapter.

Check Your Understanding Questions

Complete all the review questions listed here to test your understanding of the topics and concepts in this chapter. The appendix “Answers to ‘Check Your Understanding’ Questions” lists the answers.

1. Which information is used by routers to forward a data packet toward its destination?
 - a. source IP address
 - b. destination IP address
 - c. source data link address
 - d. destination data link address
2. A computer has to send a packet to a destination host in the same LAN. How will the packet be sent?
 - a. The packet will be sent to the default gateway first, and then, depending on the response from the gateway, it may be sent to the destination host.
 - b. The packet will be sent directly to the destination host.
 - c. The packet will first be sent to the default gateway, and then from the default gateway it will be sent directly to the destination host.
 - d. The packet will be sent only to the default gateway.
3. A router receives a packet from the Gigabit Ethernet 0/0/0 interface and determines that the packet needs to be forwarded out the Gigabit Ethernet 0/0/1 interface. What will the router do next?
 - a. route the packet out the Gigabit Ethernet 0/0/1 interface
 - b. create a new Layer 2 Ethernet frame to be sent to the destination
 - c. look into the ARP cache to determine the destination IP address
 - d. look into the routing table to determine if the destination network is in the routing table

4. Which IPv4 address can a host use to ping the loopback interface?
 - a. 126.0.0.1
 - b. 127.0.0.0
 - c. 126.0.0.0
 - d. 127.0.0.1
5. When a connectionless protocol is in use at a lower layer of the OSI model, how is missing data detected and retransmitted if necessary?
 - a. Connectionless acknowledgments are used to request retransmission.
 - b. An upper-layer connection-oriented protocol keeps track of the data received and can request retransmission from the upper-level protocol on the sending host.
 - c. Network layer IP protocols manage the communication sessions if connection-oriented transport services are not available.
 - d. The best-effort delivery process guarantees that all packets that are sent are received.
6. What was the main reason for the creation and implementation of IPv6?
 - a. to make reading a 32-bit address easier
 - b. to address the IPv4 address depletion problem
 - c. to provide more address space in the Internet Names Registry
 - d. to allow NAT support for private addressing
7. Which statement accurately describes a characteristic of IPv4?
 - a. All IPv4 addresses are assignable to hosts.
 - b. IPv4 has a 32-bit address space.
 - c. An IPv4 header has fewer fields than an IPv6 header has.
 - d. IPv4 has a 128-bit address space.
8. When a router receives an IPv6 packet, what information is examined in order to see if the packet has exceeded the number of routers that can forward the packet?
 - a. destination IP address
 - b. source IP address
 - c. hop limit
 - d. TTL

9. Which field in an IPv6 packet does a router use to determine whether the packet has expired and should be dropped?
 - a. TTL
 - b. Hop Limit
 - c. Address Unreachable
 - d. No Route to Destination
10. Which command can be used on a Windows host to display the routing table?
 - a. `netstat -s`
 - b. `show ip route`
 - c. `netstat -r`
 - d. `print route`
11. What information is added during encapsulation at OSI Layer 3?
 - a. source and destination MAC addresses
 - b. source and destination application protocols
 - c. source and destination port numbers
 - d. source and destination IP addresses
12. How does the network layer determine the MTU value?
 - a. The network layer depends on the higher-level layers to determine the MTU.
 - b. The network layer depends on the data link layer to set the MTU and adjusts the speed of transmission to accommodate it.
 - c. The network layer determines how large packets can be, based on the MTU of the data link frame.
 - d. To increase speed delivery, the network layer ignores the MTU.
13. Which characteristic describes an IPv6 enhancement over IPv4?
 - a. IPv6 is based on 128-bit flat addressing, whereas IPv4 is based on 32-bit hierarchical addressing.
 - b. The IPv6 header is simpler than the IPv4 header, which improves packet handling.
 - c. Both IPv4 and IPv6 support authentication, but only IPv6 supports privacy capabilities.
 - d. The IPv6 address space is four times bigger than the IPv4 address space.

Address Resolution

Objectives

Upon completion of this chapter, you will be able to answer the following questions:

- What are the roles of the MAC address and the IP address?
- What is the operation of IPv6 neighbor discovery?
- What is the purpose of ARP?

Key Terms

This chapter uses the following key terms. You can find the definitions in the glossary at the end of the book.

physical address page 298

logical address page 298

ARP table page 302

ARP cache page 302

Introduction (9.0)

Hosts and routers both create routing tables to ensure that they can send and receive data across networks. So how does this information get created in a routing table? As a network administrator, you could enter these MAC and IP addresses manually. But that would take a lot of time, and the likelihood of making a few mistakes is great. Are you thinking that there must be some way that this could be done automatically, by the hosts and routers themselves? Of course, you are correct! Even though this address resolution process is automatic, you must understand how it works because you may have to troubleshoot a problem or, worse, your network could be attacked by a threat actor. Are you ready to learn about address resolution? This chapter mentions several very good videos to help explain the concepts, as well as three Packet Tracer activities to cement your understanding. Why wait?

MAC and IP (9.1)

This section discusses the differences between Layer 2 data link addresses, such as Ethernet MAC addresses, and Layer 3 network address, such as IP addresses.

Destination on Same Network (9.1.1)

Sometimes a host must send a message, but it only knows the IP address of the destination device. The host needs to know the MAC address of that device, but how can it be discovered? This is where address resolution becomes critical.

There are two primary addresses assigned to a device on an Ethernet LAN:

- **Physical address (the MAC address):** This address is used for NIC-to-NIC communications on the same Ethernet network.
- **Logical address (the IP address):** This address is used to send a packet from the source device to the destination device. The destination IP address may be on the same IP network as the source, or it may be on a remote network.

Layer 2 physical addresses (that is, Ethernet MAC addresses) are used to deliver a data link frame with an encapsulated IP packet from one NIC to another NIC on the same network. If the destination IP address is on the same network, the destination MAC address will be that of the destination device.

In Figure 9-1, PC1 wants to send a packet to PC2. The figure displays the Layer 2 destination and source MAC addresses and the Layer 3 IPv4 addressing that would be included in the packet sent from PC1.

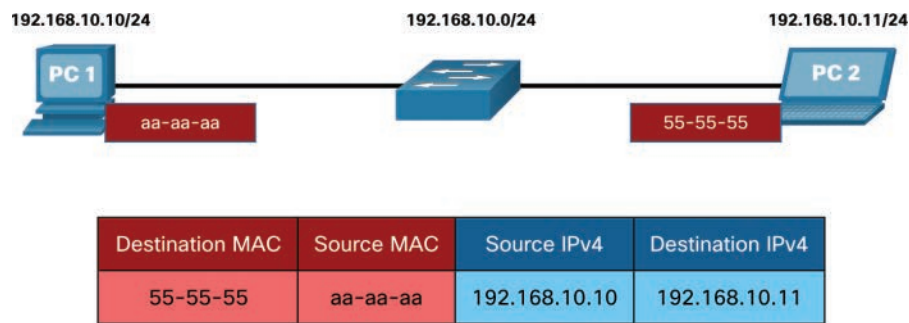


Figure 9-1 Addressing of Hosts on the Same Network

The Layer 2 Ethernet frame contains the following:

- **Destination MAC address:** This is the simplified MAC address of PC2, 55-55-55.
- **Source MAC address:** This is the simplified MAC address of the Ethernet NIC on PC1, aa-aa-aa.

The Layer 3 IP packet contains the following:

- **Source IPv4 address:** This is the IPv4 address of PC1, 192.168.10.10.
- **Destination IPv4 address:** This is the IPv4 address of PC2, 192.168.10.11.

Destination on Remote Network (9.1.2)

When the destination IP address (IPv4 or IPv6) is on a remote network, the destination MAC address is the address of the host default gateway (that is, the router interface).

In Figure 9-2, PC1 wants to send a packet to PC2. PC2 is located on a remote network. Because the destination IPv4 address is not on the same local network as PC1, the destination MAC address is that of the local default gateway on the router.

Routers examine the destination IPv4 address to determine the best path for forwarding the IPv4 packet. When a router receives the Ethernet frame, it de-encapsulates the Layer 2 information. Using the destination IPv4 address, it determines the next hop device and then encapsulates the IPv4 packet in a new data link frame for the outgoing interface.

R1 can now encapsulate the packet with new Layer 2 address information, as shown in Figure 9-3.

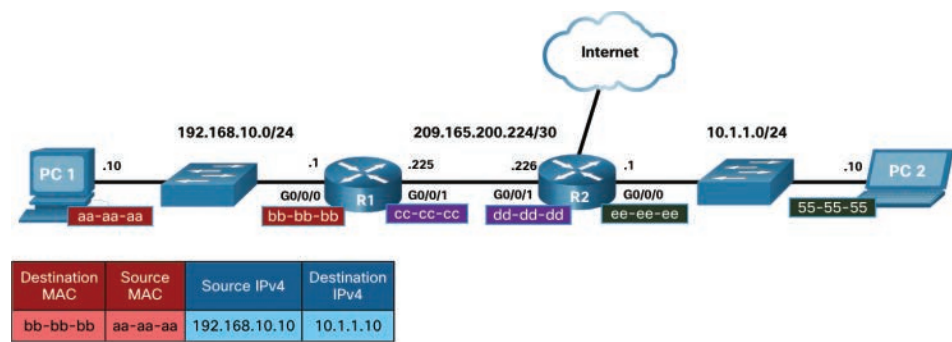


Figure 9-2 Remote Network Example: PC1 to R1

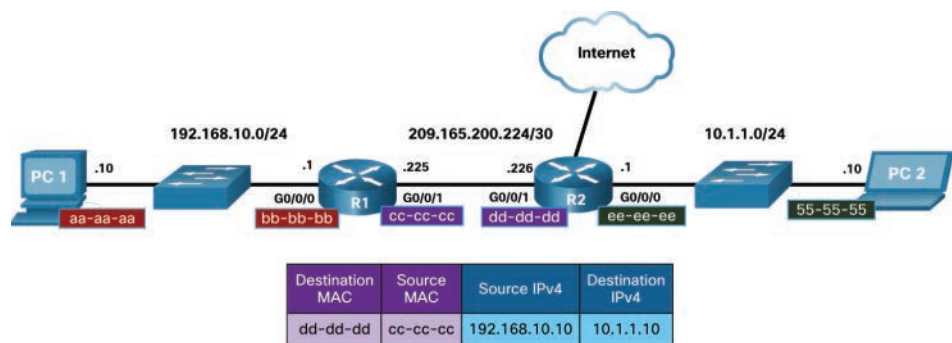


Figure 9-3 Remote Network Example: R1 to R2

The new destination MAC address would be the address of the R2 G0/0/1 interface, and the new source MAC address would be that of the R1 G0/0/1 interface.

Along each link in a path, an IP packet is encapsulated in a frame. The frame is specific to the data link technology that is associated with that link, such as Ethernet. If the next hop device is the final destination, the destination MAC address is the address of the device's Ethernet NIC, as shown in Figure 9-4.

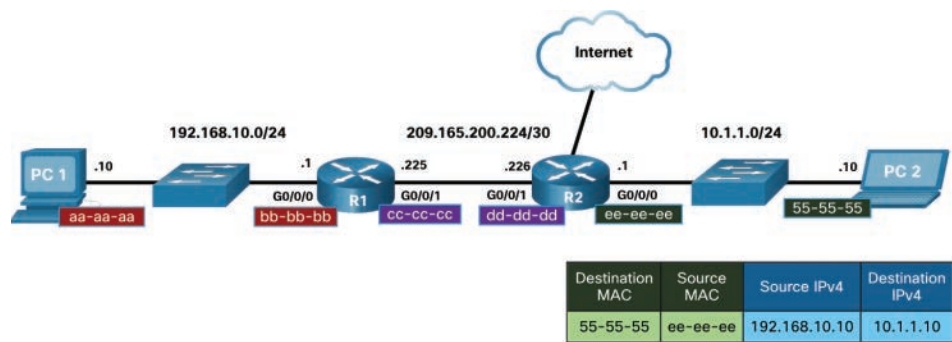


Figure 9-4 Remote Network Example: R2 to PC2

How are the IP addresses of the IP packets in a data flow associated with the MAC addresses on each link along the path to the destination? For IPv4 packets, this is done through a process called Address Resolution Protocol (ARP). For IPv6 packets, the process is ICMPv6 Neighbor Discovery (ND).

Packet Tracer
Activity

Packet Tracer—Identify MAC and IP Addresses (9.1.3)

In this Packet Tracer activity, you will complete the following objectives:

- Gather PDU Information for Local Network Communication
- Gather PDU Information for Remote Network Communication

This activity is optimized for viewing PDUs. The devices are already configured. You will gather PDU information in simulation mode and answer a series of questions about the data you collect.

Interactive
Graphic

Check Your Understanding—MAC and IP (9.1.4)

Refer to the online course to complete this activity.

ARP (9.2)

This section discusses the relationship between MAC addresses and IPv4 addresses, as well as how Address Resolution Protocol (ARP) is used to map these two addresses.

ARP Overview (9.2.1)

If your network is using the IPv4 communications protocol, Address Resolution Protocol (ARP) is what you need in order to map IPv4 addresses to MAC addresses. This section explains how ARP works.

Every IP device on an Ethernet network has a unique Ethernet MAC address. When a device sends an Ethernet Layer 2 frame, it contains these two addresses:

- **Destination MAC address:** The Ethernet MAC address of the destination device on the same local network segment. If the destination host is on another network, then the destination address in the frame would be the address of the default gateway (that is, router).
- **Source MAC address:** The MAC address of the Ethernet NIC on the source host.

Figure 9-5 illustrates the problem when sending a frame to another host on the same segment on an IPv4 network.

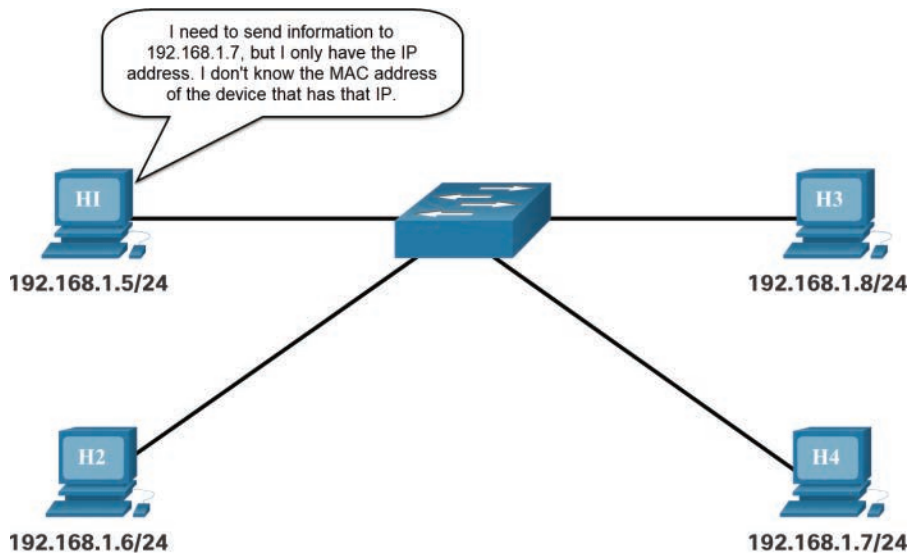


Figure 9-5 A Host Does Not Know the MAC Address for a Destination

To send a packet to another host on the same local IPv4 network, a host must know the IPv4 address and the MAC address of the destination device. Device destination IPv4 addresses are either known or resolved by device name. However, MAC addresses must be discovered.

A device uses ARP to determine the destination MAC address of a local device when it knows its IPv4 address.

ARP provides two basic functions:

- Resolving IPv4 addresses to MAC addresses
- Maintaining a table of IPv4-to-MAC address mappings

ARP Functions (9.2.2)

When a packet is sent to the data link layer to be encapsulated into an Ethernet frame, the device refers to a table in its memory to find the MAC address that is mapped to the IPv4 address. This table, which is stored temporarily in RAM, is called the *ARP table* or the *ARP cache*.

The sending device searches its ARP table for a destination IPv4 address and a corresponding MAC address. Then:

- If the packet's destination IPv4 address is on the same network as the source IPv4 address, the device searches the ARP table for the destination IPv4 address.
- If the destination IPv4 address is on a different network from the source IPv4 address, the device searches the ARP table for the IPv4 address of the default gateway.

In both cases, the search is for an IPv4 address and a corresponding MAC address for the device.

Each entry, or row, of the ARP table binds an IPv4 address with a MAC address. We call the relationship between the two values a *map*. The map simply enables you to locate an IPv4 address in the table and discover the corresponding MAC address. The ARP table temporarily saves (caches) the mapping for the devices on the LAN.

If the device locates the IPv4 address, its corresponding MAC address is used as the destination MAC address in the frame. If no entry is found, the device sends an ARP request, as shown in Figure 9-6.

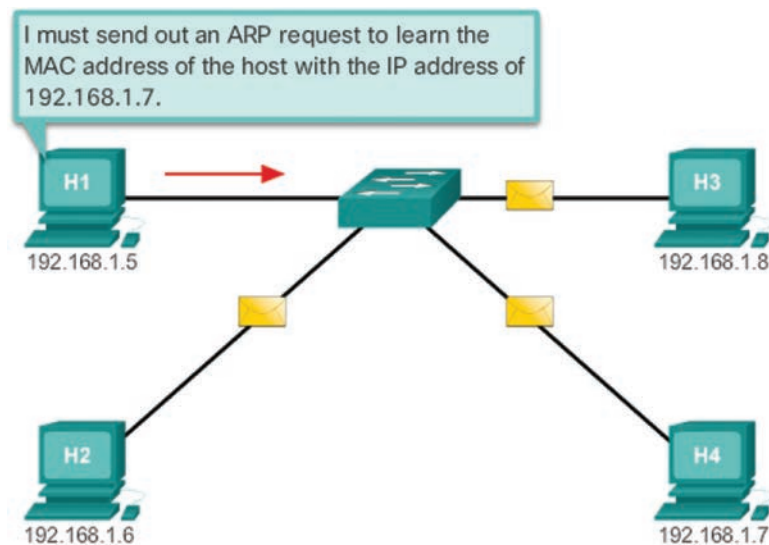


Figure 9-6 H1 Sends a Broadcast ARP Request

The destination responds with an ARP reply, as shown in Figure 9-7.

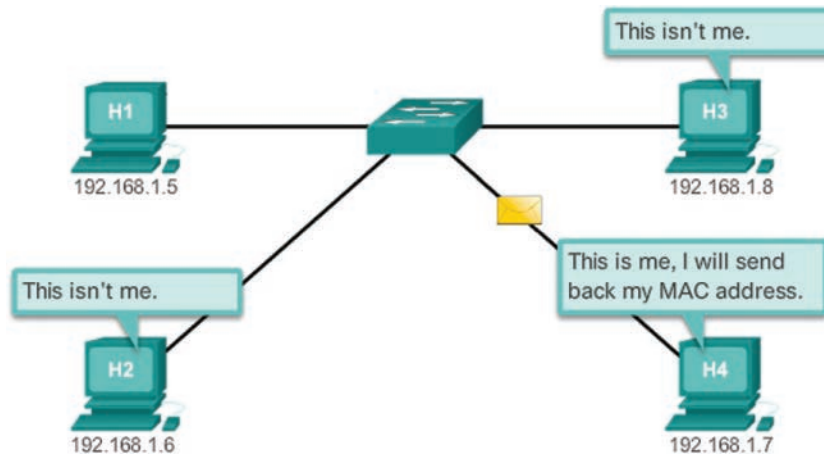


Figure 9-7 H4 Sends a Unicast ARP Reply

Video

Video—ARP Request (9.2.3)

An ARP request is sent when a device needs to determine the MAC address that is associated with an IPv4 address, and it does not have an entry for the IPv4 address in its ARP table.

ARP messages are encapsulated directly within an Ethernet frame. There is no IPv4 header. An ARP request is encapsulated in an Ethernet frame using the following header fields:

- **Destination MAC Address:** This is a broadcast address FF-FF-FF-FF-FF-FF, which requires all Ethernet NICs on the LAN to accept and process the ARP request.
- **Source MAC Address:** This is the MAC address of the sender of the ARP request.
- **Type:** ARP messages have a Type field setting of 0x806. This informs the receiving NIC that the data portion of the frame needs to be passed to the ARP process.

Because ARP requests are broadcasts, a switch floods them out all ports except for the receiving port. All Ethernet NICs on the LAN process broadcasts and must deliver the ARP request to the device's operating system for processing. Every device must process the ARP request to see if the target IPv4 address matches its own. A router does not forward broadcasts out other interfaces.

Only one device on a LAN has an IPv4 address that matches the target IPv4 address in the ARP request. All other devices do not reply.

Refer to the online course to view this video.

Video**Video—ARP Operation—ARP Reply (9.2.4)**

Only the device with the target IPv4 address associated with the ARP request responds with an ARP reply. The ARP reply is encapsulated in an Ethernet frame using the following header fields:

- **Destination MAC Address:** This is the MAC address of the sender of the ARP request.
- **Source MAC Address:** This is the MAC address of the sender of the ARP reply.
- **Type:** ARP messages have a Type field setting of 0x806. This informs the receiving NIC that the data portion of the frame needs to be passed to the ARP process.

Only the device that originally sent an ARP request receives a unicast ARP reply. After the ARP reply is received, the device adds the IPv4 address and the corresponding MAC address to its ARP table. Packets destined for that IPv4 address can then be encapsulated in frames using the corresponding MAC address.

If no device responds to an ARP request, the packet is dropped because a frame cannot be created.

Entries in the ARP table are timestamped. If a device does not receive a frame from a particular device before the timestamp expires, the entry for this device is removed from the ARP table.

In addition, static map entries can be entered in an ARP table, but this is rarely done. Static ARP table entries do not expire over time and must be manually removed.

Note: Whereas IPv4 uses ARP, IPv6 uses a similar process, known as ICMPv6 Neighbor Discovery (ND). IPv6 uses neighbor solicitation and neighbor advertisement messages, which are similar to IPv4 ARP requests and ARP replies.

Refer to the online course to view this video.

Video**Video—ARP Role in Remote Communications (9.2.5)**

When the destination IPv4 address is not on the same network as the source IPv4 address, the source device needs to send the frame to its default gateway. This is the interface of the local router. Whenever a source device has a packet with an IPv4 address on another network, it encapsulates that packet in a frame, using the destination MAC address of the router.

The IPv4 address of the default gateway is stored in the IPv4 configuration of the hosts. When a host creates a packet for a destination, it compares the destination IPv4 address and its own IPv4 address to determine if the two IPv4 addresses are located on the same Layer 3 network. If the destination host is not on its same

network, the source checks its ARP table for an entry with the IPv4 address of the default gateway. If there is not an entry in the ARP table, the source uses the ARP process to determine the MAC address of the default gateway.

Refer to the online course to view this video.

Removing Entries from an ARP Table (9.2.6)

For each device, an ARP cache timer removes ARP entries that have not been used for a specified period of time. The times differ depending on the operating system of the device. For example, newer Windows operating systems store ARP table entries for 15 to 45 seconds, as illustrated in Figure 9-8.

Commands may also be used to manually remove some or all of the entries in the ARP table. After an entry has been removed, the process for sending an ARP request and receiving an ARP reply must occur again in order to once again enter the map in the ARP table.

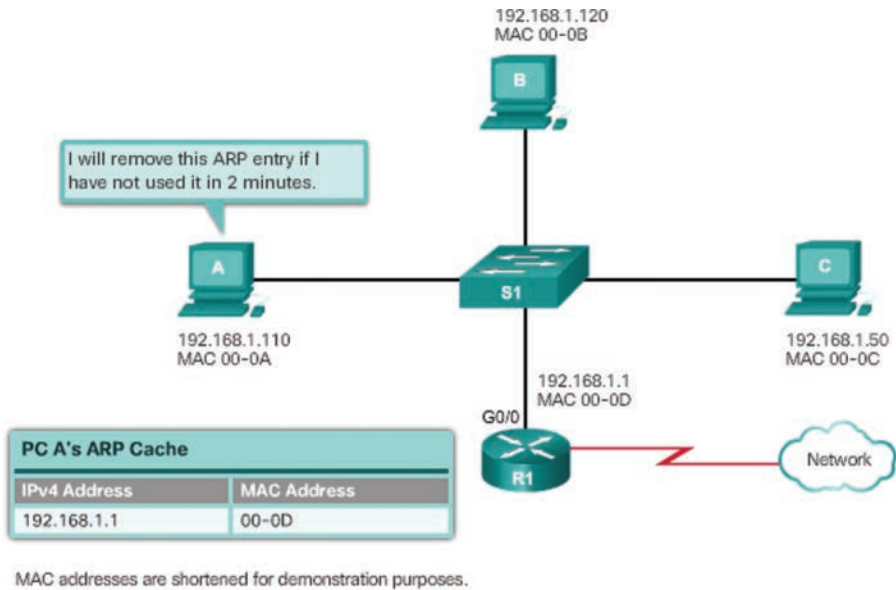


Figure 9-8 Removing MAC-to-IP Address Mappings

ARP Tables on Networking Devices (9.2.7)

On a Cisco router, the `show ip arp` command is used to display the ARP table, as shown in Example 9-1.

Example 9-1 Cisco Router ARP Table

```
R1# show ip arp
```

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	192.168.10.1	-	a0e0.af0d.e140	ARPA	GigabitEthernet0/0/0
Internet	209.165.200.225	-	a0e0.af0d.e141	ARPA	GigabitEthernet0/0/1
Internet	209.165.200.226	1	a03d.6fe1.9d91	ARPA	GigabitEthernet0/0/1

```
R1#
```

On a Windows 10 PC, the **arp -a** command is used to display the ARP table, as shown in Example 9-2.

Example 9-2 Windows 10 PC ARP Table

```
C:\Users\PC> arp -a
```

Interface: 192.168.1.124 --- 0x10

Internet Address	Physical Address	Type
192.168.1.1	c8-d7-19-cc-a0-86	dynamic
192.168.1.101	08-3e-0c-f5-f7-77	dynamic
192.168.1.110	08-3e-0c-f5-f7-56	dynamic
192.168.1.112	ac-b3-13-4a-bd-d0	dynamic
192.168.1.117	08-3e-0c-f5-f7-5c	dynamic
192.168.1.126	24-77-03-45-5d-c4	dynamic
192.168.1.146	94-57-a5-0c-5b-02	dynamic
192.168.1.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

```
C:\Users\PC>
```

ARP Issues—ARP Broadcasts and ARP Spoofing (9.2.8)

As a broadcast frame, an ARP request is received and processed by every device on the local network. On a typical business network, these broadcasts would probably have minimal impact on network performance. However, if a large number of devices were to be powered up and all start accessing network services at the same time, there could be some reduction in performance for a short period of time, as shown in Figure 9-9. After the devices send out the initial ARP broadcasts and have learned the necessary MAC addresses, any impact on the network will be minimized.

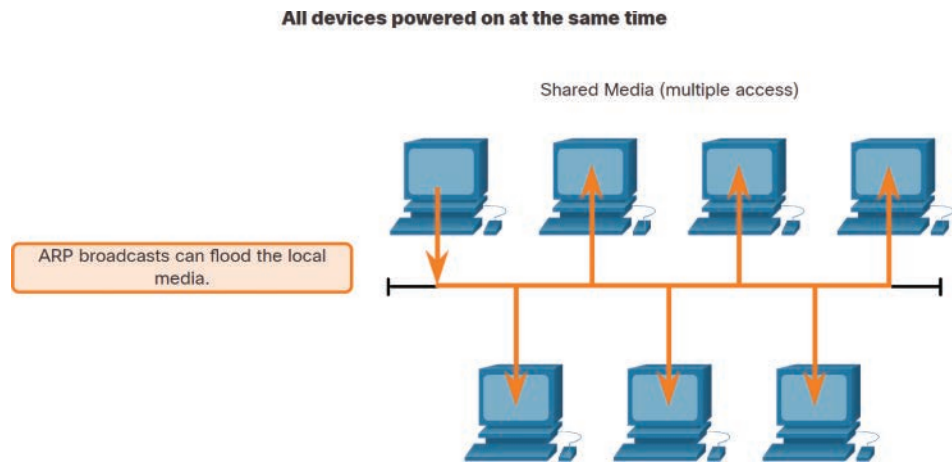
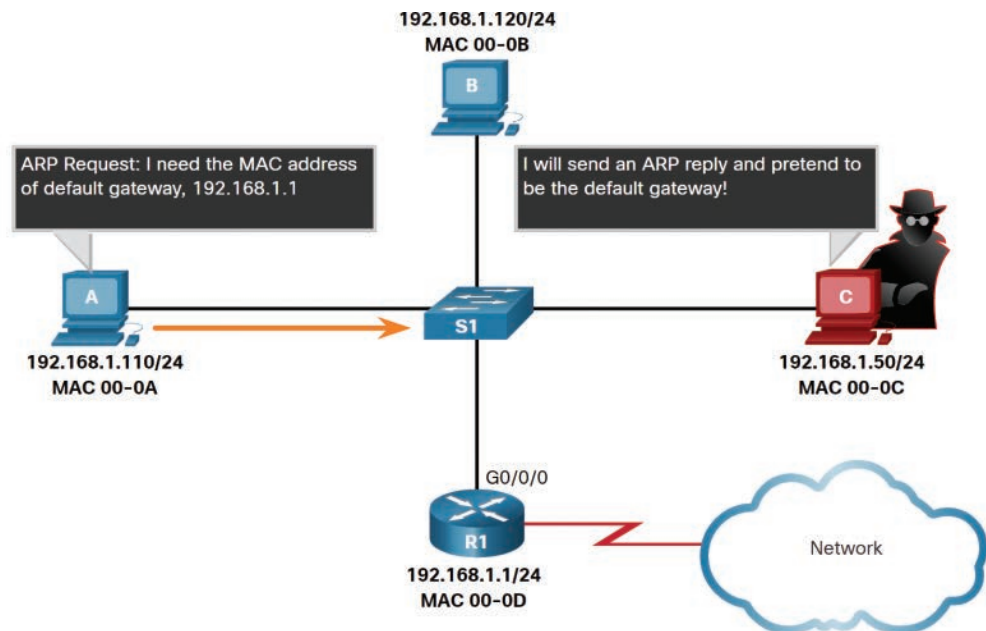


Figure 9-9 ARP Broadcasts Flooding a Network

In some cases, the use of ARP can lead to a potential security risk. A threat actor can use ARP spoofing to perform an ARP poisoning attack. This is a technique used by a threat actor to reply to an ARP request for an IPv4 address belonging to another device, such as the default gateway, as shown in Figure 9-10. The threat actor sends an ARP reply with its own MAC address. The receiver of the ARP reply adds the wrong MAC address to its ARP table and sends these packets to the threat actor.



Note: MAC addresses are shortened for demonstration purposes.

Figure 9-10 Threat Actor Spoofing an ARP Reply

Enterprise-level switches can use mitigation techniques such as dynamic ARP inspection (DAI). DAI is beyond the scope of this course.

Packet Tracer
Activity

Packet Tracer—Examine the ARP Table (9.2.9)

In this Packet Tracer activity, you will complete the following objectives:

- Examine an ARP Request
- Examine a Switch MAC Address Table
- Examine the ARP Process in Remote Communications

This activity is optimized for viewing PDUs. The devices are already configured. You will gather PDU information in simulation mode and answer a series of questions about the data you collect.

Interactive
Graphic

Check Your Understanding—ARP (9.2.10)

Refer to the online course to complete this activity.

IPv6 Neighbor Discovery (9.3)

This section discusses the relationship between MAC addresses and IPv6 addresses and how the Neighbor Discovery (ND) protocol is used to map the two addresses.

Video

Video—IPv6 Neighbor Discovery (9.3.1)

If your network is using the IPv6 communications protocol, the Neighbor Discovery protocol (ND) is what matches IPv6 addresses to MAC addresses. This section explains how ND works.

Refer to the online course to view this video.

IPv6 Neighbor Discovery Messages (9.3.2)

IPv6 Neighbor Discovery protocol is sometimes referred to as ND or NDP. In this book, we refer to it as ND. ND provides address resolution, router discovery, and redirection services for IPv6 using ICMPv6. ICMPv6 ND uses five ICMPv6 messages to perform these services:

- Neighbor Solicitation messages
- Neighbor Advertisement messages

- Router Solicitation messages
- Router Advertisement messages
- Redirect message

Neighbor Solicitation and Neighbor Advertisement messages are used for device-to-device messaging such as address resolution (similar to ARP for IPv4). Devices include both host computers and routers, as shown in Figure 9-11.

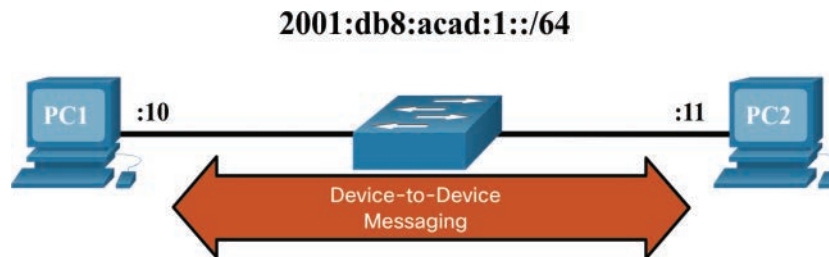


Figure 9-11 Device-to-Device Messaging

Router Solicitation and Router Advertisement messages are for messaging between devices and routers. Router discovery is typically used for dynamic address allocation and stateless address autoconfiguration (SLAAC).

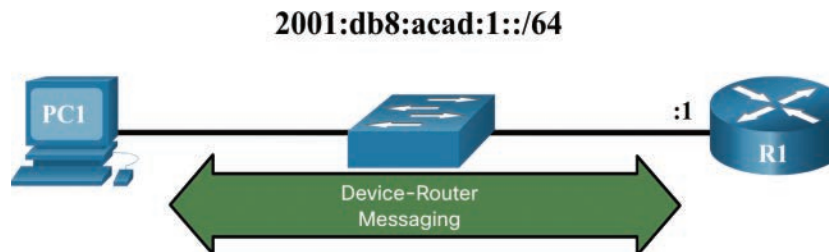


Figure 9-12 Device-Router Messaging

Note

The fifth ICMPv6 ND message is a redirect message that is used for better next hop selection. This is beyond the scope of this book.

IPv6 ND is defined in IETF RFC 4861.

IPv6 Neighbor Discovery—Address Resolution (9.3.3)

Much as with ARP for IPv4, IPv6 devices use IPv6 ND to determine the MAC address of a device that has a known IPv6 address.

ICMPv6 Neighbor Solicitation and Neighbor Advertisement messages are used for MAC address resolution. This is similar to ARP requests and ARP replies used by ARP for IPv4. For example, say that PC1 wants to ping PC2 at IPv6 address 2001:db8:acad:11. To determine the MAC address for the known IPv6 address, PC1 sends an ICMPv6 Neighbor Solicitation message, as illustrated in Figure 9-13.

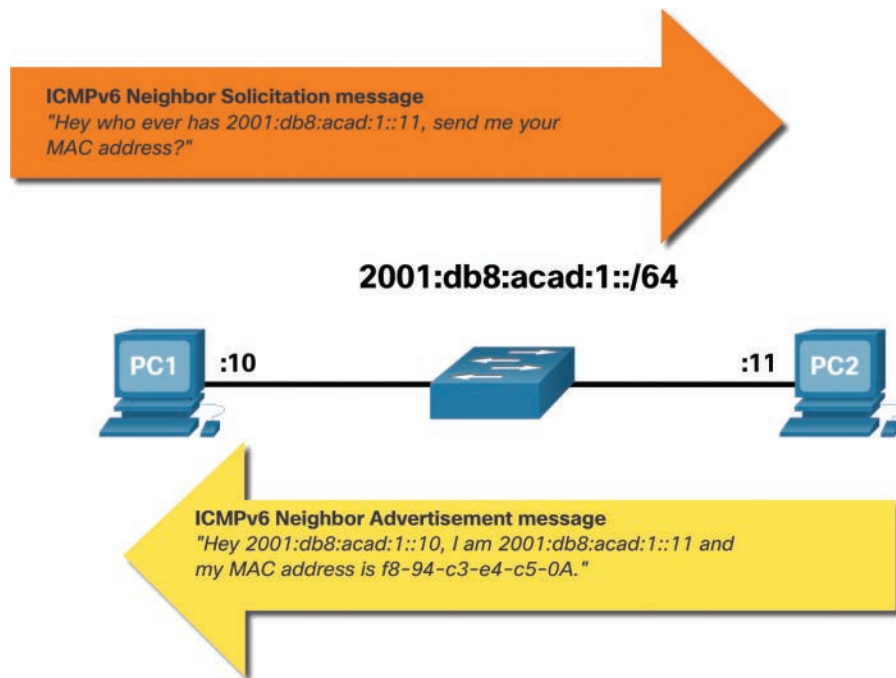


Figure 9-13 IPv6 Neighbor Discovery Process

ICMPv6 Neighbor Solicitation messages are sent using special Ethernet and IPv6 multicast addresses. This allows the Ethernet NIC of the receiving device to determine whether the Neighbor Solicitation message is for itself without having to send it to the operating system for processing.

PC2 replies to the request with an ICMPv6 Neighbor Advertisement message, which includes its MAC address.

Packet Tracer
Activity**Packet Tracer—IPv6 Neighbor Discovery (9.3.4)**

For a device to communicate with another device, the MAC address of the destination device must be known. With IPv6, a process called Neighbor Discovery is responsible for determining the destination MAC address. In this activity, you will gather PDU information in simulation mode to better understand the process. There is no Packet Tracer scoring for this activity.

Refer to the online course to complete this activity.

Interactive
Graphic**Check Your Understanding—Neighbor Discovery (9.3.5)**

Refer to the online course to complete this activity.

Summary (9.4)

The following is a summary of the topics in the chapter and their corresponding online modules.

MAC and IP

Layer 2 physical addresses (that is, Ethernet MAC addresses) are used to deliver the data link frame with the encapsulated IP packet from one NIC to another NIC on the same network. If the destination IP address is on the same network, the destination MAC address is the address of the destination device. When the destination IP address (IPv4 or IPv6) is on a remote network, the destination MAC address is the address of the host default gateway (that is, the router interface). Along each link in a path, an IP packet is encapsulated in a frame. The frame is specific to the data link technology associated with that link, such as Ethernet. If the next hop device is the final destination, the destination MAC address is the address of the device's Ethernet NIC. How are the IP addresses of the IP packets in a data flow associated with the MAC addresses on each link along the path to the destination? For IPv4 packets, this association is done through a process called ARP. For IPv6 packets, the process is ICMPv6 ND.

ARP

Every IP device on an Ethernet network has a unique Ethernet MAC address. When a device sends an Ethernet Layer 2 frame, it contains two addresses: destination MAC address and source MAC address. A device uses ARP to determine the destination MAC address of a local device when it knows its IPv4 address. ARP provides two basic functions: resolving IPv4 addresses to MAC addresses and maintaining a table of IPv4-to-MAC address mappings. The ARP request is encapsulated in an Ethernet frame using header information about the source and destination MAC addresses and type. Only one device on a LAN has an IPv4 address that matches the target IPv4 address in the ARP request. All other devices do not reply. An ARP reply contains the same header fields as a request. Only the device that originally sent the ARP request receives the unicast ARP reply. After the ARP reply is received, the device adds the IPv4 address and the corresponding MAC address to its ARP table. When the destination IPv4 address is not on the same network as the source IPv4 address, the source device needs to send the frame to its default gateway. This is the interface of the local router. For each device, an ARP cache timer removes ARP entries that have not been used for a specified period of time. Commands may also be used to manually remove some or all of the entries in the ARP table. As a broadcast frame, an ARP request is received and processed by every device on the local network, which could cause the network to slow down. A threat actor can use ARP spoofing to perform an ARP poisoning attack.

Neighbor Discovery

IPv6 does not use ARP; rather, it uses the ND protocol to resolve MAC addresses. ND provides address resolution, router discovery, and redirection services for IPv6 using ICMPv6. ICMPv6 ND uses five ICMPv6 messages to perform these services: Neighbor Solicitation, Neighbor Advertisement, Router Solicitation, Router Advertisement, and Redirect. Much as with ARP for IPv4, IPv6 devices use IPv6 ND to resolve the MAC address of a device to a known IPv6 address.

Practice

The following activities provide practice with the topics introduced in this chapter. The lab is available in the companion *Introduction to Networks Labs & Study Guide (CCNAv7)* (ISBN 9780136634454). The Packet Tracer activity instructions are also provided in the *Labs & Study Guide*. The PKA files are available in the online course.

Packet Tracer
□ **Activity**

Packet Tracer Activities

Packet Tracer 9.1.3: Identify MAC and IP Addresses

Packet Tracer 9.2.9: Examine the ARP Table

Packet Tracer 9.3.4: IPv6 Neighbor Discovery

Check Your Understanding Questions

Complete all the review questions listed here to test your understanding of the topics and concepts in this chapter. The appendix “Answers to ‘Check Your Understanding’ Questions” lists the answers.

1. When a host has an IPv4 packet sent to a host on a remote network, what address is requested in the ARP request?
 - a. the IPv4 address of the destination host
 - b. the IPv4 address of the default gateway
 - c. the MAC address of the default gateway, the router interface closest to the sending host
 - d. the MAC address of the switch port that connects to the sending host

2. How does the ARP process use an IPv4 address?
 - a. to determine the MAC address of the remote destination host
 - b. to determine the MAC address of a device on the same network
 - c. to determine the amount of time a packet takes when traveling from source to destination
 - d. to determine the network number based on the number of bits in the IP address
3. Which of the following does the ARP table in a switch map together?
 - a. Layer 3 address to a Layer 2 address
 - b. Layer 3 address to a Layer 4 address
 - c. Layer 4 address to a Layer 2 address
 - d. Layer 2 address to a Layer 4 address
4. What is one function of the ARP protocol?
 - a. obtaining an IPv4 address automatically
 - b. mapping a domain name to its IPv4 address
 - c. resolving an IPv4 address to a MAC address
 - d. maintaining a table of domain names with their resolved IPv4 addresses
5. Which router component holds the routing table, ARP cache, and running configuration file?
 - a. RAM
 - b. Flash
 - c. NVRAM
 - d. ROM
6. What type of information is contained in an ARP table?
 - a. switch ports associated with destination MAC addresses
 - b. domain name-to-IPv4 address mappings
 - c. routes to reach destination networks
 - d. IPv4 address-to-MAC address mappings

7. A cybersecurity analyst believes an attacker is spoofing the MAC address of the default gateway to perform a man-in-the-middle attack. Which command should the analyst use to view the MAC address a host is using to reach the default gateway?
 - a. `ipconfig /all`
 - b. `route print`
 - c. `netstat -r`
 - d. `arp -a`
8. What is a function of ARP?
 - a. resolving known MAC addresses to unknown IPv4 addresses
 - b. resolving known port addresses to unknown MAC addresses
 - c. resolving known MAC addresses to unknown port addresses
 - d. resolving known IPv4 addresses to unknown MAC addresses
9. What is the purpose of ARP in an IPv4 network?
 - a. to forward data onward based on the destination IP address
 - b. to obtain a specific MAC address when an IP address is known
 - c. to forward data onward based on the destination MAC address
 - d. to build the MAC address table in a switch from the information this gathered
10. Which action does a Layer 2 switch take when it receives a Layer 2 broadcast frame?
 - a. It drops the frame.
 - b. It forwards the frame out all ports except the port on which it received the frame.
 - c. It forwards the frame out all ports that are registered to forward broadcasts.
 - d. It forwards the frame out all ports.
11. Which destination MAC address is used in an ARP request?
 - a. 0.0.0.0
 - b. 255.255.255.255
 - c. FFFF.FFFF.FFFF
 - d. 127.0.0.1
 - e. 01-00-5E-00-AA-23

12. What is the destination MAC address of an ICMPv6 Neighbor Solicitation message?
 - a. multicast
 - b. unicast
 - c. broadcast
 - d. anycast

13. What does a Layer 2 switch do when the destination MAC address of a received frame is not in the MAC table?
 - a. It initiates an ARP request.
 - b. It broadcasts the frame out all ports on the switch.
 - c. It notifies the sending host that the frame cannot be delivered.
 - d. It forwards the frame out all ports except the port on which the frame was received.

14. Which two ICMPv6 messages are used during the Ethernet MAC address resolution process? (Choose two.)
 - a. Router Solicitation
 - b. Router Advertisement
 - c. Neighbor Solicitation
 - d. Neighbor Advertisement
 - e. Echo Request

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Basic Router Configuration

Objectives

Upon completion of this chapter, you will be able to answer the following questions:

- How do you configure initial settings on a Cisco IOS router?
- How do you configure two active interfaces on a Cisco IOS router?
- How do you configure devices to use the default gateway?

Introduction (10.0)

Have you ever run a relay race? The first person runs the first leg of the race and hands off the baton to the next runner, who continues forward in the second leg of the race and hands off the baton to the third runner, and so on. If the first runner does not know where to find the second runner, or if the first runner drops the baton before handing it off, then that relay team will most certainly lose the race.

Packet routing is similar to a relay. As you know, routing tables are created and used by routers to forward packets from their local networks on to other networks. But a router cannot create a routing table or forward any packets until it has been configured. If you plan to become a network administrator, you definitely must know how to do this. The good news? It is easy! This chapter includes Syntax Checker activities so that you can practice your configuration commands and see the output. There are also some Packet Tracer activities to get you started. Let's go!

Configure Initial Router Settings (10.1)

This section presents the basic configuration needed for all IOS routers.

Basic Router Configuration Steps (10.1.1)

The following tasks should be completed to configure the initial settings on a router:



Step 1. Configure the device name:

```
Router(config)# hostname hostname
```

Step 2. Secure privileged EXEC mode:

```
Router(config)# enable secret password
```

Step 3. Secure user EXEC mode:

```
Router(config)# line console 0  
Router(config-line)# password password  
Router(config-line)# login
```

Step 4. Secure remote Telnet/SSH access:

```
Router(config-line)# line vty 0 4  
Router(config-line)# password password  
Router(config-line)# login  
Router(config-line)# transport input {ssh | telnet}
```

Step 5. Secure all passwords in the config file:

```
Router(config-line)# exit  
Router(config)# service password-encryption
```

Step 6. Provide a legal notification banner:

```
Router(config)# banner motd delimiter message delimiter
```

Step 7. Save the configuration:

```
Router(config)# end
```

```
Router# copy running-config startup-config
```

Basic Router Configuration Example (10.1.2)

This section provides an example, using the topology in Figure 10-1. In this example, you will see how to configure router R1 with initial settings.

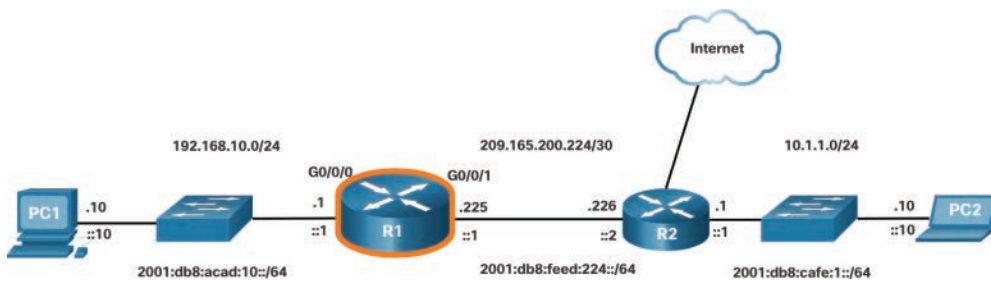


Figure 10-1 Basic Router Configuration Reference Topology

To configure the device name for R1, use the commands in Example 10-1.

Example 10-1 Configuring a Device Name

```
Router> enable
Router# configure terminal
Enter configuration commands, one per line.
End with CNTL/Z.
Router(config)# hostname R1
R1(config)#
```

Note

Notice in Example 10-1 that the router prompt now displays **hostname**.

All router access should be secured. Privileged EXEC mode provides a user with complete access to a device and its configuration. Therefore, it is the most important mode to secure.

The commands in Example 10-2 secure privileged EXEC mode and user EXEC mode, enable Telnet and SSH remote access, and encrypt all plaintext (that is, user EXEC and vty line) passwords.

Example 10-2 Securing Access to a Router

```
R1(config)# enable secret class
R1(config)#
R1(config)# line console 0
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# exit
R1(config)#
R1(config)# line vty 0 4
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# transport input ssh telnet
R1(config-line)# exit
R1(config)#
R1(config)# service password-encryption
R1(config)#
```

It is a good idea to provide a legal notification that warns users that the device should be accessed only by permitted users. Legal notification is configured as shown in Example 10-3.

Example 10-3 Configuring a Banner Warning

```
R1(config)# banner motd #
Enter TEXT message. End with a new line and the #
*****
WARNING: Unauthorized access is prohibited!
*****
#
R1(config)#
```

If the commands shown so far are all entered on a router that accidentally loses power, all this configuration is lost. For this reason, it is important to save the configuration when changes are implemented. The command shown in Example 10-4 saves the configuration to NVRAM.

Example 10-4 Saving the Running Configuration

```
R1# copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#
```

**Interactive
Graphic**
Syntax Checker—Configure Initial Router Settings (10.1.3)

Use this Syntax Checker activity to practice configuring the initial settings on a router:

- Configure the device name.
- Secure the privileged EXEC mode.
- Secure and enable remote SSH and Telnet access.
- Secure all plaintext passwords.
- Provide legal notification.

Refer to the online course to complete this activity.

**Packet Tracer
Activity**
Packet Tracer—Configure Initial Router Settings (10.1.4)

In this activity, you will perform basic router configurations. You will secure access to the CLI and console port using encrypted and plaintext passwords. You will also configure messages for users logging in to the router. These banners also warn unauthorized users that access is prohibited. Finally, you will verify and save your running configuration.

Configure Interfaces (10.2)

This section introduces the basic router interface configuration.

Configure Router Interfaces (10.2.1)

When your routers have basic configurations, the next step is to configure their interfaces. This is necessary because routers are not reachable by end devices until the interfaces are configured. There are many different types of interfaces available on Cisco routers. For example, the Cisco ISR 4321 router is equipped with two Gigabit Ethernet interfaces:

- GigabitEthernet 0/0/0 (G0/0/0)
- GigabitEthernet 0/0/1 (G0/0/1)

Configuring a router interface is similar to configuring a management SVI on a switch. Specifically, it involves issuing the following commands:

```
Router(config)# interface type-and-number
Router(config-if)# description description-text
Router(config-if)# ip address ipv4-address subnet-mask
Router(config-if)# ipv6 address ipv6-address/prefix-length
Router(config-if)# no shutdown
```

Note

When a router interface is enabled, information messages should be displayed to confirm the enabled link.

Although the **description** command is not required to enable an interface, using it is a good practice. This command can be helpful in troubleshooting on production networks because it enables you to provide information about the type of network connected. For example, for an interface that connects to an ISP or a service carrier, you can use the **description** command to enter the third-party connection and contact information.

Note

description-text is limited to 240 characters.

Using the **no shutdown** command activates the interface and is similar to powering on the interface. The interface must also be connected to another device, such as a switch or a router, for the physical layer to be active.

Note

On inter-router connections where there is no Ethernet switch, both interconnecting interfaces must be configured and enabled.

Configure Router Interfaces Example (10.2.2)

Example 10-5 shows how to enable the directly connected interfaces of R1 in Figure 10-1.

Example 10-5 Configuring the Router Interfaces with Dual Stack Addressing

```
R1> enable
R1# configure terminal
Enter configuration commands, one per line.
End with CNTL/Z.
R1(config)# interface gigabitEthernet 0/0/0
R1(config-if)# description Link to LAN
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# ipv6 address 2001:db8:acad:10::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
```



```

*Aug  1 01:43:53.435: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state
to down
*Aug  1 01:43:56.447: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state
to up
*Aug  1 01:43:57.447: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0/0, changed state to up
R1(config)#
R1(config)#
R1(config)# interface gigabitEthernet 0/0/1
R1(config-if)# description Link to R2
R1(config-if)# ip address 209.165.200.225 255.255.255.252
R1(config-if)# ipv6 address 2001:db8:feed:224::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
*Aug  1 01:46:29.170: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state
to down
*Aug  1 01:46:32.171: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state
to up
*Aug  1 01:46:33.171: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0/1, changed state to up
R1(config)#

```

Note

Notice the informational messages that says G0/0/0 and G0/0/1 are enabled.

Verify Interface Configuration (10.2.3)

Several commands can be used to verify interface configuration. The most useful of them are the **show ip interface brief** and **show ipv6 interface brief** commands, as shown in Example 10-6.

Example 10-6 Verifying the Interface Configuration

```

R1# show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0/0	192.168.10.1	YES	manual	up	up
GigabitEthernet0/0/1	209.165.200.225	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

```

R1# show ipv6 interface brief
GigabitEthernet0/0/0    [up/up]
FE80::201:C9FF:FE89:4501
2001:DB8:ACAD:10::1

```

```
GigabitEthernet0/0/1      [up/up]
    FE80::201:C9FF:FE89:4502
    2001:DB8:FEED:224::1

Vlan1                    [administratively down/down]
    unassigned

R1#
```

Configuration Verification Commands (10.2.4)

Table 10-1 describes the **show** commands that are most commonly used to verify interface configuration.

Table 10-1 Verification Commands

Commands	Output Description
show ip interface brief show ipv6 interface brief	Displays all interfaces, their IP addresses, and their current status. The configured and connected interfaces should indicate up under Status and up under Protocol. Anything else indicates a problem with either the configuration or the cabling.
show ip route show ipv6 route	Displays the contents of the IP routing tables stored in RAM.
show interfaces	Displays statistics for all interfaces on the device. However, this command displays only the IPv4 addressing information.
show ip interfaces	Displays the IPv4 statistics for all interfaces on a router.
show ipv6 interface	Displays the IPv6 statistics for all interfaces on a router.

Examples 10-7 through 10-13 show the command output for these configuration verification commands.

Example 10-7 The **show ip interface brief** Command

```
R1# show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0 192.168.10.1    YES manual up          up
GigabitEthernet0/0/1 209.165.200.225 YES manual up          up
Vlan1              unassigned      YES unset  administratively down down
R1#
```

Example 10-8 The show ipv6 interface brief Command

```

R1# show ipv6 interface brief
GigabitEthernet0/0/0          [up/up]
    FE80::201:C9FF:FE89:4501
    2001:DB8:ACAD:10::1
GigabitEthernet0/0/1          [up/up]
    FE80::201:C9FF:FE89:4502
    2001:DB8:FEED:224::1
Vlan1                          [administratively down/down]
    unassigned
R1#

```

Example 10-9 The show ip route Command

```

R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set
  192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
  209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.200.224/30 is directly connected, GigabitEthernet0/0/1
L       209.165.200.225/32 is directly connected, GigabitEthernet0/0/1
R1#

```

Example 10-10 The show ipv6 route Command

```

R1# show ipv6 route
IPv6 Routing Table - default - 5 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
       NDr - Redirect, RL - RPL, O - OSPF Intra, OI - OSPF Inter
       OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1
       ON2 - OSPF NSSA ext 2, a - Application

```

```

C    2001:DB8:ACAD:10::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L    2001:DB8:ACAD:10::1/128 [0/0]
    via GigabitEthernet0/0/0, receive
C    2001:DB8:FEED:224::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L    2001:DB8:FEED:224::1/128 [0/0]
    via GigabitEthernet0/0/1, receive
L    FF00::/8 [0/0]
    via Null0, receive
R1#

```

Example 10-11 The show interfaces Command

```

R1# show interfaces gig0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Hardware is ISR4321-2x1GE, address is a0e0.af0d.e140 (bia a0e0.af0d.e140)
  Description: Link to LAN
  Internet address is 192.168.10.1/24
  MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive not supported
  Full Duplex, 100Mbps, link type is auto, media type is RJ45
  output flow-control is off, input flow-control is off
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:01, output 00:00:35, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    1180 packets input, 109486 bytes, 0 no buffer
    Received 84 broadcasts (0 IP multicasts)
    0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
    0 watchdog, 1096 multicast, 0 pause input
    65 packets output, 22292 bytes, 0 underruns
    0 output errors, 0 collisions, 2 interface resets
    11 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    1 lost carrier, 0 no carrier, 0 pause output
    0 output buffer failures, 0 output buffers swapped out
R1#

```

Example 10-12 The show ip interface Command

```
R1# show ip interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Internet address is 192.168.10.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing Common access list is not set
  Outgoing access list is not set
  Inbound Common access list is not set
  Inbound access list is not set
  Proxy ARP is enabled
  Local Proxy ARP is disabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachable are always sent
  ICMP mask replies are never sent
  IP fast switching is enabled
  IP Flow switching is disabled
  IP CEF switching is enabled
  IP CEF switching turbo vector
  IP Null turbo vector
  Associated unicast routing topologies:
    Topology "base", operation state is UP
  IP multicast fast switching is enabled
  IP multicast distributed fast switching is disabled
  IP route-cache flags are Fast, CEF
  Router Discovery is disabled
  IP output packet accounting is disabled
  IP access violation accounting is disabled
  TCP/IP header compression is disabled
  RTP/IP header compression is disabled
  Probe proxy name replies are disabled
  Policy routing is disabled
  Network address translation is disabled
  BGP Policy Mapping is disabled
  Input features: MCI Check
  IPv4 WCCP Redirect outbound is disabled
  IPv4 WCCP Redirect inbound is disabled
  IPv4 WCCP Redirect exclude is disabled
R1#
```

Example 10-13 The show ipv6 interface Command

```
R1# show ipv6 interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::868A:8DFF:FE44:49B0
  No Virtual link-local address(es):
  Description: Link to LAN
  Global unicast address(es):
    2001:DB8:ACAD:10::1, subnet is 2001:DB8:ACAD:10::/64
  Joined group address(es):
    FF02::1
    FF02::1:FF00:1
    FF02::1:FF44:49B0
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND NS retransmit interval is 1000 milliseconds
R1#
```

**Interactive
Graphic****Syntax Checker—Configure Interfaces (10.2.5)**

Use this Syntax Checker activity to practice configuring the GigabitEthernet 0/0 interface on a router:

- Describe the link as Link to LAN.
- Configure the IPv4 address as 192.168.10.1 with the subnet mask 255.255.255.0.
- Configure the IPv6 address as 2001:db8:acad:10::1 with the /64 prefix length.
- Activate the interface.

Refer to the online course to complete this activity.

Configure the Default Gateway (10.3)

To send a packet outside the local network, a device needs to know where to forward the packet. For an end device, this is generally called the *default gateway*. This section introduces the concept and use of the default gateway.

Default Gateway on a Host (10.3.1)

If a local network has only one router, that router is the gateway router, and all hosts and switches on the network must be configured with this information. If a local network has multiple routers, one of them must be designated as the default gateway router. This section explains how to configure the default gateway on hosts and switches.

For an end device to communicate over a network, it must be configured with the correct IP address information, including the default gateway address. The default gateway is used only when the host wants to send a packet to a device on another network. The default gateway address is generally the router interface address attached to the local network of the host. The IP address of the host device and the router interface address must be in the same network.

For example, say that an IPv4 network topology consists of a router interconnecting two separate LANs. G0/0/0 is connected to network 192.168.10.0, while G0/0/1 is connected to network 192.168.11.0. Each host device is configured with the appropriate default gateway address.

In Figure 10-2, if PC1 sends a packet to PC2, the default gateway is not used. Instead, PC1 addresses the packet with the IPv4 address of PC2 and forwards the packet directly to PC2 through the switch.

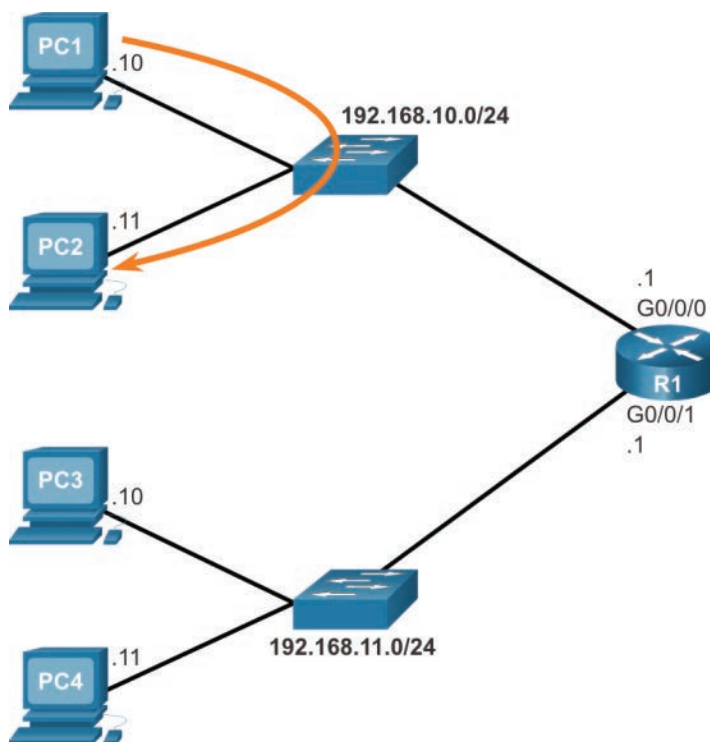


Figure 10-2 No Default Gateway Needed

What if PC1 sent a packet to PC3? PC1 would address the packet with the IPv4 address of PC3 but would forward the packet to its default gateway, which is the G0/0/0 interface of R1. The router accepts the packet and accesses its routing table to determine that G0/0/1 is the appropriate exit interface based on the destination address. R1 then forwards the packet out the appropriate interface to reach PC3, as shown in Figure 10-3.

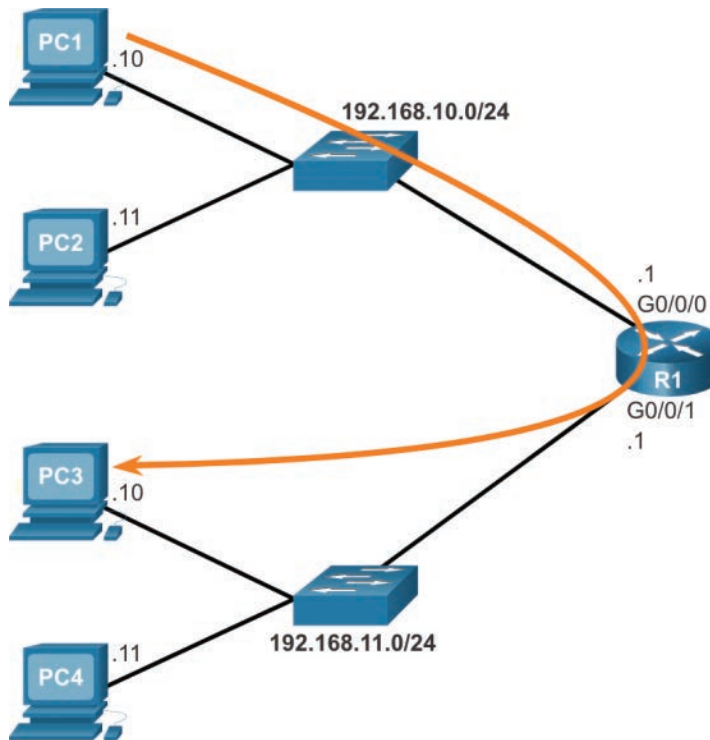


Figure 10-3 Default Gateway Is Required

The same process would occur on an IPv6 network, although this is not shown in Figures 10-2 and 10-3. Devices would use the IPv6 address of the local router as their default gateway.

Default Gateway on a Switch (10.3.2)

A switch that interconnects client computers is typically a Layer 2 device. A Layer 2 switch does not require an IP address to function properly. However, IP settings can be configured on a switch to give an administrator remote access to the switch.

To connect to and manage a switch over a local IP network, that switch must have a switch virtual interface (SVI) configured. The SVI is configured with an IPv4 address

and subnet mask on the local LAN. The switch must also have a default gateway address configured to remotely manage the switch from another network.

The default gateway address is typically configured on all devices that will communicate beyond their local network.

To configure an IPv4 default gateway on a switch, use the **ip default-gateway** *ip-address* global configuration command, where *ip-address* is the IPv4 address of the local router interface connected to the switch.

Figure 10-4 shows an administrator establishing a remote connection to switch S1 on another network and executing the **show running-config** command.

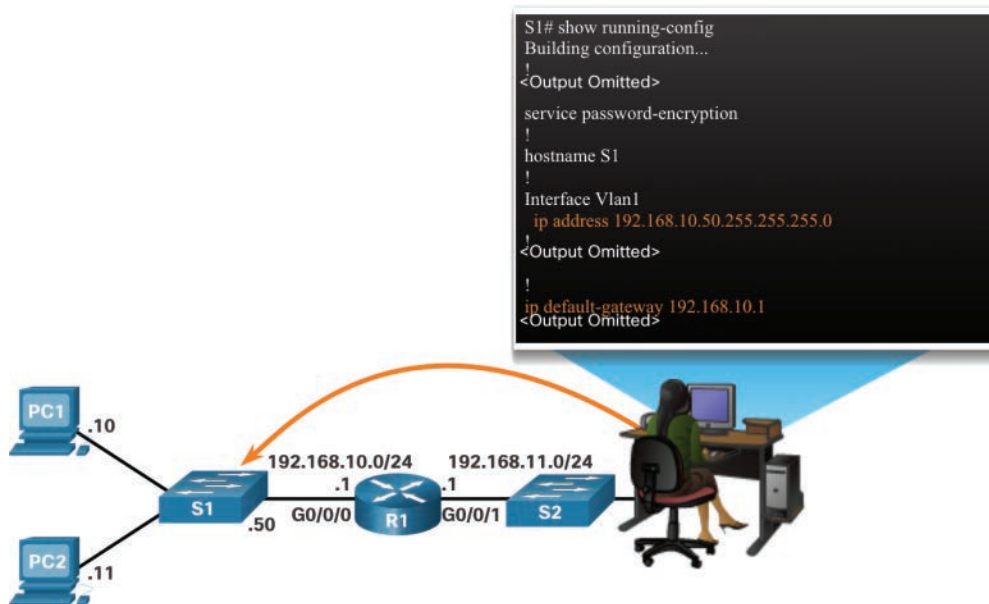


Figure 10-4 Remotely Accessing a Switch

In Figure 10-4, the administrator host would use its default gateway to send the packet to the G0/0/1 interface of R1. R1 would forward the packet to S1 out its G0/0/0 interface. Because the packet source IPv4 address is from another network, S1 would require a default gateway to forward the packet to the G0/0/0 interface of R1. Therefore, S1 must be configured with a default gateway to be able to reply and establish an SSH connection with the administrative host.

Note

Packets originating from host computers connected to the switch must already have the default gateway address configured on their host computer operating systems.

A workgroup switch can also be configured with an IPv6 address on an SVI. However, the switch does not require the IPv6 address of the default gateway to be configured manually. The switch automatically receives its default gateway from the ICMPv6 Router Advertisement message from the router.

**Interactive
Graphic****Syntax Checker—Configure the Default Gateway (10.3.3)**

Use this Syntax Checker activity to practice configuring the default gateway of a Layer 2 switch.

Refer to the online course to complete this activity.

**Packet Tracer
Activity****Packet Tracer—Connect a Router to a LAN (10.3.4)**

In this activity, you will use various **show** commands to display the current state of a router. You will then use the addressing table to configure router Ethernet interfaces. Finally, you will use commands to verify and test your configurations.

**Packet Tracer
Activity****Packet Tracer—Troubleshoot Default Gateway Issues (10.3.5)**

For a device to communicate across multiple networks, it must be configured with an IP address, a subnet mask, and a default gateway. The default gateway is used when the host wants to send a packet to a device on another network. The default gateway address is generally the router interface address attached to the local network to which the host is connected. In this activity, you will finish documenting the network. You will then verify the network documentation by testing end-to-end connectivity and troubleshooting issues. The troubleshooting method you will use consists of the following steps:

- Step 1.** Verify the network documentation and use tests to isolate problems.
 - Step 2.** Determine an appropriate solution for a given problem.
 - Step 3.** Implement the solution.
 - Step 4.** Test to verify that the problem is resolved.
 - Step 5.** Document the solution.
-

Summary (10.4)

The following is a summary of the topics in the chapter and their corresponding online modules.

Configure Initial Router Settings

The following tasks should be completed to configure the initial settings on a router:

- Step 1.** Configure the device name.
- Step 2.** Secure privileged EXEC mode.
- Step 3.** Secure user EXEC mode.
- Step 4.** Secure remote Telnet/SSH access.
- Step 5.** Secure all passwords in the config file.
- Step 6.** Provide a legal notification banner.
- Step 7.** Save the configuration.

Configure Interfaces

For routers to be reachable, the router interfaces must be configured. The Cisco ISR 4321 router is equipped with two Gigabit Ethernet interfaces: GigabitEthernet 0/0/0 (G0/0/0) and GigabitEthernet 0/0/1 (G0/0/1). Configuring a router interface is very similar to configuring a management SVI on a switch. Using the **no shutdown** command activates the interface. The interface must also be connected to another device, such as a switch or a router, for the physical layer to be active. Several commands can be used to verify interface configuration, including **show ip interface brief**, **show ipv6 interface brief**, **show ip route**, **show ipv6 route**, **show interfaces**, **show ip interface**, and **show ipv6 interface**.

Configure the Default Gateway

For an end device to communicate over the network, it must be configured with the correct IP address information, including the default gateway address. The default gateway address is generally the router interface address for the router that is attached to the local network of the host. The IP address of the host device and the router interface address must be in the same network. To connect to and manage a switch over a local IP network, it must have a switch virtual interface (SVI) configured. The SVI is configured with an IPv4 address and subnet mask on the local LAN. The switch must also have a default gateway address configured to remotely manage the switch from another network. To configure an IPv4 default gateway on a switch,

use the **ip default-gateway ip-address** global configuration command. Use the IPv4 address of the local router interface that is connected to the switch.

Video**Video—Network Device Differences: Part 1 (10.4.1)**Video**Video—Network Device Differences: Part 2 (10.4.2)**Packet Tracer
Activity**Packet Tracer—Basic Device Configuration (10.4.3)**

Your network manager is impressed with your performance in your job as a LAN technician. She would like you to now demonstrate your ability to configure a router connecting two LANs. Your tasks include configuring basic settings on a router and a switch using Cisco IOS. You will then verify your configurations as well as the configurations on existing devices by testing end-to-end connectivity.

**Lab—Build a Switch and Router Network (10.4.4)**

In this lab, you will complete the following objectives:

- Part 1: Set Up the Topology and Initialize Devices
 - Part 2: Configure Devices and Verify Connectivity
 - Part 3: Display Device Information
-

Practice

The following activities provide practice with the topics introduced in this chapter. The lab is available in the companion *Introduction to Networks Labs & Study Guide (CCNAv7)* (ISBN 9780136634454). The Packet Tracer activity instructions are also provided in the *Labs & Study Guide*. The PKA files are available in the online course.

**Lab**

Lab 10.4.4: Build a Switch and Router Network

Packet Tracer
Activity**Packet Tracer Activities**

Packet Tracer 10.1.4: Configure Initial Router Settings

Packet Tracer 10.3.4: Connect a Router to a LAN

Packet Tracer 10.3.5: Troubleshoot Default Gateway Issues

Packet Tracer 10.4.3: Basic Device Configuration

Check Your Understanding Questions

Complete all the review questions listed here to test your understanding of the topics and concepts in this chapter. The appendix “Answers to ‘Check Your Understanding’ Questions” lists the answers.

1. A router boots without any preconfigured commands. What is the reason for this?
 - a. The IOS image is corrupt.
 - b. Cisco IOS is missing from flash memory.
 - c. The configuration file is missing from NVRAM.
 - d. The POST process has detected hardware failure.
2. Which command is used to encrypt all passwords in a router configuration file?
 - a. Router_A(config)# **enable secret** <password>
 - b. Router_A(config)# **service password-encryption**
 - c. Router_A(config)# **enable password** <password>
 - d. Router_A(config)# **encrypt password**
3. Company policy requires you to use the most secure method to safeguard access to the privileged EXEC and configuration mode on the routers. The privileged EXEC password is **trustknow1**. Which of the following router commands achieves the goal of providing the highest level of security?
 - a. **secret password trustknow1**
 - b. **enable password trustknow1**
 - c. **service password-encryption**
 - d. **enable secret trustknow1**

4. What will be the command prompt from the router after the command `router(config)# hostname portsmouth` is entered?
 - a. `portsmouth#`
 - b. `portsmouth(config)#`
 - c. invalid input detected
 - d. `hostname portsmouth#`
 - e. `? command not recognized`
`Router(config)#`
5. An administrator is configuring a new router to permit out-of-band management access. Which set of commands will allow the required login using the password `cisco`?
 - a. `Router(config)# line vty 0 4`
`Router(config-line) password manage`
`Router(config-line) exit`
`Router(config)# enable password cisco`
 - b. `Router(config)# line vty 0 4`
`Router(config-line) password cisco`
`Router(config-line) login`
 - c. `Router(config)# line console 0`
`Router(config-line) password cisco`
`Router(config-line) login`
 - d. `Router(config)# line console 0`
`Router(config-line) password cisco`
`Router(config-line) exit`
`Router(config)# service password-encryption`
6. Which command can be used on a Cisco router to display all interfaces, the IPv4 address assigned, and the current status?
 - a. `show ip interface brief`
 - b. `ping`
 - c. `show ip route`
 - d. `show interface fa0/1`

7. Which CLI mode allows users to access all device commands, such as those used for configuration, management, and troubleshooting?
 - a. user EXEC mode
 - b. privileged EXEC mode
 - c. global configuration mode
 - d. interface configuration mode
8. What is the purpose of the startup configuration file on a Cisco router?
 - a. It facilitates the basic operation of the hardware components of a device.
 - b. It contains the commands that are used to initially configure a router on startup.
 - c. It contains the configuration commands that the router IOS is currently using.
 - d. It provides a limited backup version of IOS, in case the router cannot load the full-featured IOS.
9. Which characteristic describes the default gateway of a host computer?
 - a. the logical address of the router interface on the same network as the host computer
 - b. the physical address of the switch interface connected to the host computer
 - c. the physical address of the router interface on the same network as the host computer
 - d. the logical address assigned to the switch interface connected to the router
10. What is the purpose of the **banner motd** command?
 - a. It configures a message that identifies printed documents to LAN users.
 - b. Routers use it to communicate the status of their links with one another.
 - c. It provides an easy way of communicating with any user attached to a router's LAN.
 - d. It provides a way to make announcements to those who log in to a router.
11. A technician is configuring a router to allow for all forms of management access. As part of each different type of access, the technician is trying to type the command **login**. Which configuration mode should be used to do this task?
 - a. user executive mode
 - b. global configuration mode
 - c. console line and vty line configuration modes
 - d. privileged EXEC mode

12. What is stored in the NVRAM of a Cisco router?
- a. Cisco IOS
 - b. the running configuration
 - c. the bootup instructions
 - d. the startup configuration
13. Which statement regarding the **service password-encryption** command is true?
- a. It is configured in privileged EXEC mode.
 - b. It encrypts only line mode passwords.
 - c. When the **service password-encryption** command is entered, all plaintext passwords are encrypted.
 - d. To see the passwords encrypted by the **service password-encryption** command in plaintext, issue the **no service password-encryption** command.

IPv4 Addressing

Objectives

Upon completion of this chapter, you will be able to answer the following questions:

- What is the structure of an IPv4 address, including the network portion, the host portion, and the subnet mask?
- What are the characteristics and uses of unicast, broadcast, and multicast IPv4 addresses?
- What are public, private, and reserved IPv4 addresses?
- How does subnetting a network enable better communication?
- How do you calculate IPv4 subnets for a /24 prefix?
- How do you calculate IPv4 subnets for /16 and /8 prefixes?
- Given a set of requirements for subnetting, how do you implement an IPv4 addressing scheme?
- How do you create a flexible addressing scheme using variable-length subnet masking (VLSM)?
- How do you implement a VLSM addressing scheme?

Key Terms

This chapter uses the following key terms. You can find the definitions in the glossary at the end of the book.

octet boundary page 364

intranet page 375

DMZ page 375

variable-length subnet masking (VLSM) page 381

Introduction (1 1.0)

Plenty of networks are still using IPv4 addressing today, while organizations are making the transition to IPv6. So it is still very important for network administrators to know everything they can about IPv4 addressing. This chapter covers the fundamental aspects of IPv4 addressing in detail. It covers how to segment a network into subnets and how to use variable-length subnet masking (VLSM) as part of an overall IPv4 addressing scheme. Subnetting is like cutting a pie into smaller and smaller pieces. Subnetting may seem overwhelming at first, but we show you some tricks to help you along the way. This chapter includes several videos, activities to help you practice subnetting, Packet Tracer activities, and labs. Once you get the hang of Pv4 addressing, you'll be on your way to network administration!

IPv4 Address Structure (1 1.1)

This section presents the IPv4 address structure.

Network and Host Portions (1 1.1.1)

An IPv4 address is a 32-bit hierarchical address that is made up of a network portion and a host portion. When determining the network portion versus the host portion, you must look at the 32-bit stream, as shown in Figure 11-1.

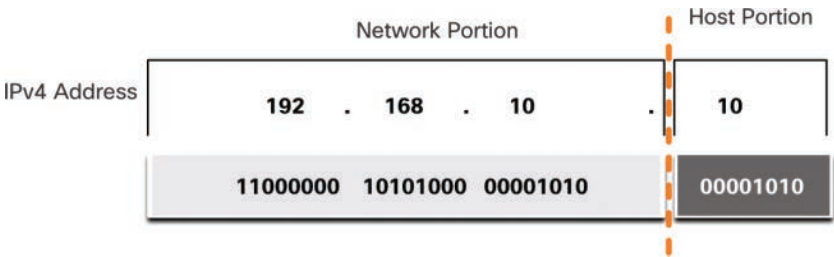


Figure 11-1 Network and Host Portions of an IPv4 Address

The bits in the network portion of the address must be identical for all devices that reside in the same network. The bits in the host portion of the address must be unique to identify a specific host within a network. If two hosts have the same bit pattern in the specified network portion of the 32-bit stream, those two hosts reside in the same network.

But how do hosts know which portion of the 32 bits identifies the network and which identifies the host? That is the role of the subnet mask.

The Subnet Mask (11.1.2)

As shown in Figure 11-2, assigning an IPv4 address to a host requires the following:

- **IPv4 address:** This is the unique IPv4 address of the host.
- **Subnet mask:** This is used to identify the network/host portion of the IPv4 address.

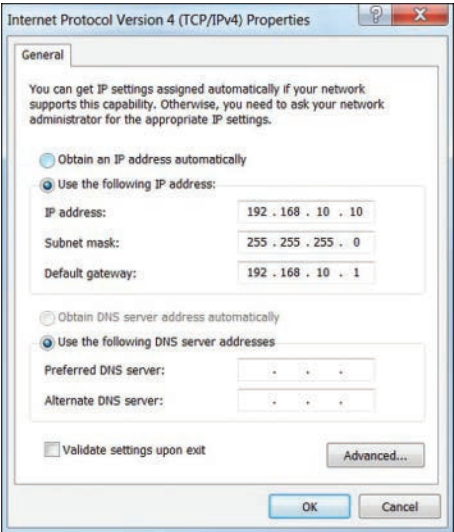


Figure 11-2 IPv4 Addressing on a Windows PC

Note

A default gateway IPv4 address is required to reach remote networks, and DNS server IPv4 addresses are required to translate domain names to IPv4 addresses.

The IPv4 subnet mask is used to differentiate the network portion from the host portion of an IPv4 address. When an IPv4 address is assigned to a device, the subnet mask is used to determine the network address of the device. The network address represents all the devices on the same network.

Figure 11-3 displays the 32-bit subnet mask in dotted decimal and binary formats.

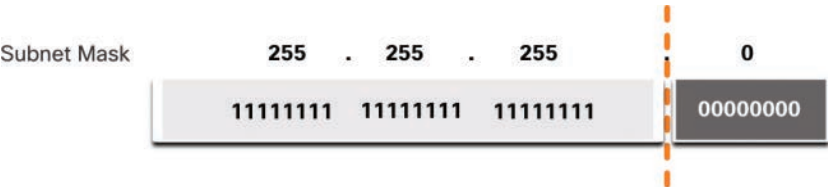


Figure 11-3 32-Bit Subnet Mask

Notice that the subnet mask is a consecutive sequence of 1 bits followed by a consecutive sequence of 0 bits.

To identify the network and host portions of an IPv4 address, the subnet mask is compared to the IPv4 address bit for bit, from left to right, as shown in Figure 11-4.

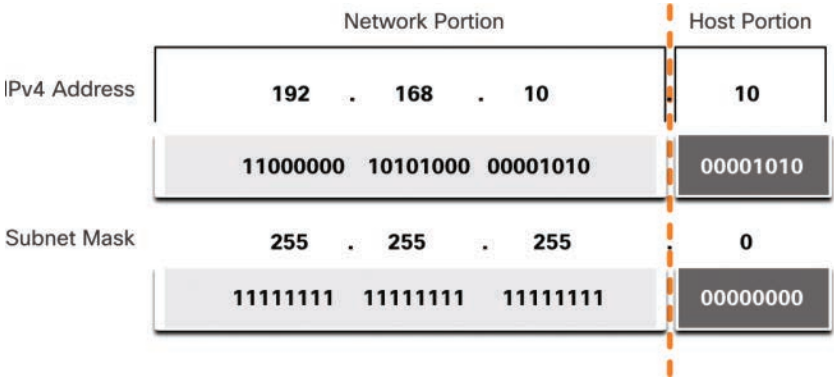


Figure 11-4 Subnet Mask Compared to IPv4 Address

Note that the subnet mask does not actually contain the network or host portion of an IPv4 address; it just tells the computer where to look for the part of the IPv4 address that is the network portion and where to look for the host portion. The process used to identify the network portion and host portion is called ANDing.

The Prefix Length (11.1.3)

Expressing network addresses and host addresses by using dotted decimal subnet mask addresses can be cumbersome. Fortunately, there is an alternative method of identifying a subnet mask: a method called the prefix length.

The prefix length is the number of bits set to 1 in the subnet mask. It is written in “slash notation,” with a forward slash (/) followed by the number of bits set to 1. To figure out the prefix length, count the number of bits in the subnet mask and pre-pend it with a slash. Table 11-1 provides some examples. The first column lists various subnet masks that can be used with a host address. The second column displays the converted 32-bit binary address. The last column displays the resulting prefix length.

Table 11-1 Comparing the Subnet Mask and Prefix Length

Subnet Mask	32-Bit Address	Prefix Length
255.0.0.0	11111111.00000000.00000000.00000000	/8
255.255.0.0	11111111.11111111.00000000.00000000	/16
255.255.255.0	11111111.11111111.11111111.00000000	/24

Subnet Mask	32-Bit Address	Prefix Length
255.255.255.128	11111111.11111111.11111111.10000000	/25
255.255.255.192	11111111.11111111.11111111.11000000	/26
255.255.255.224	11111111.11111111.11111111.11100000	/27
255.255.255.240	11111111.11111111.11111111.11110000	/28
255.255.255.248	11111111.11111111.11111111.11111000	/29
255.255.255.252	11111111.11111111.11111111.11111100	/30

Note

A network address is also referred to as a *prefix* or *network prefix*. Therefore, the prefix length is the number of 1 bits in the subnet mask.

When representing an IPv4 address using a prefix length, the IPv4 address is written followed by the prefix length with no spaces. For example, 192.168.10.10 255.255.255.0 would be written as 192.168.10.10/24. Later in this chapter, you'll learn more about using various types of prefix lengths. For now, we focus on the /24 (that is, 255.255.255.0) prefix.

Determining the Network: Logical AND (11.1.4)

A logical AND is one of three Boolean operations used in Boolean or digital logic. The other two are OR and NOT. The AND operation is used in determining the network address.

Logical AND compares two bits and produces a result, as shown here:

- 1 AND 1 = 1
- 0 AND 1 = 0
- 1 AND 0 = 0
- 0 AND 0 = 0

Note that only a 1 AND 1 produces 1. Any other combination results in 0.

Note

In digital logic, 1 represents true, and 0 represents false. When using an AND operation, both input values must be true (1) for the result to be true (1).

To identify the network address of an IPv4 host, the IPv4 address is logically ANDed, bit by bit, with the subnet mask. ANDing between the address and the subnet mask yields the network address.

To illustrate how AND is used to discover a network address, consider a host with IPv4 address 192.168.10.10 and subnet mask 255.255.255.0, as shown in Figure 11-5:

- **IPv4 host address (192.168.10.10):** Display the IPv4 address of the host in dotted decimal and binary formats.
- **Subnet mask (255.255.255.0):** Display the subnet mask of the host in dotted decimal and binary formats.
- **Network address (192.168.10.0):** The logical AND operation between the IPv4 address and subnet mask results in an IPv4 network address shown in dotted decimal and binary formats.

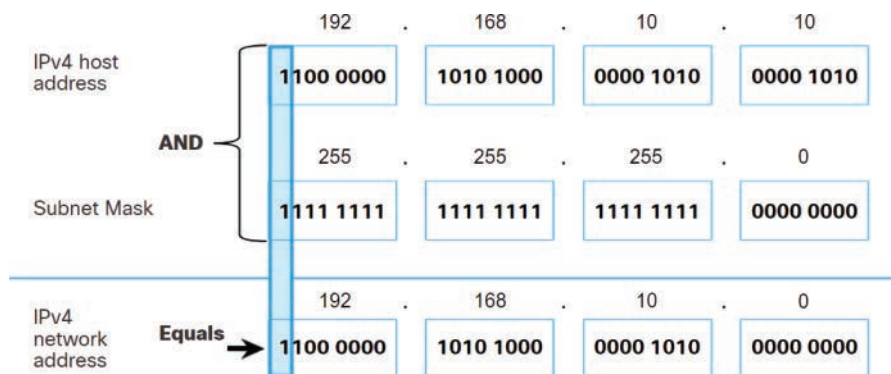


Figure 11-5 ANDing Example

Using the first sequence of bits as an example, notice that the AND operation is performed on the 1 bit of the host address with the 1 bit of the subnet mask. This results in a 1 bit for the network address: 1 AND 1 = 1.

The AND operation between an IPv4 host address and subnet mask results in the IPv4 network address for this host. In this example, the AND operation between the host address 192.168.10.10 and the subnet mask 255.255.255.0 (/24), results in the IPv4 network address 192.168.10.0/24. This is an important IPv4 operation, as it tells the host what network it belongs to.

Video

Video—Network, Host, and Broadcast Addresses (11.1.5)

Refer to the online course to view this video.

Network, Host, and Broadcast Addresses (11.1.6)

Within each network are three types of IP addresses:

- Network address
- Host addresses
- Broadcast address

The following sections examine these three types of addresses, using the topology in Figure 11-6.

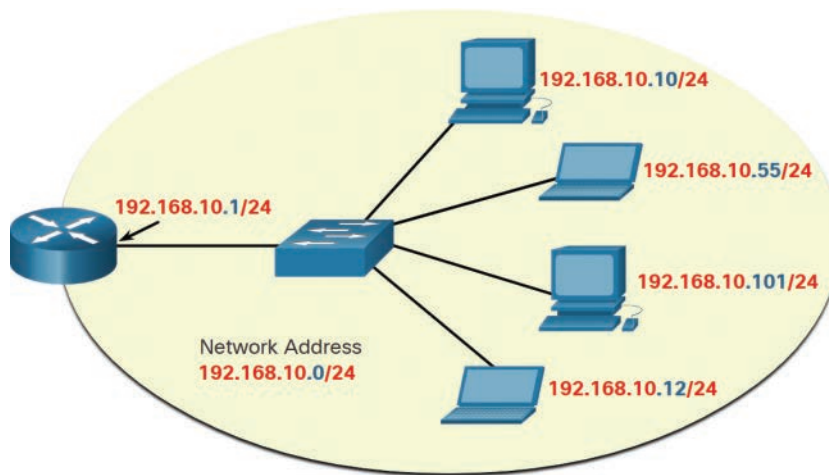


Figure 11-6 Network Address and Host Addresses Example

Network Address

A network address is an address that represents a specific network. A device belongs to this network if it meets three criteria:

- It has the same subnet mask as the network address.
- It has the same network bits as the network address, as indicated by the subnet mask.
- It is located in the same broadcast domain as other hosts with the same network address.

A host determines its network address by performing an AND operation between its IPv4 address and its subnet mask.

As shown in Table 11-2, the network address has all 0 bits in the host portion, as determined by the subnet mask. In this example, the network address is 192.168.10.0/24. A network address cannot be assigned to a device.

Table 11-2 Network, Host, and Broadcast Addresses

	Network Portion			Host Portion	Host Bits
Subnet mask: 255.255.255.0 or /24	255	255	255	0	
	11111111	11111111	11111111	00000000	
Network address: 192.168.10.0 or /24	192	168	10	0	All 0s
	11000000	10100000	00001010	00000000	
First address: 192.168.10.1 or /24	192	168	10	1	All 0s and a 1
	11000000	10100000	00001010	00000001	
Last address: 192.168.10.254 or /24	192	168	10	254	All 1s and a 0
	11000000	10100000	00001010	11111110	
Broadcast address: 192.168.10.255 or /24	192	168	10	255	All 1s
	11000000	10100000	00001010	11111111	

Host Addresses

Host addresses are addresses that can be assigned to devices such as host computers, laptops, smartphones, web cameras, printers, routers, and so on. The host portion of the address is the bits indicated by 0 bits in the subnet mask. A host address can have any combination of bits in the host portion except for all 0 bits (which would be a network address) or all 1 bits (which would be a broadcast address).

All devices in the same network must have the same subnet mask and the same network bits. Only the host bits differ and must be unique.

In Table 11-2, notice that there is a first host address, and there is a last host address:

- **First host address:** The first host in a network has all 0 bits, with the last (rightmost) bit as a 1 bit. In this example, it is 192.168.10.1/24.
- **Last host address:** The last host in a network has all 1 bits, with the last (rightmost) bit as a 0 bit. In this example, it is 192.168.10.254/24.

Any addresses between and including the first and last host addresses—in this case, 192.168.10.1/24 through 192.168.10.254/24—can be assigned to devices on the network.

Broadcast Address

A broadcast address is an address that is used to reach all devices on the IPv4 network. As shown in Table 11-2, the network broadcast address has all 1 bits in the host portion, as determined by the subnet mask. In this example, the network address is 192.168.10.255/24. A broadcast address cannot be assigned to a device.

Interactive Graphic

Activity—ANDing to Determine the Network Address (11.1.7)

Use the ANDing process to determine the network address (in binary and decimal formats).

Refer to the online course to complete this activity.

Interactive Graphic

Check Your Understanding—IPv4 Address Structure (11.1.8)

Refer to the online course to complete this activity.

IPv4 Unicast, Broadcast, and Multicast (11.2)

In IPv4 data networks, communication can take place as unicast, broadcast, or multicast. This section discusses these three methods of communication in IPv4.

Unicast (11.2.1)

In the previous section, you learned about the structure of an IPv4 address; each has a network portion and a host portion. There are different ways to send a packet from a source device, and these different transmissions affect the destination IPv4 addresses.

Unicast transmission refers to one device sending a message to one other device in one-to-one communications.

A unicast packet has a destination IP address that is a unicast address, which goes to a single recipient. A source IP address can only be a unicast address because the packet can only originate from a single source—regardless of whether the destination IP address is a unicast, broadcast, or multicast address.

Figure 11-7 shows an example of unicast transmission.

Note

In this book, all communication between devices is unicast unless otherwise noted.

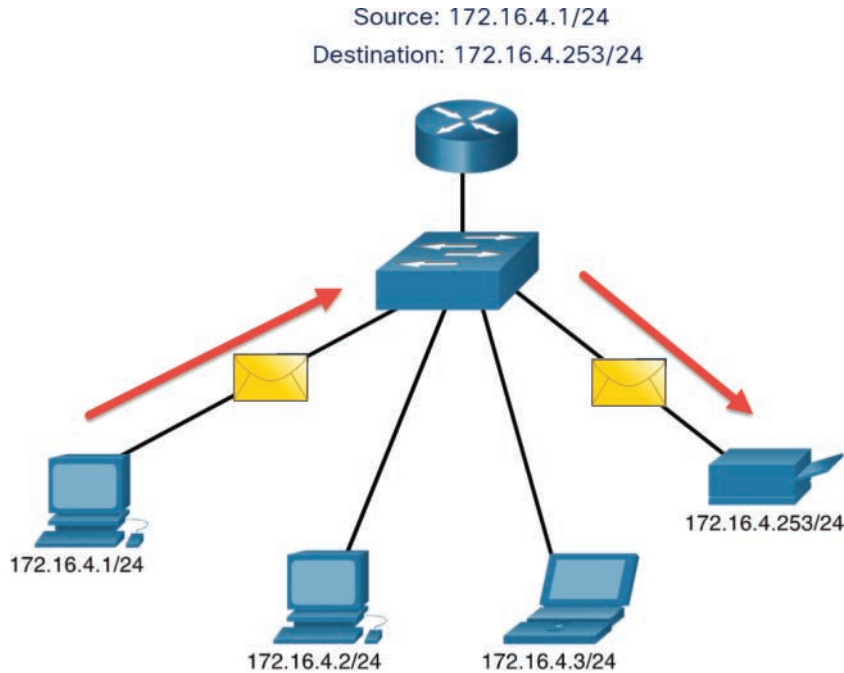


Figure 11-7 Unicast Transmission

IPv4 unicast host addresses are in the address range 1.1.1.1 to 223.255.255.255. However, within this range are many addresses that are reserved for special purposes. These special-purpose addresses are discussed later in this chapter.

Broadcast (11.2.2)

Broadcast transmission refers to a device sending a message to all the devices on a network in one-to-all communications.

A broadcast packet has a destination IP address with all 1s in the host portion, or 32 1 bits.

Note

IPv4 uses broadcast packets. However, there are no broadcast packets with IPv6.

A broadcast packet must be processed by all devices in the same broadcast domain. A broadcast domain identifies all hosts on the same network segment. A broadcast may be directed or limited. A directed broadcast is sent to all hosts on a specific

network. For example, say that a host on the 172.16.4.0/24 network sends a packet to 172.16.4.255. A limited broadcast is sent to 255.255.255.255. By default, routers do not forward broadcasts.

Figure 11-8 shows an example of a limited broadcast transmission.

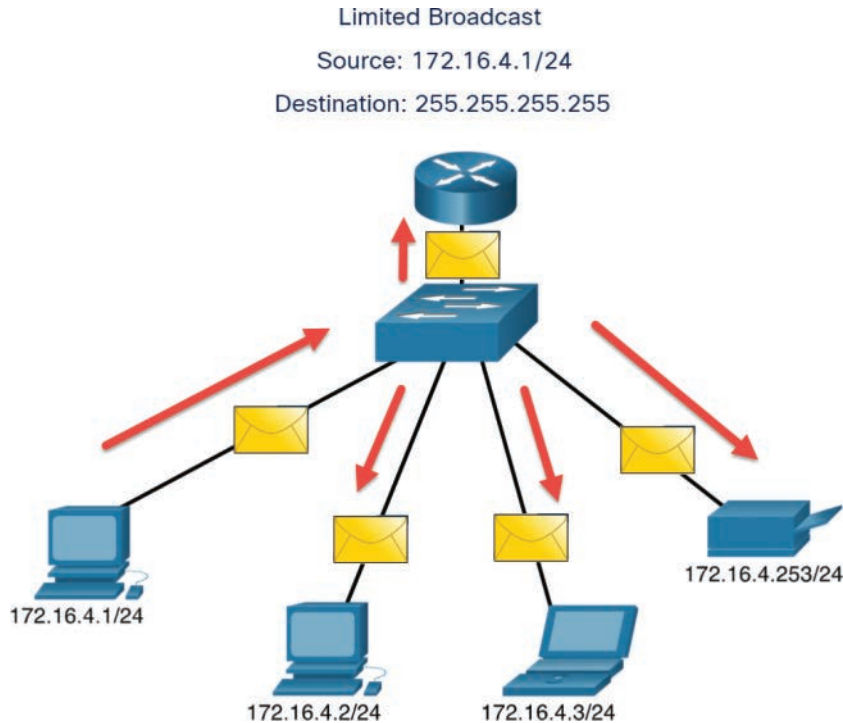


Figure 11-8 Broadcast Transmission

Broadcast packets use resources on the network and make every receiving host on the network process the packet. Therefore, broadcast traffic should be limited so that it does not adversely affect the performance of the network or devices. Because routers separate broadcast domains, subdividing networks can improve network performance by eliminating excessive broadcast traffic.

IP Directed Broadcasts

In addition to the 255.255.255.255 broadcast address, there is a broadcast IPv4 address for each network. This address, called a *directed broadcast*, uses the highest address in the network, which is the address where all the host bits are 1s. For example, the directed broadcast address for 192.168.1.0/24 is 192.168.1.255. This address

allows communication to all the hosts in that network. To send data to all the hosts in a network, a host can send a single packet that is addressed to the broadcast address of the network.

A device that is not directly connected to the destination network forwards an IP directed broadcast in the same way it would forward unicast IP packets destined to a host on that network. When a directed broadcast packet reaches a router that is directly connected to the destination network, that packet is broadcast on the destination network.

Note

Because of security concerns and prior abuse from malicious users, directed broadcasts are turned off by default starting with Cisco IOS Release 12.0 with the global configuration command `no ip directed-broadcasts`.

Multicast (11.2.3)

Multicast transmission reduces traffic by allowing a host to send a single packet to a selected set of hosts that subscribe to a multicast group.

A multicast packet is a packet with a destination IP address that is a multicast address. IPv4 has reserved the 224.0.0.0 to 239.255.255.255 addresses as a multicast range.

Hosts that receive particular multicast packets are called *multicast clients*. The multicast clients use services requested by a client program to subscribe to the multicast group.

Each multicast group is represented by a single IPv4 multicast destination address. When an IPv4 host subscribes to a multicast group, the host processes packets addressed to this multicast address and packets addressed to its uniquely allocated unicast address.

Routing protocols such as OSPF use multicast transmissions. For example, routers enabled with OSPF communicate with each other using the reserved OSPF multicast address 224.0.0.5. Only devices enabled with OSPF process these packets with 224.0.0.5 as the destination IPv4 address. All other devices ignore these packets.

Figure 11-9 illustrates clients accepting multicast packets.

Interactive Graphic

Activity—Unicast, Broadcast, or Multicast (11.2.4)

Refer to the online course to complete this activity.

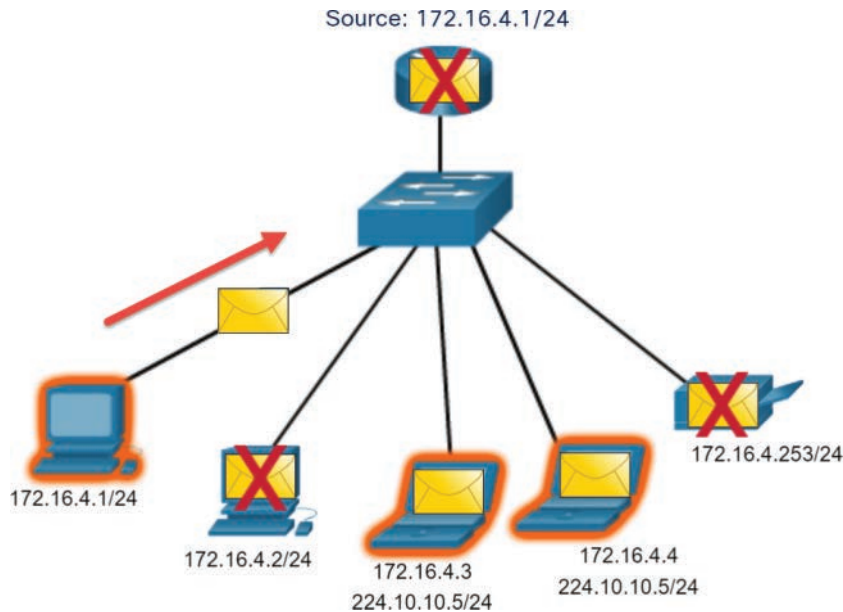


Figure 11-9 Multicast Transmission

Types of IPv4 Addresses (11.3)

This section discusses the different types of IPv4 addresses, including public, private, and legacy classful addresses.

Public and Private IPv4 Addresses (11.3.1)

Just as there are different ways to transmit an IPv4 packet, there are also different types of IPv4 addresses. Some IPv4 addresses cannot be used to go out to the internet, and others are specifically allocated for routing to the internet. Some are used to verify a connection, and others are self-assigned. As a network administrator, you will eventually become very familiar with the types of IPv4 addresses, but for now, you should at least know what they are and when to use them.

Public IPv4 addresses are addresses that are globally routed between internet service provider (ISP) routers. However, not all available IPv4 addresses can be used on the internet. There are blocks of addresses called private addresses that are used by most organizations to assign IPv4 addresses to internal hosts.

In the mid-1990s, with the introduction of the World Wide Web (WWW), the private IPv4 addresses in Table 11-3 were introduced to deal with the depletion of IPv4 address space. Private IPv4 addresses are not unique and can be used internally within any network.

Note

The long-term solution to IPv4 address depletion is IPv6.

Table 11-3 The Private Address Blocks

Network Address and Prefix	RFC 1918 Private Address Range
10.0.0.0/8	10.0.0.0–10.255.255.255
172.16.0.0/12	172.16.0.0–172.31.255.255
192.168.0.0/16	192.168.0.0–192.168.255.255

Note

Private addresses are defined in RFC 1918 and sometimes referred to as RFC 1918 address space.

Routing to the Internet (11.3.2)

Most internal networks, from large enterprises to home networks, use private IPv4 addresses for addressing all internal devices (in intranets), including hosts and routers. However, private addresses are not globally routable.

In Figure 11-10, customer networks 1, 2, and 3 are sending packets outside their internal networks. These packets have a source IPv4 address that is a private address and a destination IPv4 address that is public (globally routable). Packets with private addresses must be filtered (discarded) or have their addresses translated to public addresses before being forwarded to an ISP.

Before the ISP can forward this packet, it must translate the source IPv4 address, which is a private address, to a public IPv4 address using Network Address Translation (NAT). NAT is used to translate between private IPv4 and public IPv4 addresses. This is usually done on the router that connects the internal network to the ISP network. Private IPv4 addresses in the organization's intranet are translated to public IPv4 addresses before routing to the internet.

Note

Although a device with a private IPv4 address is not directly accessible from another device across the internet, the IETF does not consider private IPv4 addresses and NAT to be effective security measures.

Organizations that have resources available to the internet, such as a web server, also have devices that have public IPv4 addresses. As shown in Figure 11-11, this part of the network is known as the DMZ (demilitarized zone). The router in the figure not only performs routing, it also performs NAT and acts as a firewall for security.

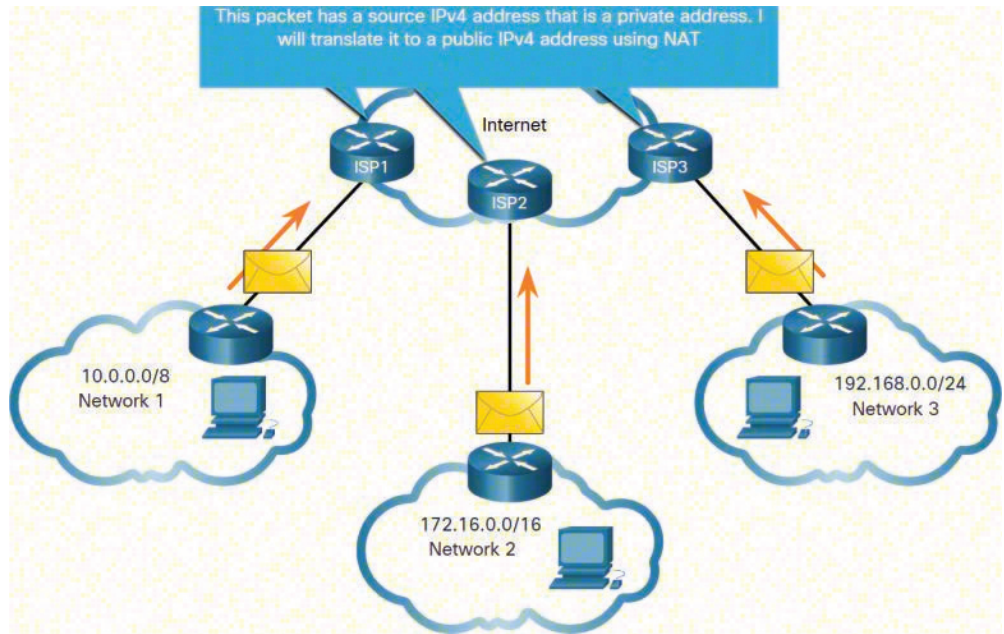


Figure 11-10 Private IPv4 Addresses Translated to Public IPv4 Addresses

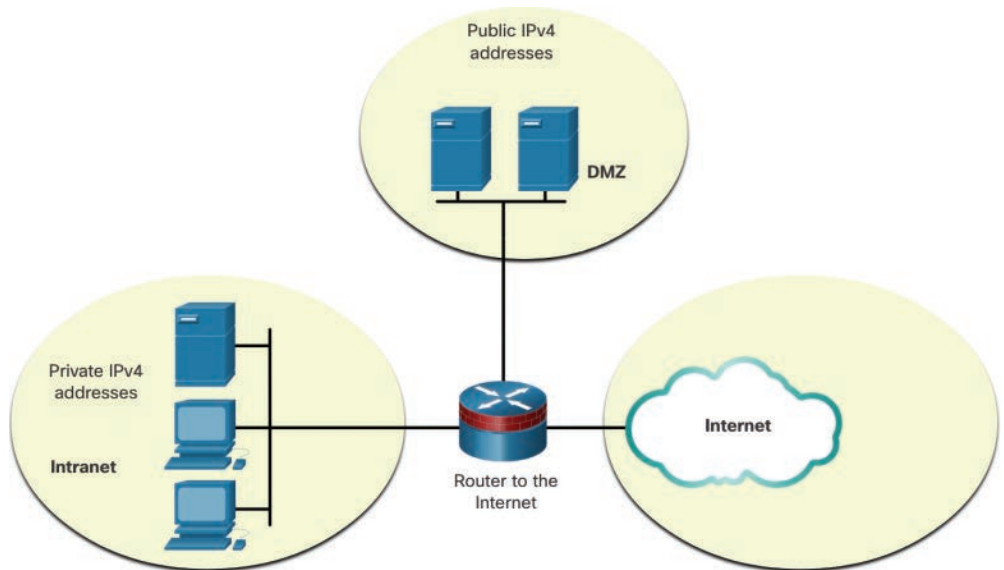


Figure 11-11 Example of a DMZ with Public IPv4 Addressing

Note

Private IPv4 addresses are commonly used for educational purposes to ensure that the addresses used are not public IPv4 addresses that belong to organizations.

**Interactive
Graphic****Activity—Pass or Block IPv4 Addresses (11.3.3)**

Refer to the online course to complete this activity.

Special Use IPv4 Addresses (11.3.4)

Certain addresses, such as the network address and broadcast address, cannot be assigned to hosts. In addition, special addresses can be assigned to hosts but with restrictions on how those hosts can interact within the network.

Loopback Addresses

A loopback address (in the range 127.0.0/8 or 127.0.0.1 to 127.255.255.254, though more commonly identified as only 127.0.0.1) is a special address that a host uses to direct traffic to itself. For example, the loopback address can be used on a host to test whether the TCP/IP configuration is operational, as shown in Example 11-1. Notice how the 127.0.0.1 loopback address replies to the **ping** command. Also notice that any address within this block will loop back to the local host, which is shown with the second **ping**.

Example 11-1 Pinging the Loopback Interface

```
C:\Users\NetAcad> ping 127.0.0.1
Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Users\NetAcad> ping 127.1.1.1
Pinging 127.1.1.1 with 32 bytes of data:
Reply from 127.1.1.1: bytes=32 time<1ms TTL=128
Reply from 127.1.1.1: bytes=32 time<1ms TTL=128
Reply from 127.1.1.1: bytes=32 time<1ms TTL=128
Reply from 127.1.1.1: bytes=32 time<1ms TTL=128
Ping statistics for 127.1.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Users\NetAcad>
```


Link-Local Addresses

Link-local addresses (in the range 169.254.0.0/16 or 169.254.0.1 to 169.254.255.254) are more commonly known as the Automatic Private IP Addressing (APIPA) addresses, or self-assigned addresses. They are used by a Windows DHCP client to self-configure in the event that there are no DHCP servers available. Link-local addresses can be used in a peer-to-peer connection but are not commonly used for this purpose.

Legacy Classful Addressing (11.3.5)

In 1981, IPv4 addresses were assigned using classful addressing, as defined in RFC 790 (<https://tools.ietf.org/html/rfc790>). A customer was allocated a network address based on one of three classes: A, B, or C. The RFC divided the unicast ranges into specific classes, as follows:

- **Class A (0.0.0.0/8 to 127.0.0.0/8):** Designed to support extremely large networks with more than 16 million host addresses. Class A used a fixed /8 prefix with the first octet to indicate the network address and the remaining three octets for host addresses (for more than 16 million host addresses per network).
- **Class B (128.0.0.0/16 to 191.255.0.0/16):** Designed to support the needs of moderate to large networks with up to approximately 65,000 host addresses. Class B used a fixed /16 prefix with the two high-order octets to indicate the network address and the remaining two octets for host addresses (for more than 65,000 host addresses per network).
- **Class C (192.0.0.0/24 to 223.255.255.0/24):** Designed to support small networks with a maximum of 254 hosts. Class C used a fixed /24 prefix with the first three octets to indicate the network and the remaining octet for the host addresses (for only 254 host addresses per network).

Note

There is also a Class D multicast block consisting of 224.0.0.0 to 239.0.0.0 and a Class E experimental address block consisting of 240.0.0.0 to 255.0.0.0.

At the time, with a limited number of computers using the internet, classful addressing was an effective means of allocating addresses. As shown in Figure 11-12, Class A and B networks have a very large number of host addresses, and Class C networks have very few host addresses. Class A networks accounted for 50% of the IPv4 networks, which meant that most of the available IPv4 addresses went unused.

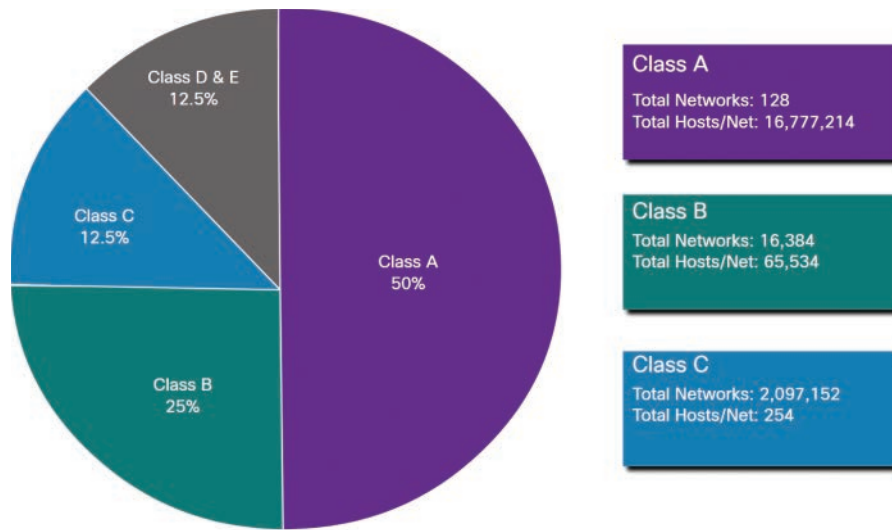


Figure 11-12 Classful Addressing

In the mid-1990s, with the introduction of the World Wide Web (WWW), classful addressing was deprecated to more efficiently allocate the limited IPv4 address space. Classful address allocation was replaced with classless addressing, which is used today. Classless addressing ignores the rules of classes (A, B, C). Public IPv4 network addresses (network addresses and subnet masks) are allocated based on the number of addresses that can be justified.

Assignment of IP Addresses (11.3.6)

Public IPv4 addresses are addresses that are globally routed over the internet. Public IPv4 addresses must be unique.

Both IPv4 and IPv6 addresses are managed by the Internet Assigned Numbers Authority (IANA). IANA manages and allocates blocks of IP addresses to the Regional Internet Registries (RIRs). The five RIRs are shown in Figure 11-13.

RIRs are responsible for allocating IP addresses to ISPs that provide IPv4 address blocks to organizations and smaller ISPs. Organizations can also get their addresses directly from an RIR (subject to the policies of that RIR).

The five RIRs are as follows:

- **AfriNIC (African Network Information Centre):** Africa region
- **APNIC (Asia Pacific Network Information Centre):** Asia/Pacific region
- **ARIN (American Registry for Internet Numbers):** North America region

- **LACNIC (Regional Latin-American and Caribbean IP Address Registry):** Latin America and some Caribbean islands
- **RIPE NCC (Réseaux IP Européens Network Coordination Centre):** Europe, the Middle East, and Central Asia



Figure 11-13 Five Regional Internet Registries

**Interactive
Graphic**

Activity—Public or Private IPv4 Address (11.3.7)

Refer to the online course to complete this activity.

**Interactive
Graphic**

Check Your Understanding—Types of IPv4 Addresses (11.3.8)

Refer to the online course to complete this activity.

Network Segmentation (11.4)

This section discusses network segmentation and the reasons we divide larger networks into smaller networks known as subnets.

Broadcast Domains and Segmentation (11.4.1)

Have you ever received an email that was addressed to every person at your work or school? That was a broadcast email, and hopefully, it contained information that

each of you needed to know. But often a broadcast is not really pertinent to everyone on the mailing list. Sometimes, only a segment of the population needs to read the information sent as a broadcast.

In an Ethernet LAN, devices use broadcasts and Address Resolution Protocol (ARP) to locate other devices. ARP sends Layer 2 broadcasts to a known IPv4 address on the local network to discover the associated MAC address. Devices on Ethernet LANs also locate other devices using services. A host typically acquires its IPv4 address configuration by using the Dynamic Host Configuration Protocol (DHCP), which sends broadcasts on the local network to locate a DHCP server.

A switch propagates a broadcast out all interfaces except the interface on which it was received. For example, if a switch in Figure 11-14 were to receive a broadcast, it would forward it to the other switches and other users connected in the network.

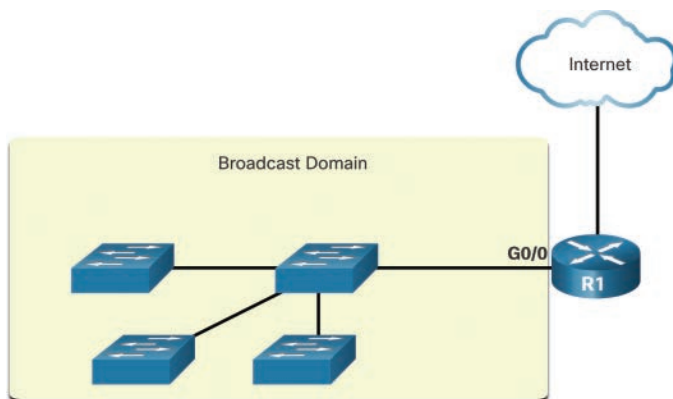


Figure 11-14 Broadcast Domain with Four Switches

Routers do not propagate broadcasts. When a router receives a broadcast, it does not forward it out other interfaces. For instance, when R1 receives a broadcast on its Gigabit Ethernet 0/0 interface, it does not forward out another interface.

Therefore, each router interface connects to a broadcast domain, and broadcasts are propagated only within that specific broadcast domain.

Problems with Large Broadcast Domains (11.4.2)

A large broadcast domain is a network that connects many hosts. A problem with a large broadcast domain is that these hosts can generate excessive broadcasts and can negatively affect the network. In Figure 11-15, LAN 1 connects 400 users that could generate an excess amount of broadcast traffic. The significant amount of traffic this setup can cause results in slow network operations and also in slow device operations because a device must accept and process each broadcast packet.

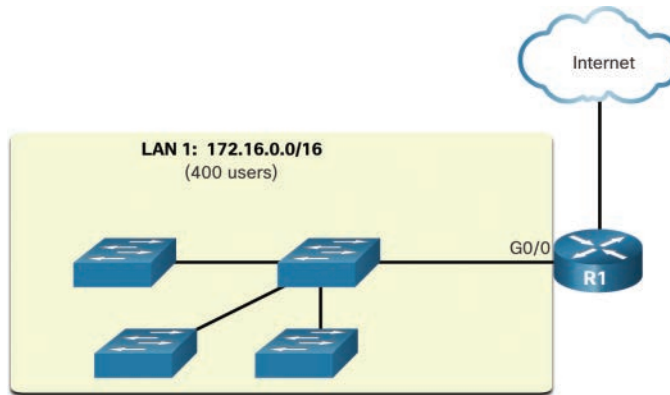


Figure 11-15 Large Broadcast Domain

The solution is to reduce the size of the network to create smaller broadcast domains in a process called *subnetting*. These smaller network spaces are called subnets.

In Figure 11-16, the 400 users in LAN 1 with network address 172.16.0.0/16 have been divided into two subnets of 200 users each: 172.16.0.0/24 and 172.16.1.0/24. Now broadcasts are propagated only within the smaller broadcast domains. Therefore, a broadcast in LAN 1 would not propagate to LAN 2.

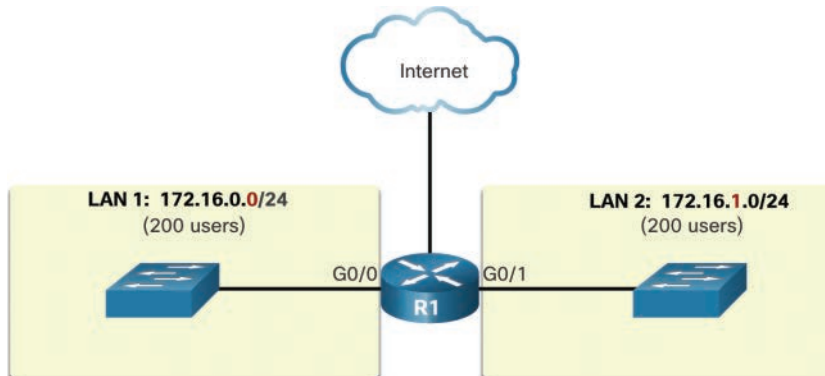


Figure 11-16 Segmenting a Large Broadcast Domain

Notice that the prefix length has changed from a single /16 network to two /24 networks. This is the basis of subnetting: using host bits to create additional subnets.

Note

The terms *subnet* and *network* are often used interchangeably. In most cases, a network is a subnet of some larger address block.

Reasons for Segmenting Networks (11.4.3)

Subnetting reduces overall network traffic and improves network performance. It also enables an administrator to implement security policies such as which subnets are allowed or not allowed to communicate together. In addition, subnetting reduces the number of devices affected by abnormal broadcast traffic due to misconfigurations, hardware/software problems, or malicious intent.

There are various ways of using subnets to help manage network devices, as shown in Figures 11-17 through 11-19.

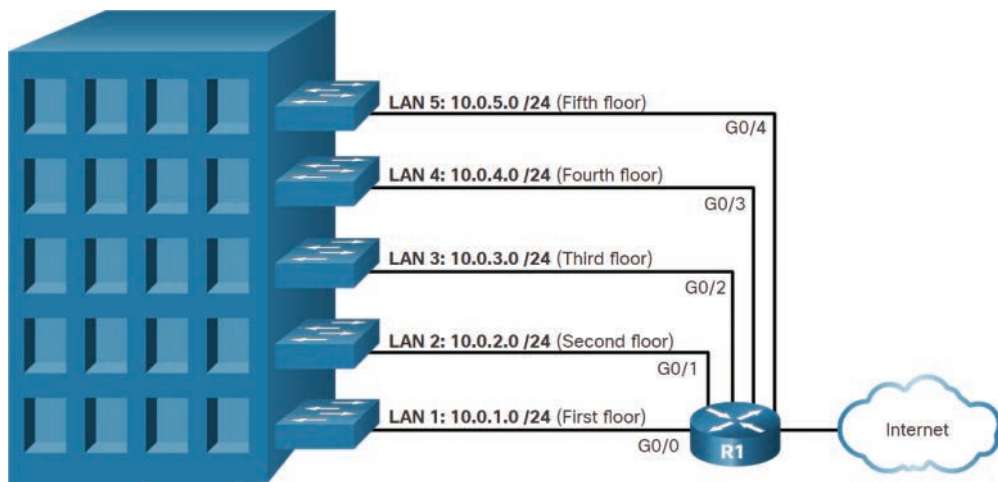


Figure 11-17 Subnetting by Location

Network administrators can create subnets using any other division that makes sense for the network. Notice in Figures 11-17 through 11-19 that the subnets use longer prefix lengths to identify networks.

Understanding how to subnet networks is a fundamental skill that all network administrators must develop. Various methods have been created to help understand this process. Although it may be a little overwhelming at first, pay close attention to the detail and, with practice, subnetting will become easier.

Check Your Understanding—Network Segmentation (11.4.4)

Refer to the online course to complete this activity.

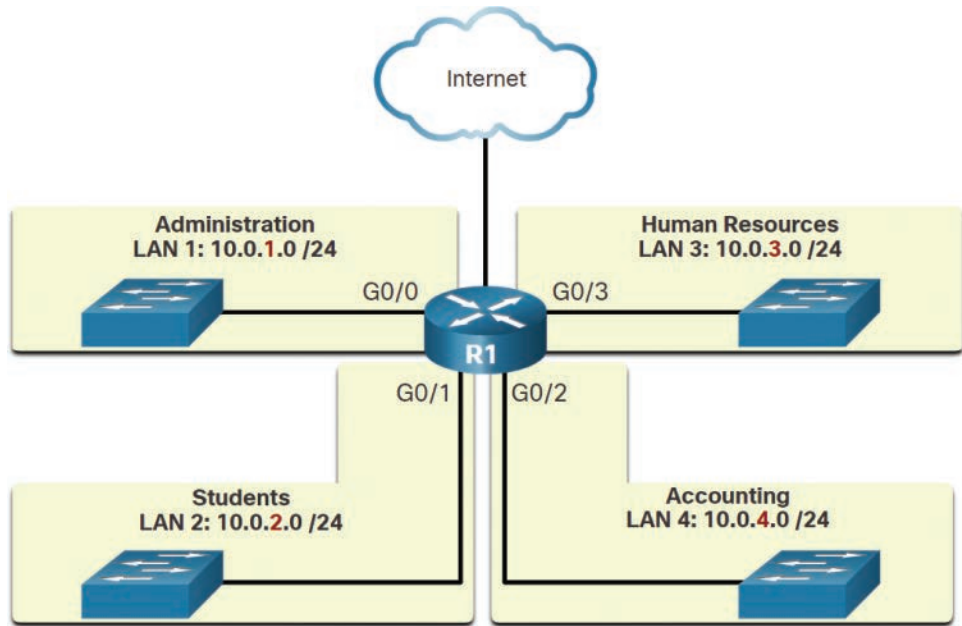


Figure 11-18 Subnetting by Group or Function

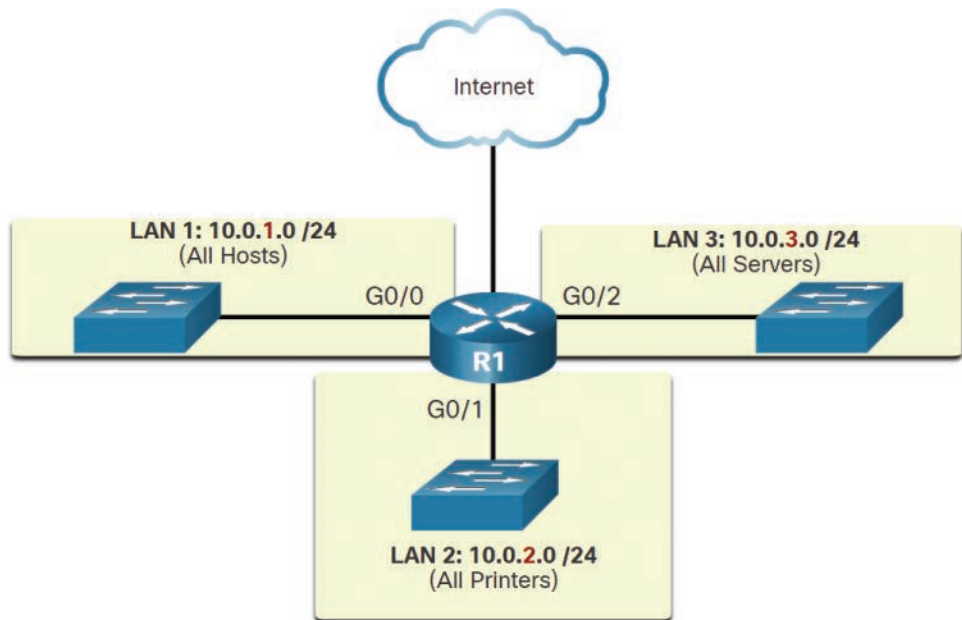


Figure 11-19 Subnetting by Device Type

Subnet an IPv4 Network (11.5)

Without subnetting, the performance of an IPv4-based network would quickly decrease as the number of hosts increased. Proper subnetting allows better control of network traffic and greatly improves network efficiency.

Subnet on an Octet Boundary (11.5.1)

In the previous section, you learned several good reasons for segmenting a network. You also learned that segmenting a network is called *subnetting*. Subnetting is a critical skill to have when administering an IPv4 network. It is a bit daunting at first, but it gets much easier with practice.

IPv4 subnets are created by using one or more of the host bits as network bits. This process involves extending the subnet mask to borrow some of the bits from the host portion of the address to create additional network bits. The more host bits that are borrowed, the more subnets that can be defined. The more bits that are borrowed to increase the number of subnets, the lower the number of hosts per subnet.

Networks are most easily subnetted at an *octet boundary*: /8, /16, or /24. Table 11-4 identifies these prefix lengths. Notice that using longer prefixes decreases the number of hosts per subnet.

Table 11-4 Subnet Masks on Octet Boundaries

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	Number of Hosts
/8	255.0.0.0	nnnnnnnnn.hhhhhhhh.hhhhhhhh.hhhhhhhh 11111111.00000000.00000000.00000000	16,777,214
/16	255.255.0.0	nnnnnnnnn.nnnnnnnnn.hhhhhhhh.hhhhhhhh 11111111.11111111.00000000.00000000	65,534
/24	255.255.255.0	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.hhhhhhhh 11111111.11111111.11111111.00000000	254

To understand how subnetting on the octet boundary can be useful, consider the following example: Say that an enterprise has chosen the private address 10.0.0.0/8 as its internal network address. That network address can connect 16,777,214 hosts in one broadcast domain. Obviously, having more than 16 million hosts on a single subnet is not ideal.

The enterprise could further subnet the 10.0.0.0/8 address at the octet boundary /16, as shown in Table 11-5. This would enable the enterprise to define up to 256 subnets

(that is, 10.0.0.0/16 to 10.255.0.0/16), and each subnet would be capable of connecting 65,534 hosts. Notice that the first two octets identify the network portion of the address, whereas the last two octets are for host IP addresses.

Table 11-5 Subnetting Network 10.0.0.0/8 Using a /16 Prefix

Subnet Address (256 Possible Subnets)	Host Range (65,534 Possible Hosts per Subnet)	Broadcast
10.0.0.0/16	10.0.0.1–10.0.255.254	10.0.255.255
10.1.0.0/16	10.1.0.1–10.1.255.254	10.1.255.255
10.2.0.0/16	10.2.0.1–10.2.255.254	10.2.255.255
10.3.0.0/16	10.3.0.1–10.3.255.254	10.3.255.255
10.4.0.0/16	10.4.0.1–10.4.255.254	10.4.255.255
10.5.0.0/16	10.5.0.1–10.5.255.254	10.5.255.255
10.6.0.0/16	10.6.0.1–10.6.255.254	10.6.255.255
10.7.0.0/16	10.7.0.1–10.7.255.254	10.7.255.255
...	...	...
10.255.0.0/16	10.255.0.1–10.255.255.254	10.255.255.255

Alternatively, the enterprise could choose to subnet the 10.0.0.0/8 network at the /24 octet boundary, as shown in Table 11-6. This would enable the enterprise to define 65,536 subnets, each capable of connecting 254 hosts. The /24 boundary is very popular in subnetting because it accommodates a reasonable number of hosts and conveniently subnets at the octet boundary.

Table 11-6 Subnetting Network 10.0.0.0/8 Using a /24 Prefix

Subnet Address (65,536 Possible Subnets)	Host Range (254 Possible Hosts per Subnet)	Broadcast
10.0.0.0/24	10.0.0.1–10.0.0.254	10.0.0.255
10.0.1.0/24	10.0.1.1–10.0.1.254	10.0.1.255
10.0.2.0/24	10.0.2.1–10.0.2.254	10.0.2.255
...	...	...
10.0.255.0/24	10.0.255.1–10.0.255.254	10.0.255.255
10.1.0.0/24	10.1.0.1–10.1.0.254	10.1.0.255
10.1.1.0/24	10.1.1.1–10.1.1.254	10.1.1.255

Subnet Address (65,536 Possible Subnets)	Host Range (254 Possible Hosts per Subnet)	Broadcast
10.1.2.0/24	10.1.2.1–10.1.2.254	10.1.2.255
...	...	...
10.100.0.0/24	10.100.0.1–10.100.0.254	10.100.0.255
...	...	...
10.255.255.0/24	10.255.255.1–10.255.255.254	10.255.255.255

Subnet Within an Octet Boundary (11.5.2)

The examples shown thus far have borrowed host bits from the common /8, /16, and /24 network prefixes. However, subnets can borrow bits from any host bit position to create other masks.

For instance, a /24 network address is commonly subnetted using longer prefixes by borrowing bits from the fourth octet. This provides an administrator with additional flexibility when assigning network addresses to a smaller number of end devices.

Table 11-7 shows six ways to subnet a /24 network.

Table 11-7 Subnetting a /24 Network

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = Network, h = Host)	Number of Subnets	Number of Hosts
/25	255.255.255.128	n n n n n n n n . n n n n n n n n . n n n n n n n n . n h h h h h h h 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 0 0 0 0 0 0 0	2	126
/26	255.255.255.192	n n n n n n n n . n n n n n n n n . n n n n n n n n . n n h h h h h h 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 0 0 0 0 0 0	4	62
/27	255.255.255.224	n n n n n n n n . n n n n n n n n . n n n n n n n n . n n n h h h h h 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 0 0 0 0 0	8	30
/28	255.255.255.240	n n n n n n n n . n n n n n n n n . n n n n n n n n . n n n n h h h h 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 0 0 0 0	16	14
/29	255.255.255.248	n n n n n n n n . n n n n n n n n . n n n n n n n n . n n n n n h h h 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 0 0 0	32	6
/30	255.255.255.252	n n n n n n n n . n n n n n n n n . n n n n n n n n . n n n n n n h h 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 0 0	64	2

For each bit borrowed from the fourth octet, the number of subnetworks available is doubled, and the number of host addresses per subnet is reduced:

- **/25 row:** Borrowing 1 bit from the fourth octet creates 2 subnets supporting 126 hosts each.
- **/26 row:** Borrowing 2 bits creates 4 subnets supporting 62 hosts each.
- **/27 row:** Borrowing 3 bits creates 8 subnets supporting 30 hosts each.
- **/28 row:** Borrowing 4 bits creates 16 subnets supporting 14 hosts each.
- **/29 row:** Borrowing 5 bits creates 32 subnets supporting 6 hosts each.
- **/30 row:** Borrowing 6 bits creates 64 subnets supporting 2 hosts each.

Video

Video—The Subnet Mask (11.5.3)

Refer to the online course to view this video.

Video

Video—Subnet with the Magic Number (11.5.4)

Refer to the online course to view this video.

Packet Tracer Activity

Packet Tracer—Subnet an IPv4 Network (11.5.5)

In this activity, starting from a single network address and network mask, you will subnet the Customer network into multiple subnets. The subnetting scheme should be based on the number of host computers required in each subnet, as well as other network considerations, such as future network host expansion.

After you have created a subnetting scheme and completed the table by filling in the missing host and interface IP addresses, you will configure the host PCs, switches, and router interfaces.

After the network devices and host PCs have been configured, you will use the **ping** command to test for network connectivity.

Subnet a Slash 16 and a Slash 8 Prefix (11.6)

This section discusses and gives examples of subnetting networks that have /16 and /8 prefix lengths.

Create Subnets with a Slash 16 Prefix (11.6.1)

Some subnetting is easier than other subnetting. This section explains how to create subnets that each have the same number of hosts.

In a situation requiring a larger number of subnets, an IPv4 network is required that has more hosts bits available to borrow. For example, the network address 172.16.0.0 has a default mask of 255.255.0.0, or /16. This address has 16 bits in the network portion and 16 bits in the host portion. The 16 bits in the host portion are available to borrow for creating subnets. Table 11-8 highlights all the possible scenarios for subnetting a /16 prefix.

Table 11-8 Subnet a /16 Network

Prefix Length	Subnet Mask	Network Address (n = Network, h = Host)	Number of Subnets	Number of Hosts
/17	255.255.128.0	nnnnnnnnn.nnnnnnnnn.nhshhshhshh.hhhshhshhshh 11111111.11111111.10000000.00000000	2	32766
/18	255.255.192.0	nnnnnnnnn.nnnnnnnnn.nnshhshhshh.hhhshhshhshh 11111111.11111111.11000000.00000000	4	16382
/19	255.255.224.0	nnnnnnnnn.nnnnnnnnn.nnnshhshhshh.hhhshhshhshh 11111111.11111111.11100000.00000000	8	8190
/20	255.255.240.0	nnnnnnnnn.nnnnnnnnn.nnnnshhshhshh.hhhshhshhshh 11111111.11111111.11110000.00000000	16	4094
/21	255.255.248.0	nnnnnnnnn.nnnnnnnnn.nnnnnshhshh.hhhshhshhshh 11111111.11111111.11111000.00000000	32	2046
/22	255.255.252.0	nnnnnnnnn.nnnnnnnnn.nnnnnnshhshh.hhhshhshhshh 11111111.11111111.11111100.00000000	64	1022
/23	255.255.254.0	nnnnnnnnn.nnnnnnnnn.nnnnnnnshhshh.hhhshhshhshh 11111111.11111111.11111110.00000000	128	510
/24	255.255.255.0	nnnnnnnnn.nnnnnnnnn.nnnnnnnnnshhshhshhshh 11111111.11111111.11111111.00000000	256	254
/25	255.255.255.128	nnnnnnnnn.nnnnnnnnn.nnnnnnnnnshhshhshhshh 11111111.11111111.11111111.10000000	512	126
/26	255.255.255.192	nnnnnnnnn.nnnnnnnnn.nnnnnnnnnnnshhshhshhshh 11111111.11111111.11111111.11000000	1024	62
/27	255.255.255.224	nnnnnnnnn.nnnnnnnnn.nnnnnnnnnnnnshhshhshhshh 11111111.11111111.11111111.11100000	2048	30

Prefix Length	Subnet Mask	Network Address (n = Network, h = Host)	Number of Subnets	Number of Hosts
/28	255.255.255.240	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnhhhh 11111111.11111111.11111111.11110000	4096	14
/29	255.255.255.248	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnhhhh 11111111.11111111.11111111.11111000	8192	6
/30	255.255.255.252	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnnhhh 11111111.11111111.11111111.11111100	16384	2

Although you do not need to memorize this table, you do need to have a good understanding of how each value in the table is generated. Do not let the size of the table intimidate you. It is big because it has 8 additional bits that can be borrowed, and, therefore, the numbers of subnets and hosts are simply larger.

Create 100 Subnets with a Slash 16 Prefix (11.6.2)

Consider a large enterprise that requires at least 100 subnets and that has chosen the private address 172.16.0.0/16 as its internal network address.

When borrowing bits from a /16 address, start borrowing bits in the third octet, going from left to right. Borrow a single bit at a time until the number of bits necessary to create 100 subnets is reached.

Figure 11-20 displays the number of subnets that can be created when borrowing bits from the third octet and the fourth octet. Notice that there are now up to 14 host bits that can be borrowed.

To satisfy the requirement of 100 subnets for the enterprise, 7 bits (that is, $2^7 = 128$ subnets) would need to be borrowed (for a total of 128 subnets), as shown in Figure 11-21.

Recall that the subnet mask must change to reflect the borrowed bits. In this example, when 7 bits are borrowed, the mask is extended 7 bits into the third octet. In decimal, the mask is represented as 255.255.254.0, or a /23 prefix, because the third octet is 11111110 in binary, and the fourth octet is 00000000 in binary.

Figure 11-22 shows the resulting subnets, from 172.16.0.0/23 up to 172.16.254.0/23.

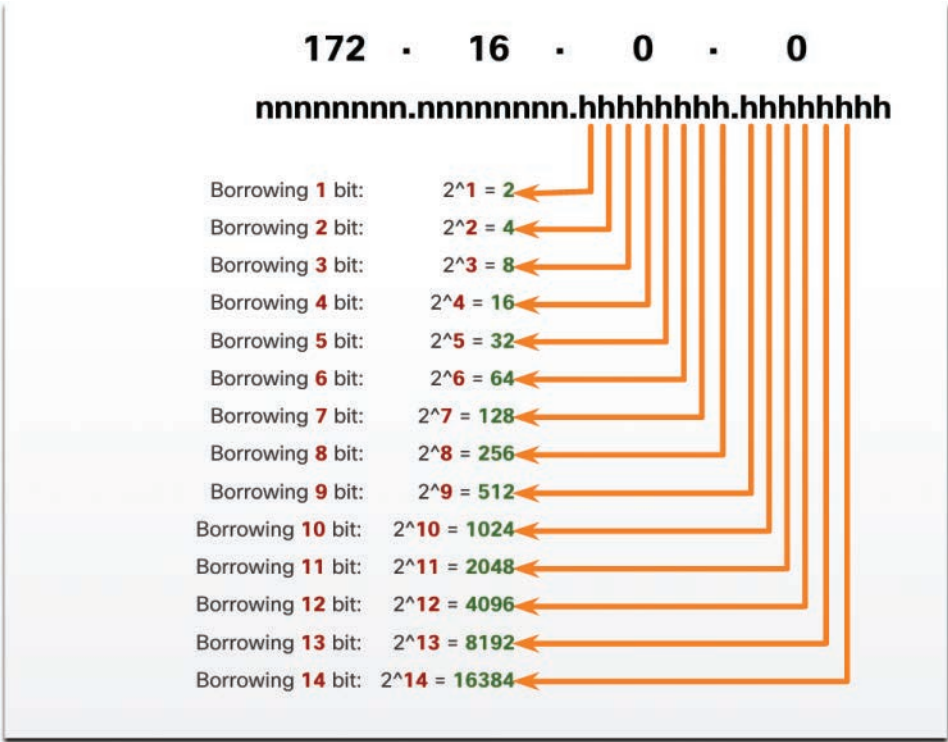


Figure 11-20 Number of Subnets Created

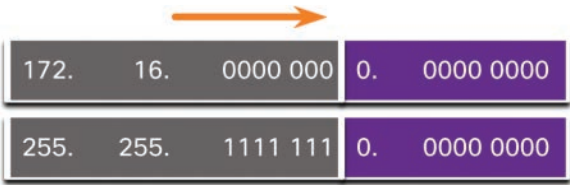
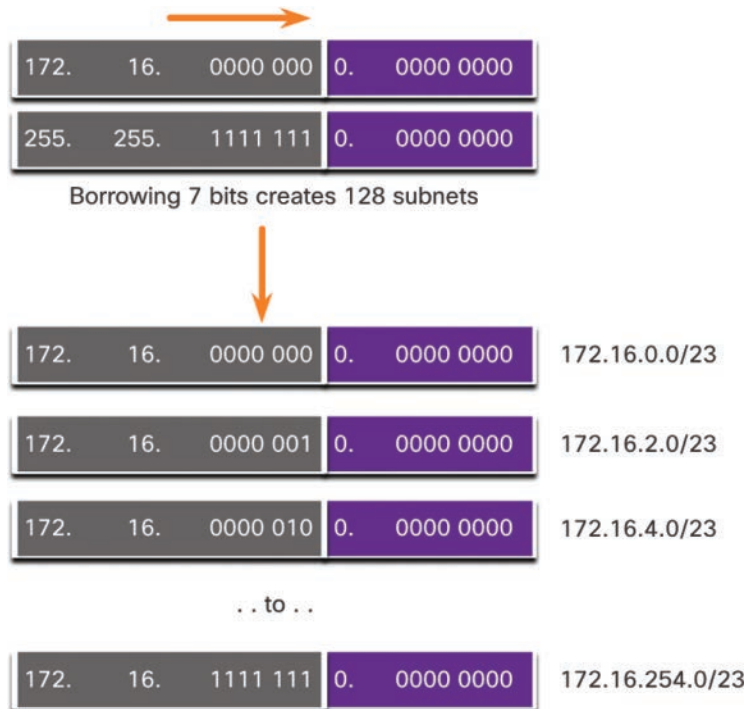
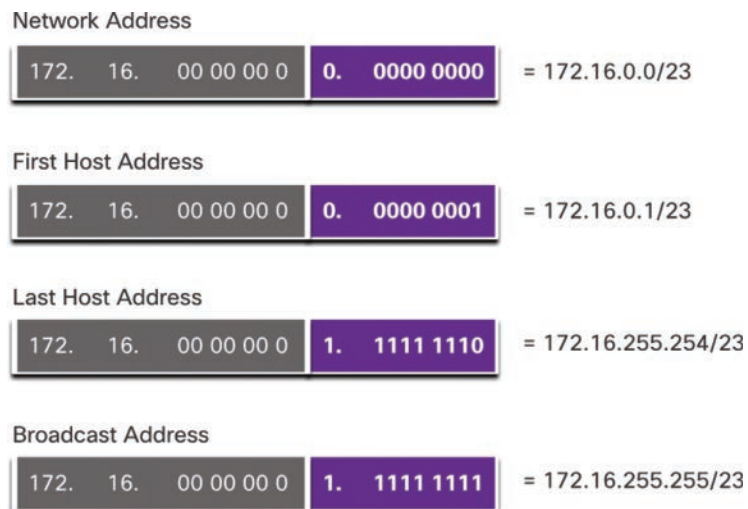


Figure 11-21 172.16.0.0/23 Network

After borrowing 7 bits for the subnet, there is 1 host bit remaining in the third octet, and there are 8 host bits remaining in the fourth octet, for a total of 9 bits not borrowed. 2^9 results in 512 total host addresses. The first address is reserved for the network address, and the last address is reserved for the broadcast address, so subtracting for these two addresses ($2^9 - 2$) leaves 510 available host addresses for each /23 subnet.

As shown in Figure 11-23, the first host address for the first subnet is 172.16.0.1, and the last host address is 172.16.1.254.

**Figure 11-22** Resulting /23 Subnets**Figure 11-23** Address Range for the 172.16.0.0/23 Subnet

Create 1000 Subnets with a Slash 8 Prefix (1 1.6.3)

Some organizations, such as small service providers or large enterprises, may need even more than 100 subnets. For example, a small ISP may need 1000 subnets for its clients. Each client needs plenty of space in the host portion to create its own subnets.

Say that an ISP has a network address 10.0.0.0 255.0.0.0, or 10.0.0.0/8. This means there are 8 bits in the network portion and 24 host bits available to borrow toward subnetting. Therefore, the small ISP will subnet the 10.0.0.0/8 network.

To create subnets, you must borrow bits from the host portion of the IPv4 address of the existing internetwork. Starting from the left at the first available host bit, borrow a single bit at a time until you reach the number of bits necessary to create 1000 subnets. As shown in Figure 11-24, you need to borrow 10 bits to create 1024 subnets ($2^{10} = 1024$). You end up borrowing 8 bits from the second octet and 2 additional bits from the third octet.

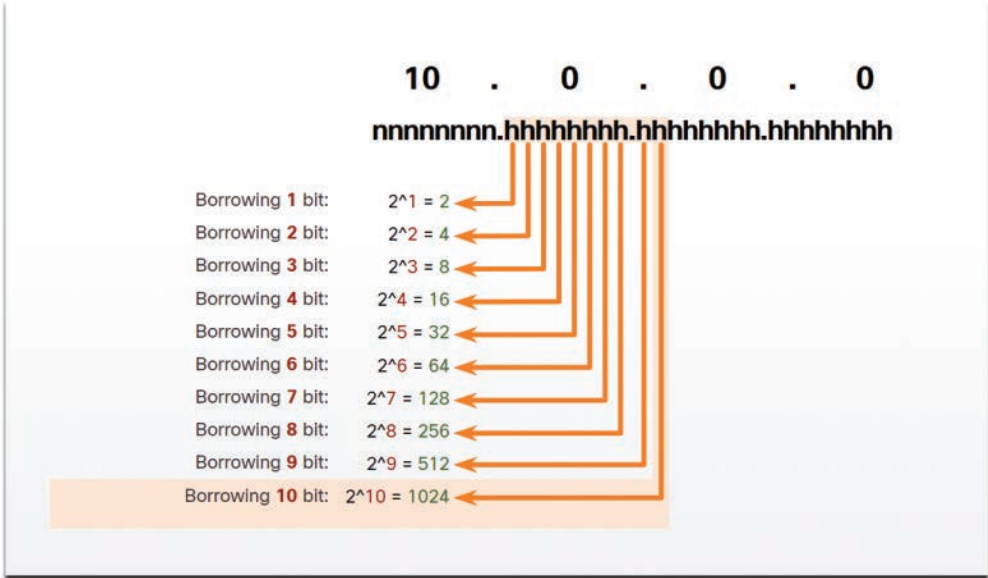
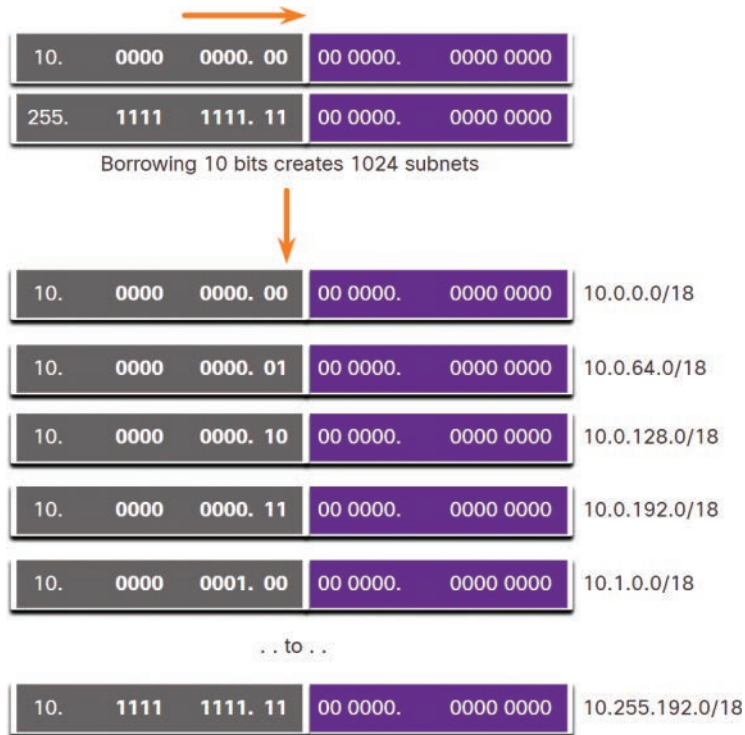


Figure 11-24 Number of Subnets Created

Figure 11-25 shows the network address and the resulting subnet mask, which converts to 255.255.192.0, or 10.0.0.0/18.

Figure 11-26 displays the subnets resulting from borrowing 10 bits, creating subnets from 10.0.0.0/18 to 10.255.128.0/18.

**Figure 11-25** 10.0.0.0/18 Network**Figure 11-26** Resulting /18 Subnets

Borrowing 10 bits to create the subnets leaves 14 host bits for each subnet. Subtracting 2 hosts per subnet (1 for the network address and 1 for the broadcast address) leaves $2^{14} - 2 = 16,382$ hosts per subnet. This means that each of the 1000 subnets can support up to 16,382 hosts.

Figure 11-27 shows the specifics of the first subnet.

Video

Video—Subnet Across Multiple Octets (11.6.4)

Refer to the online course to view this video.

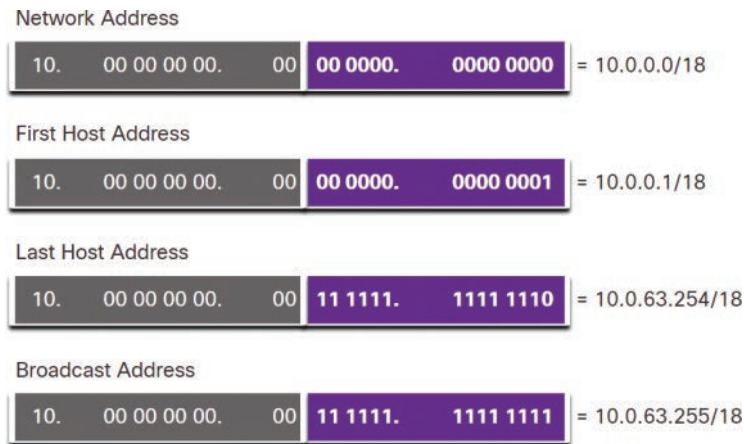


Figure 11-27 Address Range for the 10.0.0.0/18 Subnet

**Interactive
Graphic**

Activity—Calculate the Subnet Mask (11.6.5)

Refer to the online course to complete this activity.



Lab—Calculate IPv4 Subnets (11.6.6)

In this lab, you will complete the following objectives:

- Part 1: Determine IPv4 Address Subnetting
 - Part 2: Calculate IPv4 Address Subnetting
-

Subnet to Meet Requirements (11.7)

This section discusses the differences between subnetting areas of a network that use private IPv4 address space and areas that use public IPv4 address space. Although the technique of subnetting is the same, there are some important considerations.

Subnet Private Versus Public IPv4 Address Space (11.7.1)

Your organization's network may use both public and private IPv4 addresses. This affects how you will subnet your network.

Figure 11-28 shows a typical enterprise network, which includes the following components:

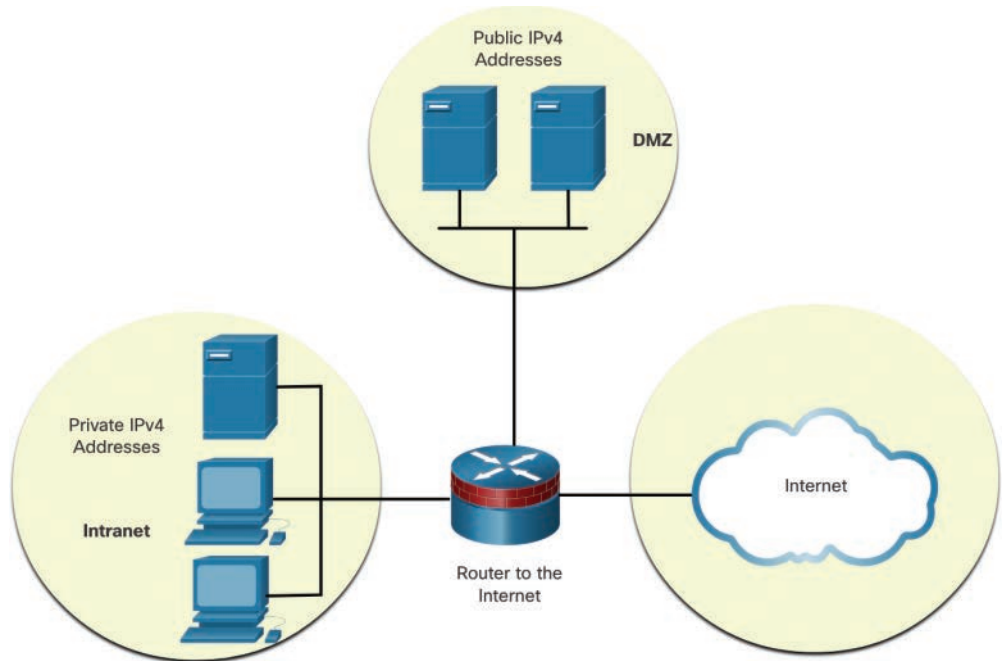


Figure 11-28 Intranet and DMZ in an Enterprise Network

- **Intranet:** This is the internal part of a company's network, accessible only within the organization. Devices in the intranet use private IPv4 addresses.
- **DMZ:** This is part of the company's network containing resources available to the internet, such as a web server. Devices in the DMZ use public IPv4 addresses.

The intranet and the DMZ have unique subnetting requirements and challenges.

The intranet uses private IPv4 addressing space. This means the organization can use any of the private IPv4 network addresses, including the 10.0.0.0/8 prefix, with 24 host bits and more than 16 million hosts. Using a network address with 24 host bits makes subnetting easier and more flexible. This includes subnetting on an octet boundary using a subnet mask of /16 or /24.

For example, the private IPv4 network address 10.0.0.0/8 can be subnetted using a /16 mask. As shown in Table 11-9, this results in 256 subnets, with 65,534 hosts per subnet. If an organization has a need for fewer than 200 subnets, allowing for some growth, this gives each subnet more than enough host addresses.

Table 11-9 Subnetting Network 10.0.0.0/8 Using a /16 Prefix

Subnet Address (256 Possible Subnets)	Host Range (65,534 Possible Hosts per Subnet)	Broadcast
10.0.0.0/16	10.0.0.1–10.0.255.254	10.0.255.255
10.1.0.0/16	10.1.0.1–10.1.255.254	10.1.255.255
10.2.0.0/16	10.2.0.1–10.2.255.254	10.2.255.255
10.3.0.0/16	10.3.0.1–10.3.255.254	10.3.255.255
10.4.0.0/16	10.4.0.1–10.4.255.254	10.4.255.255
10.5.0.0/16	10.5.0.1–10.5.255.254	10.5.255.255
10.6.0.0/16	10.6.0.1–10.6.255.254	10.6.255.255
10.7.0.0/16	10.7.0.1–10.7.255.254	10.7.255.255
...	...	...
10.255.0.0/16	10.255.0.1–10.255.255.254	10.255.255.255

Another option using the 10.0.0.0/8 private IPv4 network address is to subnet using a /24 mask. As shown in Table 11-10, this results in 65,536 subnets, with 254 hosts per subnet. If an organization needs more than 256 subnets, then a /24 mask can be used, with 254 hosts per subnet.

Table 11-10 Subnetting Network 10.0.0.0/8 Using a /24 Prefix

Subnet Address (65,536 Possible Subnets)	Host Range (254 Possible Hosts per Subnet)	Broadcast
10.0.0.0/24	10.0.0.1–10.0.0.254	10.0.0.255
10.0.1.0/24	10.0.1.1–10.0.1.254	10.0.1.255
10.0.2.0/24	10.0.2.1–10.0.2.254	10.0.2.255
...	...	...
10.0.255.0/24	10.0.255.1–10.0.255.254	10.0.255.255
10.1.0.0/24	10.1.0.1–10.1.0.254	10.1.0.255
10.1.1.0/24	10.1.1.1–10.1.1.254	10.1.1.255
10.1.2.0/24	10.1.2.1–10.1.2.254	10.1.2.255
...	...	...

Subnet Address (65,536 Possible Subnets)	Host Range (254 Possible Hosts per Subnet)	Broadcast
10.100.0.0/24	10.100.0.1–10.100.0.254	10.100.0.255
...	...	...
10.255.255.0/24	10.255.255.1–10.255.255.254	10.255.255.255

The 10.0.0.0/8 network can also be subnetted using any other number of prefix lengths, such as /12, /18, /20, and so on, which gives the network administrator a wide variety of options. Using a 10.0.0.0/8 private IPv4 network address makes subnet planning and implementation easy.

What About the DMZ?

Because the devices in the DMZ need to be publicly accessible from the internet, these devices require public IPv4 addresses. The depletion of public IPv4 address space became an issue beginning in the mid-1990s. Since 2011, IANA and four out of the five RIRs have run out of IPv4 address space. Although organizations are making the transition to IPv6, the remaining IPv4 address space remains severely limited. This means an organization must maximize its own limited number of public IPv4 addresses; the network administrator must therefore subnet the network’s public address space into subnets with different subnet masks in order to minimize the number of unused host addresses per subnet. This is known as variable-length subnet masking (VLSM).

Minimize Unused Host IPv4 Addresses and Maximize Subnets (11.7.2)

To minimize the number of unused host IPv4 addresses and maximize the number of available subnets, there are two considerations when planning subnets: the number of host addresses required for each network and the number of individual subnets needed.

Table 11-11 displays the specifics for subnetting a /24 network. Notice that there is an inverse relationship between the number of subnets and the number of hosts. The more bits that are borrowed to create subnets, the fewer host bits remain available. If more host addresses are needed, more host bits are required, resulting in fewer subnets.

The number of host addresses required in the largest subnet determines how many bits must be left in the host portion. Recall that two of the addresses cannot be used, so the usable number of addresses can be calculated as $2^n - 2$.

Table 11-11 Subnetting a /24 Network

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = Network, h = Host)	Number of Subnets	Number of Hosts per Subnet
/25	255.255.255.128	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nhhhhhhh 11111111.11111111.11111111.10000000	2	126
/26	255.255.255.192	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnhhhhhh 11111111.11111111.11111111.11000000	4	62
/27	255.255.255.224	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnhhhhh 11111111.11111111.11111111.11100000	8	30
/28	255.255.255.240	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnhhhh 11111111.11111111.11111111.11110000	16	14
/29	255.255.255.248	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnnhhh 11111111.11111111.11111111.11111000	32	6
/30	255.255.255.252	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnnnh 11111111.11111111.11111111.11111100	64	2

Network administrators must devise a network addressing scheme that accommodates the maximum number of hosts for each network and the number of subnets. The addressing scheme should allow for growth in both the number of host addresses per subnet and the total number of subnets.

Example: Efficient IPv4 Subnetting (11.7.3)

In this example, an ISP has allocated a corporate headquarters the public network address 172.16.0.0/22 (with 10 host bits). As shown in Figure 11-29, this address provides 1022 host addresses.

Note

172.16.0.0/22 is part of the IPv4 private address space; it is not an actual public IPv4 address.

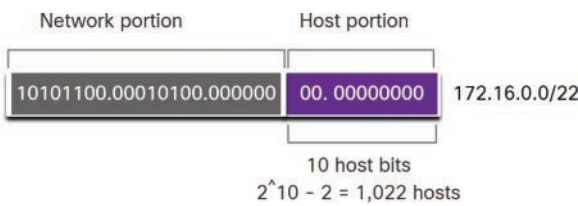


Figure 11-29 Network Address

The corporate headquarters has a DMZ and four branch offices, each needing its own public IPv4 address space. Corporate headquarters needs to make the best use of its limited IPv4 address space.

The topology shown in Figure 11-30 consists of five sites: a corporate office and four branch sites. Each site requires internet connectivity and, therefore, five internet connections. This means that the organization requires 10 subnets from the company's 172.16.0.0/22 public address. The largest subnet requires 40 addresses.

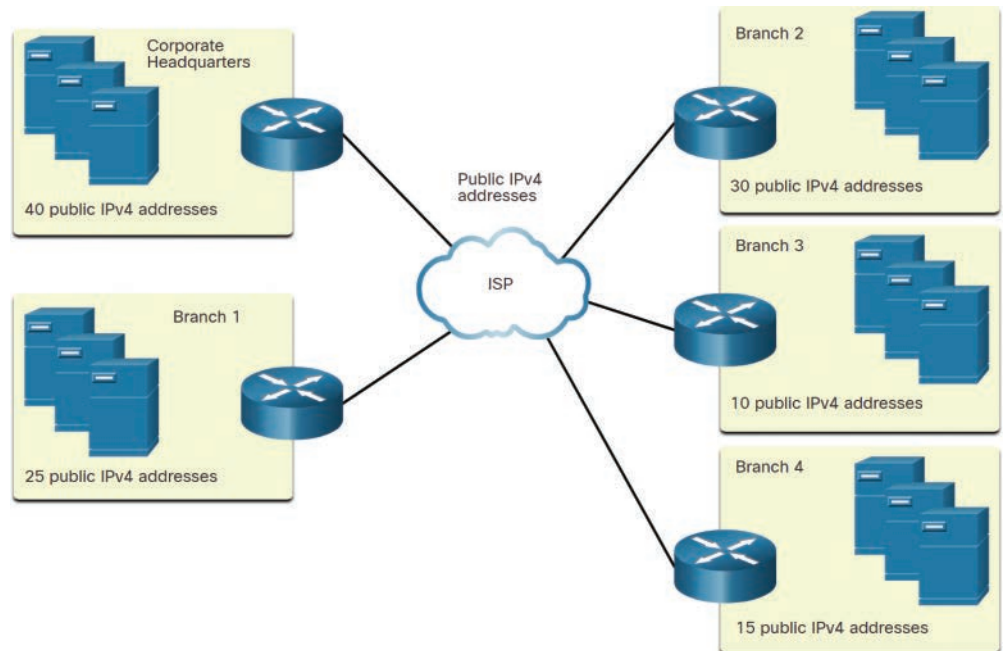


Figure 11-30 Corporate Topology with Five Sites

The 172.16.0.0/22 network address has 10 host bits, as shown in Figure 11-31. Because the largest subnet requires 40 hosts, the administrator needs a minimum of 6 host bits to provide addressing for 40 hosts. (This is determined by using the formula $2^6 - 2 = 62$ hosts.)

Using the formula for determining subnets results in 16 subnets (that is, $2^4 = 16$). The internetwork in this example requires 10 subnets, so this will meet the requirement and allow for some additional growth.

In this case, the first 4 host bits can be used to allocate subnets. This means 2 bits from the third octet and 2 bits from the fourth octet will be borrowed. When 4 bits are borrowed from the 172.16.0.0/22 network, the new prefix length is /26, with a subnet mask of 255.255.255.192.

	Network portion	Host portion	Dotted Decimal
	10101100.00010000.000000	00.00 000000	172.16.0.0/22
0	10101100.00010000.000000	00.00 000000	172.16.0.0/26
1	10101100.00010000.000000	00.01 000000	172.16.0.64/26
2	10101100.00010000.000000	00.10 000000	172.16.0.128/26
3	10101100.00010000.000000	00.11 000000	172.16.0.192/26
4	10101100.00010000.000000	01.00 000000	172.16.1.0/26
5	10101100.00010000.000000	01.01 000000	172.16.1.64/26
6	10101100.00010000.000000	01.10 000000	172.16.1.128/26
Nets 7 - 13 not shown			
14	10101100.00010000.000000	11.10 000000	172.16.3.128/26
15	10101100.00010000.000000	11.11 000000	172.16.3.192/26

4-bits borrowed from host portion to create subnets

Figure 11-31 Subnet Scheme

As shown in Figure 11-32, the subnets can be assigned to each location and router-to-ISP connections.

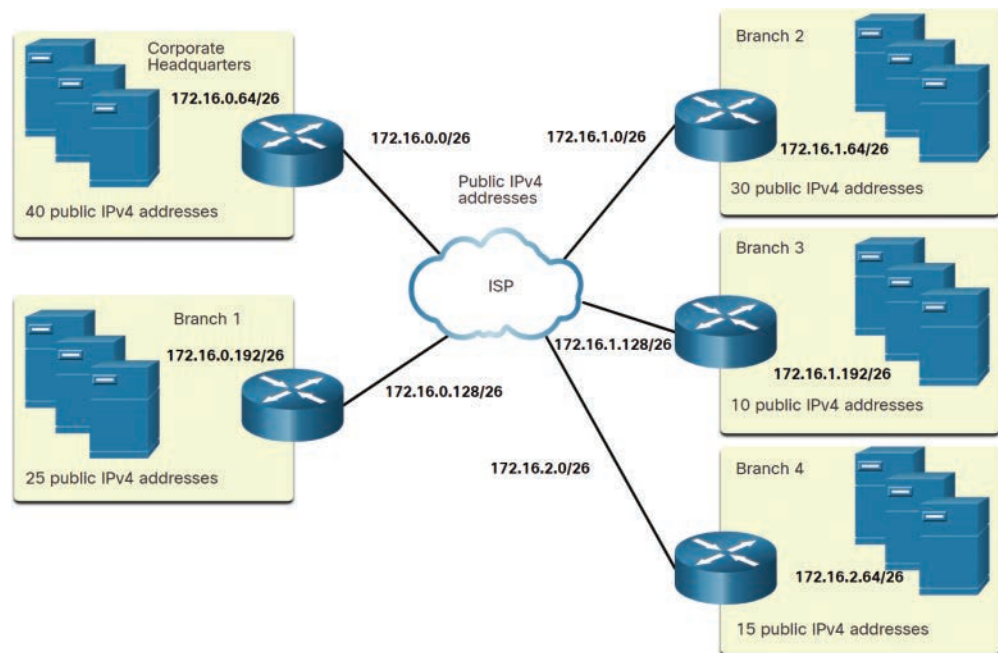


Figure 11-32 Subnet Assignments to Each Site and the ISP

**Interactive
Graphic****Activity—Determine the Number of Bits to Borrow (11.7.4)**

Refer to the online course to complete this activity.

**Packet Tracer
Activity****Packet Tracer—Subnetting Scenario (11.7.5)**

In this activity, you need to subnet the network address 192.168.100.0/24 and provide the IP addressing for the network shown in the topology. Each LAN in the network requires enough space for at least 25 addresses; this includes end devices as well as the switch and the router. The connection between R1 to R2 will require an IP address for each end of the link.

VLSM (11.8)

This section discusses a technique called *variable-length subnet masking (VLSM)* that can be used to subnet a subnet. VLSM is typically used to help conserve IPv4 address space.

Video**Video—VLSM Basics (11.8.1)**

As mentioned in the previous section, public and private addresses affect the way you subnet a network. There are also other issues that affect subnetting schemes. A standard /16 subnetting scheme creates subnets that each have the same number of hosts. Not every subnet you create will need this many hosts, and many IPv4 addresses will be unused. Perhaps you will need one subnet that contains many more hosts. This is why variable-length subnet masking (VLSM) was developed.

Refer to the online course to view this video.

Video**Video—VLSM Example (11.8.2)**

Refer to the online course to view this video.

IPv4 Address Conservation (11.8.3)

Due to the depletion of public IPv4 address space, making the most out of the available host addresses is a primary concern when subnetting IPv4 networks.

Note

The larger IPv6 address allows for much easier address planning and allocation than IPv4 allows. Conserving IPv6 addresses is not an issue. This is one of the driving forces for transitioning to IPv6.

Using traditional subnetting, the same number of addresses is allocated for each subnet. If all the subnets have the same requirements for the number of hosts, or if conserving IPv4 address space is not an issue, these fixed-size address blocks are efficient. However, with public IPv4 addresses, that is typically not the case. For example, the topology shown in Figure 11-33 requires seven subnets: one for each of the four LANs and one for each of the three connections between the routers.

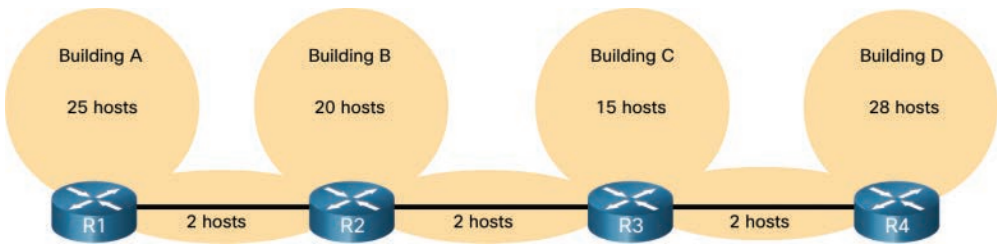


Figure 11-33 Topology Example for IPv4 Addressing

Using traditional subnetting with the address 192.168.20.0/24, 3 bits can be borrowed from the host portion in the last octet to meet the subnet requirement of seven subnets. As shown in Figure 11-34, borrowing 3 bits creates eight subnets and leaves 5 host bits with 30 usable hosts per subnet. This scheme creates the needed subnets and meets the host requirement of the largest LAN.

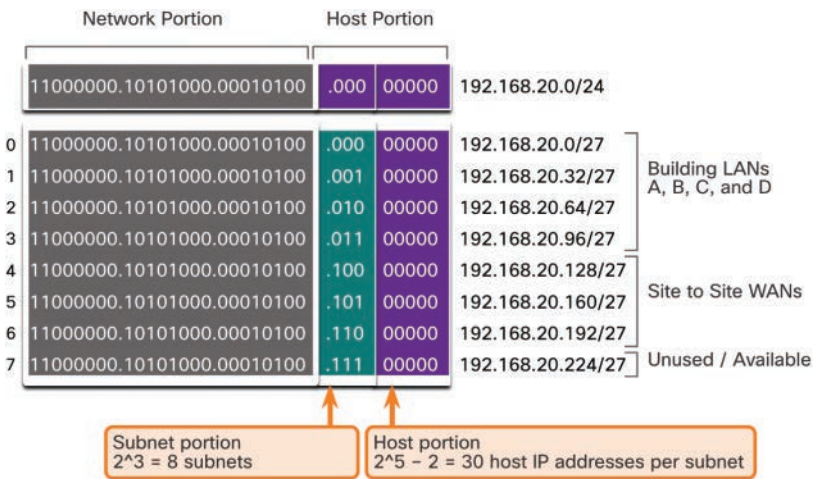


Figure 11-34 Basic Subnetting Scheme

These seven subnets could be assigned to the LAN and WAN networks, as shown in Figure 11-35.

Although this traditional subnetting meets the needs of the largest LAN and divides the address space into an adequate number of subnets, it results in significant waste of unused addresses.

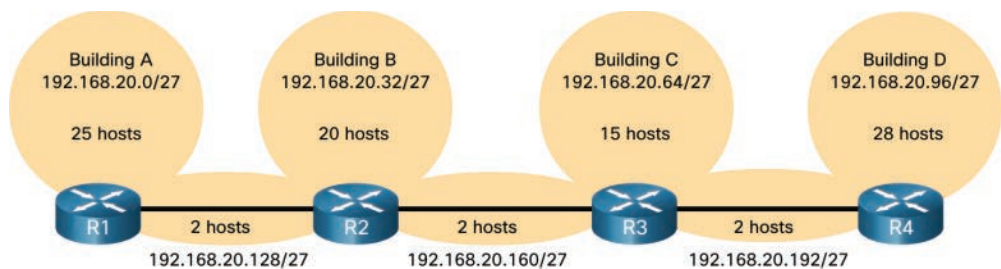


Figure 11-35 IPv4 Addresses Assigned with a /27 Subnet Mask

For example, only two addresses are needed in each subnet for the three WAN links. Because each subnet has 30 usable addresses, there are 28 unused addresses in each of these subnets. As shown in Figure 11-36, this results in 84 unused addresses (that is, 28×3).

	Network Portion	Host Portion	Dotted Decimal
4	11000000.10101000.00010100	.100 00000	192.168.20.128/27
5	11000000.10101000.00010100	.101 00000	192.168.20.160/27
6	11000000.10101000.00010100	.110 00000	192.168.20.192/27

Host portion
 $2^5 - 2 = 30$ host IP addresses per subnet
 $30 - 2 = 28$
Each WAN subnet wastes 28 addresses
 $28 \times 3 = 84$
84 addresses are unused

Figure 11-36 Unused Addresses on WAN Subnets

Furthermore, this scheme limits future growth by reducing the total number of subnets available. This inefficient use of addresses is characteristic of traditional subnetting. Applying a traditional subnetting scheme to this scenario is not very efficient and is wasteful.

Variable-length subnet masking (VLSM) was developed to avoid wasting addresses by making it possible to subnet a subnet.

VLSM (11.8.4)

In all the previous subnetting examples, the same subnet mask was applied for all the subnets. This means that each subnet has the same number of available host addresses.

As illustrated on the left side of Figure 11-37, traditional subnetting creates subnets of equal size. Each subnet in a traditional scheme uses the same subnet mask. As shown in the right side of the figure, VLSM allows a network space to be divided into unequal parts. With VLSM, the subnet mask varies depending on how many bits have been borrowed for a particular subnet—hence the *variable* part of the VLSM.

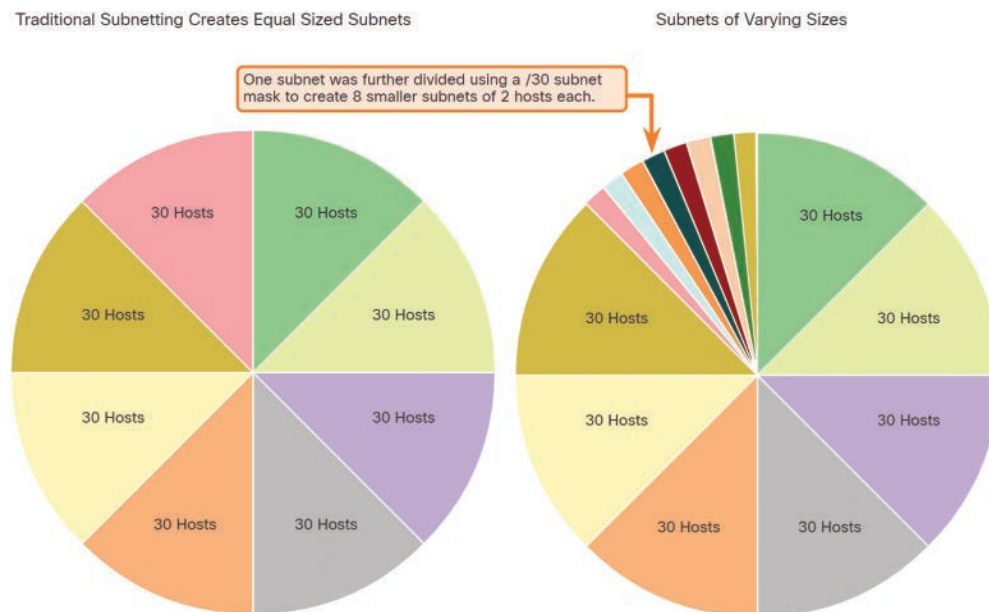


Figure 11-37 Traditional Subnetting Versus VLSM

VLSM is just subnetting a subnet. The same topology used previously is shown in Figure 11-38. In this case, we again use the 192.168.20.0/24 network and subnet it for seven subnets: one for each of the four LANs and one for each of the three connections between the routers.

Figure 11-39 shows network 192.168.20.0/24 subnetted into eight equal-sized subnets with 30 usable host addresses per subnet. Four subnets are used for the LANs, and three subnets could be used for the connections between the routers.

However, the connections between the routers require only 2 host addresses per subnet (1 host address for each router interface). Currently all subnets have 30 usable host addresses per subnet. To avoid wasting 28 addresses per subnet, VLSM can be used to create smaller subnets for the inter-router connections.

To create smaller subnets for the inter-router links, one of the subnets will be divided. In this example, the last subnet, 192.168.20.224/27, will be further subnetted. Figure 11-40 shows the last subnet subnetted further by using the subnet mask 255.255.255.252, or /30.

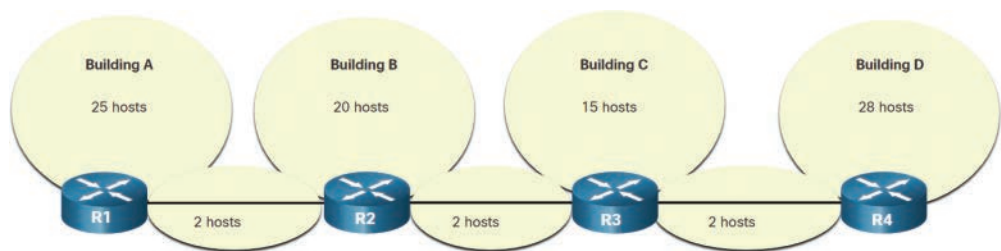


Figure 11-38 Topology Example for IPv4 Addressing

	Network portion	Host portion	Dotted Decimal	
	11000000.10101000.00010100	.00000000	192.168.20.0/24	
0	11000000.10101000.00010100	.000 00000	192.168.20.0/27	LAN's A, B, C, D
1	11000000.10101000.00010100	.001 00000	192.168.20.32/27	
2	11000000.10101000.00010100	.010 00000	192.168.20.64/27	
3	11000000.10101000.00010100	.011 00000	192.168.20.96/27	
4	11000000.10101000.00010100	.100 00000	192.168.20.128/27	Unused / Available
5	11000000.10101000.00010100	.101 00000	192.168.20.160/27	
6	11000000.10101000.00010100	.110 00000	192.168.20.192/27	
7	11000000.10101000.00010100	.111 00000	192.168.20.224/27	

Subnet 7 will be subnetted further.

Figure 11-39 Basic Subnetting Scheme

Why /30? Recall that when the number of needed host addresses is known, the formula $2^n - 2$ (where n equals the number of host bits remaining) can be used. To provide two usable addresses, 2 host bits must be left in the host portion.

Because there are 5 host bits in the subnetted 192.168.20.224/27 address space, 3 more bits can be borrowed, leaving 2 bits in the host portion. The calculations at this point are exactly the same as those used for traditional subnetting. The bits are borrowed, and the subnet ranges are determined. Figure 11-41 shows how the four /27 subnets have been assigned to the LANs and three of the /30 subnets have been assigned to the inter-router links.

This VLSM subnetting scheme reduces the number of addresses per subnet to a size appropriate for the networks that require fewer subnets. Subnetting subnet 7 for inter-router links allows subnets 4, 5, and 6 to be available for future networks and makes five additional subnets available for inter-router connections.

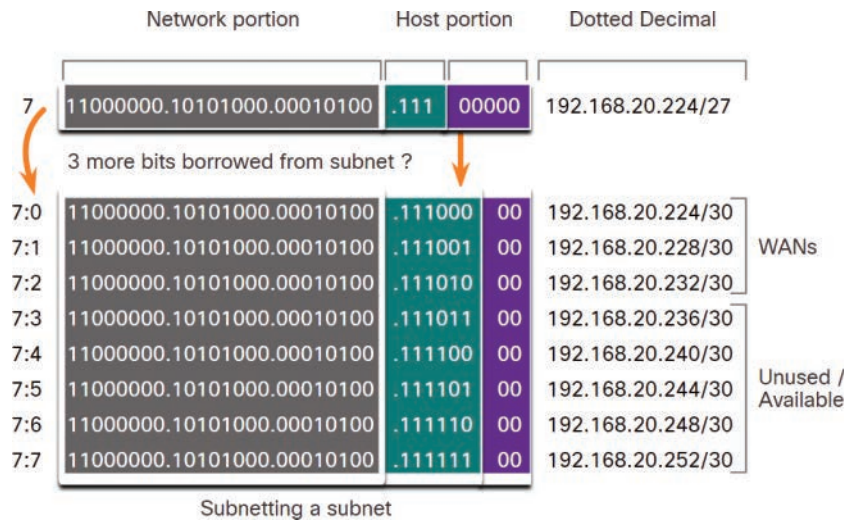


Figure 11-40 VLSM Subnetting Scheme

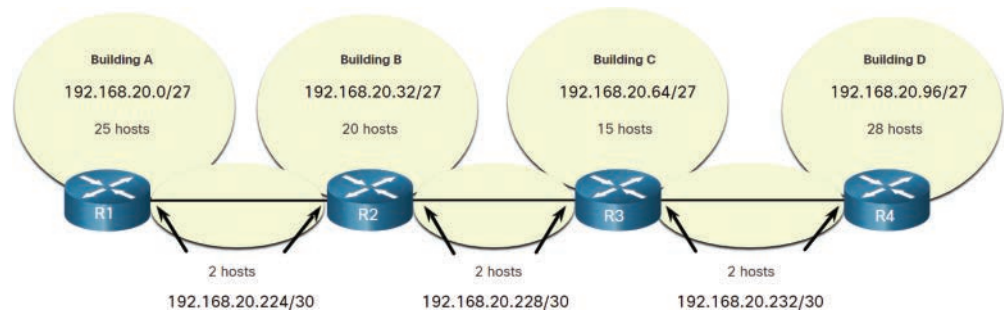


Figure 11-41 VLSM Addressing Scheme Assigned to Networks

Note

When using VLSM, always begin by satisfying the host requirements of the largest subnet. Continue subnetting until the host requirements of the smallest subnet are satisfied.

VLSM Topology Address Assignment (11.8.5)

Using the VLSM subnets, the LAN and inter-router networks can be addressed without unnecessary waste.

Figure 11-42 shows the network address assignments and the IPv4 addresses assigned to the router interfaces.

Using a common addressing scheme, the first host IPv4 address for each subnet is assigned to the LAN interface of the router. Hosts on each subnet will have a host IPv4

address from the range of host addresses for that subnet and an appropriate mask. Hosts will use the address of the attached router LAN interface as the default gateway address.

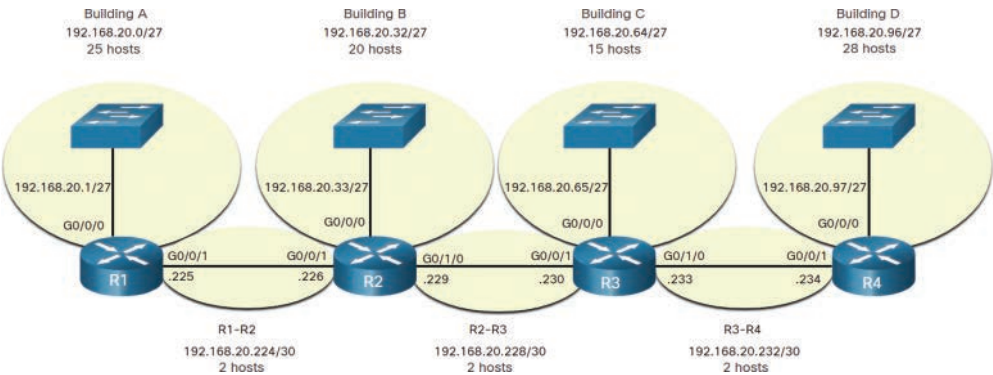


Figure 11-42 IPv4 Addresses Assigned to Interfaces

Table 11-12 shows the network addresses and the range of host addresses for each network. The default gateway address is displayed for each of the four LANs.

Table 11-12 VLSM Addressing Table

	Network Address	Range of Host Addresses	Default Gateway Address
Building A	192.168.20.0/27	192.168.20.1/27–192.168.20.30/27	192.168.20.1/27
Building B	192.168.20.32/27	192.168.20.33/27–192.168.20.62/27	192.168.20.33/27
Building C	192.168.20.64/27	192.168.20.65/27–192.168.20.94/27	192.168.20.65/27
Building D	192.168.20.96/27	192.168.20.97/27–192.168.20.126/27	192.168.20.97/27
R1–R2	192.168.20.224/30	192.168.20.225/30–192.168.20.226/30	
R2–R3	192.168.20.228/30	192.168.20.229/30–192.168.20.230/30	
R3–R4	192.168.20.232/30	192.168.20.233/30–192.168.20.234/30	

Interactive
Graphic

Activity—VLSM Practice (11.8.6)

Refer to the online course to complete this activity.

Structured Design (11.9)

To accommodate all the current and future devices that need IP address, it is necessary to develop a plan and an addressing schema that meets the requirements of the network.

IPv4 Network Address Planning (11.9.1)

Before you start subnetting, you should develop an IPv4 addressing schema for your entire network. You must determine how many subnets you need, how many hosts a particular subnet requires, what devices are part of the subnet, which parts of your network use private addresses and which use public, and many other factors. A good addressing scheme allows for growth. A good addressing scheme is also the sign of a good network administrator.

Planning IPv4 network subnets requires you to examine both the needs of an organization's network usage and how the subnets will be structured. Performing a network requirement study is the starting point. This means looking at the entire network—both the intranet and the DMZ—and determining how each area will be segmented. The address plan includes determining where address conservation is needed (usually in the DMZ) and where there is more flexibility (usually in the intranet).

Where address conservation is required, the plan should determine how many subnets are needed and how many hosts per subnet are needed. As discussed earlier, conservation is usually required for public IPv4 address space within the DMZ, and it can often be addressed by using VLSM.

Address conservation is usually less of an issue in the corporate intranet than in the DMZ. This is largely due to the fact that private IPv4 addressing, including 10.0.0.0/8, provides more than 16 million host IPv4 addresses.

For most organizations, private IPv4 addresses allow for more than enough internal (intranet) addresses. For many larger organizations and ISPs, even private IPv4 address space is not large enough to accommodate the internal needs. This is another reason organizations are transitioning to IPv6.

For intranets that use private IPv4 addresses and DMZs that use public IPv4 addresses, address planning and assignment are important.

An address plan should typically include a determination of the needs of each subnet in terms of size. How many hosts will there be per subnet? The address plan also needs to include how host addresses will be assigned, which hosts will require static IPv4 addresses, and which hosts can use DHCP for obtaining their addressing information. This will also help prevent duplication of addresses, while allowing for monitoring and management of addresses for performance and security reasons.

Knowing your IPv4 address requirements will help you determine the range, or ranges, of host addresses to implement and help ensure that there are enough addresses to cover your network needs.

Device Address Assignment (11.9.2)

Within a network, different types of devices require addresses:

- **End-user clients:** Most networks allocate IPv4 addresses to client devices dynamically, using Dynamic Host Configuration Protocol (DHCP). This reduces the burden on network support staff and virtually eliminates entry errors. With DHCP, addresses are only leased for a period of time, and they can be reused when the lease expires. This is an important feature for networks that support transient users and wireless devices. Changing the subnetting scheme means that the DHCP server needs to be reconfigured, and the clients must renew their IPv4 addresses. IPv6 clients can obtain address information by using DHCPv6 or SLAAC.
- **Servers and peripherals:** Each server or peripheral should have a predictable static IP address. Use a consistent numbering system for these devices.
- **Servers that are accessible from the internet:** Any server that needs to be publicly available on the internet must have a public IPv4 address, most often accessed using NAT. In some organizations, internal servers (which are not publicly available) must be made available to remote users. In most cases, these servers are assigned private addresses internally, and the user is required to create a virtual private network (VPN) connection to access the server. This has the same effect as the user accessing the server from a host within the intranet.
- **Intermediary devices:** These devices are assigned addresses for network management, monitoring, and security. Because network administrators need to know how to communicate with intermediary devices, they should have predictable, statically assigned addresses.
- **The gateway:** Routers and firewall devices have an IP address assigned to each interface that serves as the gateway for the hosts in that network. Typically, the router interface uses either the lowest or highest address in the network.

When developing an IP addressing scheme, it is generally recommended that you follow a set pattern for allocating addresses to the various types of devices. Having such conventions benefits administrators when adding and removing devices and when filtering traffic based on IP address, and it also simplifies documentation.

Packet Tracer

□ Activity

Packet Tracer—VLSM Design and Implementation Practice (11.9.3)

In this activity, you are given a /24 network address to use to design a VLSM addressing scheme. Based on a set of requirements, you will assign subnets and addressing, configure devices, and verify connectivity.

Summary (11.10)

The following is a summary of the topics in the chapter and their corresponding online modules.

IPv4 Addressing Structure

An IPv4 address is a 32-bit hierarchical address that is made up of a network portion and a host portion. The bits in the network portion of the address must be identical for all devices that reside in the same network. The bits in the host portion of the address must be unique to identify a specific host within a network. A host requires a unique IPv4 address and a subnet mask to show the network/host portions of the address. The prefix length is the number of bits set to 1 in the subnet mask. It is written in “slash notation,” which is a / followed by the number of bits set to 1. Logical AND is the comparison of 2 bits. Only a 1 AND 1 produces 1, and any other combination results in 0. Within each network are network addresses, host addresses, and a broadcast address.

IPv4 Unicast, Broadcast, and Multicast

Unicast transmission refers to a device sending a message to one other device in one-to-one communications. A unicast packet is a packet with a destination IP address that is a unicast address, which is the address of a single recipient. Broadcast transmission refers to a device sending a message to all the devices on a network in one-to-all communications. A broadcast packet has a destination IP address with all ones (1s) in the host portion, or 32 one (1) bits. Multicast transmission reduces traffic by allowing a host to send a single packet to a selected set of hosts that subscribe to a multicast group. A multicast packet is a packet with a destination IP address that is a multicast address. IPv4 has reserved the 224.0.0.0 to 239.255.255.255 addresses as a multicast range.

Types of IPv4 Addresses

Public IPv4 addresses are globally routed between ISP routers. Not all available IPv4 addresses can be used on the internet. Blocks of addresses called private addresses are used by most organizations to assign IPv4 addresses to internal hosts. Most internal networks use private IPv4 addresses for addressing all internal devices (in intranets); however, these private addresses are not globally routable. A host can use the loopback address to direct traffic back to itself. Link-local addresses are more commonly known as APIPA addresses, or self-assigned addresses. In 1981, IPv4 addresses were assigned using classful addressing: A, B, or C. Public IPv4 addresses must be unique and are globally routed over the internet. Both IPv4 and IPv6 addresses are managed by IANA, which allocates blocks of IP addresses to the RIRs.

Network Segmentation

In an Ethernet LAN, devices locate other devices by using ARP. A switch propagates a broadcast out all interfaces except the interface on which it was received. Routers do not propagate broadcasts; instead, each router interface connects a broadcast domain, and broadcasts are propagated only within that specific domain. A large broadcast domain is a network that connects many hosts. A problem with a large broadcast domain is that the hosts can generate excessive broadcasts and negatively affect the network. The solution is to reduce the size of the network to create smaller broadcast domains in a process called subnetting. These smaller network spaces are called subnets. Subnetting reduces overall network traffic and improves network performance. An administrator may subnet by location, between networks, or by device type.

Subnet an IPv4 Network

IPv4 subnets are created by using one or more of the host bits as network bits. This is done by extending the subnet mask to borrow some of the bits from the host portion of the address to create additional network bits. The more host bits that are borrowed, the more subnets that can be defined. The more bits that are borrowed to increase the number of subnets, the smaller the number of hosts per subnet. Networks are most easily subnetted at the octet boundaries: /8, /16, and /24. Subnets can borrow bits from any host bit position to create other masks.

Subnet a /16 and a /8 Prefix

A situation requiring a larger number of subnets calls for an IPv4 network that has more host bits available to borrow. To create subnets, you must borrow bits from the host portion of the IPv4 address of the existing internetwork. Starting from the left with the first available host bit, borrow a single bit at a time until you reach the number of bits necessary to create the number of subnets required. When borrowing bits from a /16 address, start borrowing bits in the third octet, going from left to right. The first address is reserved for the network address, and the last address is reserved for the broadcast address.

Subnet to Meet Requirements

A typical enterprise network contains an intranet and a DMZ, both of which have subnetting requirements and challenges. An intranet uses private IPv4 addressing space. The 10.0.0.0/8 network can also be subnetted using any other number of prefix lengths, such as /12, /18, /20, and so on, which means the network administrator has many options. Because devices in the DMZ need to be publicly accessible from the internet, these devices require public IPv4 addresses. An organization must maximize its own limited supply of public IPv4 addresses. To reduce the number of unused

host addresses per subnet, a network administrator must subnet the public address space into subnets with different subnet masks. This is known as variable-length subnet masking (VLSM). Administrators must consider how many host addresses are required for each network and how many subnets are needed.

Variable-Length Subnet Masking

Traditional subnetting might meet an organization's needs for its largest LAN and divide the address space into an adequate number of subnets. But it is also likely to result in significant waste of unused addresses. VLSM allows a network space to be divided into unequal parts. With VLSM, the subnet mask varies depending on how many bits have been borrowed for a particular subnet—hence the *variable* part of the VLSM. VLSM is just subnetting a subnet. When using VLSM, always begin by satisfying the host requirements of the largest subnet. Continue subnetting until the host requirements of the smallest subnet are satisfied. A subnet always needs to be started on an appropriate bit boundary.

Structured Design

A network administrator should study the network requirements to better plan how IPv4 network subnets will be structured. This means looking at the entire network—both the intranet and the DMZ—and determining how each area will be segmented. The address plan includes determining where address conservation is needed (usually within the DMZ) and where there is more flexibility (usually within the intranet). Where address conservation is required, the plan should determine how many subnets are needed and how many hosts per subnet are needed. As discussed earlier, conservation is usually required for public IPv4 address space within the DMZ, and it can often be addressed by using VLSM. The address plan includes how host addresses will be assigned, which hosts will require static IPv4 addresses, and which hosts can use DHCP for obtaining their addressing information. In a network, different types of devices require addresses: end-user clients, servers and peripherals, servers that are accessible from the internet, intermediary devices, and gateways. When developing an IP addressing scheme, it is generally recommended that you follow a set pattern for allocating addresses to the various types of devices. Having such conventions benefits administrators when adding and removing devices and when filtering traffic based on IP address, and it also simplifies documentation.



Packet Tracer—Design and Implement a VLSM Addressing Scheme (11.10.1)

In this activity, you will design a VLSM addressing scheme based on a network address and host requirements. You will configure addressing on routers, switches, and network hosts:

1. Design a VLSM IP addressing scheme based on the given requirements.

2. Configure addressing on network devices and hosts.
 3. Verify IP connectivity.
 4. Troubleshoot connectivity issues, as required.
-



Lab—Design and Implement a VLSM Addressing Scheme (11.10.2)

In this lab, use the 192.168.33.128/25 network address to develop an addressing scheme for the network displayed in the topology diagram. Use VLSM to meet the IPv4 addressing requirements. After you have designed the VLSM address scheme, you will configure the interfaces on the routers with the appropriate IP address information. The future LANs at BR2 need to have addresses allocated, but no interfaces will be configured at this time.

Practice

The following activities provide practice with the topics introduced in this chapter. The lab is available in the companion *Introduction to Networks Labs & Study Guide (CCNA v7)* (ISBN 9780136634454). The Packet Tracer activity instructions are also provided in the *Labs & Study Guide*. The PKA files are available in the online course.



Labs

Lab 11.6.6: Calculate IPv4 Subnets

Lab 11.10.2: Design and Implement a VLSM Addressing Scheme



Packet Tracer Activities

Packet Tracer 11.5.5: Subnet an IPv4 Network

Packet Tracer 11.7.5: Subnetting Scenario

Packet Tracer 11.9.3: VLSM Design and Implementation Practice

Packet Tracer 11.10.1: Design and Implement a VLSM Addressing Scheme

Check Your Understanding Questions

Complete all the review questions listed here to test your understanding of the topics and concepts in this chapter. The appendix “Answers to ‘Check Your Understanding’ Questions” lists the answers.

1. What is the prefix length notation for the subnet mask 255.255.255.224?
 - a. /25
 - b. /26
 - c. /27
 - d. /28
2. How many valid host addresses are available on an IPv4 subnet that is configured with a /26 mask?
 - a. 254
 - b. 190
 - c. 192
 - d. 62
 - e. 64
3. Which subnet mask would be used if 5 host bits are available?
 - a. 255.255.255.0
 - b. 255.255.255.128
 - c. 255.255.255.224
 - d. 255.255.255.240
4. A network administrator subnets the 192.168.10.0/24 network into subnets with /26 masks. How many equal-sized subnets are created?
 - a. 1
 - b. 2
 - c. 4
 - d. 8
 - e. 16
 - f. 64
5. What subnet mask is represented by the slash notation /20?
 - a. 255.255.255.248
 - b. 255.255.224.0
 - c. 255.255.240.0
 - d. 255.255.255.0
 - e. 255.255.255.192

6. Which statement is true about variable-length subnet masking?
 - a. All the subnets are equally sized.
 - b. The sizes of subnets may be different, depending on requirements.
 - c. Subnets may only be subnetted one additional time.
 - d. Bits are returned, rather than borrowed, to create additional subnets.
7. Why does a Layer 3 device perform the ANDing process on a destination IPv4 address and subnet mask?
 - a. to identify the broadcast address of the destination network
 - b. to identify the host address of the destination host
 - c. to identify faulty frames
 - d. to identify the network address of the destination network
8. How many usable IPv4 addresses are available on the 192.168.1.0/27 network?
 - a. 256
 - b. 254
 - c. 62
 - d. 30
 - e. 16
 - f. 32
9. Which subnet mask would be used if exactly 4 host bits are available?
 - a. 255.255.255.224
 - b. 255.255.255.128
 - c. 255.255.255.240
 - d. 255.255.255.248
10. Which of the following are components of an IPv4 address? (Choose two.)
 - a. subnet portion
 - b. network portion
 - c. logical portion
 - d. host portion
 - e. physical portion
 - f. broadcast portion

11. If a network device has a mask of /30, how many IPv4 addresses are available for hosts on this network?
 - a. 64
 - b. 8
 - c. 2
 - d. 32
 - e. 16
 - f. 4
12. What does the IPv4 address 172.17.4.250/24 represent?
 - a. network address
 - b. multicast address
 - c. host address
 - d. broadcast address
13. If a network device has a mask of /28, how many IP addresses are available for hosts on this network?
 - a. 256
 - b. 254
 - c. 62
 - d. 32
 - e. 16
 - f. 14
14. What is the purpose of the subnet mask in conjunction with an IPv4 address?
 - a. to uniquely identify a host on a network
 - b. to identify whether the address is public or private
 - c. to determine the subnet to which the host belongs
 - d. to mask the IP address to outsiders
15. A network administrator is variably subnetting a network. The smallest subnet has a mask of 255.255.255.224. How many usable host addresses will this subnet provide?
 - a. 2
 - b. 6
 - c. 14
 - d. 30
 - e. 62

IPv6 Addressing

Objectives

Upon completion of this chapter, you will be able to answer the following questions:

- Why is IPv6 addressing needed?
- How are IPv6 addresses represented?
- What are the types of IPv6 network addresses?
- How do you configure static global unicast and link-local IPv6 network addresses?
- How do you configure global unicast addresses dynamically?
- How do you configure link-local addresses dynamically?
- How do you identify IPv6 addresses?
- How do you implement a subnetted IPv6 addressing scheme?

Key Terms

This chapter uses the following key terms. You can find the definitions in the glossary at the end of the book.

dual stack page 399

tunneling page 400

Network Address Translation 64 (NAT64) page 401

preferred format page 402

global unicast address (GUA) page 408

link-local address (LLA) page 408

global routing prefix page 410

subnet ID page 410

interface ID page 410

Router Advertisement (RA) message page 417

Router Solicitation (RS) message page 417

stateless address autoconfiguration (SLAAC) page 418

stateless DHCPv6 page 419

stateful DHCPv6 page 420

Extended Unique Identifier (EUI) page 422

well-known IPv6 multicast address page 430

solicited-node multicast address page 432

Introduction (12.0)

It is a great time to be (or become) a network administrator! Why? In many networks, you will find both IPv4 and IPv6 working together. After the hard work of learning to subnet an IPv4 network, you may find that subnetting an IPv6 network is much easier. You probably didn't expect that, did you? A Packet Tracer at the end of this chapter will give you the opportunity to subnet an IPv6 network. Go ahead, jump in!

IPv4 Issues (12.1)

This section examines the reasons for the migration to IPv6.

Need for IPv6 (12.1.1)

You already know about the shortage of IPv4 addresses. That is why you need to learn about IPv6.

IPv6 is designed to be the successor to IPv4. IPv6 has a larger 128-bit address space, providing 340 undecillion (that is, 340 followed by 36 zeros) possible addresses. However, IPv6 is more than just larger and more addresses.

When the IETF began its development of a successor to IPv4, it took the opportunity to fix the limitations of IPv4 and include enhancements. One example is Internet Control Message Protocol version 6 (ICMPv6), which includes address resolution and address autoconfiguration not found in ICMP for IPv4 (ICMPv4).

The depletion of IPv4 address space has been the motivating factor for moving to IPv6. Africa, Asia, and other areas of the world have become more connected to the internet, and there are not enough IPv4 addresses to accommodate this growth. As shown in Figure 12-1, four out of the five RIRs have run out of IPv4 addresses.

IPv4 has a theoretical maximum of 4.3 billion addresses. Private addresses in combination with Network Address Translation (NAT) have been instrumental in slowing the depletion of IPv4 address space. However, NAT is problematic for many applications, creates latency, and has limitations that severely impede peer-to-peer communications.

With the ever-increasing number of mobile devices, mobile providers have been leading the way with the transition to IPv6. The top two mobile providers in the United States report that over 90% of their traffic now occurs over IPv6.

Most top ISPs and content providers such as YouTube, Facebook, and Netflix have also made the transition. Many companies, including Microsoft, Facebook, and LinkedIn, are transitioning to IPv6-only networks internally. In 2018, broadband ISP Comcast reported IPv6 deployment of over 65%, and British Sky Broadcasting reported deployment of over 86%.

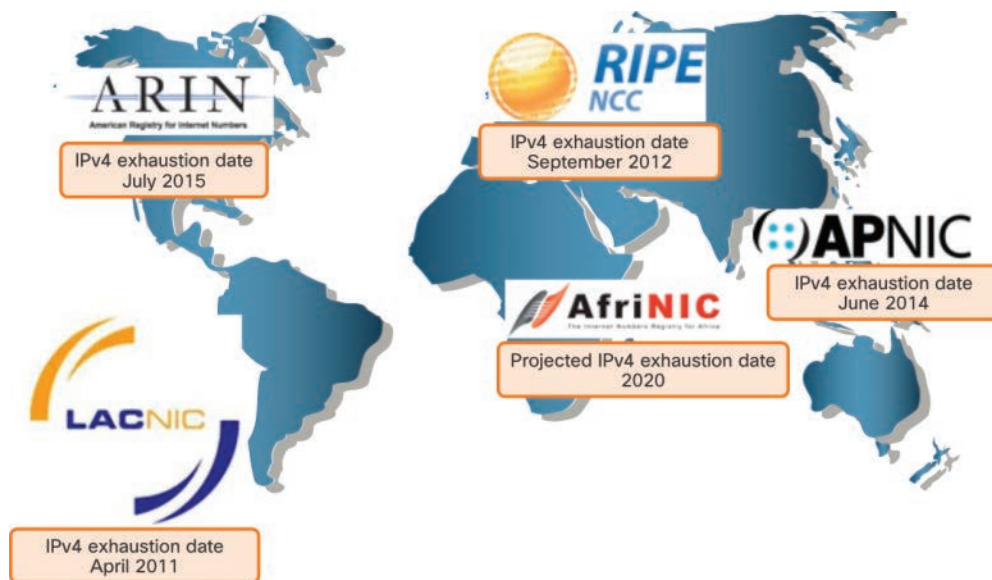


Figure 12-1 RIR IPv4 Exhaustion Dates

Internet of Things

The internet of today is significantly different from the internet of past decades. The internet of today is more than email, web pages, and file transfers between computers. The evolving internet is becoming an Internet of Things (IoT). No longer are the only devices accessing the internet computers, tablets, and smartphones. Sensor-equipped, internet-ready devices include everything from automobiles and biomedical devices to household appliances and natural ecosystems.

With an increasing internet population, a limited IPv4 address space, issues with NAT and the IoT, the time has come to transition to IPv6 in earnest.

IPv4 and IPv6 Coexistence (12.1.2)

There is no specific date for moving to an IPv6-only internet. IPv4 and IPv6 will coexist in the near future, and the transition will take several years. The IETF has created various protocols and tools to help network administrators migrate their networks to IPv6. The migration techniques can be divided into three categories: dual stack, tunneling, and translation.

Dual Stack

Dual stack allows IPv4 and IPv6 to coexist on the same network segment. Dual stack devices run the IPv4 and IPv6 protocol stacks simultaneously, as shown in Figure 12-2. With dual stack, also known as *native IPv6*, the customer network has an IPv6 connection to the ISP and is able to access content found on the internet over IPv6.

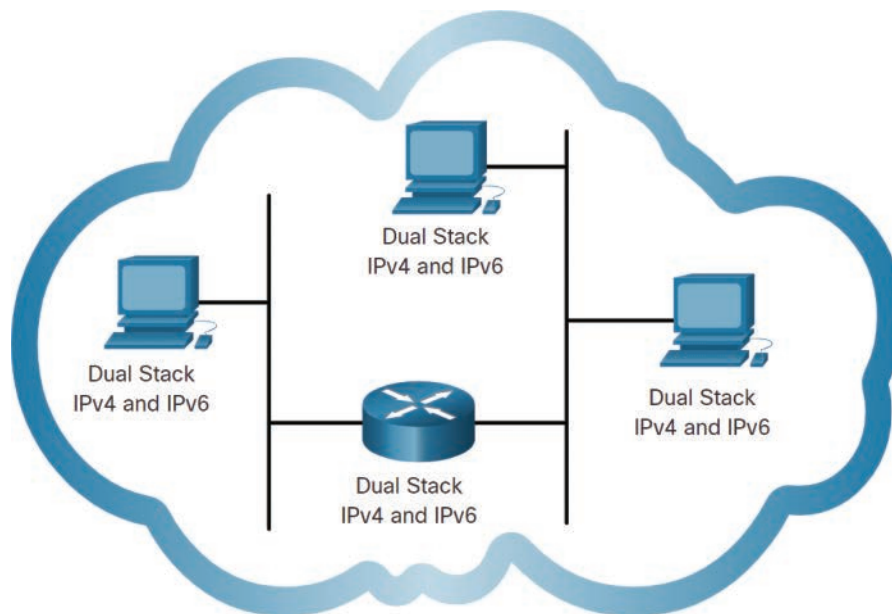


Figure 12-2 Dual Stack Topology

Tunneling

Tunneling is a method of transporting an IPv6 packet over an IPv4 network. The IPv6 packet is encapsulated inside an IPv4 packet, similar to other types of data, as shown in Figure 12-3.

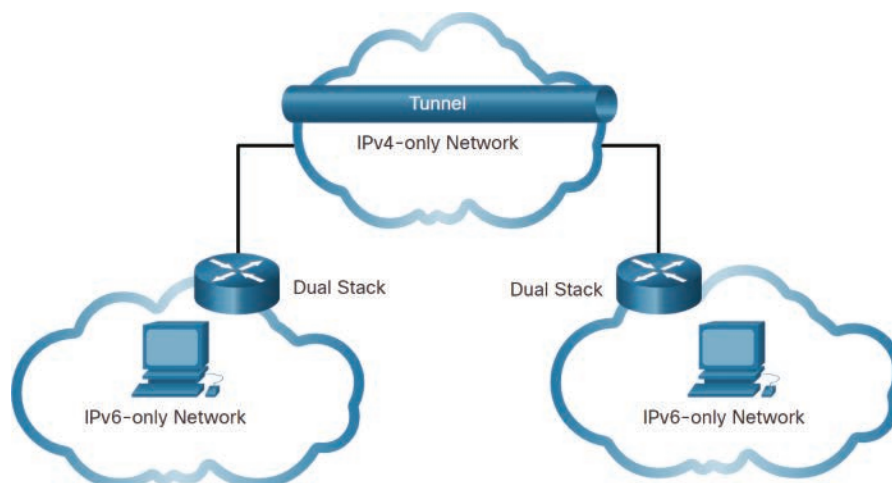


Figure 12-3 Tunneling Topology

Translation

Network Address Translation 64 (NAT64) allows IPv6-enabled devices to communicate with IPv4-enabled devices using a translation technique similar to NAT for IPv4. An IPv6 packet is translated to an IPv4 packet, and an IPv4 packet is translated to an IPv6 packet, as shown in Figure 12-4.

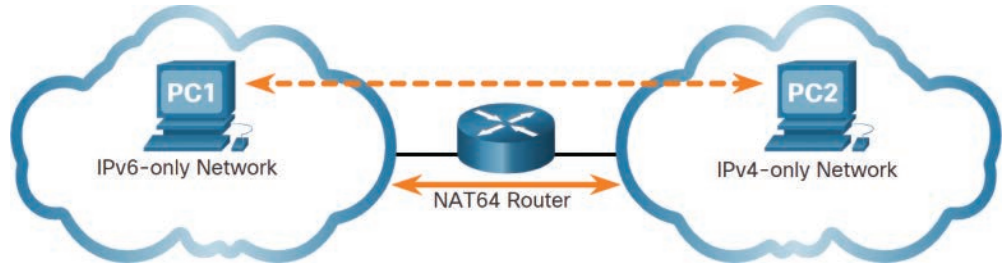


Figure 12-4 NAT64 Topology

Note

Tunneling and translation are for transitioning to native IPv6 and should only be used where needed. The goal should be native IPv6 communications from source to destination.

Interactive Graphic

Check Your Understanding—IPv4 Issues (12.1.3)

Refer to the online course to complete this activity.

IPv6 Address Representation (12.2)

This section discusses the representation of IPv6 addresses.

IPv6 Addressing Formats (12.2.1)

The first step in learning about IPv6 in networks is to understand the way an IPv6 address is written and formatted. IPv6 addresses are much larger than IPv4 addresses, which is why we are unlikely to run out of them.

An IPv6 address is 128 bits in length and written as a string of hexadecimal values. Every 4 bits is represented by a single hexadecimal digit, for a total of 32 hexadecimal values, as shown in Figure 12-5. IPv6 addresses are not case-sensitive and can be written in either lowercase or uppercase.

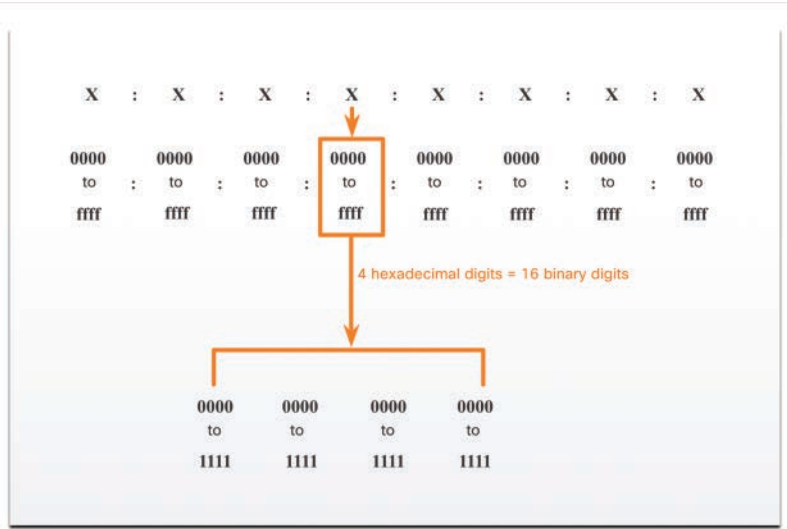


Figure 12-5 16-Bit Segments, or Hextets

Preferred Format

Figure 12-5 shows the preferred format for writing an IPv6 address: $x:x:x:x:x:x$, with each x consisting of 4 hexadecimal values. The term *octet* refers to the 8 bits of an IPv4 address. In IPv6, *hextet* is the unofficial term used to refer to a segment of 16 bits, or 4 hexadecimal values. Each x is a single hextet, which is 16 bits or 4 hexadecimal digits.

When you use *preferred format*, you write an IPv6 address using all 32 hexadecimal digits. Even though this format is referred to as *preferred*, it is not necessarily the *ideal* method for representing an IPv6 address. The following sections describe two rules that help reduce the number of digits needed to represent an IPv6 address.

Example 12-1 shows several IPv6 addresses in preferred format.

Example 12-1 IPv6 Address Preferred Format

```
2001 : 0db8 : 0000 : 1111 : 0000 : 0000 : 0000 : 0200
2001 : 0db8 : 0000 : 00a3 : abcd : 0000 : 0000 : 1234
2001 : 0db8 : 000a : 0001 : c012 : 9aff : fe9a : 19ac
2001 : 0db8 : aaaa : 0001 : 0000 : 0000 : 0000 : 0000
fe80 : 0000 : 0000 : 0000 : 0123 : 4567 : 89ab : cdef
fe80 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0001
fe80 : 0000 : 0000 : 0000 : c012 : 9aff : fe9a : 19ac
fe80 : 0000 : 0000 : 0000 : 0123 : 4567 : 89ab : cdef
0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0001
0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000
```

Rule 1—Omit Leading Zeros (12.2.2)

The first rule to help reduce the number of digits in IPv6 addresses is to omit any leading 0s (zeros) in any hextet. Here are four examples of ways to omit leading 0s:

- 01ab can be represented as 1ab
- 09f0 can be represented as 9f0
- 0a00 can be represented as a00
- 00ab can be represented as ab

This rule only applies only to leading 0s—not to trailing 0s, which would make addresses ambiguous. For example, if both leading and trailing 0s could be omitted, the hextet abc could be either 0abc or abc0, but these do not represent the same value. Table 12-1 shows examples of omitting leading 0s.

Table 12-1 Omitting Leading 0s

Type	Format
Preferred	2001:0db8:0000:1111:0000:0000:0200
No leading 0s	2001: db8: 0:1111: 0: 0: 0: 200
Preferred	2001:0db8:0000:00a3:ab00:0ab0:00ab:1234
No leading 0s	2001: db8: 0: a3:ab00: ab0: ab:1234
Preferred	2001:0db8:000a:0001:c012:90ff:fe90:0001
No leading 0s	2001: db8: a: 1:c012:90ff:fe90: 1
Preferred	2001:0db8:aaaa:0001:0000:0000:0000:0000
No leading 0s	2001: db8:aaaa: 1: 0: 0: 0: 0
Preferred	fe80:0000:0000:0000:0123:4567:89ab:cdef
No leading 0s	fe80: 0: 0: 0: 123:4567:89ab:cdef

Type	Format
Preferred	fe80 : 0000 : 0000 : 0000 : 0000 : 0000 : 0001
No leading 0s	fe80 : 0 : 0 : 0 : 0 : 0 : 0 : 1
Preferred	0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0001
No leading 0s	0 : 0 : 0 : 0 : 0 : 0 : 0 : 1
Preferred	0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000
No leading 0s	0 : 0 : 0 : 0 : 0 : 0 : 0 : 0

Rule 2—Double Colon (12.2.3)

The second rule to help reduce the number of digits in IPv6 addresses is that a double colon (::) can replace any single, contiguous string of one or more 16-bit hextets consisting of all 0s. For example, 2001:db8:cafe:1:0:0:0:1 (leading 0s omitted) could be represented as 2001:db8:cafe:1::1. The double colon (::) is used in place of the three all-0 hextets (0:0:0).

The double colon (::) can be used only once in an address; if it could be used more than once, the address could expand to more than one possible result. Using the double colon rule and also omitting leading 0s—commonly known as compressed format—often allows you to greatly reduce the number of characters in an IPv6 address.

Here is an example of the incorrect use of the double colon: 2001:db8::abcd::1234. The double colon is used twice in this example, which you know violates the rule. This incorrect compressed format address could be expanded multiple ways:

- 2001:db8::abcd:0000:0000:1234
- 2001:db8::abcd:0000:0000:0000:1234
- 2001:db8:0000:abcd::1234
- 2001:db8:0000:0000:abcd::1234

If an address has more than one contiguous string of all-0 hextets, best practice is to use the double colon (::) on the longest string. If the strings are equal, the first string should use the double colon (::). Table 12-2 shows examples of omitting leading 0s and all 0 segments.

Table 12-2 Omitting Leading 0s and All 0 Segments

Type	Format
Preferred	2001 : 0db8 : 0000 : 1111 : 0000 : 0000 : 0200
Compressed with spaces	2001 : db8 : 0 : 1111 : : 200
Compressed	2001:db8:0:1111::200
Preferred	2001 : 0db8 : 0000 : 0000 : ab00 : 0000 : 0000
Compressed with spaces	2001 : db8 : 0 : 0 : ab00 ::
Compressed	2001:db8:0:0:ab00::
Preferred	2001 : 0db8 : aaaa : 0001 : 0000 : 0000 : 0000
Compressed with spaces	2001 : db8 : aaaa : 1 ::
Compressed	2001:db8:aaaa:1::
Preferred	fe80 : 0000 : 0000 : 0000 : 0123 : 4567 : 89ab : cdef
Compressed with spaces	fe80 : : 123 : 4567 : 89ab : cdef
Compressed	fe80::123:4567:89ab:cdef
Preferred	fe80 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0001
Compressed with spaces	fe80 : : : 1
Compressed	fe80::0
Preferred	0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0001
Compressed with spaces	: : : 1
Compressed	::1
Preferred	0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000
Compressed with spaces	::
Compressed	::

**Interactive
Graphic****Activity—IPv6 Address Representation (12.2.4)**

Refer to the online course to complete this activity.

IPv6 Address Types (12.3)

This section introduces the different types and uses of IPv6 addresses.

Unicast, Multicast, Anycast (12.3.1)

As with IPv4, there are different types of IPv6 addresses. In fact, there are three broad categories of IPv6 addresses:

- **Unicast:** An IPv6 unicast address uniquely identifies an interface on an IPv6-enabled device.
- **Multicast:** An IPv6 multicast address is used to send a single IPv6 packet to multiple destinations.
- **Anycast:** An IPv6 anycast address is any IPv6 unicast address that can be assigned to multiple devices. A packet sent to an anycast address is routed to the nearest device that has that address. Anycast addresses are beyond the scope of this book.

Unlike IPv4, IPv6 does not have a broadcast address. However, there is an IPv6 all-nodes multicast address that essentially does the same thing.

IPv6 Prefix Length (12.3.2)

The prefix, or network portion, of an IPv4 address can be identified by a dotted decimal subnet mask or prefix length (slash notation). For example, the IPv4 address 192.168.1.10 with dotted decimal subnet mask 255.255.255.0 is equivalent to 192.168.1.10/24.

In IPv6 it is only called the *prefix length*. IPv6 does not use the dotted decimal subnet mask notation. As in IPv4, the prefix length is represented in slash notation and is used to indicate the network portion of an IPv6 address.

The prefix length can range from 0 to 128. The recommended IPv6 prefix length for LANs and most other types of networks is /64, as shown in Figure 12-6.

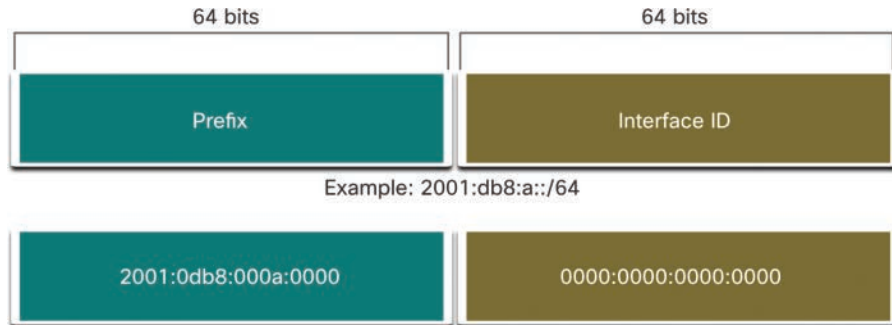


Figure 12-6 IPv6 Prefix Length

It is strongly recommended to use a 64-bit interface ID for most networks. This is because stateless address autoconfiguration (SLAAC) uses 64 bits for the interface ID. It also makes subnetting easier to create and manage.

Types of IPv6 Unicast Addresses (12.3.3)

An IPv6 unicast address uniquely identifies an interface on an IPv6-enabled device. A packet sent to a unicast address is received by the interface assigned that address. Much as with IPv4, a source IPv6 address must be a unicast address. The destination IPv6 address can be either a unicast or multicast address. Figure 12-7 shows the different types of IPv6 unicast addresses.

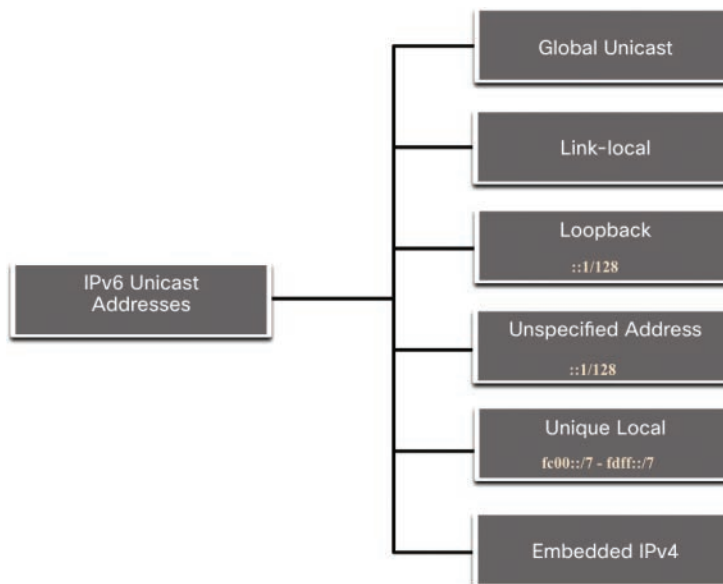


Figure 12-7 IPv6 Unicast Addresses

Unlike an IPv4 device, which has only a single address, an IPv6 address typically has two unicast addresses:

- **Global unicast address (GUA):** A GUA is similar to a public IPv4 address. It is a globally unique, internet-routable address. GUAs can be configured statically or assigned dynamically.
- **Link-local address (LLA):** An LLA is required for every IPv6-enabled device. LLAs are used to communicate with other devices on the same local link. With IPv6, the term *link* refers to a subnet. An LLA is confined to a single link. The uniqueness of an LLA must only be confirmed on that link because LLAs are not routable beyond the link. In other words, routers do not forward packets with link-local source or destination addresses.

A Note About the Unique Local Address (12.3.4)

Unique local addresses (in the range fc00::/7 to fdff::/7) are not yet commonly implemented. Therefore, this chapter only covers GUA and LLA configuration. However, unique local addresses may eventually be used to address devices that should not be accessible from the outside, such as internal servers and printers.

The IPv6 unique local addresses have some similarities to RFC 1918 private addresses for IPv4, but there are significant differences:

- Unique local addresses are used for local addressing within a site or between a limited number of sites.
- Unique local addresses can be used for devices that will never need to access another network.
- Unique local addresses are not globally routed or translated to global IPv6 addresses.

Note

Many sites also use the private nature of RFC 1918 addresses to attempt to secure or hide a network from potential security risks. However, this was never the intended use of these technologies, and the IETF has always recommended that sites take the proper security precautions on their internet-facing routers.

IPv6 GUA (12.3.5)

IPv6 global unicast addresses (GUAs) are globally unique and routable on the IPv6 internet. These addresses are equivalent to public IPv4 addresses. The Internet Committee for Assigned Names and Numbers (ICANN), the operator for IANA, allocates

IPv6 address blocks to the five RIRs. Currently, only GUAs that start with 001 or 2000::/3 are being assigned, as shown in Figure 12-8.

Figure 12-8 shows the range of values for the first hextet, where the first hexadecimal digit for currently available GUAs begins with a 2 or a 3. This is only one-eighth of the total available IPv6 address space, excluding only a very small portion for other types of unicast and multicast addresses.

Note

The 2001:db8::/32 address has been reserved for documentation purposes, including use in examples.

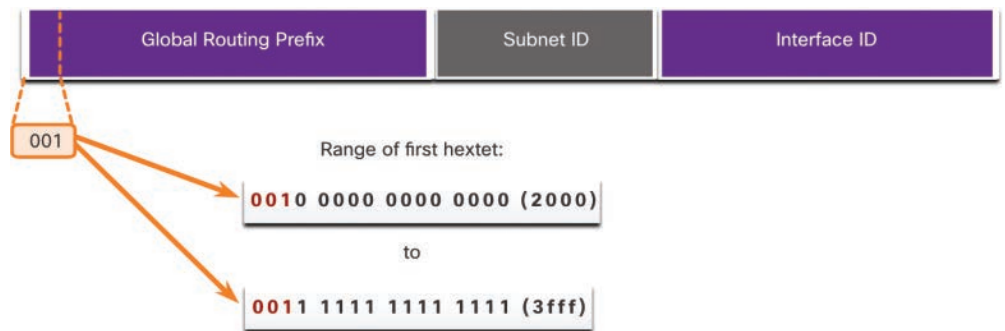


Figure 12-8 Range of First Hextet Values for GUAs

Figure 12-9 shows the structure and range of a GUA.

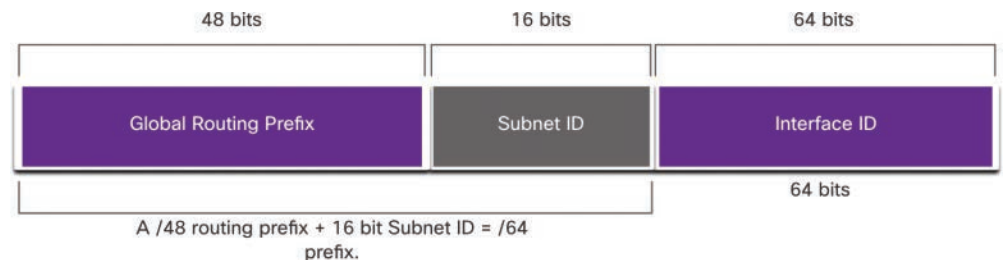


Figure 12-9 IPv6 Address with a /48 Global Routing Prefix and /64 Prefix

IPv6 GUA Structure (12.3.6)

As shown in Figure 12-9, a GUA has three parts:

- Global routing prefix
- Subnet ID
- Interface ID

Global Routing Prefix

The *global routing prefix* is the prefix, or network, portion of an address that is assigned by a provider, such as an ISP, to a customer or site. For example, it is common for an ISP to assign a /48 global routing prefix to its customers. The global routing prefix typically varies depending on the policies of the ISP.

Figure 12-9 shows a GUA using a /48 global routing prefix, which is a commonly assigned global routing prefix (and is therefore what most of the examples throughout this book use).

For example, the IPv6 address 2001:db8:acad::/48 has a global routing prefix that indicates that the first 48 bits (3 hextets: 2001:db8:acad); which is how the ISP knows this prefix (network) as the global routing prefix. The double colon (::) following the /48 prefix length means the rest of the address contains all 0s. The size of the global routing prefix determines the size of the subnet ID.

Subnet ID

The *subnet ID* field is the area between the global routing prefix and the interface ID. Unlike IPv4, where you must borrow bits from the host portion to create subnets, IPv6 was designed with subnetting in mind. An organization uses the subnet ID to identify subnets within its site. The larger the subnet ID, the more subnets available.

Note

Many organizations receive a /32 global routing prefix. If you use the recommended /64 prefix in order to create a 64-bit interface ID you're left with a 32-bit subnet ID. This means an organization with a /32 global routing prefix and a 32-bit subnet ID will have 4.3 billion subnets, each with 18 quintillion devices per subnet. That is as many subnets as there are public IPv4 addresses!

The IPv6 address in Figure 12-9 has a /48 global routing prefix, which is common among many enterprise networks. This makes it especially easy to examine the different parts of the address. Using a typical /64 prefix length, the first four hextets are for the network portion of the address, and the fourth hextet indicates the subnet ID. The remaining four hextets indicate the interface ID.

Interface ID

The IPv6 *interface ID* is equivalent to the host portion of an IPv4 address. The term *interface ID* is used because a single host may have multiple interfaces, each with

one or more IPv6 addresses. Figure 12-9 shows an example of the structure of an IPv6 GUA. It is strongly recommended that you use /64 subnets in most cases, which creates a 64-bit interface ID. A 64-bit interface ID allows for 18 quintillion devices or hosts per subnet.

A /64 subnet or prefix (global routing prefix + subnet ID) leaves 64 bits for the interface ID. This is recommended to allow SLAAC-enabled devices to create their own 64-bit interface IDs. It also makes developing an effective IPv6 addressing plan simple.

Note

Unlike in IPv4, in IPv6, the all-0s and all-1s host addresses can be assigned to a device. The all-1s address can be used because broadcast addresses are not used in IPv6. The all-0s address can also be used, but it is reserved as a subnet-router anycast address and should be assigned only to routers.

IPv6 LLA (12.3.7)

An IPv6 link-local address (LLA) enables a device to communicate with other IPv6-enabled devices on the same link and only on that link (subnet). Packets with a source or destination LLA cannot be routed beyond the link from which the packets originated.

The GUA is not a requirement. However, every IPv6-enabled network interface must have an LLA.

If an LLA is not configured manually on an interface, the device will automatically create its own without communicating with a DHCP server. IPv6-enabled hosts create an IPv6 LLA even if the device has not been assigned a global unicast IPv6 address. This allows IPv6-enabled devices to communicate with other IPv6-enabled devices on the same subnet, including the default gateway (router).

IPv6 LLAs are in the fe80::/10 range. The /10 indicates that the first 10 bits are 1111 1110 10xx xxxx. The first hextet has a range of 1111 1110 1000 0000 (fe80) to 1111 1110 1011 1111 (febf).

Figure 12-10 shows an example of communication using IPv6 LLAs. The PC is able to communicate directly with the printer using the LLAs.

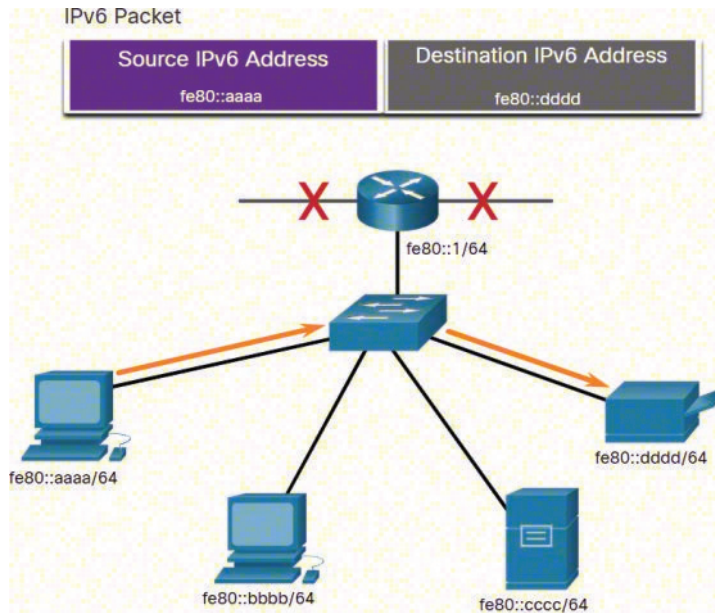


Figure 12-10 IPv6 Link-Local Communications

Figure 12-11 shows some of the uses for IPv6 LLAs:

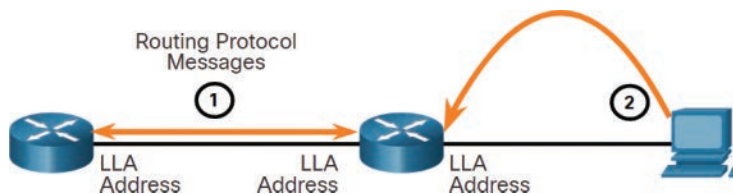


Figure 12-11 Example of Using IPv6 LLAs

1. Routers use the LLAs of neighbor routers to send routing updates.
2. Hosts use the LLA of a local router as the default gateway.

Note

Typically, it is the LLA of the router, and not the GUA, that is used as the default gateway for other devices on the link.

There are two ways that a device can obtain an LLA:

- **Statically:** This means the device is manually configured.
- **Dynamically:** This means the device creates its own interface ID by using randomly generated values or using the Extended Unique Identifier (EUI) method, which uses the client MAC address along with additional bits.

Interactive
Graphic

Check Your Understanding—IPv6 Address Types (12.3.8)

Refer to the online course to complete this activity.

GUA and LLA Static Configuration (12.4)

This section discusses the static configuration of IPv6 global unicast (GUA) and link-local addresses.

Static GUA Configuration on a Router (12.4.1)

IPv6 GUAs are the same as public IPv4 addresses. They are globally unique and routable on the IPv6 internet. An IPv6 LLA makes it possible for two IPv6-enabled devices to communicate with each other on the same link (subnet). It is easy to statically configure IPv6 GUAs and LLAs on routers to aid in creating an IPv6 network. This section teaches you how to do just that.

Most IPv6 configuration and verification commands in Cisco IOS are similar to their IPv4 counterparts. In many cases, the only difference is the use of **ipv6** in place of **ip** within the commands.

For example, the Cisco IOS command to configure an IPv4 address on an interface is **ip address ip-address subnet-mask**. In contrast, the command to configure an IPv6 GUA on an interface is **ipv6 address ipv6-address/prefix-length**. Notice that there is no space between *ipv6-address* and *prefix-length*.

Figure 12-12 shows the topology used in the configuration example in this section. The topology includes these IPv6 subnets:

- 2001:db8:acad:1::/64
- 2001:db8:acad:2::/64
- 2001:db8:acad:3::/64

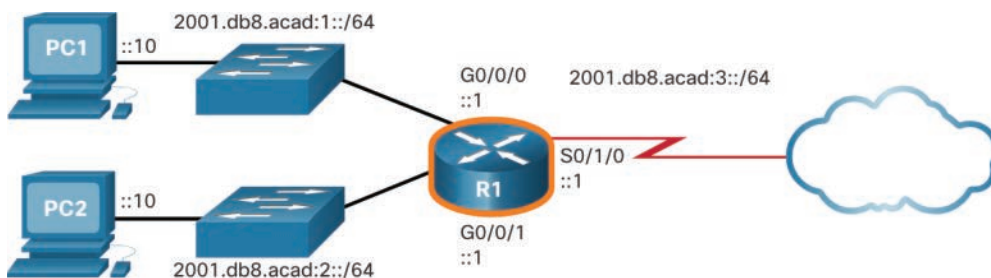


Figure 12-12 IPv6 Addressing Topology

Example 12-2 shows the commands required to configure the IPv6 GUA on the GigabitEthernet 0/0/0, GigabitEthernet 0/0/1, and Serial 0/1/0 interfaces of R1.

Example 12-2 IPv6 GUA Configuration on Router R1

```
R1(config)# interface gigabitethernet 0/0/0
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface gigabitethernet 0/0/1
R1(config-if)# ipv6 address 2001:db8:acad:2::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface serial 0/1/0
R1(config-if)# ipv6 address 2001:db8:acad:3::1/64
R1(config-if)# no shutdown
```

Static GUA Configuration on a Windows Host (12.4.2)

Manually configuring the IPv6 address on a host is similar to configuring an IPv4 address.

As shown in Figure 12-13, the default gateway address configured for PC1 is 2001:db8:acad:1::1. This is the GUA of the R1 GigabitEthernet interface on the same network. Alternatively, the default gateway address can be configured to match the LLA of the GigabitEthernet interface. Using the LLA of the router as the default gateway address is considered best practice, but either configuration will work.

Just as with IPv4, with IPv6, statically configuring addresses on clients does not scale to larger environments. For this reason, most network administrators enable dynamic assignment of IPv6 addresses.

There are two ways in which a device can obtain an IPv6 GUA automatically:

- Stateless address autoconfiguration (SLAAC)
- Stateful DHCPv6

SLAAC and DHCPv6 are covered in the next section.

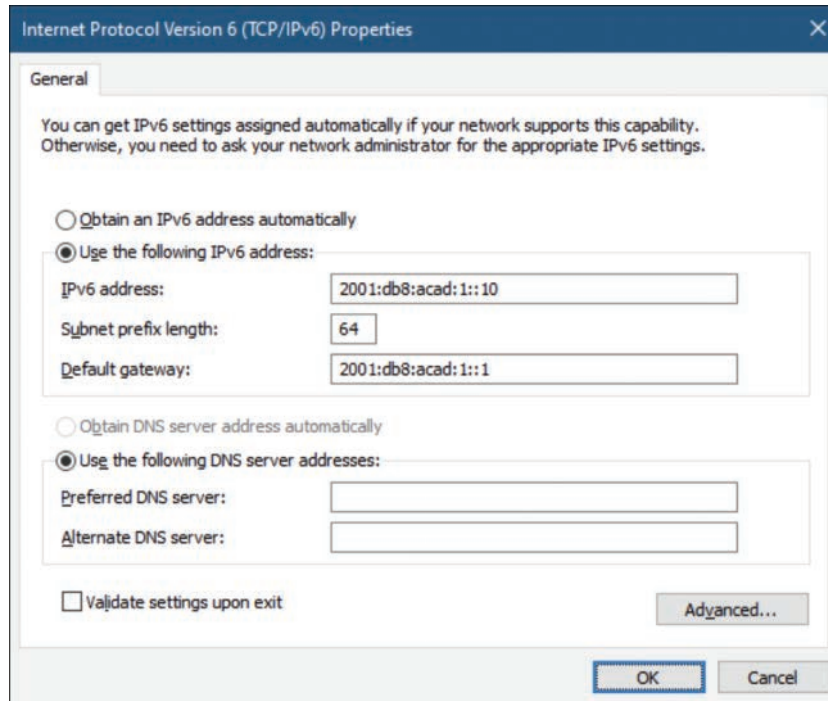


Figure 12-13 Manually Configuring IPv6 Addressing on a Windows Host

Note

When DHCPv6 or SLAAC is used, the LLA of the router is automatically specified as the default gateway address.

Static Configuration of a Link-Local Unicast Address (12.4.3)

By configuring the LLA manually, you can create an address that is recognizable and easier to remember. Typically, it is only necessary to create recognizable LLAs on routers. Recognizability is beneficial because router LLAs are used as default gateway addresses and in routing advertisement messages.

LLAs can be configured manually using the **ipv6 address ipv6-link-local-address link-local** command. When an address begins with a hextet in the range fe80 to febf, the **link-local** parameter must follow the address.

Figure 12-14 shows an example of a topology with an LLA on each interface.

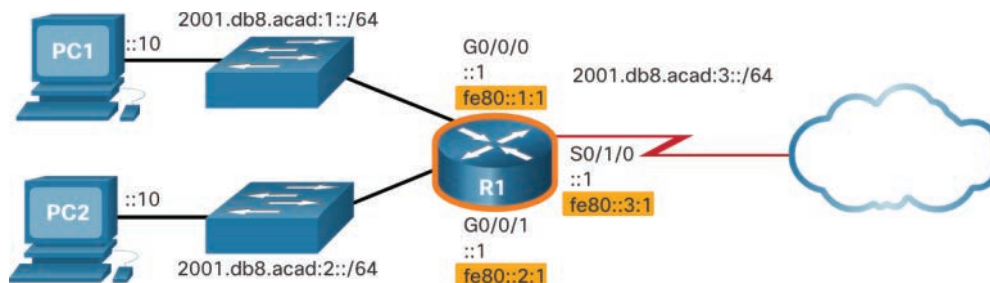


Figure 12-14 IPv6 Addressing Topology with LLAs

Example 12-3 shows the configuration of an LLA on router R1.

Example 12-3 R1 Static LLA Configuration

```
R1(config)# interface gigabitethernet 0/0/0
R1(config-if)# ipv6 address fe80::1:1 link-local
R1(config-if)# exit
R1(config)# interface gigabitethernet 0/0/1
R1(config-if)# ipv6 address fe80::1:2 link-local
R1(config-if)# exit
R1(config)# interface serial 0/1/0
R1(config-if)# ipv6 address fe80::1:3 link-local
R1(config-if)# exit
```

LLAs may be statically configured to be more easily recognizable as belonging to router R1. In Example 12-3, each of the interfaces of router R1 has been configured with an LLA that begins with `fe80::1:n` and a unique rightmost digit *n*. The 1 represents router R1.

If the topology included router R2 and you wanted to follow the same naming convention as for router R1, you would configure R2's three interfaces with the LLAs `fe80::2:1`, `fe80::2:2`, and `fe80::2:3`.

Note

Exactly the same LLA could be configured on each link, as long as it is unique on that link. This is because an LLA only has to be unique on its link. However, common practice is to create a different LLA on each interface of the router to make it easy to identify the router and the specific interface.

Syntax Checker—GUA and LLA Static Configuration (12.4.4)

Refer to the online course to complete this activity.

Dynamic Addressing for IPv6 GUAs (12.5)

This section discusses the different ways a device can automatically create or receive an IPv6 GUA.

RS and RA Messages (12.5.1)

If you do not want to statically configure IPv6 GUAs, don't worry. Most devices obtain their IPv6 GUAs dynamically. This section explains how this process works when using *Router Advertisement (RA) messages* and *Router Solicitation (RS) messages*. This section gets rather technical, but when you understand the difference between the three methods that a router advertisement can use, as well as how the EUI-64 process for creating an interface ID differs from a randomly generated process, you will have made a huge leap in your IPv6 expertise!

For the GUA, a device obtains the address dynamically through Internet Control Message Protocol version 6 (ICMPv6) messages. IPv6 routers send out ICMPv6 RA messages every 200 seconds to all IPv6-enabled devices on the network. An RA message is also sent in response to a host sending an ICMPv6 RS message, which is a request for an RA message. As shown in Figure 12-15, these messages work as follows:

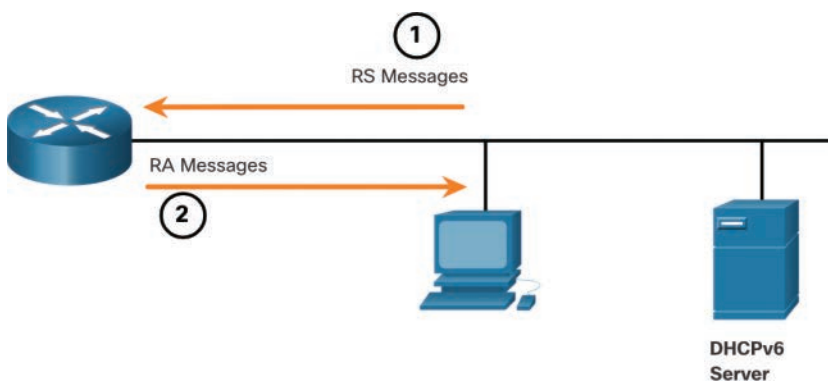


Figure 12-15 ICMPv6 RS and RA Messages

1. RS messages are sent to all IPv6 routers by hosts requesting addressing information.
2. RA messages are sent to all IPv6 nodes. If Method 1 (SLAAC only) is used—as described in the next section—the RA includes network prefix, prefix length, and default gateway information.

RA messages are on IPv6 router Ethernet interfaces. A router must be enabled for IPv6 routing, which is not enabled by default. To enable a router as an IPv6 router, the **ipv6 unicast-routing** global configuration command must be used.

The ICMPv6 RA message is a suggestion to a device on how to obtain an IPv6 GUA. The ultimate decision is up to the device operating system. The ICMPv6 RA message includes the following:

- **Network prefix and prefix length:** This tells the device which network it belongs to.
- **Default gateway address:** This is an IPv6 LLA, the source IPv6 address of the RA message.
- **DNS addresses and domain name:** These are the addresses of DNS servers and a domain name.

There are three methods for RA messages:

- **Method 1: SLAAC:** “I have everything you need, including the prefix, prefix length, and default gateway address.”
- **Method 2: SLAAC with a stateless DHCPv6 server:** “Here is my information, but you need to get other information, such as DNS addresses, from a stateless DHCPv6 server.”
- **Method 3: Stateful DHCPv6 (no SLAAC):** “I can give you your default gateway address. You need to ask a stateful DHCPv6 server for all your other information.”

Method 1: SLAAC (12.5.2)

With *stateless address autoconfiguration (SLAAC)*, a device creates its own GUA without the services of DHCPv6. Using SLAAC, devices rely on the ICMPv6 RA messages of the local router to obtain the necessary information.

By default, the RA message suggests that the receiving device use the information in the RA message to create its own IPv6 GUA and all other necessary information. The services of a DHCPv6 server are not required.

SLAAC is stateless, which means there is no central server (for example, a stateful DHCPv6 server) allocating GUAs and keeping a list of devices and their addresses. With SLAAC, the client device uses the information in the RA message to create its own GUA. As shown in Figure 12-16, the two parts of the address are created:

- **Prefix:** The prefix is advertised in the RA message.
- **Interface ID:** The interface ID uses the EUI-64 process or generates a random 64-bit number, depending on the device operating system.

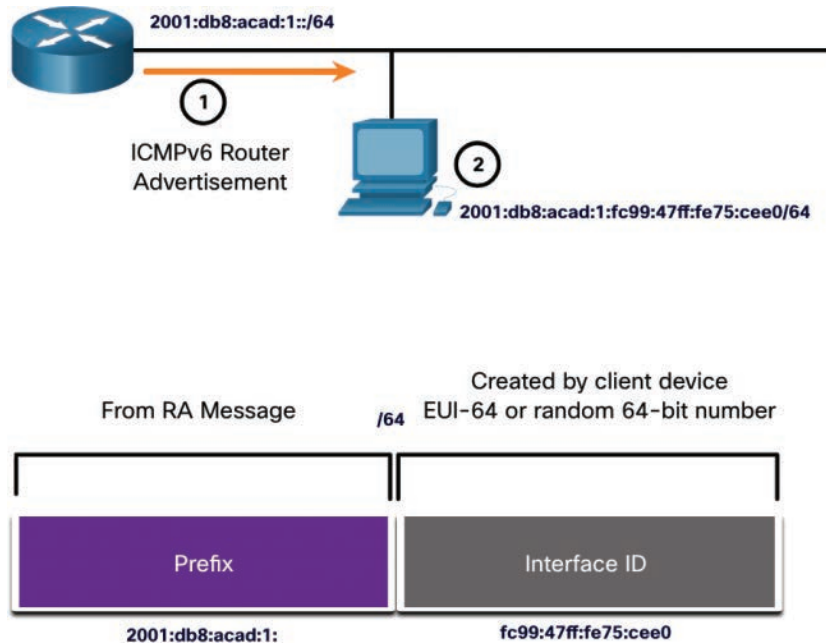


Figure 12-16 SLAAC Example

This is the process shown in Figure 12-16:

1. The router sends an RA message with the prefix for the local link.
2. The PC uses SLAAC to obtain a prefix from the RA message and creates its own interface ID.

Method 2: SLAAC and Stateless DHCPv6 (12.5.3)

A router interface can be configured to send a router advertisement using SLAAC and *stateless DHCPv6*.

As shown in Figure 12-17, with this method, the RA message suggests devices use the following:

- SLAAC to create its own IPv6 GUA
- The router LLA, which is the RA source IPv6 address, as the default gateway address
- A stateless DHCPv6 server to obtain other information, such as a DNS server address and a domain name

Note

A stateless DHCPv6 server distributes DNS server addresses and domain names. It does not allocate GUAs.

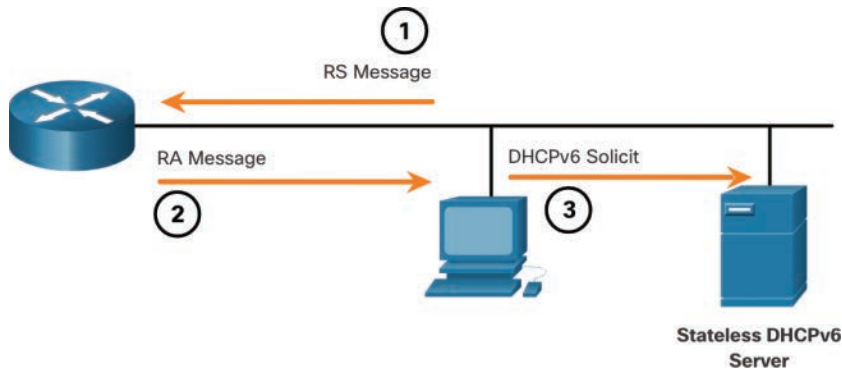


Figure 12-17 SLAAC and Stateless DHCPv6 Example

This is the process shown in Figure 12-17:

1. The PC sends an RS message to all IPv6 routers: “I need addressing information.”
2. The router sends an RA message to all IPv6 nodes with Method 2 (SLAAC and DHCPv6) specified: “Here are your prefix, prefix length, and default gateway information, but you will need to get DNS information from a DHCPv6 server.”
3. The PC sends a DHCPv6 Solicit message to all DHCPv6 servers: “I used SLAAC to create my IPv6 address and get my default gateway address, but I need other information from a stateless DHCPv6 server.”

Method 3: Stateful DHCPv6 (12.5.4)

A router interface can be configured to send an RA message using stateful DHCPv6 only.

Stateful DHCPv6 is similar to DHCP for IPv4. A device can automatically receive its addressing information—including a GUA, prefix length, and the addresses of DNS servers—from a stateful DHCPv6 server.

As shown in Figure 12-18, with this method, the RA message suggests that devices use the following:

- The router LLA, which is the RA source IPv6 address, for the default gateway address
- A stateful DHCPv6 server to obtain a GUA, DNS server address, domain name, and other necessary information

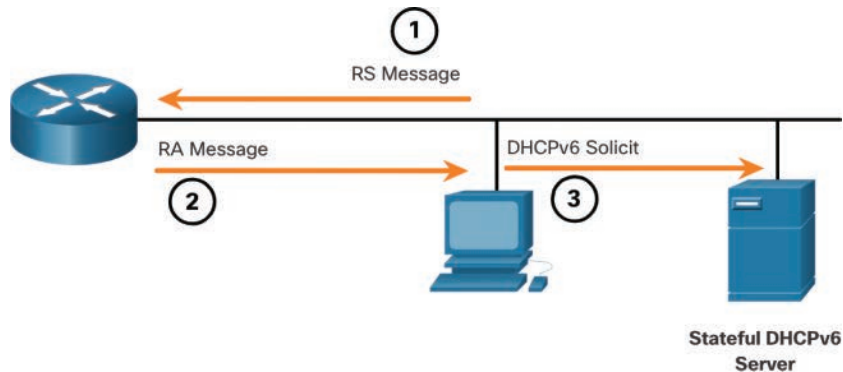


Figure 12-18 Stateful DHCPv6 Example

This is the process shown in Figure 12-18:

1. The PC sends an RS message to all IPv6 routers: “I need addressing information.”
2. The router sends an RA message to all IPv6 nodes with Method 3 (stateful DHCPv6) specified: “I am your default gateway, but you need to ask a stateful DHCPv6 server for your IPv6 address and other addressing information.”
3. The PC sends a DHCPv6 Solicit message to all DHCPv6 servers: “I received my default gateway address from the RA message, but I need an IPv6 address and all other addressing information from a stateful DHCPv6 server.”

A stateful DHCPv6 server allocates and maintains a list of which device receives which IPv6 address. DHCP for IPv4 is also stateful.

Note

The default gateway address can only be obtained dynamically from the RA message. The stateless or stateful DHCPv6 server does not provide the default gateway address.

EUI-64 Process vs. Randomly Generated (12.5.5)

When an RA message is either SLAAC or SLAAC with stateless DHCPv6, the client must generate its own interface ID. The client knows the prefix portion of the address from the RA message but must create its own interface ID. The interface ID can be created using the EUI-64 process or a randomly generated 64-bit number, as shown in Figure 12-19.

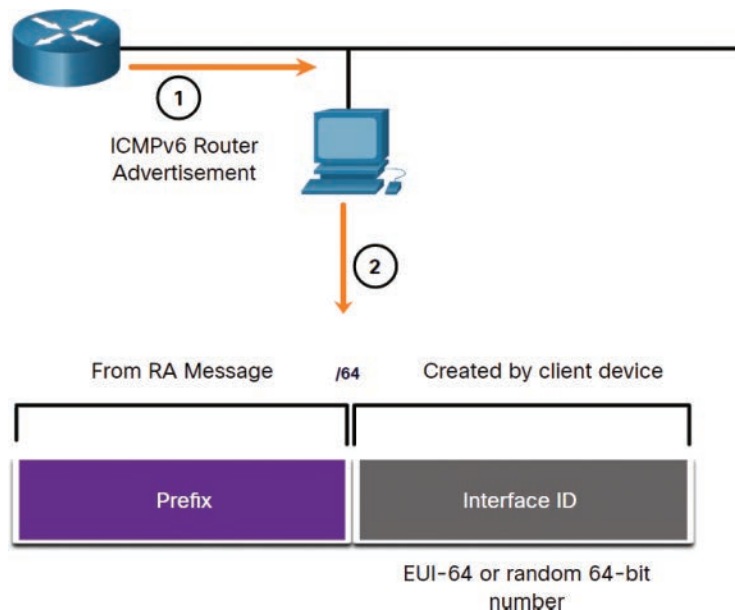


Figure 12-19 Dynamically Creating an Interface ID

This is the process shown in Figure 12-19:

1. The router sends an RA message.
2. The PC uses the prefix in the RA message and uses either EUI-64 or a random 64-bit number to generate an interface ID.

EUI-64 Process (12.5.6)

The IEEE defined the *Extended Unique Identifier (EUI)*, or modified EUI-64, process. This process uses the 48-bit Ethernet MAC address of a client and inserts another 16 bits in the middle of the 48-bit MAC address to create a 64-bit interface ID.

Ethernet MAC addresses are usually represented in hexadecimal and are made up of two parts:

- **Organizationally unique identifier (OUI):** The OUI is a 24-bit (6 hexadecimal digits) vendor code assigned by the IEEE.
- **Device identifier:** The device identifier is a unique 24-bit (6 hexadecimal digits) value within a common OUI.

An EUI-64 interface ID is represented in binary and is made up of three parts:

- 24-bit OUI from the client MAC address, but the 7th bit (the universal/local [U/L] bit) is reversed. This means that if the 7th bit is a 0, it becomes a 1 and vice versa.

- The inserted 16-bit value fffe (in hexadecimal).
- 24-bit device identifier from the client MAC address.

The EUI-64 process is illustrated in Figure 12-20, using the R1 GigabitEthernet MAC address fc99:4775:cee0.

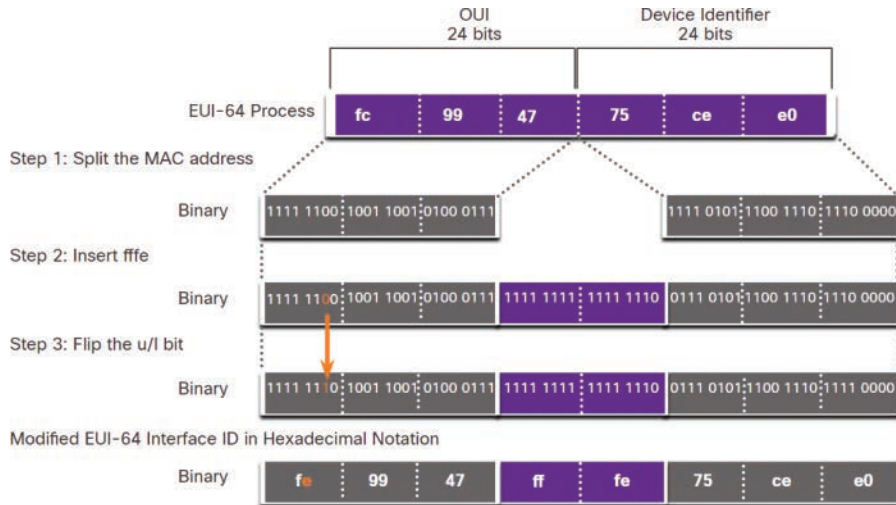


Figure 12-20 The EUI-64 Process

This is the process shown in Figure 12-20:

- Step 1.** Divide the MAC address between the OUI and device identifier.
- Step 2.** Insert the hexadecimal value fffe, which in binary is 1111 1111 1111 1110.
- Step 3.** Convert the first 2 hexadecimal values of the OUI to binary and flip the U/L bit (bit 7). In this example, the 0 in bit 7 is changed to a 1.

The result is the EUI-64 generated interface ID fe99:47ff:fe75:cee0.

Note

The use of the U/L bit and the reasons for reversing its value are discussed in RFC 5342.

The output in Example 12-4 for the **ipconfig** command shows the IPv6 GUA being dynamically created using SLAAC and the EUI-64 process. An easy way to identify that an address was probably created using EUI-64 is the presence of fffe in the middle of the interface ID.

The advantage of EUI-64 is that the Ethernet MAC address can be used to determine the interface ID. It also allows network administrators to easily track an IPv6 address to an end device by using the unique MAC address. However, this has caused privacy

concerns among many users, who have worried that their packets could be traced to an actual physical computer. Due to these concerns, a randomly generated interface ID may be used instead.

Example 12-4 EUI-64 Generated Interface ID

```
C:\> ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2001:db8:acad:1:fc99:47ff:fe75:cee0
    Link-local IPv6 Address . . . . . : fe80::fc99:47ff:fe75:cee0
    Default Gateway . . . . . : fe80::1
C:\>
```

Randomly Generated Interface IDs (12.5.7)

Depending on the operating system, a device may use a randomly generated interface ID instead of using the MAC address and the EUI-64 process. Beginning with Windows Vista, Windows uses a randomly generated interface ID instead of one created with EUI-64. Windows XP and previous Windows operating systems used EUI-64.

After the interface ID is established, either through the EUI-64 process or through random generation, it can be combined with an IPv6 prefix in the RA message to create a GUA, as shown in Example 12-5.

Example 12-5 Random 64-Bit Generated Interface ID

```
C:\> ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2001:db8:acad:1:50a5:8a35:a5bb:66e1
    Link-local IPv6 Address . . . . . : fe80::50a5:8a35:a5bb:66e1
    Default Gateway . . . . . : fe80::1
C:\>
```

Note

To ensure the uniqueness of any IPv6 unicast address, the client may use a process known as duplicate address detection (DAD). DAD is similar to a client ARP request for its own address. If there is no reply, then the address is unique.

**Interactive
Graphic**
Check Your Understanding—Dynamic Addressing for IPv6 GUAs (12.5.8)

Refer to the online course to complete this activity.

Dynamic Addressing for IPv6 LLAs (12.6)

This section discusses how a device automatically creates an IPv6 link-local address. Regardless of how you create your LLAs (and your GUAs), it is important that you verify all IPv6 address configuration. This section explains dynamically generated LLAs and IPv6 configuration verification.

Dynamic LLAs (12.6.1)

Every IPv6 device must have an IPv6 LLA. As with IPv6 GUAs, you can also create LLAs dynamically.

Figure 12-21 shows an LLA dynamically created using the fe80::/10 prefix and the interface ID using the EUI-64 process, or a randomly generated 64-bit number.

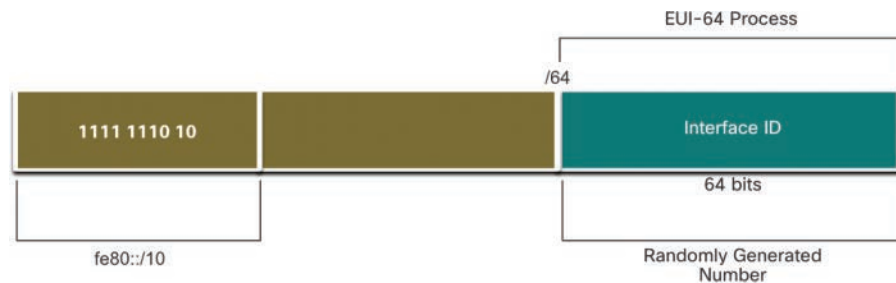


Figure 12-21 Dynamic Creation of an LLA

Dynamic LLAs on Windows (12.6.2)

Operating systems such as Windows typically use the same method for both a SLAAC-created GUA and a dynamically assigned LLA. The highlighted areas in Examples 12-6 and 12-7 repeat configurations shown earlier in this chapter to illustrate.

Example 12-6 EUI-64 Generated Interface ID

```

C:\> ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2001:db8:acad:1:fc99:47ff:fe75:cee0
    Link-local IPv6 Address . . . . . : fe80::fc99:47ff:fe75:cee0
    Default Gateway . . . . . : fe80::1

C:\>

```

Example 12-7 Random 64-Bit Generated Interface ID

```

C:\> ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2001:db8:acad:1:50a5:8a35:a5bb:66e1
    Link-local IPv6 Address . . . . . : fe80::50a5:8a35:a5bb:66e1
    Default Gateway . . . . . : fe80::1

C:\>

```

Dynamic LLAs on Cisco Routers (12.6.3)

A Cisco router automatically creates an IPv6 LLA whenever a GUA is assigned to the interface. By default, Cisco IOS routers use EUI-64 to generate the interface IDs for all LLAs on IPv6 interfaces. For serial interfaces, a router uses the MAC address of an Ethernet interface. Recall that an LLA must be unique only on that link or network. However, a drawback to using a dynamically assigned LLA is its long interface ID, which makes it challenging to identify and remember assigned addresses. Example 12-8 displays the MAC address on the GigabitEthernet 0/0/0 interface of router R1. This address is used to dynamically create the LLA on the same interface and also for the Serial 0/1/0 interface.

To make it easier to recognize and remember these addresses on routers, it is common to statically configure IPv6 LLAs on routers.

Example 12-8 IPv6 LLA Using EUI-64 on Router R1

```

R1# show interface gigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
    Hardware is ISR4221-2x1GE, address is 7079.b392.3640 (bia 7079.b392.3640)
(Output omitted)

```

```

R1# show ipv6 interface brief
GigabitEthernet0/0/0    [up/up]
    FE80::7279:B3FF:FE92:3640
    2001:DB8:ACAD:1::1
GigabitEthernet0/0/1    [up/up]
    FE80::7279:B3FF:FE92:3641
    2001:DB8:ACAD:2::1
Serial10/1/0            [up/up]
    FE80::7279:B3FF:FE92:3640
    2001:DB8:ACAD:3::1
Serial10/1/1            [down/down]
    unassigned
R1#

```

Verify IPv6 Address Configuration (12.6.4)

Figure 12-22 shows the topology used in the example in this section.

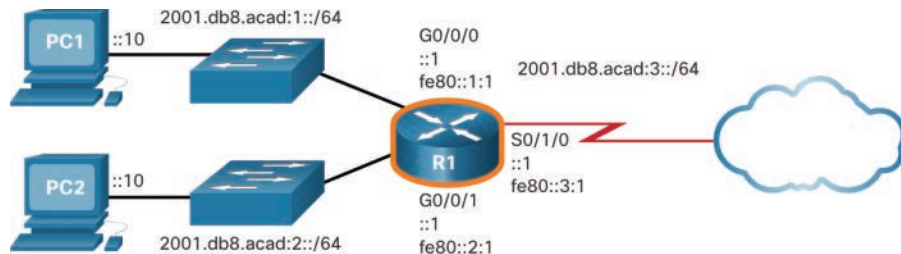


Figure 12-22 IPv6 Addressing Topology

The `show ipv6 interface brief` command in Example 12-9 displays the MAC address of the Ethernet interfaces. EUI-64 uses this MAC address to generate the interface ID for the LLA. In addition, the `show ipv6 interface brief` command displays abbreviated output for each of the interfaces. The [up/up] output on the same line as the interface indicates the Layer 1/Layer 2 interface state. This is the same as the Status and Protocol columns in the equivalent IPv4 command.

Example 12-9 The show ipv6 interface brief Command on R1

```

R1# show ipv6 interface brief
GigabitEthernet0/0/0    [up/up]
    FE80::1:1
    2001:DB8:ACAD:1::1
GigabitEthernet0/0/1    [up/up]
    FE80::1:2
    2001:DB8:ACAD:2::1
Serial0/1/0              [up/up]
    FE80::1:3
    2001:DB8:ACAD:3::1
Serial0/1/1              [down/down]
    unassigned
R1#

```

Notice that each interface has two IPv6 addresses. The second address for each interface is the GUA that was configured. The first address—the one that begins with fe80—is the link-local unicast address for the interface. Recall that the LLA is automatically added to the interface when a GUA is assigned.

Also notice that the R1 Serial 0/1/0 LLA is the same as its GigabitEthernet 0/0/0 interface. Serial interfaces do not have Ethernet MAC addresses, so Cisco IOS uses the MAC address of the first available Ethernet interface. This is possible because a link-local interface only has to be unique on the link.

As shown in Example 12-10, the **show ipv6 route** command can be used to verify that IPv6 networks and specific IPv6 interface addresses have been installed in the IPv6 routing table. The **show ipv6 route** command displays only IPv6 networks, not IPv4 networks.

Example 12-10 The show ipv6 route Command on R1

```

R1# show ipv6 route
IPv6 Routing Table - default - 7 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

C    2001:DB8:ACAD:1::/64 [0/0]
      via GigabitEthernet0/0/0, directly connected
L    2001:DB8:ACAD:1::1/128 [0/0]
      via GigabitEthernet0/0/0, receive
C    2001:DB8:ACAD:2::/64 [0/0]
      via GigabitEthernet0/0/1, directly connected

```



```

L   2001:DB8:ACAD:2::1/128 [0/0]
    via GigabitEthernet0/0/1, receive
C   2001:DB8:ACAD:3::/64 [0/0]
    via Serial0/1/0, directly connected
L   2001:DB8:ACAD:3::1/128 [0/0]
    via Serial0/1/0, receive
L   FF00::/8 [0/0]
    via Null0, receive
R1#

```

Within the routing table, a C next to a route indicates a directly connected network. When the router interface is configured with a GUA and is in the up/up state, the IPv6 prefix and prefix length are added to the IPv6 routing table to indicate a connected route.

Note

The L indicates a local route, the specific IPv6 address assigned to the interface. This is not an LLA. LLAs are not included in the routing table of a router because they are not routable addresses.

The IPv6 GUA configured on the interface is also installed in the routing table as a local route. The local route has a /128 prefix. A router uses local routes in the routing table to efficiently process packets with the router interface address as the destination address.

The **ping** command for IPv6 is identical to the command used with IPv4, except that an IPv6 address is used. As shown in Example 12-11, the command is used to verify Layer 3 connectivity between R1 and PC1. When pinging an LLA from a router, Cisco IOS prompts the user for the exit interface. Because the destination LLA can be on one or more of its links or networks, the router needs to know which interface to send the ping to.

Example 12-11 The ping Command on R1

```

R1# ping 2001:db8:acad:1::10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:1::10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R1#

```

**Interactive
Graphic****Syntax Checker—Verify IPv6 Address Configuration (12.6.5)**

Refer to the online course to complete this activity.

**Packet Tracer
Activity****Packet Tracer—Configure IPv6 Addressing (12.6.6)**

In this activity, you will practice configuring IPv6 addresses on a router, servers, and clients. You will also practice verifying your IPv6 addressing implementation.

IPv6 Multicast Addresses (12.7)

This section introduces the two types of IPv6 multicast addresses: well-known multicast and solicited-node multicast addresses.

Assigned IPv6 Multicast Addresses (12.7.1)

Earlier in this chapter, you learned that there are three broad categories of IPv6 addresses: unicast, anycast, and multicast. This section goes into more detail about multicast addresses.

IPv6 multicast addresses are similar to IPv4 multicast addresses. Recall that a multicast address is used to send a single packet to one or more destinations (multicast group). IPv6 multicast addresses have the prefix `ff00::/8`.

Note

Multicast addresses can only be destination addresses and not source addresses.

There are two types of IPv6 multicast addresses:

- Well-known multicast addresses
- Solicited-node multicast addresses

Well-Known IPv6 Multicast Addresses (12.7.2)

Well-known IPv6 multicast addresses are assigned. Assigned multicast addresses are reserved multicast addresses for predefined groups of devices. An assigned multicast address is a single address used to reach a group of devices running a common protocol or service. Assigned multicast addresses are used with specific protocols such as DHCPv6.

These are two common IPv6 assigned multicast groups:

- **ff02::1 all-nodes multicast group:** This is a multicast group that all IPv6-enabled devices join. A packet sent to this group is received and processed by all IPv6 interfaces on the link or network. This has the same effect as a broadcast address in IPv4. Figure 12-23 shows an example of communication using the all-nodes multicast address. An IPv6 router sends ICMPv6 RA messages to the all-nodes multicast group.
- **ff02::2 all-routers multicast group:** This is a multicast group that all IPv6 routers join. A router becomes a member of this group when it is enabled as an IPv6 router with the `ipv6 unicast-routing` global configuration command. A packet sent to this group is received and processed by all IPv6 routers on the link or network.

The fourth digit in the address refers to the scope. A 2 for the scope indicates that these addresses have link-local scope, which means that packets with this destination address are not be routed off this link or network.

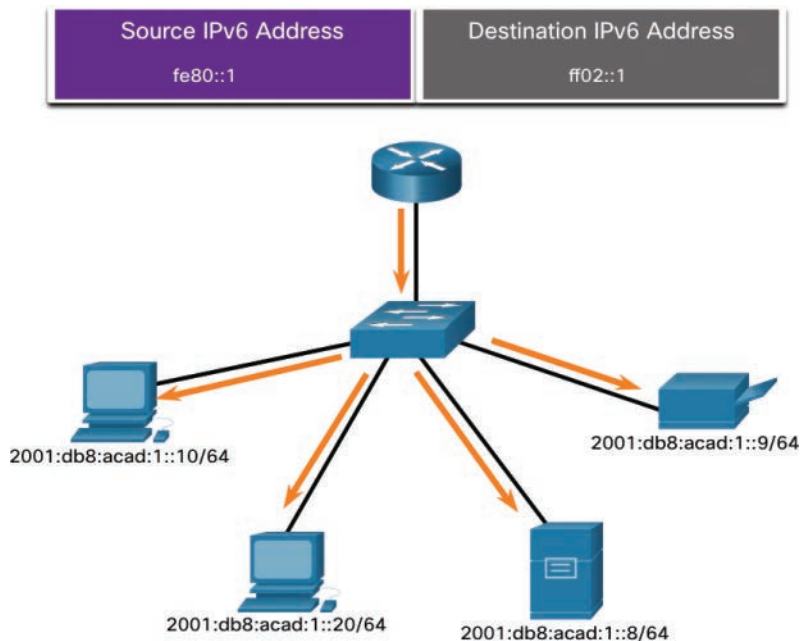


Figure 12-23 IPv6 All-Nodes Multicast: RA Message

IPv6-enabled devices send ICMPv6 RS messages to the all-routers multicast address. An RS message requests an RA message from the IPv6 router to assist the device in its address configuration. The IPv6 router responds with an RA message, as shown in Figure 12-23.

Solicited-Node IPv6 Multicast Addresses (12.7.3)

A *solicited-node multicast address* is similar to the all-nodes multicast address. The advantage of a solicited-node multicast address is that it is mapped to a special Ethernet multicast address. This allows the Ethernet NIC to filter the frame by examining the destination MAC address without sending it to the IPv6 process to see if the device is the intended target of the IPv6 packet, as shown in Figure 12-24.

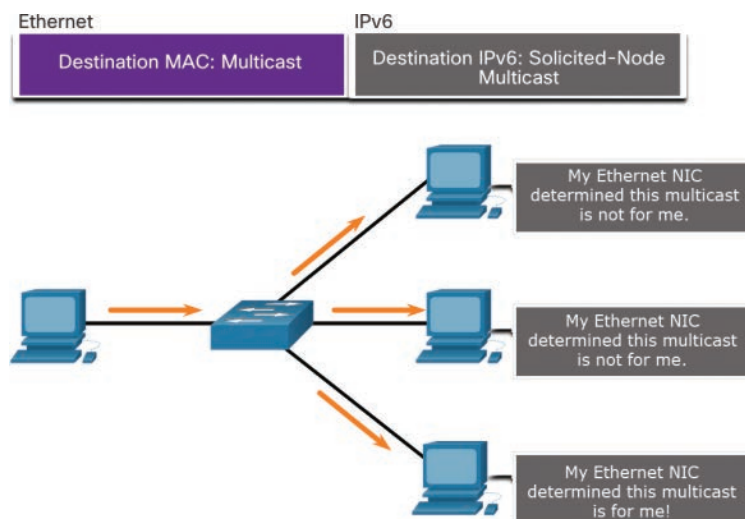


Figure 12-24 Solicited-Node IPv6 Multicast Example



Lab—Identify IPv6 Addresses (12.7.4)

In this lab, you will complete the following objectives:

- Part 1: Identify the Different Types of IPv6 Addresses
- Part 2: Examine a Host IPv6 Network Interface and Address
- Part 3: Practice IPv6 Address Abbreviation

Subnet an IPv6 Network (12.8)

This section discusses basic IPv6 subnetting.

Subnet Using the Subnet ID (12.8.1)

As mentioned at the beginning of this chapter, you can subnet an IPv6 network—and doing so is a bit easier than subnetting an IPv4 network. This section describes the process.

Recall that with IPv4, you must borrow bits from the host portion to create subnets. This system is awkward because subnetting was an afterthought with IPv4. However, IPv6 was designed with subnetting in mind. A separate subnet ID field in the IPv6 GUA is used to create subnets. As shown in Figure 12-25, the subnet ID field is the area between the global routing prefix and the interface ID.

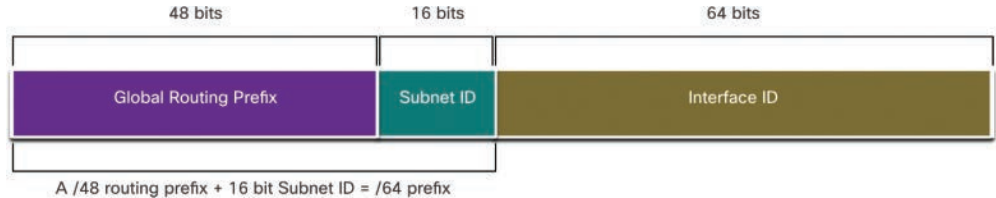


Figure 12-25 GUA with a 16-Bit Subnet ID

The benefit of a 128-bit address is that it can support more than enough subnets and hosts per subnet for each network—so address conservation is not an issue. For example, if the global routing prefix is /48, and you use the typical 64 bits for the interface ID, you end up with a 16-bit subnet ID:

- **16-bit subnet ID:** Creates up to 65,536 subnets
- **64-bit interface ID:** Supports up to 18 quintillion host IPv6 addresses per subnet (that is, 18,000,000,000,000,000,000)

Note

Subnetting into the 64-bit interface ID (or host portion) is also possible, but it is rarely required.

IPv6 subnetting is also easier to implement than IPv4 because there is no conversion to binary required. To determine the next available subnet, just count up in hexadecimal.

IPv6 Subnetting Example (12.8.2)

To see how IPv6 subnetting works, say that an organization has been assigned the 2001:db8:acad::/48 global routing prefix with a 16-bit subnet ID. This would allow the organization to create 65,536 /64 subnets, as shown in Figure 12-26. Notice that the global routing prefix is the same for all the subnets. Only the subnet ID hextet is incremented in hexadecimal for each subnet.

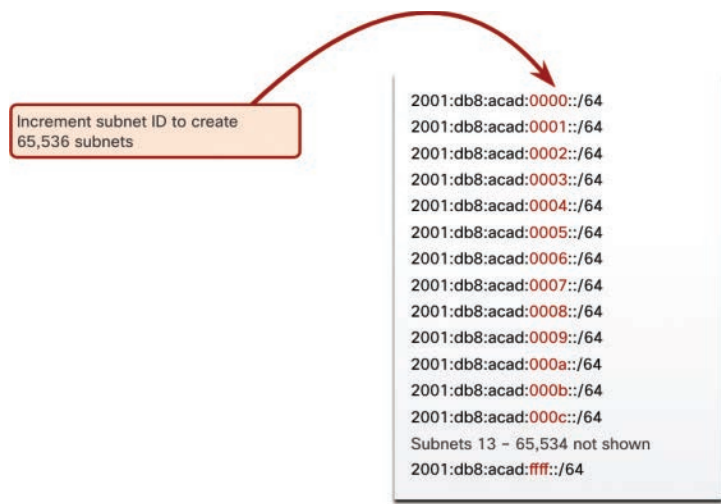


Figure 12-26 IPv6 Subnetting Example

IPv6 Subnet Allocation (12.8.3)

Because there are more than 65,536 subnets to choose from, the task of a network administrator is to design a logical scheme to address the network.

The topology shown in Figure 12-27 requires five subnets: one for each LAN as well as one for the serial link between R1 and R2. Unlike for IPv4, with IPv6, the serial link subnet has the same prefix length as the LANs. Although this may seem to “waste” addresses, address conservation is not a concern when using IPv6.

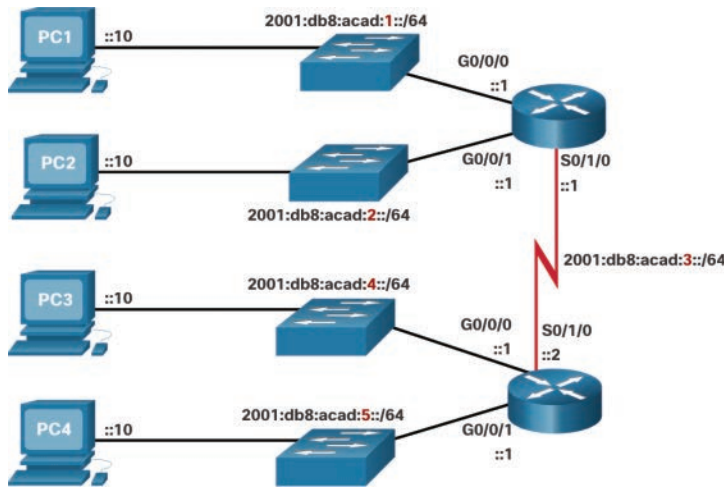


Figure 12-27 IPv6 Subnetting Example Topology

As shown in Figure 12-28, the five IPv6 subnets are allocated with the subnet ID field 0001 through 0005 for this example. Each /64 subnet provides more addresses than will ever be needed.

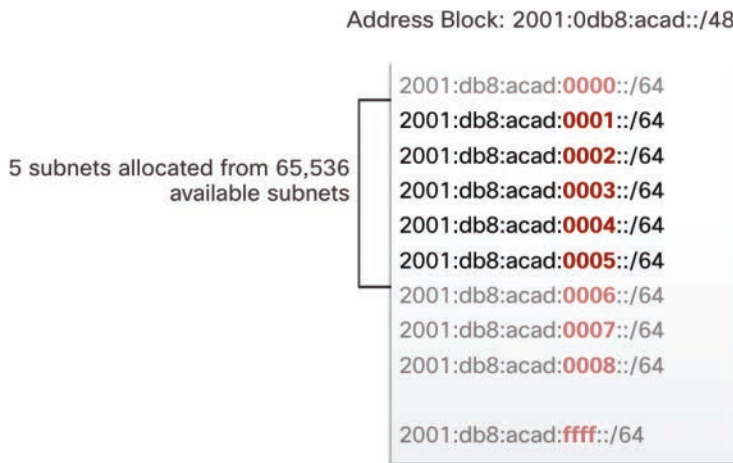


Figure 12-28 Five Allocated Subnets

Router Configured with IPv6 Subnets (12.8.4)

Configuring a router with IPv6 subnets is similar to the process for IPv4. Example 12-12 shows each of the router interfaces configured to be on a different IPv6 subnet.

Example 12-12 IPv6 Address Configuration on Router R1

```
R1(config)# interface gigabitethernet 0/0/0
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface gigabitethernet 0/0/1
R1(config-if)# ipv6 address 2001:db8:acad:2::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface serial 0/1/0
R1(config-if)# ipv6 address 2001:db8:acad:3::1/64
R1(config-if)# no shutdown
```

Check Your Understanding—Subnet an IPv6 Network (12.8.5)

Refer to the online course to complete this activity.

Summary (12.9)

The following is a summary of the topics in the chapter and their corresponding online modules.

IPv4 Issues

IPv4 has a theoretical maximum of 4.3 billion addresses. The use of private addresses in combination with NAT have helped to slow the depletion of IPv4 address space. With an increasing internet population, a limited IPv4 address space, issues with NAT and the IoT—the transition to IPv6 has begun. IPv4 and IPv6 will coexist in the near future, and the complete transition will take several years. The IETF has created various protocols and tools to help network administrators migrate their networks to IPv6. The migration techniques can be divided into three categories: dual stack, tunneling, and translation.

IPv6 Address Representation

An IPv6 address is 128 bits in length and written as a string of hexadecimal values. Every 4 bits is represented by a single hexadecimal digit, for a total of 32 hexadecimal values. The preferred format for writing an IPv6 address is $x:x:x:x:x:x:x$, with each x consisting of 4 hexadecimal values (for example, 2001:0db8:0000:1111:0000:0000:0200). Two rules help reduce the number of digits needed to represent an IPv6 address. The first rule is to omit any leading 0s (zeros) in any hextet (for example, 2001:db8:0:1111:0:0:0:200). The second rule is that a double colon (::) can replace any single, contiguous string of one or more 16-bit hextets consisting of all 0s (for example, 2001:db8:0:1111::200).

IPv6 Address Types

There are three types of IPv6 addresses: unicast, multicast, and anycast. IPv6 does not use the dotted decimal subnet mask notation. As with IPv4, the prefix length is represented in slash notation and is used to indicate the network portion of an IPv6 address. An IPv6 unicast address uniquely identifies an interface on an IPv6-enabled device. IPv6 addresses typically have two unicast addresses: GUA and LLA. IPv6 unique local addresses have a number of uses: They are used for local addressing within a site or between a limited number of sites, they can be used for devices that will never need to access another network, and they are not globally routed or translated to a global IPv6 address. IPv6 global unicast addresses (GUAs) are globally unique and routable on the IPv6 internet. These addresses are equivalent to public IPv4 addresses. A GUA has three parts: a global routing prefix, a subnet ID, and an interface ID. An IPv6 link-local address (LLA) enables a device to communicate with

other IPv6-enabled devices on the same link and only on that link (subnet). A device can obtain an LLA either statically or dynamically.

GUA and LLA Static Configuration

The Cisco IOS command to configure an IPv4 address on an interface is **ip address *ip-address subnet-mask***. The command to configure an IPv6 GUA on an interface is **ipv6 address *ipv6-address/prefix-length***. Just as with IPv4, configuring static addresses on clients does not scale to larger environments. For this reason, most network administrators in an IPv6 network enable dynamic assignment of IPv6 addresses. Configuring the LLA manually lets you create an address that is recognizable and easier to remember. Typically, it is only necessary to create recognizable LLAs on routers. LLAs can be configured manually using the **ipv6 address *ipv6-link-local-address* link-local** command.

Dynamic Addressing for IPv6 GUAs

A device obtains a GUA dynamically through ICMPv6 messages. IPv6 routers send out ICMPv6 RA messages every 200 seconds to all IPv6-enabled devices on the network. An RA message is also sent in response to a host sending an ICMPv6 RS message, which is a request for an RA message. The ICMPv6 RA message includes the network prefix and prefix length, default gateway address, and DNS addresses and domain name. RA messages have three methods: SLAAC, SLAAC with a stateless DHCPv6 server, and stateful DHCPv6 (no SLAAC). With SLAAC, the client device uses the information in the RA message to create its own GUA because the message contains the prefix and the interface ID. With SLAAC with stateless DHCPv6, the RA message suggests that devices use SLAAC to create their own IPv6 GUA, use the router LLA as the default gateway address, and use a stateless DHCPv6 server to obtain other necessary information. With stateful DHCPv6, the RA suggests that devices use the router LLA as the default gateway address and the stateful DHCPv6 server to obtain a GUA, a DNS server address, the domain name, and all the other necessary information. The interface ID can be created using the EUI-64 process or a randomly generated 64-bit number. The EUI process uses the 48-bit Ethernet MAC address of the client and inserts another 16 bits in the middle of the MAC address to create a 64-bit interface ID. Depending on the operating system, a device may use a randomly generated interface ID.

Dynamic Addressing for IPv6 LLAs

Every IPv6 device must have an IPv6 LLA. An LLA can be configured manually or created dynamically. Operating systems such as Windows typically use the same method for a SLAAC-created GUA and a dynamically assigned LLA. A Cisco router

automatically creates an IPv6 LLA whenever a GUA is assigned to the interface. By default, Cisco IOS routers use EUI-64 to generate the interface IDs for all LLAs on IPv6 interfaces. For serial interfaces, a router uses the MAC address of an Ethernet interface. To make it easier to recognize and remember these addresses on routers, it is common to statically configure IPv6 LLAs on routers. To verify IPv6 address configuration, use the following three commands: **show ipv6 interface brief**, **show ipv6 route**, and **ping**.

IPv6 Multicast Addresses

There are two types of IPv6 multicast addresses: well-known multicast addresses and solicited-node multicast addresses. Assigned multicast addresses are reserved multicast addresses for predefined groups of devices. Well-known multicast addresses are assigned. Two common IPv6 assigned multicast groups are the ff02::1 all-nodes multicast group and the ff02::2 all-routers multicast group. A solicited-node multicast address is similar to the all-nodes multicast address. The advantage of a solicited-node multicast address is that it is mapped to a special Ethernet multicast address.

Subnet an IPv6 Network

IPv6 was designed with subnetting in mind. A separate subnet ID field in the IPv6 GUA is used to create subnets. The subnet ID field is the area between the global routing prefix and the interface ID. The benefit of a 128-bit address is that it can support more than enough subnets and hosts per subnet for each network. Address conservation is not an issue. For example, if the global routing prefix is /48, and you use the typical 64 bits for the interface ID, you end up with a 16-bit subnet ID:

- **16-bit subnet ID:** Creates up to 65,536 subnets
- **64-bit interface ID:** Supports up to 18 quintillion host IPv6 addresses per subnet (that is, 18,000,000,000,000,000,000)

Because there are more than 65,536 subnets to choose from, the task of a network administrator is to design a logical scheme to address the network. Address conservation is not a concern when using IPv6. Much as when configuring IPv4, with IPv6, each router interface can be configured to be on a different IPv6 subnet.



Packet Tracer—Implement a Subnetted IPv6 Addressing Scheme (12.9.1)

Your network administrator wants you to assign five /64 IPv6 subnets to the network shown in the topology. Your job is to determine the IPv6 subnets, assign IPv6 addresses to the routers, and set the PCs to automatically receive IPv6 addressing. Your final step is to verify connectivity between IPv6 hosts.



Lab—Configure IPv6 Addresses on Network Devices (12.9.2)

In this lab, you will complete the following objectives:

- Part 1: Set Up Topology and Configure Basic Router and Switch Settings
 - Part 2: Configure IPv6 Addresses Manually
 - Part 3: Verify End-to-End Connectivity
-

Practice

The following activities provide practice with the topics introduced in this chapter. The lab is available in the companion *Introduction to Networks Labs & Study Guide (CCNAv7)* (ISBN 9780136634454). The Packet Tracer activity instructions are also provided in the *Labs & Study Guide*. The PKA files are available in the online course.



Labs

Lab 12.7.4: Identify IPv6 Addresses

Lab 12.9.2: Configure IPv6 Addresses on Network Devices



Packet Tracer Activities

Packet Tracer 12.6.6: Configure IPv6 Addressing

Packet Tracer 12.9.1: Implement a Subnetted IPv6 Addressing Scheme

Check Your Understanding Questions

Complete all the review questions listed here to test your understanding of the topics and concepts in this chapter. The appendix “Answers to ‘Check Your Understanding’ Questions” lists the answers.

1. What is indicated by a successful ping to the ::1 IPv6 address?
 - a. The host is cabled correctly.
 - b. The default gateway address is configured correctly.
 - c. All hosts on the local link are available.
 - d. The link-local address is correctly configured.
 - e. IP is properly installed on the host.

2. What is the most compressed representation of the IPv6 address 2001:0db8:0000:abcd:0000:0000:0000:0001?
 - a. 2001:0db8:abcd::1
 - b. 2001:db8:0:abcd::1
 - c. 2001:0db8:abcd:1
 - d. 2001:0db8:0000:abcd::1
 - e. 2001:db8::abcd:0:1
3. What is the purpose of the command **ping ::1**?
 - a. It tests the internal configuration of an IPv6 host.
 - b. It tests the broadcast capability of all hosts on the subnet.
 - c. It tests the multicast connectivity to all hosts on the subnet.
 - d. It tests the reachability of the default gateway for the network.
4. At a minimum, which address is required on IPv6-enabled interfaces?
 - a. link-local
 - b. unique local
 - c. site local
 - d. global unicast
5. What is the interface ID of the IPv6 address 2001:db8::1000:a9cd:47ff:fe57:fe94/64?
 - a. fe94
 - b. fe57:fe94
 - c. 47ff:fe57:fe94
 - d. a9cd:47ff:fe57:fe94
 - e. 1000:a9cd:47ff:fe57:fe94
6. What are the three parts of an IPv6 global unicast address? (Choose three.)
 - a. an interface ID that is used to identify the local network for a particular host
 - b. a global routing prefix that is used to identify the network portion of the address that has been provided by the ISP
 - c. a subnet ID that is used to identify networks inside the local enterprise site
 - d. a global routing prefix that is used to identify the portion of the network address provided by a local administrator
 - e. an interface ID that is used to identify the local host on the network

7. What is the most compressed format possible for the IPv6 address 2001:0db8:0000:ab00:0000:0000:1234?
 - a. 2001:db8:0:ab00::1234
 - b. 2001:db8:0:ab::1234
 - c. 2001:db8:0000:ab::1234
 - d. 2001:db8:0:ab:0::1234
8. What is the prefix associated with the IPv6 address 2001:db8:d15:ea:cc44::1/64?
 - a. 2001::/64
 - b. 2001:db8::/64
 - c. 2001:db8:d15:ea::/64
 - d. 2001:db8:d15:ea:cc44::/64
9. What type of address is automatically assigned to an interface when IPv6 is enabled on that interface?
 - a. global unicast
 - b. link-local
 - c. loopback
 - d. unique local
10. Which IPv6 network prefix is only intended for local links and cannot be routed?
 - a. 2001::/3
 - b. fc00::/7
 - c. fe80::/10
 - d. ff00::/12
11. Your organization is issued the IPv6 prefix 2001:0:130f::/48 by your service provider. With this prefix, how many bits are available for your organization to create /64 subnetworks if interface ID bits are not borrowed?
 - a. 8
 - b. 16
 - c. 80
 - d. 128

12. What is the network address for the IPv6 address 2001:db8:aa04:b5::1/64?
 - a. 2001::/64
 - b. 2001:db8::/64
 - c. 2001:db8:aa04::/64
 - d. 2001:db8:aa04:b5::/64
13. Which type of IPv6 address is not routable and is used only for communication on a single subnet?
 - a. global unicast address
 - b. link-local address
 - c. loopback address
 - d. unique local address
 - e. unspecified address
14. Which address type is not supported in IPv6?
 - a. private
 - b. multicast
 - c. unicast
 - d. broadcast
15. What is the minimum configuration for a router interface that is enabled for IPv6?
 - a. to have a link-local IPv6 address
 - b. to have both IPv4 and IPv6 addresses
 - c. to have a self-generated loopback address
 - d. to have both a link-local address and a global unicast address
 - e. to have only an automatically generated multicast address

Objectives

Upon completion of this chapter, you will be able to answer the following questions:

- How is ICMP used to test network connectivity?
- How do you use the **ping** and **tracert** utilities to test network connectivity?

Introduction (13.0)

Imagine that you have an intricate model train set. Your tracks and trains are all connected and powered up and ready to go. You throw the switch. The train goes halfway around the track and stops. You know right away that the problem is most likely located where the train has stopped, so you look there first. It is not as easy to visualize problems with a network. Fortunately, there are tools to help you locate problem areas in a network—and they work with both IPv4 and IPv6 networks! You will be happy to know that this chapter has a couple Packet Tracer activities to help you practice using these tools, so let's get testing!

ICMP Messages (13.1)

In this section, you will learn about the different types of Internet Control Message Protocol (ICMP) messages and the tools that are used to send them.

ICMPv4 and ICMPv6 Messages (13.1.1)

Although IP is only a best-effort protocol, the TCP/IP suite does provide for error messages and informational messages when communicating with another IP device. These messages are sent using the services of ICMP. The purpose of these messages is to provide feedback about issues related to the processing of IP packets under certain conditions—not to make IP reliable. ICMP messages are not required and are often not allowed in a network for security reasons.

ICMP is available for both IPv4 and IPv6. ICMPv4 is the messaging protocol for IPv4. ICMPv6 provides the same services for IPv6 but includes additional functionality. In this book, the term *ICMP* is used to refer to both ICMPv4 and ICMPv6.

The types of ICMP messages and the reasons they are sent are extensive. The ICMP messages common to both ICMPv4 and ICMPv6 and discussed in this chapter include

- Host reachability (Echo Request and Echo Reply) messages
- Destination Unreachable or Service Unreachable messages
- Time Exceeded messages

Host Reachability (13.1.2)

ICMP Echo Request and Echo Reply messages can be used to test the reachability of a host on an IP network. The local host sends an ICMP Echo Request to a host. If

the host is available, the destination host responds with an Echo Reply, as shown in Figure 13-1. This use of the ICMP Echo messages is the basis of the **ping** utility.

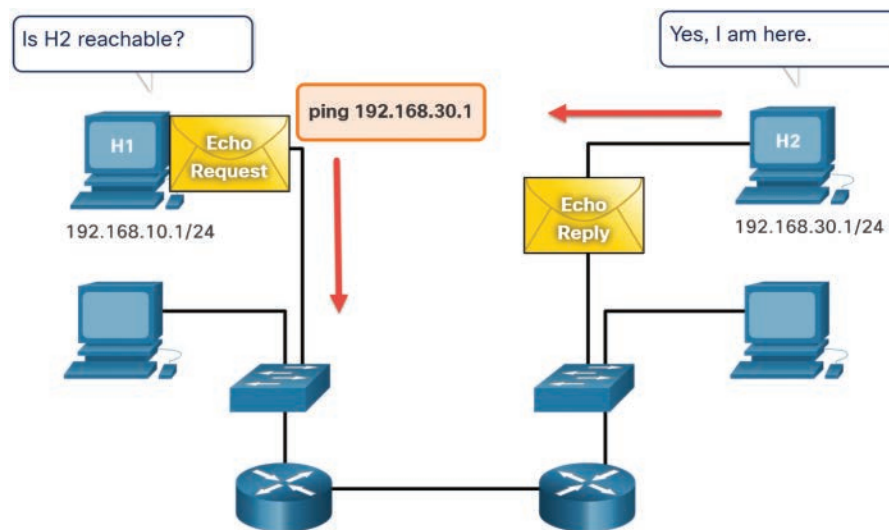


Figure 13-1 Echo Request and Echo Reply

Destination or Service Unreachable (13.1.3)

When a host or gateway receives a packet that it cannot deliver, it can use an ICMP Destination Unreachable message to notify the source that the destination or service is unreachable. This message includes a code that indicates why the packet could not be delivered.

Some of the Destination Unreachable codes for ICMPv4 are as follows:

- 0: Net unreachable
- 1: Host unreachable
- 2: Protocol unreachable
- 3: Port unreachable

Some of the Destination Unreachable codes for ICMPv6 are as follows:

- 0: No route to destination
- 1: Communication with the destination is administratively prohibited (for example, by a firewall)
- 2: Beyond scope of the source address

- 3: Address unreachable
- 4: Port unreachable

Time Exceeded (13.1.4)

A router uses an ICMPv4 Time Exceeded message to indicate that a packet cannot be forwarded because the Time-to-Live (TTL) field of the packet has been decremented to 0. If a router receives a packet and decrements the TTL field in the IPv4 packet to 0, it discards the packet and sends a Time Exceeded message to the source host.

ICMPv6 also sends a Time Exceeded message if the router cannot forward an IPv6 packet because the packet has expired. Instead of using the IPv4 TTL field, ICMPv6 uses the IPv6 Hop Limit field to determine if a packet has expired.

Note

Time Exceeded messages are used by the **tracert** tool.

ICMPv6 Messages (13.1.5)

The ICMPv6 informational and error messages are very similar to the control and error messages implemented with ICMPv4. However, ICMPv6 has new features and improved functionality not found in ICMPv4. ICMPv6 messages are encapsulated in IPv6.

ICMPv6 includes four new messages as part of the Neighbor Discovery Protocol (ND or NDP).

Messages between an IPv6 router and an IPv6 device, including dynamic address allocation, are as follows:

- Router Solicitation (RS) message
- Router Advertisement (RA) message

Messages between IPv6 devices, including duplicate address detection and address resolution, are as follows:

- Neighbor Solicitation (NS) message
- Neighbor Advertisement (NA) message

Note

ICMPv6 ND also includes the redirect message, which has a similar function to the redirect message used in ICMPv4.

RA messages are sent by IPv6-enabled routers every 200 seconds to provide addressing information to IPv6-enabled hosts. An RA message can include addressing information for the host, such as the prefix, prefix length, DNS address, and domain name. A host using stateless address autoconfiguration (SLAAC) sets its default gateway to the link-local address of the router that sent the RA.

In Figure 13-2, R1 sends an RA message to ff02::1, the all-nodes multicast address, which will reach PC1.

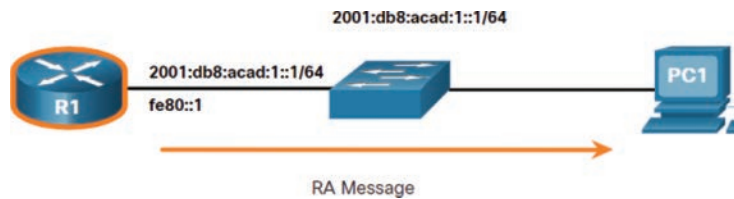


Figure 13-2 RA Message

An IPv6-enabled router also sends out an RA message in response to an RS message. In Figure 13-3, PC1 sends a RS message to determine how to receive its IPv6 address information dynamically. R1 replies to the RS with an RA message:

1. PC1 sends the RS message “Hi, I just booted up. Is there an IPv6 router on the network? I need to know how to get my IPv6 address information dynamically.”
2. R1 replies with the RA message “Hi, all IPv6-enabled devices. I’m R1, and you can use SLAAC to create an IPv6 global unicast address. The prefix is 2001:db8:acad:1::/64. By the way, use my link-local address fe80::1 as your default gateway.”

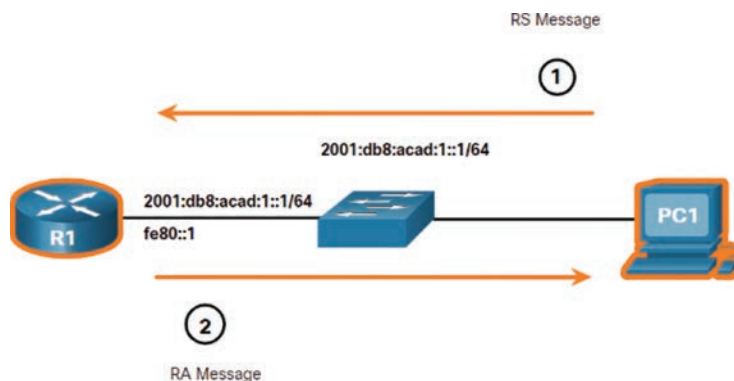


Figure 13-3 RS Message

When a device is assigned a global IPv6 unicast or link-local unicast address, it may perform duplicate address detection (DAD) to ensure that the IPv6 address is unique. To check the uniqueness of an address, the device sends an NS message with its own IPv6 address as the targeted IPv6 address, as shown in Figure 13-3.

If another device on the network has this address, it responds with an NA message that notifies the sending device that the address is in use. If a corresponding NA message is not returned within a certain amount of time, the unicast address is unique and acceptable for use.

Note

DAD is not required, but RFC 4861 recommends that DAD be performed on unicast addresses.

In Figure 13-4, PC1 sends this NS message to check the uniqueness of an address: “Will whoever has the IPv6 address 2001:db8:acad:1::10 send me your MAC address?”

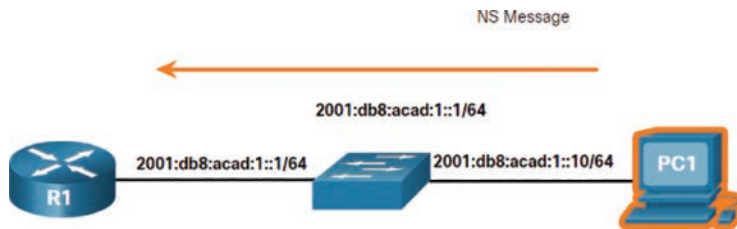


Figure 13-4 NS Message

Address resolution is used when a device on the LAN knows the IPv6 unicast address of a destination but does not know its Ethernet MAC address. To determine the MAC address for the destination, the device sends an NS message to the solicited node address. The message includes the known (targeted) IPv6 address. The device that has the targeted IPv6 address responds with an NA message containing its Ethernet MAC address.

In Figure 13-5, R1 sends an NS message to 2001:db8:acad:1::10, asking for its MAC address:

1. R1 sends the address resolution NS message “Will whoever has the IPv6 address 2001:db8:acad:1::10 send me your MAC address?”
2. PC1 replies with the NA message “I’m 2001:db8:acad:1::10, and my MAC address is 00:aa:bb:cc:dd:ee.”

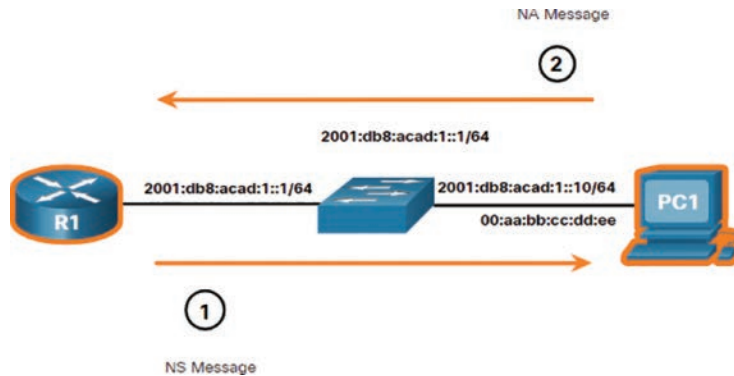


Figure 13-5 NA Message

Interactive
Graphic

Check Your Understanding—ICMP Messages (13.1.6)

Refer to the online course to complete this activity.

Ping and Traceroute Tests (13.2)

This section discusses two important tools that are used to verify Layer 3 connectivity: **ping** and **traceroute**.

Ping—Test Connectivity (13.2.1)

In this section, you will learn about the situations in which the **ping** and **traceroute** (**tracert**) tools are used and how to use them. **ping** is an IPv4 and IPv6 testing utility that uses ICMP Echo Request and Echo Reply messages to test connectivity between hosts.

To test connectivity to another host on a network, an Echo Request is sent to the host address by using the **ping** command. If the host at the specified address receives the Echo Request, it responds with an Echo Reply. As each Echo Reply is received, **ping** provides feedback on the time between when the Echo Request was sent and when the Echo Reply was received. This can be a measure of network performance.

ping has a timeout value for the reply. If a reply is not received within the timeout period, **ping** provides a message indicating that a response was not received. This may indicate that there is a problem, but it could also indicate that security features blocking **ping** messages have been enabled on the network. It is common for the first **ping** to time out if address resolution (ARP or ND) needs to be performed before the ICMP Echo Request is sent.

After all the requests are sent, the **ping** utility provides a summary that includes the success rate and average round-trip time to the destination.

You can use **ping** to perform the following types of connectivity tests:

- Pinging the local loopback
- Pinging the default gateway
- Pinging the remote host

Ping the Loopback (13.2.2)

ping can be used to test the internal configuration of IPv4 or IPv6 on the local host. To perform this test, you **ping** the local loopback address 127.0.0.1 for IPv4 (:::1 for IPv6), as shown in Figure 13-6.

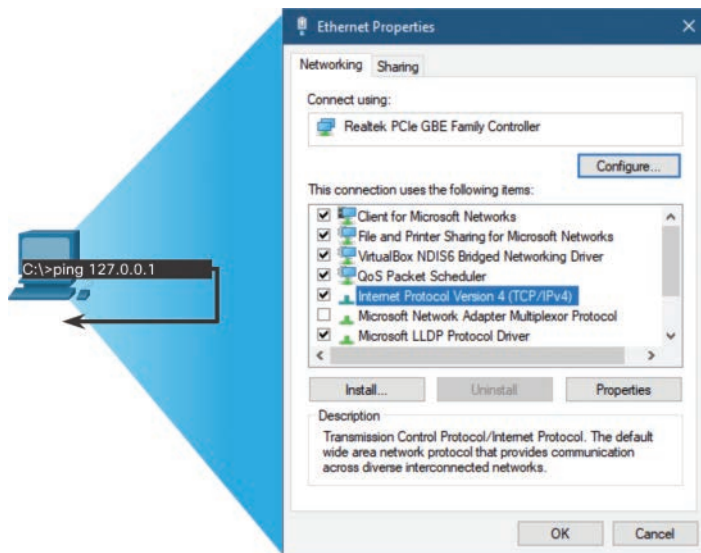


Figure 13-6 Pinging the Loopback on a Windows Host

A response from 127.0.0.1 for IPv4 or from :::1 for IPv6 indicates that IP is properly installed on the host. This response comes from the network layer. This response is not, however, an indication that the addresses, masks, or gateways are properly configured. It also does not indicate anything about the status of the lower layer of the network stack. It simply tests IP down through the network layer of IP. An error message indicates that TCP/IP is not operational on the host.

Ping the Default Gateway (13.2.3)

You can use **ping** to test the ability of a host to communicate on the local network. This is generally done by pinging the IP address of the default gateway of the host, as shown in Figure 13-7. A successful **ping** to the default gateway indicates that the host

and the router interface serving as the default gateway are both operational on the local network.

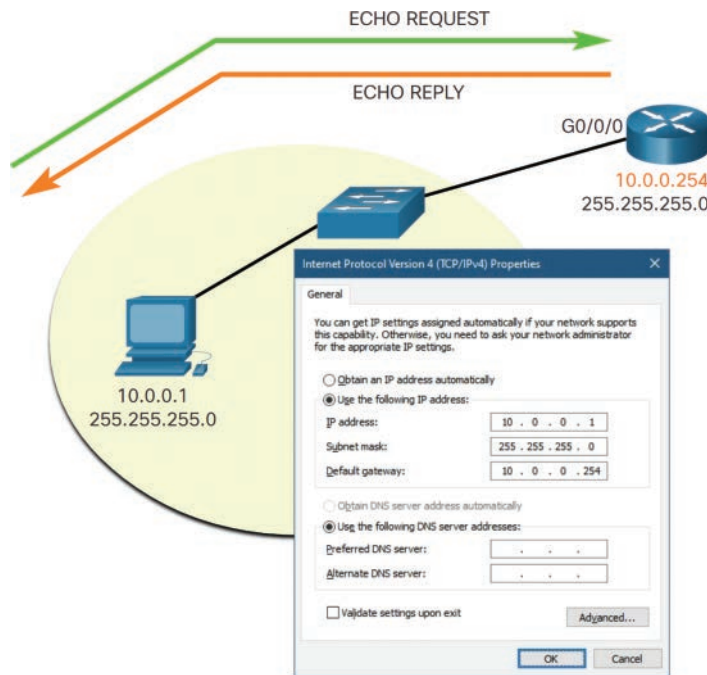


Figure 13-7 Pinging the Default Gateway

For this test, the default gateway address is most often used because the router is normally always operational. If the default gateway address does not respond, a **ping** can be sent to the IP address of another host on the local network that is known to be operational.

If either the default gateway or another host responds, this confirms that the local host can successfully communicate over the local network. If the default gateway does not respond but another host does, this could indicate a problem with the router interface serving as the default gateway. One possibility is that the wrong default gateway address might have been configured on the host. Another possibility is that the router interface may be fully operational but might have security applied to it that prevents it from processing or responding to **ping** requests.

Ping a Remote Host (13.2.4)

ping can be used to test the ability of a local host to communicate across an internetwork. The local host can ping an operational IPv4 host of a remote network, as shown in Figure 13-8. The router uses its IP routing table to forward the packets.

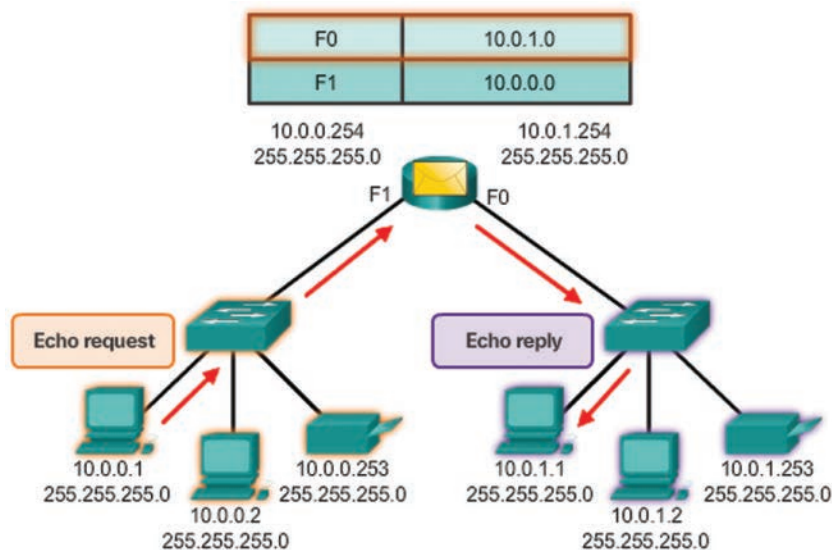


Figure 13-8 Testing Connectivity to a Remote LAN

If this **ping** is successful, the operation of a large piece of the internetwork can be verified. A successful **ping** across the internetwork confirms communication on the local network, the operation of the router serving as the default gateway, and the operation of all other routers that might be in the path between the local network and the network of the remote host.

In addition, the functionality of the remote host can be verified. If the remote host could not communicate outside its local network, it would not have responded.

Note

Many network administrators limit or prohibit the entry of ICMP messages into the corporate network; therefore, the lack of a **ping** response could be due to security restrictions.

Traceroute—Test the Path (13.2.5)

ping is used to test connectivity between two hosts but does not provide information about the details of devices between the hosts. **traceroute** (**tracert**) is a utility that generates a list of hops that were successfully reached along the path. This list can provide important verification and troubleshooting information. If the data reaches the destination, then the trace lists the interface of every router in the path between the hosts. If the data fails at some hop along the way, the address of the last router that responded to the trace can provide an indication of where the problem or security restrictions are occurring.

Round-Trip Time (RTT)

Using **tracert** provides round-trip time for each hop along the path and indicates whether a hop fails to respond. The round-trip time is the time a packet takes to reach the remote host and for the response from the host to return. An asterisk (*) is used to indicate a lost packet or a packet that does not receive a reply. This information can be used to locate a problematic router in the path or may indicate that the router is configured not to reply. If the display shows high response times or data losses from a particular hop, this is an indication that the resources of the router or its connections may be stressed.

IPv4 TTL and IPv6 Hop Limit

tracert makes use of a function of the TTL field in IPv4 and the Hop Limit field in IPv6 in the Layer 3 headers, along with the ICMP Time Exceeded message.

The first sequence of messages sent from **tracert** have a TTL field value of 1. This causes the TTL to time out the IPv4 packet at the first router. This router then responds with an ICMPv4 Time Exceeded message. **tracert** now has the address of the first hop.

tracert then progressively increments the TTL field (2, 3, 4, and so on) for each sequence of messages. This provides the trace with the address of each hop as the packets time out further down the path. The TTL field continues to be increased until the destination is reached, or it is incremented to a predefined maximum.

After the final destination is reached, the host responds with either an ICMP Port Unreachable message or an ICMP Echo Reply message instead of the ICMP Time Exceeded message.

Interactive Graphic

Go to the online course to view an animation of how **tracert** takes advantage of TTL.

Packet Tracer Activity

Packet Tracer—Verify IPv4 and IPv6 Addressing (13.2.6)

IPv4 and IPv6 can coexist on the same network. From the command prompt of a PC, there are some differences in the way commands are issued and in the way output is displayed.

Packet Tracer Activity

Packet Tracer—Use Ping and Traceroute to Test Network Connectivity (13.2.7)

There are connectivity issues in this activity. In addition to gathering and documenting information about the network, you will locate the problems and implement acceptable solutions to restore connectivity.

Summary (13.3)

The following is a summary of the topics in the chapter and their corresponding online modules.

ICMP Messages

The TCP/IP suite provides for error messages and informational messages when communicating with other IP devices. These messages are sent using ICMP. The purpose of these messages is to provide feedback about issues related to the processing of IP packets under certain conditions. The ICMP messages common to both ICMPv4 and ICMPv6 are Host Reachability, Destination Unreachable or Service Unreachable, and Time Exceeded. An ICMP Echo message tests the reachability of a host on an IP network. The local host sends an ICMP Echo Request to a host. If the host is available, the destination host responds with an Echo Reply. This is the basis of the **ping** utility. When a host or gateway receives a packet that it cannot deliver, it can use an ICMP Destination Unreachable message to notify the source. This message includes a code that indicates why the packet could not be delivered. A router uses an ICMPv4 Time Exceeded message to indicate that a packet cannot be forwarded because the Time-to-Live (TTL) field of the packet was decremented to 0. If a router receives a packet and decrements the TTL field to zero, it discards the packet and sends a Time Exceeded message to the source host. ICMPv6 also sends a Time Exceeded in this situation. ICMPv6 uses the IPv6 Hop Limit field to determine whether the packet has expired. Time Exceeded messages are used by the **traceroute** tool. The messages between an IPv6 router and an IPv6 device using dynamic address allocation include RS and RA. The messages between IPv6 devices include the redirect messages (similar to those in IPv4): NS and NA.

Ping and Traceroute Testing

ping (used by IPv4 and IPv6) uses ICMP Echo Request and Echo Reply messages to test connectivity between hosts. To test connectivity to another host on a network, an Echo Request is sent to the host address, using the **ping** command. If the host at the specified address receives the Echo Request, it responds with an Echo Reply. As each Echo Reply is received, **ping** provides feedback on the time between when the request was sent and when the reply was received. After all the requests are sent, the **ping** utility provides a summary that includes the success rate and average round-trip time to the destination. **ping** can be used to test the internal configuration of IPv4 or IPv6 on the local host. You can **ping** the local loopback address 127.0.0.1 for IPv4 or ::1 for IPv6. You can also use **ping** to test the ability of a host to communicate on the local network, by pinging the IP address of the default gateway of the host. A successful **ping** to the default gateway indicates that the host and the router interface

serving as the default gateway are both operational on the local network. **ping** can also be used to test the ability of a local host to communicate across an internet-work. The local host can ping an operational IPv4 host of a remote network. **trace-route (tracert)** generates a list of hops that were successfully reached along the path. This list provides verification and troubleshooting information. If the data reaches the destination, then the trace lists the interface of every router in the path between the hosts. If the data fails at some hop along the way, the address of the last router that responded to the trace can provide an indication of where the problem or security restrictions are occurring. The round-trip time is the time a packet takes to reach the remote host and for the response from the host to return. **tracert** makes use of a function of the TTL field in IPv4 and the Hop Limit field in IPv6 in the Layer 3 headers, along with the ICMP Time Exceeded message.

Packet Tracer
Activity

Packet Tracer—Use ICMP to Test and Correct Network Connectivity (13.3.1)

In this Packet Tracer activity, you will use ICMP to test network connectivity and locate network problems. You will also correct simple configuration issues and restore connectivity to the network:

- Use ICMP to locate connectivity issues.
- Configure network devices to correct connectivity issues.



Lab—Use Ping and Traceroute to Test Network Connectivity (13.3.2)

In this lab, you will complete the following objectives:

- Part 1: Build and Configure the Network
- Part 2: Use **ping** Command for Basic Network Testing
- Part 3: Use **tracert** and **tracert** Commands for Basic Network Testing
- Part 4: Troubleshoot the Topology

Practice

The following activities provide practice with the topics introduced in this chapter. The lab is available in the companion *Introduction to Networks Labs & Study Guide (CCNAv7)* (ISBN 9780136634454). The Packet Tracer activity instructions are also provided in the *Labs & Study Guide*. The PKA files are available in the online course.

**Lab**

Lab 13.3.2: Use Ping and Traceroute to Test Network Connectivity

Packet Tracer
Activity**Packet Tracer Activities**

Packet Tracer 13.2.6: Verify IPv4 and IPv6 Addressing

Packet Tracer 13.2.7: Use **ping** and **tracert** to Test Network Connectivity

Packet Tracer 13.3.1: Use ICMP to Test and Correct Network Connectivity

Check Your Understanding Questions

Complete all the review questions listed here to test your understanding of the topics and concepts in this chapter. The appendix “Answers to ‘Check Your Understanding’ Questions” lists the answers.

1. A user calls to report that a PC cannot access the internet. The network technician asks the user to issue the command **ping 127.0.0.1** in a command prompt window. The user reports that the result is four positive replies. What conclusion can be drawn, based on this connectivity test?
 - a. The PC can access the network. The problem exists beyond the local network.
 - b. The IP address obtained from the DHCP server is correct.
 - c. The PC can access the Internet. However, the web browser may not work.
 - d. The TCP/IP implementation is functional.
2. Which command can be used to test connectivity between two devices using Echo Request and Echo Reply messages?
 - a. **netstat**
 - b. **ipconfig**
 - c. **icmp**
 - d. **ping**
3. What IPv6 field does a router use to determine that a packet has expired?
 - a. TTL field
 - b. CRC field
 - c. Hop Limit field
 - d. Time Exceeded field

4. Which protocol provides feedback from the destination host to the source host about errors in packet delivery?
 - a. ARP
 - b. BOOTP
 - c. DNS
 - d. ICMP
5. Which utility uses Internet Control Messaging Protocol (ICMP)?
 - a. RIP
 - b. DNS
 - c. ping
 - d. NTP
6. A network administrator can successfully ping the server at www.cisco.com but cannot ping the company web server located at an ISP in another city. Which tool or command would help identify the specific router where the packet was lost or delayed?
 - a. ipconfig
 - b. netstat
 - c. telnet
 - d. traceroute
7. Which protocol does IPv6 use to provide address resolution and dynamic address allocation information?
 - a. ICMPv4
 - b. NDP
 - c. ARP
 - d. DHCP
8. What message can a host send to check the uniqueness of an IPv6 address before using that address?
 - a. Neighbor Solicitation
 - b. ARP Request
 - c. Echo Request
 - d. Router Solicitation

9. A technician is troubleshooting a network where it is suspected that a defective node in the network path is causing packets to be dropped. The technician only has the IP address of the endpoint device and does not have any details about the intermediate devices. What Windows command can the technician use to identify the faulty node?
- a. **tracert**
 - b. **ping**
 - c. **ipconfig /flushdns**
 - d. **ipconfig /displaydns**
10. A user who is unable to connect to the file server contacts the help desk. The help desk technician asks the user to ping the IP address of the default gateway that is configured on the workstation. What is the purpose of this **ping** command?
- a. to obtain a dynamic IP address from the server
 - b. to request that the gateway forward the connection request to the file server
 - c. to test that the host has the capability to reach hosts on other networks
 - d. to resolve the domain name of the file server to its IP address
11. What does the Windows **tracert** command do that the **ping** command does not when these commands are used on a workstation?
- a. The **tracert** command reaches the destination faster.
 - b. The **tracert** command shows the information of routers in the path.
 - c. The **tracert** command sends one ICMP message to each hop in the path.
 - d. The **tracert** command is used to test the connectivity between two devices.
12. Which ICMP message does the **tracert** utility use during the process of finding the path between two end hosts?
- a. Redirect
 - b. **ping**
 - c. Time Exceeded
 - d. Destination Unreachable

13. Which two things can be determined by using the **ping** command?
(Choose two.)
- a. the number of routers between the source and the destination device
 - b. the IP address of the router nearest the destination device
 - c. the average time it takes a packet to reach the destination and for the response to return to the source
 - d. the reachability of the destination device through the network
 - e. the average time it takes each router in the path between the source and the destination to respond
14. Which statement describes a characteristic of the **tracert** utility?
- a. It sends four Echo Request messages.
 - b. It utilizes the ICMP Source Quench messages.
 - c. It is primarily used to test connectivity between two hosts.
 - d. It identifies the routers in the path from a source host to a destination host.

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Transport Layer

Objectives

Upon completion of this chapter, you will be able to answer the following questions:

- What is the purpose of the transport layer in managing the transportation of data in end-to-end communication?
- What are the characteristics of TCP?
- What are the characteristics of UDP?
- How do TCP and UDP use port numbers?
- How do the TCP session establishment and termination processes facilitate reliable communication?
- How are TCP protocol data units transmitted and acknowledged to guarantee delivery?
- What are the operations of transport layer protocols in supporting end-to-end communication?

Key Terms

This chapter uses the following key terms. You can find the definitions in the glossary at the end of the book.

Transmission Control Protocol (TCP)
page 462

User Datagram Protocol (UDP) page 462

segment page 463

connection-oriented protocol page 471

stateful page 471

port number page 476

socket page 477

socket pair page 477

three-way handshake page 483

initial sequence number (ISN) page 487

expectational acknowledgment page 488

selective acknowledgment (SACK) page 489

window size page 490

Introduction (14.0)

The transport layer is where, as the name implies, data is transported from one host to another. This is where your network really gets moving! The transport layer uses two protocols: TCP and UDP. To understand TCP, imagine getting a registered letter in the mail: You have to sign for it before the mail carrier will let you have it. This slows down the process a bit, but the sender knows for certain that you received the letter and when you received it. To understand UDP, imagine a regular, stamped letter: If it arrives in your mailbox, it is probably intended for you, but it might actually be for someone else who does not live at your address. Also, it may not arrive in your mailbox at all, and the sender cannot be sure you received it. Regardless of UDP's weaknesses, there are times when it is the protocol that is needed. This chapter dives into how TCP and UDP work in the transport layer. Later in this chapter, you will be able to view several videos to help understand these processes.

Transportation of Data (14.1)

As discussed in earlier chapters, for communication to occur between a source and a destination, a set of rules, or protocols, must be followed. This section focuses on the protocols at the transport layer.

Role of the Transport Layer (14.1.1)

Application layer programs generate data that must be exchanged between source and destination hosts. The transport layer is responsible for logical communications between applications running on different hosts. This may include processes such as establishing a temporary session between two hosts and the reliable transmission of information for an application.

As shown in Figure 14-1, the transport layer is the link between the application layer and the lower layers that are responsible for network transmission.

The transport layer has no knowledge of the destination host type, the type of media over which the data must travel, the path taken by the data, the congestion on a link, or the size of the network.

The transport layer includes two protocols:

- *Transmission Control Protocol (TCP)*
- *User Datagram Protocol (UDP)*

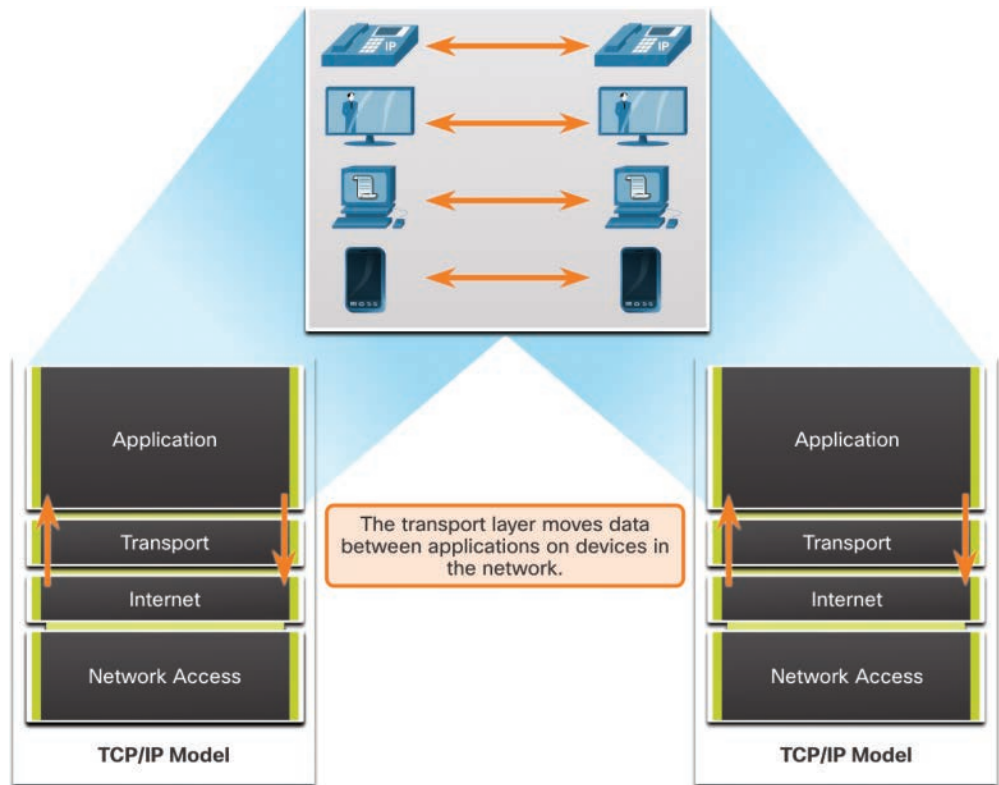


Figure 14-1 The Transport Layer in the TCP/IP Model

Transport Layer Responsibilities (14.1.2)

The transport layer has many responsibilities. At the transport layer, each set of data flowing between a source application and a destination application is known as a *conversation* and is tracked separately. It is the responsibility of the transport layer to maintain and track all the conversations. As illustrated in Figure 14-2, a host may have multiple applications that are communicating across the network simultaneously.

Most networks have a limitation on the amount of data that can be included in a single packet. Therefore, data must be divided into manageable pieces.

It is the transport layer's responsibility to divide the application data into appropriately sized blocks. Depending on the transport layer protocol used, the transport layer blocks are called either *segments* or *datagrams*. Figure 14-3 illustrates the transport layer using different blocks for each conversation.

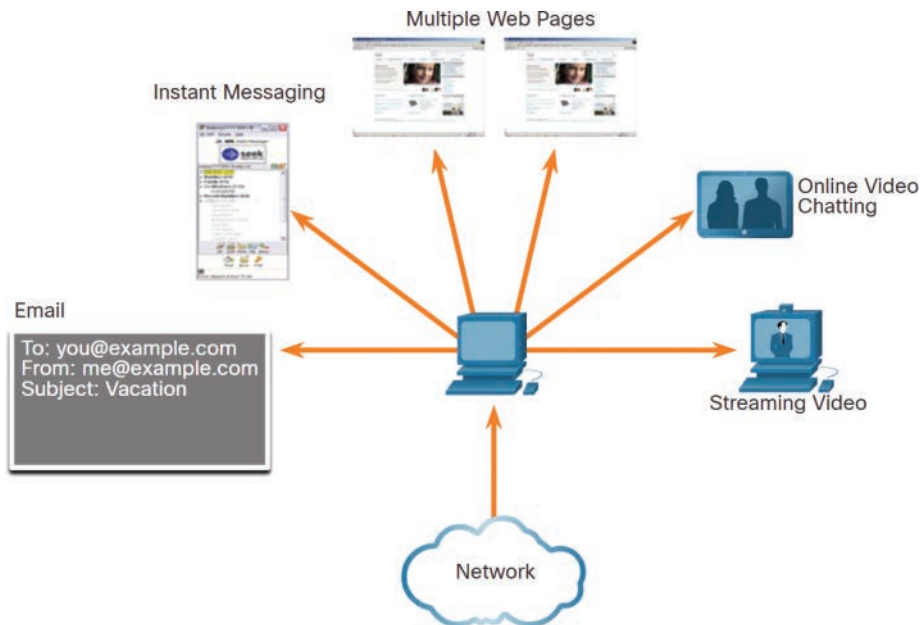


Figure 14-2 Tracking Individual Conversations

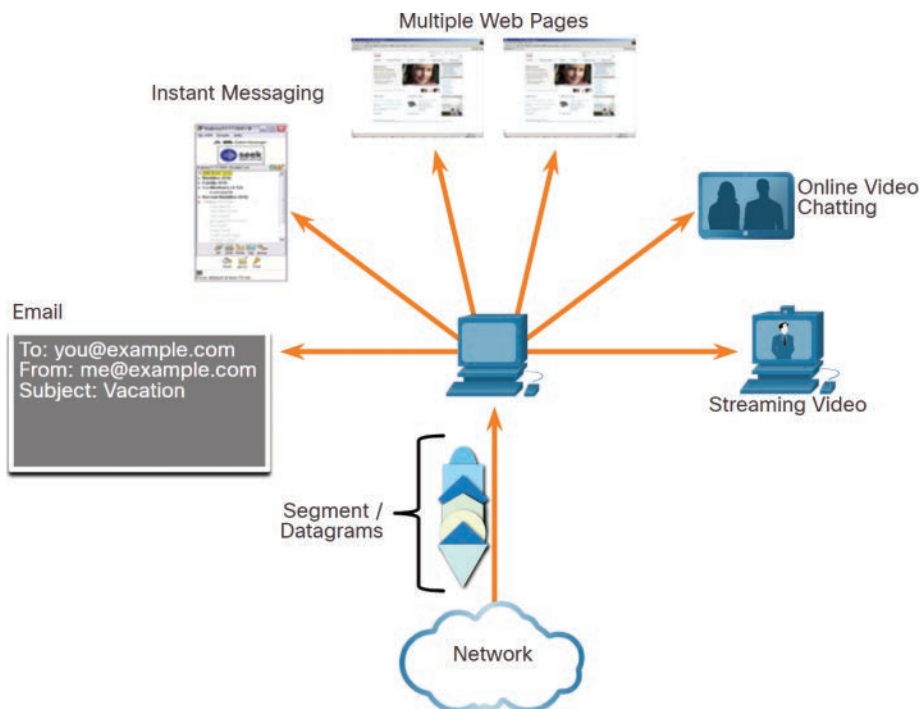


Figure 14-3 Segmenting Data and Reassembling Segments

The transport layer divides data into smaller blocks (segments or datagrams) that are easier to manage and transport.

A transport layer protocol adds to each block of data header information containing binary data organized into several fields. The values in these fields enable various transport layer protocols to perform different functions in managing data communication.

For instance, the receiving host uses the header information to reassemble the blocks of data into a complete data stream for the receiving application layer program, as shown in Figure 14-4.

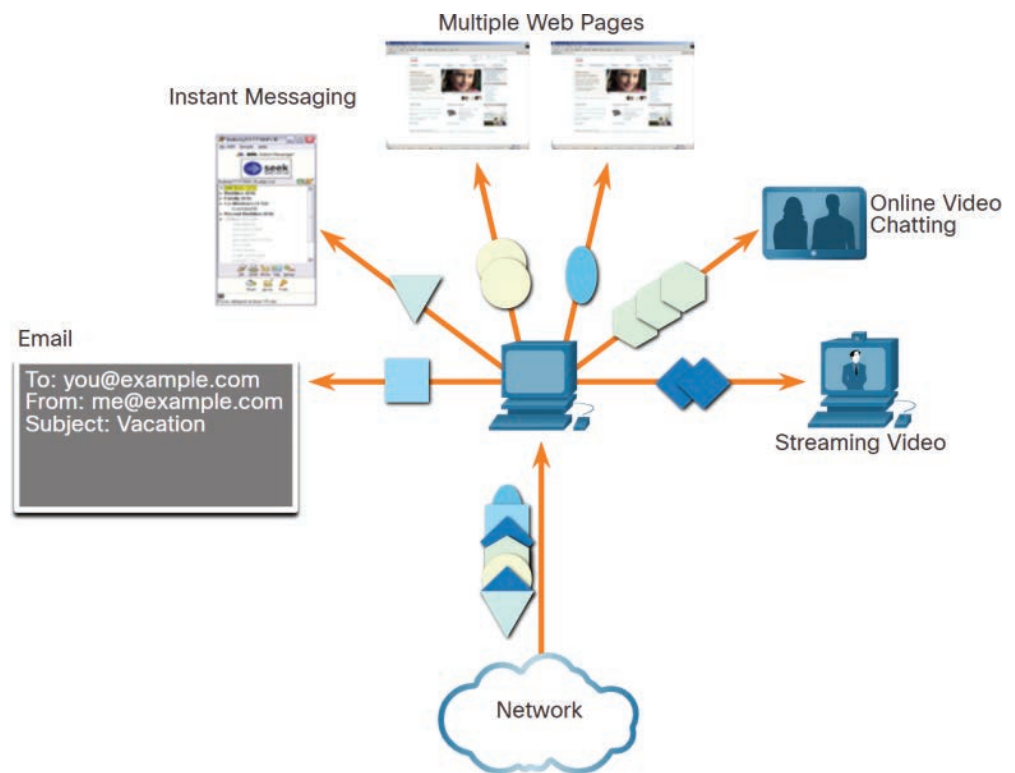


Figure 14-4 Adding Header Information

The transport layer ensures that even with multiple applications running on a device, all applications receive the correct data.

The transport layer must be able to separate and manage multiple communications with different transport requirement needs. To pass data streams to the proper applications, the transport layer identifies the target application by using an identifier called a *port number*. As illustrated in Figure 14-5, each software process that needs to access the network is assigned a port number unique to that host.

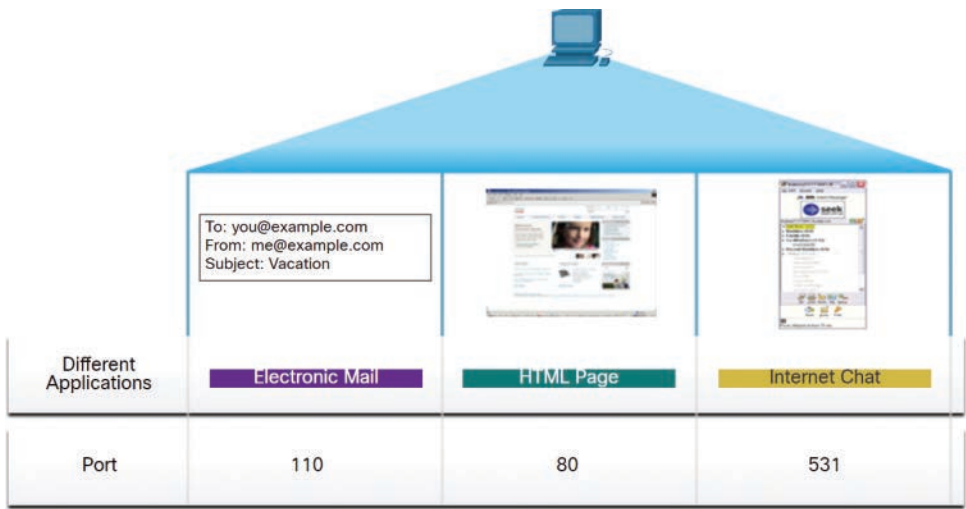


Figure 14-5 Identifying the Applications

Sending some types of data (for example, a streaming video) across a network as one complete communication stream can consume all the available bandwidth. This would prevent other communication conversations from occurring at the same time. It would also make error recovery and retransmission of damaged data difficult.

As shown in Figure 14-6, the transport layer uses segmentation and multiplexing to enable different communication conversations to be interleaved on the same network. To prevent problems, error checking can be performed on the data in a segment to determine whether the segment was altered during transmission.

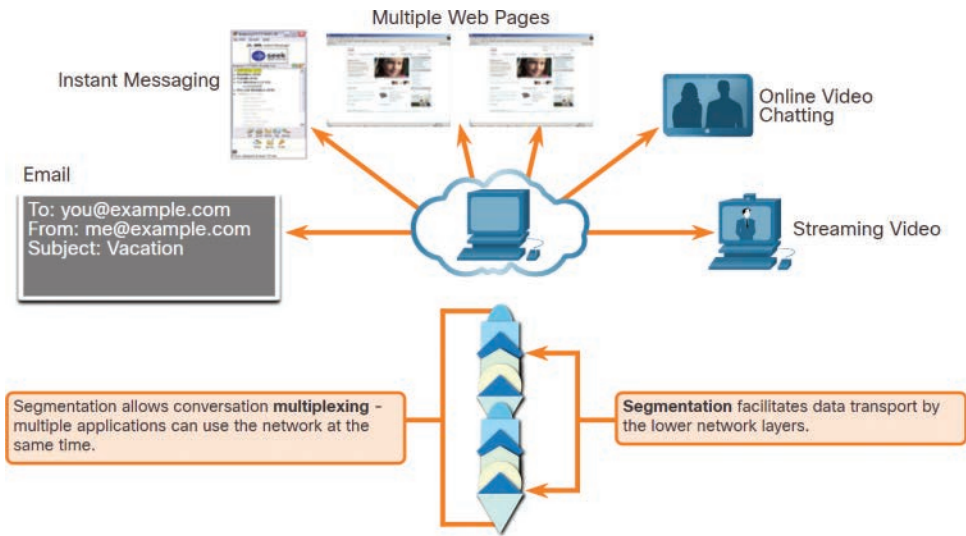


Figure 14-6 Conversation Multiplexing

Transport Layer Protocols (14.1.3)

IP is concerned only with the structure, addressing, and routing of packets. IP does not specify how the delivery or transportation of the packets takes place.

Transport layer protocols specify how to transfer messages between hosts and are responsible for managing the reliability requirements of a conversation. The transport layer includes the TCP and UDP protocols.

Different applications have different transport reliability requirements. Therefore, TCP/IP provides two transport layer protocols, as shown in Figure 14-7.

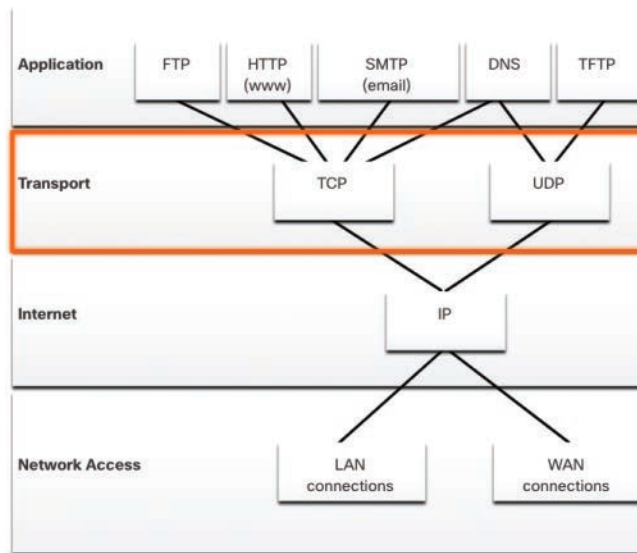


Figure 14-7 Transport Layer Protocols

Transmission Control Protocol (TCP) (14.1.4)

IP is concerned only with the structure, addressing, and routing of packets, from original sender to final destination. IP is not responsible for guaranteeing delivery or determining whether a connection between the sender and receiver needs to be established.

TCP is considered a reliable, full-featured transport layer protocol, and it ensures that all of the data arrives at the destination. TCP includes fields that ensure the delivery of the application data. These fields require additional processing by the sending and receiving hosts.

Note

TCP divides data into segments.

TCP transport is analogous to the process of sending packages that are tracked from source to destination. If a shipping order is broken up into several packages, a customer can check online to see the order in which the packages will be delivered.

TCP provides reliability and flow control using these basic operations:

- It numbers and tracks data segments transmitted to a specific host from a specific application.
- It acknowledges received data.
- It retransmits any unacknowledged data after a certain amount of time.
- It sequences data that might arrive in the wrong order.
- It sends data at an efficient rate that is acceptable to the receiver.

In order to maintain the state of a conversation and track the information, TCP must first establish a connection between the sender and the receiver. This is why TCP is known as a connection-oriented protocol.

Interactive
Graphic

Go to the online course to view an animation of TCP segments and acknowledgments being transmitted between sender and receiver.

User Datagram Protocol (UDP) (14.1.5)

UDP is a simpler transport layer protocol than TCP. It does not provide reliability and flow control, which means it requires fewer header fields. Because the sender and the receiver UDP processes do not have to manage reliability and flow control, UDP datagrams can be processed faster than TCP segments. UDP provides the basic functions for delivering datagrams between the appropriate applications, with very little overhead and data checking.

Note

UDP divides data into datagrams, which are also referred to as segments.

UDP is a connectionless protocol. Because UDP does not provide reliability or flow control, it does not require an established connection. Because UDP does not track information sent or received between the client and server, UDP is also known as a *stateless* protocol.

UDP is also known as a *best-effort* delivery protocol because it provides no acknowledgment that the data is received at the destination. With UDP, there are no transport layer processes that inform the sender of a successful delivery.

The UDP process is like the process of placing a regular, nonregistered, letter in the mail. The sender of the letter is not aware of the availability of the receiver to receive

the letter. Also, the post office is not responsible for tracking the letter or informing the sender if the letter does not arrive at the final destination.

**Interactive
Graphic**

Go to the online course to view an animation of UDP segments being transmitted from sender to receiver.

The Right Transport Layer Protocol for the Right Application (14.1.6)

Some applications can tolerate some data loss during transmission over the network, but they cannot accept delays in transmission. For these applications, UDP is a better choice than TCP because it requires less network overhead. UDP is preferable for applications such as voice over IP (VoIP). With VoIP, acknowledgments and retransmissions would slow down delivery and make a voice conversation unacceptable.

UDP is also used by request-and-reply applications where the data is minimal, and retransmission can be done quickly. For example, Domain Name System (DNS) uses UDP for this type of transaction. The client requests IPv4 and IPv6 addresses for a known domain name from a DNS server. If the client does not receive a response in a predetermined amount of time, it simply sends the request again.

For example, if one or two segments of a live video stream fail to arrive, there is a momentary disruption in the stream. This may appear as distortion in the image or sound, but it may not be noticeable to the user. If the destination device had to account for lost data, the stream could be delayed while the device waited for retransmissions, and this would cause the image or sound to be greatly degraded. In this case, it is better to render the best media possible with the segments received and forgo reliability.

For other applications, it is important that all the data arrives and that it can be processed in its proper sequence. For these types of applications, TCP is used as the transport protocol. For example, applications such as databases, web browsers, and email clients require that all data that is sent arrives at the destination in its original condition. Any missing data could corrupt a communication, making it either incomplete or unreadable. For example, it is important when accessing banking information over the web to make sure all the information is sent and received correctly.

Application developers must choose which transport protocol type is appropriate based on the requirements of the applications. Video may be sent over TCP or UDP. Applications that stream stored audio and video typically use TCP. In such a case, the application uses TCP to perform buffering, bandwidth probing, and congestion control in order to provide a better user experience.

Real-time video and voice usually use UDP, but they may also use TCP or both UDP and TCP. A video conferencing application may use UDP by default, but because many firewalls block UDP, the application may also be sent over TCP.

Applications that stream stored audio and video use TCP. For example, if your network suddenly cannot support the bandwidth needed to watch an on-demand movie, the application pauses the playback. During the pause, you might see a “buffering...” message while TCP works to reestablish the stream. When all the segments are in order and a minimum level of bandwidth is restored, your TCP session resumes, and the movie begins playing again.

Figure 14-8 summarizes the differences between UDP and TCP.

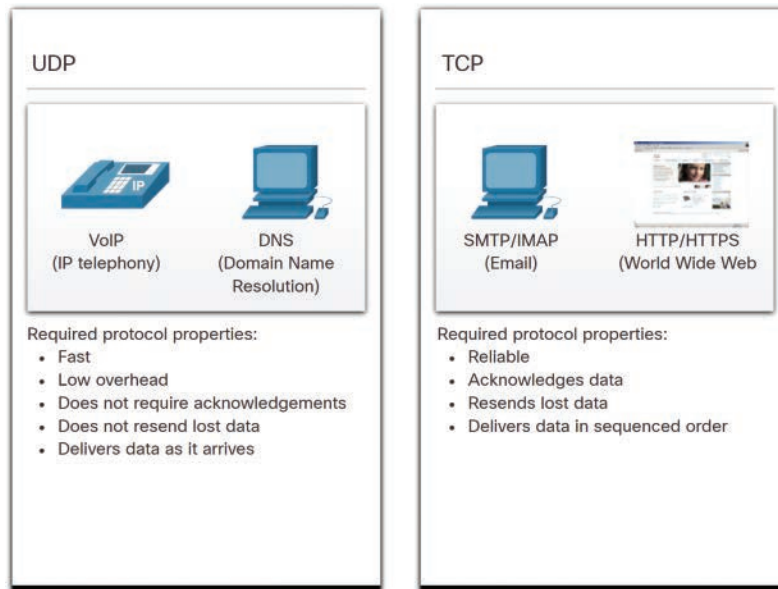


Figure 14-8 Properties of UDP and TCP

**Interactive
Graphic**

Check Your Understanding—Transportation of Data (14.1.7)

Refer to the online course to complete this activity.

TCP Overview (14.2)

TCP and UDP are transport layer protocols, and it is up to a developer to determine which of these protocols best matches the requirements of the application being developed. TCP establishes a connection that provides reliability and flow control.

TCP Features (14.2.1)

In the previous section, you learned that TCP and UDP are the two transport layer protocols. This section gives more details about what TCP does and when it is a good idea to use it instead of UDP.

To understand the differences between TCP and UDP, it is important to understand how each protocol implements specific reliability features and how each protocol tracks conversations.

In addition to supporting the basic functions of data segmentation and reassembly, TCP also provides the following services:

- **Establishes a session:** TCP is a *connection-oriented protocol* that negotiates and establishes a permanent connection (or session) between source and destination devices prior to forwarding any traffic. Through session establishment, the devices negotiate the amount of traffic that can be forwarded at a given time, and the communication data between the two can be closely managed.
- **Ensures reliable delivery:** As a segment is transmitted over a network, it might for many reasons become corrupted or lost completely. TCP ensures that each segment that is sent by the source arrives at the destination.
- **Provides same-order delivery:** Because networks may provide multiple routes that can have different transmission rates, data can arrive in the wrong order. By numbering and sequencing the segments, TCP ensures that segments are reassembled in the proper order.
- **Supports flow control:** Network hosts have limited resources (that is, memory and processing power). When TCP is aware that these resources are overtaxed, it can request that the sending application reduce the rate of data flow. TCP does this by regulating the amount of data the source transmits. This process, called *flow control*, can prevent the need for retransmission of the data when the resources of the receiving host are overwhelmed.

For more information on TCP, search the internet for RFC 793.

TCP Header (14.2.2)

TCP is a *stateful* protocol, which means it keeps track of the state of a communication session. To track the state of a session, TCP records which information it has sent and which information has been acknowledged. The stateful session begins with the session establishment and ends with the session termination.

A TCP segment adds 20 bytes (that is, 160 bits) of overhead when encapsulating the application layer data. Figure 14-9 shows the fields in a TCP header.

TCP Header Fields (14.2.3)

Table 14-1 describes the fields in the TCP header.

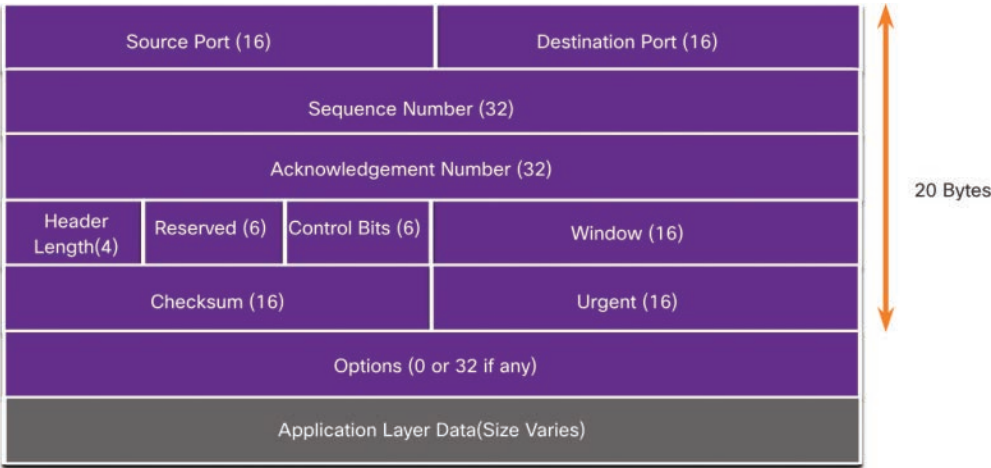


Figure 14-9 Fields of the TCP Header

Table 14-1 Details of the TCP Header Fields

TCP Header Field	Description
Source Port	A 16-bit field used to identify the source application by port number
Destination Port	A 16-bit field used to identify the destination application by port number
Sequence Number	A 32-bit field used for data reassembly purposes
Acknowledgment Number	A 32-bit field used to indicate that data has been received and the next byte expected from the source
Header Length	A 4-bit field known as <i>data offset</i> that indicates the length of the TCP segment header
Reserved	A 6-bit field that is reserved for future use
Control Bits	A 6-bit field used that includes bit codes, or flags, which indicate the purpose and function of the TCP segment
Window Size	A 16-bit field used to indicate the number of bytes that can be accepted at one time
Checksum	A 16-bit field used for error checking of the segment header and data
Urgent	A 16-bit field used to indicate whether the contained data is urgent

Applications That Use TCP (14.2.4)

TCP provides a good example of how the different layers of the TCP/IP protocol suite have specific roles. TCP handles all tasks associated with dividing the data

stream into segments, providing reliability, controlling data flow, and reordering segments. TCP frees the application from having to manage any of these tasks. Applications, like those shown in Figure 14-10, can simply send the data stream to the transport layer and use the services of TCP.

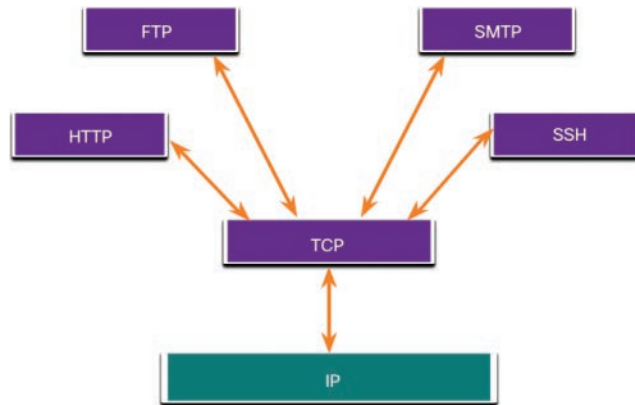


Figure 14-10 Applications That Use TCP

**Interactive
Graphic**

Check Your Understanding—TCP Overview (14.2.5)

Refer to the online course to complete this activity.

UDP Overview (14.3)

The reliability and flow control features provided by TCP come with additional overhead related to connection establishment and tracking whether or not segments were received. When such overhead creates unnecessary delay, it is time to use UDP, which is a simpler transport layer protocol than TCP. UDP is used by many applications and protocols, including delay-sensitive VoIP applications and simple reply and request protocols (that is, DNS and DHCP).

UDP Features (14.3.1)

This section covers UDP, including what it does and when it is a good idea to use it instead of TCP. UDP is a best-effort transport protocol. It is a lightweight transport protocol that offers the same data segmentation and reassembly as TCP, but without TCP's reliability and flow control. UDP is such a simple protocol that it is usually described in terms of what it does not do compared to TCP.

UDP's features include the following:

- Data is reconstructed in the order in which it is received.
- Any segments that are lost are not re-sent.

- There is no session establishment.
- The sender is not informed about resource availability.

For more information on UDP, search the internet for UDP RFCs.

UDP Header (14.3.2)

UDP is a stateless protocol, which means neither the client nor the server tracks the state of a communication session. If reliability is required when using UDP as the transport protocol, it must be handled by the application.

One of the most important requirements for delivering live video and voice over a network is that the data must continue to flow quickly. Live video and voice applications can tolerate some data loss with minimal or no noticeable effect; they are perfectly suited to UDP.

The blocks of communication in UDP are called *datagrams*, or *segments*. The transport layer protocol sends datagrams in a best-effort manner.

The UDP header is far simpler than the TCP header because it has only four fields and requires 8 bytes (that is, 64 bits). Figure 14-11 shows the fields in a UDP header.

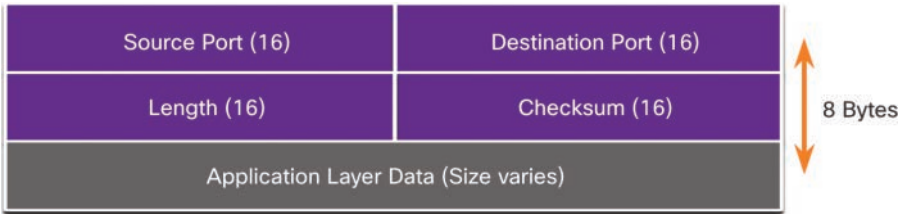


Figure 14-11 Fields of the UDP Header

UDP Header Fields (14.3.3)

Table 14-2 describes the fields in the UDP header.

Table 14-2 Details of the UDP Header Fields

UDP Header Field	Description
Source Port	A 16-bit field used to identify the source application by port number
Destination Port	A 16-bit field used to identify the destination application by port number
Length	A 16-bit field that indicates the length of the UDP datagram header
Checksum	A 16-bit field used for error checking of the datagram header and data

Applications that use UDP (14.3.4)

Three types of applications are best suited for UDP:

- **Live video and multimedia applications:** These applications can tolerate some data loss but require little or no delay. Examples include VoIP and live streaming video.
- **Simple request and reply applications:** Applications with simple transactions where a host sends a request and may or may not receive a reply. Examples include DNS and DHCP.
- **Applications that handle reliability themselves:** Unidirectional communications where flow control, error detection, acknowledgments, and error recovery are not required or can be handled by the application. Examples include SNMP and TFTP.

Figure 14-12 identifies applications that use UDP.

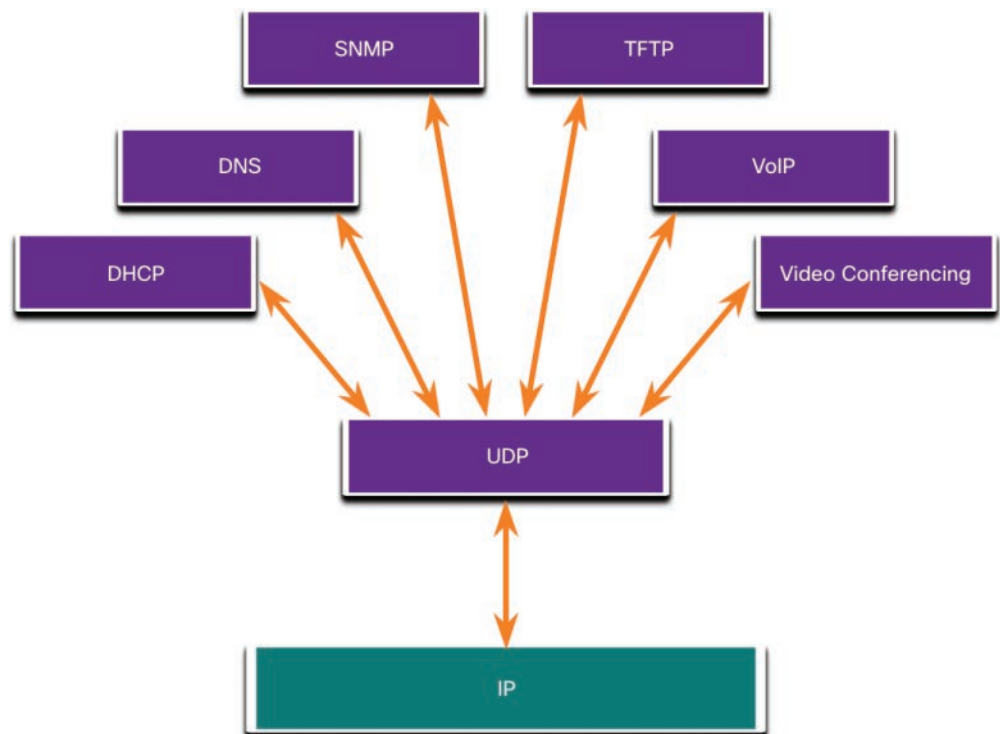


Figure 14-12 Applications That Use UDP

Although DNS and SNMP use UDP by default, they can both also use TCP. DNS uses TCP if the DNS request or DNS response is more than 512 bytes, such as when

a DNS response includes many name resolutions. Similarly, under some circumstances, a network administrator may want to configure SNMP to use TCP.

**Interactive
Graphic****Check Your Understanding—UDP Overview (14.3.5)**

Refer to the online course to complete this activity.

Port Numbers (14.4)

This section covers how both TCP and UDP use port numbers to identify the proper application layer process.

Multiple Separate Communications (14.4.1)

As you have learned, there are some situations in which TCP is the right protocol for the job, and there are other situations in which UDP should be used. No matter what type of data is being transported, the TCP and UDP transport layer protocols use *port numbers* to manage multiple, simultaneous conversations. As shown in Figure 14-13, the TCP and UDP header fields identify a source application port number and destination application port number.



Figure 14-13 Source and Destination Port Fields

The source port number is associated with the originating application on the local host, whereas the destination port number is associated with the destination application on the remote host.

For instance, say that a host is initiating a web page request from a web server. When the host initiates the web page request, the source port number is dynamically generated by the host to uniquely identify the conversation. Each request generated by a host uses a different dynamically created source port number. This process allows multiple conversations to occur simultaneously.

In the request, the destination port number is what identifies the type of service being requested of the destination web server. For example, when a client specifies port 80 in the destination port, the server that receives the message knows that web services are being requested.

A server can offer more than one service simultaneously, such as web services on port 80 and File Transfer Protocol (FTP) connection establishment on port 21.

Socket Pairs (14.4.2)

The source and destination ports are placed within a segment. The segments are then encapsulated within an IP packet. The IP packet contains the IP addresses of the source and destination. The combination of the source IP address and source port number or the destination IP address and destination port number is known as a *socket*.

In the example in Figure 14-14, the PC is simultaneously requesting FTP and web services from the destination server.

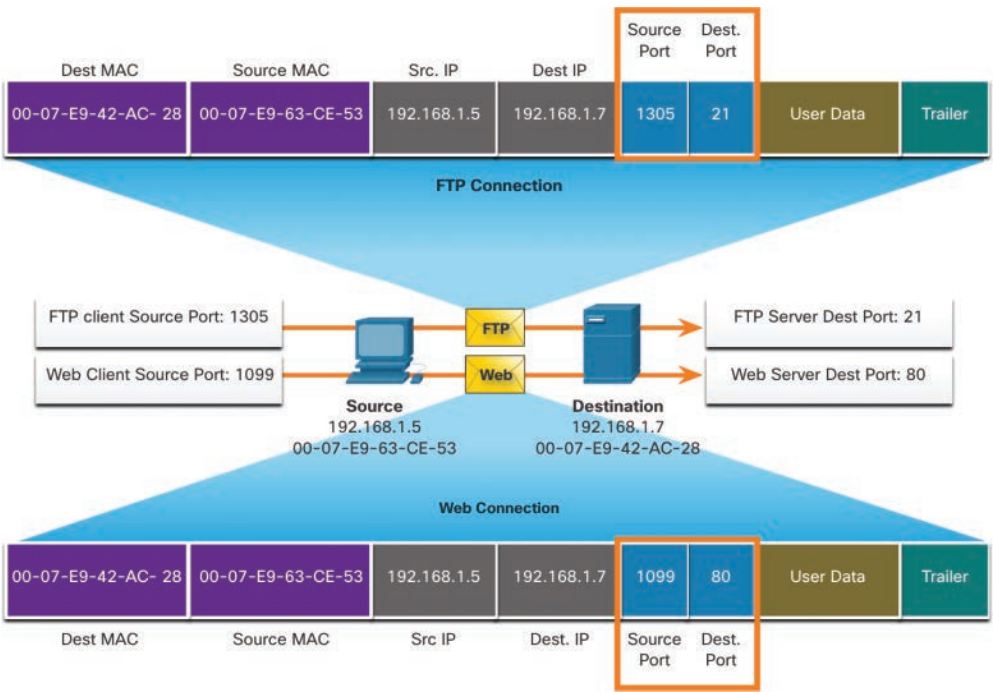


Figure 14-14 Host Sending Multiple Simultaneous Communications

In the example, the FTP request generated by the PC includes the Layer 2 MAC addresses and the Layer 3 IP addresses. The request also identifies the source port number 1305 (which is dynamically generated by the host) and the destination port number 21, for the FTP services. The host also has requested a web page from the server by using the same Layer 2 and Layer 3 addresses. However, it is using the source port number 1099 (which is dynamically generated by the host) and the destination port number 80 for web services.

The socket is used to identify the server and service being requested by the client. A client socket might look like this, with 1099 representing the source port number: 192.168.1.5:1099. The socket on a web server might be 192.168.1.7:80. Together, these two sockets combine to form a *socket pair*: 192.168.1.5:1099, 192.168.1.7:80.

Sockets enable multiple processes running on a client to distinguish themselves from each other; they also make it possible for multiple connections to a server process to be distinguished from each other.

The source port number acts as a return address for the requesting application. The transport layer keeps track of this port and the application that initiated the request so that when a response is returned, it can be forwarded to the correct application.

Port Number Groups (14.4.3)

The Internet Assigned Numbers Authority (IANA) is the standards organization responsible for assigning various addressing standards, including the 16-bit port numbers. The 16 bits used to identify the source and destination port numbers provide a range of ports from 0 through 65535.

IANA has divided the range of numbers into the three port groups shown in Table 14-3.

Table 14-3 Details of Port Number Groups

Port Group	Number Range	Description
Well-known ports	0 to 1023	These port numbers are reserved for common or popular services and applications such as web browsers, email clients, and remote access clients.
		Defined well-known ports for common server applications enable clients to easily identify the associated service required.
Registered ports	1024 to 49,151	These port numbers are assigned by IANA to a requesting entity to use with specific processes or applications.
		These processes are primarily individual applications that a user has chosen to install rather than common applications that would receive well-known port numbers. For example, Cisco has registered port 1812 for its RADIUS server authentication process.
Private and/or dynamic ports	49,152 to 65,535	These ports are also known as <i>ephemeral ports</i> .
		The client's OS usually assigns port numbers dynamically when a connection to a service is initiated.
		The dynamic port is then used to identify the client application during communication.

Note

Some client operating systems may use registered port numbers instead of dynamic port numbers for assigning source ports.

Table 14-4 displays some common well-known port numbers and their associated applications.

Table 14-4 Well-Known Port Numbers

Port Number	Protocol	Application
20	TCP	File Transfer Protocol (FTP)—data
21	TCP	File Transfer Protocol (FTP)—control
22	TCP	Secure Shell (SSH)
23	TCP	Telnet
25	TCP	Simple Mail Transfer Protocol (SMTP)
53	UDP, TCP	Domain Name Service (DNS)
67	UDP	Dynamic Host Configuration Protocol (DHCP)—server
68	UDP	Dynamic Host Configuration Protocol (DHCP)—client
69	UDP	Trivial File Transfer Protocol (TFTP)
80	TCP	Hypertext Transfer Protocol (HTTP)
110	TCP	Post Office Protocol version 3 (POP3)
143	TCP	Internet Message Access Protocol (IMAP)
161	UDP	Simple Network Management Protocol (SNMP)
443	TCP	Hypertext Transfer Protocol Secure (HTTPS)

Some applications can use both TCP and UDP. For example, DNS uses UDP when clients send requests to a DNS server. However, communication between two DNS servers always uses TCP.

Search the IANA website for *port registry* to view the full list of port numbers and associated applications.

The netstat Command (14.4.4)

Unexplained TCP connections can pose major security threats. They can indicate that something or someone is connected to the local host. Sometimes it is necessary to know which active TCP connections are open and running on a networked host.

netstat is an important network utility that can be used to verify such connections. As shown in Example 14-1, the output of the command **netstat** lists the protocols in use, their local addresses and port numbers, their foreign addresses and port numbers, and their connection states.

Example 14-1 The **netstat** Command on a Windows Host

C:\> netstat				
Active Connections				
Proto	Local Address	Foreign Address	State	
TCP	192.168.1.124:3126	192.168.0.2:netbios-ssn	ESTABLISHED	
TCP	192.168.1.124:3158	207.138.126.152:http	ESTABLISHED	
TCP	192.168.1.124:3159	207.138.126.169:http	ESTABLISHED	
TCP	192.168.1.124:3160	207.138.126.169:http	ESTABLISHED	
TCP	192.168.1.124:3161	sc.msn.com:http	ESTABLISHED	
TCP	192.168.1.124:3166	www.cisco.com:http	ESTABLISHED	

By default, the **netstat** command attempts to resolve IP addresses to domain names and port numbers to well-known applications. The **-n** option can be used to display IP addresses and port numbers in their numerical form.

Interactive
Graphic

Check Your Understanding—Port Numbers (14.4.5)

Refer to the online course to complete this activity.

TCP Communication Process (14.5)

TCP is considered a stateful protocol because it establishes a session between the source and the destination and keeps track of the data within that session. This section covers how TCP establishes such a connection to ensure reliability and flow control.

TCP Server Processes (14.5.1)

You already know the fundamentals of TCP. Understanding the role of port numbers will help you to grasp the details of the TCP communication process. In this section, you will also learn about the TCP three-way handshake and session termination processes.

Each application process running on a server is configured to use a port number. The port number is either automatically assigned or configured manually by a system administrator.

An individual server cannot have two services assigned to the same port number within the same transport layer services. For example, a host running a web server application and a file transfer application cannot have both applications configured to use the same port, such as TCP port 80.

An active server application assigned to a specific port is considered open, which means that the transport layer accepts and processes segments addressed to that port. Any incoming client request addressed to the correct socket is accepted, and the data is passed to the server application. There can be many ports open simultaneously on a server, one for each active server application.

Let's look at the TCP server processes. In Figure 14-15, Client 1 is requesting web services, and Client 2 is requesting email services from the same sever.

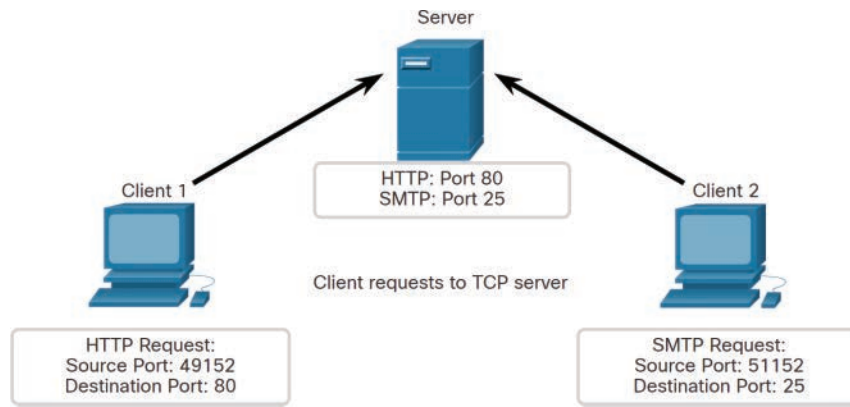


Figure 14-15 Clients Sending TCP Requests

In Figure 14-16, Client 1 is requesting web services using well-known destination port 80 (HTTP), and Client 2 is requesting email services using well-known port 25 (SMTP).

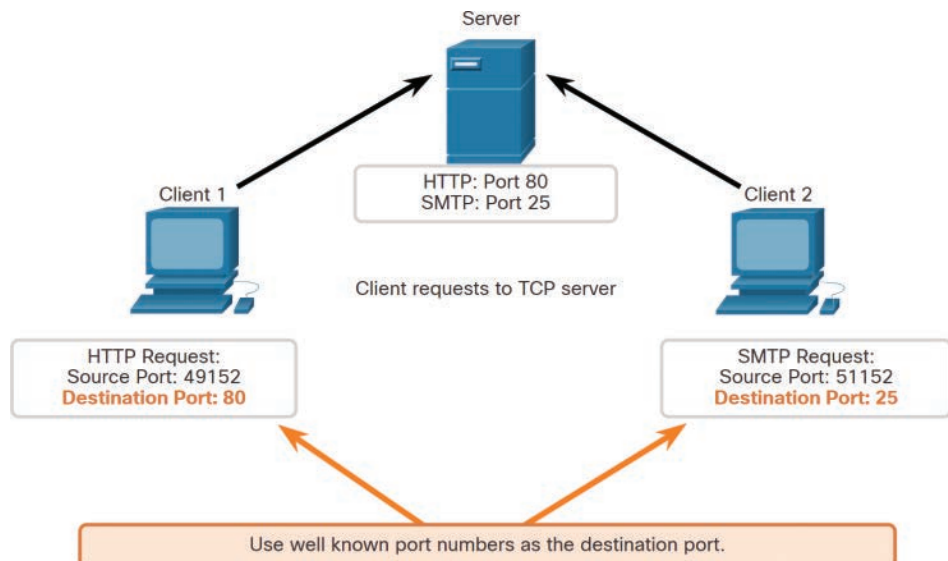


Figure 14-16 Requesting Destination Ports

A client request dynamically generates a source port number. In Figure 14-17, Client 1 is using source port 49152, and Client 2 is using source port 51152.

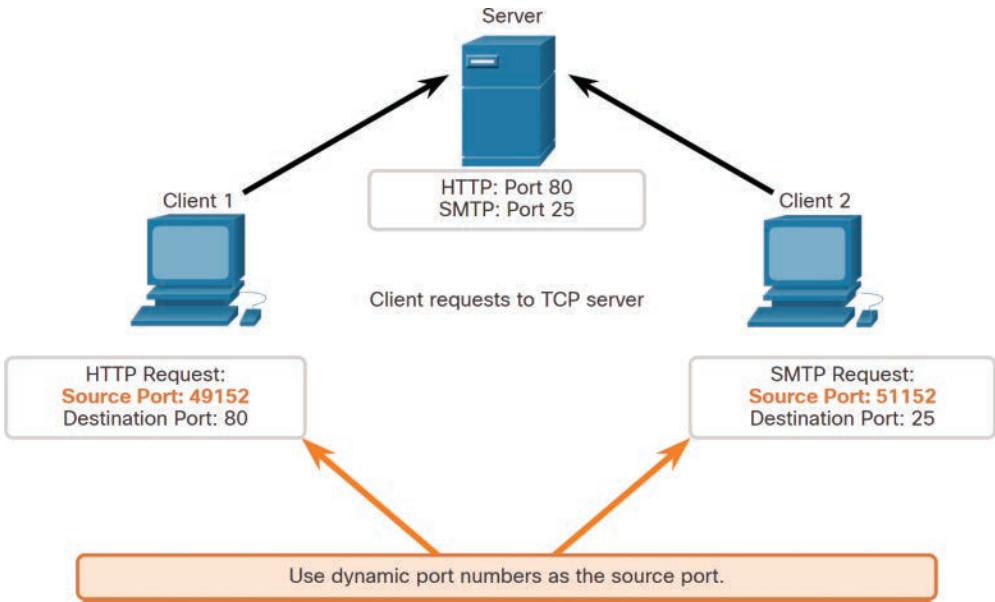


Figure 14-17 Requesting Source Ports

When the server responds to the client requests, it reverses the destination and source ports of the initial request, as shown in Figures 14-18 and 14-19. Notice in Figure 14-18 that the server response to the web request now has destination port 49152, and the email response now has destination port 51152.

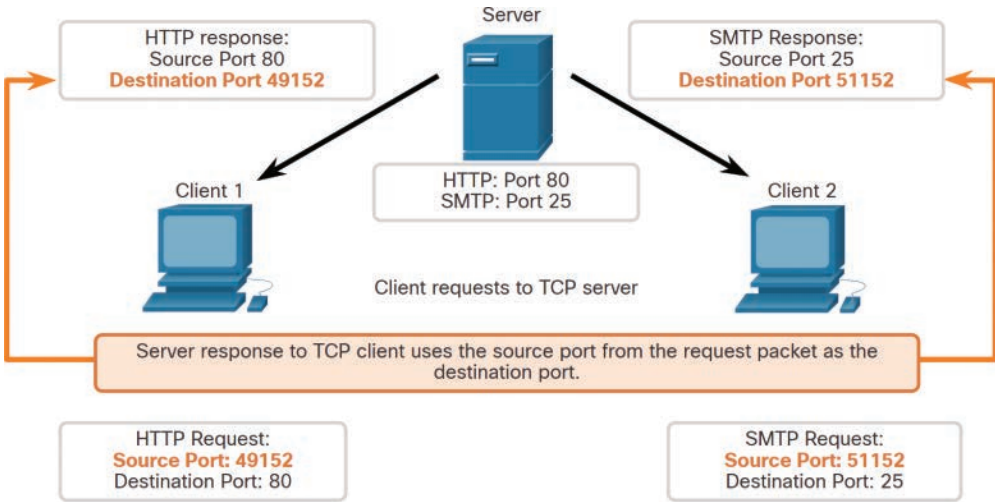


Figure 14-18 Response Destination Ports

The source port in the server response is the original destination port in the initial requests, as shown in Figure 14-19.

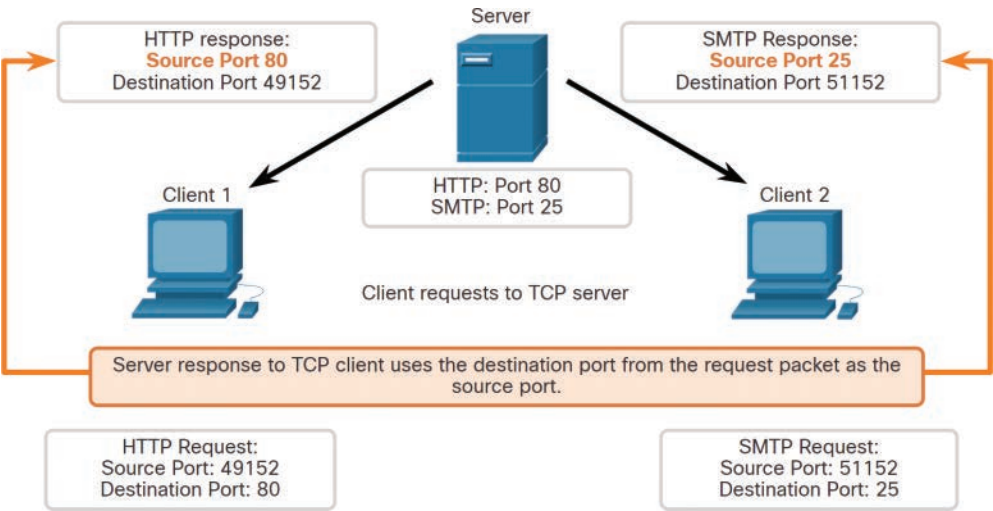


Figure 14-19 Response Source Ports

TCP Connection Establishment (14.5.2)

In some cultures, when two persons meet, they greet each other by shaking hands. Both parties understand the act of shaking hands as a signal for a friendly greeting. Connections on a network are similar. In TCP connections, the host client establishes a connection with a server by using the *three-way handshake* process.

Figure 14-20 shows the steps in the TCP connection establishment process:

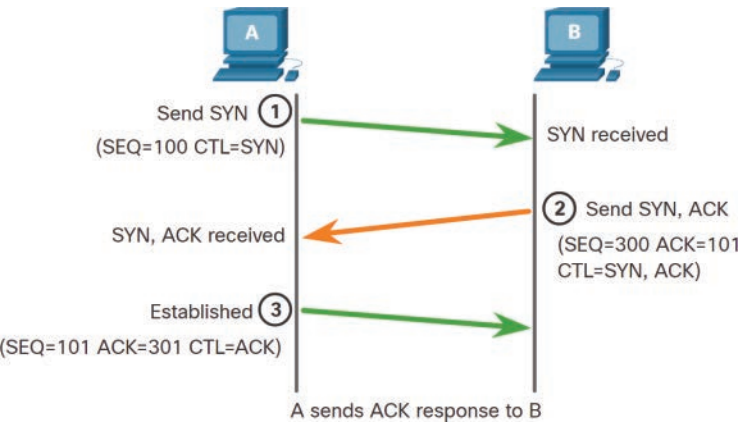


Figure 14-20 Steps in the TCP Connection Establishment Process

- Step 1.** SYN: The initiating client requests a client-to-server communication session with the server.
- Step 2.** ACK and SYN: The server acknowledges the client-to-server communication session and requests a server-to-client communication session.
- Step 3.** ACK: The initiating client acknowledges the server-to-client communication session.

The three-way handshake validates that the destination host is available to communicate. In this example, Host A has validated that Host B is available.

Session Termination (14.5.3)

To close a connection, the Finish (FIN) control flag must be set in the segment header. To end each one-way TCP session, a two-way handshake, consisting of a FIN segment and an Acknowledgment (ACK) segment, is used. Therefore, to terminate a single conversation supported by TCP, four exchanges are needed to end both sessions. Either the client or the server can initiate the termination.

In the following example, the terms *client* and *server* are used as a reference for simplicity, but any two hosts that have an open session can initiate the termination process.

Figure 14-21 shows the steps in the TCP session termination process:

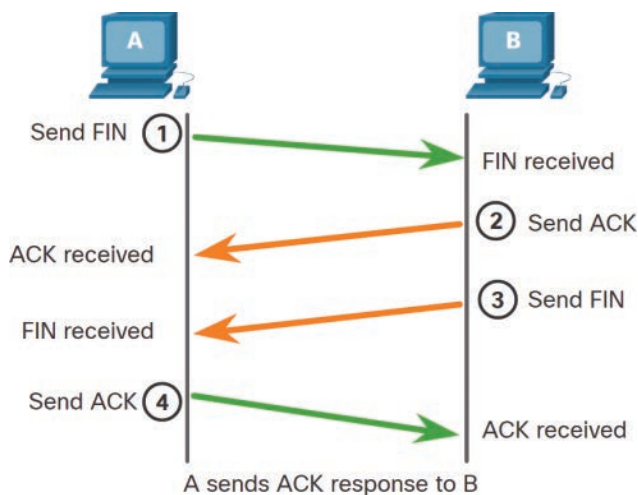


Figure 14-21 Steps in the TCP Session Termination Process

- Step 1.** FIN: When the client has no more data to send in the stream, it sends a segment with the FIN flag set.
- Step 2.** ACK: The server sends an ACK to acknowledge the receipt of the FIN to terminate the session from client to server.

- Step 3. FIN:** The server sends a FIN to the client to terminate the server-to-client session.
- Step 4. ACK:** The client responds with an ACK to acknowledge the FIN from the server.

When all segments have been acknowledged, the session is closed.

TCP Three-Way Handshake Analysis (14.5.4)

Hosts maintain state, track each data segment within a session, and exchange information about what data is received, using the information in the TCP header. TCP is a full-duplex protocol, where each connection represents two one-way communication sessions. To establish a connection, the hosts perform a three-way handshake. As shown in Figure 14-22, control bits in the TCP header indicate the progress and status of a connection.

These are the functions of the three-way handshake:

- It establishes that the destination device is present on the network.
- It verifies that the destination device has an active service and is accepting requests on the destination port number that the initiating client intends to use.
- It informs the destination device that the source client intends to establish a communication session on that port number.

After the communication is completed, the sessions are closed, and the connection is terminated. The connection and session mechanisms enable TCP reliability.

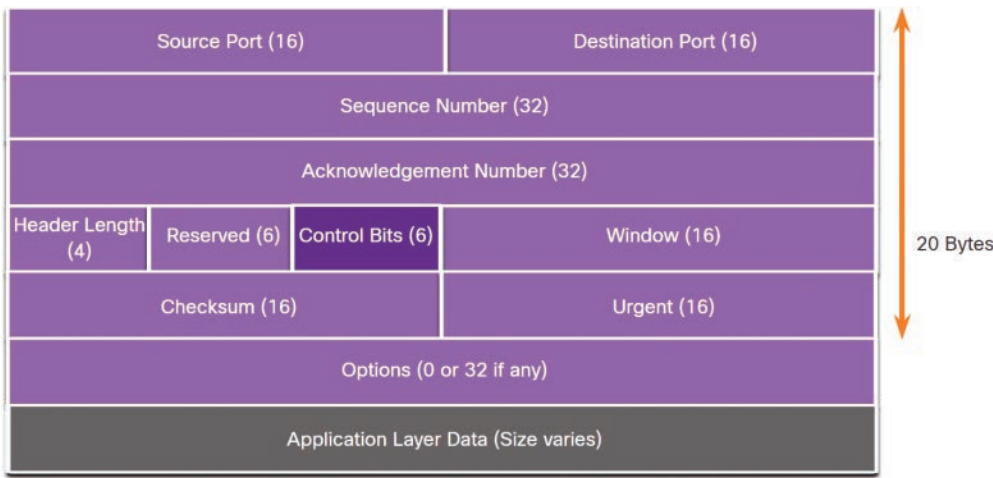


Figure 14-22 Control Bits Field

The 6 bits in the Control Bits field of the TCP segment header are also known as *flags*. A flag is a bit that is set to either on or off.

The six control bits flags are as follows:

- **URG:** Allows an application to process this data immediately.
- **ACK:** Acknowledgment flag used in connection establishment and session termination
- **PSH:** Push function
- **RST:** Flag used to reset the connection when an error or timeout occurs
- **SYN:** Flag used to synchronize sequence numbers used in connection establishment
- **FIN:** Flag indicating no more data from sender; used in session termination

Search the internet to learn more about the PSH and URG flags.

Video

Video—TCP 3-Way Handshake (14.5.5)

Refer to the online course to view this video.

Interactive Graphic

Check Your Understanding—TCP Communication Process (14.5.6)

Refer to the online course to complete this activity.

Reliability and Flow Control (14.6)

Reliability and flow control are two of the main features of TCP that are not present in UDP.

TCP Reliability—Guaranteed and Ordered Delivery (14.6.1)

The reason that TCP is the better protocol for some applications is because, unlike UDP, it re-sends dropped packets and numbers packets to indicate their proper order before delivery. TCP can also help maintain the flow of packets so that devices do not become overloaded. This section covers these features of TCP in detail.

There may be times when TCP segments do not arrive at their destination. Other times, TCP segments might arrive out of order. For the original message to be understood by the recipient, all the data must be received, and the data in the segments must be reassembled into the original order. Sequence numbers are assigned in the header of each packet to achieve this goal. The sequence number is the first data byte of a TCP segment.

During session setup, an *initial sequence number (ISN)* is set. This ISN represents the starting value of the bytes that are transmitted to the receiving application. As data is transmitted during the session, the sequence number is incremented by the number of bytes that have been transmitted. This data byte tracking enables each segment to be uniquely identified and acknowledged. Missing segments can then be identified.

The ISN does not begin at 1 but is effectively a random number. This prevents certain types of malicious attacks. For simplicity, we use an ISN of 1 for the examples in this chapter.

Segment sequence numbers indicate how to reassemble and reorder received segments, as shown in Figure 14-23.

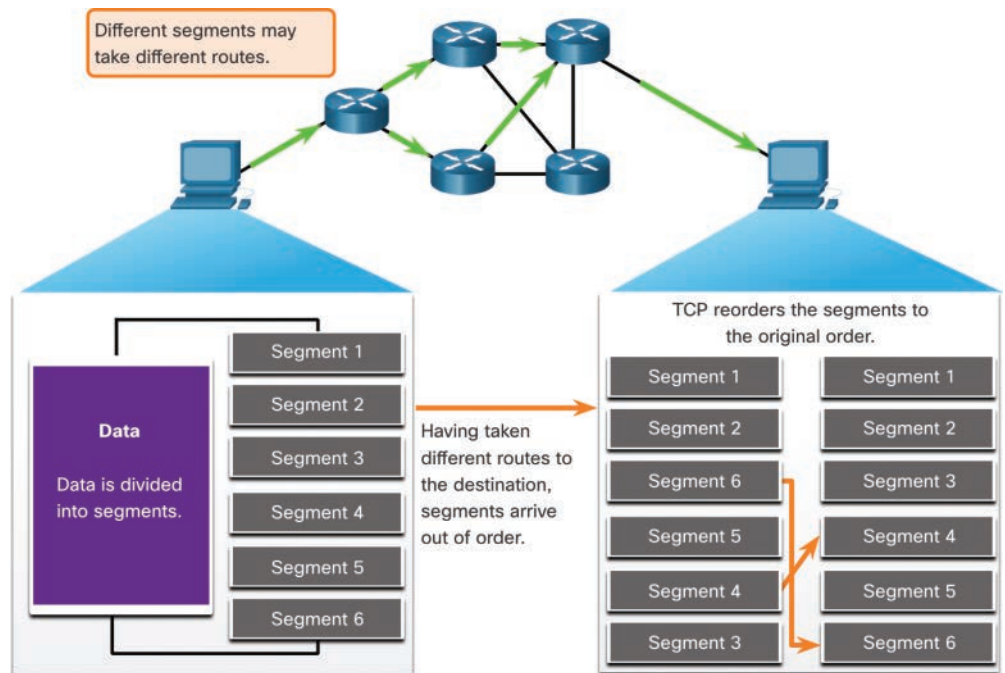


Figure 14-23 TCP Segments Are Reordered at the Destination

The receiving TCP process places the data from a segment into a receiving buffer. Segments are then placed in the proper sequence and are passed to the application layer when reassembled properly. Any segments that arrive with sequence numbers that are out of order are held for later processing. Then, when the segments with the missing bytes arrive, the segments are all processed in order.

TCP Reliability—Data Loss and Retransmission (14.6.3)

No matter how well designed a network is, data loss occasionally occurs. TCP provides methods of managing segment losses. Among these is a mechanism to retransmit segments for unacknowledged data.

The sequence (SEQ) number and acknowledgment (ACK) number are used together to confirm receipt of the bytes of data contained in the transmitted segments. The SEQ number identifies the first byte of data in the segment being transmitted. TCP uses the ACK number sent back to the source to indicate the next byte that the receiver expects to receive. This is called *expectational acknowledgment*.

Prior to later enhancements, TCP could only acknowledge the next byte expected. For example, in Figure 14-24, using segment numbers for simplicity, Host A sends segments 1 through 10 to Host B. If all the segments arrive except for segments 3 and 4, Host B replies with an acknowledgment specifying that the next segment expected is segment 3. Host A has no idea if any other segments arrived or not. Therefore, Host A re-sends segments 3 through 10. If all the re-sent segments arrive successfully, segments 5 through 10 are duplicates. This can lead to delays, congestion, and inefficiencies.

Note

For simplicity, segment numbers are used here instead of the byte numbers.

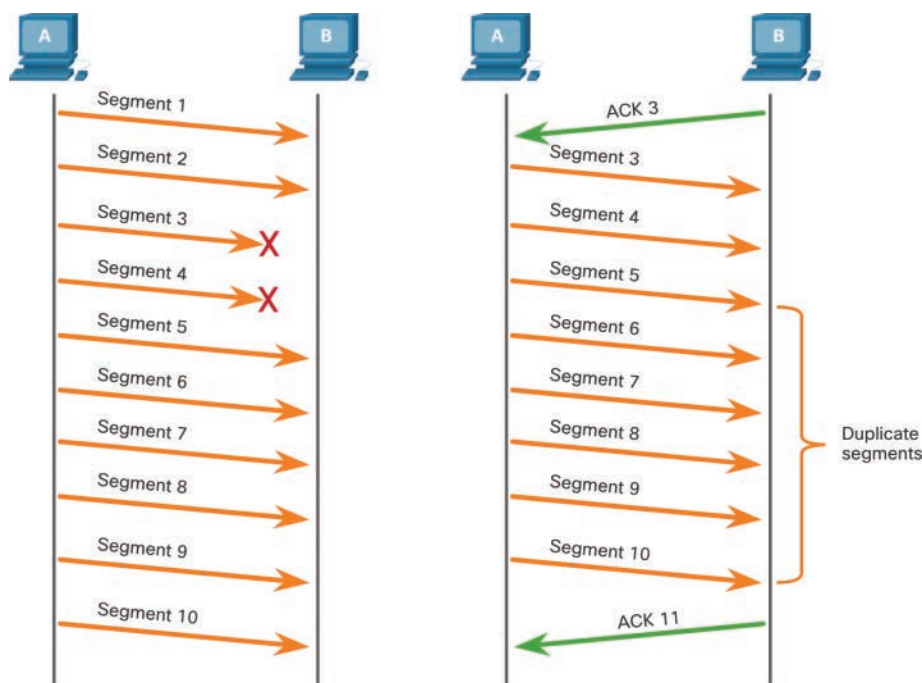


Figure 14-24 Data Retransmission

Host operating systems today typically employ an optional TCP feature called *selective acknowledgment (SACK)*, which is negotiated during the three-way handshake. If both hosts support SACK, the receiver can explicitly acknowledge which segments (bytes) were received, including any discontinuous segments. The sending host therefore needs to retransmit only the missing data. For example, in Figure 14-25, again using segment numbers for simplicity, Host A sends segments 1 through 10 to Host B. If all the segments arrive except for segments 3 and 4, Host B can acknowledge that it has received segments 1 and 2 (ACK 3) and selectively acknowledge segments 5 through 10 (SACK 5-10). Host A needs to re-send only segments 3 and 4.

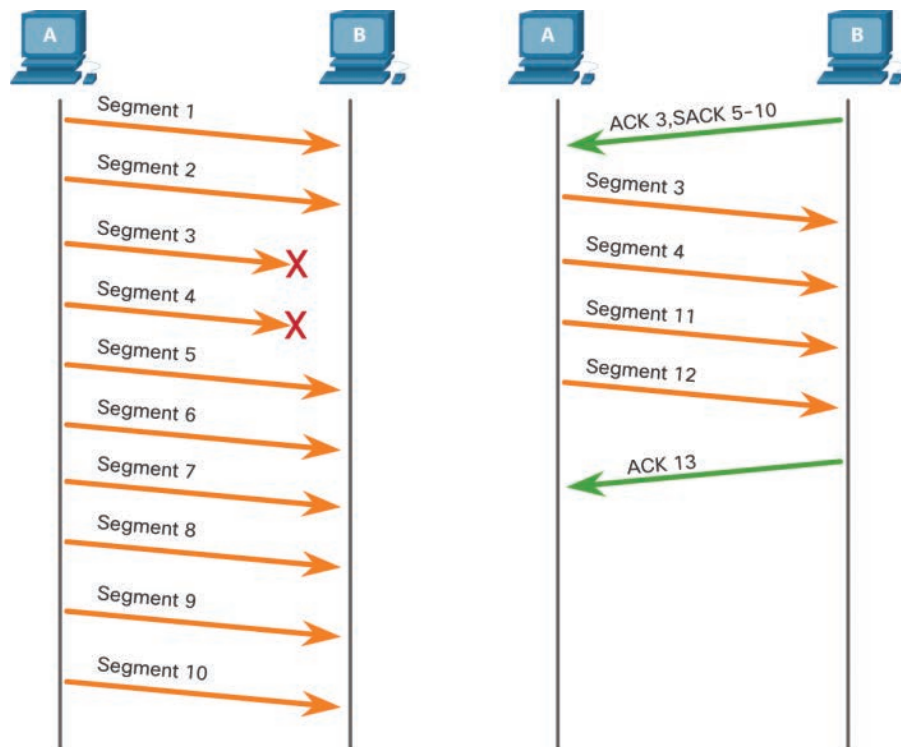


Figure 14-25 Selective Acknowledgment

Note

TCP typically sends ACKs for every other packet, but other factors beyond the scope of this section may alter this behavior.

TCP uses timers to know how long to wait before re-sending a segment.



Video—TCP Reliability—Data Loss and Retransmission (14.6.4)

Refer to the online course to view this video. Play the video and click the link to download the PDF file. The video and PDF file examine TCP data loss and retransmission.

TCP Flow Control—Window Size and Acknowledgments (14.6.5)

TCP provides mechanisms for flow control, which has to do with the amount of data the destination can receive and process reliably. Flow control helps maintain the reliability of TCP transmission by adjusting the rate of data flow between source and destination for a given session. To accomplish this, the TCP header includes a 16-bit field called the *window size*.

Figure 14-26 shows an example of window size and acknowledgments.

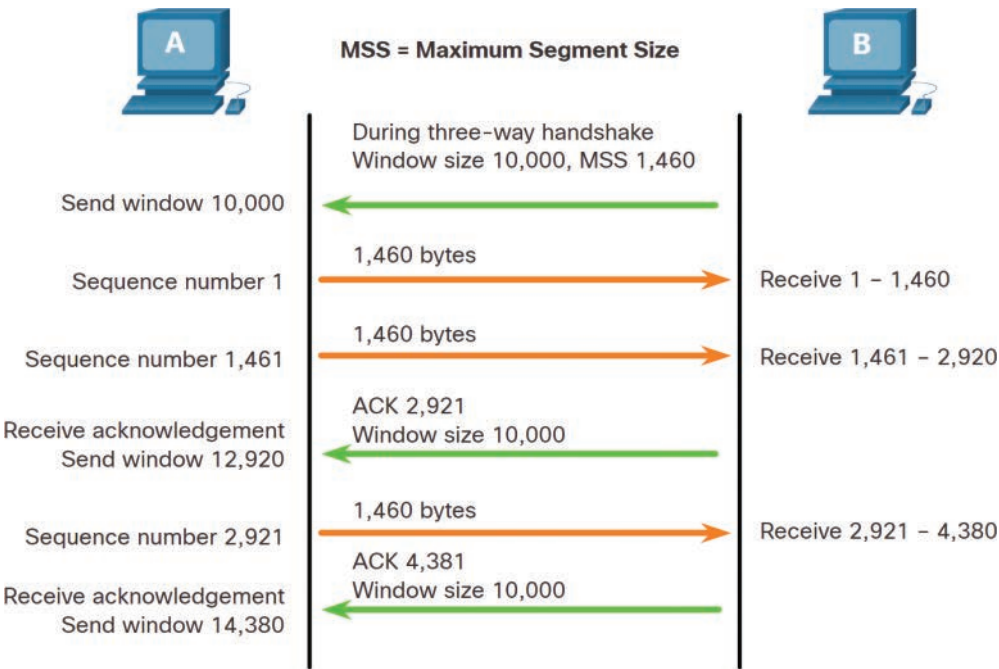


Figure 14-26 TCP Window Size Example

The window size determines the number of bytes that can be sent before an acknowledgment should be expected. The acknowledgment number is the number of the next expected byte.

The window size is the number of bytes that the destination device of a TCP session can accept and process at one time. In this example, the PC B initial window size for the

TCP session is 10,000 bytes. Starting with the first byte, byte number 1, the last byte PC A can send without receiving an acknowledgment is byte 10,000. This is known as the *send window* of PC A. The window size is included in every TCP segment, so the destination can modify the window size at any time, depending on buffer availability.

The initial window size is agreed upon when the TCP session is established during the three-way handshake. The source device must limit the number of bytes sent to the destination device, based on the window size of the destination. Only after the source device receives an acknowledgment that the bytes have been received can it continue sending more data for the session. Typically, the destination does not wait for all the bytes for its window size to be received before it replies with an acknowledgment. As the bytes are received and processed, the destination sends acknowledgments to inform the source that it can continue to send additional bytes.

For example, it is typical that PC B would not wait until all 10,000 bytes have been received before sending an acknowledgment. This means PC A can adjust its send window as it receives acknowledgments from PC B. As shown in Figure 14-26, when PC A receives an acknowledgment with the acknowledgment number 2921, which is the next expected byte, the PC A send window increments 2920 bytes. This changes the send window from 10,000 bytes to 12,920. PC A can now continue to send up to another 10,000 bytes to PC B, as long as it does not send more than its new send window of 12,920.

A destination sending acknowledgments as it processes bytes received and the continual adjustment of the source send window is known as *sliding window*. In the previous example, the send window of PC A increments, or slides over, another 2921 bytes, from 10,000 to 12,920.

If the availability of the destination's buffer space decreases, the destination may reduce its window size to inform the source to reduce the number of bytes it should send without receiving an acknowledgment.

Note

Devices today use the sliding window protocol. The receiver typically sends an acknowledgment after receiving every two segments. The number of segments received before acknowledgment occurs may vary, however. The advantage of the sliding window is that it allows the sender to continuously transmit segments, as long as the receiver is acknowledging previous segments. The details of the sliding window are beyond the scope of this book.

TCP Flow Control—Maximum Segment Size (MSS) (14.6.6)

In Figure 14-27, the source is transmitting 1460 bytes of data within each TCP segment. This is typically the maximum segment size (MSS) that the destination device

can receive. The MSS is part of the Options field in the TCP header, and it specifies the largest amount of data, in bytes, that a device can receive in a single TCP segment. The MSS does not include the TCP header. The MSS is typically included during the three-way handshake.

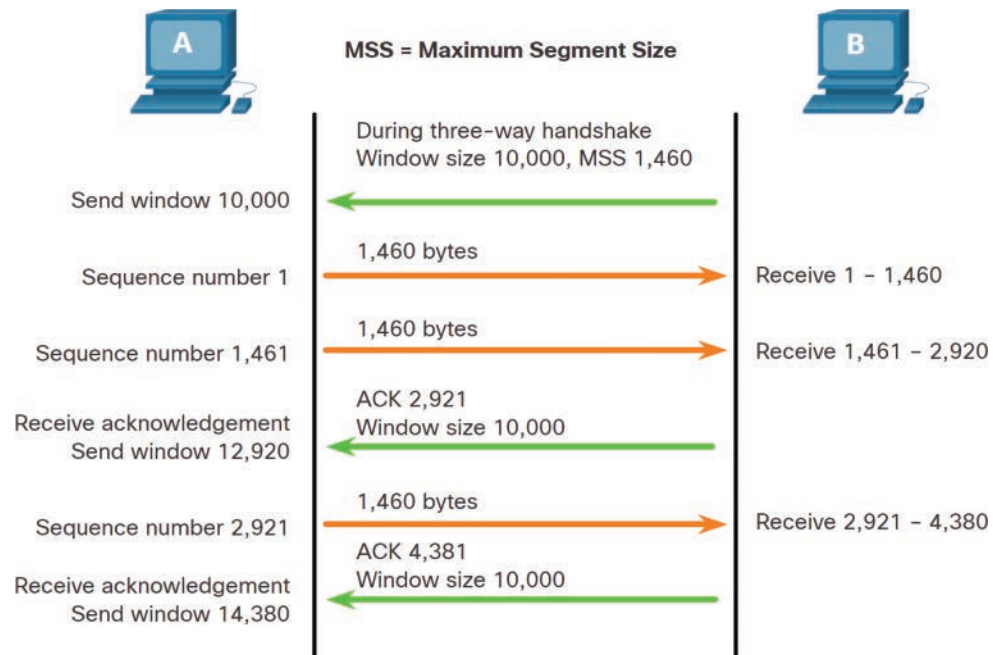


Figure 14-27 Maximum Segment Size

A common MSS when using IPv4 is 1460 bytes. A host determines the value of its MSS field by subtracting the IP and TCP headers from the Ethernet maximum transmission unit (MTU). On an Ethernet interface, the default MTU is 1500 bytes. Subtracting the IPv4 header of 20 bytes and the TCP header of 20 bytes, the default MSS size is 1460 bytes, as shown in Figure 14-28.

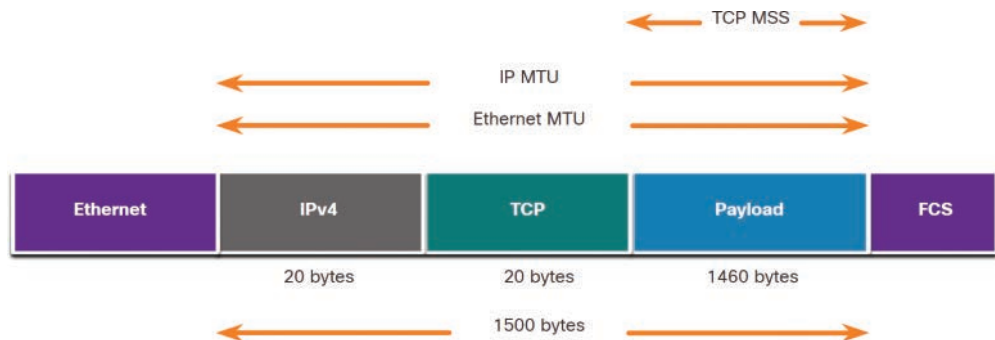


Figure 14-28 1460-Byte MSS

TCP Flow Control—Congestion Avoidance (14.6.7)

Congestion on a network results in packets being discarded by the overloaded router. When packets containing TCP segments do not reach their destination, they are left unacknowledged. By determining the rate at which TCP segments are sent but not acknowledged, the source can assume a certain level of network congestion.

Whenever there is congestion, retransmission of lost TCP segments from the source occurs. If the retransmission is not properly controlled, the additional retransmission of the TCP segments can make the congestion even worse. Not only are new packets with TCP segments introduced into the network but the feedback effect of the retransmitted TCP segments that were lost also adds to the congestion. To avoid and control congestion, TCP employs several congestion-handling mechanisms, timers, and algorithms.

If the source determines that the TCP segments are either not being acknowledged or are not being acknowledged in a timely manner, then it can reduce the number of bytes it sends before receiving an acknowledgment. In Figure 14-29, for example, PC A senses that there is congestion and, therefore, reduces the number of bytes it sends before receiving an acknowledgment from PC B.

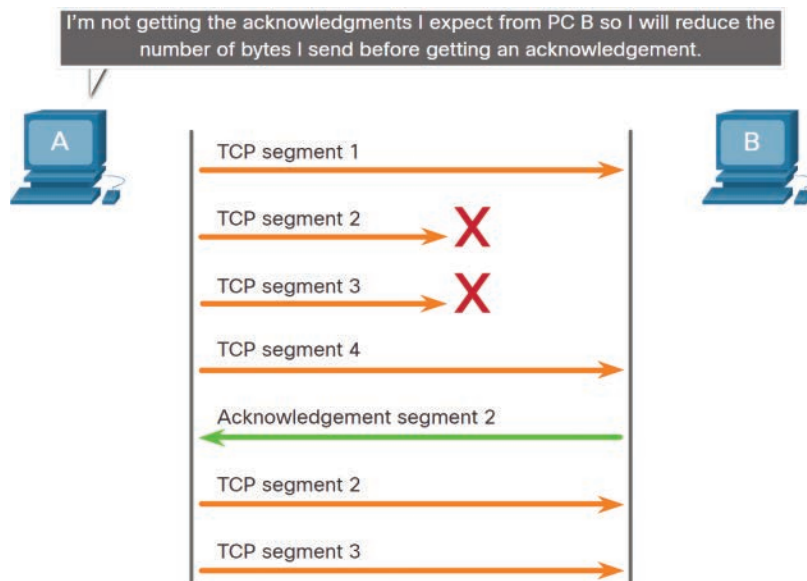


Figure 14-29 TCP Congestion Control

Notice that it is the source that is reducing the number of unacknowledged bytes it sends and not the window size determined by the destination.

Note

Explanations of congestion-handling mechanisms, timers, and algorithms are beyond the scope of this book.

**Interactive
Graphic****Check Your Understanding—Reliability and Flow Control (14.6.8)**

Refer to the online course to complete this activity.

UDP Communication (14.7)

Sometimes the reliability associated with TCP is not required or the overhead associated with providing such reliability is not suitable for the application. This is where UDP is used.

UDP Low Overhead Versus Reliability (14.7.1)

As explained earlier in this chapter, UDP is perfect for communications such as VoIP that need to be fast. This section explains in detail why UDP is perfect for some types of transmissions. As shown in Figure 14-30, UDP does not establish a connection. UDP provides low-overhead data transport because it has a small datagram header and no network management traffic.

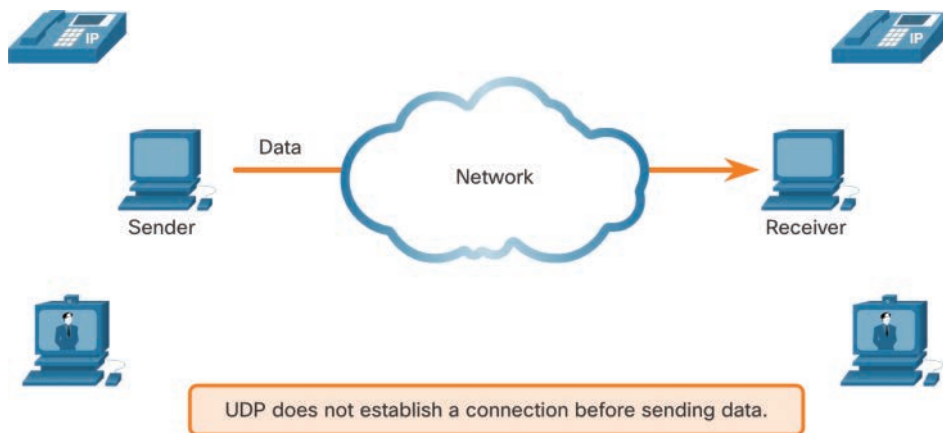


Figure 14-30 Connectionless Transport Between Sender and Receiver

UDP Datagram Reassembly (14.7.2)

Like TCP segments, UDP datagrams sent to the same destination often take different paths and arrive in the wrong order. UDP does not track sequence numbers the way TCP does. UDP has no way to reorder the datagrams into their transmission order, as shown in Figure 14-31.

Therefore, UDP simply reassembles the data in the order in which it was received and forwards it to the application. If the data sequence is important to the application, the application must identify the proper sequence and determine how the data should be processed.

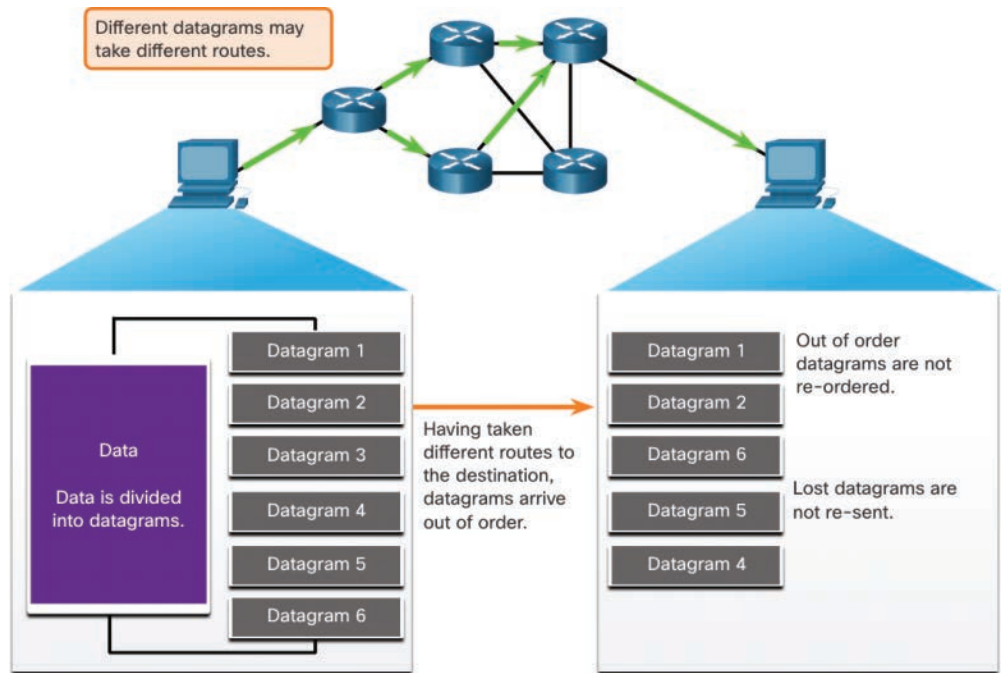


Figure 14-31 UDP: Connectionless and Unreliable

UDP Server Processes and Requests (14.7.3)

Like TCP-based applications, UDP-based server applications are assigned well-known or registered port numbers, as shown in Figure 14-32. When these applications or processes are running on a server, they accept the data matched with the assigned port number. When UDP receives a datagram destined for one of these ports, it forwards the application data to the appropriate application, based on its port number.

Note

The Remote Authentication Dial-in User Service (RADIUS) server shown in Figure 14-32 provides authentication, authorization, and accounting services to manage user access. The operation of RADIUS is beyond the scope of this book.

UDP Client Processes (14.7.4)

As with TCP, with UDP, client/server communication is initiated by a client application that requests data from a server process. The UDP client process dynamically selects a port number from the range of port numbers and uses it as the source port for the conversation. The destination port is usually the well-known or registered port number assigned to the server process.

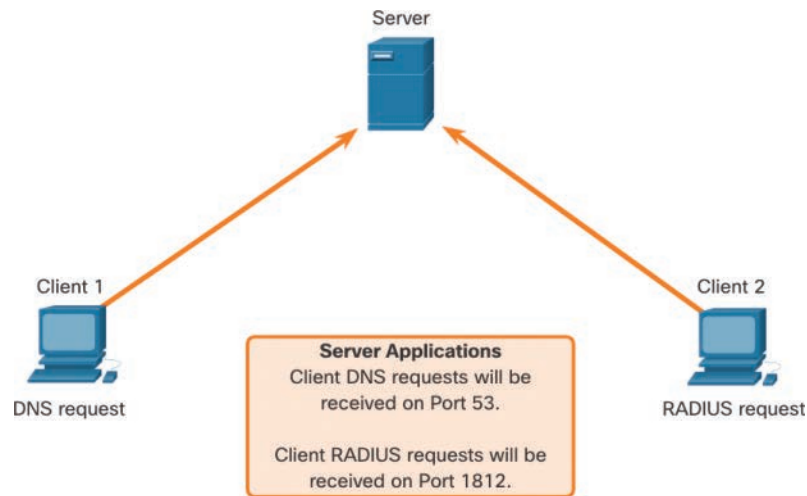


Figure 14-32 UDP Server Listening for Requests

After a client has selected the source and destination ports, the same two ports are used in the headers of all datagrams in the transaction. For the data returning to the client from the server, the source and destination port numbers in the datagram header are reversed.

Figure 14-33 is an illustration of two hosts requesting services from the DNS and RADIUS authentication server. In the figure, Client 1 is sending a DNS request, and Client 2 is requesting RADIUS authentication services of the same server.

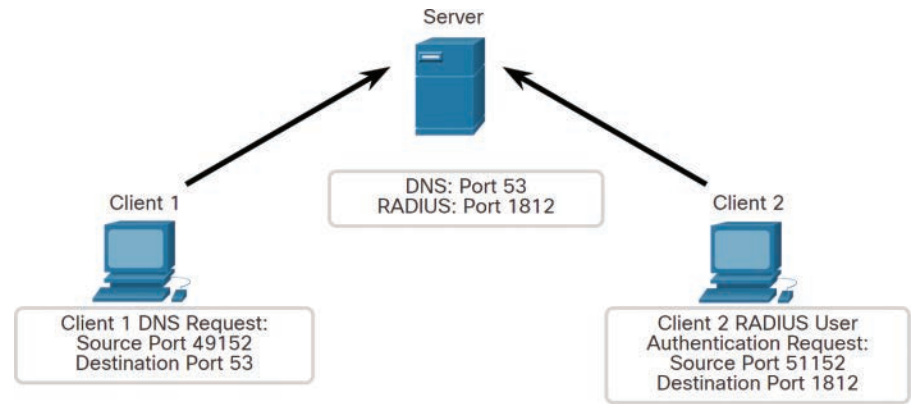


Figure 14-33 Clients Sending UDP Requests

In Figure 14-34, Client 1 is sending a DNS request using the well-known destination port 53, and Client 2 is requesting RADIUS authentication services using the registered destination port 1812.

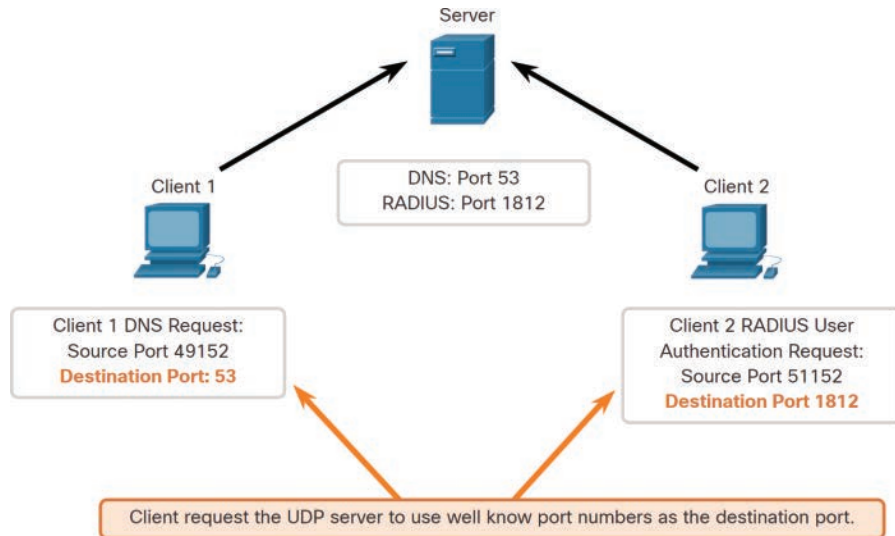


Figure 14-34 UDP Requesting Destination Ports

The requests of the clients dynamically generate source port numbers. In this case, Client 1 is using source port 49152, and Client 2 is using source port 51152, as shown in Figure 14-35.

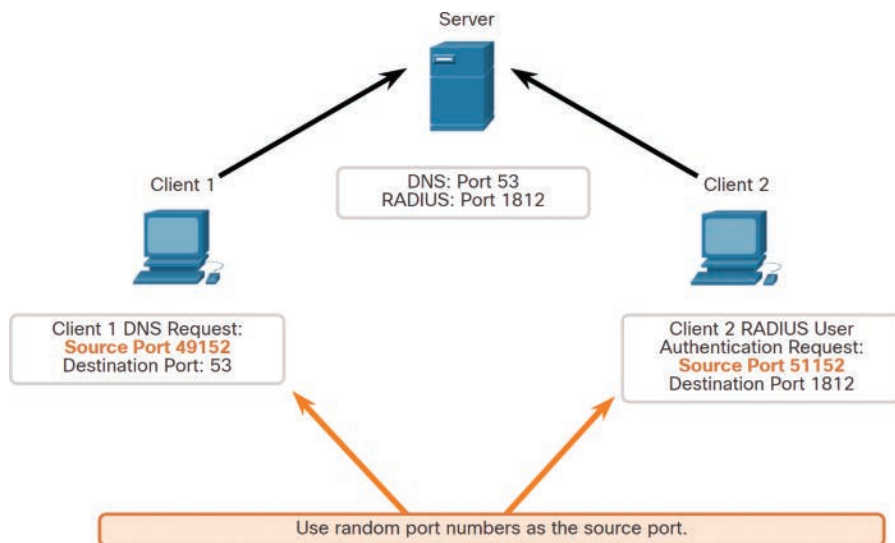


Figure 14-35 UDP Requesting Source Ports

When the server responds to the client requests, it reverses the destination and source ports of the initial request, as shown in Figures 14-36 and 14-37. The server's response to the DNS request now includes destination port 49152, and the RADIUS authentication response is now destination port 51152, as shown in Figure 14-36.

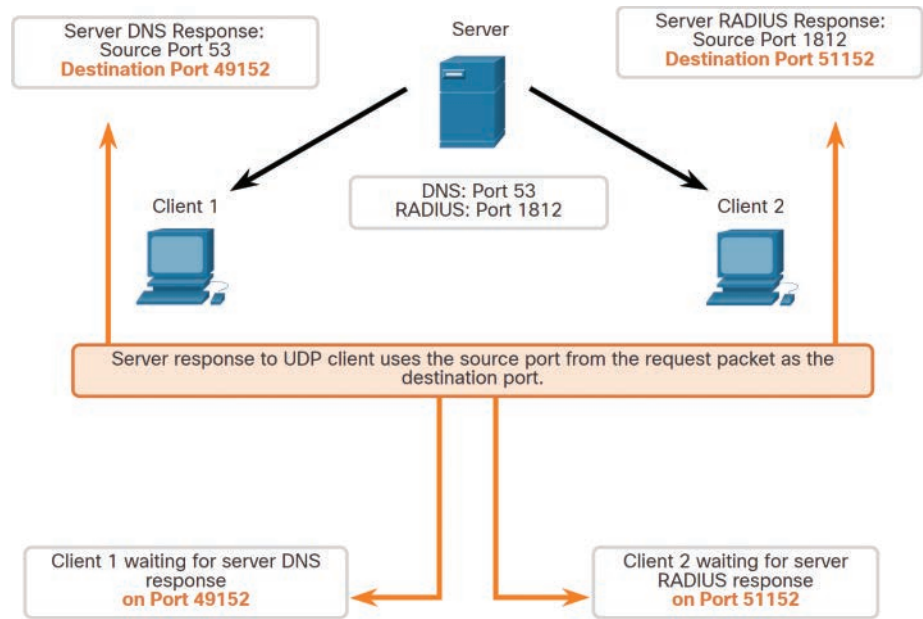


Figure 14-36 UDP Response Destination

The source ports in the server's response are the original destination ports in the initial requests, as shown in Figure 14-37.

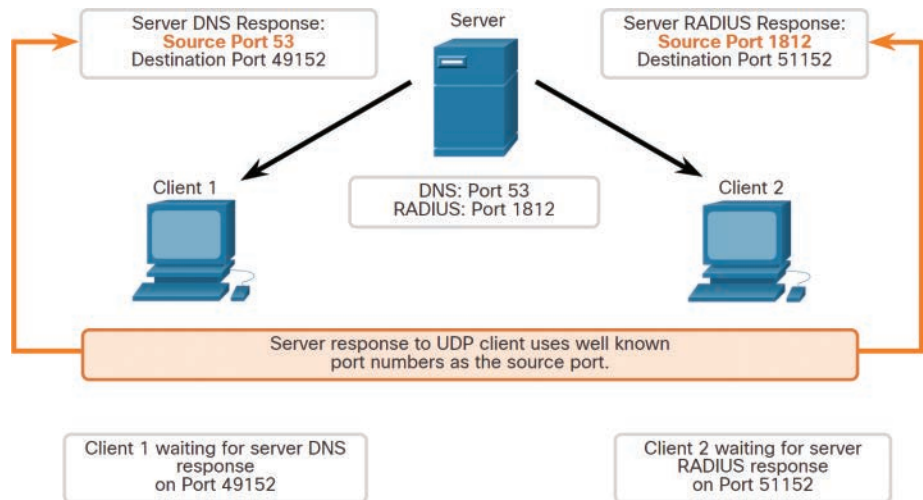


Figure 14-37 UDP Response Source Ports

Check Your Understanding—UDP Communication (14.7.5)

Refer to the online course to complete this activity.

Summary (14.8)

The following is a summary of the topics in the chapter and their corresponding online modules.

Transportation of Data

The transport layer is the link between the application layer and the lower layers that are responsible for network transmission. The transport layer is responsible for logical communications between applications running on different hosts. The transport layer includes TCP and UDP. Transport layer protocols specify how to transfer messages between hosts and is responsible for managing the reliability requirements of a conversation. The transport layer is responsible for tracking conversations (sessions), segmenting data and reassembling segments, adding header information, identifying applications, and handling conversation multiplexing. TCP is stateful and reliable, it acknowledges data, it re-sends lost data, and it delivers data in sequenced order. Use TCP for email and the web. UDP is stateless, fast, has low overhead, does not require acknowledgments, does not re-send lost data, and delivers data in the order in which it arrives. Use UDP for VoIP and DNS.

TCP Overview

TCP establishes sessions, ensures reliability, provides same-order delivery, and supports flow control. A TCP segment adds 20 bytes of overhead as header information when encapsulating the application layer data. The TCP header fields are Source Port, Destination Port, Sequence Number, Acknowledgment Number, Header Length, Reserved, Control Bits, Window Size, Checksum, and Urgent. Applications such as HTTP, FTP, SMTP, and Telnet use TCP.

UDP Overview

UDP reconstructs data in the order in which it is received, it does not re-send lost segments, it does not establish sessions, and does not inform the sender of resource availability. The UDP header fields are Source Port, Destination Port, Length, and Checksum. Applications such as DHCP, DNS, SNMP, TFTP, VoIP, and video conferencing use UDP.

Port Numbers

The TCP and UDP transport layer protocols use port numbers to manage multiple simultaneous conversations. This is why the TCP and UDP header fields identify source and destination application port numbers. The source and destination ports

are placed within the segment. The segments are then encapsulated within an IP packet. The IP packet contains the IP addresses of the source and destination. The combination of the source IP address and source port number or the destination IP address and destination port number is known as a socket. The socket is used to identify the server and service being requested by the client. There is a range of port numbers from 0 through 65535. This range is divided into groups: well-known ports, registered ports, private ports, and dynamic ports. There are a few well-known port numbers that are reserved for common applications such as FTP, SSH, DNS, and HTTP. Sometimes it is necessary to know which active TCP connections are open and running on a networked host. **netstat** is an important network utility that can be used to verify those connections.

TCP Communications Process

Each application process running on a server is configured to use a port number. The port number is either automatically assigned or configured manually by a system administrator. TCP server processes include clients sending TCP requests, requesting destination ports, requesting source ports, and responding to destination port and source port requests. To terminate a single conversation supported by TCP, four exchanges are needed to end both sessions. Either the client or the server can initiate the termination. The three-way handshake establishes that the destination device is present on the network, verifies that the destination device has an active service and is accepting requests on the destination port number that the initiating client intends to use, and informs the destination device that the source client intends to establish a communication session on that port number. The six control bits flags are URG, ACK, PSH, RST, SYN, and FIN.

Reliability and Flow Control

For the original message to be understood by the recipient, all the data must be received, and the data in these segments must be reassembled into the original order. Sequence numbers are assigned in the packet headers. No matter how well designed a network is, data loss occasionally occurs. TCP provides ways to manage segment losses. There is a mechanism to retransmit segments for unacknowledged data. Host operating systems today typically employ an optional TCP feature called selective acknowledgment (SACK), which is negotiated during the three-way handshake. If both hosts support SACK, the receiver can explicitly acknowledge which segments (bytes) were received, including any discontinuous segments. The sending host therefore needs to retransmit only the missing data. Flow control helps maintain the reliability of TCP transmission by adjusting the rate of data flow between source and destination. To accomplish this, the TCP header includes a 16-bit field called the window size. The process of the destination sending acknowledgments as it processes

bytes received and the continual adjustment of the source's send window is known as sliding window. A source might be transmitting 1460 bytes of data within each TCP segment. This is the typical MSS that a destination device can receive. To avoid and control congestion, TCP employs several congestion-handling mechanisms. It is the source that is reducing the number of unacknowledged bytes it sends and not the window size determined by the destination.

UDP Communication

UDP is a simple protocol that provides the basic transport layer functions. When UDP datagrams are sent to a destination, they often take different paths and arrive in the wrong order. UDP does not track sequence numbers the way TCP does. UDP has no way to reorder the datagrams into their transmission order. UDP simply reassembles the data in the order in which it was received and forwards it to the application. If the data sequence is important to an application, the application must identify the proper sequence and determine how the data should be processed. UDP-based server applications are assigned well-known or registered port numbers. When UDP receives a datagram destined for one of these ports, it forwards the application data to the appropriate application, based on its port number. The UDP client process dynamically selects a port number from the range of port numbers and uses it as the source port for the conversation. The destination port is usually the well-known or registered port number assigned to the server process. After a client has selected the source and destination ports, the same two ports are used in the header of all datagrams in the transaction. For the data returning to the client from the server, the source and destination port numbers in the datagram header are reversed.



Packet Tracer—TCP and UDP Communications (14.8.1)

In this activity, you will explore the functionality of the TCP and UDP protocols, multiplexing, and the function of port numbers in determining which local application requested data or is sending data.

Practice

The following activities provide practice with the topics introduced in this chapter. The lab is available in the companion *Introduction to Networks Labs & Study Guide (CCNAv7)* (ISBN 9780136634454). The Packet Tracer activity instructions are also provided in the *Labs & Study Guide*. The PKA files are available in the online course.

Packet Tracer ActivityPacket Tracer 14.8.1: TCP and UDP Communications

Check Your Understanding Questions

Complete all the review questions listed here to test your understanding of the topics and concepts in this chapter. The appendix “Answers to ‘Check Your Understanding’ Questions” lists the answers.

1. Which action is performed by a client when establishing communication with a server via the use of UDP at the transport layer?
 - a. The client sets the window size for the session.
 - b. The client sends an ISN to the server to start the three-way handshake.
 - c. The client selects its source port number.
 - d. The client sends a synchronization segment to begin the session.
2. Which transport layer feature is used to establish a connection-oriented session?
 - a. UDP ACK flag
 - b. TCP three-way handshake
 - c. UDP sequence number
 - d. TCP port number
3. What is the complete range of TCP and UDP well-known ports?
 - a. 0 to 255
 - b. 0 to 1023
 - c. 256 to 1023
 - d. 1024 to 49151
4. What is a socket?
 - a. the combination of the source and destination IP addresses and source and destination Ethernet addresses
 - b. the combination of the source IP address and port number or destination IP address and port number
 - c. the combination of the source and destination sequence numbers and acknowledgment numbers
 - d. the combination of the source and destination sequence numbers and port numbers

-
5. How does a networked server manage requests from multiple clients for different services?
 - a. The server sends all requests through the default gateway.
 - b. Each request has a combination of source and destination port numbers, coming from a unique IP address.
 - c. The server uses IP addresses to identify different services.
 - d. Each request is tracked through the physical address of the client.
 6. Which two services or protocols prefer UDP for fast transmission and low overhead? (Choose two.)
 - a. FTP
 - b. DNS
 - c. HTTP
 - d. POP3
 - e. VoIP
 7. What is the purpose of using a source port number in a TCP communication?
 - a. to notify the remote device that the conversation is over
 - b. to assemble the segments that arrived out of order
 - c. to keep track of multiple conversations between devices
 - d. to inquire about a non-received segment
 8. Which number or set of numbers represents a socket?
 - a. 01-23-45-67-89-AB
 - b. 21
 - c. 192.168.1.1:80
 - d. 10.1.1.15
 9. Which two flags in the TCP header are used in a TCP three-way handshake to establish connectivity between two network devices? (Choose two.)
 - a. ACK
 - b. FIN
 - c. PSH
 - d. RST
 - e. SYN
 - f. URG

10. What happens if part of an FTP message is not delivered to the destination?
 - a. The message is lost because FTP does not use a reliable delivery method.
 - b. The FTP source host sends a query to the destination host.
 - c. The part of the FTP message that was lost is re-sent.
 - d. The entire FTP message is re-sent.
11. What type of applications are best suited for using UDP?
 - a. applications that are sensitive to delay and can tolerate some data loss
 - b. applications that need reliable delivery
 - c. applications that require retransmission of lost segments
 - d. applications that are sensitive to packet loss
12. Network congestion has resulted in a source learning of the loss of TCP segments that were sent to the destination. What is one way that TCP addresses this?
 - a. The source decreases the amount of data that is transmits before it receives an acknowledgment from the destination.
 - b. The source decreases the window size to decrease the rate of transmission from the destination.
 - c. The destination decreases the window size.
 - d. The destination sends fewer acknowledgment messages in order to conserve bandwidth.
13. Which two operations are provided by TCP but not by UDP? (Choose two.)
 - a. identifying applications
 - b. acknowledging received data
 - c. tracking individual conversations
 - d. retransmitting any unacknowledged data
 - e. reconstructing data in the order received
14. What TCP mechanism is used to enhance performance by allowing a device to continuously send a steady stream of segments as long as the device is also receiving necessary acknowledgments?
 - a. three-way handshake
 - b. socket pair
 - c. two-way handshake
 - d. sliding window

- 15.** What is a responsibility of transport layer protocols?
- a.** providing network access
 - b.** identifying individual conversations
 - c.** determining the best path for forwarding a packet
 - d.** translating private IP addresses to public IP addresses

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Application Layer

Objectives

Upon completion of this chapter, you will be able to answer the following questions:

- How do the functions of the application layer, presentation layer, and session layer work together to provide network services to end-user applications?
- How do end-user applications operate in a peer-to-peer network?
- How do web and email protocols operate?
- How do DNS and DHCP operate?
- How do file transfer protocols operate?

Key Terms

This chapter uses the following key terms. You can find the definitions in the glossary at the end of the book.

Bootstrap Protocol (BOOTP) page 510

Simple Mail Transfer Protocol (SMTP)
page 510

Post Office Protocol (POP3) page 510

Internet Message Access Protocol (IMAP) page 510

File Transfer Protocol (FTP) page 511

Trivial File Transfer Protocol (TFTP)
page 511

Server Message Block (SMB) page 531

Introduction (15.0)

As you have learned, the transport layer is where data actually gets moved from one host to another. But before that can take place, a lot of details have to be determined so that the data transport happens correctly. This is why there is an application layer in both the OSI model and the TCP/IP model. As an example, before there was streaming video over the internet, we had to watch home movies in a variety of other ways. If you had videotaped some of your child's soccer game, and your parents, in another city, had only a video cassette player, you had to copy your video from your camera onto the right type of video cassette to send to them. If you wanted to also share the video with your brother, who had a DVD player, you had to transfer the video to a DVD and send that to him. This is what the application layer is all about: making sure that your data is in a format that the receiving device can use. Let's dive in!

Application, Presentation, and Session (15.1)

This section introduces some protocols of the TCP/IP application layer, which also relates to the top three layers of the OSI model.

Application Layer (15.1.1)

In the OSI model and the TCP/IP model, the application layer is the closest layer to the end user. As shown in Figure 15-1, it is the layer that provides the interface between the applications used to communicate and the underlying network over which messages are transmitted. Application layer protocols are used to exchange data between programs running on the source and destination hosts.

Based on the TCP/IP model, the upper three layers of the OSI model—the application, presentation, and session layers—define functions of the TCP/IP application layer.

There are many application layer protocols, and new protocols are being developed all the time. Some of the most widely known application layer protocols are Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), Trivial File Transfer Protocol (TFTP), Internet Message Access Protocol (IMAP), and Domain Name System (DNS).

Presentation and Session Layer (15.1.2)

The presentation layer has three primary functions:

- Formatting, or presenting, data at the source device into a compatible format for receipt by the destination device.
- Compressing data in a way that can be decompressed by the destination device.
- Encrypting data for transmission and decrypting data upon receipt.

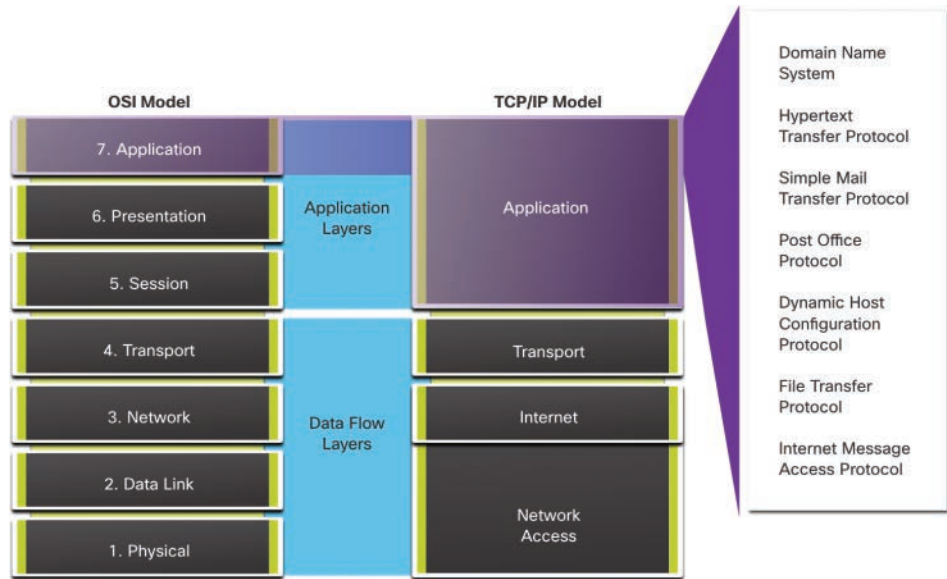


Figure 15-1 Examples of Application Layer Protocols

As shown in Figure 15-2, the presentation layer formats data for the application layer, and it sets standards for file formats. Some well-known standards for video include Matroska Video (MKV), Motion Picture Experts Group (MPG), and QuickTime Video (MOV). Some well-known graphic image formats are Graphics Interchange Format (GIF), Joint Photographic Experts Group (JPG), and Portable Network Graphics (PNG) formats.

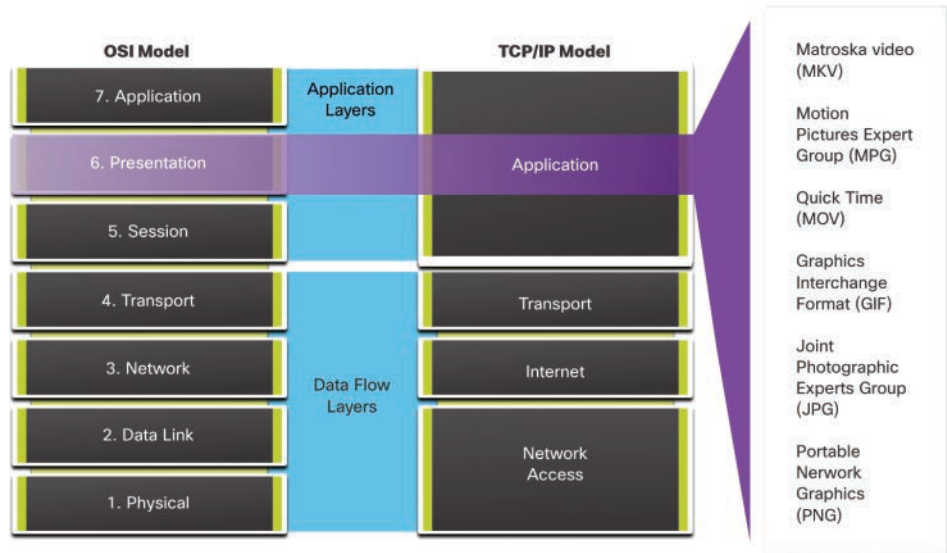


Figure 15-2 Examples of Presentation Layer Protocols

As the layer’s name implies, functions at the session layer create and maintain dialogues between source and destination applications. The session layer handles the exchange of information to initiate dialogues, keep them active, and restart sessions that are disrupted or idle for a long period of time.

TCP/IP Application Layer Protocols (15.1.3)

The TCP/IP application layer protocols specify the format and control information necessary for many common internet communication functions. Application layer protocols are used by both the source and destination devices during a communication session. For communications to be successful, the application layer protocols that are implemented on the source and destination hosts must be compatible.

Table 15-1 describes the most popular application layer protocols.

Table 15-1 TCP/IP Application Layer Protocols

Application	Protocol(s)	Port Number	Characteristics
Name system	Domain Name System (DNS)	TCP, UDP client 53	Translates domain names, such as cisco.com, into IP addresses
Host config	<i>Bootstrap Protocol (BOOTP)</i>	UDP client 68, server 67	Enables a diskless workstation to discover its own IP address, the IP address of a BOOTP server on the network, and a file to be loaded into memory to boot the machine BOOTP is being superseded by DHCP
	Dynamic Host Configuration Protocol (DHCP)	UDP client 68, server 67	Dynamically assigns IP addresses to be reused when no longer needed
Email	<i>Simple Mail Transfer Protocol (SMTP)</i>	TCP 25	Enables clients to send email to a mail server Enables servers to send email to other servers
	<i>Post Office Protocol (POP3)</i>	TCP 110	Enables clients to retrieve email from a mail server Downloads the email to the local mail application of the client
	<i>Internet Message Access Protocol (IMAP)</i>	TCP 143	Enables clients to access email stored on a mail server Maintains email on the server

Application	Protocol(s)	Port Number	Characteristics
File transfer	<i>File Transfer Protocol (FTP)</i>	TCP 20 to 21	Sets rules that enable a user on one host to access and transfer files to and from another host over a network FTP is a reliable, connection-oriented, and acknowledged file delivery protocol
	<i>Trivial File Transfer Protocol (TFTP)</i>	UDP client 69	A simple, connectionless file transfer protocol with best-effort, unacknowledged file delivery Uses less overhead than FTP
Web	Hypertext Transfer Protocol (HTTP)	TCP 80, 8080	A set of rules for exchanging text, graphic images, sound, video, and other multimedia files on the World Wide Web
	HTTP Secure (HTTPS)	TCP, UDP 443	The browser uses encryption to secure HTTP communications Authenticates the website to which you are connecting your browser

Interactive Graphic

Check Your Understanding—Application, Session, Presentation (15.1.4)

Refer to the online course to complete this activity.

Peer-to-Peer (15.2)

In the previous section, you learned that TCP/IP application layer protocols implemented on both the source and destination host must be compatible. In this section, you will learn about the client/server model and peer-to-peer networks and the processes they use, which are in the application layer.

Client-Server Model (15.2.1)

In the client/server model, the device requesting the information is called a *client*, and the device responding to the request is called a *server*. The client is a hardware/software combination that people use to directly access the resources that are stored on the server. Client and server processes are considered to be in the application

layer. The client begins an exchange by requesting data from the server, which responds by sending one or more streams of data to the client. Application layer protocols describe the format of the requests and responses between clients and servers. In addition to the actual data transfer, this exchange may also require user authentication and the identification of a data file to be transferred.

One example of a client/server network is the email service of an ISP used to send, receive, and store email. The email client on a home computer issues a request to the email server of the ISP for any unread mail. The server responds by sending the requested email to the client. Data transfer from a client to a server is referred to as an *upload*, and data from a server to a client is called a *download*.

As shown in Figure 15-3, files are downloaded from the server to the client.

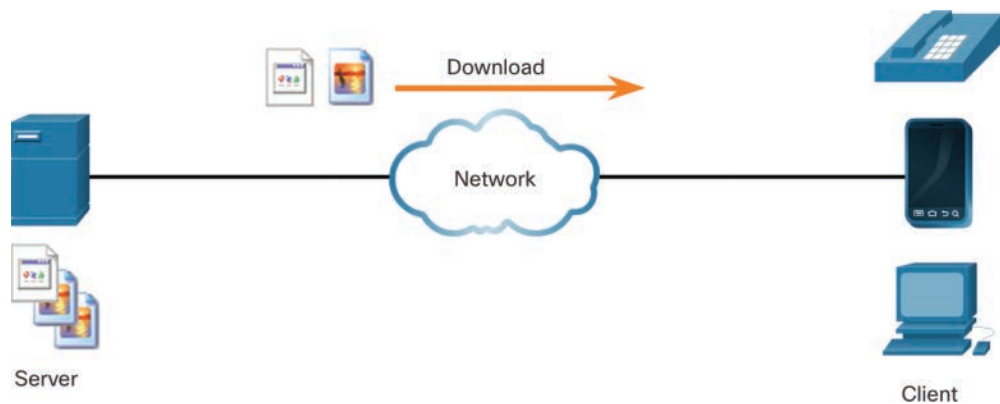


Figure 15-3 Downloading from a Server

Peer-to-Peer Networks (15.2.2)

In the peer-to-peer (P2P) networking model, the data is accessed from a peer device without the use of a dedicated server. The P2P network model involves two parts: P2P networks and P2P applications. The two parts have similar features, but in practice they work quite differently.

In a P2P network, two or more computers are connected through a network and can share resources (such as printers and files) without having a dedicated server. Every connected end device (known as a *peer*) can function as both a server and a client. One computer might assume the role of server for one transaction while simultaneously serving as a client for another. The roles of client and server are set on a per-request basis.

In addition to sharing files, a network such as this one would allow users to enable networked games or share an internet connection.

In a peer-to-peer exchange, both devices are considered equal in the communication process. Peer 1 has files that are shared with Peer 2 and can access the shared printer

that is directly connected to Peer 2 to print files. Peer 2 is sharing the directly connected printer with Peer 1 while accessing the shared files on Peer 1, as shown in Figure 15-4.

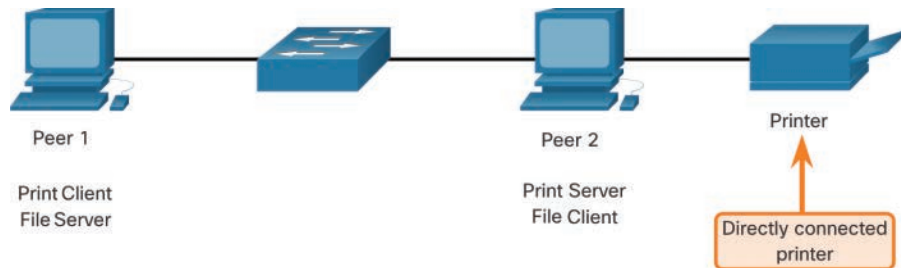


Figure 15-4 PC Operating as a Print Server

Peer-to-Peer Applications (15.2.3)

A P2P application allows a device to act as both a client and a server within the same communication, as shown in Figure 15-5. In this model, every client is a server, and every server is a client. P2P applications require that each end device provide a user interface and run a background service.

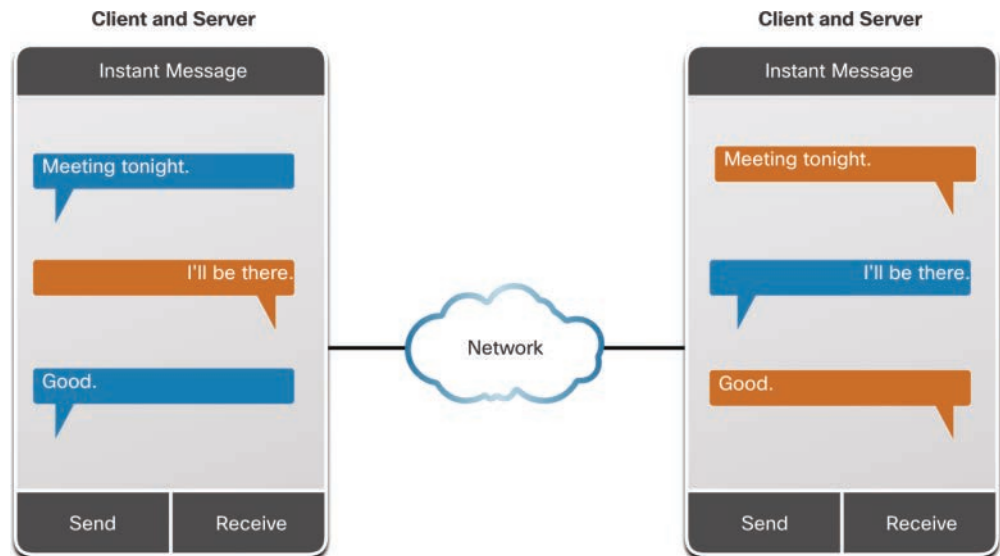


Figure 15-5 Texting as an Example of a Peer-to-Peer Application

Some P2P applications use a hybrid system in which resource sharing is decentralized, but the indexes that point to resource locations are stored in a centralized directory. In a hybrid system, each peer accesses an index server to get the location of a resource stored on another peer.

Common P2P Applications (15.2.4)

With P2P applications, each computer in the network that is running the application can act as a client or as a server for the other computers in the network that are also running the application. Common P2P networks include the following:

- BitTorrent
- Direct Connect
- eDonkey
- Freenet

Some P2P applications are based on the Gnutella protocol, and each user shares whole files with other users. As shown in Figure 15-6, Gnutella-compatible client software allows users to connect to Gnutella services over the internet and to locate and access resources shared by other Gnutella peers. Many Gnutella client applications are available, including uTorrent, BitComet, DC++, Deluge, and eMule.

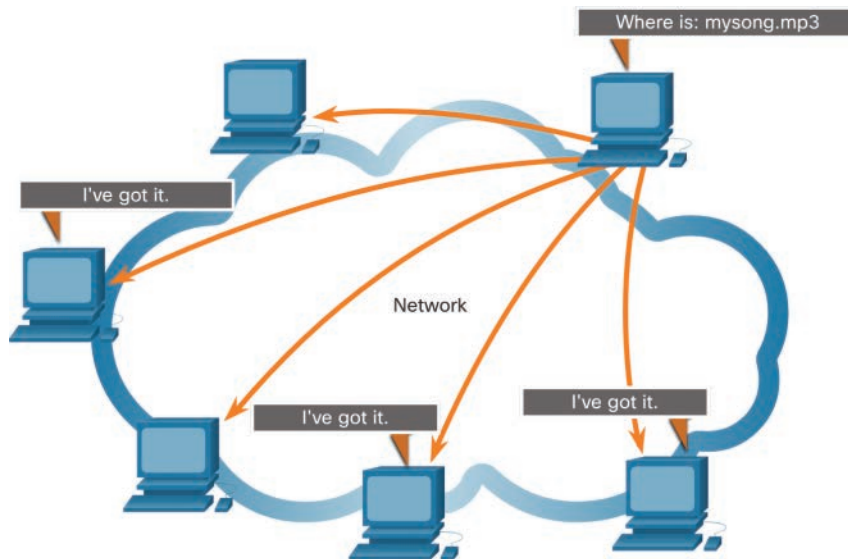


Figure 15-6 Gnutella Client Software in a P2P Network

Many P2P applications allow users to share pieces of many files with each other at the same time. Clients use a torrent file to locate other users who have pieces that they need so that they can then connect directly to them. This torrent file also contains information about tracker computers that keep track of which users have specific pieces of certain files. Clients ask for pieces from multiple users at the same time. This is known as a *swarm*, and the technology is called *BitTorrent*. BitTorrent has its own client, and there are also many other BitTorrent clients, including uTorrent, Deluge, and qBittorrent.

Note

Any type of file can be shared between users. Many of the shared files are copyrighted, meaning that only the creators have the right to use and distribute them. It is against the law to download or distribute copyrighted files without permission from the copyright holder. Copyright violation can result in criminal charges and civil lawsuits.

Interactive Graphic**Check Your Understanding—Peer-to-Peer (15.2.5)**

Refer to the online course to complete this activity.

Web and Email Protocols (15.3)

There are application layer–specific protocols that are designed for common uses such as web browsing and email. This section goes into more detail on the protocols introduced earlier in this chapter.

Hypertext Transfer Protocol and Hypertext Markup Language (15.3.1)

When a web address or uniform resource locator (URL) is typed into a web browser, the web browser establishes a connection to the web service. The web service is running on the server that is using HTTP. URLs and uniform resource identifiers (URIs) are the names most people associate with web addresses.

To better understand how a web browser and a web server interact, examine how a web page (in this case, <http://www.cisco.com/index.html>) is opened in a browser.

Step 1. As shown in Figure 15-7, the browser interprets the three parts of the URL:

- `http` (the protocol or scheme)
- `www.cisco.com` (the server name)
- `index.html` (the specific filename requested)



Figure 15-7 Step 1: Browser Interpreting the URL

Step 2. As shown in Figure 15-8, the browser checks with a name server to convert `www.cisco.com` into a numeric IP address, which it uses to connect to the server. The client initiates an HTTP request to a server by sending a GET request to the server and asks for the `index.html` file.

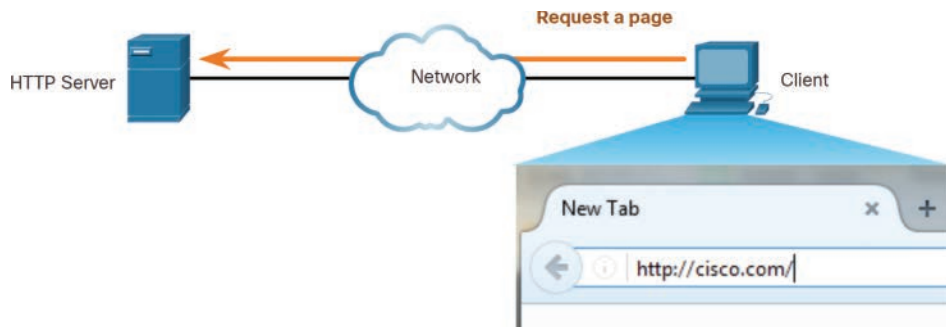


Figure 15-8 Step 2: Requesting a Web Page

Step 3. In response to the request, the server sends the HTML code for this web page to the browser, as shown in Figure 15-9.

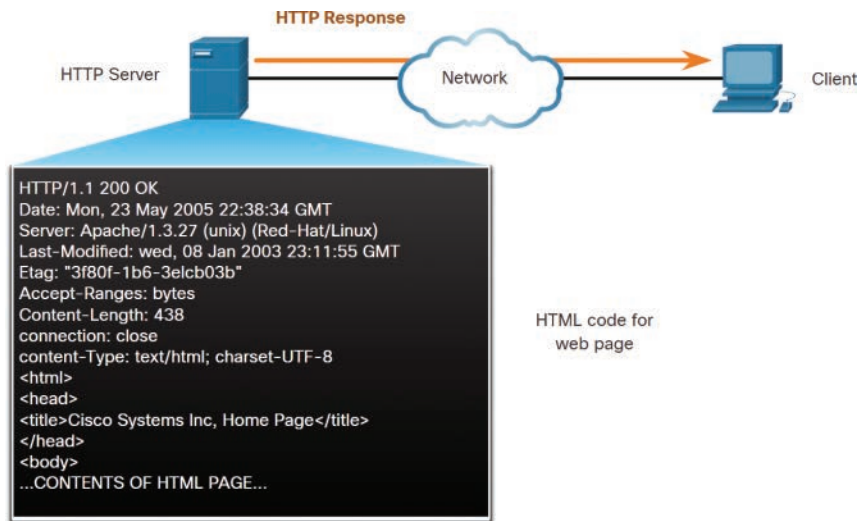


Figure 15-9 Step 3: Web Server Response

Step 4. The browser deciphers the HTML code and formats the page for the browser window, as shown in Figure 15-10.

HTTP and HTTPS (15.3.2)

HTTP is a request/response protocol. When a client, typically a web browser, sends a request to a web server, HTTP specifies the message types used for that communication. The three common message types are GET, POST, and PUT:

- **GET:** This is a client request for data. A client (web browser) sends the GET message to the web server to request HTML pages (see Figure 15-11).

- **POST:** This uploads data files such as form data to the web server.
- **PUT:** This uploads resources or content such as an image to the web server.

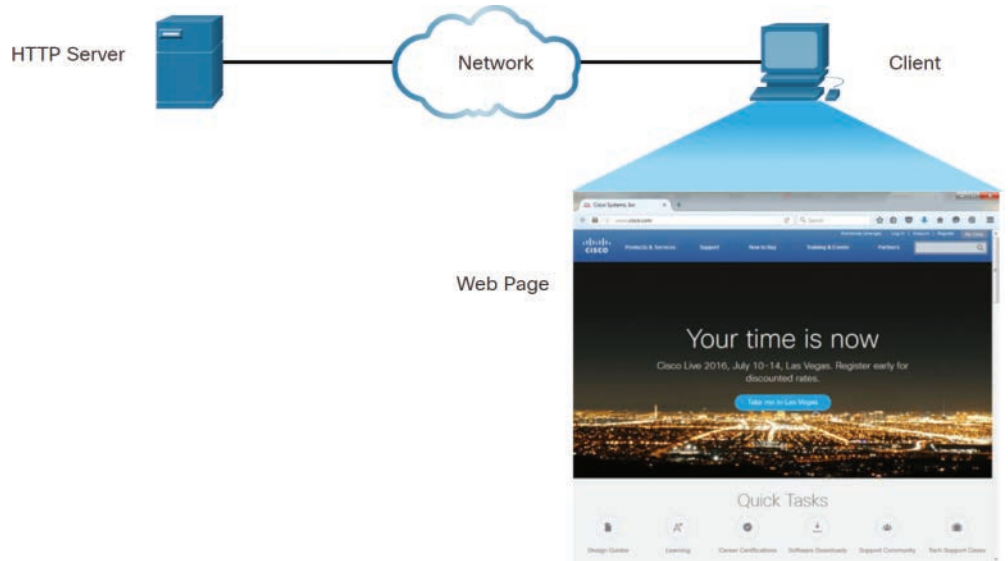


Figure 15-10 Step 4: Browser Interpreting and Displaying HTML

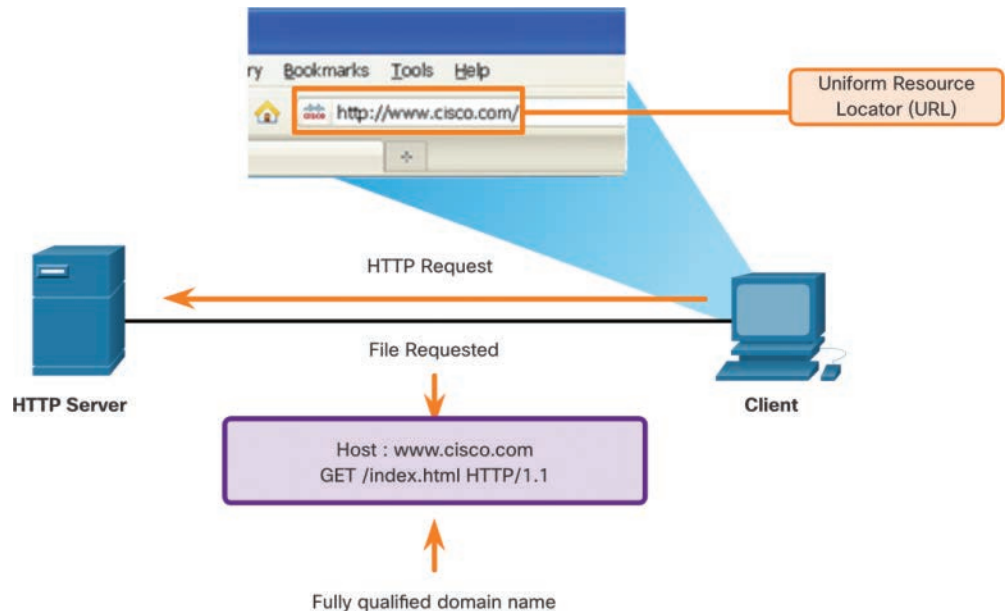


Figure 15-11 HTTP GET Message Example

Although HTTP is remarkably flexible, it is not a secure protocol. The request messages send information to the server in plaintext that can be intercepted and read. The server responses, typically HTML pages, are also unencrypted.

For secure communication across the internet, the HTTP Secure (HTTPS) protocol is used. HTTPS uses authentication and encryption to secure data as it travels between the client and server. HTTPS uses the same client request/server response process as HTTP, but the data stream is encrypted with Transport Layer Security (TLS) or its predecessor, Secure Socket Layer (SSL), before being transported across the network.

Email Protocols (15.3.3)

One of the primary services offered by an ISP is email hosting. To run on a computer or other end device, email requires several applications and services, as shown in Figure 15-12. Email is a store-and-forward method of sending, storing, and retrieving electronic messages across a network. Email messages are stored in databases on mail servers.

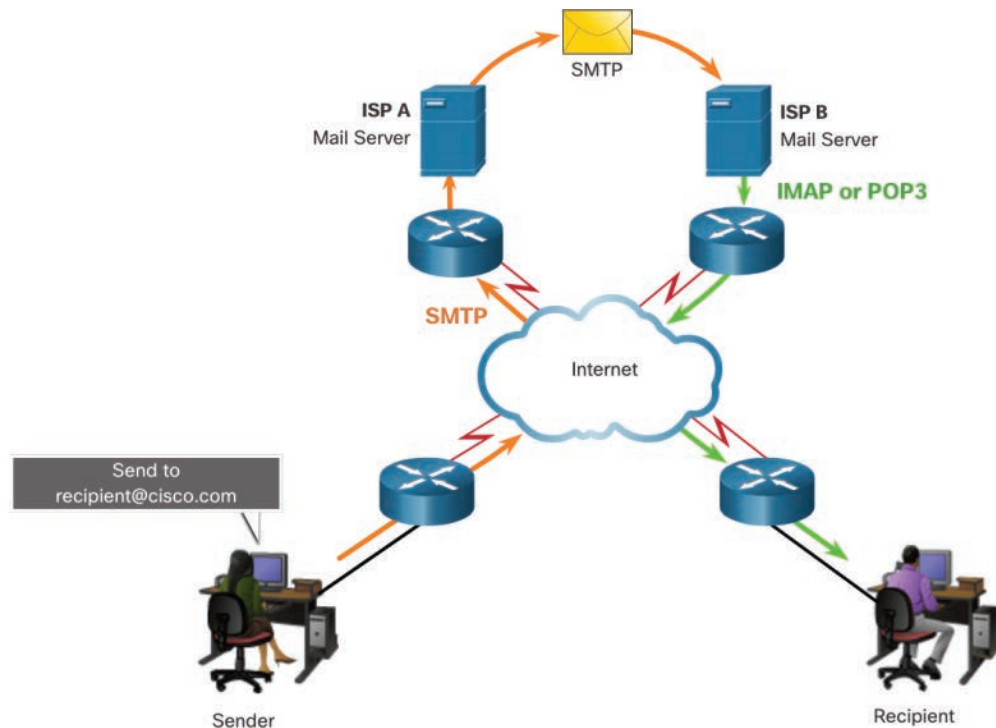


Figure 15-12 Email Protocols in Operation

Email clients communicate with mail servers to send and receive email. Mail servers communicate with other mail servers to transport messages from one domain

to another. An email client does not communicate directly with another email client when sending email. Instead, both clients rely on the mail server to transport messages.

Email supports three separate protocols for operation: Simple Mail Transfer Protocol (SMTP), Post Office Protocol (POP), and IMAP. The application layer process that sends mail uses SMTP. A client retrieves email using one of the two application layer protocols: POP or IMAP.

SMTP, POP, and IMAP (15.3.4)

The following sections describe the email protocols SMTP, POP, and IMAP.

SMTP

An SMTP message must have a message header and a message body. The message body can contain any amount of text, and the message header must have a properly formatted recipient email address and a sender address.

When a client sends email, the client SMTP process connects with a server SMTP process on well-known port 25. After the connection is made, the client attempts to send the email to the server across the connection, as shown in Figure 15-13. When the server receives the message, it either places the message in a local account, if the recipient is local, or forwards the message to another mail server for delivery.

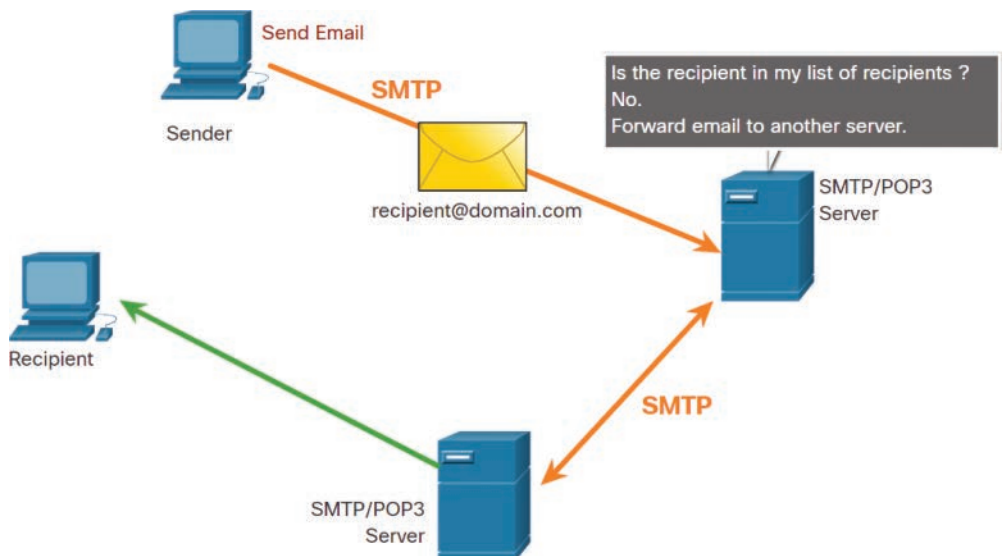


Figure 15-13 SMTP Example

The destination email server may not be online or may be busy when email messages are sent. Therefore, SMTP spools messages to be sent at a later time. The server periodically checks the queue for messages and attempts to send them again. If a message is still not delivered after a predetermined expiration time, it is returned to the sender as undeliverable.

POP

An application can use POP to retrieve mail from a mail server. With POP, mail is downloaded from the server to the client and then deleted on the server. This is the default operation of POP.

The server starts the POP service by passively listening on TCP port 110 for client connection requests. When a client wants to make use of the service, it sends a request to establish a TCP connection with the server, as shown in Figure 15-14. When the connection is established, the POP server sends a greeting. The client and POP server then exchange commands and responses until the connection is closed or aborted.

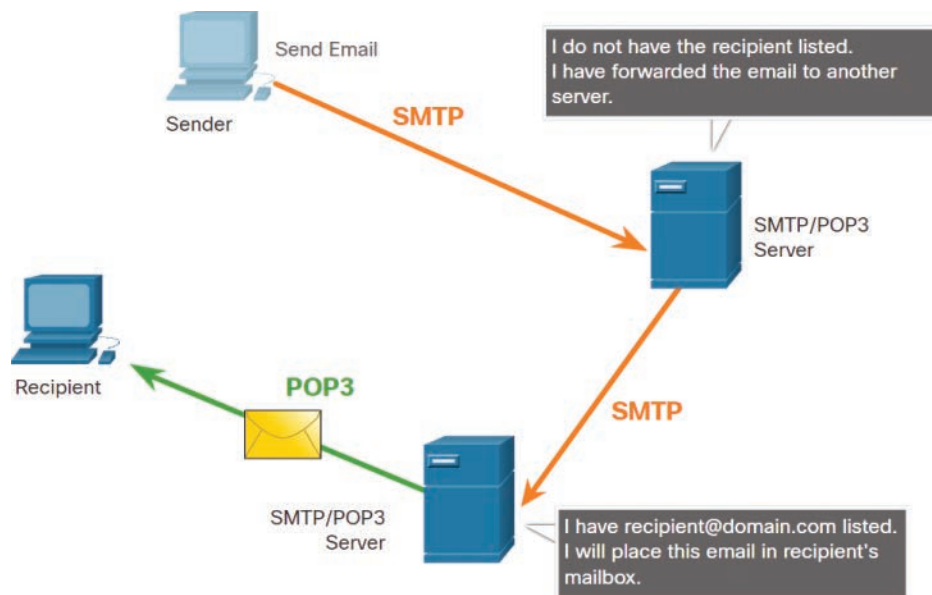


Figure 15-14 POP Example

With POP, email messages are downloaded to the client and removed from the server, so there is no centralized location where email messages are kept. Because POP does not store messages, it is not recommended for a small business that needs a centralized backup solution.

POP3 is the most commonly used version of POP.

IMAP

IMAP is another protocol that describes a method to retrieve email messages. Unlike with POP, when the user connects to an IMAP-capable server, copies of the messages are downloaded to the client application, as shown in Figure 15-15. The original messages are kept on the server until they are manually deleted. Users view copies of the messages in their email client software.

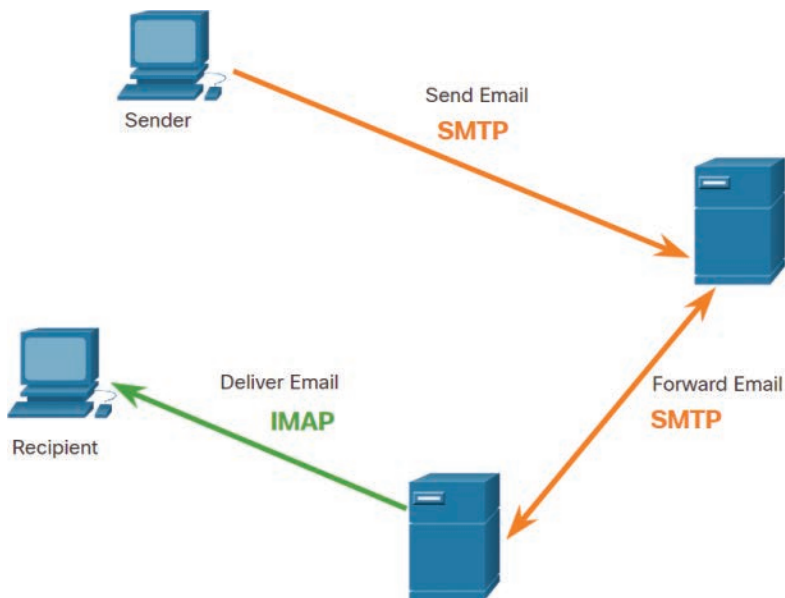


Figure 15-15 IMAP Example

Users can create a file hierarchy on the server to organize and store mail. That file structure is duplicated on the email client. When a user decides to delete a message, the server synchronizes that action and deletes the message from the server.

Interactive
Graphic

Check Your Understanding—Web and Email Protocols (15.3.5)

Refer to the online course to complete this activity.

IP Addressing Services (15.4)

Some application layer–specific protocols were designed to make it easier to obtain addresses for network devices. These services are essential because it would be very time-consuming and difficult to remember IP addresses instead of URLs or to

manually configure all the devices in a medium to large network. This section goes into more detail about the IP addressing services DNS and DHCP.

Domain Name Service (15.4.1)

In data networks, devices are labeled with numeric IP addresses to send and receive data over networks. Domain names were created to convert these numeric address into recognizable names.

On the internet, fully qualified domain names (FQDNs), such as `http://www.cisco.com`, are much easier for people to remember than `198.133.219.25`, which is the actual numeric address for the server at `www.cisco.com`. If Cisco decides to change the numeric address of `www.cisco.com`, the change is transparent to the user because the domain name remains the same. The new address is linked to the existing domain name, and connectivity is maintained.

The DNS protocol defines an automated service that matches resource names with the required numeric network addresses. It includes the format for queries, responses, and data. DNS communications use a single format called a *message*. This message format is used for all types of client queries and server responses, error messages, and the transfer of resource record information between servers.

The following are the steps in the DNS process:

- Step 1.** The user types an FQDN into a browser application Address field, as shown in Figure 15-16.

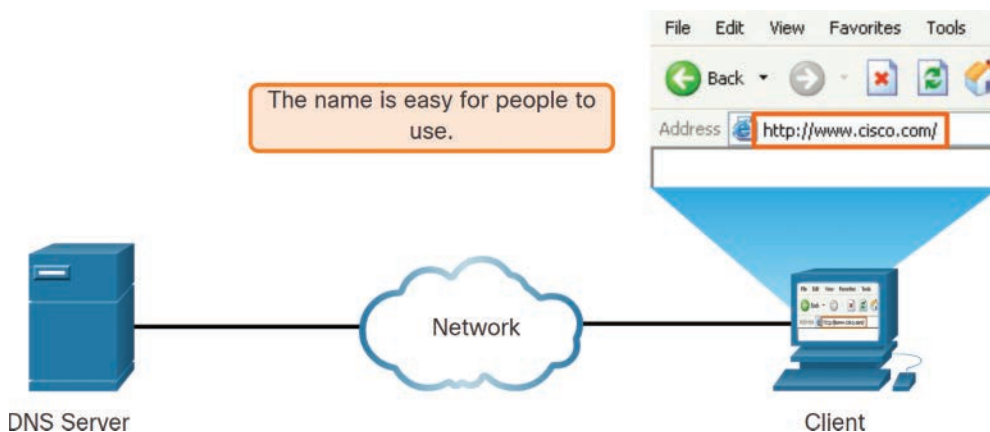


Figure 15-16 Step 1: Entering a URL in a Browser

- Step 2.** A DNS query is sent to the designated DNS server for the client computer, as shown in Figure 15-17.

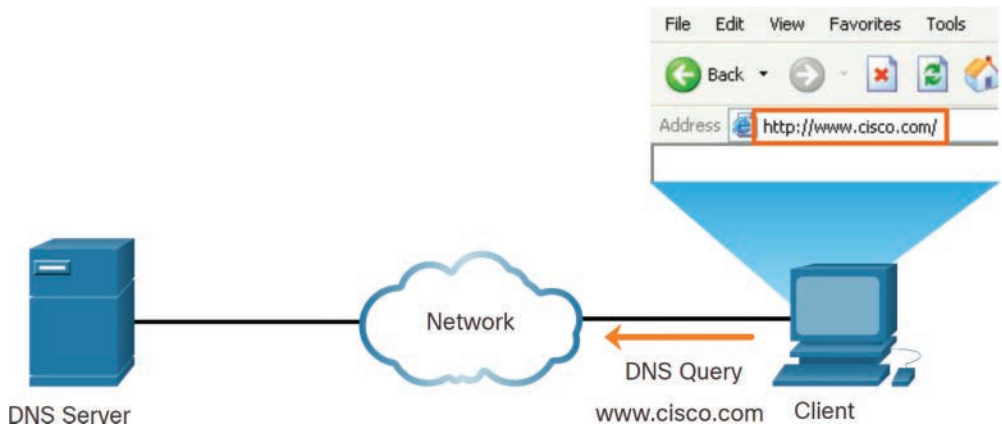


Figure 15-17 Step 2: Sending a DNS Query to the DNS Server

Step 3. The DNS server matches the FQDN with its IP address, as shown in Figure 15-18.

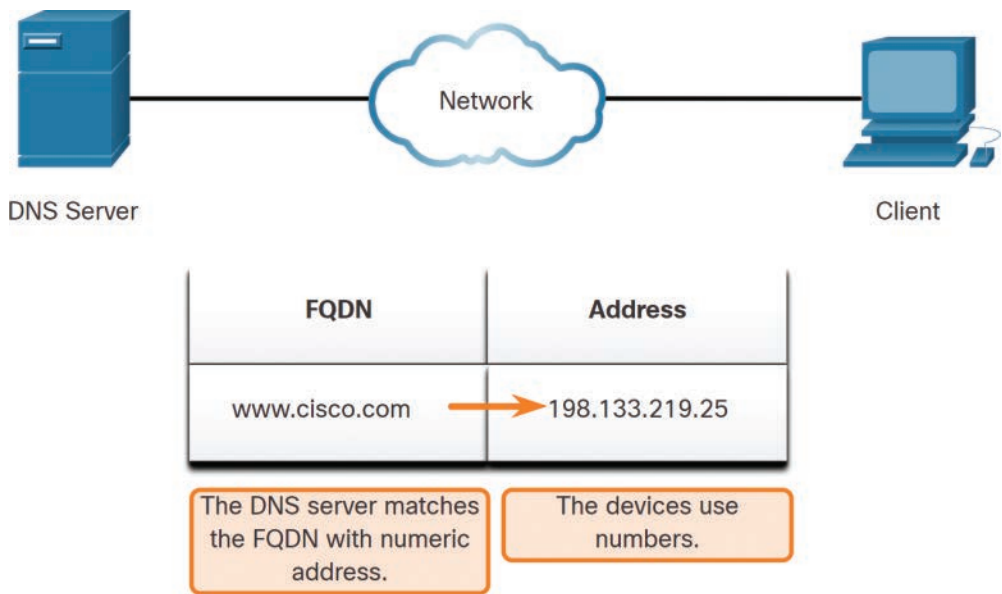


Figure 15-18 Step 3: DNS Server Matching the FQDN to an IP Address

Step 4. The DNS query response is sent back to the client with the IP address for the FQDN, as shown in Figure 15-19.

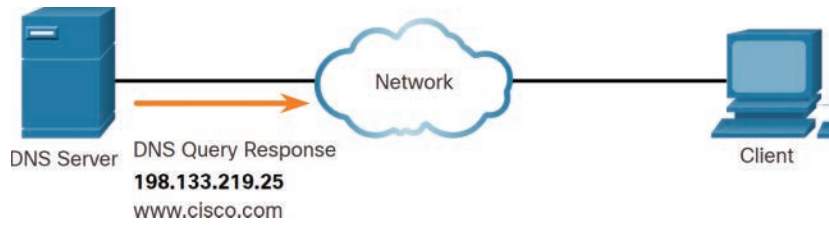


Figure 15-19 Step 4: DNS Server Responding to the DNS Query

Step 5. The client computer uses the IP address to make requests of the server, as shown in Figure 15-20.

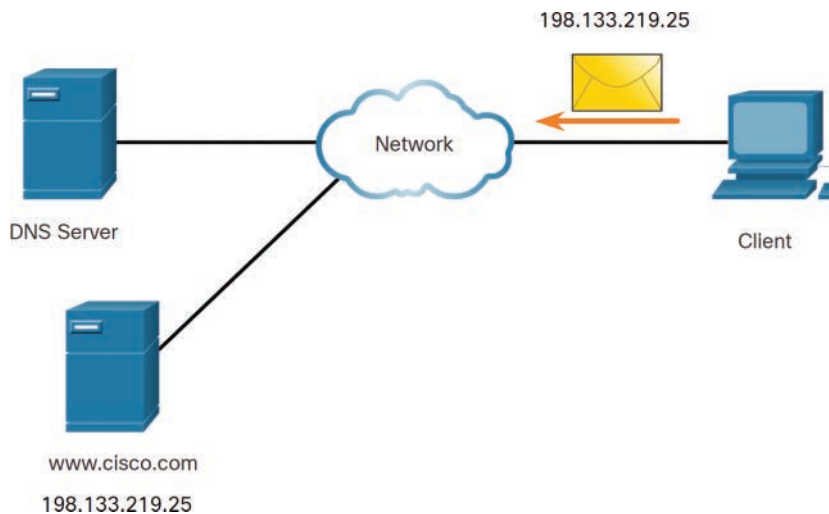


Figure 15-20 Step 5: Client Sending Web Request Using IP Address

DNS Message Format (15.4.2)

The DNS server stores different types of resource records that are used to resolve names. Each record contains the name, address, and type of record. Some of these record types are as follows:

- **A:** An end-user device IPv4 address
- **NS:** An authoritative name server
- **AAAA:** An end-user device IPv6 address; pronounced “quad-A”
- **MX:** A mail exchange record

When a client makes a query, the DNS process on the server first looks at the server’s own records to resolve the name. If it is unable to resolve the name by using its

stored records, it contacts other servers to resolve the name. After a match is found and returned to the original requesting server, the server temporarily stores the numbered address in case the same name is requested again.

The DNS client service on Windows PCs also stores previously resolved names in memory. The `ipconfig /displaydns` command displays all the cached DNS entries.

As shown in Table 15-2, DNS uses the same message format between servers, consisting of a question, an answer, an authority, and additional information for all types of client queries and server responses, error messages, and transfer of resource record information.

Table 15-2 DNS Message Sections

DNS Message Section	Description
Question	The question for the name server
Answer	Resource records answering the question
Authority	Resource records pointing toward an authority
Additional	Resource records holding additional information

DNS Hierarchy (15.4.3)

The DNS protocol uses a hierarchical system to create a database to provide name resolution, as shown in Figure 15-21. DNS uses domain names to form the hierarchy.

The naming structure is broken down into small, manageable zones. Each DNS server maintains a specific database file and is responsible for managing name-to-IP mappings for only that small portion of the entire DNS structure. When a DNS server receives a request for a name translation that is not within its DNS zone, the DNS server forwards the request to another DNS server within the proper zone for translation. DNS is scalable because hostname resolution is spread across multiple servers.

Each top-level domain represents either the type of organization or the country of origin. Examples of top-level domains include the following:

- **.com:** A business or an industry
- **.org:** A non-profit organization
- **.au:** Australia
- **.co:** Colombia

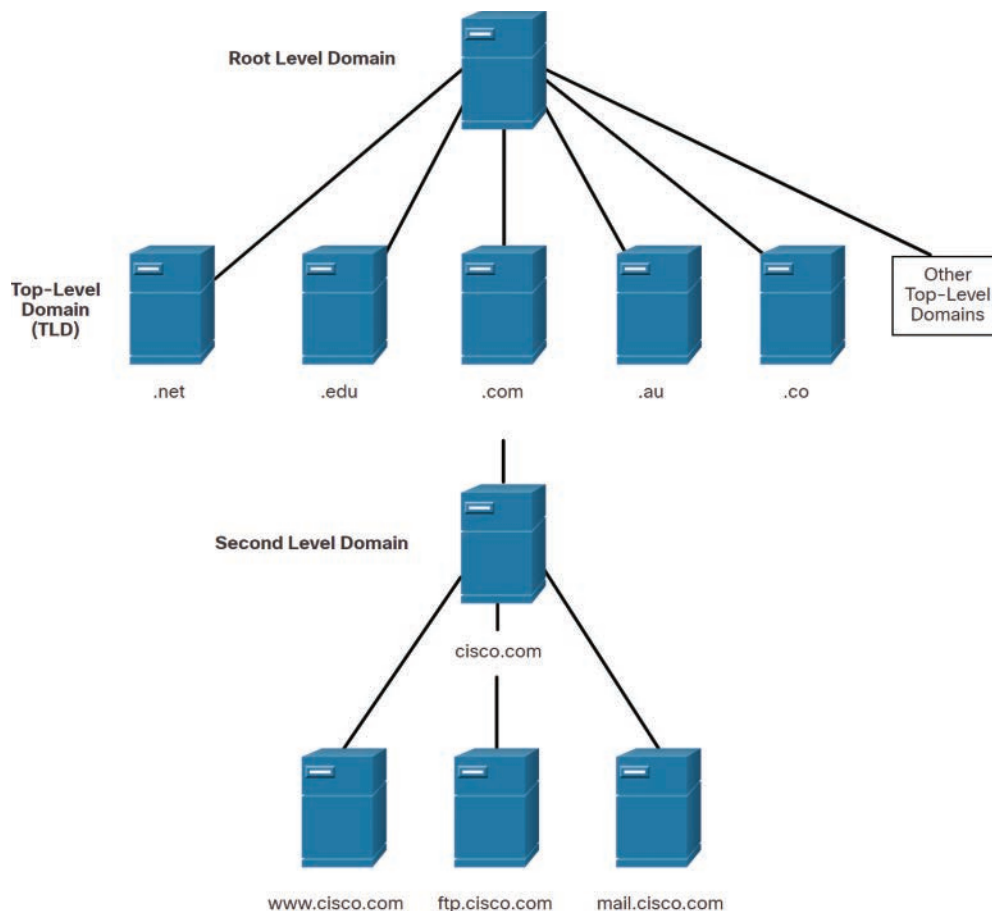


Figure 15-21 DNS Hierarchy

The nslookup Command (15.4.4)

When configuring a network device, one or more DNS server addresses are provided that the DNS client can use for name resolution. Usually the ISP provides the addresses to use for the DNS servers. When a user application requests to connect to a remote device by name, the requesting DNS client queries the name server to resolve the name to a numeric address.

Computer operating systems also have a utility called **nslookup** that allows a user to manually query the name servers to resolve a given hostname. This utility can also be used to troubleshoot name resolution issues and to verify the current status of the name servers.

When the **nslookup** command is issued, the default DNS server configured for the host is displayed, as shown in Example 15-1. The name of a host or domain can be

entered at the **nslookup** prompt. The **nslookup** utility has many options available for extensive testing and verification of the DNS process.

Example 15-1 Using the **nslookup** Command on a Windows Host

```
C:\Users> nslookup
Default Server:  dns-sj.cisco.com
Address:  171.70.168.183
> www.cisco.com
Server:  dns-sj.cisco.com
Address:  171.70.168.183
Name:    origin-www.cisco.com
Addresses:  2001:420:1101:1::a
           173.37.145.84
Aliases:  www.cisco.com
> cisco.netacad.net
Server:  dns-sj.cisco.com
Address:  171.70.168.183
Name:    cisco.netacad.net
Address:  72.163.6.223
>
```

Interactive Graphic

Syntax Checker—The **nslookup** Command (15.4.5)

Refer to the online course to complete this activity.

Dynamic Host Configuration Protocol (15.4.6)

The Dynamic Host Configuration Protocol (DHCP) for IPv4 service automates the assignment of IPv4 addresses, subnet masks, gateways, and other IPv4 networking parameters. This is referred to as *dynamic addressing*. The alternative to dynamic addressing is *static addressing*, in which the network administrator manually enters IP address information on hosts.

When a host connects to the network, the DHCP server is contacted, and an address is requested. The DHCP server chooses an address from a configured range of addresses called a *pool* and assigns (leases) it to the host.

On larger networks, or where the user population changes frequently, DHCP is preferred for address assignment. New users may arrive and need connections; others may have new computers that must be connected. Rather than use static addressing for each connection, it is more efficient to have IPv4 addresses assigned automatically using DHCP.

DHCP can allocate IP addresses for a configurable period of time, called a *lease period*. The lease period is an important DHCP setting. When the lease period

expires or the DHCP server gets a DHCPRELEASE message, the address is returned to the DHCP pool for reuse. Users can freely move from location to location and can easily reestablish network connections through DHCP.

As Figure 15-22 shows, various types of devices can be DHCP servers. The DHCP server in most medium to large networks is usually a local, dedicated PC-based server. With home networks, the DHCP server is usually located on the local router that connects the home network to the ISP.

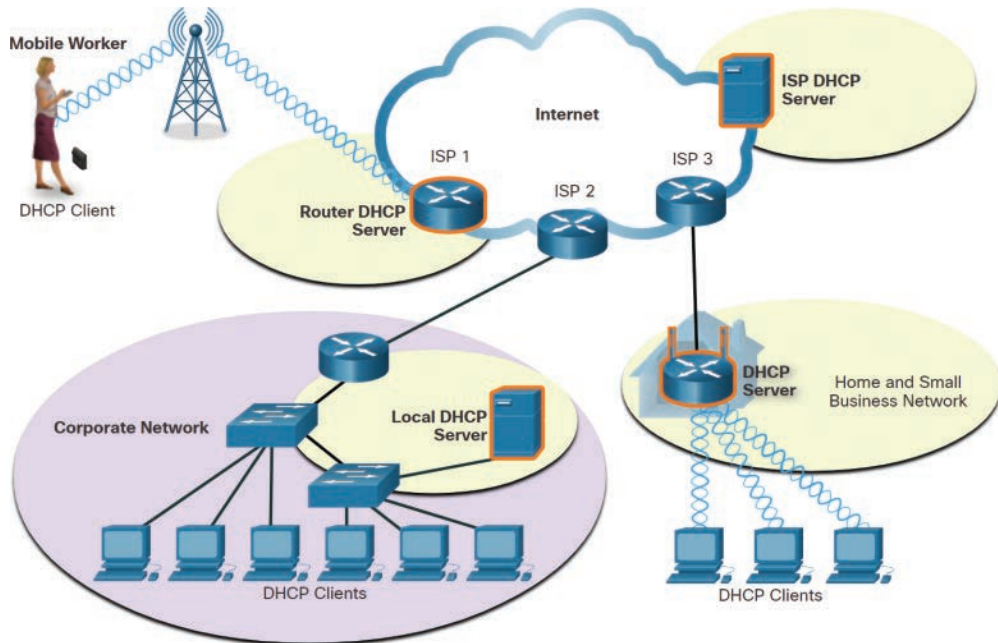


Figure 15-22 Examples of Different DHCP Servers and Clients

Many networks use both DHCP and static addressing. DHCP is used for general-purpose hosts, such as end-user devices. Static addressing is used for network devices, such as gateway routers, switches, servers, and printers.

DHCP for IPv6 (DHCPv6) provides similar services for IPv6 clients. One important difference between DHCP for IPv4 and DHCPv6 is that DHCPv6 does not provide a default gateway address. This address can only be obtained dynamically from the Router Advertisement message of the router.

DHCP Operation (15.4.7)

As shown in Figure 15-23, when an IPv4 DHCP-configured device boots up or connects to the network, the client broadcasts a DHCP discover (DHCPDISCOVER) message to identify any available DHCP servers on the network. A DHCP server replies with a DHCP offer (DHCPOFFER) message, which offers a lease to the client.

The offer message contains the IPv4 address and subnet mask to be assigned, the IPv4 address of the DNS server, and the IPv4 address of the default gateway. The lease offer also includes the duration of the lease.

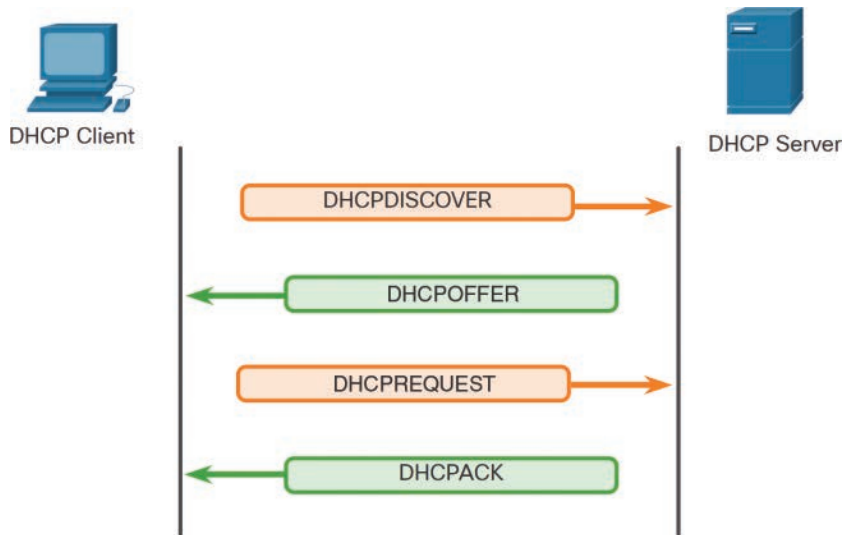


Figure 15-23 DHCP Messages

The client may receive multiple DHCPOFFER messages if the local network has more than one DHCP server. In such a case, the client must choose between the offers and sends a DHCP request (DHCPREQUEST) message that identifies the explicit server and lease offer that it is accepting. A client may also choose to request an address that it was previously allocated by the server.

If the IPv4 address requested by the client or offered by the server is still available, the server returns a DHCP acknowledgment (DHCPACK) message that acknowledges to the client that the lease has been finalized. If the offer is no longer valid, the selected server responds with a DHCP negative acknowledgment (DHCPNAK) message. If a DHCPNAK message is returned, the selection process must begin again, with a new DHCPDISCOVER message being transmitted. After the client has the lease, it must be renewed prior to the lease expiration through another DHCPREQUEST message.

The DHCP server ensures that all IP addresses are unique; that is, the same IP address cannot be assigned to two different network devices simultaneously. Most ISPs use DHCP to allocate addresses to their customers.

DHCPv6 has a set of messages that are similar to those for DHCP for IPv4. The DHCPv6 messages are SOLICIT, ADVERTISE, INFORMATION REQUEST, and REPLY.

**Lab—Observe DNS Resolution (15.4.8)**

In this lab, you will complete the following objectives:

- Part 1: Observe the DNS Conversion of a URL to an IP Address
 - Part 2: Observe DNS Lookup Using the **nslookup** Command on a Website
 - Part 3: Observe DNS Lookup Using the **nslookup** Command on Mail Servers
-

**Interactive
Graphic****Check Your Understanding—IP Addressing Services (15.4.9)**

Refer to the online course to complete this activity.

File Sharing Services (15.5)

Transferring files from one computer to another is a common process. This section introduces protocols that support file sharing.

File Transfer Protocol (15.5.1)

As you learned in previous sections, in the client/server model, the client can upload data to a server and download data from a server if both devices are using a file sharing protocol such as File Transfer Protocol (FTP). Like HTTP, email, and addressing protocols, FTP is a commonly used application layer protocol. This section discusses FTP in more detail.

FTP was developed to allow for data transfers between a client and a server. An FTP client is an application that runs on a computer that is being used to push and pull data from an FTP server.

As shown in Figure 15-24, the client establishes the first connection to the server for control traffic by using TCP port 21. The traffic consists of client commands and server replies.

The client establishes the second connection to the server for the actual data transfer, using TCP port 20. This connection is created every time there is data to be transferred.

The data transfer can happen in either direction: The client can download (pull) data from the server, or the client can upload (push) data to the server.

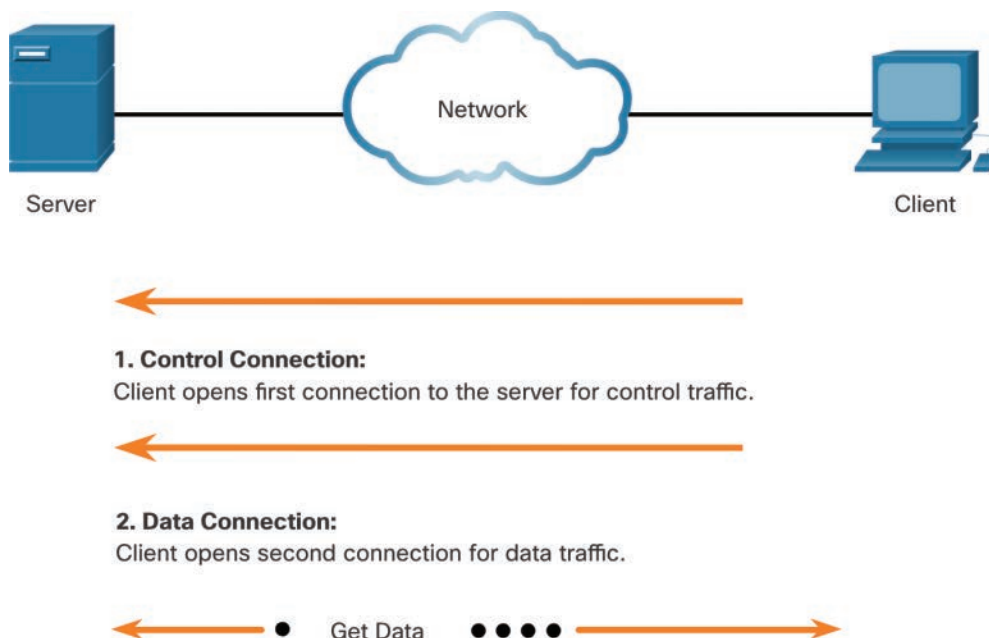


Figure 15-24 FTP Control and Data Connections

Server Message Block (15.5.2)

Server Message Block (SMB) is a client/server file sharing protocol that describes the structure of shared network resources, such as directories, files, printers, and serial ports. It is a request/response protocol. All SMB messages have a common format: a fixed-sized header followed by a variable-sized parameter and data component.

SMB functions carry out functions such as the following:

- Start, authenticate, and terminate sessions
- Control file and printer access
- Allow an application to send or receive messages to or from another device

SMB file sharing and print services have become the mainstay of Microsoft networking. With the introduction of the Windows 2000 software series, Microsoft changed the underlying structure for using SMB. In previous versions of Microsoft products, the SMB services used a non-TCP/IP protocol to implement name resolution. Windows 2000 and all subsequent Microsoft products use DNS naming, which allows TCP/IP protocols to directly support SMB resource sharing, as shown in Figure 15-25.

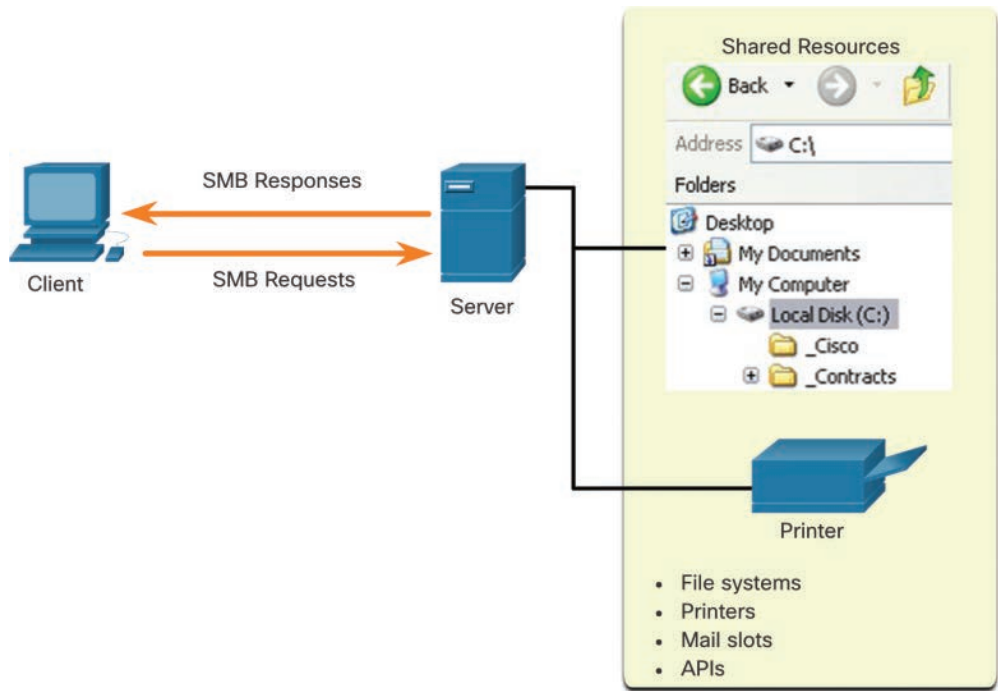


Figure 15-25 SMB Messages

Figure 15-26 shows the SMB file exchange process between Windows PCs.

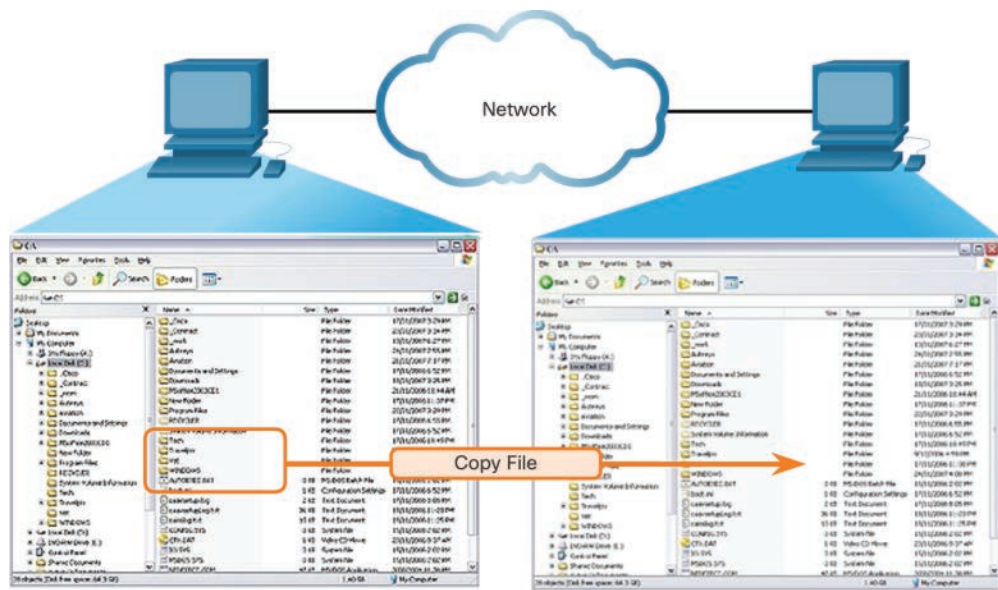


Figure 15-26 SMB File Exchange Between Windows PCs

Unlike with the file sharing supported by FTP, with SMB, clients establish long-term connections to servers. After a connection is established, the user of the client can access the resources on the server as if the resource were local to the client host.

The Linux and UNIX operating systems also provide a method of sharing resources with Microsoft networks, using a version of SMB called SAMBA. macOS also supports resource sharing with the SMB protocol.

**Interactive
Graphic****Check Your Understanding—File Sharing Services (15.5.3)**

Refer to the online course to complete this activity.

Summary

The following is a summary of the topics in the chapter and their corresponding online modules.

Application, Presentation, and Session

In the OSI model and the TCP/IP model, the application layer is the layer closest to the end user. Application layer protocols are used to exchange data between programs running on the source and destination hosts. The presentation layer has three primary functions: formatting, or presenting, data at the source device into a compatible form for receipt by the destination device; compressing data in a way that can be decompressed by the destination device; and encrypting data for transmission and decrypting data upon receipt. The session layer creates and maintains dialogues between source and destination applications. The session layer handles the exchange of information to initiate dialogues, keep them active, and restart sessions that are disrupted or idle for a long period of time. TCP/IP application layer protocols specify the format and control information necessary for many common internet communication functions. These protocols are used by both the source and destination devices during a session. The protocols implemented on the source and destination hosts must be compatible.

Peer-to-Peer

In the client/server model, the device requesting the information is called a client, and the device responding to the request is called a server. The client begins an exchange by requesting data from the server, which responds by sending one or more streams of data to the client. In a P2P network, two or more computers are connected on a network and can share resources without having a dedicated server. Every peer can function as both a server and a client. One computer might assume the role of server for one transaction while simultaneously serving as a client for another. P2P applications require that each end device provide a user interface and run a background service. Some P2P applications use a hybrid system in which resource sharing is decentralized, but the indexes that point to resource locations are stored in a centralized directory. Many P2P applications allow users to share pieces of files with each other at the same time. Clients use a small file called a torrent file to locate other users who have pieces that they need so that they can connect directly to them. This file also contains information about tracker computers that keep track of which users have what pieces of which files.

Web and Email Protocols

When a web address or URL is typed into a web browser, the web browser establishes a connection to the web service. The web service is running on the server that is using HTTP, which is a request/response protocol. When a client, typically a web

browser, sends a request to a web server, HTTP specifies the message types used for that communication. The three common message types are GET, POST, and PUT. For secure communication across the internet, HTTPS uses the same client request/server response process as HTTP, but the data stream is encrypted with SSL before being transported across the network. Email supports three separate protocols for operation: SMTP, POP, and IMAP. The application layer process that sends mail uses SMTP. A client retrieves email by using POP or IMAP. An SMTP message must have a message header and a message body. The message body can contain any amount of text, and the message header must have a properly formatted recipient email address and a sender address. An application can use POP to retrieve mail from a mail server. With POP, mail is downloaded from the server to the client and then deleted on the server. With IMAP, unlike with POP, when the user connects to an IMAP-capable server, copies of the messages are downloaded to the client application. The original messages are kept on the server until they are manually deleted.

IP Addressing Services

The DNS protocol matches resource names with the required numeric network addresses. DNS protocol communications use a message format for all types of client queries and server responses, error messages, and the transfer of resource record information between servers. DNS uses domain names to form a hierarchy. Each DNS server maintains a specific database file and is responsible for managing name-to-IP mappings for only a small portion of the entire DNS structure. Computer OSs use **nslookup** to allow the user to manually query the name servers to resolve a given hostname. DHCP for IPv4 automates the assignment of IPv4 addresses, subnet masks, gateways, and other IPv4 networking parameters. DHCPv6 provides similar services for IPv6 clients, except that it does not provide a default gateway address. When an IPv4 DHCP-configured device boots up or connects to the network, the client broadcasts a DHCP-DISCOVER message to identify any available DHCP servers on the network. A DHCP server replies with a DHCPOFFER message, which offers a lease to the client. DHCPv6 has a set of messages that are similar to those for DHCPv4. The DHCPv6 messages are SOLICIT, ADVERTISE, INFORMATION REQUEST, and REPLY.

File Sharing Services

An FTP client is an application which runs on a computer that is being used to push and pull data from an FTP server. The client establishes the first connection to the server for control traffic by using TCP port 21. The client establishes the second connection to the server for the actual data transfer by using TCP port 20. The client can download (pull) data from the server, or the client can upload (push) data to the server. The following are examples of the functions of SMB messages: start, authenticate, and terminate sessions; control file and printer access; and allow an application to send or receive messages to or from another device. Unlike with the file sharing

supported by FTP, with SMB clients establish long-term connections to servers. After a connection is established, the user of the client can access the resources on the server as if the resource were local to the client host.

Practice

The following activities provide practice with the topics introduced in this chapter. The lab is available in the companion *Introduction to Networks Labs & Study Guide (CCNAv7)* (ISBN 9780136634454). The Packet Tracer activity instructions are also provided in the *Labs & Study Guide*. The PKA files are available in the online course.



Lab

Lab 15.4.8: Observe DNS Resolution

Check Your Understanding Questions

Complete all the review questions listed here to test your understanding of the topics and concepts in this chapter. The appendix “Answers to ‘Check Your Understanding’ Questions” lists the answers.

1. Which protocol can be used to transfer messages from an email server to an email client?
 - a. SMTP
 - b. POP3
 - c. SNMP
 - d. SMB
2. When retrieving email messages, which protocol allows for easy, centralized storage and backup of emails that would be desirable for a small- to medium-sized business?
 - a. IMAP
 - b. POP
 - c. SMTP
 - d. HTTPS

3. Which application layer protocol is used to provide file sharing and print services to Microsoft applications?
 - a. HTTP
 - b. SMTP
 - c. DHCP
 - d. SMB
4. An author is uploading one chapter document from a personal computer to a file server of a book publisher. What role is the personal computer assuming in this network model?
 - a. client
 - b. master
 - c. server
 - d. slave
 - e. transient
5. Which statement is true about FTP?
 - a. A client can choose whether FTP should establish one or two connections.
 - b. A client can download data from or upload to a server.
 - c. FTP is a peer-to-peer application.
 - d. FTP does not provide reliability during data transmission.
6. A wireless host needs to request an IPv4 address. What protocol would be used to process the request?
 - a. FTP
 - b. HTTP
 - c. DHCP
 - d. ICMP
 - e. SNMP
7. Which TCP/IP model layer is closest to the end user?
 - a. application
 - b. internet
 - c. network access
 - d. transport

8. Which three protocols or standards are used at the application layer of the TCP/IP model? (Choose three.)
 - a. TCP
 - b. HTTP
 - c. MPEG
 - d. GIF
 - e. IP
 - f. UDP
9. Which protocol uses encryption?
 - a. DHCP
 - b. DNS
 - c. FTP
 - d. HTTPS
10. Why is DHCP for IPv4 preferred for use on large networks?
 - a. Large networks send more requests for domain-to-IP address resolution than do smaller networks.
 - b. DHCP uses a reliable transport protocol.
 - c. It prevents sharing of files that are copyrighted.
 - d. It is a more efficient way to manage IPv4 addresses than static address assignment.
 - e. Hosts on large networks require more IPv4 addressing configuration settings than do hosts on small networks.
11. Which two tasks can be performed by a local DNS server? (Choose two.)
 - a. providing IP addresses to local hosts
 - b. allowing data transfer between two network devices
 - c. mapping names to IP addresses for internal hosts
 - d. forwarding name resolution requests between servers
 - e. retrieving email messages
12. Which device is most likely to provide dynamic IPv4 addressing to clients on a home network?
 - a. a dedicated file server
 - b. a home router
 - c. an ISP DHCP server
 - d. a DNS server

13. What part of the URL `http://www.cisco.com/index.html` represents the top-level DNS domain?
 - a. `.com`
 - b. `www`
 - c. `http`
 - d. `index`
14. What are two characteristics of the application layer of the TCP/IP model? (Choose two.)
 - a. responsible for logical addressing
 - b. responsible for physical addressing
 - c. responsible for the creation and maintenance of dialogues between source and destination applications
 - d. closest to the end user
 - e. responsible for establishing window size
15. What message type does an HTTP client use to request data from a web server?
 - a. GET
 - b. POST
 - c. PUT
 - d. ACK

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Network Security Fundamentals

Objectives

Upon completion of this chapter, you will be able to answer the following questions:

- Why are basic security measures necessary on network devices?
- How do you identify security vulnerabilities?
- How do you identify general mitigation techniques?
- How do you configure network devices with device hardening features to mitigate security threats?

Key Terms

This chapter uses the following key terms. You can find the definitions in the glossary at the end of the book.

virus page 546

worm page 547

Trojan horse page 547

reconnaissance attack page 547

access attack page 547

denial of service (DoS) attack page 547

internet query page 548

ping sweep page 548

port scan page 548

AAA (authentication, authorization, and accounting) page 555

stateful packet inspection (SPI) page 557

brute-force attack page 560

Introduction (16.0)

You may have already set up a network, or you may be getting ready to do so. Here is something to think about: Setting up a network without securing it is like opening all the doors and windows to your home and then going on vacation. Anyone could come by, gain entry, steal or break items, or just make a mess. As news articles indicate all the time, it is possible to break into *any* network! As a network administrator, it is part of your job to make it difficult for threat actors to gain access to your network. This chapter provides an overview of the types of network attacks and what you can do to reduce a threat actor's chances of succeeding. It also has Packet Tracer activities to let you practice some basic techniques for network security. If you have a network, but it is not as secure as possible, you should read this chapter right now!

Security Threats and Vulnerabilities (16.1)

This section provides an overview the various types of network security threats and vulnerabilities.

Types of Threats (16.1.1)

Wired and wireless computer networks are essential to everyday activities. Individuals and organizations depend on their computers and networks. Intrusion by an unauthorized person can result in costly network outages and loss of work. Attacks on a network can be devastating and can result in a loss of time and money due to damage or theft of important information or assets.

Intruders can gain access to a network through software vulnerabilities, through hardware attacks, or by guessing someone's username and password. Intruders who gain access by modifying software or exploiting software vulnerabilities are called *threat actors*.

After a threat actor gains access to a network, four types of threats may arise:

- **Information theft:** This type of threat involves breaking into a computer to obtain confidential information. Information can be used or sold for various purposes. An example is stealing an organization's proprietary information, such as research and development data.
- **Data loss and manipulation:** This type of threat involves breaking into a computer to destroy or alter data records. An example of data loss is a threat actor sending a virus that reformats a computer hard drive. An example of data manipulation is breaking into a records system to change information, such as the price of an item.

- **Identity theft:** This type of threat is a form of information theft in which personal information is stolen for the purpose of taking over someone’s identity. Using this information, a threat actor can obtain legal documents, apply for credit, and make unauthorized online purchases. Identity theft is a growing problem that costs billions of dollars per year.
- **Disruption of service:** This type of threat involves preventing legitimate users from accessing services to which they are entitled. Examples include denial-of-service (DoS) attacks on servers, network devices, or network communications links.

Types of Vulnerabilities (16.1.2)

Vulnerability refers to the degree of weakness in a network or device. Some degree of vulnerability is inherent in routers, switches, desktops, servers, and even security devices. Typically, the network devices under attack are the endpoints, such as servers and desktop computers.

There are three primary sources of vulnerabilities or weaknesses: technological, configuration, and security policy. All three of these sources of vulnerabilities can leave a network or device open to various attacks, including malicious code attacks and network attacks. Tables 16-1 through 16-3 describe examples of the vulnerabilities in each category.

Table 16-1 Technological Vulnerabilities

Vulnerability	Description
TCP/IP protocol weakness	Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), and Internet Control Message Protocol (ICMP) are inherently insecure.
	Simple Network Management Protocol (SNMP) and Simple Mail Transfer Protocol (SMTP) are related to the inherently insecure structure on which TCP was designed.
Operating system weakness	Each operating system has security problems that must be addressed.
	UNIX, Linux, macOS, Mac OS X, Windows Server 2012, Windows 7, and Windows 8 are documented in the Computer Emergency Response Team (CERT) archives at http://www.cert.org .
Network equipment weakness	Various types of network equipment, such as routers, firewalls, and switches, have security weaknesses that must be recognized and protected against. Their weaknesses include password protection, lack of authentication, routing protocols, and firewall holes.

Table 16-2 Configuration Vulnerabilities

Vulnerability	Description
Unsecured user accounts	User account information may be transmitted insecurely across the network, exposing usernames and passwords to threat actors.
System accounts with easily guessed passwords	This common problem is the result of poorly created user passwords.
Misconfigured internet services	When JavaScript is turned on in a web browser, threat actors may be able to access untrusted sites. Other potential sources of weakness include misconfigured terminal services, FTP, or web servers (such as Microsoft Internet Information Services [IIS] and Apache HTTP server).
Unsecured default settings in products	Many products have default settings that create or enable holes in security.
Misconfigured network equipment	Misconfigurations of equipment can cause significant security problems. For example, misconfigured access lists, routing protocols, or SNMP community strings can create or enable holes in security.

Table 16-3 Policy Vulnerabilities

Vulnerability	Description
Lack of written security policy	A security policy cannot be consistently applied or enforced if it is not written down.
Politics	Political battles and turf wars can make it difficult to implement a consistent security policy.
Lack of authentication continuity	Poorly chosen, easily cracked, or default passwords can allow unauthorized access to a network.
Logical access controls not applied	Inadequate monitoring and auditing allow attacks and unauthorized use to continue, wasting company resources. It could result in legal action against or termination of IT technicians, IT management, or even company leaders who allow these unsafe conditions to persist.
Software and hardware installation and changes that do not follow policy	Unauthorized changes to the network topology and installation of unapproved applications create or enable holes in security.
Lack of a disaster recovery plan	Without a disaster recovery plan, chaos, panic, and confusion may occur when a natural disaster occurs or a threat actor attacks the enterprise.

Physical Security (16.1.3)

An important vulnerable area of the network to consider is the physical security of devices. If network resources can be physically compromised, a threat actor can deny the use of network resources.

The four classes of physical threats are as follows:

- **Hardware threats:** This includes physical damage to servers, routers, switches, the cabling plant, and workstations.
- **Environmental threats:** This includes temperature extremes (too hot or too cold) or humidity extremes (too damp or too dry).
- **Electrical threats:** This includes voltage spikes, insufficient supply voltage (brownouts), unconditioned power (noise), and total power loss.
- **Maintenance threats:** This includes poor handling of key electrical components (electrostatic discharge), lack of critical spare parts, poor cabling, and poor labeling.

An organization needs to create and implement a good plan for physical security to address these issues. Figure 16-1 shows an example of a physical security plan, which includes taking the following actions:

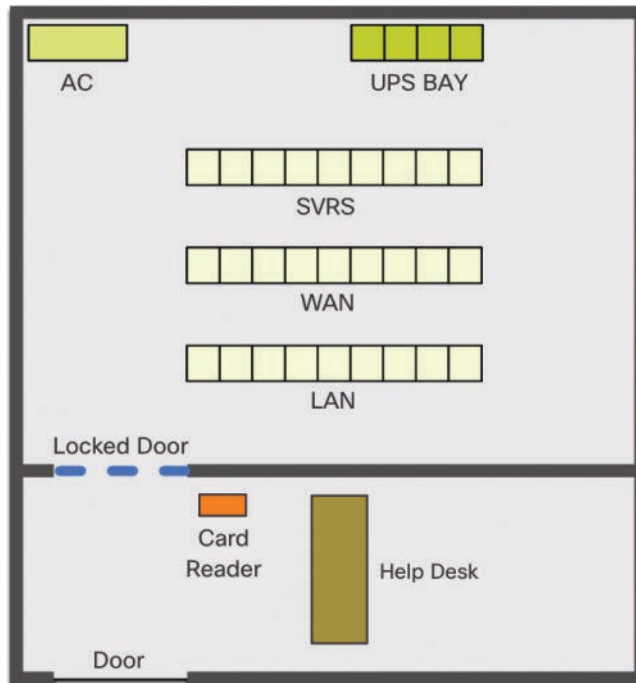


Figure 16-1 Plan Physical Security to Limit Damage to Equipment

- Secure the computer room.
- Implement physical security to limit damage to equipment.
- Lock up equipment and prevent unauthorized access from the doors, ceiling, raised floor, windows, ducts, and vents.
- Monitor and control closet entry with electronic logs.
- Use security cameras.

**Interactive
Graphic****Check Your Understanding—Security Threats and Vulnerabilities (16.1.4)**

Refer to the online course to complete this activity.

Network Attacks (16.2)

Many different types of network attacks may occur, using a variety of different methods. The previous section explains the types of network threats and the vulnerabilities that make threats possible. This section goes into more detail about how threat actors gain access to network or restrict authorized users from having access. It discusses different categories of network attacks, such as malware, reconnaissance attacks, access attacks, and denial-of-service attacks, and provides examples of each.

Types of Malware (16.2.1)

Malware is short for *malicious software*. It is code or software specifically designed to damage, disrupt, steal, or inflict “bad,” or illegitimate, action on data, hosts, or networks. Viruses, worms, and Trojan horses are types of malware.

Viruses

A computer *virus* is a type of malware that propagates by inserting a copy of itself into, and becoming part of, another program. It spreads from one computer to another, leaving infection as it travels. Viruses can range in severity from causing mildly annoying effects to damaging data or software and causing denial-of-service (DoS) conditions. Almost all viruses are attached to executable files, which means the virus may exist on a system but be inactive and unable to spread until a user runs or opens the malicious host file or program. When the host code is executed, the viral code is executed as well. Normally, the host program keeps functioning after a virus infects it. However, some viruses overwrite other programs with copies of themselves, destroying the host programs altogether. A virus spreads when the software or document it is attached to is transferred from one computer to another using the network, a disk, file sharing, or infected email attachments.

Worms

Computer *worms* are similar to viruses in that they replicate functional copies of themselves and can cause the same type of damage. In contrast to viruses, which require the spreading of an infected host file, worms are standalone software and do not require a host program or human help to propagate. A worm does not need to attach to a program to infect a host and enter a computer through a vulnerability in the system. Worms take advantage of system features to travel through the network unaided.

Trojan Horses

A *Trojan horse* is a type of malware named after the wooden horse the Greeks used to infiltrate Troy. It is a harmful piece of software that looks legitimate. Users are typically tricked into loading and executing a Trojan on their systems. After it is activated, it can achieve any number of attacks on the host, from irritating the user (by presenting excessive pop-up windows or changing the desktop) to damaging the host (deleting files, stealing data, or activating and spreading other malware, such as viruses). Trojan horses are also known to create backdoors to give malicious users access to the system.

Unlike viruses and worms, Trojan horses do not reproduce by infecting other files. They self-replicate. Trojan horses must spread through user interaction; for example, a user may need to open an email attachment or download and run a file from the internet.

Interactive Graphic

Animated Explanation of the Three Types of Malware

Go to the online course to view an animated explanation of the three types of malware.

Reconnaissance Attacks (16.2.2)

In addition to being threatened by malicious code attacks, it is also possible for networks to fall prey to various network attacks. Network attacks can be classified into three major categories:

- **Reconnaissance attacks:** These attacks involve discovery and mapping of systems, services, or vulnerabilities.
- **Access attacks:** These attacks involve unauthorized manipulation of data, system access, or user privileges.
- **Denial-of-service (DoS) attacks:** These attacks involve disabling or corruption of networks, systems, or services.

For reconnaissance attacks, external threat actors can use internet tools such as the **nslookup** and **whois** utilities, to easily determine the IP address space assigned to a

given corporation or entity. After the IP address space is determined, a threat actor can then ping the publicly available IP addresses to identify the addresses that are active. To help automate this step, a threat actor may use a ping sweep tool, such as **fping** or **gping**, to systematically ping all network addresses in a given range or subnet. This is similar to going through a section of a telephone book and calling each number to see who answers.

- **Internet queries:** The threat actor looks for initial information about a target. Various tools can be used, including Google search, the websites of organizations, whois, and more.
- **Ping sweeps:** The threat actor initiates a ping sweep to determine which IP addresses are active.
- **Port scans:** The threat actor performs a port scan on the discovered active IP addresses.

Interactive Graphic

Animations of Internet Queries, Ping Sweeps, and Port Scans

Go to the online course to view animations of internet queries, ping sweeps, and port scans.

Access Attacks (16.2.3)

Access attacks exploit known vulnerabilities in authentication services, FTP services, and web services to gain entry to web accounts, confidential databases, and other sensitive information. An access attack allows individuals to gain unauthorized access to information that they have no right to view. Access attacks can be classified into four types: password attacks, trust exploitation, port redirection, and man-in-the-middle attacks.

Password Attacks

Threat actors can implement password attacks using several different methods:

- Brute-force attacks
- Trojan horse attacks
- Packet sniffers

Trust Exploitation

In a trust exploitation attack, a threat actor uses unauthorized privileges to gain access to a system and may compromise the target. In Figure 16-2, System A

trusts System B. System B trusts everyone. The threat actor wants to gain access to System A. Therefore, the threat actor compromises System B first and then can use System B to attack System A.

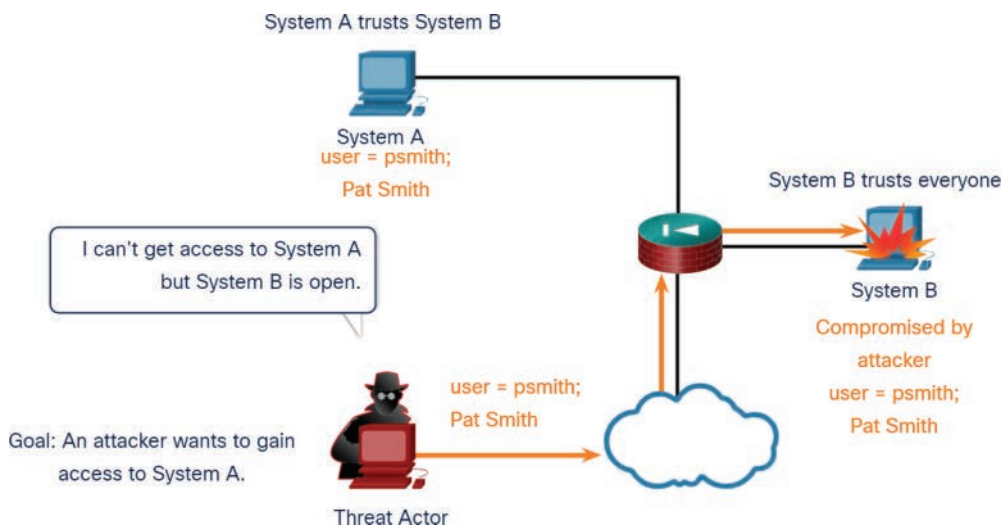


Figure 16-2 Example of a Trust Exploitation

Port Redirection

In a port redirection attack, a threat actor uses a compromised system as a base for attacks against other targets. The example in Figure 16-3 shows a threat actor using SSH (port 22) to connect to a compromised Host A. Host A is trusted by Host B and, therefore, the threat actor can use Telnet (port 23) to access it.

Man-in-the-Middle

In a man-in-the-middle attack, the threat actor is positioned in between two legitimate entities in order to read or modify the data that passes between the two parties. Figure 16-4 shows an example of a man-in-the-middle attack where the numbers relate to the following steps:

- Step 1.** When a victim requests a web page, the request is directed to the threat actor's computer.
- Step 2.** The threat actor's computer receives the request and retrieves the real page from the legitimate website.
- Step 3.** The threat actor can alter the legitimate web page and make changes to the data.
- Step 4.** The threat actor forwards the requested page to the victim.

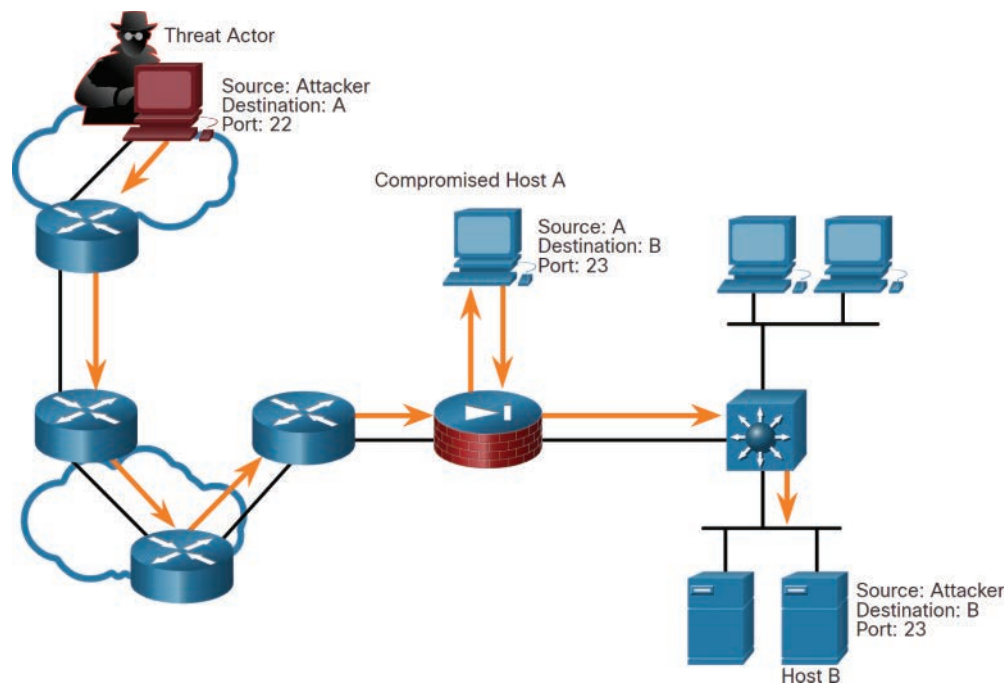


Figure 16-3 Example of Port Redirection

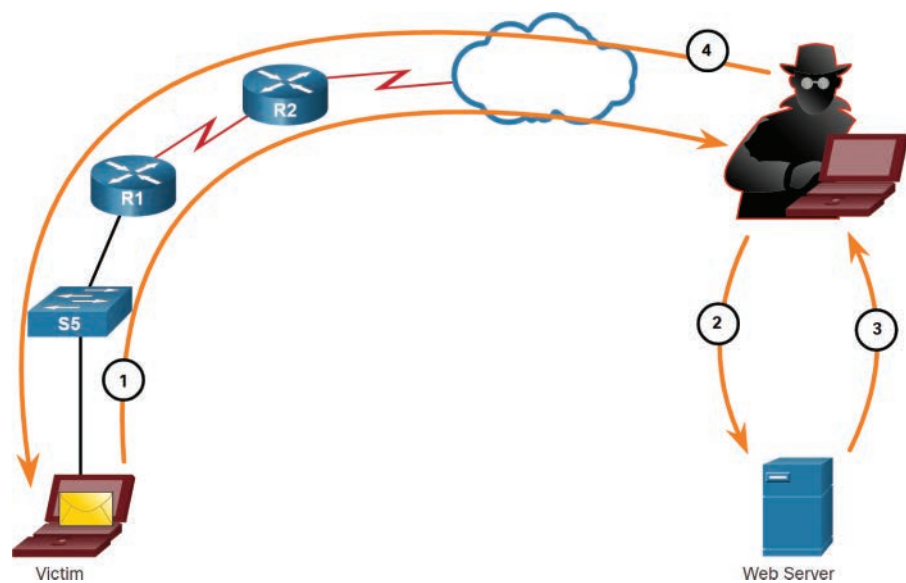


Figure 16-4 Example of a Man-in-the-Middle Attack

Denial of Service Attacks (16.2.4)

Denial-of-service (DoS) attacks are the most publicized form of attack and among the most difficult to eliminate. However, because of their ease of implementation and potentially significant damage, DoS attacks deserve special attention from security administrators.

DoS attacks take many forms. Ultimately, they prevent authorized people from using a service by consuming system resources. To help prevent DoS attacks, it is important to stay up to date with the latest security updates for operating systems and applications.

DoS Attack

DoS attacks are a major risk because they interrupt communication and cause significant loss of time and money. These attacks are relatively simple to conduct, even by an unskilled threat actor. Figure 16-5 shows an example of a DoS attack.

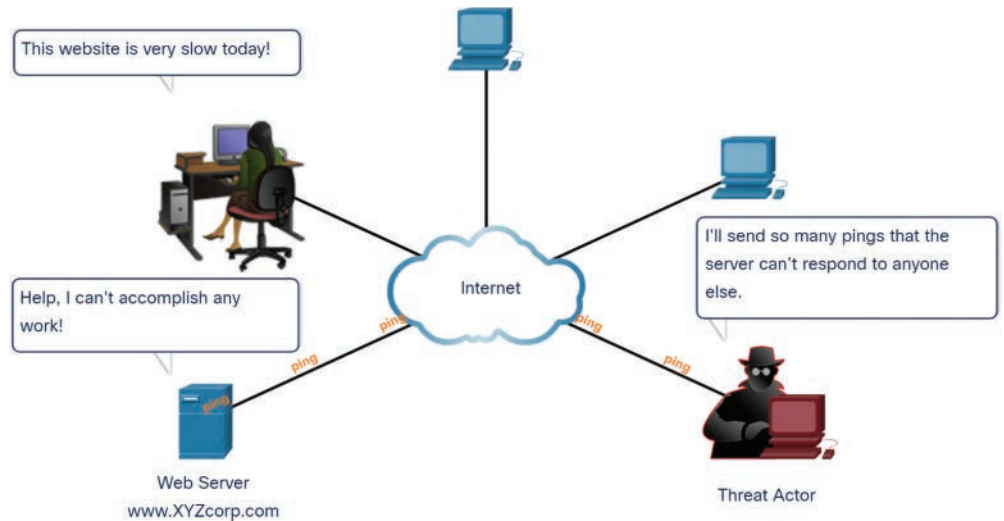


Figure 16-5 Example of a DoS Attack

DDoS Attack

A distributed denial-of-service (DDoS) attack is similar to a DoS attack, but it originates from multiple, coordinated sources. For example, in Figure 16-6, a threat actor builds a network of infected hosts, known as *zombies*, to form a *botnet*. The threat actor uses a command-and-control (CnC) program to instruct the botnet of zombies to carry out a DDoS attack.

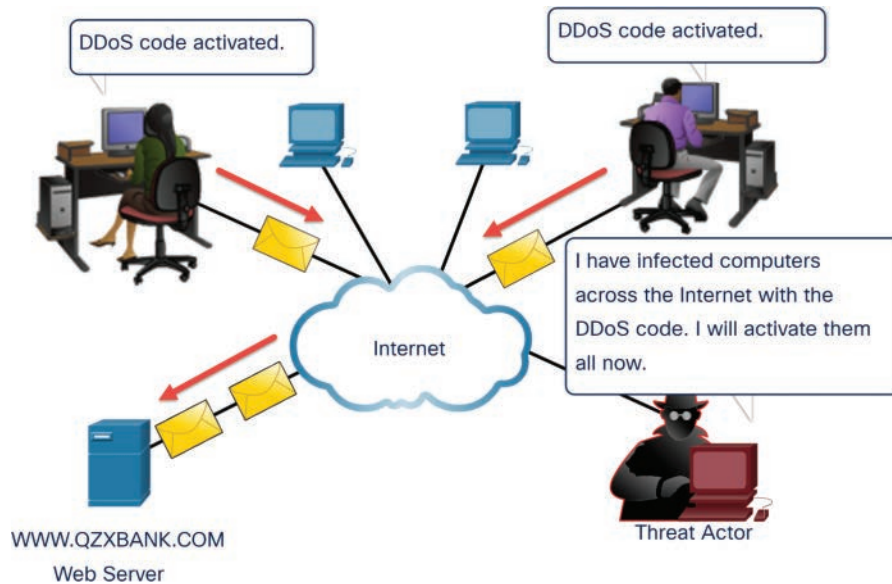


Figure 16-6 Example of a DDoS Attack

**Interactive
Graphic**

Check Your Understanding—Network Attacks (16.2.5)

Refer to the online course to complete this activity.



Lab—Research Network Security Threats (16.2.6)

In this lab, you will complete the following objectives:

- Part 1: Explore the SANS Website
- Part 2: Identify Recent Network Security Threats
- Part 3: Detail a Specific Network Security Threat

Network Attack Mitigations (16.3)

An important aspect of being a network professional is to take the necessary precautions to prevent network attacks before they happen. Now that you know more about how threat actors can break into networks, you need to understand what to do to prevent such unauthorized access. This section details several actions you can take to make a network more secure.

The Defense-in-Depth Approach (16.3.1)

To mitigate network attacks, you must first secure devices, including routers, switches, servers, and hosts. Most organizations use a defense-in-depth approach (also known as a *layered approach*) to security. This requires a combination of networking devices and services working in tandem.

Consider the network in Figure 16-7. Several security devices and services have been implemented to protect its users and assets against TCP/IP threats.

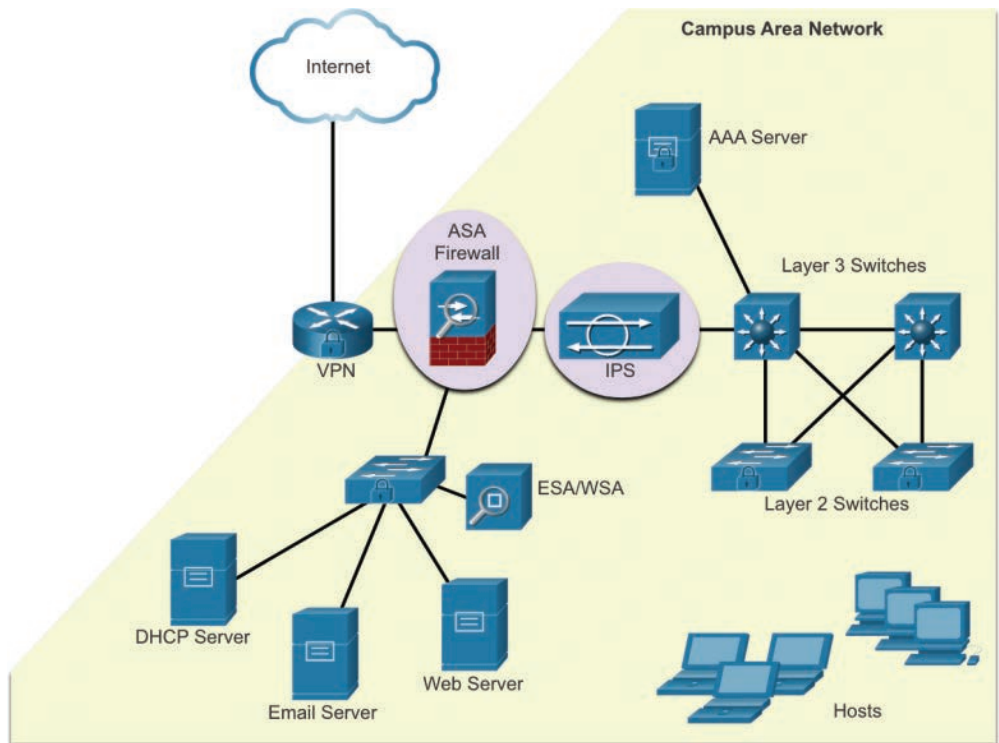


Figure 16-7 Example of a Defense-in-Depth Topology

All network devices, including the router and switches, are also hardened, as indicated by the padlocks on their respective icons. This indicates that they have been secured to prevent threat actors from gaining access and tampering with the devices.

Keep Backups (16.3.2)

Backing up device configurations and data is one of the most effective ways of protecting against data loss. A data backup stores a copy of the information on a computer to removable backup media that can be kept in a safe place. Infrastructure

devices should have backups of configuration files and IOS images on an FTP or similar file server. If a computer or a router hardware fails, the data or configuration can be restored using the backup copy.

Backups should be performed on a regular basis, as identified in the security policy. Data backups are usually stored offsite to protect the backup media in case anything happens to the main facility. Windows hosts have a backup and restore utility. It is important for users to back up their data to another drive or to a cloud-based storage provider.

Table 16-4 describes some important backup considerations.

Table 16-4 Backup Considerations

Consideration	Description
Frequency	Perform backups on a regular basis, as identified in the security policy. Full backups can be time-consuming, so perform monthly or weekly backups with frequent partial backups of changed files.
Storage	Always validate backups to ensure the integrity of the data and validate the file restoration procedures.
Security	Backups should be transported to an approved offsite storage location on a daily, weekly, or monthly rotation, as required by the security policy.
Validation	Backups should be protected using strong passwords that are required to restore the data.

Upgrade, Update, and Patch (16.3.3)

Keeping up to date with the latest developments can lead to more effective defense against network attacks. As new malware is released, enterprises need to keep current with the latest versions of antivirus software.

The most effective way to mitigate a worm attack is to download security updates from the operating system vendor and patch all vulnerable systems. Administering numerous systems involves creating a standard software image (of operating system and accredited applications that are authorized for use on client systems) that is deployed on new or upgraded systems. However, security requirements change, and deployed systems may need to have updated security patches installed.

One solution to the management of critical security patches is to make sure all end systems automatically download updates, as shown for Windows 10 in Figure 16-8, to ensure that security patches are automatically downloaded and installed without user intervention.

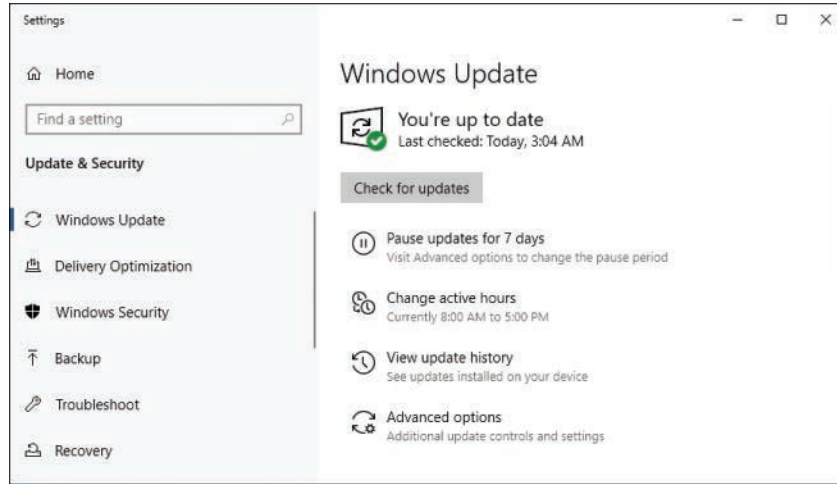


Figure 16-8 Windows 10 Update

Authentication, Authorization, and Accounting (16.3.4)

All network devices should be securely configured to provide only authorized individuals with access. *Authentication, authorization, and accounting* (AAA, or “triple A”) network security services provide the primary framework to set up access control on network devices.

AAA makes it possible to control who is permitted to access a network (authenticate) and what actions they can perform while accessing the network (authorize), as well as to make a record of what was done while they were there (accounting).

The concept of AAA is similar to the use of a credit card. The credit card identifies who can use it and how much that user can spend, and it keeps an account of what items the user spent money on, as shown in Figure 16-9.

Firewalls (16.3.5)

A firewall is one of the most effective security tools available for protecting users from external threats. A firewall protects computers and networks by preventing undesirable traffic from entering internal networks.

Network firewalls reside between two or more networks, control the traffic between them, and help prevent unauthorized access. For example, the top topology in Figure 16-10 illustrates how a firewall enables traffic from an internal network host to exit the network and return to the inside network. The bottom topology illustrates how traffic initiated by the outside network (that is, the internet) is denied access to the internal network.

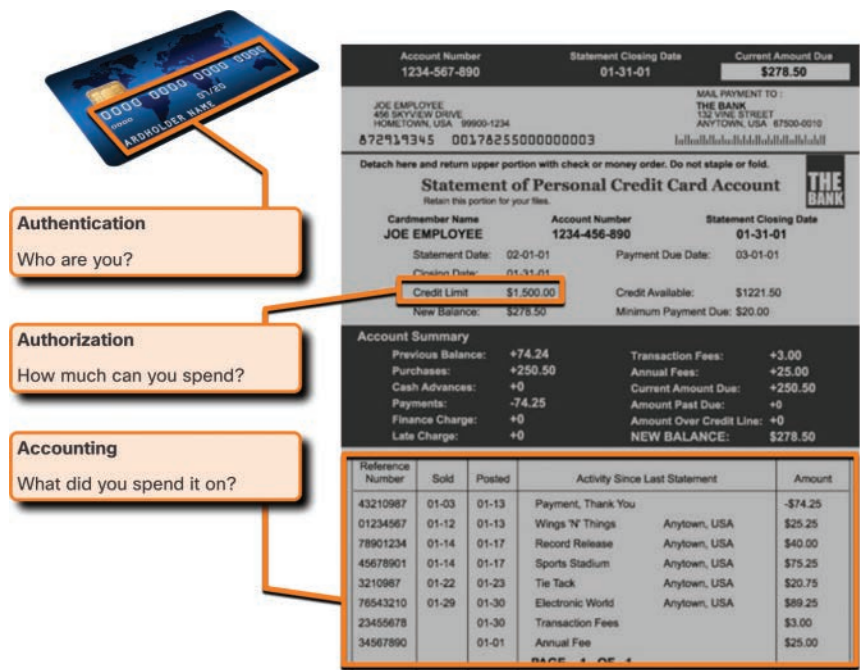


Figure 16-9 AAA Credit Card Bill Analogy

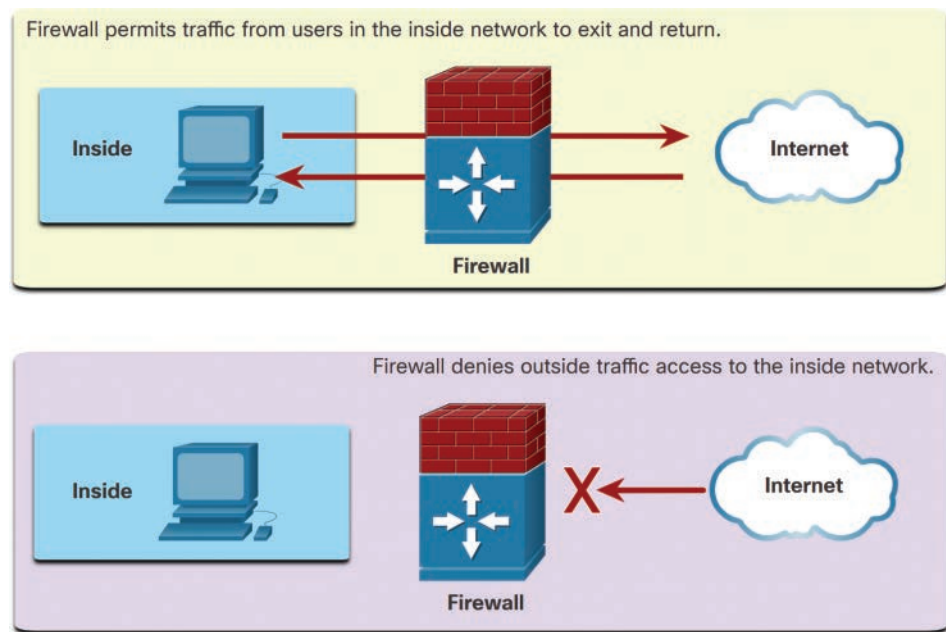


Figure 16-10 Firewall Operation

A firewall may be able to allow outside users controlled access to specific services. For example, servers accessible to outside users are usually located on a special network referred to as the demilitarized zone (DMZ), as shown in Figure 16-11. The DMZ enables a network administrator to apply specific policies for hosts connected to that network.

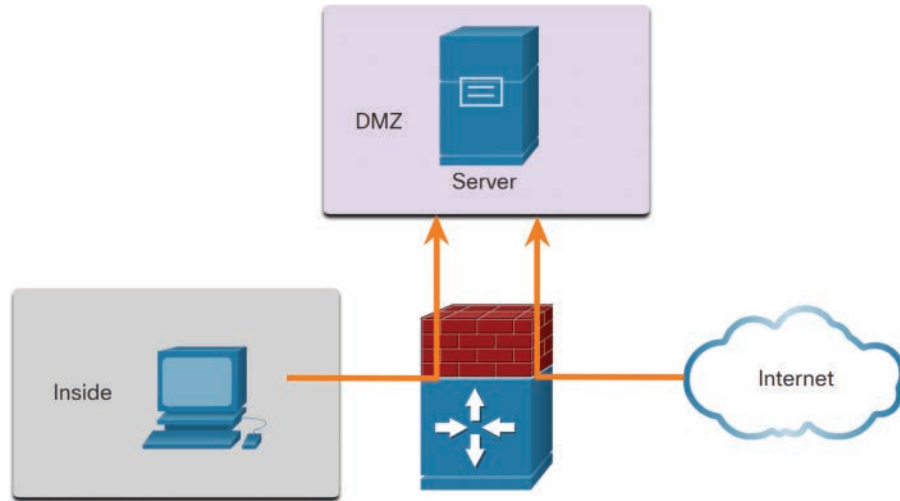


Figure 16-11 Firewall Topology with DMZ

Types of Firewalls (16.3.6)

Firewall products come packaged in various forms. These products use different techniques for determining what will be permitted or denied access to a network. They include the following:

- **Packet filtering:** Prevents or allows access based on IP addresses or MAC addresses.
- **Application filtering:** Prevents or allows access by specific application types based on port numbers.
- **URL filtering:** Prevents or allows access to websites based on specific URLs or keywords.
- **Stateful packet inspection (SPI):** Ensures that incoming packets are legitimate responses to requests from internal hosts. Unsolicited packets are blocked unless permitted specifically. SPI can also include the capability to recognize and filter out specific types of attacks, such as denial-of-service (DoS) attacks.

Endpoint Security (16.3.7)

An endpoint, or host, is an individual computer system or device that acts as a network client. Common endpoints are laptops, desktops, servers, smartphones, and tablets. Securing endpoint devices is one of the most challenging jobs of a network administrator because human nature creates complications. A company must have well-documented policies in place, and employees must be trained on the rules and proper use of the network. Policies often include the use of antivirus software and host intrusion prevention. More comprehensive endpoint security solutions rely on network access control.

Interactive Graphic

Check Your Understanding—Network Attack Mitigation (16.3.8)

Refer to the online course to complete this activity.

Device Security (16.4)

Devices on a network require special security. You probably already have a password for your computer, smartphone, or tablet. Is it as strong as it could be? Are you using other tools to enhance the security of your devices? This section discusses how to protect network devices, including end devices and intermediary devices, with proper security measures.

Cisco AutoSecure (16.4.1)

When a new operating system is installed on a device, the security settings are set to the default values. In most cases, this level of security is inadequate. The Cisco AutoSecure feature can be used to assist in securing Cisco routers, as shown in Example 16-1.

Example 16-1 Configuring Cisco AutoSecure

```
Router# auto secure
      --- AutoSecure Configuration ---
*** AutoSecure configuration enhances the security of
the router but it will not make router absolutely secure
from all security attacks ***
```

In addition, there are some simple security guidelines apply to most operating systems:

- Default usernames and passwords should be changed immediately.
- Access to system resources should be restricted to only the individuals who are authorized to use those resources.

- Any unnecessary services and applications should be turned off and uninstalled when possible.

Often, devices shipped from the manufacturer have been sitting in a warehouse for a period of time and do not have the most up-to-date patches installed. It is important to update any software and install any security patches prior to implementation.

Passwords (16.4.2)

To protect network devices, it is important to use strong passwords. Here are standard guidelines to follow:

- Use a password length of at least 8 characters—and preferably 10 or more characters. A longer password is a more secure password.
- Make passwords complex. Include a mix of uppercase and lowercase letters, numbers, symbols, and spaces, if allowed.
- Avoid passwords based on repetition, common dictionary words, letter or number sequences, usernames, relative or pet names, biographical information such as birthdates, ID numbers, ancestor names, or other easily identifiable pieces of information.
- Deliberately misspell a password. For example, Smith = Smyth = 5mYth or Security = 5ecur1ty.
- Change passwords often. This way, if a password is unknowingly compromised, the window of opportunity for the threat actor to use the password is limited.
- Do not write down passwords in obvious places such as on the desk or monitor.

Tables 16-5 and 16-6 show examples of strong and weak passwords.

Table 16-5 Weak Passwords

Weak Password	Why It Is Weak
secret	Simple dictionary password
smith	Maiden name of mother
toyota	Make of a car
bob1967	Name and birthday of the user
Blueleaf23	Simple words and numbers

Table 16-6 Strong Passwords

Strong Password	Why It Is Strong
b67n42d39c	Combines alphanumeric characters
12^h u4@1p7	Combines alphanumeric characters and symbols and includes a space

On Cisco routers, leading spaces are ignored in passwords, but spaces after the first character are not. Therefore, one method to create a strong password is to use spaces in a phrase consisting of many words. This is called a passphrase. A passphrase is often easier to remember than a simple password. It is also longer and harder to guess.

Additional Password Security (16.4.3)

Strong passwords are useful only if they are secret. Several steps can help ensure that passwords remain secret on a Cisco router and switch, including these:

- Encrypt all plaintext passwords.
- Set a minimum acceptable password length.
- Deter brute-force password guessing attacks.
- Disable an inactive privileged EXEC mode access after a specified amount of time.

As shown in the sample configuration in Example 16-2, the **service password-encryption** global configuration command prevents unauthorized individuals from viewing plaintext passwords in the configuration file. This command encrypts all plaintext passwords. Notice in the example that the password `cisco` has been encrypted as `03095A0F034F`. (Keep in mind that `cisco` would not be a secure password; it is used here for illustration only.)

To ensure that all configured passwords are a minimum of a specified length, use the **security passwords min-length *length*** command in global configuration mode. In Example 16-2, any new password configured would need to have a minimum length of eight characters.

Threat actors may use password cracking software to conduct a *brute-force attack* on a network device. Such an attack repeatedly attempts to guess the valid passwords until one works. Use the **login block-for *number-of attempts* *attempts* within *seconds*** global configuration command to deter this type of attack. In Example 16-2, the **login block-for 120 attempts 3 within 60** command blocks vty login attempts for 120 seconds if there are three failed login attempts within 60 seconds.

Network administrators might become distracted and accidentally leave a privileged EXEC mode session open on a terminal. This could enable an internal threat actor access to change or erase the device configuration.

By default, Cisco routers log out an EXEC session after 10 minutes of inactivity. However, you can reduce this setting by using the `exec-timeout minutes seconds` line console configuration command. This command can be applied on line console, auxiliary, and vty lines. In Example 16-2, `exec-timeout 5 30` tells the Cisco device to automatically disconnect an inactive user on a vty line after the user has been idle for 5 minutes and 30 seconds.

Example 16-2 Configuring Additional Password Security on a Cisco Router

```
Router(config)# service password-encryption
Router(config)# security password min-length 8
Router(config)# login block-for 120 attempts 3 within 60
Router(config)# line vty 0 4
Router(config-line)# password cisco
Router(config-line)# exec-timeout 5 30
Router(config-line)# transport input ssh
Router(config-line)# end
Router#
Router# show running-config | section line vty
line vty 0 4
  password 7 03095A0F034F
  exec-timeout 5 30
  login
Router#
```

Enable SSH (16.4.4)

Telnet simplifies remote device access, but it is not secure. Data contained in a Telnet packet is transmitted unencrypted. For this reason, it is highly recommended to enable Secure Shell (SSH) on devices for secure remote access.

It is possible to configure a Cisco device to support SSH by using the following six steps:



- Step 1.** Configure a unique device hostname other than the default.
- Step 2.** Configure the IP domain name of the network by using the global configuration mode command `ip domain name name`.

- Step 3.** Generate a key to encrypt SSH traffic by using the global configuration command **crypto key generate rsa general-keys modulus *bits***. The modulus *bits* determines the size of the key and can be configured to a value between 360 bits and 2048 bits. The larger the bit value, the more secure the key. However, with larger bit values, it also takes longer to encrypt and decrypt information. The minimum recommended modulus length is 1024 bits.
- Step 4.** Verify or create a local database entry by using the **username** global configuration command. In the example, the parameter **secret** is used so that the password will be encrypted using MD5.
- Step 5.** Use the **login local** line configuration command to authenticate the vty line against the local database.
- Step 6.** Enable vty inbound SSH sessions. By default, no input session is allowed on vty lines. You can specify multiple input protocols including Telnet and SSH by using the **transport input {ssh | telnet}** command.

In Example 16-3, router R1 is configured in the span.com domain. This information is used along with the bit value specified in the **crypto key generate rsa general-keys modulus** command to create an encryption key. Next, a local database entry for a user named Bob is created. Finally, the vty lines are configured to authenticate against the local database and to accept only incoming SSH sessions.

Example 16-3 Configuring SSH Access on a Cisco Router

```
Router# configure terminal
Router(config)# hostname R1
R1(config)# ip domain name span.com
R1(config)# crypto key generate rsa general-keys modulus 1024
The name for the keys will be: R1.span.com % The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
Dec 13 16:19:12.079: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#
R1(config)# username Bob secret cisco
R1(config)# line vty 0 4
R1(config-line)# login local
R1(config-line)# transport input ssh
R1(config-line)# exit
R1(config)#
```

Disable Unused Services (16.4.5)

Cisco routers and switches start with a list of active services that may or may not be required in the network. It is a best practice to disable any unused services to preserve system resources, such as CPU cycles and RAM, and prevent threat actors from exploiting these services. The type of services that are on by default vary depending on the IOS version. For example, IOS XE typically has only HTTPS and DHCP ports open. You can verify this with the **show ip ports all** command, as shown in Example 16-4.

Example 16-4 Showing Open Ports on IOS XE

```
Router# show ip ports all
Proto Local Address      Foreign Address    State      PID/Program Name
TCB   Local Address      Foreign Address    (state)
tcp   :::443              :::*               LISTEN     309/[IOS]HTTP CORE
tcp   *:443               *:                 LISTEN     309/[IOS]HTTP CORE
udp   *:67                0.0.0.0:          387/[IOS]DHCPD Receive
Router#
```

IOS versions prior to IOS XE use the **show control-plane host open-ports** command. You might see this command on older devices. The output is similar to the output shown in Example 16-4. However, notice that this older router has an insecure HTTP server and Telnet running. Both of these services should be disabled. As shown in Example 16-5, you can disable HTTP with the **no ip http server** global configuration command. You disable Telnet by specifying only SSH in the line configuration command: **transport input ssh**.

Example 16-5 Showing Open Ports on IOS Versions Prior to IOS XE

```
Router# show control-plane host open-ports
Active internet connections (servers and established)
Prot      Local Address      Foreign Address    Service      State
tcp       *:23               *:0                Telnet       LISTEN
tcp       *:80               *:0                HTTP CORE    LISTEN
udp       *:67               *:0                DHCPD Receive LISTEN
Router# configure terminal
Router(config)# no ip http server
Router(config)# line vty 0 15
Router(config-line)# transport input ssh
```

Packet Tracer
Activity**Packet Tracer—Configure Secure Passwords and SSH (16.4.6)**

The network administrator has asked you to prepare RTA and SW1 for deployment. Before they can be connected to the network, security measures must be enabled.

**Lab—Configure Network Devices with SSH (16.4.7)**

In this lab, you will complete the following objectives:

- Part 1: Configure Basic Device Settings
 - Part 2: Configure the Router for SSH Access
 - Part 3: Configure the Switch for SSH Access
 - Part 4: SSH from the CLI on the Switch
-

Summary

The following is a summary of the topics in the chapter and their corresponding online modules.

Security Threats and Vulnerabilities

Attacks on a network can be devastating and can result in lost time and money due to damage or theft of important information or assets. Intruders who gain access by modifying software or exploiting software vulnerabilities are threat actors. After a threat actor gains access to a network, four types of threats may arise: information theft, data loss and manipulation, identity theft, and disruption of service. There are three primary vulnerabilities or weaknesses: technological, configuration, and security policy. The four classes of physical threats are hardware, environmental, electrical, and maintenance.

Network Attacks

Malware, which is short for malicious software, is code or software specifically designed to damage, disrupt, steal, or inflict “bad,” or illegitimate, action on data, hosts, or networks. Viruses, worms, and Trojan horses are types of malware. Network attacks can be classified into three major categories: reconnaissance, access, and denial of service. The four classes of physical threats are hardware, environmental, electrical, and maintenance. The three types of reconnaissance attacks are internet queries, ping sweeps, and port scans. The four types of access attacks are password (brute-force, Trojan horse, packet sniffers), trust exploitation, port redirection, and man-in-the-middle attacks. The two types of service disruption attacks are DoS and DDoS.

Network Attack Mitigation

To mitigate network attacks, you must first secure devices, including routers, switches, servers, and hosts. Most organizations employ a defense-in-depth approach to security. This requires a combination of networking devices and services working together. Several security devices and services are implemented to protect an organization’s users and assets against TCP/IP threats: VPN, ASA firewall, IPS, ESA/WSA, and AAA server. Infrastructure devices should have backups of configuration files and IOS images on an FTP or similar file server. If a computer’s or a router’s hardware fails, the data or configuration can be restored using the backup copy. The most effective way to mitigate a worm attack is to download security updates from the operating system vendor and patch all vulnerable systems. To manage critical security patches, make sure all end systems automatically download updates.

AAA makes it possible to control who is permitted to access a network (authenticate), what they can do while they are there (authorize), and what actions they perform while accessing the network (accounting). Network firewalls reside between two or more networks, control the traffic between them, and help prevent unauthorized access. Servers accessible to outside users are usually located on a special network referred to as the DMZ. Firewalls use various techniques for determining what is permitted or denied access to a network, including packet filtering, application filtering, URL filtering, and SPI. Securing endpoint devices is critical to network security. A company must have well-documented policies in place, which may include the use of antivirus software and host intrusion prevention. More comprehensive endpoint security solutions rely on network access control.

Device Security

When a new OS is installed on a device, the security settings are set to the default values. This level of security is inadequate. For Cisco routers, the Cisco AutoSecure feature can be used to assist in securing a system. For most OSs, default usernames and passwords should be changed immediately, access to system resources should be restricted to only the individuals authorized to use those resources, and any unnecessary services and applications should be turned off and uninstalled when possible. To protect network devices, it is important to use strong passwords. A passphrase is often easier to remember than a simple password; it is also longer and harder to guess. For routers and switches, encrypt all plaintext passwords, set a minimum acceptable password length, deter brute-force password guessing attacks, and disable inactive privileged EXEC mode access after a specified amount of time. Configure appropriate devices to support SSH and disable unused services.

Packet Tracer
 Activity

Packet Tracer—Secure Network Devices (16.5.1)

In this activity, you will configure a router and a switch based on a list of requirements.



Lab—Secure Network Devices (16.5.2)

In this lab, you will complete the following objectives:

- Part 1: Configure Basic Device Settings
 - Part 2: Configure Basic Security Measures on the Router
 - Part 3: Configure Basic Security Measures on the Switch
-

Practice

The following activities provide practice with the topics introduced in this chapter. The lab is available in the companion *Introduction to Networks Labs & Study Guide (CCNAv7)* (ISBN 9780136634454). The Packet Tracer activity instructions are also provided in the *Labs & Study Guide*. The PKA files are available in the online course.



Labs

Lab 16.2.6: Research Network Security Threats

Lab 16.4.7: Configure Network Devices with SSH

Lab 16.5.2: Secure Network Devices



Packet Tracer Activities

Packet Tracer 16.4.6: Configure Secure Passwords and SSH

Packet Tracer 16.5.1: Secure Network Devices

Check Your Understanding Questions

Complete all the review questions listed here to test your understanding of the topics and concepts in this chapter. The appendix “Answers to ‘Check Your Understanding’ Questions” lists the answers.

1. Which component is designed to protect against unauthorized communications to and from a computer?
 - a. security center
 - b. port scanner
 - c. antimalware
 - d. antivirus
 - e. firewall
2. Which command blocks login attempts on RouterA for a period of 30 seconds if there are 2 failed login attempts within 10 seconds?
 - a. RouterA(config)# login block-for 10 attempts 2 within 30
 - b. RouterA(config)# login block-for 30 attempts 2 within 10
 - c. RouterA(config)# login block-for 2 attempts 30 within 10
 - d. RouterA(config)# login block-for 30 attempts 10 within 2

3. What is the purpose of the network security accounting function?
 - a. to require users to prove who they are
 - b. to determine which resources a user can access
 - c. to keep track of the actions of users
 - d. to provide challenge-and-response questions
4. What type of attack may involve the use of tools such as **nslookup** and **fping**?
 - a. access attack
 - b. reconnaissance attack
 - c. denial-of-service attack
 - d. worm attack
5. Which benefit does SSH offer over Telnet for remotely managing a router?
 - a. encryption
 - b. TCP usage
 - c. authorization
 - d. connections via multiple vty lines
6. What is one of the most effective security tools available for protecting users from external threats?
 - a. firewall
 - b. router that run AAA services
 - c. path server
 - d. password encryption
7. Which type of network threat is intended to prevent authorized users from accessing resources?
 - a. DoS attack
 - b. access attack
 - c. reconnaissance attack
 - d. trust exploitation

8. Which three services are provided by the AAA framework? (Choose three.)
 - a. accounting
 - b. automation
 - c. authorization
 - d. authentication
 - e. availability
 - f. autoconfiguration
9. Which malicious code attack is self-contained and tries to exploit a specific vulnerability in a system?
 - a. virus
 - b. worm
 - c. Trojan horse
 - d. maintenance
10. Some routers and switches in a wiring closet malfunctioned after an air conditioning unit failed. What type of threat does this situation describe?
 - a. configuration
 - b. environmental
 - c. electrical
 - d. maintenance
11. What does the term *vulnerability* mean?
 - a. a weakness that makes a target susceptible to an attack
 - b. a computer that contains sensitive information
 - c. a method of attack to exploit a target
 - d. a known target or victim machine
 - e. a potential threat a hacker creates
12. What three configuration steps must be performed to implement SSH access to a router? (Choose three.)
 - a. a password on the console line
 - b. an IP domain name
 - c. a user account
 - d. an enable mode password
 - e. a unique hostname
 - f. an encrypted password

13. What is the objective of a network reconnaissance attack?
 - a. discover and map systems
 - b. manipulate data without authorization to do so
 - c. disable network systems or services
 - d. deny access to resources by legitimate users
14. For security reasons, a network administrator needs to ensure that local computers cannot ping each other. Which settings can accomplish this task?
 - a. smartcard settings
 - b. firewall settings
 - c. MAC address settings
 - d. file system settings
15. A network administrator establishes a connection to a switch through SSH. What characteristic uniquely describes the SSH connection?
 - a. out-of-band access to a switch through the use of a virtual terminal with password authentication
 - b. remote access to the switch through the use of a telephone dialup connection
 - c. on-site access to a switch through the use of a directly connected PC and a console cable
 - d. remote access to a switch where data is encrypted during the session
 - e. direct access to the switch through the use of a terminal emulation program

Build a Small Network

Objectives

Upon completion of this chapter, you will be able to answer the following questions:

- What devices are used in a small network?
- What protocols and applications are used in a small network?
- How does a small network serve as the basis of larger networks?
- How do you use the output of the **ping** and **tracert** commands to verify connectivity and establish relative network performance?
- How do you use host and IOS commands to acquire information about the devices in a network?
- What are common network troubleshooting methodologies?
- How do you troubleshoot issues with devices in a network?

Key Terms

This chapter uses the following key terms. You can find the definitions in the glossary at the end of the book.

redundancy page 576

protocol analyzer page 583

quality of service (QoS) page 577

Introduction (17.0)

Hooray! You have come to the final chapter in this book. You now have most of the foundational knowledge needed to set up your own network. Where do you go from here? You build a network, of course. Once you build a network, you need to verify that it is working, and you may also need to troubleshoot some common network problems. This chapter has labs and Packet Tracer activities to help you practice your new skills as a network administrator. Let's get going!

Devices in a Small Network (17.1)

The number and type of network devices in a small network often differ from those in larger networks, but networks of all sizes must be able to provide many of the same services.

Small Network Topologies (17.1.1)

The majority of businesses are small; therefore, it is not surprising that the majority of business networks are also small.

A small network design is usually simple. Compared to a larger network, a small network has significantly fewer devices and types of devices. For instance, refer to the sample small business network shown in Figure 17-1.

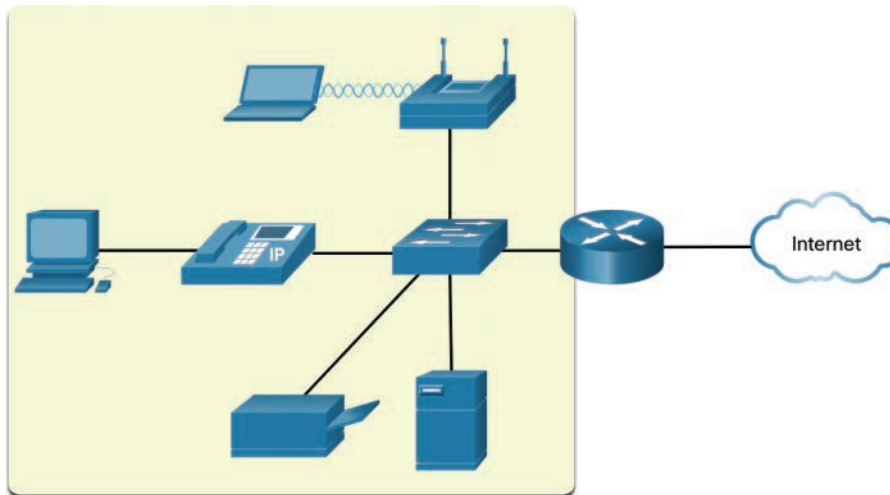


Figure 17-1 Small Business Network Topology

This small network requires a router, a switch, and a wireless access point to connect wired and wireless users, an IP phone, a printer, and a server. A small network typically has a single WAN connection provided by a DSL, cable, or Ethernet connection.

A large network requires an IT department to maintain, secure, and troubleshoot network devices and to protect organizational data. Managing a small network requires many of the same skills required for managing a larger one. Small networks are managed by a local IT technician or by a contracted professional.

Device Selection for a Small Network (17.1.2)

Like large networks, small networks require planning and design to meet user requirements. Planning ensures that all requirements, cost factors, and deployment options are given due consideration.

One of the first design considerations is the type of intermediary devices to use to support the network. The following sections describe the factors that must be considered when selecting network devices.

Cost

The cost of a switch or router is determined by its capacity and features, such as the number and types of ports available and the backplane speed. Other factors that influence the cost are network management capabilities, embedded security technologies, and optional advanced switching technologies. The expense of cable runs required to connect every device on the network must also be considered. Another key element affecting cost considerations is the amount of redundancy to incorporate into the network.

Speed and Types of Ports/Interfaces

Choosing the number and type of ports on a router or switch is a critical decision. Newer computers have built-in 1 Gbps NICs. Some servers may even have 10 Gbps ports. Although it is more expensive, choosing Layer 2 devices that can accommodate increased speeds allows the network to evolve without requiring replacement of central devices.

Expandability

Networking devices are available in fixed and modular physical configurations. A fixed configuration device has a specific number and type of ports or interfaces and cannot be expanded. A modular device has expansion slots for adding new modules as requirements evolve. Switches are available with additional ports for

high-speed uplinks. Routers can be used to connect different types of networks. Care must be taken to select the appropriate modules and interfaces for the specific media.

Operating System Features and Services

Network devices must have operating systems that can support the organization's requirements, such as the following:

- Layer 3 switching
- Network Address Translation (NAT)
- Dynamic Host Configuration Protocol (DHCP)
- Security
- Quality of service (QoS)
- Voice over IP (VoIP)

IP Addressing for a Small Network (17.1.3)

When implementing a network, it is important to create an IP addressing scheme and use it. Every host or device in an internetwork must have a unique address.

Devices that factor into the IP addressing scheme include the following:

- End-user devices, including the number of devices and the connection types (i.e., wired, wireless, remote access)
- Servers and peripherals devices (for example, printers and security cameras)
- Intermediary devices, including switches and access points

It is recommended that you plan, document, and maintain an IP addressing scheme based on device type. The use of a planned IP addressing scheme makes it easier to identify a type of device and to troubleshoot problems, as, for instance, when troubleshooting network traffic issues with a protocol analyzer.

For example, consider the topology of a small to medium-sized organization in Figure 17-2.

The organization requires three user LANs (that is, 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24). The organization has decided to implement a consistent IP addressing scheme for each 192.168.x.0/24 LAN, using the plan shown in Table 17-1.

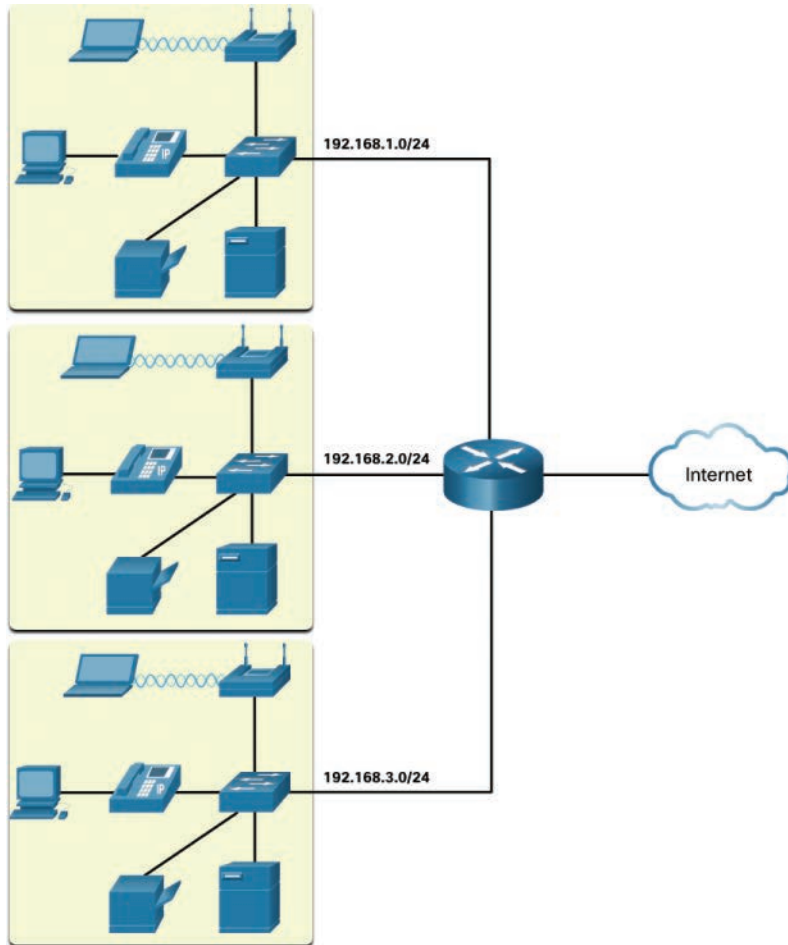


Figure 17-2 Small to Medium-Sized Organization Topology

Table 17-1 Example of a Consistent IPv4 Addressing Scheme

Device Type	Assignable IP Address Range	Summarized As
Default gateway (router)	192.168.x.1–192.168.x.2	192.168.x.0/30
Switches (max 2)	192.168.x.5–192.168.x.6	192.168.x.4/30
Access points (max 6)	192.168.x.9–192.168.x.14	192.168.x.8/29
Servers (max 6)	192.168.x.17–192.168.x.22	192.168.x.16/29
Printers (max 6)	192.168.x.25–192.168.x.30	192.168.x.24/29
IP phones (max 6)	192.168.x.33–192.168.x.38	192.168.x.32/29

Device Type	Assignable IP Address Range	Summarized As
Wired devices (max 62)	192.168.x.65–192.168.x.126	192.168.x.64/26
Wireless devices (max 62)	192.168.x.193–192.168.x.254	192.168.x.192/26

Figure 17-3 shows an example of the 192.168.2.0/24 network devices with assigned IP addresses using the predefined IP addressing scheme.

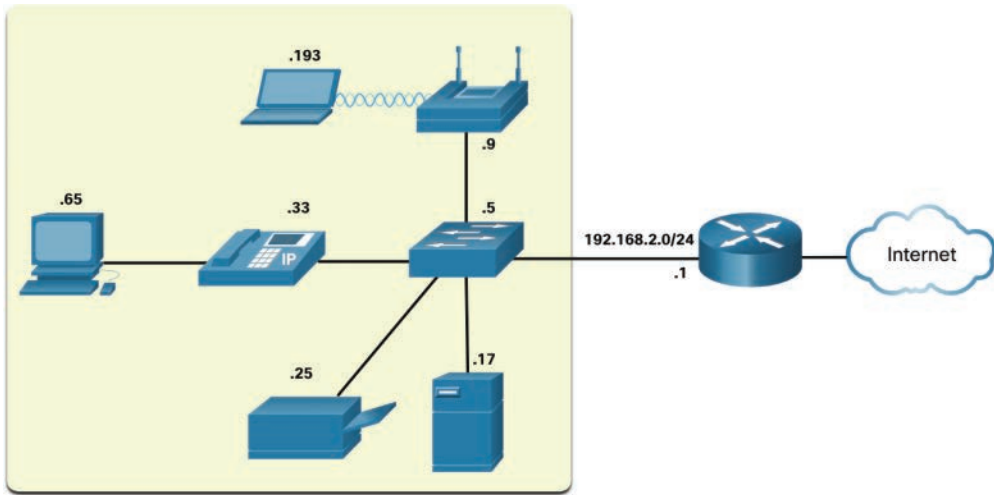


Figure 17-3 Small Business Topology with Addressing Assigned

For instance, the default gateway IP address is 192.168.2.1/24, the switch IP address is 192.168.2.5/24, the server IP address is 192.168.2.17/24, and so on.

Notice that the assignable IP address ranges were deliberately allocated on subnet-boundary boundaries to simplify summarization of the group type. For instance, assume that another switch with IP address 192.168.2.6 is added to the network. To identify all switches in a network policy, the administrator could specify the summarized network address 192.168.x.4/30.

Redundancy in a Small Network (17.1.4)

An important part of network design is reliability. Even a small business often relies heavily on its network for business operation. A network failure can be very costly.

To maintain a high degree of reliability, *redundancy* is required in the network design. Redundancy helps to eliminate single points of failure. There are many ways to accomplish redundancy in a network. *Redundancy* can be accomplished by installing duplicate equipment, and it can also be accomplished by supplying duplicate network links for critical areas, as shown in Figure 17-4.

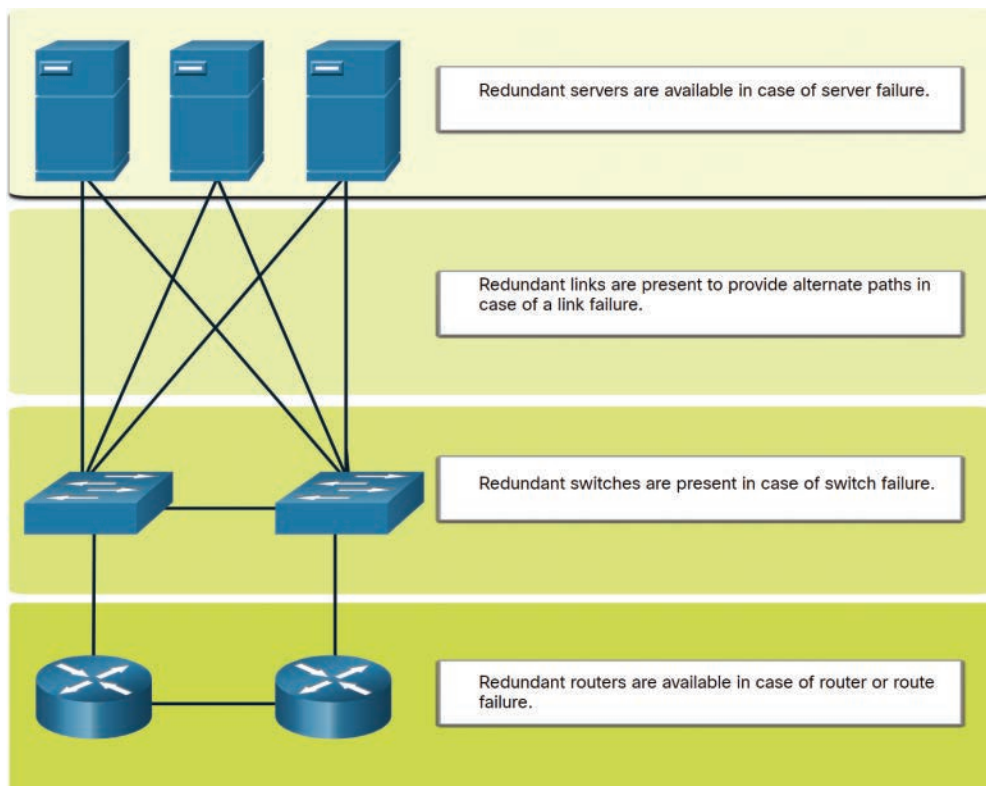


Figure 17-4 Small Network with Redundant Devices and Links

A small network typically provides a single exit point toward the internet through one or more default gateways. If the router fails, the entire network loses connectivity to the internet. For this reason, it may be advisable for a small business to pay for a second service provider as backup.

Traffic Management (17.1.5)

The goal for a good network design—even in a small network—is to enhance the productivity of the employees and minimize network downtime. The network administrator should consider the various types of traffic and their treatment in the network design.

The routers and switches in a small network should be configured to support real-time traffic, such as voice and video, in an appropriate manner relative to other data traffic. In fact, a good network design implements *quality of service (QoS)* to classify traffic carefully according to priority during times of congestion, as shown in Figure 17-5.

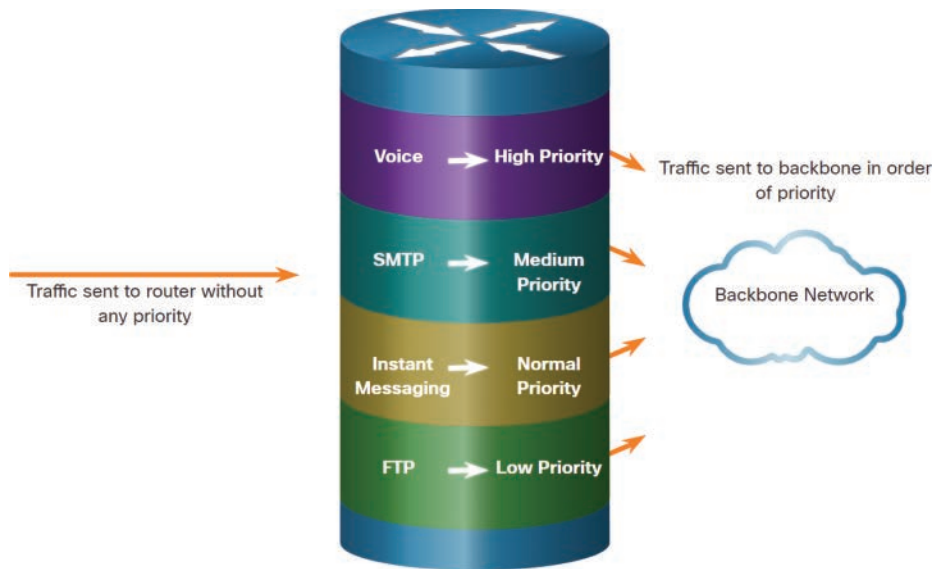


Figure 17-5 QoS Priority Queuing

**Interactive
Graphic**

Check Your Understanding—Devices in a Small Network (17.1.6)

Refer to the online course to complete this activity.

Small Network Applications and Protocols (17.2)

Along with considering the network devices in a small network, it is important to examine the applications and services that it must support.

Common Applications (17.2.1)

The previous section discusses the components of a small network, as well as some of the design considerations. These considerations are necessary when you are just setting up a network. After you have set it up, the network still needs certain types of applications and protocols in order to work.

A network is only as useful as the applications that are on it. There are two forms of software programs or processes that provide access to the network: network applications and application layer services.

Network Applications

Applications are the software programs used to communicate over a network. Some end-user applications are network aware, meaning that they implement application

layer protocols and are able to communicate directly with the lower layers of the protocol stack. Email clients and web browsers are examples of this type of application.

Application Layer Services

Other programs may need the assistance of application layer services to use network resources such as file transfer or network print spooling. Although they are transparent to an employee, these services are the programs that interface with the network and prepare the data for transfer. Different types of data—whether text, graphics, or video—require different network services to ensure that they are properly prepared for processing by the functions occurring at the lower layers of the OSI model.

Each application or network service uses protocols, which define the standards and data formats to be used. Without protocols, the data network would not have a common way to format and direct data. In order to understand the function of various network services, it is necessary to become familiar with the underlying protocols that govern their operation.

On a Windows PC, you use Task Manager to view the currently running applications, processes, and services, as shown in Figure 17-6.

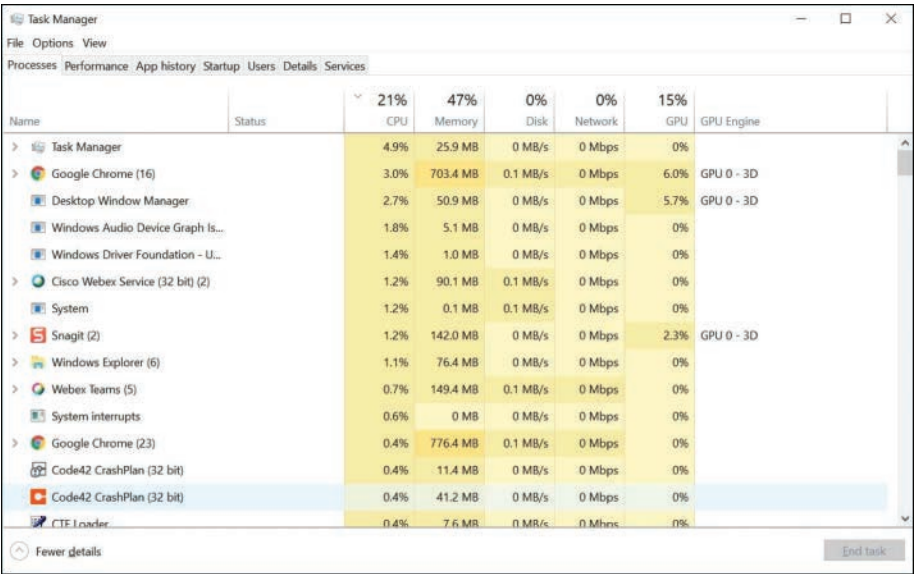


Figure 17-6 Windows Task Manager

Common Protocols (17.2.2)

Most of a technician’s work, in either a small network or a large network, is in some way involved with network protocols. Network protocols support the applications and services used by employees in a small network.

Network administrators commonly require access to network devices and servers. The two most common remote access solutions are Telnet and Secure Shell (SSH). SSH is a secure alternative to Telnet. When connected, administrators can access the SSH server device as though they were logged in locally.

SSH is used to establish a secure remote access connection between an SSH client and other SSH-enabled devices:

- **Network device:** The network device (for example, router, switch, access point) must support SSH to provide remote access SSH server services to clients.
- **Server:** The server (for example, web server, email server) must support remote access SSH server services to clients.

Network administrators must also support common network servers and their required related network protocols, as shown in Figure 17-7 and described in the list that follows:

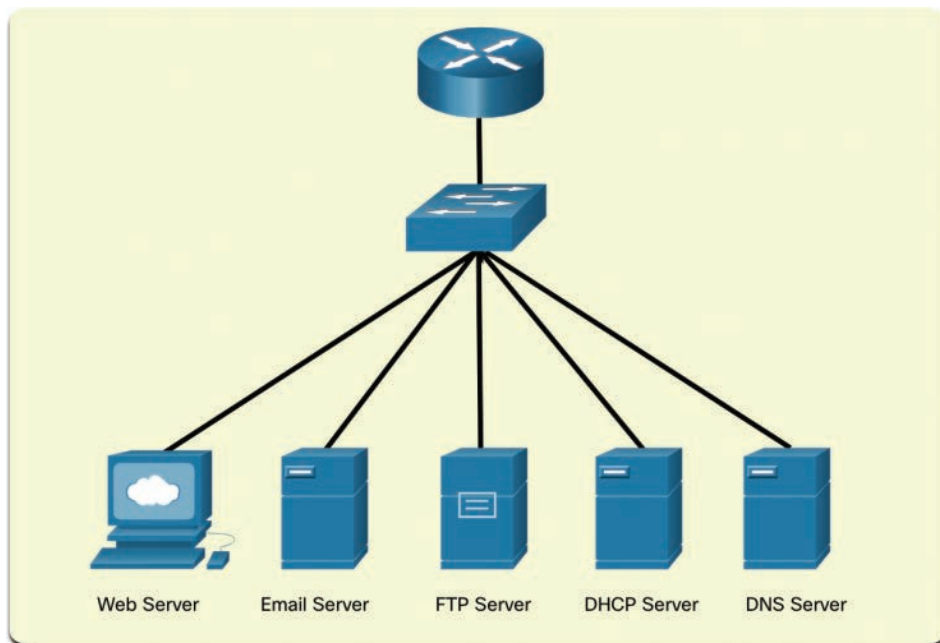


Figure 17-7 Common Network Servers

- **Web server:**
 - Web clients and web servers exchange web traffic by using Hypertext Transfer Protocol (HTTP).
 - Web clients and web servers exchange secure web communication by using Hypertext Transfer Protocol Secure (HTTPS).

■ Email server:

- Email servers and clients use Simple Mail Transfer Protocol (SMTP) to send emails.
- Email clients use Post Office Protocol (POP3) or Internet Message Access Protocol (IMAP) to retrieve email.
- Recipients are specified using the *user@xyz.xxx* format.

■ FTP server:

- File Transfer Protocol (FTP) allows files to be downloaded and uploaded between a client and an FTP server.
- FTP Secure (FTPS) and Secure FTP (SFTP) are used for secure FTP file exchange.

■ DHCP server:

- Clients use Dynamic Host Configuration Protocol (DHCP) to acquire an IP configuration (that is, IP address, subnet mask, default gateway and more) from a DHCP server.

■ DNS server:

- Domain Name System (DNS) resolves a domain name to an IP address (for example, *cisco.com* = 72.163.4.185).
- DNS provides the IP address of a website (that is, domain name) to a requesting host.

Note

A server could provide multiple network services. For instance, a server could be an email server, an FTP server, and an SSH server.

These network protocols comprise the fundamental toolset of a network professional. Each of these network protocols defines the following:

- Processes on either end of a communication session
- Types of messages
- Syntax of messages
- Meaning of informational fields
- How messages are sent and the expected responses
- Interaction with the next lower layer

Many companies have established a policy of using secure versions (for example, SSH, SFTP, HTTPS) of these protocols whenever possible.

Voice and Video Applications (17.2.3)

Businesses today are increasingly using IP telephony and streaming media to communicate with customers and business partners. Many organizations are enabling their employees to work remotely. Many of their users require access to corporate software and files, as well as support for voice and video applications.

A network administrator must ensure that the proper equipment is installed in the network and that the network devices are configured to ensure priority delivery.

The following are factors that a small network administrator must consider when supporting real-time applications:

- **Infrastructure:**

- The network infrastructure must support the real-time applications.
- Existing devices and cabling must be tested and validated.
- Newer networking products may be required.

- **VoIP:**

- VoIP devices convert analog telephone signals into digital IP packets.
- Typically, VoIP is less expensive than an IP telephony solution, but the quality of communications does not meet the same standards.
- Small network voice and video over IP can be solved using Skype and non-enterprise versions of Cisco WebEx.

- **IP telephony:**

- An IP phone performs voice-to-IP conversion with the use of a dedicated server for call control and signaling.
- Many vendors provide small business IP telephony solutions such as the Cisco Business Edition 4000 Series products.

- **Real-time applications:**

- The network must support quality-of-service (QoS) mechanisms to minimize latency issues for real-time streaming applications.
- Real-Time Transport Protocol (RTP) and Real-Time Transport Control Protocol (RTCP) are two protocols that support this requirement.

Check Your Understanding—Small Network Applications and Protocols (17.2.4)

Refer to the online course to complete this activity.

Scale to Larger Networks (17.3)

Networks support the business and must be able to grow as the business grows.

Small Network Growth (17.3.1)

If your network is for a small business, presumably, you want that business to grow, and you want your network to grow along with it. This is called *scaling* a network, and there are some best practices for doing this.

Growth is a natural process for many small businesses, whose networks must grow accordingly. Ideally, a network administrator has enough lead time to make intelligent decisions about growing the network in alignment with the growth of the company.

To scale a network, several elements are required:

- **Network documentation:** Physical and logical topology
- **Device inventory:** List of devices that use or comprise the network
- **Budget:** Itemized IT budget, including the fiscal year equipment purchasing budget
- **Traffic analysis:** Documentation of protocols, applications, and services and their respective traffic requirements

These elements are used to inform the decision making that accompanies the scaling of a small network.

Protocol Analysis (17.3.2)

As a network grows, it becomes important to determine how to manage network traffic. It is important to understand the type of traffic that is crossing the network as well as the current traffic flow. Several network management tools can be used for this purpose. However, a simple *protocol analyzer* such as Wireshark can also be used.

For instance, running Wireshark on several key hosts can reveal the types of network traffic flowing through the network. Figure 17-8 shows Wireshark protocol hierarchy statistics for a Windows host on a small network. The screen capture reveals that the host is using IPv6 and IPv4 protocols. The IPv4-specific output also reveals that the host has used DNS, SSL, HTTP, ICMP, and other protocols.

To determine traffic flow patterns, it is important to do the following:

- Capture traffic during peak utilization times to get a good representation of the different traffic types.
- Perform the capture on different network segments and devices as some traffic will be local to a particular segment.

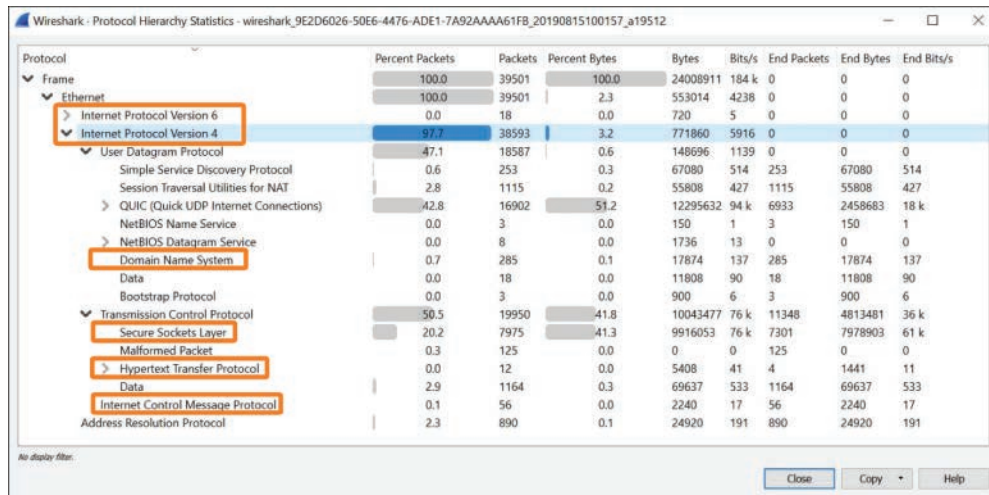


Figure 17-8 Wireshark Capture Showing Packet Statistics

Information gathered by the protocol analyzer is evaluated based on the source and destination of the traffic, as well as the type of traffic being sent. This analysis can be used to make decisions on how to manage the traffic more efficiently. This can be done by reducing unnecessary traffic flows or changing flow patterns altogether by moving a server, for example.

Sometimes, simply relocating a server or service to another network segment improves network performance and accommodates the growing traffic needs. At other times, optimizing network performance requires major network redesign and intervention.

Employee Network Utilization (17.3.3)

In addition to understanding changing traffic trends, a network administrator must be aware of how network use is changing. Many operating systems provide built-in tools to display such information. For example, a Windows host provides tools such as Task Manager, Event Viewer, and Data Usage. Such tools can be used to capture “snapshots” of information such as the following:

- OS and OS version
- CPU utilization
- RAM utilization
- Drive utilization
- Non-network applications
- Network applications

Documenting snapshots for employees in a small network over a period of time is very useful in identifying evolving protocol requirements and associated traffic flows. A shift in resource utilization may require the network administrator to adjust network resource allocations accordingly.

The Windows 10 Data Usage tool is especially useful for determining which applications are using network services on a host. The Data Usage tool is accessed by selecting **Settings > Network & Internet > Data usage > network interface** (from the last 30 days).

The example in Figure 17-9 shows the applications running on a remote user's Windows 10 host using the local Wi-Fi network connection.

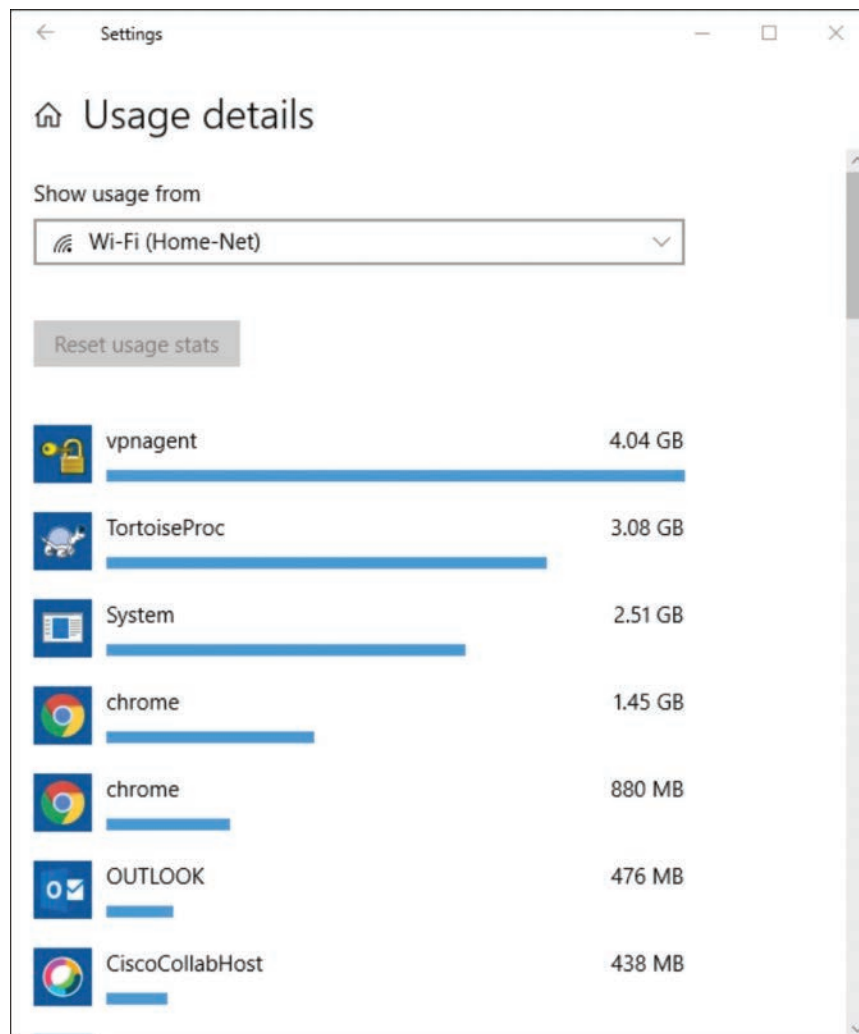


Figure 17-9 Windows 10 Usage Details for a Wi-Fi Network Connection

Interactive
Graphic

Check Your Understanding—Scale to Larger Networks (17.3.4)

Refer to the online course to complete this activity.

Verify Connectivity (17.4)

After a network has been implemented, a network administrator must be able to test the network connectivity to ensure that it is operating appropriately. In addition, it is a good idea for the network administrator to document the network.

Verify Connectivity with Ping (17.4.1)

Whether a network is small and new, and even if you are scaling an existing network, you will always want to be able to verify that your components are properly connected to each other and to the internet. This section discusses some utilities that you can use to ensure that a network is connected.

Using the **ping** command is the most effective way to quickly test Layer 3 connectivity between a source IP address and a destination IP address. This command also displays various round-trip time statistics.

Specifically, the **ping** command uses the Internet Control Message Protocol (ICMP) Echo Request (ICMP Type 8) and Echo Reply (ICMP Type 0) messages. The **ping** command is available in most operating systems, including Windows, Linux, macOS, and Cisco IOS.

On a Windows 10 host, the **ping** command sends four consecutive ICMP Echo Request messages and expects four consecutive ICMP Echo Reply messages from the destination.

For example, say that PC A pings PC B. As shown in Figure 17-10, the PC A Windows host sends four consecutive ICMP Echo Request messages to PC B (that is, 10.1.1.10).

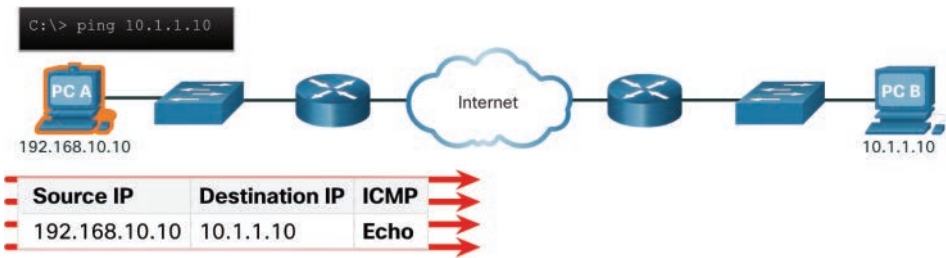


Figure 17-10 PC A Pinging PC B

The destination host receives and processes the ICMP Echo Request, sometimes referred to as an ICMP Echo. As shown in Figure 17-11, PC B responds by sending four ICMP Echo Reply messages to PC A.

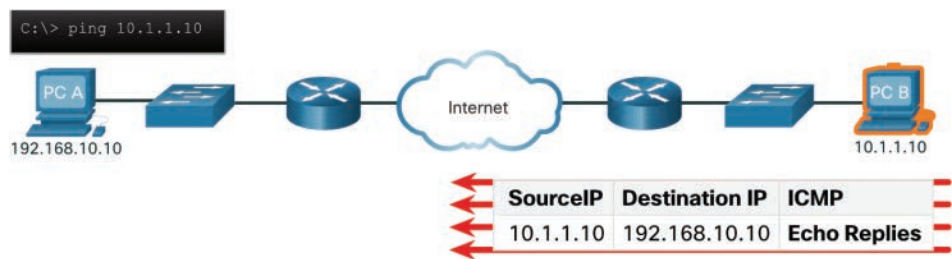


Figure 17-11 PC B Responding to PC A

As shown in the command output in Example 17-1, PC A has received Echo Reply messages from PC B, verifying the Layer 3 network connection. This output validates Layer 3 connectivity between PC A and PC B.

Example 17-1 ping Output on PC A

```
C:\Users\PC-A> ping 10.1.1.10
Pinging 10.1.1.10 with 32 bytes of data:
Reply from 10.1.1.10: bytes=32 time=47ms TTL=51
Reply from 10.1.1.10: bytes=32 time=60ms TTL=51
Reply from 10.1.1.10: bytes=32 time=53ms TTL=51
Reply from 10.1.1.10: bytes=32 time=50ms TTL=51
Ping statistics for 10.1.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 47ms, Maximum = 60ms, Average = 52ms
C:\Users\PC-A>
```

ping command output in Cisco IOS varies from ping command output on a Windows host. For instance, the IOS ping sends five ICMP Echo messages, as shown in Example 17-2.

Example 17-2 ping Output on R1

```
R1# ping 10.1.1.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R1#
```

Notice the !!!!! output characters. The IOS ping command displays one ! output character for each ICMP Echo Reply received. Table 17-2 lists the most common output characters from the ping command.

Table 17-2 IOS ping Indicators

Element	Description
!	■ Exclamation point indicates successful receipt of an Echo Reply message. ■ It validates a Layer 3 connection between the source and the destination.
.	■ A period means that time expired while waiting for an Echo Reply message. ■ This indicates that a connectivity problem occurred somewhere along the path.
U	■ Uppercase U indicates that a router along the path responded with an ICMP Type 3 “destination unreachable” error message. ■ Possible reasons include the router not knowing the direction to the destination network or being unable to find the host on the destination network.

Note

Other possible ping replies include Q, M, ?, and &. However, these replies are beyond the scope of this chapter.

Extended Ping (17.4.2)

A standard **ping** uses the IP address of the interface closest to the destination network as the source of the **ping**. The source IP address of the **ping 10.1.1.10** command on R1 would be that of the G0/0/0 interface (that is, 209.165.200.225), as illustrated in Figure 17-12.

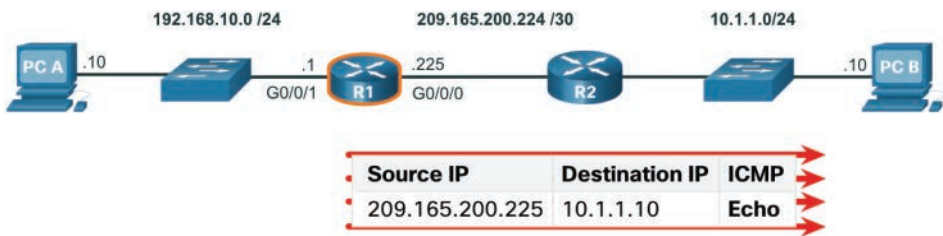


Figure 17-12 R1 Pinging PC B Using the Exit Interface as the Source IPv4 Address

Cisco IOS offers an “extended” mode of the **ping** command. This mode enables the user to create special types of pings by adjusting parameters related to the command operation.

You enter an extended **ping** in privileged EXEC mode by typing **ping** without a destination IP address. You are then given several prompts to customize the extended **ping**.

Note

Press Enter to accept the indicated default values.

For example, say that you wanted to test connectivity from the R1 LAN (that is, 192.168.10.0/24) to the 10.1.1.0 LAN. This could be verified from PC A. However, an extended **ping** could be configured on R1 to specify a different source address.

As illustrated in Figure 17-13, the source IP address of the extended **ping** command on R1 could be configured to use the G0/0/1 interface IP address (that is, 192.168.10.1).

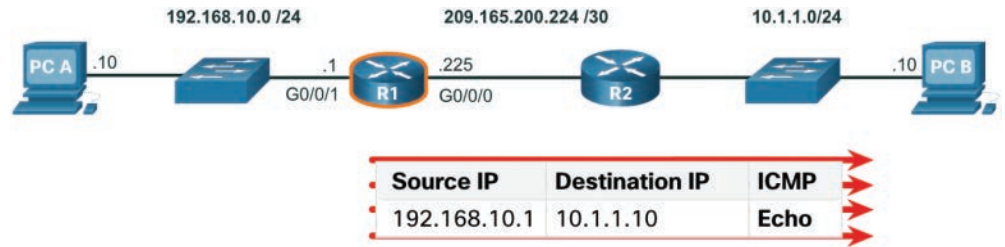


Figure 17-13 R1 Pinging PC B Using an Extended **ping**

Example 17-3 shows the configuration of an extended **ping** on R1, with the source IP address of the G0/0/1 interface (that is, 192.168.10.1).

Example 17-3 Extended **ping** on R1

```
R1# ping
Protocol [ip]:
Target IP address: 10.1.1.10
Repeat count [5]:
Datagram size [100]:
Timeout in seconds [2]:
Extended commands [n]: y
Ingress ping [n]:
Source address or interface: 192.168.10.1
DSCP Value [0]:
Type of service [0]:
Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0x0000ABCD]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]:
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.10.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R1#
```

Note

The **ping ipv6** command is used for IPv6 extended pings.

Verify Connectivity with Traceroute (17.4.3)

The **ping** command is useful for quickly determining whether there is a Layer 3 connectivity problem. However, it does not identify where a problem is located along the path.

traceroute can help locate Layer 3 problem areas in a network. It returns a list of hops as a packet is routed through a network. It could be used to identify the point along the path where the problem can be found.

The syntax of the **traceroute** command varies between operating systems, as illustrated in Figure 17-14.

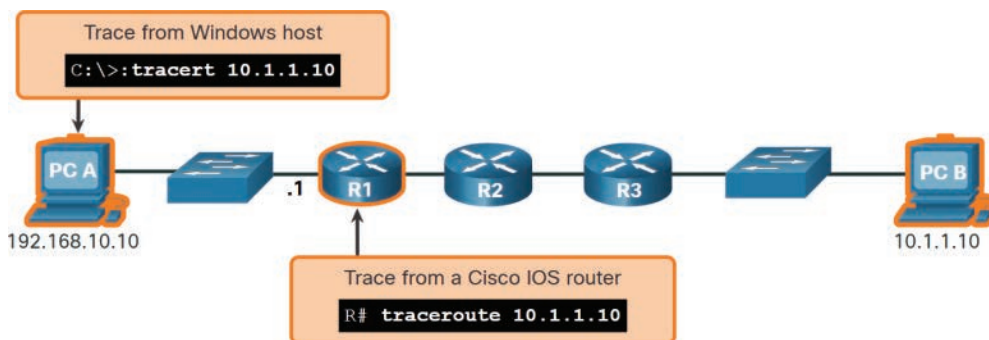


Figure 17-14 Windows and Cisco IOS Trace Commands

Example 17-4 shows the output of the **tracert** command on a Windows 10 host.

Example 17-4 The **tracert** Command on PC A

```
C:\Users\PC-A> tracert 10.1.1.10
Tracing route to 10.1.10 over a maximum of 30 hops:
  1    2 ms    2 ms    2 ms  192.168.10.1
  2    *        *        *    Request timed out.
  3    *        *        *    Request timed out.
  4    *        *        *    Request timed out.
^C
C:\Users\PC-A>
```

Note

Use Ctrl+C to interrupt **tracert** in Windows.

The only successful response in Example 17-4 was from the gateway on R1. Trace requests to the next hop timed out, as indicated by the asterisk (*), meaning that the next hop router did not respond. The timed out requests indicate that there is a failure in the internetwork beyond the LAN or that these routers have been configured to not respond to Echo Request messages used in the trace. In this example, there appears to be a problem between R1 and R2.

The output of the Cisco IOS **traceroute** command differs from the output of the Windows **tracert** command. The topology in Figure 17-15 provides an example.

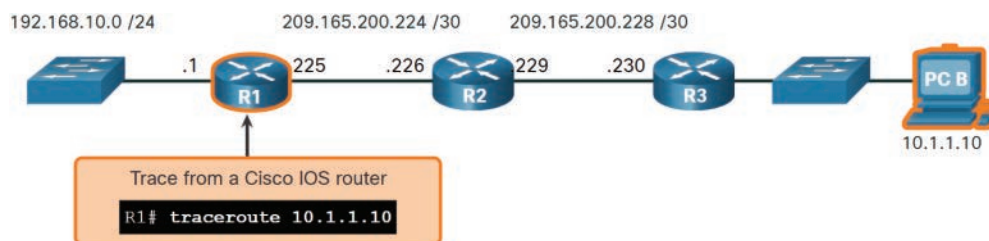


Figure 17-15 Cisco IOS **traceroute** Command

Example 17-5 shows sample output of the **traceroute** command on R1. In this example, the output validates that the **traceroute** command could successfully reach PC B.

Example 17-5 The **traceroute** Command on R1

```
R1# traceroute 10.1.1.10
Type escape sequence to abort.
Tracing the route to 10.1.1.10
VRF info: (vrf in name/id, vrf out name/id)
  1 209.165.200.226 1 msec 0 msec 1 msec
  2 209.165.200.230 1 msec 0 msec 1 msec
  3 10.1.1.10 1 msec 0 msec
R1#
```

Timeouts indicate potential problems. For instance, if the 10.1.1.10 host were not available, the **traceroute** command would display the output shown in Example 17-6.

Example 17-6 Host Unreachable Output for the **traceroute** Command on R1

```
R1# traceroute 10.1.1.10
Type escape sequence to abort.
Tracing the route to 10.1.1.10
VRF info: (vrf in name/id, vrf out name/id)
  0  209.165.200.226 1 msec 0 msec 1 msec
  1  209.165.200.230 1 msec 0 msec 1 msec
  2  * * *
  3  * * *
  4  *
  5
```

Use Ctrl+Shift+6 to interrupt a **traceroute** command in Cisco IOS.

Note

The Windows implementation of **traceroute** (**tracert**) sends ICMP Echo Request messages. Cisco IOS and Linux use UDP with an invalid port number. The final destination returns an ICMP port unreachable message.

Extended Traceroute (17.4.4)

Like the extended **ping** command, there is also an extended **traceroute** command. An extended **traceroute** command allows an administrator to adjust parameters related to the command operation. This is helpful for locating problems when troubleshooting routing loops, determining the next hop router, or determining where packets are getting dropped or denied by a router or firewall.

The Windows **tracert** command allows for the input of several parameters through options at the command line. However, this additional input is not guided, as it is for the extended **traceroute** IOS command. Example 17-7 shows the available options for the Windows **tracert** command.

The Cisco IOS extended **traceroute** option enables the user to create a special type of trace by adjusting parameters related to the command operation. Extended **traceroute** is entered in privileged EXEC mode by typing **traceroute** without a destination IP address. IOS guides you through the command options by presenting a number of prompts related to the setting of all the different parameters.

Note

Press Enter to accept the indicated default values.

Example 17-7 The Options for the **tracert** Command on PC A

```

C:\Users\PC-A> tracert /?
Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
              [-R] [-S srcaddr] [-4] [-6] target_name

Options:
    -d                Do not resolve addresses to hostnames.
    -h maximum_hops   Maximum number of hops to search for target.
    -j host-list       Loose source route along host-list (IPv4-only).
    -w timeout         Wait timeout milliseconds for each reply.
    -R                Trace round-trip path (IPv6-only).
    -S srcaddr         Source address to use (IPv6-only).
    -4                Force using IPv4.
    -6                Force using IPv6.

C:\Users\PC-A>

```

For example, say that you want to test connectivity to PC B from the R1 LAN. Although this could be verified from PC A, an extended **tracert** could be configured on R1 to specify a different source address, as shown in Figure 17-16.

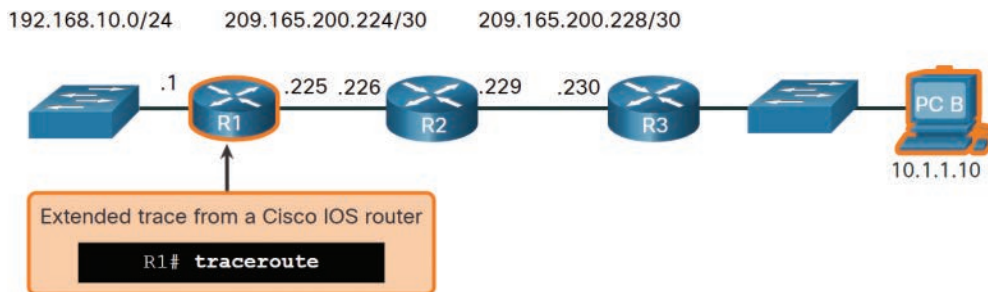


Figure 17-16 Cisco IOS Extended **traceroute** Command

As illustrated in Example 17-8, the source IP address of the extended **traceroute** command on R1 could be configured to use the R1 LAN interface IP address (that is, 192.168.10.1).

Network Baseline (17.4.5)

One of the most effective tools for monitoring and troubleshooting network performance is a network baseline. Creating an effective network performance baseline is accomplished over a period of time. Measuring performance at

varying times and loads will assist in creating a better picture of overall network performance.

Example 17-8 Extended **traceroute** Command on R1

```
R1# traceroute
Protocol [ip]:
Target IP address: 10.1.1.10
Ingress traceroute [n]:
Source address: 192.168.10.1
DSCP Value [0]:
Numeric display [n]:
Timeout in seconds [3]:
Probe count [3]:
Minimum Time to Live [1]:
Maximum Time to Live [30]:
Port Number [33434]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Type escape sequence to abort.
Tracing the route to 192.168.10.10
VRF info: (vrf in name/id, vrf out name/id)
  1 209.165.200.226 1 msec 1 msec 1 msec
  2 209.165.200.230 0 msec 1 msec 0 msec
  3  *
    10.1.1.10 2 msec 2 msec
R1#
```

The output derived from network commands contributes data to the network baseline. One method for starting a baseline is to copy and paste the results from an executed **ping**, **traceroute**, or other relevant commands into a text file. Such a text file can be timestamped with the date and saved into an archive for later retrieval and comparison with other similar files.

Among items to consider are error messages and the response times from host to host. If there is a considerable increase in response times, there may be a latency issue to address.

For example, the **ping** output in Example 17-9 was captured and pasted into a text file.

Example 17-9 ping on August 19, 2019, at 08:14:43

```
C:\Users\PC-A> ping 10.1.1.10
Pinging 10.1.1.10 with 32 bytes of data:
Reply from 10.1.1.10: bytes=32 time<1ms TTL=64
Reply from 10.1.1.10: bytes=32 time<1ms TTL=64
Reply from 10.1.1.10: bytes=32 time<1ms TTL=64
Reply from 10.1.1.10: bytes=32 time<1ms TTL=64
Ping statistics for 10.1.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Users\PC-A>
```

Notice that the **ping** round-trip times are less than 1 ms.

A month later, the **ping** is repeated and captured, as shown in Example 17-10.

Example 17-10 ping on September 19, 2019, at 10:18:21

```
C:\Users\PC-A> ping 10.1.1.10
Pinging 10.1.1.10 with 32 bytes of data:
Reply from 10.1.1.10: bytes=32 time=50ms TTL=64
Reply from 10.1.1.10: bytes=32 time=49ms TTL=64
Reply from 10.1.1.10: bytes=32 time=46ms TTL=64
Reply from 10.1.1.10: bytes=32 time=47ms TTL=64
Ping statistics for 10.1.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 46ms, Maximum = 50ms, Average = 48ms
C:\Users\PC-A>
```

Notice that this time the **ping** round-trip times are much longer, indicating a potential problem.

Corporate networks should have extensive baselines—more extensive than we can describe in this book. Professional-grade software tools are available for storing and maintaining baseline information. In this chapter, we cover a few basic techniques and discuss the purpose of baselines.

Cisco's best practices for baseline processes can be found by searching the internet for "Baseline Process Best Practices."



Lab—Test Network Latency with Ping and Traceroute (17.4.6)

In this lab, you will complete the following objectives:

- Part 1: Use Ping to Document Network Latency
- Part 2: Use Traceroute to Document Network Latency

Host and IOS Commands (17.5)

In addition to the **show** commands, a number of additional commands are available on hosts and network devices.

IP Configuration on a Windows Host (17.5.1)

If you have used any of the tools in the previous section to verify connectivity and found that some part of your network is not working as it should, you can use some commands to troubleshoot your devices. Host and IOS commands can help you determine if a problem is with the IP addressing of your devices, which is a common network problem.

Checking the IP addressing on host devices is a common practice in networking for verifying and troubleshooting end-to-end connectivity. In Windows 10, you can access the IP address details from the Network and Sharing Center, as shown in Figure 17-17, to quickly view four important settings: address, mask, router, and DNS.

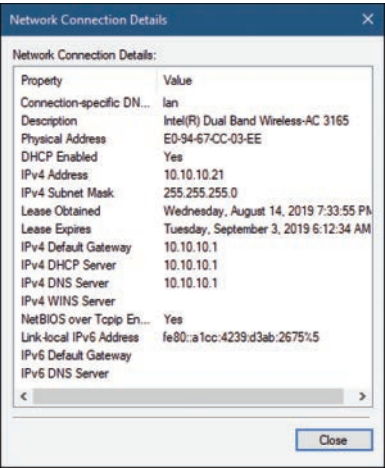


Figure 17-17 Windows 10 Network Connection Details

However, network administrators typically view the IP addressing information on a Windows host by issuing the **ipconfig** command at the command line of a Windows computer, as shown in Example 17-11.

Example 17-11 Verifying the IP Configuration on a Windows Host

```

C:\Users\PC-A> ipconfig
Windows IP Configuration
(Output omitted)
Wireless LAN adapter Wi-Fi:
    Connection-specific DNS Suffix . . :
    Link-local IPv6 Address . . . . . : fe80::a4aa:2dd1:ae2d:a75e%16
    IPv4 Address. . . . . : 192.168.10.10
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.10.1
(Output omitted)

```

You can use the **ipconfig /all** command to view the MAC address, as well as a number of details regarding the Layer 3 addressing of the device, as shown in Example 17-12.

Example 17-12 Verifying Complete Addressing Information on a Windows Host

```

C:\Users\PC-A> ipconfig /all
Windows IP Configuration
    Host Name . . . . . : PC-A-00H20
    Primary Dns Suffix . . . . . : cisco.com
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : cisco.com
(Output omitted)
Wireless LAN adapter Wi-Fi:
    Connection-specific DNS Suffix . . :
    Description . . . . . : Intel(R) Dual Band Wireless-AC 8265
    Physical Address. . . . . : F8-94-C2-E4-C5-0A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::a4aa:2dd1:ae2d:a75e%16(Preferred)
    IPv4 Address. . . . . : 192.168.10.10(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : August 17, 2019 1:20:17 PM
    Lease Expires . . . . . : August 18, 2019 1:20:18 PM
    Default Gateway . . . . . : 192.168.10.1
    DHCP Server . . . . . : 192.168.10.1
    DHCPv6 IAID . . . . . : 100177090
    DHCPv6 Client DUID. . . . . : 00-01-00-01-21-F3-76-75-54-E1-AD-DE-DA-9A
    DNS Servers . . . . . : 192.168.10.1
    NetBIOS over Tcpip. . . . . : Enabled

```

If a host is configured as a DHCP client, the IP address configuration can be renewed by using the **ipconfig /release** and **ipconfig /renew** commands, as shown in Example 17-13.

Example 17-13 Releasing and Renewing the IP Configuration on a Windows Host

```
C:\Users\PC-A> ipconfig /release
(Output omitted)
Wireless LAN adapter Wi-Fi:
    Connection-specific DNS Suffix . . : 
    Link-local IPv6 Address . . . . . : fe80::a4aa:2dd1:ae2d:a75e%16
    Default Gateway . . . . . : 
(Output omitted)
C:\Users\PC-A> ipconfig /renew
(Output omitted)
Wireless LAN adapter Wi-Fi:
    Connection-specific DNS Suffix . . : 
    Link-local IPv6 Address . . . . . : fe80::a4aa:2dd1:ae2d:a75e%16
    IPv4 Address. . . . . : 192.168.1.124
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1
(Output omitted)
C:\Users\PC-A>
```

The DNS client service on Windows PCs also optimizes the performance of DNS name resolution by storing previously resolved names in memory. The **ipconfig /displaydns** command displays all the cached DNS entries on a Windows computer system, as shown in Example 17-14.

Example 17-14 Verifying the DNS Information Stored on a Windows Host

```
C:\Users\PC-A> ipconfig /displaydns
Windows IP Configuration
(Output omitted)
    netacad.com
    -----
    Record Name . . . . . : netacad.com
    Record Type . . . . . : 1
    Time To Live . . . . . : 602
    Data Length . . . . . : 4
    Section . . . . . : Answer
    A (Host) Record . . . : 54.165.95.219
(Output omitted)
```

IP Configuration on a Linux Host (17.5.2)

How you verify IP settings in the GUI on a Linux machine depends on the Linux distribution (distro) and desktop interface. Figure 17-18 shows the Connection Information dialog box on the Ubuntu distro running the Gnome desktop.



Figure 17-18 Linux Ubuntu Connection Information

On the command line, network administrators use the `ifconfig` command to display the status of the currently active interfaces and their IP configuration, as shown in Example 17-15.

Example 17-15 Verifying the IP Configuration on a Linux Host

```
[analyst@secOps ~]$ ifconfig
enp0s3      Link encap:Ethernet  HWaddr 08:00:27:b5:d6:cb
            inet addr: 10.0.2.15  Bcast:10.0.2.255  Mask: 255.255.255.0
            inet6 addr: fe80::57c6:ed95:b3c9:2951/64 Scope:Link
            UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
            RX packets:1332239 errors:0 dropped:0 overruns:0 frame:0
            TX packets:105910 errors:0 dropped:0 overruns:0 carrier:0
            collisions:0 txqueuelen:1000
            RX bytes:1855455014 (1.8 GB)  TX bytes:13140139 (13.1 MB)

lo: flags=73  mtu 65536
            inet 127.0.0.1  netmask 255.0.0.0
            inet6 ::1  prefixlen 128  scopeid 0x10
            loop txqueuelen 1000  (Local Loopback)
            RX packets 0  bytes 0 (0.0 B)
            RX errors 0  dropped 0  overruns 0  frame 0
            TX packets 0  bytes 0 (0.0 B)
            TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

The Linux **ip address** command is used to display addresses and their properties. It can also be used to add or delete IP addresses.

Note

The output displayed may vary depending on the Linux distribution.

IP Configuration on a macOS Host (17.5.3)

In the GUI of a Mac host, open **Network Preferences > Advanced** to get the IP addressing information, as shown in Figure 17-19.

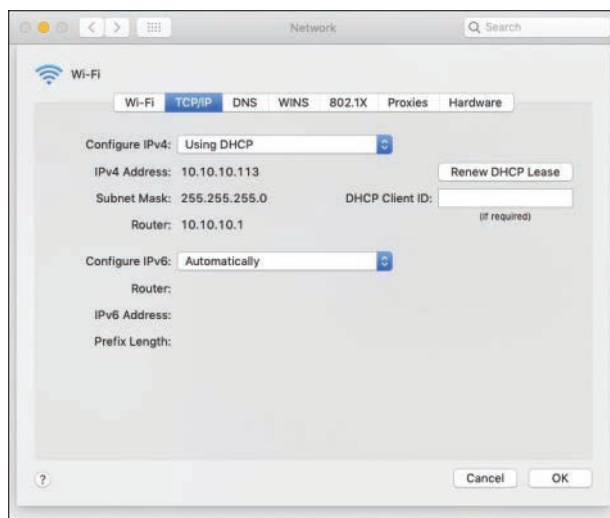


Figure 17-19 Configuration Information on a macOS Host

However, the **ifconfig** command can also be used to verify the interface IP configuration, as shown in Example 17-16.

Example 17-16 Verifying the IP Configuration on a macOS Host

```
MacBook-Air:~ Admin$ ifconfig en0
en0: flags=8863 mtu 1500
    ether c4:b3:01:a0:64:98
    inet6 fe80::c0f:1bf4:60b1:3adb%en0 prefixlen 64 secured scopeid 0x5
    inet 10.10.10.113 netmask 0xffffffff broadcast 10.10.10.255
    nd6 options=201
    media: autoselect
    status: active
MacBook-Air:~ Admin$
```

Other useful macOS commands to verify the host IP settings include **networksetup -listallnetworkservices** and **networksetup -getinfo <network service>**. The **networksetup -listallnetworkservices** command is shown in Example 17-17.

Example 17-17 Verifying Other Network Configuration Information on a macOS Host

```
MacBook-Air:~ Admin$ networksetup -listallnetworkservices
An asterisk (*) denotes that a network service is disabled.
iPhone USB
Wi-Fi
Bluetooth PAN
Thunderbolt Bridge
MacBook-Air:~ Admin$
MacBook-Air:~ Admin$ networksetup -getinfo Wi-Fi
DHCP Configuration
IP address: 10.10.10.113
Subnet mask: 255.255.255.0
Router: 10.10.10.1
Client ID:
IPv6: Automatic
IPv6 IP address: none
IPv6 Router: none
Wi-Fi ID: c4:b3:01:a0:64:98
MacBook-Air:~ Admin$
```

The arp Command (17.5.4)

The **arp** command is executed from the Windows, Linux, or Mac command prompt. This command lists all devices currently in the ARP cache of the host, as well as each device's IPv4 address, physical address, and the type of addressing (static/dynamic). For instance, refer to the topology in Figure 17-20.

Example 17-18 displays the output of the **arp -a** command on the Windows PC-A host.

Example 17-18 The ARP Table on a Windows Host

```
C:\Users\PC-A> arp -a
Interface: 192.168.93.175 --- 0xc
    Internet Address      Physical Address      Type
    -----
    10.0.0.2               d0-67-e5-b6-56-4b    dynamic
    10.0.0.3               78-48-59-e3-b4-01    dynamic
    10.0.0.4               00-21-b6-00-16-97    dynamic
    10.0.0.254             00-15-99-cd-38-d9    dynamic
```

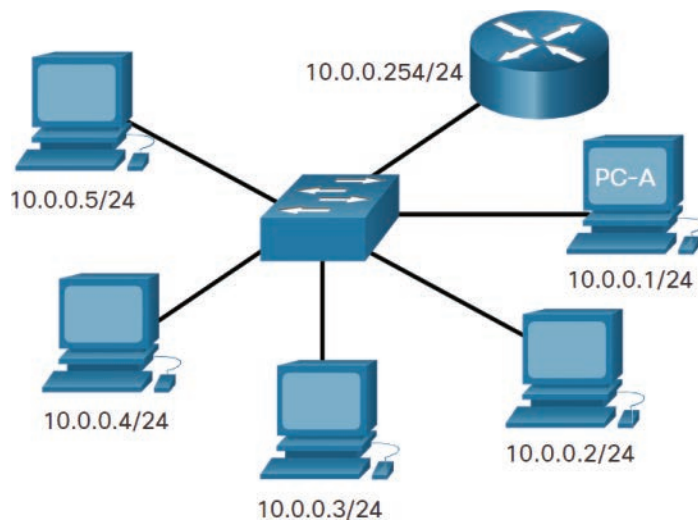


Figure 17-20 ARP Example Topology

The **arp -a** command displays the known IP address and MAC address binding. Notice that IP address 10.0.0.5 is not included in Example 17-18. This is because the ARP cache only displays information from devices that have been recently accessed.

To ensure that the ARP cache is populated, you can ping a device so that it has an entry in the ARP table. For instance, if PC-A pings 10.0.0.5, the ARP cache then contains an entry for that IP address.

A network administrator who wants to repopulate the cache with updated information can clear the cache by using the **netsh interface ip delete arp** command.

Note

You may need administrator access on the host to be able to use the **netsh interface ip delete arp** command.

Common show Commands Revisited (17.5.5)

In the same way that commands and utilities are used to verify a host configuration, commands can be used to verify the interfaces of intermediary devices. Cisco IOS provides commands to verify the operation of router and switch interfaces.

The Cisco IOS CLI **show** commands display relevant information about the configuration and operation of a device. Network technicians use **show** commands extensively for viewing configuration files, checking the status of device interfaces and processes, and verifying a device's operational status. The status of nearly every process or function of a router can be displayed by using a **show** command.

Table 17-3 lists commonly used **show** commands and when to use them.

Table 17-3 Common **show** Commands

Command	When It Is Useful
<code>show running-config</code>	To verify the current configuration and settings
<code>show interfaces</code>	To verify the interface status and see if there are any error messages
<code>show ip interface</code>	To verify the Layer 3 information of an interface
<code>show arp</code>	To verify the list of known hosts on the local Ethernet LANs
<code>show ip route</code>	To verify the Layer 3 routing information
<code>show protocols</code>	To verify which protocols are operational
<code>show version</code>	To verify the memory, interfaces, and licenses of the device

Example 17-19 through 17-25 display output from each of these **show** commands.

Note

The output of some commands has been edited to focus on pertinent settings and to reduce content.

As shown in Example 17-19, the **show running-config** command verifies the current configuration and settings.

Example 17-19 The **show running-config** Command

```
R1# show running-config
(Output omitted)
!
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname R1
!
interface GigabitEthernet0/0/0
description Link to R2
ip address 209.165.200.225 255.255.255.252
negotiation auto
```

```
!  
interface GigabitEthernet0/0/1  
  description Link to LAN  
  ip address 192.168.10.1 255.255.255.0  
  negotiation auto  
!  
router ospf 10  
  network 192.168.10.0 0.0.0.255 area 0  
  network 209.165.200.224 0.0.0.3 area 0  
!  
banner motd ^C Authorized access only! ^C  
!  
line con 0  
  password 7 14141B180F0B  
  login  
line vty 0 4  
  password 7 00071A150754  
  login  
  transport input telnet ssh  
!  
end  
R1#
```

As shown in Example 17-20, the **show interfaces** command verifies the interface status and displays any error messages.

Example 17-20 The show interfaces Command

```
R1# show interfaces  
GigabitEthernet0/0/0 is up, line protocol is up  
  Hardware is ISR4321-2x1GE, address is a0e0.af0d.e140 (bia a0e0.af0d.e140)  
  Description: Link to R2  
  Internet address is 209.165.200.225/30  
  MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,  
    reliability 255/255, txload 1/255, rxload 1/255  
  Encapsulation ARPA, loopback not set  
  Keepalive not supported  
  Full Duplex, 100Mbps, link type is auto, media type is RJ45  
  output flow-control is off, input flow-control is off  
  ARP type: ARPA, ARP Timeout 04:00:00  
  Last input 00:00:01, output 00:00:21, output hang never  
  Last clearing of "show interface" counters never  
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0  
  Queueing strategy: fifo
```



```

Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  5127 packets input, 590285 bytes, 0 no buffer
    Received 29 broadcasts (0 IP multicasts)
      0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
    0 watchdog, 5043 multicast, 0 pause input
  1150 packets output, 153999 bytes, 0 underruns
    0 output errors, 0 collisions, 2 interface resets
    0 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    1 lost carrier, 0 no carrier, 0 pause output
    0 output buffer failures, 0 output buffers swapped out
GigabitEthernet0/0/1 is up, line protocol is up

(Output omitted)

```

As shown in Example 17-21, the **show ip interface** command verifies the Layer 3 information of an interface.

Example 17-21 The **show ip interface** Command

```

R1# show ip interface
GigabitEthernet0/0/0 is up, line protocol is up
  Internet address is 209.165.200.225/30
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Multicast reserved groups joined: 224.0.0.5 224.0.0.6
  Outgoing Common access list is not set
  Outgoing access list is not set
  Inbound Common access list is not set
  Inbound access list is not set
  Proxy ARP is enabled
  Local Proxy ARP is disabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachable are always sent
  ICMP mask replies are never sent

```

```
IP fast switching is enabled
IP Flow switching is disabled
IP CEF switching is enabled
IP CEF switching turbo vector
IP Null turbo vector
Associated unicast routing topologies:
    Topology "base", operation state is UP
IP multicast fast switching is enabled
IP multicast distributed fast switching is disabled
IP route-cache flags are Fast, CEF
Router Discovery is disabled
IP output packet accounting is disabled
IP access violation accounting is disabled
TCP/IP header compression is disabled
RTP/IP header compression is disabled
Probe proxy name replies are disabled
Policy routing is disabled
Network address translation is disabled
BGP Policy Mapping is disabled
Input features: MCI Check
IPv4 WCCP Redirect outbound is disabled
IPv4 WCCP Redirect inbound is disabled
IPv4 WCCP Redirect exclude is disabled
GigabitEthernet0/0/1 is up, line protocol is up

(Output omitted)
```

As shown in Example 17-22, the **show arp** command 2 verifies the list of known hosts on the local Ethernet LANs.

Example 17-22 The show arp Command

```
R1# show arp
Protocol Address      Age (min)  Hardware Addr  Type   Interface
-----
Internet 192.168.10.1      -         a0e0.af0d.e141 ARPA   GigabitEthernet0/0/1
Internet 192.168.10.10    95        c07b.bcc4.a9c0 ARPA   GigabitEthernet0/0/1
Internet 209.165.200.225  -         a0e0.af0d.e140 ARPA   GigabitEthernet0/0/0
Internet 209.165.200.226  138      a03d.6fe1.9d90 ARPA   GigabitEthernet0/0/0
R1#
```

As shown in Example 17-23, the **show ip route** command verifies the Layer 3 routing information.

Example 17-23 The show ip route Command

```

R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
Gateway of last resort is 209.165.200.226 to network 0.0.0.0
O*E2  0.0.0.0/0 [110/1] via 209.165.200.226, 02:19:50, GigabitEthernet0/0/0
      10.0.0.0/24 is subnetted, 1 subnets
O      10.1.1.0 [110/3] via 209.165.200.226, 02:05:42, GigabitEthernet0/0/0
      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.10.0/24 is directly connected, GigabitEthernet0/0/1
L      192.168.10.1/32 is directly connected, GigabitEthernet0/0/1
      209.165.200.0/24 is variably subnetted, 3 subnets, 2 masks
C      209.165.200.224/30 is directly connected, GigabitEthernet0/0/0
L      209.165.200.225/32 is directly connected, GigabitEthernet0/0/0
O      209.165.200.228/30
      [110/2] via 209.165.200.226, 02:07:19, GigabitEthernet0/0/0
R1#

```

As shown in Example 17-24, the **show protocols** command verifies which protocols are operational.

Example 17-24 The show protocols Command

```

R1# show protocols
Global values:
  Internet Protocol routing is enabled
GigabitEthernet0/0/0 is up, line protocol is up
  Internet address is 209.165.200.225/30
GigabitEthernet0/0/1 is up, line protocol is up
  Internet address is 192.168.10.1/24
Serial0/1/0 is down, line protocol is down
Serial0/1/1 is down, line protocol is down
GigabitEthernet0 is administratively down, line protocol is down
R1#

```

As shown in Example 17-25, the **show version** command verifies the memory, interfaces, and licenses of a device.

Example 17-25 The show version Command

```
R1# show version
Cisco IOS XE Software, Version 03.16.08.S - Extended Support Release
Cisco IOS Software, ISR Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version
  15.5(3)S8, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2018 by Cisco Systems, Inc.
Compiled Wed 08-Aug-18 10:48 by mcpre

(Output omitted)

ROM: IOS-XE ROMMON
R1 uptime is 2 hours, 25 minutes
Uptime for this control processor is 2 hours, 27 minutes
System returned to ROM by reload
System image file is "bootflash:/isr4300-universalk9.03.16.08.S.155-3.S8-ext.SPA.
  bin"
Last reload reason: LocalSoft

(Output omitted)

Technology Package License Information:
-----
Technology      Technology-package      Technology-package
              Current        Type                Next reboot
-----
appxk9          appxk9                 RightToUse          appxk9
uck9            None                   None                 None
securityk9      securityk9             Permanent           securityk9
ipbase          ipbasek9               Permanent           ipbasek9
cisco ISR4321/K9 (1RU) processor with 1647778K/6147K bytes of memory.
Processor board ID FLM2044W0LT
2 Gigabit Ethernet interfaces
2 Serial interfaces
32768K bytes of non-volatile configuration memory.
4194304K bytes of physical memory.
3207167K bytes of flash memory at bootflash:.
978928K bytes of USB flash at usb0:.
Configuration register is 0x2102
R1#
```

The show cdp neighbors Command (17.5.6)

In addition to the commands discussed so far in this chapter, several other IOS commands are useful. Cisco Discovery Protocol (CDP) is a Cisco-proprietary protocol that runs at the data link layer. Because CDP operates at the data link layer, two or more Cisco network devices, such as routers that support different network layer protocols, can learn about each other even if Layer 3 connectivity has not been established.

When a Cisco device boots, CDP starts by default. CDP automatically discovers neighboring Cisco devices running CDP, regardless of which Layer 3 protocol or suites are running. CDP exchanges hardware and software device information with its directly connected CDP neighbors.

CDP provides the following information about each CDP neighbor device:

- **Device identifiers:** The configured hostname of a switch, router, or other device
- **Address list:** Up to one network layer address for each protocol supported
- **Port identifier:** The name of the local and remote ports, in the form of an ASCII character string, such as FastEthernet 0/0
- **Capabilities list:** Capabilities, such as whether a specific device is a Layer 2 switch or a Layer 3 switch
- **Platform:** The hardware platform of a device (for example, a Cisco 1841 series router)

Refer to the topology in Figure 17-21 and the **show cdp neighbors** command output in Example 17-26.



Figure 17-21 CDP Neighbors Example Topology

Example 17-26 The show cdp neighbors Command

```
R3# show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID         Local Intrfce   Holdtme    Capability   Platform   Port ID
S3                 Gig 0/0/1      122        S I          WS-C2960+  Fas 0/5

Total cdp entries displayed : 1
R3#
```

The output in Example 17-26 shows that the R3 GigabitEthernet 0/0/1 interface is connected to the FastEthernet 0/5 interface of S3, which is a Cisco Catalyst 2960+ switch. Notice that R3 has not gathered information about S4. This is because CDP can only discover directly connected Cisco devices. S4 is not directly connected to R3 and therefore is not listed in the output.

The **show cdp neighbors detail** command reveals the IP address of a neighboring device. This command reveals the IP address of the neighbor regardless of whether you can ping that neighbor. This command is very helpful when two Cisco routers cannot route across their shared data link. The **show cdp neighbors detail** command helps determine whether one of the CDP neighbors has an IP configuration error.

As helpful as CDP is, it can also be a security risk because it can provide useful network infrastructure information to threat actors. For example, by default, many IOS versions send CDP advertisements out all enabled ports. However, best practice suggests that CDP should be enabled only on interfaces that are connecting to other infrastructure Cisco devices. CDP advertisements should be disabled on user-facing ports.

Because some IOS versions send out CDP advertisements by default, it is important to know how to disable CDP. To disable CDP globally, use the global configuration command **no cdp run**. To disable CDP on an interface, use the interface command **no cdp enable**.

The show ip interface brief Command (17.5.7)

One of the most frequently used commands is the **show ip interface brief** command. This command provides more abbreviated output than the **show ip interface** command. It provides a summary of the key information for all the network interfaces on a router.

In Example 17-27, the **show ip interface brief** output displays all interfaces on the router, the IP address assigned to each interface, if any, and the operational status of the interface.

Example 17-27 The **show ip interface brief** Command on a Router

```
R1# show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0/0	209.165.200.225	YES	manual	up	up
GigabitEthernet0/0/1	192.168.10.1	YES	manual	up	up
Serial0/1/0	unassigned	NO	unset	down	down
Serial0/1/1	unassigned	NO	unset	down	down
GigabitEthernet0	unassigned	YES	unset	administratively down	down

```
R1#
```

Verify Switch Interfaces

The **show ip interface brief** command can be used to verify the status of the switch interfaces, as shown in Example 17-28.

Example 17-28 The **show ip interface brief** Command on a Switch

S1# show ip interface brief						
Interface	IP-Address	OK?	Method	Status		Protocol
Vlan1	192.168.254.250	YES	manual	up		up
FastEthernet0/1	unassigned	YES	unset	down		down
FastEthernet0/2	unassigned	YES	unset	up		up
FastEthernet0/3	unassigned	YES	unset	up		up

The output in Example 17-28 shows that the VLAN 1 interface is assigned the IPv4 address 192.168.254.250, has been enabled, and is operational. The output also shows that the FastEthernet0/1 interface is down. This indicates that either no device is connected to the interface or the device that is connected has a network interface that is not operational. The output also shows that the FastEthernet0/2 and FastEthernet0/3 interfaces are operational. This is indicated by both the status and protocol being shown as up.



Video—The show version Command (17.5.8)

The **show version** command can be used to verify and troubleshoot some of the basic hardware and software components used during the boot process. Click Play to view a video from earlier in the course, which reviews an explanation of the **show version** command.

Refer to the online course to view this video.



Packet Tracer—Interpret show Command Output (17.5.9)

This activity is designed to reinforce the use of router **show** commands. In it you will examine the output of several **show** commands.

Troubleshooting Methodologies (17.6)

In this chapter, you have learned about some utilities and commands that you can use to help identify problem areas in a network. This is an important part of

troubleshooting. There are many ways to troubleshoot a network problem. This section details a structured troubleshooting process that can help you to become a better network administrator. It also provides a few more commands to help you resolve problems. Network troubleshooting is a critical skill for any network professional.

Basic Troubleshooting Approaches (17.6.1)

Network problems can be simple or complex, and they can result from a combination of hardware, software, and connectivity issues. Technicians must be able to analyze problems and determine the cause of an error before they can resolve the network issue. This process is called *troubleshooting*.

A common and efficient troubleshooting methodology is based on the scientific method. Table 17-4 shows the six main steps in the troubleshooting process.

Table 17-4 Six Troubleshooting Steps

Step	Description
Step 1. Identify the problem.	This is the first step in the troubleshooting process.
	Although tools can be used in this step, a conversation with the user is often very helpful.
Step 2. Establish a theory of probable causes.	After the problem is identified, try to establish a theory of probable causes.
	This step often yields more than a few probable causes to the problem.
Step 3. Test the theory to determine the cause.	Based on the probable causes, test your theories to determine which one is the cause of the problem.
	A technician often applies a quick procedure to see if it solves the problem.
	If a quick procedure does not correct the problem, you might need to research the problem further to establish the exact cause.
Step 4. Establish a plan of action and implement the solution.	After you have determined the exact cause of the problem, establish a plan of action to resolve the problem and implement the solution.

Step	Description
Step 5. Verify the solution and implement preventive measures.	After you have corrected the problem, verify full functionality. If applicable, implement preventive measures.
Step 6. Document findings, actions, and outcomes.	In the final step of the troubleshooting process, document your findings, actions, and outcomes. This is very important for future reference.

To assess a problem, determine how many devices on the network are experiencing the problem. If there is a problem with one device on the network, start the troubleshooting process at that device. If there is a problem with all devices on the network, start the troubleshooting process at the device where all other devices are connected. You should develop a logical and consistent method for diagnosing network problems by eliminating one problem at a time.

Resolve or Escalate? (17.6.2)

In some situations, it might not be possible to resolve a problem immediately. A problem should be escalated when it requires a manager decision, some specific expertise, or a network access level unavailable to the troubleshooting technician.

For example, after troubleshooting, say that a technician concludes that a router module should be replaced. This problem should be escalated for manager approval. The manager may have to escalate the problem again as it may require the approval of the financial department before a new module can be purchased.

A company policy should clearly state when and how a technician should escalate a problem.

The debug Command (17.6.3)

OS processes, protocols, mechanisms, and events generate messages to communicate their status. These messages can provide valuable information when troubleshooting or verifying system operations. The IOS **debug** command allows an administrator to display these messages in real time for analysis. It is a very important tool for monitoring events on a Cisco IOS device.

All **debug** commands are entered in privileged EXEC mode. Cisco IOS allows for narrowing the output of **debug** to include only the relevant feature or subfeature. This is important because debugging output is assigned high priority in the CPU process, and it can render the system unusable. For this reason, use **debug** commands only to troubleshoot specific problems.

For example, to monitor the status of ICMP messages in a Cisco router, use **debug ip icmp**, as shown in Example 17-29.

Example 17-29 Using a debug Command to Monitor ICMP

```
R1# debug ip icmp
ICMP packet debugging is on
R1#
R1# ping 10.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R1#
*Aug 20 14:18:59.605: ICMP: echo reply rcvd, src 10.1.1.1, dst
    209.165.200.225, topology BASE, dscp 0 topoid 0
*Aug 20 14:18:59.606: ICMP: echo reply rcvd, src 10.1.1.1, dst
    209.165.200.225, topology BASE, dscp 0 topoid 0
*Aug 20 14:18:59.608: ICMP: echo reply rcvd, src 10.1.1.1, dst
    209.165.200.225, topology BASE, dscp 0 topoid 0
*Aug 20 14:18:59.609: ICMP: echo reply rcvd, src 10.1.1.1, dst
    209.165.200.225, topology BASE, dscp 0 topoid 0
*Aug 20 14:18:59.611: ICMP: echo reply rcvd, src 10.1.1.1, dst
    209.165.200.225, topology BASE, dscp 0 topoid 0
R1#
```

To list a brief description of all the debugging command options, use the **debug ?** command in privileged EXEC mode at the command line.

To turn off a specific debugging feature, add the **no** keyword in front of the **debug** command:

```
Router# no debug ip icmp
```

Alternatively, you can enter the **undebug** form of the command in privileged EXEC mode:

```
Router# undebug ip icmp
```

To turn off all active **debug** commands at once, use the **undebug all** command:

```
Router# undebug all
```

Be cautious when using some **debug** command. Commands such as **debug all** and **debug ip packet** generate a substantial amount of output and can use a lot of system resources. A router could get so busy displaying **debug** messages that it would not have enough processing power to perform its network functions or even listen to commands to turn off debugging. For this reason, using these command options is not recommended and should be avoided.

The terminal monitor Command (17.6.4)

Connections to grant access to the IOS command-line interface can be established in two ways:

- **Locally:** Local connections (that is, console connection) require physical access to the router or switch console port, using a rollover cable.
- **Remotely:** Remote connections require the use of Telnet or SSH to establish a connection to an IP configured device.

Certain IOS messages are automatically displayed on a console connection but not on a remote connection. For instance, **debug** output is displayed by default on console connections. However, **debug** output is not automatically displayed on remote connections. This is because **debug** messages are log messages, which are prevented from being displayed on vty lines.

In Example 17-30, for instance, the user established a remote connection using Telnet from R2 to R1. The user then issued the **debug ip icmp** command. However, the command failed to display **debug** output.

Example 17-30 Demonstrating No Terminal Output for a **debug** Command

```
R2# telnet 209.165.200.225
Trying 209.165.200.225 ... Open
  Authorized access only!
User Access Verification
Password:
R1> enable
Password:
R1# debug ip icmp
ICMP packet debugging is on
R1# ping 10.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R1#
```

To display log messages on a terminal (virtual console), use the **terminal monitor** privileged EXEC command. To stop logging messages on a terminal, use the **terminal no monitor** privileged EXEC command.

For instance, in Example 17-31, notice that the **terminal monitor** command has been entered, and the **ping** command displays the **debug** output.

Example 17-31 Enabling and Verifying Terminal Monitoring

```
R1# terminal monitor
R1# ping 10.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R1#
*Aug 20 16:03:49.735: ICMP: echo reply rcvd, src 10.1.1.1, dst
209.165.200.225, topology BASE, dscp 0 topoid 0
**Aug 20 16:03:49.737: ICMP: echo reply rcvd, src 10.1.1.1, dst
209.165.200.225, topology BASE, dscp 0 topoid 0
**Aug 20 16:03:49.738: ICMP: echo reply rcvd, src 10.1.1.1, dst
209.165.200.225, topology BASE, dscp 0 topoid 0
**Aug 20 16:03:49.740: ICMP: echo reply rcvd, src 10.1.1.1, dst
209.165.200.225, topology BASE, dscp 0 topoid 0
**Aug 20 16:03:49.741: ICMP: echo reply rcvd, src 10.1.1.1, dst
209.165.200.225, topology BASE, dscp 0 topoid 0
R1# no debug ip icmp
ICMP packet debugging is off
R1#
```

Note

The intent of the **debug** command is to capture live output for a short period of time (that is, a few seconds to a minute or so). Always disable **debug** when it is not required.

Interactive Graphic

Check Your Understanding—Troubleshooting Methodologies (17.6.5)

Refer to the online course to complete this activity.

Troubleshooting Scenarios (17.7)

This section focuses on a variety of troubleshooting scenarios.

Duplex Operation and Mismatch Issues (17.7.1)

Many common network problems can be identified and resolved with little effort. Now that you have the tools and know the process for troubleshooting a network, this section reviews some common networking issues that you are likely to find as a network administrator.

In data communications, *duplex* refers to the direction of data transmission between two devices. There are two duplex communication modes:

- **Half-duplex:** Communication is restricted to the exchange of data in one direction at a time.
- **Full-duplex:** Communications are permitted to be sent and received simultaneously.

Figure 17-22 illustrates how each duplex method operates.

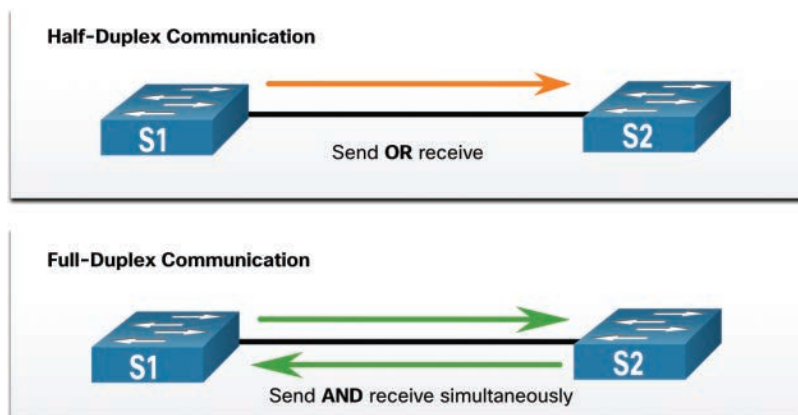


Figure 17-22 Duplex Operation

Interconnected Ethernet interfaces must operate in the same duplex mode for best communication performance and to avoid inefficiency and latency on the link.

The Ethernet autonegotiation feature facilitates configuration, minimizes problems, and maximizes link performance between two interconnecting Ethernet links. The connected devices first announce their supported capabilities and then choose the highest performance mode supported by both ends. For example, the switch and router in Figure 17-23 have successfully autonegotiated full-duplex mode.

If one of the two connected devices is operating in full-duplex and the other is operating in half-duplex, a duplex mismatch occurs. While data communication can occur through a link with a duplex mismatch, link performance is very poor.

Duplex mismatches are typically caused by misconfigured interfaces or, in rare instances, by failed autonegotiation. Duplex mismatches may be difficult to troubleshoot as the communication between devices still occurs.

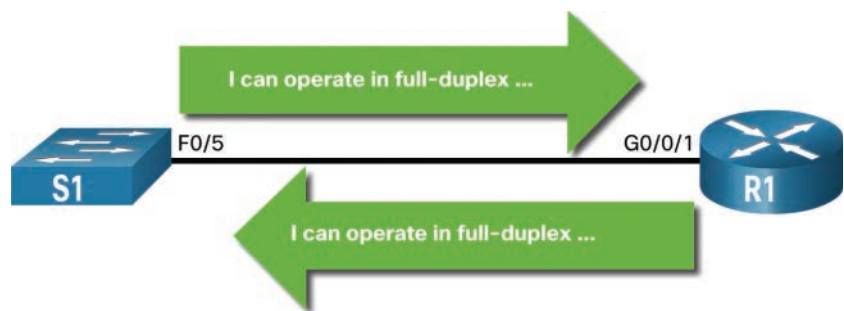


Figure 17-23 Duplex Autonegotiation

IP Addressing Issues on IOS Devices (17.7.2)

IP address-related problems are likely to prevent remote network devices from communicating. Because IP addresses are hierarchical, any IP address assigned to a network device must conform to that range of addresses in that network. Wrongly assigned IP addresses create a variety of issues, including IP address conflicts and routing problems.

Two common causes of incorrect IPv4 assignment are manual assignment mistakes and DHCP-related issues.

Network administrators often have to manually assign IP addresses to devices such as servers and routers. If a mistake is made during the assignment, communications issues with the device are very likely to occur.

On an IOS device, use the **show ip interface** command or the **show ip interface brief** command to verify what IPv4 addresses are assigned to the network interfaces. For example, issuing the **show ip interface brief** command as shown in Example 17-32 would validate the interface status on R1.

Example 17-32 Using show ip interface brief to Verify IPv4 Addressing on a Cisco Device

```
R1# show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0/0	209.165.200.225	YES	manual	up	up
GigabitEthernet0/0/1	192.168.10.1	YES	manual	up	up
Serial0/1/0	unassigned	NO	unset	down	down
Serial0/1/1	unassigned	NO	unset	down	down
GigabitEthernet0	unassigned	YES	unset	administratively down	down

```
R1#
```

IP Addressing Issues on End Devices (17.7.3)

In Windows-based machines, when a device cannot contact a DHCP server, Windows automatically assigns an address belonging to the 169.254.0.0/16 range. This automatic addressing, called Automatic Private IP Addressing (APIPA), is designed to facilitate communication within the local network. Think of it as Windows saying, “I will use this address from the 169.254.0.0/16 range because I could not get any other address.”

Often, a computer with an APIPA address cannot communicate with other devices in the network because those devices most likely do not belong to the 169.254.0.0/16 network. Such a situation indicates an automatic IPv4 address assignment problem that should be fixed.

Note

Other operating systems, such as Linux and macOS, do not assign an IPv4 address to the network interface if communication with a DHCP server fails.

Most end devices are configured to rely on a DHCP server for automatic IPv4 address assignment. If a device is unable to communicate with the DHCP server, the server cannot assign an IPv4 address for the specific network, and the device is unable to communicate.

To verify the IP addresses assigned to a Windows-based computer, use the **ipconfig** command, as shown in Example 17-33.

Example 17-33 Using **ipconfig** to Verify IPv4 Addressing on a Windows Host

```
C:\Users\PC-A> ipconfig

Windows IP Configuration

(Output omitted)

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . . : 
    Link-local IPv6 Address . . . . . : fe80::a4aa:2dd1:ae2d:a75e%16
    IPv4 Address. . . . . : 192.168.10.10
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.10.1

(Output omitted)
```

Default Gateway Issues (17.7.4)

The default gateway for an end device is the closest networking device that can forward traffic to other networks. If a device has an incorrect or nonexistent default

gateway address, it is unable to communicate with devices in remote networks. Because the default gateway is the path to remote networks, its address must belong to the same network as the end device.

The address of the default gateway can be manually set or obtained from a DHCP server. Much like IPv4 addressing issues, default gateway problems can be related to misconfiguration (in the case of manual assignment) or DHCP problems (if automatic assignment is in use).

To solve misconfigured default gateway issues, ensure that the device has the correct default gateway configured. If the default address was manually set but is incorrect, you can simply replace it with the proper address. If the default gateway address was automatically set, you need to ensure that the device can communicate with the DHCP server. It is also important to verify that the proper IPv4 address and subnet mask were configured on the interface of the router and that the interface is active.

To verify the default gateway on a Windows-based computer, use the **ipconfig** command, as shown in Example 17-34.

Example 17-34 Using **ipconfig** to Verify the Default Gateway on a Windows Host

```
C:\Users\PC-A> ipconfig
Windows IP Configuration
(Output omitted)
Wireless LAN adapter Wi-Fi:
    Connection-specific DNS Suffix . . :
    Link-local IPv6 Address . . . . . : fe80::a4aa:2dd1:ae2d:a75e%16
    IPv4 Address. . . . . : 192.168.10.10
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.10.1
(Output omitted)
```

On a router, use the **show ip route** command to list the routing table and verify that the default gateway, known as a default route, has been set. This route is used when the destination address of the packet does not match any other routes in its routing table.

The output in Example 17-35 verifies that R1 has a default gateway (that is, gateway of last resort) configured, and it is pointing to IP address 209.168.200.226.

In Example 17-35, the first highlighted line basically states that the gateway to any (that is, 0.0.0.0) should be sent to IP address 209.165.200.226. The second highlighted line shows how R1 learned about the default gateway. In this case, R1 received the information from another OSPF-enabled router.

Example 17-35 Using **show ip route** to Verify the Default Gateway on a Router

```

R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.226 to network 0.0.0.0
O*E2 0.0.0.0/0 [110/1] via 209.165.200.226, 02:19:50, GigabitEthernet0/0/0
    10.0.0.0/24 is subnetted, 1 subnets
O      10.1.1.0 [110/3] via 209.165.200.226, 02:05:42, GigabitEthernet0/0/0
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.10.0/24 is directly connected, GigabitEthernet0/0/1
L      192.168.10.1/32 is directly connected, GigabitEthernet0/0/1
    209.165.200.0/24 is variably subnetted, 3 subnets, 2 masks
C      209.165.200.224/30 is directly connected, GigabitEthernet0/0/0
L      209.165.200.225/32 is directly connected, GigabitEthernet0/0/0
O      209.165.200.228/30
        [110/2] via 209.165.200.226, 02:07:19, GigabitEthernet0/0/0
R1#

```

Troubleshooting DNS Issues (17.7.5)

Domain Name System (DNS) is an automated service that matches a website name, such as `www.cisco.com`, with an IP address. Although DNS resolution is not crucial to device communication, it is very important to the end user.

It is common for users to mistakenly relate the operation of an internet link to the availability of DNS. User complaints such as “the network is down” or “the internet is down” are often caused by unreachable DNS servers. While packet routing and all other network services are still operational, DNS failures often lead the user to the wrong conclusion. If a user types in a domain name such as `www.cisco.com` in a web browser and the DNS server is unreachable, the name will not be translated into an IP address, and the website will not display.

DNS server addresses can be manually or automatically assigned. Network administrators are often responsible for manually assigning DNS server addresses on servers and other devices, while DHCP is used to automatically assign DNS server addresses to clients.

Although it is common for companies and organizations to manage their own DNS servers, any reachable DNS server can be used to resolve names. Small office and home office (SOHO) users often rely on the DNS server maintained by their ISP for name resolution. ISP-maintained DNS servers are assigned to SOHO customers via DHCP. In addition, Google maintains a public DNS server that can be used by anyone; it is very useful for testing. The IPv4 address of Google’s public DNS server is `8.8.8.8`, and `2001:4860:4860::8888` is its IPv6 DNS address.

Cisco offers OpenDNS, which provides secure DNS service by filtering phishing and some malware sites. You can change your DNS addresses to 208.67.222.222 and 208.67.220.220 in the Preferred DNS server and Alternate DNS server fields. Advanced features such as web content filtering and security are available to families and businesses.

Use the **ipconfig /all** command, as shown in Example 17-36, to verify which DNS server a Windows computer is using.

Example 17-36 Using ipconfig /all to Verify the DNS Server in Use on a Windows Host

```
C:\Users\PC-A> ipconfig /all
(Output omitted)
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . . . : 
    Description . . . . . : Intel(R) Dual Band Wireless-AC 8265
    Physical Address. . . . . : F8-94-C2-E4-C5-0A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::a4aa:2dd1:ae2d:a75e%16(Preferred)
    IPv4 Address. . . . . : 192.168.10.10(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : August 17, 2019 1:20:17 PM
    Lease Expires . . . . . : August 18, 2019 1:20:18 PM
    Default Gateway . . . . . : 192.168.10.1
    DHCP Server . . . . . : 192.168.10.1
    DHCPv6 IAID . . . . . : 100177090
    DHCPv6 Client DUID. . . . . : 00-01-00-01-21-F3-76-75-54-E1-AD-DE-DA-9A
    DNS Servers . . . . . : 208.67.222.222
    NetBIOS over Tcpip. . . . . : Enabled
(Output omitted)
```

The **nslookup** command is another useful DNS troubleshooting tool for PCs. With **nslookup**, a user can manually place DNS queries and analyze the DNS responses. The **nslookup** command in Example 17-37 shows the output for a query for www.cisco.com. Notice that you can also simply enter an IP address, and **nslookup** will resolve the name.

Note

For various reasons, it is not always possible to type an IP address in **nslookup** and receive the domain name. One of the most common reasons for this is that most websites run on servers that support multiple sites.

Example 17-37 Using **nslookup** on a Windows Host to Verify DNS Information

```

C:\Users\PC-A> nslookup
Default Server:  Home-Net
Address:  192.168.1.1
> cisco.com
Server:  Home-Net
Address:  192.168.1.1
Non-authoritative answer:
Name:     cisco.com
Addresses: 2001:420:1101:1::185
           72.163.4.185
> 8.8.8.8
Server:  Home-Net
Address:  192.168.1.1
Name:     dns.google
Address:  8.8.8.8
>
> 208.67.222.222
Server:  Home-Net
Address:  192.168.1.1
Name:     resolver1.opendns.com
Address:  208.67.222.222
>

```

**Lab—Troubleshoot Connectivity Issues (17.7.6)**

In this lab, you will complete the following objectives:

- Identify the Problem
- Implement Network Changes
- Verify Full Functionality
- Document Findings and Configuration Changes

Packet Tracer
☐ Activity

Packet Tracer—Troubleshoot Connectivity Issues (17.7.7)

The objective of this Packet Tracer activity is to troubleshoot and resolve connectivity issues, if possible. Otherwise, the issues should be clearly documented so they can be escalated.

Summary (17.8)

The following is a summary of the topics in the chapter and their corresponding online modules.

Devices in a Small Network

A small network typically has a single WAN connection provided by DSL, cable, or an Ethernet connection. A small network is managed by a local IT technician or by a contracted professional. Factors to consider when selecting network devices for a small network are cost, speed, types of ports/interfaces, expandability, and OS features and services. When implementing a network, create an IP addressing scheme and use it on end devices, servers and peripherals, and intermediary devices. Redundancy can be accomplished by installing duplicate equipment, but it can also be accomplished by supplying duplicate network links for critical areas. The routers and switches in a small network should be configured to support real-time traffic, such as voice and video, in an appropriate manner relative to other data traffic. In fact, a good network design implements quality of service (QoS) to classify traffic carefully according to priority.

Small Network Applications and Protocols

Two forms of software programs or processes provide access to a network: network applications and application layer services. Some end-user applications implement application layer protocols and are able to communicate directly with the lower layers of the protocol stack. Email clients and web browsers are examples of this type of application. Other programs may need the assistance of application layer services to use network resources such as file transfer or network print spooling. These are the programs that interface with the network and prepare the data for transfer. The two most common remote access solutions are Telnet and Secure Shell (SSH). SSH is a secure alternative to Telnet. Network administrators must also support common network servers and their required related network protocols, such as web servers, email servers, FTP servers, DHCP servers, and DNS servers. Businesses today are increasingly using IP telephony and streaming media to communicate with customers and business partners. These are real-time applications. The network infrastructure must therefore support VoIP, IP telephony, and other real-time applications.

Scale to Larger Networks

To scale a network, several elements are required: network documentation, device inventory, budget, and traffic analysis. It is important to know the type of traffic that is crossing the network as well as the current traffic flow. Capture traffic during peak

utilization times to get a good representation of the different traffic types and perform the capture on different network segments and devices as some traffic will be local to a particular segment. Network administrators must know how network use is changing. Usage details of employee computers can be captured in a “snapshot” with tools such as the Windows Task Manager, Event Viewer, and Data Usage.

Verify Connectivity

Using the **ping** command is the most effective way to quickly test Layer 3 connectivity between a source IP address and a destination IP address. This command also displays various round-trip time statistics. Cisco IOS offers an “extended” mode of the **ping** command, which lets the user create special types of pings by adjusting parameters related to the command operation. Extended **ping** is entered in privileged EXEC mode by typing **ping** without a destination IP address. **tracert** can help locate Layer 3 problem areas in a network. A trace returns a list of hops as a packet is routed through a network. It is used to identify the point along the path where a problem can be found. In Windows, the command is **tracert**, whereas in Cisco IOS the equivalent command is **tracert**. There is also an extended **tracert** command. It allows an administrator to adjust parameters related to the command operation. The output derived from network commands contributes data to the network baseline. One method for starting a baseline is to copy and paste the results from an executed **ping**, **tracert**, or other relevant commands into a text file. Such text files can be timestamped with the date and saved into an archive for later retrieval and comparison.

Host and IOS Commands

Network administrators view IP addressing information (address, mask, router, and DNS) on a Windows host by issuing the **ipconfig** command. Other necessary commands are **ipconfig /all**, **ipconfig /release** and **ipconfig /renew**, and **ipconfig /displaydns**. Verifying IP settings by using the GUI on a Linux machine differs depending on the Linux distribution (distro) and desktop interface. Necessary commands are **ifconfig** and **ip address**. In the GUI of a Mac host, open **Network Preferences > Advanced** to get the IP addressing information. Other IP addressing commands for Mac are **ifconfig**, **networksetup -listallnetworkservices**, and **networksetup -getinfo <network service>**. The **arp** command, which is executed from the Windows, Linux, or Mac command prompt, lists all devices currently in the ARP cache of the host and includes the IPv4 address, physical address, and type of addressing (static/dynamic) for each device. The **arp -a** command displays the known IP address and MAC address binding. Common **show** commands are **show running-config**, **show interfaces**, **show ip address**, **show arp**, **show ip route**, **show protocols**, and **show version**. The **show cdp neighbor** command provides

the following information about each CDP neighbor device: identifiers, address list, port identifier, capabilities list, and platform. The **show cdp neighbors detail** command helps determine if one of the CDP neighbors has an IP configuration error. The **show ip interface brief** command output displays all interfaces on the router, the IP address assigned to each interface, if any, and the operational status of the interface.

Troubleshooting Methodologies

- Step 1.** Identify the problem.
- Step 2.** Establish a theory of probable causes.
- Step 3.** Test the theory to determine the cause.
- Step 4.** Establish a plan of action and implement the solution.
- Step 5.** Verify the solution and implement preventive measures.
- Step 6.** Document findings, actions, and outcomes.

A problem should be escalated when it requires the decision of a manager, some specific expertise, or network access level unavailable to the troubleshooting technician. OS processes, protocols, mechanisms, and events generate messages to communicate their status. The IOS **debug** command allows an administrator to display these messages in real time for analysis. To display log messages on a terminal (virtual console), use the **terminal monitor** privileged EXEC command.

Troubleshooting Scenarios

There are two duplex communication modes: half-duplex and full-duplex. If one of two connected devices is operating in full-duplex and the other is operating in half-duplex, a duplex mismatch occurs. While data communication can occur through a link with a duplex mismatch, link performance is very poor.

Incorrectly assigned IP addresses create a variety of issues, including IP address conflicts and routing problems. Two common causes of incorrect IPv4 assignment are manual assignment mistakes and DHCP-related issues. Typically, an end device is configured to rely on a DHCP server for automatic IPv4 address assignment. If a device is unable to communicate with the DHCP server, then the server cannot assign an IPv4 address for the specific network, and the device is unable to communicate.

The default gateway for an end device is the closest networking device that can forward traffic to other networks. If a device has an incorrect or nonexistent default gateway address, it is unable to communicate with devices in remote networks. Because the default gateway is the path to remote networks, its address must belong to the same network as the end device.

DNS failures often lead users to conclude that the network is down. If a user types in a domain name such as `www.cisco.com` in a web browser, and the DNS server is unreachable, the name is not translated into an IP address, and the website does not appear.



Lab—Design and Build a Small Business Network (17.8.1)

In this lab, you will design and build a network.



Packet Tracer—Skills Integration Challenge (17.8.2)

In this Packet Tracer activity, you will use all the skills you have acquired throughout this book.



Packet Tracer—Troubleshooting Challenge (17.8.3)

In this Packet Tracer activity, you will troubleshoot and resolve a number of issues in an existing network.

Practice

The following activities provide practice with the topics introduced in this chapter. The lab is available in the companion *Introduction to Networks Labs & Study Guide (CCNA v7)* (ISBN 9780136634454). The Packet Tracer activity instructions are also provided in the *Labs & Study Guide*. The PKA files are available in the online course.



Labs

Lab 17.4.6: Test Network Latency with Ping and Traceroute

Lab 17.7.6: Troubleshoot Connectivity Issues

Lab 17.8.1: Design and Build a Small Network



Packet Tracer Activities

Packet Tracer 17.5.9: Interpret **show** Command Output

Packet Tracer 17.7.7: Troubleshoot Connectivity Issues

Packet Tracer 17.8.2: Skills Integration Challenge

Packet Tracer 17.8.3: Troubleshooting Challenge

Check Your Understanding Questions

Complete all the review questions listed here to test your understanding of the topics and concepts in this chapter. The appendix “Answers to ‘Check Your Understanding’ Questions” lists the answers.

1. Which network design consideration would be more important to a large corporation than to a small business?
 - a. Internet router
 - b. firewall
 - c. low port density switch
 - d. redundancy
2. A newly hired network technician is given the task of ordering new hardware for a small business with a large growth forecast. Which primary factor should the technician be concerned with when choosing the new devices?
 - a. devices with a fixed number and type of interfaces
 - b. devices that have support for network monitoring
 - c. redundant devices
 - d. devices with support for modularity
3. What type of traffic would most likely have the highest priority through a network?
 - a. FTP
 - b. instant messaging
 - c. voice
 - d. SNMP
4. A network technician is investigating network connectivity from a PC to a remote host with the address 10.1.1.5. Which command, when issued on a Windows PC, displays the path to the remote host?
 - a. `trace 10.1.1.5`
 - b. `tracert 10.1.1.5`
 - c. `tracert 10.1.1.5`
 - d. `ping 10.1.1.5`
5. A user is unable to reach the website when typing `http://www.cisco.com` in a web browser. However, she can reach the same site by typing `http://72.163.4.185`. What is the issue?
 - a. default gateway
 - b. DHCP

- c. DNS
 - d. TCP/IP protocol stack
- 6. Where are Cisco IOS **debug** output messages sent by default?
 - a. memory buffers
 - b. vty lines
 - c. syslog server
 - d. console line
- 7. Which element of scaling a network involves identifying the physical and logical topologies?
 - a. traffic analysis
 - b. network documentation
 - c. device inventory
 - d. cost analysis
- 8. What mechanism can be implemented in a small network to help minimize network latency for real-time streaming applications?
 - a. QoS
 - b. PoE
 - c. AAA
 - d. ICMP
- 9. Which process failed if a computer cannot access the internet and received the IPv4 address 169.254.142.5?
 - a. IP
 - b. DNS
 - c. DHCP
 - d. HTTP
- 10. A small company has only one router as the exit point to its ISP. Which solution could be adopted to maintain connectivity if the router itself, or its connection to the ISP, fails?
 - a. Activate another router interface that is connected to the ISP so the traffic can flow through it.
 - b. Have a second router that is connected to another ISP.
 - c. Purchase a second least-cost link from another ISP to connect to this router.
 - d. Add more interfaces to the router that is connected to the internal network.

11. When should an administrator establish a network baseline?
 - a. when the traffic is at peak in the network
 - b. when there is a sudden drop in traffic
 - c. at the lowest point of traffic in the network
 - d. at regular intervals over a period of time
12. Which two traffic types require delay-sensitive delivery? (Choose two.)
 - a. email
 - b. web
 - c. FTP
 - d. voice
 - e. video
13. A network technician suspects that a particular network connection between two Cisco switches is experiencing a duplex mismatch. Which command would the technician use to see the Layer 1 and Layer 2 details of a switch port?
 - a. `show interfaces`
 - b. `show running-config`
 - c. `show ip interface brief`
 - d. `show mac address-table`
14. Which statement is true about CDP on a Cisco device?
 - a. The `show cdp neighbor detail` command reveals the IP address of a neighbor only if there is Layer 3 connectivity.
 - b. To disable CDP globally, the `no cdp enable` command must be used in interface configuration mode.
 - c. CDP can be disabled globally or on a specific interface.
 - d. Because it runs at the data link layer, CDP can only be implemented in switches.
15. What factor should be considered in the design of a small network when devices are being chosen?
 - a. cost of devices
 - b. redundancy
 - c. traffic analysis
 - d. ISP

Answers to “Check Your Understanding” Questions

Chapter 1

1. C. Spyware is a type of software that is installed on a user's device to collect information about the user.
2. C. An organization may use an extranet to provide secure and safe access to individuals who work for a different organization but require access to the organization's data.
3. C. BYOD gives end users the freedom to use personal devices to access information and communicate across a business or campus network.
4. C. Home users, remote workers, and small offices typically require a connection to an ISP (internet service provider) to access the internet.
5. B. A wireless internet service provider (WISP) is an ISP that connects subscribers to a designated access point or hotspot using wireless technologies similar to those in home wireless local-area networks (WLANs). WISPs are most commonly found in rural environments where DSL or cable services are not available.
6. B. A scalable network expands quickly to support new users and applications.
7. D. A fault-tolerant network limits the number of affected devices during a failure. It is built to allow quick recovery when such a failure occurs.
8. B and D. A scalable network expands quickly to support new users and applications. It does this without degrading the performance of services that are being accessed by existing users. Scalability is typically required in networks with wireless and mobile devices, where the number of devices may increase at any time.
9. A. A router is a device that interconnects multiple networks and is responsible for forwarding messages between them.
10. B and C. Cellular and satellite communications are wireless technologies that do not require physical cables.
11. B. The internet is the communications network for accessing ecommerce websites.
12. D. BYOD gives end users the freedom to use personal devices to access information and communicate across a business or campus network. The trend has been to allow employees access to internal network services and information from personal devices.
13. C. A virtual private network (VPN) provides secure access to an organization's network for remote workers.
14. C. The internet is a worldwide collection of interconnected networks (internetworks, or internet for short).
15. A and D. Users interface with a network using an end device. An end device is either the source or destination of a message transmitted over the network.

Chapter 2

1. A. The running configuration file reflects the current configuration. Modifying a running configuration affects the operation of a Cisco device immediately.

2. C and E. User EXEC mode has limited capabilities, but it is useful for basic operations. It allows only a limited number of basic monitoring commands but does not allow the execution of any commands that might change the configuration of the device. User EXEC mode is identified by the CLI prompt that ends with the `>` symbol.
3. B. Higher configuration modes, such as global configuration mode, can only be reached from privileged EXEC mode. When this mode is configured, the **enable secret** password is required to enter privileged EXEC mode.
4. A. VLAN 1 is not a physical interface but a virtual one. VLAN 1 is the default VLAN on a Cisco switch.
5. A, C, and E. The guidelines for configuring a hostname are:
 - Start with a letter
 - Include no spaces
 - End with a letter or digit
 - Use only letters, digits, and dashes
 - Be fewer than 64 characters in length
6. B. The portion of the OS that interacts directly with computer hardware is known as the kernel.
7. D. The switch will boot up in user EXEC mode. Without a console password previously set on the switch, the user will be in user EXEC mode.
8. D. Most commands entered in IOS take effect immediately, including the configuration of an IP address on an interface.
9. D. RAM is volatile memory and loses all information when the device is restarted.
10. C. The command **copy startup-config running-config** copies the startup configuration file from NVRAM into the running configuration file in RAM.
11. A. A server using DHCP can be used to automatically assign IP address information to a host.
12. B and D. Context-sensitive help provides:
 - Commands available in each command mode
 - Commands that start with specific characters or group of characters
 - Arguments and keywords that are available to particular commands
13. C. The startup configuration file on a router is stored in NVRAM and retains information when power is lost.

Chapter 3

1. A, C, and D. IANA, IEEE, and IETF are standards organizations. TCP/IP and OSI are protocol suites and models, and MAC is a sublayer for LANs and WLANs.
2. A. Broadcast communication is used to send a message to all devices in a LAN; it is one-to-all communication. Unicast is for one-to-one communication, and multicast is for one-to-group communication. Allcast is not a type of communication.
3. A. Encoding is the process of converting information into another acceptable form for transmission.
4. D. Broadcast communication is used to send a message to all devices in a LAN; it is one-to-all communications. Unicast is one-to-one communication, and multicast is one-to-group communication.

5. A and D. One benefit of a layered model is assisting in protocol design because protocols that operate at a specific layer have defined information that they act upon and a defined interface to the layers above and below. Another benefit is preventing technology or capability changes in one layer from affecting other layers above and below.
6. C. Protocols are the rules that govern communications.
7. B. IP addressing is used to deliver data to a device on the same network and to devices on other networks. The destination IP address is the ultimate destination of the message.
8. C. The term *protocol data unit (PDU)* is used to describe data at different layers of a networking model. For example, a frame is the PDU used at Layer 2 of the OSI model.
9. C and D. ICMP and IP are protocols at the internet layer of the TCP/IP model. POP and BOOTP are application layer protocols, and Ethernet is an access layer protocol.
10. D. The transport layer is responsible for segmenting data when transmitting and reassembling data when it is received.
11. B. Multicast is one-to-group communication. Broadcast is one-to-all communication, and unicast is one-to-one communication.
12. B. Encoding is the process of converting information into another acceptable form for transmission. Decoding reverses this process to interpret the information.
13. D. An IP packet is encapsulated in an Layer 2 frame to be transmitted over the physical medium. Layer 2 is for NIC-to-NIC communication on the same network.
14. C. Encapsulation is the process of prepending the proper protocol header at the next, lower layer.

15. B. To request a web page from a server, the web client prepares the HTTP request, then encapsulates it in a TCP header, then encapsulates it in an IP header, and finally encapsulates it in an Ethernet header and trailer.

Chapter 4

1. B. All data between two devices must be transmitted over network media, which is the purpose of the physical layer. It provides a means of representing and transporting the bits over a physical medium as a series of signals such as different voltage levels or electromagnetic frequencies.
2. D. One strand is used for sending, and the other strand is used for receiving.
3. B. Crosstalk is a disturbance caused by the electric or magnetic fields of a signal on one wire affecting the signal on an adjacent wire.
4. B. Cable designers have discovered that they can limit the negative effect of crosstalk by varying the number of twists per wire pair. Cancellation can reduce crosstalk: Two wires in an electrical circuit with magnetic fields exactly opposite each other are placed close together, and the two magnetic fields cancel each other out and also cancel out any outside EMI and RFI signals.
5. D. A straight-through cable is used to connect “unlike” devices, such as a PC (computer) and a switch.
6. C. Bandwidth is the amount of data that can be transmitted over a specific amount of time, usually measured in bits per second.
7. D. Encoding is a method of converting a stream of data bits into a predefined “code.” This process helps distinguish data bits from control bits.

8. A. Cancellation occurs when pairing wires in a circuit. When two wires in an electrical circuit with magnetic fields exactly opposite each other are placed close together, their magnetic fields are exactly opposite of each other. Therefore, the two magnetic fields cancel each other out and also cancel out any outside EMI and RFI signals.
9. A. A microwave oven uses the same frequency as many WLANs and may cause interference.
10. D. Throughput is the measure of the transfer of bits across media over a given period of time. Due to a number of factors, throughput is usually less than the specified bandwidth.
11. D. Fiber-optic cables can transmit signals with less attenuation than can copper, which allows the signal to travel farther.
12. B. IEEE is the organization that oversees WLAN standards (IEEE 802.11).
13. C. WLANs allow devices to be mobile without losing connectivity.
14. C. Signal distortion by the NIC occurs at the physical layer.
15. B. A rollover cable is used to connect a device to a Cisco console port.
5. A and D. IPv6 addresses are 128 bits represented using up to 32 hexadecimal numbers. IPv4 addresses are 32 bits represented in dotted decimal notation.
6. B. An IPv4 address is 32 bits represented in dotted decimal notation.
7. C. 203.0.113.211
8. A. 149
9. A. **0x3** converts to **0011**, and **0xF** converts to **1111**. Therefore, **0x3F** in binary is **0011 1111**, which equals to **63** in decimal.
10. A. **00001010** is **10** in decimal, **01100100** is **100** in decimal, **00010101** is **21** in decimal, **00000001** is **1** in decimal.
11. C. **0xC** converts to **1100**, and **0x9** converts to **1001**. Therefore, **0xC9** in binary is **1100 1001**, which is **201** in decimal.
12. A. There are 16 hexadecimal numbers: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, and F.
13. C. **0xC** converts to **1100**, and **0xA** converts to **1010**. Therefore, **0xCA** is **1100 1010** in binary.
14. A. An IPv4 address is 32 bits represented in dotted decimal notation.

Chapter 5

1. C. 10101101
2. D. **11101100** is **236** in decimal, **00010001** is **17** in decimal, **00001100** is **12** in decimal, and **00001010** is **10** in decimal.
3. D. An IPv6 address is 128 bits represented using up to 32 hexadecimal numbers.
4. A. 11101000

Chapter 6

1. B. Ethernet MAC addresses are used to uniquely identify and address devices on an Ethernet LAN. Every device on an Ethernet LAN has an Ethernet NIC with a unique MAC address.
2. A and C. Both the IEEE and ITU define standards that apply to the data link layer. IANA is responsible for allocating IP address space, domain names, and port numbers.

ISOC oversees the Internet Architecture Board (IAB), and EIA helps define electrical characteristics for data transmission.

3. C. The data link layer encapsulates data, usually Layer 3 packets, to be transmitted over different types of media such as Ethernet or Wi-Fi.
4. D. Logical topologies typically include the devices and types of networks used to transfer data between devices, such as an Ethernet LAN.
5. C. In a star topology, a central device is used to connect all end devices.
6. C. In a mesh network or full-mesh network, all end devices (or nodes) are connected to all other end devices.
7. B. Half-duplex is a type of transmission that can occur in either direction, but the data can flow in only one direction at a time.
8. C. The LLC is the sublayer used to identify which Layer 3 or network layer protocol, such as IPv6, is encapsulated within the Layer 2 frame.
9. A. CSMA/CD is used by legacy Ethernet hubs. Because today's Ethernet LANs use full-duplex Ethernet switches, CSMA/CD is not required and is not used.
10. C and E. The two sublayers of the data link layer are LLC and MAC. The LLC sublayer takes the network protocol data, which is typically an IPv4 or IPv6 packet, and adds Layer 2 control information to help deliver the packet to the destination node. The MAC sublayer controls the NIC and other hardware that is responsible for sending and receiving data on the wired or wireless LAN/MAN medium.
11. C. Wireless networks (specified in IEEE 802.11) use CSMA/CA to manage access to the shared wireless medium.
12. B and C. The data link layer encapsulates Layer 3 packets (that is, IPv4 or IPv6 packets) into a Layer 2 data link frame such as Ethernet. The data link layer is also responsible for accessing the media, such as Ethernet or a wireless LAN, and for performing error detection.
13. C. Ethernet is a data link layer protocol that uses the MAC address to transmit an Ethernet frame from one Ethernet NIC to another Ethernet NIC on the same network.
14. C. CSMA/CD is used to contend for access on a shared, half-duplex media. The use of full-duplex switches means that Ethernet NICs can operate in full-duplex and no longer have to contend for access.

Chapter 7

1. C. An Ethernet switch examines the destination MAC address and looks for a matching entry in its MAC address table to make a forwarding decision.
2. C. An Ethernet switch examines the destination MAC address and looks for a matching entry in its MAC address table to make a forwarding decision.
3. B. The LLC communicates with upper-layer protocols, such as IPv4 or IPv6.
4. C. The first 3 bytes of the MAC address, known as the OUI (organizationally unique identifier), are assigned to the vendor.
5. A. Cisco switches drop runt frames.

6. B and E. The minimum size of an Ethernet frame is 64 bytes. The expected maximum size of an Ethernet frame is 1518 bytes. The maximum might be higher, such as when VLAN tagging is required.
7. D. An Ethernet switch builds its MAC address table by examining and recording the source MAC addresses and incoming port numbers.
8. A and D. IEEE 802.3 is the IEEE standard for Ethernet. It uses unique MAC addresses to determine the Ethernet frame sent to and processed by the correct end device.
9. A. Although there are some situations in which duplicates may purposely occur, the vast majority of Ethernet MAC addresses are globally unique.
10. D. 01-00-5E means that this is a multicast Ethernet frame, and it is meant for devices belonging to this specific multicast group indicated by the following 24 bits.
11. A. The host will discard the frame. The destination MAC address is at the beginning of the frame so the device can immediately determine whether it is the destination of the frame.
12. D. Auto-MDIX allows the use of either a straight-through or crossover cable on a port.
13. A and C. The MAC sublayer is responsible for accessing the media, such as CSMA/CD with legacy hubs. It also adds framing, including both a header and trailer, to the encapsulated data.
14. D. 01-00-5E means that this is a multicast Ethernet frame, and it is meant for devices belonging to this specific multicast group indicated by the following 24 bits.

Chapter 8

1. B. A router examines a packet's destination IP address to find the best match in the router's routing table.
2. B. If the destination IP address is on the same network as the sending host, then the packet is forwarded directly to the destination host. The default gateway is used only if the destination IP address is not on the same network segment as the sending host.
3. B. The router removes Layer 2 frames when the packet is received. The router encapsulates the packet in a new data link frame when forwarding the packet out the appropriate interface.
4. D. 127.0.0.1 is an IPv4 loopback address.
5. B. The upper-layer transport protocol TCP is responsible for reliability and for retransmitting any data that was not received.
6. B. IPv6 was developed primarily due to the exhaustion of IPv4 address space.
7. B. An IPv4 address is 32 bits.
8. C. A router examines a packet's destination IP address to find the best match in the router's routing table. The information in the routing table determines how to forward the packet.
9. B. When a router receives an IPv6 packet, it decrements the Hop Limit field by 1. If the Hop Limit equals 0, the router drops the packet.
10. C. The `netstat -r` command displays a routing table on a Windows host.
11. D. The OSI Layer 3 protocols IPv4 and IPv6 include the source and destination IP addresses.

- 12. C. The network layer MTU is based on the size of the MTU of the data link frame. IP was designed to be transmitted over many different lower-layer technologies.
- 13. B. The IPv6 header is a fixed length and has fewer fields than the IPv4 header. The IPv6 header does not include any fields for fragmentation or a payload length, which makes the processing of IPv6 packets more efficient.

Chapter 9

- 1. C. The ARP request contains the known IPv4 address of the default gateway and requests the MAC address associated with this IPv4 address.
- 2. B. The ARP process maps known IPv4 addresses to MAC addresses on the same network.
- 3. A. The ARP table maps Layer 3 IPv4 addresses to Layer 2 MAC addresses.
- 4. C. The ARP process maps known IPv4 addresses to MAC addresses on the same network.
- 5. A. The routing table, ARP cache, and running configuration file are all maintained in RAM memory.
- 6. D. ARP tables contain mappings of IPv4 addresses to MAC addresses.
- 7. D. `arp -a` shows the MAC address used to reach a specific IPv4 address. The analyst can examine this information to see if it is the correct MAC address of the default gateway.
- 8. D. The ARP process maps known IPv4 addresses to unknown MAC addresses on the same network and stores this information in the local ARP table.
- 9. B. The ARP process maps known IPv4 addresses to MAC addresses on the same network and stores this information in the local ARP table. This information is then used to forward the Ethernet frame.
- 10. B. On a Layer 2 Ethernet switch, broadcast frames are forwarded out all ports except the incoming ports. Broadcast frames are not forwarded by routers.
- 11. C. ARP requests are sent as Ethernet broadcasts.
- 12. A. ICMPv6 Neighbor Solicitation messages are sent using special Ethernet and IPv6 multicast addresses. This allows the Ethernet NIC of the receiving device to determine whether the Neighbor Solicitation message is for itself without having to send it to the operating system for processing.
- 13. D. On a Layer 2 Ethernet switch, if the destination MAC address is not in the MAC address table, the frame is forwarded out all ports except the incoming ports. These frames are called unknown unicasts.
- 14. C and D. A Neighbor Solicitation message in IPv6 is the equivalent of an ARP request in IPv4, and a Neighbor Advertisement message in IPv6 is the equivalent of an ARP reply in IPv4.

Chapter 10

- 1. C. Without a startup-config file stored in NVRAM, the router will boot up without any preconfigured commands.
- 2. B. The `service password-encryption` command encrypts all passwords in the

running-config and startup-config files. By default, this is not strong encryption, but it can be enough to keep someone from learning passwords by just seeing a configuration file.

3. D. The **enable secret** *<password>* command is used to securely safeguard the password used to go from user mode to privileged EXEC mode.
4. B. This command will change the router prompt from router to portsmouth.
5. C. The console port is used to access the router using out-of-band management. Both the **password** and **login** commands are required.
6. A. The **show ip interface brief** command gives a summary of all the router interfaces, any IPv4 addresses configured on those interfaces, and the current status of each interface.
7. B. Privileged EXEC mode is required for all configuration commands and many of the management and troubleshooting commands. Privileged EXEC mode gives access to all IOS commands.
8. B. The startup configuration file contains all previously saved configuration commands and is used by the IOS on startup.
9. A. The IP address of the default gateway is the IP address of the local router on the same network as the host.
10. D. The **banner motd** command is used to make announcements—such as some sort of statement about “authorized access only”—before someone logs in to the router.
11. C. The technician could use any line configuration mode, such as console or vty, to configure the appropriate access commands.
12. D. The startup-config file is store in NVRAM (nonvolatile RAM) and is used during startup.
13. C. The **service password-encryption** command encrypts all passwords in the running configuration file that were previously stored in plaintext.

Chapter 11

1. C. 255.255.255.224 contains 27 continuous 1 bits.
2. D. A /26 mask means 26 1 bits indicating the network portion of the address, which leaves 6 bits for the host portion. $2^6 - 2$ bits (for the network and broadcast addresses) equals 62 valid host addresses.
3. C. 255.255.255.224, or /27, contains 27 continuous 1 bits. This leaves 5 bits for host addresses.
4. C. A 192.168.10.0/24 network subnetted using a /26 prefix extends the subnet mask 2 bits, from /24 to /26. $2^2 = 4$ subnets.
5. C. /20 is 20 continuous 1 bits, calculated as follows:

$255 = 8$ 1 bits
 $255 = 8$ 1 bits
 $240 = 4$ 1 bits
 $0 = 0$ 1 bits
 Total = 20 1 bits
6. B. VLSM allows for the option to subnet any subnet further as long as there are enough host bits to do so. This makes possible a variety of sizes of subnets or networks.
7. D. The ANDing process on an IPv4 address and the subnet mask determines the network portion of an address.

8. D. A network address with a /27 prefix length contains 27 continuous 1 bits. This leaves 5 bits for host addresses, or $2^5 - 2$ (for the network and broadcast addresses), which equals 30 valid hosts addresses.
9. C. 255.255.255.240 is 28 continuous 1 bits. This leaves 4 host bits.
10. B and D. The two portions of an IPv4 address are the network portion and the host portion, as determined by the subnet mask.
11. C. A /30 prefix length means 30 1 bits indicating the network portion of the address, which leaves 2 bits for the host portion. $2^2 - 2$ bits (for the network and broadcast addresses) equals 2 valid host addresses.
12. C. An AND operation between the IPv4 address 172.17.4.250 and subnet mask 255.255.255.0 (/24), results in a network address of 172.17.4.0 (all 0 bits in the host portion). The broadcast address for this network is 172.17.4.255 (all 1 bits in the host portion). This means 172.17.4.250 is neither a network address nor a broadcast address and is therefore a host address.
13. F. A /28 prefix length means 28 continuous 1 bits, which indicates the network portion of the address. This leaves 4 bits for the host portion. $2^4 - 2$ bits (for the network and broadcast addresses) equals 14 valid host addresses.
14. C. The ANDing process is used between an IPv4 address and its subnet mask to determine the network address for that device.
15. D. The 255.255.255.224 subnet mask or a /27 prefix length results in 28 continuous 1 bits, which indicates the network portion of the address. This leaves 5 bits for the host portion. $2^5 - 2$ bits (for the network and broadcast addresses) equals 30 valid host addresses.

Chapter 12

1. E. Pinging a loopback address verifies that IP is working on the local host. Most host operating systems—including Windows, macOS, Linux, iOS, and Android—have both IPv4 and IPv6 installed by default.
2. B. Leading 0s are omitted, and a single contiguous string of all-0 hextets can be replaced with a double colon (::).
3. A. ::1 is an IPv6 loopback address. Pinging a loopback address helps verify the internal configuration of IP on the host.
4. A. For any device to be enabled for IPv6, the interface must have a link-local address.
5. D. A /64 prefix length indicates that the first 64 bits, 2001:db8::1000, indicate the network address. This leaves 64 bits for the interface ID (or 4 hextets): a9cd:47ff:fe57:fe94.
6. B, C and E. An IPv6 GUA has three parts: (1) the global routing prefix allocated by the ISP or provider of the IPv6 address; (2) a subnet ID, which is the bits between the global routing prefix and the interface ID; and (3) the interface ID, which is typically 64 bits and is highly recommended for compatibility with SLAAC.
7. A. Leading 0s are omitted, and a single contiguous string of all-0 hextets can be replaced with a double colon (::).
8. C. With a /64 prefix length, the first 64 bits (or first 4 hextets)—in this case, 2001:db8:d15:ea—indicate the network address.
9. B. When a device is enabled for IPv6 on an interface, that interface automatically assigns itself a link-local address. Most host operating systems—including Windows, macOS, Linux, iOS, and Android—have both IPv4

and IPv6 installed by default. This means they have, at minimum, an IPv6 link-local address.

10. C. Link-local addresses are only for communications on the local link or network and are not routable off that link.
11. B. With a /48 global routing prefix and a /64 prefix length, you are left with 16 bits between the global routing prefix and interface ID for the subnet ID. Subtracting 48 (global routing prefix) from 64 (the prefix length) yields the subnet ID.
12. D. With a /64 prefix length, the first 64 bits (or first 4 hextets)—in this case, 2001:db8:aa04:b5—indicates the network address.
13. B. Link-local addresses are only for communications on the local link or network and are not routable off that link.
14. D. IPv6 does not have a broadcast. IPv6 does include an all-IPv6-device multicast address.
15. A. Like any other IPv6 device, a Cisco router's interface must have a link-local address to be enabled for IPv6. It does not have to have a global unicast address, but it must have a link-local address.
4. D. ICMP provides information and error messaging.
5. C. The **ping** utility uses ICMP Echo Request and Echo Reply messages.
6. D. The **tracert** (or **tracert** in Windows) command is used to determine where a packet might be dropped or delayed by a router. This command displays the IP addresses of the routers in the path that successfully received the packet(s).
7. B. ICMPv6 NDP (Neighbor Discovery Protocol) is used to provide address resolution for a known IPv6 address and unknown MAC address using Neighbor Solicitation and Neighbor Advertisement messages. Router Solicitation and Router Advertisement messages are used for dynamic address allocation information for IPv6.
8. A. An IPv6 host can send a Neighbor Solicitation message to see if its IPv6 address is unique before using it. The NS message includes the IPv6 address the device wants to use. If the device does not receive a Neighbor Advertisement in response, it can assume that its IPv6 address is unique. This is an optional process, but most operating systems implement it.
9. A. The technician can use the Windows **tracert** command to determine the last router in the path that successfully received the packets.

Chapter 13

1. D. A successful **ping** to the loopback address verifies that the TCP/IP stack is functional.
2. D. The **ping** command uses ICMP Echo Request and Echo Reply messages to test connectivity.
3. C. A router decrements the IPv6 Hop Limit field by 1 and drops the packet when the field is 0.
10. C. A successful **ping** to the default gateway indicates that the device can reach the router used to forward packets to other networks.
11. B. The **ping** command only verifies connectivity. The **tracert** command (or **tracert** in Windows) verifies connectivity and displays information about the routers in the path.

12. C. A router uses the ICMP Time Exceeded message when it has decremented an IPv4 TTL or IPv6 Hop Limit field to 0. **tracert** uses the source IP address of the ICMP Time Exceeded message sent by the router to determine the router’s IP address.
13. C and D. The **ping** command verifies that the destination IP address is reachable and displays the average round-trip time between the source and destination.
14. D. The **tracert** utility identifies the routers in the path to the destination. When a router receives an IP packet from **tracert**, it decrements the IPv4 TTL or IPv6 Hop Limit field by 1. When the field’s value is 0, the router returns an ICMP Time Exceeded message to the source. **tracert** uses the source IP address of the ICMP Time Exceeded message sent by the router to determine the router’s IP address.
5. B. A server receives segments from clients, each with a source port number and a destination port number. Along with the source IP address of the client, the server can uniquely identify each service request.
6. B and E. DNS and VoIP commonly use UDP as the transport protocol. DNS uses UDP because it is a transaction-based application. VoIP uses UDP to avoid unnecessary delays and because it can tolerate some data loss.
7. C. TCP tracks multiple conversations to ensure reliability and flow control.
8. C. 192.168.1.1:80 is an example of a socket using a destination IPv4 address and well-known destination port 80 (HTTP).
9. A and E. The ACK and SYN flags are used in the TCP three-way handshake in the following order: (1) SYN, (2) SYN, ACK, (3) ACK.

10. C. FTP uses TCP, so any segments not received are re-sent.

11. A. UDP is best suited for applications that are sensitive to delay and can tolerate some loss of data.

12. A. If the source determines that the TCP segments are either not being acknowledged or are not being acknowledged in a timely manner, it can reduce the number of bytes it sends before receiving an acknowledgment.

13. B and D. TCP acknowledges data received and retransmits any unacknowledged data.

14. D. Sliding window allows a device to continuously send segments as long the destination is continuously sending acknowledgments as it processes the bytes received.

15. B. Both TCP and UDP identify individual conversations using port numbers and, more specifically, socket pairs.

Chapter 14

1. C. The client selects a source port number to uniquely identify this process for this device.
2. B. The TCP three-way handshake is used to establish a connection-oriented session between a client and a server.
3. B. TCP and UDP port numbers 0 to 1023 are reserved for well-known network applications.
4. B. A socket is the combination of source IP address and source port number or destination IP address and destination port number. The two combinations together are known as a socket pair.

Chapter 15

1. B. POP3 transfers messages from an email server to an email client.
2. A. IMAP is used to store email messages on an email server, and the client can read, delete, and send messages from any location.
3. D. SMB is a Microsoft file sharing and print services application protocol.
4. A. The author would be using a client version of file transfer software, using a protocol such as FTP. The file server would be using the server version of the same application.
5. B. FTP uses TCP for reliability and flow control. FTP allows a client device to download or upload data to an FTP file server.
6. C. IPv4 devices obtain their IPv4 addressing information from a DHCP for IPv4 server.
7. A. Users interact with a network by using a client network application such as a web browser. These applications use application layer protocols such as HTTP or HTTPS.
8. B, C, and D. The application layer of the TCP/IP model is the equivalent of the application, presentation, and session layers of the OSI model. HTTP, MPEG, and GIF are all used at this layer. TCP and UDP are transport layer protocols, and IP is an internet layer protocol.
9. D. HTTPS uses TLS (Transport Layer Security) or its predecessor, SSL (Secure Sockets Layer), for encryption.
10. D. DHCP for IPv4 is more efficient and less error prone than assigning IPv4 address information statically on client devices. Static IPv4 address assignment is still done

on devices where the addresses need to be consistent, such as servers and intermediary devices.

11. C and D. DNS is used to map a domain name to an IP address. DNS is available for both IPv4 and IPv6.
12. B. Typically, a home router provides private IPv4 addressing to clients using DHCP. It also performs NAT between the private IPv4 addresses and a public IPv4 address.
13. A. The top-level domain (TLD) is .com.
14. C and D. The application layer creates dialogues between source and destination applications, typically clients and servers (for example, HTTP GET request and HTTP response). Users interact with the network by using a client network application such as a web browser. These applications use application layer protocols such as HTTP or HTTPS.
15. A. A client's browser uses an HTTP or HTTPS GET message to request data from a web server.

Chapter 16

1. E. A firewall protects computers and networks by preventing undesirable traffic from entering internal networks and protecting devices from sending or receiving undesirable messages.
2. B. The syntax for the **login block-for** command is **login block-for seconds attempts tries within seconds**.
3. C. The network security accounting function tracks the actions of users and stores that information in a log file.

4. B. **nslookup** and **fping** are commands commonly used in reconnaissance attacks, to aid in the discovery and mapping of systems, services, or vulnerabilities.
5. A. Telnet sends passwords in plaintext, whereas SSH encrypts both the username and the password.
6. A. A firewall protects computers and networks by preventing undesirable traffic from entering internal networks and protecting devices from sending or receiving undesirable messages.
7. A. A DoS (denial-of-service) attack disables a server with bogus message in order to prevent the device from servicing authorized users.
8. A, C, and D. AAA makes it possible to control who is permitted to access a network (authenticate), what they can do while they are there (authorize), and what actions they perform while accessing the network (accounting).
9. B. Computer worms are similar to viruses in that they replicate functional copies of themselves and can cause the same type of damage. In contrast to viruses, which require the spreading of an infected host file, worms are standalone software and do not require a host program or human help to propagate.
10. B. Failure of an air conditioning unit causing a networking device to overheat would be an environmental situation.
11. A. A vulnerability such as an unpatched security issue or a weak password makes a device susceptible to an attack.
12. B, C, and E. To require an IP domain name, use the **ip domain name** global configuration command. Unless an authentication server is used, there must be a local database username entry; you add this by using the **username** global configuration command. A device must have a unique hostname other than the default.
13. A. A reconnaissance attack aids in the discovery and mapping of systems, services, or vulnerabilities.
14. B. Firewall settings on network devices or end devices can prevent ICMP Echo Request and Echo Reply messages from being forwarded or received.
15. D. Unlike Telnet, which sends the passwords in plaintext, SSH encrypts the username and password.

Chapter 17

1. D. Redundancy might not be critical to the business operations in a smaller size business, but it can be very important in a large corporation.
2. D. Devices with support for modularity allow for device expansion. Ensuring modularity can be less expensive than purchasing additional devices to accommodate additional growth.
3. C. Voice traffic is more delay sensitive than FTP, instant messaging, and SNMP, and it benefits from QoS during times of congestion.
4. C. Microsoft Windows uses the **tracert** command in place of **traceroute**.
5. C. If a remote device can be reached directly by using the IP address but not its domain name, the issue may be that the DNS server is unreachable or is unable to resolve the name. It is not always possible to reach a

website by using its IP address, however, because most websites run on servers that support many different websites.

6. D. IOS messages are sent to the console by default.
7. B. An important aspect of network documentation includes both physical and logical topologies.
8. A. QoS can be used to minimize latency by giving priority to certain types of traffic during times of congestion.
9. C. If a computer has an IPv4 address beginning with 169.254.x.x, it most likely means that it was unable to receive an IPv4 address using DHCP.
10. B. A redundant router and a redundant connection to a different ISP can help provide reliability in the event that there is a failure with one of the routers, a failure with connection to the ISP, or an issue with the ISP.
11. D. A network baseline is established at regular intervals over a period of time. It is best practice to take these measurements during different times of the day and week in order to capture a more realistic measure of network traffic.
12. D and E. Real-time voice and video traffic are more sensitive to delay than are mail, web, and FTP traffic.
13. A. The **show interfaces** command provides Layer 1 and Layer 2 information, including addressing, duplex, and bandwidth information.
14. C. CDP can be disabled globally or on a specific interface. Many network administrators disable CDP on all or parts of their network for security reasons and then enable it for troubleshooting.
15. A. When choosing devices, many small networks look at cost. ISP is not related to choosing devices. Redundancy and traffic analysis may be less of a consideration and can still be implemented in parts of the network, independent of the devices currently installed.

Key Terms Glossary

This glossary defines the terms and abbreviations listed at the beginning of each chapter in the book.

A

AAA (authentication, authorization, and accounting) The primary framework to set up access control on a network device.

access attack A type of attack that exploits known vulnerabilities in authentication services, FTP services, and web services to gain entry to web accounts, confidential databases, and other sensitive information. An access attack allows individuals to gain unauthorized access to information that they have no right to view.

access method A set of rules used by LAN hardware to direct traffic on the network. It determines which host or device uses the LAN next.

access point (AP) A device that connects to a wireless router and is used to extend the reach of a wireless network.

acknowledgment A notification sent from one network device to another to acknowledge that some event (for example, receipt of a message) has occurred.

Address Resolution Protocol (ARP) An internet protocol used to map an IP address to a MAC address. Defined in RFC 826.

adjacency table A table in a router that contains a list of the relationships formed

between selected neighboring routers and end nodes for the purpose of exchanging routing information. Adjacency is based on the use of a common media segment.

alternating current An electrical current that changes direction at a uniformly repetitious rate. This type of electricity typically is provided by a utility company and is accessed through wall sockets.

American National Standards Institute (ANSI) A private nonprofit organization that oversees development of standards in the United States.

American Standard Code for Information Interchange (ASCII) An 8-bit code (7 bits plus parity) for character representation.

analog telephone A type of telephone that can transmit data over standard voice telephone lines for internet access. This type of service uses an analog modem to place a telephone call to another modem at a remote site. This method of connection is known as dialup.

AND (logical) One of three basic binary logic operations. ANDing yields the following results: 1 AND 1 = 1, 1 AND 0 = 0, 0 AND 1 = 0, 0 AND 0 = 0.

ARP cache Logical storage in a host's RAM for ARP entries. *See also* ARP table.

ARP table Logical storage in a host's RAM for ARP entries. *See also* ARP cache.

assigned multicast Reserved IPv6 multicast addresses for predefined groups of devices.

asymmetric switching A switching technique that allows for different data rates on different ports.

authentication, authorization, and accounting

See AAA (authentication, authorization, and accounting).

automatic medium-dependent interface crossover (auto-MDIX) A feature on a switch port or hub port that detects the type of cable used between switches or hubs. Once the cable type is detected, the port is connected and configured accordingly. With auto-MDIX, a crossover cable or a straight-through cable can be used for connections to a copper 10/100/1000 port on the switch, regardless of the type of device on the other end of the connection.

availability The assurance of timely and reliable access to data services for authorized users. Network firewall devices, along with desktop and server anti-virus software, can ensure system reliability and the robustness to detect, repel, and cope with breaches of network security. Building fully redundant network infrastructures, with few single points of failure, can reduce the impact of these threats.

B

baby giant frame An Ethernet frame with more than 1500 bytes of data. Also known as a jumbo frame.

bandwidth The rated throughput capacity of a given network medium or protocol. Bandwidth is listed as available or consumed data communication resources expressed in bits per second.

best effort Describes the agreement or the attempt to fulfill expectations or the requirements of a standard.

best-effort delivery Describes a network system that does not use a sophisticated acknowledgment system to guarantee reliable delivery of information.

Binary Number expressed using the base-2 number system.

Bluetooth (IEEE 802.15) A wireless personal area network (WPAN) standard that uses a device pairing process to communicate over distances from 1 to 100 meters.

Bootstrap Protocol (BOOTP) A protocol used by a network node to determine the IP address of its Ethernet interfaces in order to facilitate network booting.

bring your own device (BYOD) A policy that allows end users to use personal tools to access information and communicate across a business or campus network.

broadcast A form of transmission in which one device transmits to all devices within the network or on another network.

broadcast address A special address reserved for sending a message to all stations. Generally, a broadcast address is a MAC destination address of all ones. *Compare with* multicast address and unicast address.

brute-force attack An attempt to access usernames or passwords by trial and error.

buffered memory A memory chip that has a control chip built into the module. The control chip assists the memory controller in managing large quantities of RAM.

burned-in address (BIA) The MAC address that is permanently assigned to a LAN interface or NIC. It is called burned-in because the address is burned into a chip on the card, and the address cannot be changed. Also called universally administered address (UAA).

bus topology A network topology in which all end systems are chained to each other and terminated in some form on each end. Infrastructure devices such as switches are not required to interconnect the end devices. Legacy Ethernet networks were often bus topologies using coax cables because it was inexpensive and easy to set up.

C

cable connection The point at which a cable connects to the device.

cable internet A form of internet service that uses coaxial cable lines originally designed to carry cable television and connects an end user's computer to the cable company.

cable tester A testing device used to check for wiring shorts, faults, or wires connected to the wrong pins.

Carrier Sense Multiple Access (CSMA)

A media-access mechanism in which devices ready to transmit data first check the channel for a carrier. If no carrier is sensed for a specific period of time, a device can transmit. *See also* CSMA/CA and CSMA/CD.

Carrier Sense Multiple Access/Collision Avoidance (CSMA/CA) A media-access mechanism that regulates the transmission of data onto a network medium. CSMA/CA is similar to CSMA/CD except that devices first request the right to send, in order to avoid collisions. CSMA/CA is used in 802.11 WLANs.

Carrier Sense Multiple Access/Collision Detect (CSMA/CD)

A media-access mechanism that requires a node wishing to transmit to listen for a carrier signal before trying to send. If a carrier is sensed, the node waits for the transmission in progress to finish before initiating its own transmission. If a collision occurs and is detected, the sending node uses the backoff algorithm before retransmitting.

cellular connection Cellular internet access that uses a cell phone network to connect. Wherever a user can get a cellular signal, the user can get cellular internet access. Performance is limited by the capabilities of the phone and the cell tower to which it is connected.

channel A communication path over a medium used to transport information from a sender to a receiver. Multiple channels can be multiplexed over a single cable.

circuit switched A switching system in which a dedicated physical circuit path exists between sender and receiver for the duration of the call. Used heavily in the telephone company network.

Cisco Express Forwarding (CEF) A Layer 3 switching method that speeds up packet forwarding by decoupling the usual strict interdependence between Layer 2 and Layer 3 decision making. The forwarding decision information is stored in several data structures for CEF switching. This forwarding information can be rapidly referenced to expedite packet forwarding decisions.

Cisco Internetwork Operating System (IOS)

Generic term for the collection of network operating systems used by Cisco networking devices.

classful addressing A type of addressing in which a unicast IP address has three parts: a network part, a subnet part, and a host part. The term *classful* refers to the fact that the classful network rules are first applied to the address, and then the rest of the address can be separated into a subnet and host part to perform subnetting. Originally, IPv4 addresses were divided into five classes: Class A, Class B, Class C, Class D, and Class E. Classful addressing is not generally practiced in current network implementations.

classless addressing An IPv4 addressing scheme that uses a subnet mask that does not follow classful addressing limitations. It provides increased flexibility when dividing ranges of IP addresses into separate networks. Classless addressing is considered the best in current network implementations. *See also* variable length subnet masking (VLSM).

client A network device that accesses a service on another computer remotely through a network.

client/server A computer system setup in which tasks are distributed between a service provider (server) and a service user, such as a workstation (client). The server is used to store the applications and data, and the majority of the computer processing is done on the server.

cloud computing Computing resources (hardware and software) delivered as a service over a network. A company uses the hardware and software in the cloud, and a service fee is charged.

cloud storage Online storage that is accessed via the internet.

coaxial cable (coax) Cable consisting of a hollow outer cylindrical conductor that surrounds a single inner wire conductor. Two types of coaxial cable are currently used in LANs: 50-ohm cable, which is used for digital signaling, and 75-ohm cable, which is used for analog signaling.

collaboration A process in which more than one person works on a document or documents in real time across a network.

collision fragment Any frame less than 64 bytes in length. These frames are automatically discarded by receiving stations. Also called runt frame.

command-line interface (CLI) A user interface to a computer operating system or application that depends on textual commands being entered by the user.

communication Transmission and receipt of information.

communities Groups of people who share common experiences and hobbies and who exchange ideas and information. Communities allow for social interaction that is independent of location or time zone.

confidentiality The state of ensuring that only intended and authorized recipients—individuals, processes, or devices—can access and read data. Confidentiality is accomplished by having a strong system for user authentication, enforcing passwords that are difficult to guess, and requiring users to change passwords frequently. Encrypting data so that only the intended recipient can read it is also part of confidentiality.

congested A condition in which a network has more bits to transmit than the bandwidth of the communication channel can deliver.

congestion Traffic in excess of network capacity.

connection oriented Term used to describe data transfer that requires the establishment of a virtual circuit.

connection-oriented protocol A protocol that requires the establishment of a virtual circuit.

connectionless Term used to describe data transfer without the existence of a virtual circuit.

connectivity The state of being connected or interconnected to another device.

console Term used to describe data transfer that requires the establishment of a virtual circuit.

content addressable memory (CAM) table Memory that is accessed based on its contents rather than on its memory address. Also known as associative memory.

contention-based access method A nondeterministic method of networking, which means any device can try to transmit data across the shared medium whenever it has data to send.

converged data network A network that aggregates various forms of traffic, such as voice, video, and data, on the same network infrastructure.

core The light transmission element at the center of optical fiber.

crimper A tool used to attach connectors to wires to make a cable.

crosstalk A source of interference that occurs when cables are bundled together for long lengths, in which the signal from one cable leaks out and enters adjacent cables. *See also* electromagnetic interference (EMI).

CSMA/Collision Avoidance (CSMA/CA) *See* Carrier Sense Multiple Access/Collision Avoidance (CSMA/CA).

CSMA/Collision Detect (CSMA/CD) *See* Carrier Sense Multiple Access/Collision Detect (CSMA/CD).

custom cloud A cloud built to meet the needs of a specific industry, such as healthcare or media. Custom clouds can be private or public.

cut-through switching A frame forwarding method that forwards a frame before it is entirely received. At a minimum, the destination address of the frame must be read before the frame can be forwarded.

cyclic redundancy check (CRC) A type of hash function (one-way encryption) that is used to produce a small, fixed-size checksum of a block of data, such as a packet or a computer file. A CRC is computed and appended before transmission or storage and verified afterward by the recipient to confirm that no changes have occurred in transit. It is an error-checking technique in which the frame recipient calculates a remainder by dividing frame contents by a prime binary divisor and comparing the calculated remainder to a value stored in the frame by the sending node.

D

daemon A computer program that runs in the background and is usually initiated as a process. Daemons often support server processes.

data center A facility that houses computer systems and associated components, including redundant data communications connections, high-speed virtual servers, redundant storage systems, and security devices.

data network Infrastructure historically used by businesses to record and manage business systems. Data networks have evolved to enable the transmission of many different types of information services, including email, video, messaging, and telephone.

datagram A logical grouping of information sent as a network layer unit over a transmission medium without prior establishment of a virtual circuit. IP datagrams are the primary information units in the internet. Frames, messages, packets, and segments are also terms for datagrams. *See also* protocol data unit (PDU).

decapsulation A process by which an end device, after it receives data over some transmission medium, examines the headers and trailers at each successively higher layer, and eventually hands the data to the correct application. Sometimes called de-encapsulation.

decoding To convert from one form to another.

de-encapsulation *See* decapsulation.

default gateway A device on a network that serves as an access point to other networks. A default gateway is used by a host to forward IP packets that have destination addresses outside the local subnet. A router interface typically is

used as the default gateway. When a computer needs to send a packet to another subnet, it sends the packet to its default gateway. Also known as the default router.

default route A route that needs zero (no) bits to match with the destination IP address of the packet.

denial of service (DoS) An attack that consumes system resources in order to prevent authorized people from using a service. To help prevent DoS attacks, it is important to stay up to date with the latest security updates for operating systems and applications.

destination The device that is the intended recipient of the message.

destination port number A UDP or TCP port number associated with the destination application on the remote device.

dial-up telephone connection An inexpensive communications option that uses any phone line and a modem. The low bandwidth provided by a dial-up modem connection is not sufficient for large data transfer, although it is useful for mobile access while traveling.

digital camera An input device that captures images and videos that can be stored, displayed, printed, or altered.

Digital Subscriber Line (DSL) An always-on internet service that provides high bandwidth and high availability. Voice and data signals are carried on different frequencies on the copper telephone wires. In general, small office and home office users connect using Asymmetrical DSL (ADSL), which means that the download speed is faster than the upload speed.

directed broadcast A term that describes IPv4 packets sent to all hosts in a particular network. In a directed broadcast, a single copy of the packet is routed to the specified network, where it is broadcast to all hosts on that network.

directly connected network A network that is connected to a router's physical Ethernet or serial interfaces.

DMZ (demilitarized zone) An area of an internal network where resources are available to the internet, such as a web server, and where devices have IPv6 addresses and public IPv4 addresses accessible through the internet.

Domain Name System (DNS) An internet-wide system by which a hierarchical set of DNS servers collectively hold all the name-to-IP address mappings, and DNS servers refer users to the correct DNS server to successfully resolve a DNS name.

dotted decimal The representation of an IPv4 address using four decimal numbers separated by periods.

dual stack A term for a device that is enabled for both IPv4 and IPv6 protocols.

duplex A setting used for communications on a network. *See also* half duplex and full duplex.

duplex multimode LC connector A fiber connector that accepts both the transmitting and receiving fibers in a single connector.

Dynamic Host Configuration Protocol (DHCP) A protocol used to dynamically assign IP configurations to hosts. The services defined by the protocol are used to request and assign an IP address, a default gateway, and a DNS server address to a network host.

dynamic routing protocols Protocols such as EIGRP and OSPF that are used to access remote networks.

E

electromagnetic interference (EMI) Interference by magnetic signals caused by the flow of electricity. EMI can cause reduced data integrity and increased error rates on transmission channels. Electrical currents create magnetic fields, which in turn cause other electrical currents in nearby wires, and the induced electrical currents can interfere with proper operation of the other wire.

enable password An unencrypted password used to limit access to privileged EXEC mode from IOS user EXEC mode.

enable secret An encrypted password used to limit access to privileged EXEC mode from IOS user EXEC mode.

encapsulation The process by which a device adds networking headers and trailers to data from an application for the eventual transmission of the data onto a transmission medium.

encoding A process by which bits are represented on a medium.

end device Either the source or destination of a message transmitted over a network.

EtherChannel A logical interface on a Cisco device associated with a bundle of routed ports in order to aggregate bandwidth.

Ethernet A baseband LAN specification invented by Xerox Corporation and developed jointly by Xerox, Intel, and Digital Equipment Corporation. Ethernet networks use CSMA/CD and run over a variety of cable types at 10 Mbps. Ethernet is similar to the IEEE 802.3 series.

EUI-64 (Extended Unique Identifier-64)

A process for creating an IPv6 interface ID by using the 48-bit Ethernet MAC address, inserting fffe in the middle, and flipping the seventh bit.

expectational acknowledgment An acknowledgment used by TCP where the ACK number is sent back to the source to indicate the next byte that the receiver expects to receive.

extended star topology A hierarchical star topology with devices connected to a central device and additional devices connected to those devices.

Extended Unique Identifier (EUI-64) *See* EUI-64 (Extended Unique Identifier-64).

extranet Part of a network that provides secure and safe access to individuals who work for a different organization but require access to the organization's data.

F

fast-forward switching A type of switching that offers a low level of latency by immediately forwarding a packet after reading the destination address.

fault tolerant network A term for limiting the impact of a failure so that the fewest number of devices are affected and for the shortest time.

fiber-optic cable A physical medium that uses glass or plastic threads to transmit data. A fiber-optic cable consists of a bundle of these threads, each of which is capable of transmitting data into light waves.

fiber optics A technology that uses light to transmit data.

File Transfer Protocol (FTP) An application protocol that is part of the TCP/IP protocol stack and that is used for transferring files between network nodes. FTP is defined in RFC 959.

File Transfer Protocol Secure (FTPS) An encrypted version of FTP.

firewall A hardware or software device that protects a computer or a network by preventing undesirable traffic from entering internal networks.

firmware Permanent software programmed into ROM memory.

flow control The management of data flow between devices in a network. It is used to prevent too much data from arriving before a device can handle it, causing data overflow.

Forwarding Information Base (FIB) A data structure that contains all the known routes. Conceptually, the FIB is similar to a routing table. A networking device uses the FIB lookup table to make destination-based switching decisions.

fragmentation The division of IP datagrams to meet the MTU requirements of a Layer 2 protocol.

fragment-free switching A type of switching in which a switch stores the first 64 bytes of

the frame before forwarding. It can be viewed as a compromise between store-and-forward switching and fast-forward switching.

full-duplex Bidirectional communication in which both devices can transmit and receive on the media at the same time.

G

gateway Normally, a relatively general term that refers to different kinds of networking devices. Historically, when routers were created, they were called gateways.

global configuration mode A mode used to configure global parameters or enter other configuration submodes, such as interface, router, and line configuration submodes.

global routing prefix An IPv6 prefix, or network, portion of an address that is assigned by the provider, such as an ISP, to a customer or site.

global unicast address (GUA) An IPv6 address similar to a public IPv4 address. It is a globally unique, internet-routable address. Global unicast addresses can be configured statically or assigned dynamically.

goodput Application-level throughput. It is the number of useful bits per unit of time from a certain source address to a certain destination, excluding protocol overhead and excluding retransmitted data packets.

graphical user interface (GUI) A user-friendly interface that uses graphical images and widgets, along with text, to indicate the information and actions available to a user when interacting with a computer.

H

half-duplex Unidirectional communication in which devices can transmit and receive on the media but cannot do so simultaneously.

hexadecimal (base 16) A number system using the digits 0 through 9, with their usual meaning, plus the letters A through F to represent hexadecimal digits with values of 10 to 15. The rightmost digit counts ones, the next counts multiples of 16, and $16^2 = 256$.

hextet The unofficial term used to refer to a segment of 16 bits or 4 hexadecimal values. For IPv6 addressing, each digit is a single hextet, 16 bits, or 4 hexadecimal digits.

host address The IPv4 address of a network host. A network layer address.

HTTP (Hypertext Transfer Protocol) A protocol that provides a set of rules for exchanging text, graphic images, sound, video, and other multimedia files on the World Wide Web.

HTTPS (Hypertext Transfer Protocol Secure) A set of rules for exchanging text, graphic images, sound, and video on the World Wide Web. HTTPS adds encryption and authentication services using Secure Sockets Layer (SSL) protocol or the newer Transport Layer Security (TLS) protocol.

hub A device that extends the reach of a network by regenerating the electrical signal. It also receives data on one port and then sends it out to all other active ports. Hubs are legacy devices and should not be used in today's networks. Hubs do not segment network traffic.

hybrid cloud A cloud made up of two or more clouds (for example, part custom, part public), where each part remains a distinctive object, but the two are connected using a single architecture.

I

initial sequence number (ISN) A randomly chosen number that is used to begin tracking the flow of data from the client to the server for a session. The ISN in the header of each segment is increased by one for each byte of data sent from the client to the server as the data conversation continues.

Institute of Electrical and Electronics Engineers (IEEE) An international, nonprofit organization for the advancement of technology related to electricity. IEEE maintains the standards defining many LAN protocols.

Integrated Services Digital Network (ISDN) A broadband standard that uses multiple channels to send voice, video, and data over normal telephone wires.

integrity The assurance that information has not been altered in transmission from origin to destination. Data integrity can be compromised when information has been corrupted—willfully or accidentally. Data integrity is made possible by requiring validation of the sender as well as using mechanisms to validate that the packet has not changed during transmission.

interface A specialized port on a networking device that connects to individual networks. Because routers are used to interconnect networks, the ports on a router are referred to as network interfaces.

interface ID The host portion of an IPv6 global unicast address.

intermediary device A device that connects end devices to the network and can connect multiple individual networks to form an internetwork.

International Organization for Standardization (ISO) An international standards body that defines many networking standards and that created the OSI model.

International Telecommunications Union (ITU) A United Nations (UN) agency responsible for issues that concern information and communication technologies.

internet A network that combines enterprise networks, individual users, and ISPs into a single global IP network.

Internet Assigned Numbers Authority (IANA) An organization that assigns the numbers important to the proper operation of TCP/IP and the internet, including assigning globally unique IP addresses.

Internet Control Message Protocol (ICMP) A protocol that is part of the TCP/IP internet layer and that defines protocol messages used to inform network engineers of how well an internetwork is working. For example, the **ping** command sends ICMP messages to determine whether a host can send packets to another host.

Internet Message Access Protocol (IMAP) A protocol that describes a method to retrieve email messages. Copies of the messages are downloaded to the client application, but the original messages are kept on the server until manually deleted.

internet query A query that searches the internet, including Google search, the websites of organizations, whois, and more.

internet service provider (ISP) A company that helps create the internet by providing connectivity to enterprises and individuals, as well as interconnecting to other ISPs to create connectivity to all other ISPs.

intranet A private connection of LANs and WANs that belongs to an organization and that is designed to be accessible only by the organization's members, employees, or others with authorization.

intrusion detection system (IDS) A system that passively monitors traffic on a network.

intrusion prevention system (IPS) A system that monitors incoming and outgoing traffic, looking for malware, network attack signatures, and more. If it recognizes a threat, it can immediately stop it.

IPv4 address A 32-bit number, written in dotted decimal notation, used by the IPv4 protocol to uniquely identify an interface connected to an IP network. It is also used as a destination address in an IP header to allow routing. As a source address, it enables a computer to receive a packet and to know to which IP address a response should be sent.

IPv6 address A 126-bit address written in hexadecimal used by the IPv6 protocol. IPv6 addresses are the successor of IPv4 addresses.

J-K-L

jacket The outer part of a fiber-optical cable, which protects the cable from abrasion, moisture, and other contaminants.

jumbo frame An Ethernet frame with more than 1500 bytes of data. Also known as a baby giant frame.

kernel The portion of the operating system that interacts directly with computer hardware.

latency Refers to the amount of time, including delays, for data to travel from one given point to another.

LDAP (Lightweight Directory Access Protocol) A protocol used to maintain user identity directory information that can be shared across networks and systems.

limited broadcast A broadcast that is sent to a specific network or series of networks.

line of sight wireless An always-on service that uses radio signals for transmitting data and internet access. A clear path between the transmission tower and customer is required.

link-local IPv4 address An IPv4 address in the range 169.254.1.0 to 169.254.254.255. Communication using such an address has a TTL of 1 and is limited to the local network.

link-local IPv6 address An IPv6 address used to communicate with other devices on the same local link. With IPv6, the term link refers to a subnet. A link-local address is confined to a single link. Its uniqueness must only be confirmed on that link because it is not routable beyond the link.

local-area network (LAN) A network infrastructure that provides access to users and end devices in a small geographic area, which is typically an enterprise, a home, or a small business network owned and managed by an individual or an IT department.

logical address An address that is used to send a packet from a source device to a destination device on the same network or a different network. Typically an IP address.

Logical Link Control (LLC) The IEEE 802.2 standard that defines the upper sublayer of the Ethernet Layer 2 specifications (and other LAN standards).

logical topology diagram A map of the devices on a network, representing how the devices communicate with each other. It identifies the devices, ports, and addressing scheme.

loopback A special reserved IPv4 address, 127.0.0.1, or IPv6 address, ::1, that can be used to test TCP/IP applications. Packets sent to 127.0.0.1 (or ::1) by a computer never leave the computer or even require a working NIC. Instead, the packet is processed by IP at the lowest layer and is then sent back up the TCP/IP stack to another application on that same computer.

loopback adapter A device that tests the basic functionality of computer ports. The adapter is specific to the port being tested.

loopback interface A virtual interface that can be used to connect or identify a device using an IP address.

LTE A designation for a 4G technology that meets the 4G speed standards.

M

MAC address table On a switch, a table that lists all known MAC addresses and the bridge/switch port that should be used to forward frames sent to each MAC address.

Manchester encoding Use of a line code in which each bit of data is signified by at least one voltage level transition.

maximum transmission unit (MTU) The largest IP packet size allowed to be sent out a particular interface. Ethernet interfaces default to an MTU of 1500 because the data field of a standard Ethernet frame should be limited to 1500 bytes, and the IP packet sits inside the Ethernet frame's data field. The Gigabit Ethernet standard supports jumbo frames, which can be as large as 9216 bytes, including tagging.

Media Access Control (MAC) The lower of the two sublayers of the IEEE standard for Ethernet. It is also the name of that sublayer (as defined by the IEEE 802.3 subcommittee).

media independent A term that describes the networking layers whose processes are not affected by the media being used. In Ethernet, these are all the layers from the LLC sublayer of the data link layer upward.

medium to large network A network used by a corporation or school that has many locations with hundreds or thousands of interconnected computers.

metropolitan-area network (MAN) A network that spans a large campus or a city.

modem A device that converts signals produced by one type of device to a form compatible with another device, often used to connect a home or small office to the internet.

multicast A message sent to selected hosts that are part of a group. A single packet is copied by the network and sent to a specific subset of network addresses. These addresses are specified

in the destination address field. *Compare with* broadcast and unicast.

multimeter A device that measures AC/DC voltage, electric current, and other electrical characteristics and that can be used to test the integrity of circuits and the quality of electricity in computer components.

multimode fiber (MMF) Optical fiber that consists of a larger core and uses LED emitters to send light pulses.

multiplexing A process in which multiple digital data streams are combined into one signal.

N

Neighbor Advertisement message An ICMPv6 message sent by a device in response to an ICMPv6 Neighbor Solicitation message; it contains the IPv6 address and the corresponding MAC address.

Neighbor Discovery (ND) A protocol that provides address resolution, router discovery, and redirection services for IPv6 using ICMPv6.

Neighbor Solicitation message An ICMPv6 message sent by a device when it knows the IPv6 address but needs the corresponding MAC address.

NetBIOS (NetBT) A system through which older computer applications can communicate over large TCP/IP networks.

network address A dotted decimal number defined by IPv4 to represent a network or subnet. It represents the network in which hosts reside. Also called a network number or network ID.

Network Address Translation (NAT) A technique used to translate IP addresses to different addresses that is commonly used to translate RFC 1918 addresses that are not routed on the internet to public domain addresses that can be routed on the internet.

Network Address Translation 64 (NAT64)

A technique that allows IPv6-enabled devices to communicate with IPv4-enabled devices using a translation technique similar to NAT for IPv4. An IPv6 packet is translated to an IPv4 packet and vice versa.

network architecture A collection of technologies that support the infrastructure and the programmed services and rules, or protocols, that move data across a network.

network attached storage (NAS) Servers that are connected to a network to provide file-level data storage to clients using a centralized storage location.

network infrastructure The architecture defining the connections within a network; refers to the physical hardware and connections used to transmit data.

network interface card (NIC) Computer hardware, typically used for LANs, that allows a computer to connect to some networking cable. The NIC can then send and receive data over the cable at the direction of the computer.

next hop The next gateway to which a Layer 3 packet is delivered in order to reach the destination.

nibble boundary The point between nibbles, which are each 4 bits or 1 hexadecimal digit. By borrowing bits from the interface ID, the best practice is to subnet on a nibble boundary.

noise Interference, such as EMI or RFI, that causes unclear power and may cause errors in a computer system.

nonreturn to zero (NRZ) A line code in which 1s are represented by one significant condition and 0s are represented by another.

nonvolatile memory Memory whose contents are not erased when the computer is powered off.

Non-Volatile Memory Express (NVMe) A specification that was developed specifically to allow computers to take greater advantage of the features of SSDs by providing a standard interface between SSDs, the PCIe bus, and operating systems.

nonvolatile RAM (NVRAM) RAM that does not lose its contents when the device is powered off.

nslookup A service or a program used to look up information in Domain Name System (DNS).

O

octet A group of 8 binary bits. It is similar to, but not the same as, a byte. One application in computer networking is to use octets to divide IPv4 addresses into four components.

octet boundary The part of an IPv4 address that falls between octets.

organizationally unique identifier (OUI)

The first half of a MAC address. Manufacturers must ensure that the value of the OUI has been registered with the IEEE. This value identifies the manufacturer of any Ethernet NIC or interface.

output device A hardware device that takes the data processed from input and passes on the information for use.

overhead Resources used to manage or operate a network. Overhead consumes bandwidth and reduces the amount of application data that can be transported across the network.

P

packet switched A network architecture that routes packets along the path perceived as the most efficient and allows a communications channel to be shared by multiple connections.

parallel port A port that has a 25-pin receptacle used to connect various peripheral devices.

peer-to-peer (P2P) A type of networking in which each device serves as both a client and a server portion of an application. P2P also describes a small local network where a host can play the role of a client and/or a server.

peer-to-peer file sharing A system that allows people to share files with each other without having to store and download them from a central server. The user joins a P2P network by simply installing the P2P software. P2P file sharing has not been embraced by everyone. Many people are concerned about violating the laws of copyrighted materials.

personal-area network (PAN) A network that connects devices, such as mice, keyboards, printers, smartphones, and tablets, within the range of an individual person.

physical address An address used for NIC-to-NIC communications on the same Ethernet network.

physical media The cabling and connectors used to interconnect network devices.

physical port A connector or an outlet on a networking device where the media are connected to an end device or another networking device.

physical topology The arrangement of the nodes in a network and the physical connections between them. It provides a representation of how the media are used to connect the devices.

physical topology diagram A diagram that identifies the physical locations of intermediary devices and cable installation.

ping A troubleshooting tool used to verify network connectivity by sending a packet to a specific IP address and waiting for the reply.

ping sweep The process of systematically pinging all network addresses in a given range or subnet. This is similar to going through a section of a telephone book and calling each number to see who answers.

POP3 See Post Office Protocol 3 (POP3).

port (1) With Ethernet hub and switch hardware, another name for an interface, which is a physical connector in the switch into which a cable can be connected. (2) With TCP and UDP, a software function that uniquely identifies a software process on a computer that uses TCP or UDP. (3) With PCs, a physical connector on a PC, such as a parallel port or a USB port.

port number A TCP or UDP field used to identify the source or destination application.

port scan A method of determining what TCP or UDP ports are open or listening on a remote device.

Post Office Protocol (POP) A protocol that allows a computer to retrieve email from a server.

Post Office Protocol 3 (POP3) A protocol used by email clients to retrieve messages from an email server.

power over Ethernet (PoE) The powering of network devices over Ethernet cable. PoE is defined by two different standards: IEEE 802.3af and Cisco.

powerline networking An emerging trend for home networking that uses existing electrical wiring to connect devices.

power-on self-test (POST) The hardware check that the basic input/output system (BIOS) performs on the main components of a computer at boot.

preferred format The IPv6 address format $x:x:x:x:x:x:x$, with each x consisting of four hexadecimal values.

prefix length In IP subnetting, the portion of a set of IP addresses whose values must be identical for the addresses to be in the same subnet.

private address As defined in RFC 1918, an IP address that does not have to be globally unique because the address exists inside packets only when the packets are inside a single private IP internetwork. Private IP addresses are popularly used in most companies today, with NAT translating the private IP addresses into globally unique IP addresses.

private cloud A repository of cloud-based applications and services intended for a specific organization or entity, such as the government.

privileged executive (EXEC) mode An IOS administrative level mode that supports access to configuration and management commands.

protocol A written specification that defines what tasks a service or device should perform. Each protocol defines messages, often in the form of headers, plus the rules and processes by which the messages are used to achieve some stated purpose.

protocol analyzer A network monitoring device that gathers information regarding the status of a network and devices attached to it. Also known as a network analyzer or packet sniffer.

protocol data unit (PDU) A generic term that refers to the data, headers, and trailers about which a particular networking layer is concerned.

protocol suite A delineation of networking protocols and standards into different categories, called layers, along with definitions of which sets of standards and protocols need to be implemented to create products that can be used to create a working network.

proxy server A computer system that has the authority to act as another computer to function as a relay between client and server.

public address An IP address that has been registered with IANA or one of its member agencies to guarantee that the address is globally unique. Globally unique public IP addresses can be used for packets sent through the internet.

public cloud Cloud-based applications and services made available to the general population.

Q–R

quality of service (QoS) A control mechanism that can provide different priorities to different users or data flows or guarantee a certain level of

performance to a data flow in accordance with requests from the application program.

queuing In routing and switching, a backlog of packets or frames waiting to be forwarded out an interface.

radio frequency interference (RFI) Noise that interferes with information being transmitted across unshielded copper cabling.

random-access memory (RAM) Also known as read/write memory, memory that can have new data written to it and that can have stored data read from it. RAM is the main working area, or temporary storage, used by a CPU for most processing and operations. A drawback of RAM is that it requires electrical power to maintain data storage. If the computer is turned off or loses power, all data stored in RAM is lost unless the data was previously saved to disk. Memory boards with RAM chips plug into the motherboard.

read-only memory (ROM) Nonvolatile memory located on the motherboard and other circuit boards that contain instructions that can be directly accessed by a CPU.

real-time traffic Data traffic that carries signal output as it happens or as fast as possible. Real-time traffic is sensitive to latency and jitter.

reconnaissance attack An attack that is used to discover and map systems, services, or vulnerabilities.

redundancy In internetworking, a network architecture designed to eliminate network downtime caused by a single point of failure. Redundancy includes the replication of devices, services, or connections that support operations even in the occurrence of a failure.

reference model A conceptual framework to help understand and implement the relationships between various protocols.

Regional Internet Registry (RIR) One of the five organizations responsible for allocating IP addresses within particular geographic regions.

remote network An IP network that can be reached by forwarding a packet to a router.

repeater A device that regenerates weak signals to extend the distance a signal can travel.

Request for Comments (RFC) A series of documents and memoranda encompassing new protocols, research, innovations, and methodologies applicable to internet technologies. RFCs are developed by the IETF for the TCP/IP protocol suite.

response timeout The amount of time a service waits on a response before taking some action. A protocol defines how long a service waits and what action is taken if a response timeout occurs.

ring topology A physical network topology in which each system is connected to its respective neighbors, forming a ring. The ring does not need to be terminated, unlike in the bus topology. Legacy Fiber Distributed Data Interface (FDDI) and Token Ring networks used ring topologies.

RJ-11 connector A physical network interface used to connect a computer to a standard telephone line.

RJ-45 connector A standardized physical network interface for connecting telecommunications or data equipment. The most common twisted-pair connector is an 8-position, 8-contact (8P8C) modular plug and jack.

ROM See read-only memory (ROM).

round-trip time (RTT) The time required for a networking PDU to be sent and received and a response PDU to be sent and received. In other words, the time between when a device sends data and when the same device receives a response.

router A network layer device that forwards data packets between networks. Routers use IP addresses to forward traffic to other networks.

Router Advertisement (RA) message An ICMPv6 message sent by a router to provide addressing information to hosts using SLAAC.

Router Solicitation message An ICMPv6 message sent by devices to request an ICMPv6 Router Advertisement message.

routing The process by which a router receives an incoming frame, discards the data link header and trailer, makes a forwarding decision based on the destination IP address, adds a new data link layer header and trailer based on the outgoing interface, and forwards the new frame out the outgoing interface.

runt frame Any frame less than 64 bytes in length. A runt frame is automatically discarded by a receiving station. Also called a collision fragment.

S

satellite connection Internet access provided using satellites and satellite dishes to serve areas that would otherwise have no internet connectivity at all. A satellite dish requires a clear line of sight to the satellite.

scalable network A network that can expand quickly to support new users and applications

without impacting the performance of the service being delivered to existing users.

Secure Shell (SSH) A protocol that provides a secure remote connection to a host through a TCP application.

segment (1) A collision domain that is a section of a LAN that is bound by bridges, routers, or switches. (2) In a LAN using a bus topology, a continuous electrical circuit that may be connected to other such segments with repeaters. (3) With TCP, to accept a large piece of data from an application and break it into smaller pieces. (4) With TCP, one of the smaller pieces of data that results from the segmentation process.

segmentation In TCP, the process of breaking a large chunk of data into small enough pieces to fit within a TCP segment without breaking any rules about the maximum amount of data allowed in a segment.

selective acknowledgment (SACK) An optional TCP feature that makes it possible for the destination to acknowledge bytes in discontinuous segments. With SACK, the source host only needs to retransmit the specific unacknowledged data rather than retransmitting all data since the last acknowledged data.

sequence number Information placed in a data header to ensure correct sequencing of the arriving data.

server Computer hardware or software that is used by multiple concurrent users or provides services to many users. For example, a web server consists of web server software running on some computer.

Server Message Block (SMB) An application level network protocol mainly applied to shared

access to files, printers, serial ports, and miscellaneous communications between nodes on a network.

session A related set of communications transactions between two or more network devices.

shell The portion of the operating system that interfaces with applications and the user.

shielded twisted-pair (STP) cable A type of network cabling that includes twisted-pair wires, with shielding around each pair of wires, as well as another shield around all wires in the cable.

Simple Mail Transfer Protocol (SMTP) An application layer protocol that provides electronic mail services to transfer mail from client to server and between servers.

single-mode fiber (SMF) Optical fiber that consists of a very small core and uses laser technology to send a single ray of light in data transmission.

SLAAC *See* stateless address autoconfiguration (SLAAC).

slash notation A method of expressing a network prefix that uses a forward slash (/) followed by the network prefix—for example, 192.168.254.0/24, where the /24 represents the 24-bit network prefix in slash format.

small office/home office (SOHO) network

A network in which computers can connect to a corporate network or access centralized, shared resources.

smart home technology Technology that is integrated into everyday appliances to allow them to interconnect with other devices, making them more “smart,” or automated.

SMB/CIFS Protocols that allow for sharing of files, printers, and other resources between nodes on a network. CIFS is a dialect of SMB.

SNMP (Simple Network Management Protocol) A protocol that enables network administrators to monitor network operations from centralized monitoring stations.

socket A logical communications endpoint within a network device. A socket is typically represented by a Layer 3 address and a Layer 4 port number.

socket pair The combination of the source IP address and source port number or the destination IP address and destination port number.

socket type A connector on a motherboard that houses a CPU and forms the electrical interface and contact with the CPU.

Solicitation (RS) message *See* Router Solicitation message.

solicited node multicast address The IPv6 multicast address associated with an IPv6 unicast address that is mapped to a special Ethernet multicast address.

source The originator of a message.

source IP address The IP address of the originating host that is placed into an IP packet header.

source port number The port number associated with the originating application on a local device.

spoofing A process in which a person or program masquerades as another to gain access to data and a network.

SSH File Transfer Protocol (SFTP) An extension to Secure Shell (SSH) protocol that can be used to establish a secure file transfer session.

standard An agreed-upon set of rules.

star topology A physical topology in which a central device or central site interconnects other devices or sites.

stateful A term that refers to tracking of actual conversations and their state of the communication session for a protocol, such as TCP.

stateful DHCPv6 Similar to DHCP for IPv4, a type of DHCP that provides IPv6 address, prefix length, and other information, such as the DNS server and domain name. It does not provide a default gateway address.

stateful packet inspection (SPI) A process in which incoming packets must be legitimate responses to requests from internal hosts, and unsolicited packets are blocked unless specifically permitted. SPI can also include the capability to recognize and filter out specific types of attacks, such as denial-of-service (DoS) attacks.

stateless address autoconfiguration (SLAAC) A plug-and-play IPv6 feature that enables devices to connect themselves to the network without any configuration and without any servers (like DHCP servers).

stateless DHCPv6 A type of DHCP that provides information other than the IPv6 address and prefix length, such as DNS server and domain name. It does not provide a default gateway address.

static route A remote network in a routing table that has been manually entered into the table by a network administrator.

store-and-forward switching A frame forwarding method that receives an entire frame and computes the CRC. CRC uses a mathematical formula, based on the number of bits (1s) in the frame, to determine whether the received frame has an error. If the CRC is valid, the switch looks up the destination address, which determines the outgoing interface. Then the frame is forwarded out the correct port.

subnet A group of IP addresses that have the same value in the first part of the IP addresses, for the purpose of allowing routing to identify the group by that initial part of the addresses. IP addresses in the same subnet typically sit on the same network medium and are not separated from each other by any routers. IP addresses on different subnets are typically separated from one another by at least one router. Subnet is short for subnetwork.

subnet ID Part of the IPv6 global unicast address used by an organization to identify subnets within its site. The larger the subnet ID, the more subnets available.

subnet mask A dotted decimal number that helps identify the structure of IPv4 addresses. The mask represents the network and subnet parts of related IPv4 addresses with binary 1s and the host part of related IPv4 addresses with binary 0s.

subnetwork *See* subnet.

switch Hardware that microsegments a LAN and that connects multiple devices on a network by receiving data and using filtering and forwarding to send the data to the intended destination device.

switch fabric The integrated circuits and the accompanying machine programming in a switch

that allow the data paths through the switch to be controlled.

switched virtual interface (SVI) A virtual interface for which there is no associated physical hardware on the device. An SVI is created in software. The virtual interfaces are used as a means to remotely manage a switch over a network. They are also used for routing between VLANs.

syslog A protocol that allows networking devices to send their system messages across the network to syslog servers.

system speaker A case speaker that a motherboard uses to indicate the computer's status during POST.

T

TCP/IP model A conceptual framework that consists of layers that perform functions necessary to prepare data for transmission over a network.

Telecommunications Industry Association/Electronic Industries Association (TIA/EIA) An organization that develops standards that relate to telecommunications technologies. Together, the TIA and the Electronic Industries Alliance (EIA) have formalized standards, such as EIA/TIA-232, for the electrical characteristics of data transmission.

Telnet A non-secure network service that supports CLI access to a remote host. It also can be used to verify the application layer software between source and destination stations.

terminal emulation A network application in which a computer runs software that makes it appear to a remote host as a directly attached terminal.

test-net address The IPv4 address block 192.0.2.0 to 192.0.2.255 (192.0.2.0/24) that is set aside for teaching and learning purposes. These addresses can be used in documentation and network examples.

three-way handshake The process used by TCP to establish a session.

throughput The actual data transfer rate between two computers at some point in time. Throughput is impacted by the slowest-speed link used to send data between the two computers, as well as myriad variables that might change during the course of a day.

Time-to-Live (TTL) A field in the IP header that prevents a packet from indefinitely looping around an IP internetwork. A router decrements the TTL field each time it forwards a packet, and if it decrements the TTL to 0, the router discards the packet, which prevents it from looping forever.

topology The arrangement networking components or nodes. Examples include star, extended star, ring, and mesh.

traceroute (tracert) A command on many computer operating systems that discovers the IP addresses and possibly hostnames of the routers used by the network when sending a packet from one computer to another.

traffic prioritization A quality of service (QoS) process in which frames are forwarded in priority order based on their marking.

Transmission Control Protocol (TCP) A Layer 4 protocol of the TCP/IP model that lets applications guarantee delivery of data across a network.

Trivial File Transfer Protocol (TFTP) A protocol similar to FTP that enables the transfer of

files from one computer to another over a network. TFTP is supported by UDP, whereas FTP is supported by TCP.

Trojan horse A type of malware named after the wooden horse the Greeks used to infiltrate Troy. It is a harmful piece of software that looks legitimate. Users are typically tricked into loading and executing Trojans on their systems.

troubleshooting The systematic process used to locate the cause of a fault in a computer system and correct the relevant hardware and software issues.

troubleshooting process steps The systematic approach to locating the cause of a fault in a computer system and correcting the relevant hardware and software issues by identifying the problem, establishing a theory of probable cause, testing the theory to determine the cause, establishing a plan of action, verifying full system functionality, and documenting the issue.

tunneling The process of encapsulating an IP packet inside another IP packet.

twisted-pair A type of cable that consists of a pair of insulated wires wrapped together in a regular spiral pattern to control the effects of electrical noise.

U

unicast A type of message sent to a single network destination. *Compare with* broadcast and multicast.

unique local address An IPv6 address that is similar to an RFC 1918 private address for IPv4. Unique local addresses are used for local

addressing within a site or between a limited number of sites. These addresses should not be routable in the global IPv6 internet. Unique local addresses are in the range FC00::/7 to FDFF::/7.

unknown unicast An Ethernet frame that does not have an entry in the switch's MAC address table for the destination MAC address.

unshielded twisted-pair (UTP) cable A general type of cable, with the cable holding twisted pairs of copper wires and the cable itself having little shielding.

unspecified address An IPv6 all-0s address represented in the compressed format as ::/128 or just ::. It cannot be assigned to an interface and is only to be used as a source address in an IPv6 packet. An unspecified address is used as a source address when a device does not yet have a permanent IPv6 address or when the source of the packet is irrelevant to the destination.

User Datagram Protocol (UDP) A connectionless transport layer protocol in the TCP/IP protocol stack. UDP is a simple protocol that exchanges datagrams without acknowledgments or guaranteed delivery, requiring that error processing and retransmission be handled at a higher layer.

user executive (EXEC) mode The limited CLI mode where the commands available to the user are a subset of those available at the privileged level. In general, use the user EXEC commands to temporarily change terminal settings, perform basic tests, and list system information.

user password A password that allows access to the BIOS based on a defined level, such as full access, limited access, view only access, or no access.

V

variable-length subnet masking (VLSM) A process that makes it possible to specify a different subnet mask for the same network number on different subnets. VLSM can help optimize available address space.

virtual circuit A logical connection created within a network between two network devices.

virtual classroom A logical classroom environment created as a collaboration space without physical restraints.

virtual local-area network (VLAN) A network of end devices that behave as if they are connected to the same network segment, even though they might be physically located on different segments of a LAN. VLANs are configured through software on the switch and router (IOS on Cisco routers and switches).

virtual terminal (vty) A text-based logical interface on an IOS device. It is accessed using Telnet or SSH to perform administrative tasks. A vty line is also called a virtual type terminal.

virtualization The creation of a virtual version of something, such as a hardware platform, an operating system (OS), a storage device, or a network resource. As an example, a virtual machine consists of a set of files and programs running on an actual physical system.

virus A type of malware that propagates by inserting a copy of itself into, and becoming part of, another program. It spreads from one computer to another, leaving infections as it travels.

voice over IP (VoIP) Voice data encapsulated in an IP packet that allows it to traverse already implemented IP networks without needing its own network infrastructure.

volatile memory Memory whose contents are erased every time the computer is powered off.

W

well-known multicast address An assigned multicast address that is a reserved multicast address for a predefined group of devices.

well-known multicast IPv6 address A predefined IPv6 multicast address used to reach a group of devices running a common protocol or service.

wide-area network (WAN) A network infrastructure that provides access to other networks over a wide geographic area, which is typically owned and managed by a telecommunications service provider.

Wi-Fi (IEEE 802.11) A wireless LAN (WLAN) technology that uses a contention-based protocol known as CSMA/CA. The wireless NIC must first listen before transmitting to determine if the radio channel is clear. If another wireless device is transmitting, the NIC must wait until the channel is clear. Wi-Fi, which is a trademark of the Wi-Fi Alliance, is used with certified WLAN devices based on the IEEE 802.11 standards.

Wi-Fi analyzer A mobile tool for auditing and troubleshooting wireless networks.

WiMAX (IEEE 802.16) Worldwide Interoperability for Microwave Access, a wireless standard that uses a point-to-multipoint topology to provide wireless broadband access.

window size In the TCP header that is set in a sent segment, the maximum amount of unacknowledged data the host is willing to receive

before the other sending host must wait for an acknowledgment. Used for flow control.

wireless access point (WAP) A network device that provides connectivity of wireless clients to connect to a data network. A wireless AP uses radio waves to communicate with the wireless NICs in the devices and other wireless access points.

wireless internet service provider (WISP) An ISP that connects subscribers to a designated access point or hotspot using wireless technologies similar to those found in home wireless local-area networks (WLANs).

wireless LAN (WLAN) A network that is similar to a LAN but that wirelessly connects users and devices in a small geographic area instead of using a wired connection. A WLAN uses radio waves to transmit data between wireless devices.

wireless mesh network (WMN) A technology that uses multiple access points to extend a WLAN.

wireless network interface card (NIC) A device that connects a computer to a network using radio frequencies.

wireless router A device that connects multiple wireless devices to a network and may include a switch to connect wired hosts.

worm Malware that is similar to a virus in that it replicates functional copies of itself and can cause the same type of damage. In contrast to viruses, which require the spreading of an infected host file, worms are standalone software and do not require a host program or human help to propagate. A worm does not need to attach to a program to infect a host but can enter a computer through a vulnerability in the system.

X-Y-Z

Zigbee (IEEE 802.15.4) A specification used for low-data-rate, low-power communications. It is intended for applications that require short

ranges, low data rates, and long battery life. Zigbee is typically used for industrial and Internet of Things (IoT) environments such as wireless light switches and medical device data collection.

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